Astrophysics

Sky, stars and the universe from six principles

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Preface

Six principles

Astrophysics can look like a long list of formulas. Yet almost every result about stars, gas and the universe follows from six general principles, applied with ordinary classical and quantum physics. A formula that has been forgotten can always be rebuilt from them.

P1 **Conservation.** What is in a region changes only by what crosses its boundary and what acts on it.

P2 **Gravity of a sphere.** Inside a spherical body, only the mass inside a radius pulls.

P3 **Counting states.** Pressure, energy density and populations follow from counting quantum states and how full they are.

P4 **Balance of rates.** In a steady state gains match losses. In equilibrium each process matches its reverse.

P5 **Random walk.** Distance grows as the square root of the number of steps, so scattering things flow down their gradients.

P6 **Compare two scales.** A ratio of two times or two lengths decides which physics matters.

Each of them is stated and derived in Chapter 9.

Orders of magnitude

Quantity	Value
Solar mass, $M_\odot$	1.99×10^{30} kg
Solar radius, $R_\odot$	6.957×10^{5} km
Solar luminosity, $L_\odot$	3.83×10^{26} W
Astronomical Unit, 1 AU	1.496×10^{8} km
Light-year, 1 ly	9.461×10^{12} km
Parsec, 1 pc	3.086×10^{13} km = 3.26156 ly

Table 1: Units of mass, length, distance and power.

- **Solar System.** 8 planets, 9 dwarf planets, 468 known moons, about 1.5 million catalogued asteroids, and a vast reservoir of comets in the Kuiper belt (30–50 AU) and the Oort cloud (2000–200,000 AU, about 3 ly at its outer edge). The total mass is about $1.0014\,M_\odot$. The nearest star, Proxima Centauri, lies 4.24 ly away.
- **The Milky Way.** 10^{11} to 4×10^{11} stars, a total mass of about $1.15 \times 10^{12}\,M_\odot$, a central black hole of about $4.3 \times 10^{6}\,M_\odot$, and a radius of about 27 kpc. The Sun lies about 8 kpc from the centre, the frame used for the (U, V, W) velocities of Chapter 3.

- **The observable universe.** About 2×10^{12} galaxies within a diameter of about 93 Gly. The mean density of all its matter and energy is about $8.5 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$, the weight of about five protons per cubic metre. Ordinary matter makes up 4.9% of it, about one proton in every four cubic metres. Dark matter makes up 26.8% and dark energy 68.3%.

The messengers of astrophysics

Information reaches us from the cosmos through four channels. They are electromagnetic waves, gravitational waves, neutrinos and cosmic rays. They are produced by different processes, travel and are absorbed differently, and so carry different information about the same source. Seeing one event through more than one channel is multi-messenger astronomy. The neutron-star merger GW170817 was detected both in gravitational waves and across the electromagnetic spectrum. Light still carries almost everything known about stars and galaxies, from its passage through the atmosphere and the interstellar medium to the physics of the bodies that emit it. Neutrinos carry away most of the energy of a collapsing stellar core (Chapter 13).

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Chapter 1

Time Systems and the Rotating Earth

An astronomer who points a telescope, predicts where a planet will be, or times the pulses of a pulsar needs two numbers that turn out to be one. The Earth turns east at a steady rate, so the whole sky seems to turn west at the same rate. How far the sky has turned is a time, and the time on a clock tied to the sky is an angle on the sky. One turn of 360° takes 24 hours of this clock, so

$$1^{\mathrm{h}} \equiv 15^\circ, \qquad 1^{\mathrm{m}} \equiv 15' = 0.25^\circ, \qquad 1^{\mathrm{s}} \equiv 15'' = 0.004167^\circ. \tag{1.1}$$

Our observatory is the Earth, and it does three things at once. It spins once a day, it goes round the Sun once a year, and its axis slowly swings round a cone once in about 26 000 years. Each motion gives a clock, and each clock differs from the others by an amount that follows from geometry. The grid of coordinates on the sky (Chapter 2) is built on these clocks.

1.1 The Celestial Sphere

We can tell the direction of a star at once, but not its distance. So every star can be drawn as a point on a large sphere centred on us, of any radius we like. This is the **celestial sphere**. A planetarium dome is a good picture of it. The stars painted on the dome look right whatever its size, because only the directions matter.

Two features of the sphere are fixed by the Earth's spin alone, wherever the observer stands. The **celestial poles** are the two points where the Earth's spin axis, extended, meets the sphere. The **celestial equator** is the great circle where the plane of the Earth's equator meets the sphere.

The sky turns rigidly about the poles once per sidereal day (Section 1.2). The observer's **celestial meridian** is the great circle through both poles and the **zenith**, the point straight overhead. As the sky turns, every point of the celestial equator crosses the meridian once per turn. So naming the point of the equator that sits on the meridian now is the same as reading a clock. This one fact makes a sidereal clock possible.

A direction on the sphere is named by two angles, built like latitude and longitude on the Earth. The **declination** δ is the angle north (+) or south (−) of the celestial equator. The **right ascension** α is the angle measured east along the celestial equator from a chosen zero point. Stars are so far away that both angles hardly change in a lifetime, so a catalogue of (α, δ) stays useful for years.

The zero point is the **vernal equinox** $\hat{\gamma}$. It is the direction, among the stars, in which the Sun crosses the celestial equator from south to north each March. Section 1.6 shows why the Sun crosses the equator at all. Section 1.8 shows why $\hat{\gamma}$, and with it every (α, δ), is not quite fixed. Chapter 2 builds the full (α, δ) system and its link to the observer's horizon. A sidereal clock needs only what is given here.

1.2 Two Clocks: the Solar Day and the Sidereal Day

We know that the Earth turns once a day. That is enough for daily life, where noon comes round every 24 hours.

But "once" needs a reference. One turn relative to the Sun and one turn relative to the stars are not the same, because the Sun itself moves against the stars as the Earth goes round its orbit.

Look at the hands of a clock. The minute hand starts on top of the hour hand at noon. After 60 minutes it is back at the top, but the hour hand has moved on to the 1. The minute hand needs a little more than five extra minutes to catch it. It laps the hour hand every 65.45 minutes, not every 60.

The observer's meridian is the minute hand. The direction to the Sun is the hour hand, which creeps east by about a degree a day as the Earth moves along its orbit. One turn relative to the stars is the **sidereal day**. One turn relative to the Sun, from one noon to the next, is the **solar day**. The solar day is the time the meridian takes to lap the Sun.

The clock in numbers. The minute hand turns at $1/60$ of a turn per minute and the hour hand at $1/720$. The gap between them grows at $1/60 - 1/720 = 11/720$ of a turn per minute, so it closes a full turn in $720/11 = 65.45$ minutes.

The picture is two runners on a circular track, both going the same way. The faster one laps the slower one, and the lapping time is longer than her own lap time.

The same lapping gives the synodic month. The Moon goes round the Earth in 27.32 days against the stars but takes 29.53 days from one full Moon to the next, because the Sun has moved on meanwhile. It also gives the time between two closest approaches of two planets, and in sound it is the beat between two tones close in pitch. Here the ratio of the two periods, a year to a day, is about 366. It is far from one, so the solar and sidereal days differ by only one part in 366 (P6). Next we find the two days and the exact size of their difference.

The two days. The **solar day** is the time between two transits of the Sun across the meridian, from one local noon to the next. Its average over the year, the **mean solar day**, is defined to be exactly

$$24^{\mathrm{h}} = 1440^{\mathrm{m}} = 86\,400\,\mathrm{s}.$$

The **sidereal day** is the time between two transits of the vernal equinox $\hat{\gamma}$. In mean solar time it lasts

$$T_{\mathrm{sid}} \simeq 23^{\mathrm{h}}56^{\mathrm{m}}04.0905^{\mathrm{s}} = 86\,164.0905\,\mathrm{s}, \tag{1.2}$$

which we derive below.

A first estimate. The Earth goes round its orbit in the same sense as it spins, both anticlockwise seen from above the north pole (Figure 1.1). In one day it moves along its orbit by

$$\frac{360^\circ}{365.2422\ \mathrm{days}} = 0.985647^\circ\ \text{per day}.$$

After one turn relative to the stars, the meridian points the same way in space as the day before (dashed arrow). The Sun now lies 0.9856° further east, so the Earth must turn this extra angle before the meridian reaches it again (solid arrow). At 360° per 1440^{m} of solar time, that is 4^{m} per degree, and the extra angle takes

$$0.985647^\circ \times 4^{\mathrm{m}}/^\circ = 3.94259^{\mathrm{m}} = 3^{\mathrm{m}}56.555^{\mathrm{s}}.$$

This estimate is slightly off, because the extra angle is turned at the Earth's spin rate and not at the Sun's rate of 360° per solar day. The lapping calculation gets it right.

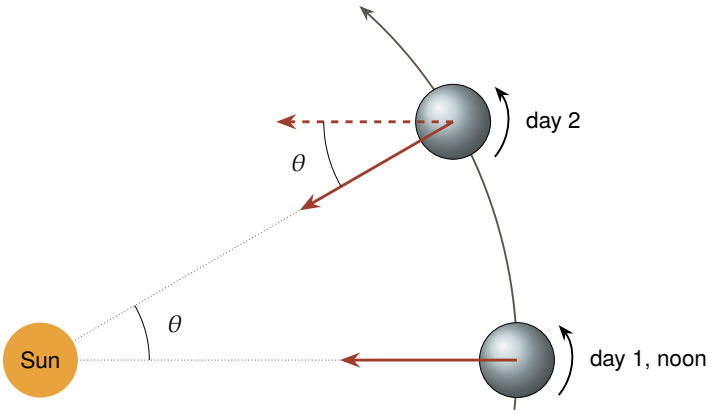


Figure 1.1: The Earth on two successive days, seen from above the north pole. Day 1 is at local noon. The red arrows show the direction of the observer's meridian. On day 2 the dashed arrow points the same way in space as on day 1, and the solid arrow points at the Sun. The angle θ moved along the orbit in one day is drawn much larger than its true value of about 1°.

Lapping, as a rotation. Put the z-axis along the Earth's spin axis, and let

$$R_z(\varphi) = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.3}$$

turn a vector by the angle φ anticlockwise about it, seen from the north. Let $\omega_\oplus = 2\pi/T_{\text{sid}}$ be the Earth's spin rate against the equinox, and let $n = 2\pi/T_{\text{yr}}$ be the mean rate at which the Sun's direction turns east, where T_{yr} is the year measured against the equinox. This n is the mean motion of the Earth's orbit, the same n as in Eq. (6.49). The meridian direction turns as $R_z(\omega_\oplus t)$ and the mean Sun's direction as $R_z(nt)$. To see the meridian from the Sun, we turn everything back by nt,

$$\begin{aligned} R_z(-nt)\,R_z(\omega_\oplus t) &\overset{(1)}{=} \begin{pmatrix} \cos nt & \sin nt & 0 \\ -\sin nt & \cos nt & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\omega_\oplus t & -\sin\omega_\oplus t & 0 \\ \sin\omega_\oplus t & \cos\omega_\oplus t & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &\overset{(2)}{=} \begin{pmatrix} \cos(\omega_\oplus - n)t & -\sin(\omega_\oplus - n)t & 0 \\ \sin(\omega_\oplus - n)t & \cos(\omega_\oplus - n)t & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= R_z\big((\omega_\oplus - n)t\big). \end{aligned} \tag{1.4}$$

(1) Eq. (1.3) with $\varphi = -nt$ and $\varphi = \omega_\oplus t$.

(2) Matrix product, then $\cos x \cos y + \sin x \sin y = \cos(x - y)$ and $\sin x \cos y - \cos x \sin y = \sin(x - y)$.

Seen from the Sun, the meridian turns at $\omega_\oplus - n$, and so

$$\omega_\oplus - n = \frac{2\pi}{T_{\text{sol}}}, \tag{1.5}$$

where T_{sol} is the mean solar day. This is the clock-hand rule. The rates subtract because the two motions go the same way.

Days per year. Writing each rate as 2π over its period,

$$\frac{1}{T_{\text{sol}}} = \frac{1}{T_{\text{sid}}} - \frac{1}{T_{\text{yr}}}. \tag{1.6}$$

We multiply by T_{yr}. Then T_{yr}/T_{sol} is the number of solar days in a year, 365.2422, and T_{yr}/T_{sid} the number of sidereal days,

Result 1.1

$$\frac{T_{yr}}{T_{sid}} = \frac{T_{yr}}{T_{sol}} + 1 = 365.2422 + 1 = 366.2422 \textit{ sidereal days per year}. \tag{1.7}$$

In one year the Earth turns once more against the stars than against the Sun. The orbit itself supplies the extra turn.

The length of the sidereal day. Dividing the two counts,

$$\begin{aligned} T_{sid} &\overset{(1)}{=} T_{sol}\,\frac{365.2422}{366.2422} \\ &= 86\,400\,\mathrm{s} \times 0.99726957 = 86\,164.0905\,\mathrm{s}, \end{aligned} \tag{1.8}$$

(1) Eq. (1.7), the same year counted in the two kinds of day.

which is Eq. (1.2). The difference of the two days is

$$\boxed{T_{sol} - T_{sid} = \frac{T_{sol}}{366.2422} = 235.909\,\mathrm{s} = 3^{m}55.909^{s}.}$$

It depends only on how many days there are in a year, not on the size or shape of the orbit. The first estimate, $3^{m}56.555^{s}$, is larger by the factor 366.2422/365.2422 = 1.0027379. That number is not wrong. It is the amount by which a sidereal clock gains on a solar clock in one solar day, counted in sidereal seconds, and it returns in Eq. (1.45).

The year in Eq. (1.6) is the one measured against the equinox, so T_{sid} is the turn against the equinox. The equinox drifts slowly west (Section 1.8). Measured against the distant stars instead, with the sidereal year of 365.256363 days in place of 365.2422, the turn takes about 0.009 s longer. The difference matters only for the most precise work.

Why the night sky changes with the seasons. A sidereal clock gains about $3^{m}56^{s}$ on a solar clock each day. That adds up to about 2^{h} a month and a full 24^{h} in a year. A star that crosses the meridian at midnight on 1 October crosses it at 22:00 on 1 November, and at noon on 1 April, when it is lost in daylight. This is why each season has its own constellations.

1.3 Three Years: Tropical, Sidereal and Anomalistic

The lapping rule of Eq. (1.6) turned the two references for a day into two kinds of day.

A year has the same problem. We count it as one trip of the Sun round the sky, but we must say where the trip starts and ends. The equinox, the stars and the point of closest approach to the Sun all give different answers, because two of them move.

Think of a lap of a running track where the finish line is moved a little each lap. If the line is moved back toward the runner, she crosses it before a full lap and her lap time is short. If it is moved forward, she must run past her last lap to reach it, and her lap time is long.

The Sun's yearly trip has one fixed reference, the distant stars, and two moving ones. The equinox moves west, toward the oncoming Sun, by 50.29″ a year (Section 1.8). The perihelion, the point of the orbit closest to the Sun, moves east, ahead of the Sun, by about 11.6″ a year,

pulled by the other planets. So the year to the equinox is shorter than the year to the stars, and the year to perihelion is longer.

In numbers, the Sun covers $360^\circ = 1\,296\,000''$ in one sidereal year of 365.256 days. Shortening the trip by $50.29''$ saves

$$\frac{50.29''}{1\,296\,000''} \times 365.256 \text{ d} = 0.014174 \text{ d} = 20.41 \text{ min}.$$

The lapping rule is at work again. A reference that moves changes the rate at which the moving body reaches it, by adding or subtracting the reference's own rate. The three years are listed in Table 1.1.

Year	Measured from	Length (days)	Minus tropical
Tropical	equinox to equinox, the cycle of the seasons	365.242190	0
Sidereal	a fixed direction among the stars back to itself	365.256363	$+20^{\text{m}}24.5^{\text{s}}$
Anomalistic	perihelion to perihelion	365.259636	$+25^{\text{m}}07.3^{\text{s}}$

Table 1.1: The three astronomical years.

The tropical year. The **tropical year** runs from one passage of the Sun through the equinox $\hat{\gamma}$ to the next. It sets the seasons and the calendar. Let the Sun move at the mean rate 360° per sidereal year against the stars, while the equinox moves toward it at $50.29''$ per year. The Sun reaches the equinox when it has gained a full turn on it,

$$\begin{aligned} \text{tropical year} &\overset{(1)}{=} \frac{360^\circ}{360^\circ/\text{sidereal year} + 50.29''/\text{year}} \\ &\overset{(2)}{=} \text{sidereal year} \times \left(1 - \frac{50.29''}{1\,296\,000''}\right) \\ &= 365.256363 \text{ d} - 0.014174 \text{ d} = 365.242189 \text{ d}. \end{aligned} \tag{1.9}$$

(1) The rates add, since the Sun and the equinox move toward each other (the lapping rule, Eq. (1.5), with the sign of n reversed).

(2) $1/(1+x) \simeq 1 - x$ for $x = 50.29''/1\,296\,000'' = 3.9 \times 10^{-5}$.

This agrees with the measured value in Table 1.1 to 10^{-6} days. The difference from the sidereal year is 20.41 minutes, which is $20^{\text{m}}24.5^{\text{s}}$.

The sidereal year. The **sidereal year** is one full turn of the Earth round the Sun against the distant stars, 365.256363 days. It is the true orbital period of Eq. (6.49).

The anomalistic year. The **anomalistic year** runs from one perihelion passage to the next. The perihelion moves ahead of the Earth at about $11.6''$ per year, so the rates now subtract, and the same steps give

$$\text{anomalistic year} - \text{sidereal year} \simeq \frac{11.6''}{1\,296\,000''} \times 365.256 \text{ d} = 0.00327 \text{ d} = 4.71 \text{ min}.$$

The table gives $365.259636 - 365.256363 = 0.003273$ days, or 4.71 minutes. The two years differ by the time the Earth needs to catch up with its own perihelion.

The Gregorian calendar. The Julian calendar took the year to be exactly 365.25 days, with a leap year every fourth year. That is longer than the tropical year by 0.0078 days, or 11.2 minutes, each year, which adds up to one day in 128 years. By 1582 the March equinox had drifted from 21 March, its date in the year 325, to 11 March. Pope Gregory XIII dropped ten days from the calendar and changed the rule. A year divisible by 4 is a leap year, except a century year not divisible by 400. So 1600 and 2000 were leap years, while 1700, 1800, 1900 and 2100 are not. Of every 400 years, 97 are leap years, and the mean year is

$$365 + \frac{97}{400} = 365.2425 \text{ days}.$$

This exceeds the tropical year by 0.00031 days a year, one day in about 3200 years.

1.4 Longitude and Local Solar Time

The mean solar day of Section 1.2 gives the length of a day. It does not say when noon is.

Noon comes when the Sun crosses the local meridian, and the Sun reaches different meridians at different moments. A town to the east has its noon first.

Two friends talk on the phone, one in Kolkata and one in Mumbai. The friend in Kolkata sees the Sun set about an hour earlier. Their clocks show the same time, because India keeps one time across the country, but the Sun does not.

The Earth turns east relative to the Sun at the rate $\omega_\oplus - n$ of Eq. (1.5), 360° per mean solar day,

$$\omega_\oplus - n = \frac{360^\circ}{24^\mathrm{h}} = 15^\circ \text{ per hour} = 1^\circ \text{ per } 4^\mathrm{m} = 1' \text{ per } 4^\mathrm{s}.$$

So each degree of longitude moves the time of noon by four minutes. Kolkata (88.4°E) and Mumbai (72.9°E) are 15.5° apart, which is 62 minutes between their noons. Their sunsets differ by between about 55 and 70 minutes over the year, because the two cities are also at different latitudes.

Picture the line of noon sweeping west across the globe like the shadow of a slowly turning gate, one degree every four minutes. Time zones are the same rule with the meridian of each zone fixed by law. It gives the time of local noon from the longitude.

Two meridians. Let longitude λ be measured east from the Greenwich meridian. The Sun crosses the meridian at λ_1 earlier than the one at λ_2 by

$$\Delta t = \frac{\lambda_1 - \lambda_2}{15^\circ \text{ per hour}}. \tag{1.10}$$

The observer further east sees the Sun cross first.

Civil time. Daily life cannot run on a clock that differs from town to town. A country instead keeps the **standard time** of one chosen meridian, at longitude λ_ref. An observer at longitude λ_obs is east of it if $\lambda_\mathrm{obs} > \lambda_\mathrm{ref}$, and then has local noon before the standard clock shows 12:00. By Eq. (1.10),

Result 1.2

$$\textit{local mean solar time} = \textit{standard time} + \frac{\lambda_\mathrm{obs} - \lambda_\mathrm{ref}}{15^\circ \textit{ per hour}}, \tag{1.11}$$

for an observer on either side of the reference meridian. This is *mean* solar time, the time of a Sun that crosses each meridian at the same moment every day. The real Sun crosses up to 16 minutes earlier or later (Section 1.7).

Worked example 1.1: Local mean solar time at IISER Thiruvananthapuram

IISER TVM is at longitude 77.1382°E. Indian Standard Time is the mean time of the meridian at 82.5°E, which passes near Prayagraj. Then

$$\frac{\lambda_{\text{obs}} - \lambda_{\text{ref}}}{15^\circ \text{ per hour}} = \frac{77.1382^\circ - 82.5^\circ}{15^\circ \text{ per hour}} = -5.3618^\circ \times 4^{\text{m}}/^\circ = -21.4472^{\text{m}} = -21^{\text{m}}26.83^{\text{s}}.$$

When the clock reads 12:00:00 IST, the local mean solar time at IISER TVM is 11:38:33. Local mean noon comes at 12:21:27 IST.

1.5 Sidereal Time and the Hour Angle

Solar time follows the Sun, and it is what the observer's day runs on.

A telescope follows stars, not the Sun. A star rises about four minutes earlier each solar day (Section 1.2), so a solar clock cannot say directly where a star is.

Sit on a merry-go-round that faces a ring of shops. Whichever shop is straight ahead tells you how far the ride has turned. Read the shop name and you know the time on the ride's own clock.

The sky is the ring of shops, and right ascension is written along its equator. The right ascension that sits on the meridian now is the reading of the sky's clock. This is sidereal time.

In numbers, the star Sirius has right ascension $6^{\text{h}}45^{\text{m}}$. When the sidereal clock reads $6^{\text{h}}45^{\text{m}}$, Sirius is on the meridian. Two sidereal hours later it is 30° west of the meridian.

Picture the meridian as a fixed pointer and the sky turning past it with the numbers 0^{h} to 24^{h} painted along the equator. The same idea, a clock read off the position of a rotating body, is how a sundial works and how pulsar timing works. It gives the sidereal time at any longitude and the angle of any star from the meridian.

Local sidereal time. The **local sidereal time** is the right ascension on the observer's meridian,

$$\text{LST} \equiv \alpha_{\text{Meridian}}. \tag{1.12}$$

The **Greenwich sidereal time** GST is the same at the Greenwich meridian.

From Greenwich to any longitude. Use the sky's frame, with x toward $\hat{\gamma}$ and z toward the north celestial pole. The point of the celestial equator on the Greenwich meridian has right ascension GST, so it lies along $R_z(\text{GST})\,\hat{x}$. The observer's meridian is a further λ east, so its point of the equator is

$$\begin{aligned} R_z(\text{GST})\,R_z(\lambda)\,\hat{x} &\overset{(1)}{=} R_z(\text{GST}+\lambda)\,\hat{x} \\ &\overset{(2)}{=} \begin{pmatrix} \cos(\text{GST}+\lambda) \\ \sin(\text{GST}+\lambda) \\ 0 \end{pmatrix}. \end{aligned} \tag{1.13}$$

(1) Two turns about the same axis add, as in Eq. (1.4).
(2) The first column of Eq. (1.3).

A point of the equator at angle α from $\hat{x}$ has right ascension α, so

Result 1.3

$$\mathrm{LST} = \mathrm{GST} + \lambda, \tag{1.14}$$

with λ turned into time by Eq. (1.1). Only the longitude enters. Observers on the same meridian, at any latitude, share one sidereal time.

The hour angle. A star has its own right ascension α, and the meridian has right ascension LST. Both are angles along the celestial equator, so their difference is how far the sky has turned since the star was on the meridian. This is the **hour angle**,

Result 1.4

$$H \equiv \mathrm{LST} - \alpha. \tag{1.15}$$

- $H = 0^{\mathrm{h}}$. The object is on the meridian, at its highest point (culmination, or transit).
- $H > 0$. The object has crossed the meridian and is moving toward the western horizon.
- $H < 0$. The object is in the eastern sky and has not yet reached the meridian.

For Sirius in the example above, LST $= 8^{\mathrm{h}}45^{\mathrm{m}}$ gives $H = +2^{\mathrm{h}}$. The hour angle grows at one sidereal hour per sidereal hour, for every star alike.

Solar time as an hour angle. The same definition works for the Sun. Its hour angle is $H_\odot = \mathrm{LST} - \alpha_\odot$, and it is zero at local noon. The time on a sundial, the **apparent solar time**, is this hour angle shifted so that noon reads 12:00,

$$\mathrm{AST} = H_\odot + 12^{\mathrm{h}}. \tag{1.16}$$

At Greenwich apparent noon $H_\odot = 0$ at the Greenwich meridian, so GST $= \alpha_\odot$ at that moment.

1.6 The Sun's Coordinates through the Year

A star keeps its (α, δ) for years, and the sidereal clock tells when it crosses the meridian.

The Sun does not keep its coordinates. It moves about a degree a day against the stars, and it also climbs north and sinks south over the year, which is why noon is higher in summer.

Drive round a circular track whose surface is tilted, high on one side and low on the other. Seen from the middle, the car rises above a level line for half the lap and falls below it for the other half. At the two points where the track crosses the level line, the car is level.

The track is the **ecliptic**, the Sun's yearly path among the stars. It is the plane of the Earth's orbit seen from the Earth. The level line is the celestial equator. The Earth's spin axis is tilted from the normal to its orbit by the **obliquity of the ecliptic**, $\epsilon \simeq 23.44°$, so the ecliptic and the equator are two great circles tilted by ϵ. They cross at two points, the equinoxes. The Sun's declination follows from one rotation by ϵ.

In numbers, the Sun's angle along the ecliptic from $\hat{\gamma}$, its **ecliptic longitude** $\lambda_\odot$, is 90° at the June solstice. There it is as far north as it gets, at $\delta_\odot = +23.44°$. Halfway, at $\lambda_\odot = 45°$ in early May, it is at 16.3°, as computed below.

Picture two hoops of the same size sharing a centre, one tilted by 23.44° against the other, with a bead running round the tilted one. The same rotation, applied to any star, turns ecliptic coordinates into equatorial ones (Eq. (2.39)). It gives $(\alpha_\odot, \delta_\odot)$ for any $\lambda_\odot$.

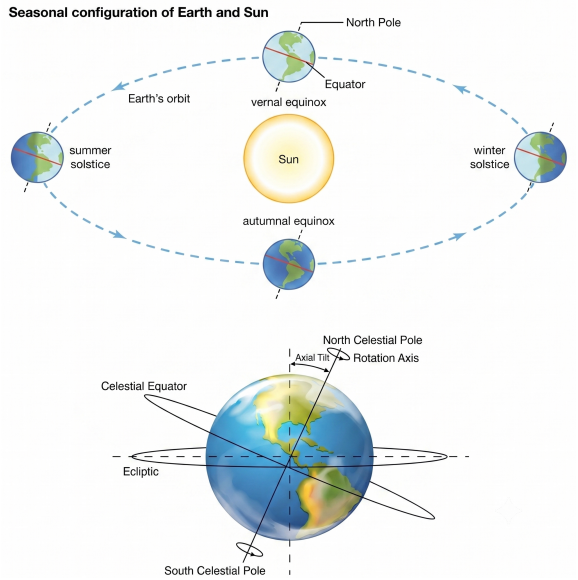


Figure 1.2: The Earth at the two equinoxes and the two solstices on its orbit (top), and the tilt of its axis between the ecliptic and the celestial equator (bottom).

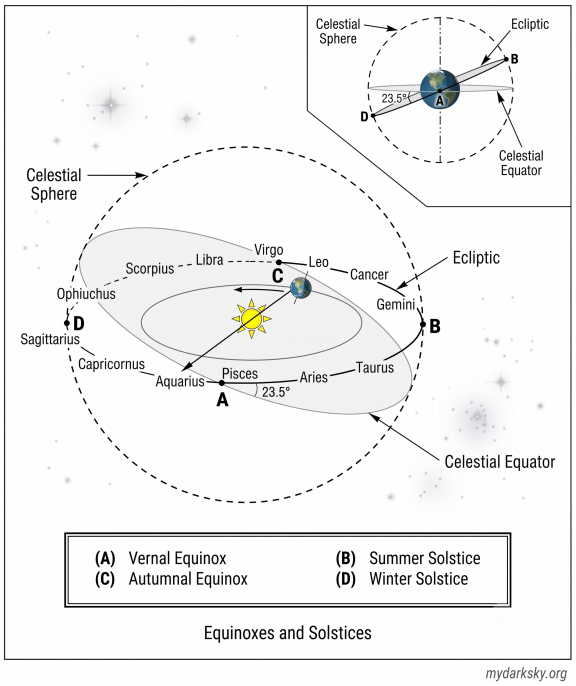


Figure 1.3: The ecliptic and the celestial equator on the celestial sphere, with the equinoxes (A, C), the solstices (B, D) and the constellations along the ecliptic.

One rotation about the line of equinoxes. Use the ecliptic frame, with x toward $\hat{\gamma}$ and z toward the north pole of the ecliptic. The Sun lies in the ecliptic plane, along $(\cos\lambda_\odot, \sin\lambda_\odot, 0)$. The equatorial frame shares the x-axis, since $\hat{\gamma}$ lies on both circles, and it is tilted about that axis by ϵ. In equatorial coordinates the north pole of the ecliptic is at $(0, -\sin\epsilon, \cos\epsilon)$, toward $\alpha = 18^{\mathrm{h}}$. The rotation that carries the ecliptic pole $(0,0,1)$ there is a turn by ϵ about x, so the Sun's equatorial unit vector is

$$\begin{pmatrix} \cos\delta_\odot \cos\alpha_\odot \\ \cos\delta_\odot \sin\alpha_\odot \\ \sin\delta_\odot \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\epsilon & -\sin\epsilon \\ 0 & \sin\epsilon & \cos\epsilon \end{pmatrix} \begin{pmatrix} \cos\lambda_\odot \\ \sin\lambda_\odot \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\lambda_\odot \\ \cos\epsilon\, \sin\lambda_\odot \\ \sin\epsilon\, \sin\lambda_\odot \end{pmatrix}. \tag{1.17}$$

(1) The left side is a unit vector at $(\alpha_\odot, \delta_\odot)$, written as in Eq. (2.2). The matrix turns by ϵ about the x-axis, and it sends $(0,0,1)$ to $(0, -\sin\epsilon, \cos\epsilon)$, the ecliptic pole.

The third row, and the ratio of the second row to the first, give

$$\boxed{\sin\delta_\odot = \sin\epsilon\, \sin\lambda_\odot, \qquad \tan\alpha_\odot = \cos\epsilon\, \tan\lambda_\odot.} \tag{1.18}$$

The declination swings between $-\epsilon$ and $+\epsilon$ and depends only on the tilt and the season. The right ascension keeps step with $\lambda_\odot$ at the four points where $\tan\lambda_\odot$ is 0 or infinite, and runs ahead of it or behind it in between. This mismatch is half of the equation of time (Section 1.7).

In early May, $\lambda_\odot = 45°$, and $\sin\delta_\odot = 0.3978 \times 0.7071 = 0.2813$, so $\delta_\odot = 16.3°$.

The four seasonal points. Setting $\lambda_\odot = 0°, 90°, 180°, 270°$ in Eq. (1.18) gives Table 1.2. The last column uses GST $= \alpha_\odot$ at Greenwich apparent noon, from Section 1.5.

Point	Date (about)	$\lambda_\odot$	$\alpha_\odot$	$\delta_\odot$	GST at Greenwich noon
March (vernal) equinox	20 March	$0°$	0^h	$0°$	0^h
June solstice	21 June	$90°$	6^h	$+23.44°$	6^h
September equinox	23 September	$180°$	12^h	$0°$	12^h
December solstice	21 December	$270°$	18^h	$-23.44°$	18^h

Table 1.2: The Sun's coordinates at the equinoxes and solstices, and the Greenwich sidereal time at Greenwich apparent noon.

1.7 The Equation of Time and the Analemma

A clock keeps mean solar time, in which every day is exactly 24 hours (Section 1.2).

A sundial does not agree with it. In mid-February the shadow of a sundial reaches noon about 14 minutes after the clock does, and in early November about 16 minutes before. The days measured by the real Sun are not all equal.

Two runners go round a track. One runs at a steady pace. The other runs faster on one side of the track and slower on the other, but finishes each lap together with the first. The gap between them grows and shrinks over the lap and never adds up.

The steady runner is the **mean Sun**, an imagined Sun that moves along the celestial equator at a uniform rate. The other is the real Sun. It moves along the ecliptic at a varying speed, because the Earth's orbit is an ellipse, and its motion along the tilted ecliptic counts unevenly in right ascension. The **equation of time** is the gap between the two, read as a time.

In numbers, the eccentricity of the Earth's orbit is $e = 0.0167$. Kepler's second law makes the Sun run ahead of the mean Sun by up to $2e = 0.0334$ radians, which is $1.91°$. At four minutes per degree, that is 7.66 minutes.

Picture a figure eight drawn by the Sun in the sky. Photograph the Sun at the same clock time every few days for a year. It moves up and down with its declination and left and right with the equation of time. The first of the two effects is angular momentum conservation (P1) through Kepler's second law, and the second is the rotation of Eq. (1.17). Together they give the gap between sundial and clock on any day of the year.

Definition. Apparent solar time is the Sun's hour angle, Eq. (1.16). **Mean solar time** MST is the hour angle of the mean Sun, shifted in the same way. The mean Sun's right ascension α_{mean} grows uniformly, and it equals the **mean longitude** $\bar{\lambda}_\odot$, the ecliptic longitude the Sun would have if it moved at the uniform rate n. The difference of the two times is

Result 1.5

$$\text{EoT} \equiv \text{AST} - \text{MST} = \alpha_{\text{mean}} - \alpha_\odot . \tag{1.19}$$

The last step follows from Eq. (1.15), since the two hour angles are LST $-\,\alpha_\odot$ and LST $-\,\alpha_{\text{mean}}$, read at the same LST.

Splitting into two parts. We add and subtract the Sun's true ecliptic longitude $\lambda_\odot$,

$$\begin{aligned} \text{EoT} &\overset{(1)}{=} \bar{\lambda}_\odot - \alpha_\odot \\ &= \underbrace{\left(\bar{\lambda}_\odot - \lambda_\odot\right)}_{\text{uneven speed}} + \underbrace{\left(\lambda_\odot - \alpha_\odot\right)}_{\text{tilt}} . \end{aligned} \tag{1.20}$$

(1) $\alpha_{\text{mean}} = \bar{\lambda}_\odot$, from the definition of the mean Sun.

The first part would vanish on a circular orbit. The second would vanish if the axis were not tilted.

Uneven speed, from Kepler's second law. The Sun's yearly path is the Earth's orbit seen from the other end, so it has the same eccentricity e, true anomaly v and mean anomaly ℓ as the Earth's orbit (Chapter 6). Both $\lambda_\odot$ and $\bar{\lambda}_\odot$ are measured from $\hat{\gamma}$, and they differ from v and ℓ by the same fixed angle, the longitude of perihelion. So $\bar{\lambda}_\odot - \lambda_\odot = \ell - v$. To find $v - \ell$ we write the rate of the true anomaly with angular momentum conservation (P1),

$$\begin{aligned}
\frac{\mathrm{d}v}{\mathrm{d}t} &\overset{(1)}{=} \frac{L}{\mu r^2} \\
&\overset{(2)}{=} \frac{\sqrt{GMa(1-e^2)}\,(1+e\cos v)^2}{a^2(1-e^2)^2} \\
&\overset{(3)}{=} n\,\frac{(1+e\cos v)^2}{(1-e^2)^{3/2}} \\
&\overset{(4)}{\simeq} n\,(1+2e\cos v).
\end{aligned} \tag{1.21}$$

(1) Eq. (6.21). The true anomaly differs from the polar angle ϕ by a constant, so they share a rate.
(2) The orbit $r = a(1-e^2)/(1+e\cos v)$ from Eq. (6.27), whose semi-latus rectum $L^2/(GM\mu^2)$ equals $a(1-e^2)$, so $L/\mu = \sqrt{GMa(1-e^2)}$.
(3) Kepler's third law, $GM = n^2a^3$, Eq. (6.49).
(4) Keep only the first order in e, since $e^2 = 2.8 \times 10^{-4}$.

The mean anomaly is $\ell = n(t - t_0)$, with t_0 the time of perihelion, where $v = \ell = 0$. Separating the variables and integrating from perihelion,

$$\begin{aligned}
\ell &\overset{(1)}{=} \int_0^v \frac{dv'}{1+2e\cos v'} \\
&\overset{(2)}{\simeq} \int_0^v (1-2e\cos v')\,dv' \\
&= v - 2e\sin v.
\end{aligned} \tag{1.22}$$

(1) $n\,dt = dv/(1+2e\cos v)$ from the last line of the previous chain.
(2) $1/(1+x) \simeq 1-x$ for small x, again to first order in e.

So $v - \ell \simeq 2e\sin v \simeq 2e\sin\ell$, where replacing v by ℓ in the small term changes it only at order e^2. The Sun runs ahead of the mean Sun from perihelion to aphelion and behind it in the other half year.

Tilt, from the rotation. Eq. (1.18) gives $\tan\alpha_\odot = \cos\epsilon\,\tan\lambda_\odot$. To find $\lambda_\odot - \alpha_\odot$ we take its tangent,

$$\begin{aligned}
\tan(\lambda_\odot - \alpha_\odot) &\overset{(1)}{=} \frac{\tan\lambda_\odot - \tan\alpha_\odot}{1 + \tan\lambda_\odot\tan\alpha_\odot} \\
&\overset{(2)}{=} \frac{(1-\cos\epsilon)\tan\lambda_\odot}{1 + \cos\epsilon\tan^2\lambda_\odot} \\
&\overset{(3)}{=} \frac{(1-\cos\epsilon)\sin\lambda_\odot\cos\lambda_\odot}{\cos^2\lambda_\odot + \cos\epsilon\sin^2\lambda_\odot} \\
&\overset{(4)}{=} \frac{(1-\cos\epsilon)\sin 2\lambda_\odot}{(1+\cos\epsilon) + (1-\cos\epsilon)\cos 2\lambda_\odot} \\
&\overset{(5)}{=} \frac{\tan^2(\epsilon/2)\,\sin 2\lambda_\odot}{1 + \tan^2(\epsilon/2)\cos 2\lambda_\odot}.
\end{aligned} \tag{1.23}$$

(1) The tangent of a difference.
(2) $\tan\alpha_\odot = \cos\epsilon\,\tan\lambda_\odot$.
(3) Multiply the top and the bottom by $\cos^2\lambda_\odot$.
(4) Multiply the top and the bottom by 2, and use $2\sin\lambda\cos\lambda = \sin 2\lambda$, $2\cos^2\lambda = 1 + \cos 2\lambda$ and $2\sin^2\lambda = 1 - \cos 2\lambda$.
(5) Divide the top and the bottom by $1 + \cos\epsilon$, and use $(1-\cos\epsilon)/(1+\cos\epsilon) = \tan^2(\epsilon/2)$.

For $\epsilon = 23.44°$, $\tan^2(\epsilon/2) = 0.0430$ is small, and expanding the arctangent to first order gives

$$\lambda_\odot - \alpha_\odot \simeq \tan^2(\epsilon/2)\,\sin 2\lambda_\odot.$$

The next term, $-\frac{1}{2}\tan^4(\epsilon/2)\sin 4\lambda_\odot$, is at most 0.21 minutes. The tilt term vanishes at the equinoxes and at the solstices. Near an equinox the ecliptic crosses the equator at an angle, so only the fraction $\cos\epsilon$ of the Sun's motion counts in right ascension, and $\alpha_\odot$ falls behind. Near a solstice the ecliptic runs parallel to the equator but closer to the pole, where an hour of right ascension is a shorter arc, so $\alpha_\odot$ catches up.

The equation of time. With both parts in Eq. (1.20), and each radian turned into $4 \times 57.296 = 229.2$ minutes,

$$\begin{aligned}
\mathrm{EoT} &\overset{(1)}{=} -(v - \ell) + (\lambda_\odot - \alpha_\odot) \\
&\overset{(2)}{\simeq} -2e\sin\ell + \tan^2(\epsilon/2)\,\sin 2\lambda_\odot \\
&\overset{(3)}{=} -7.66^{\mathrm{m}}\sin\ell + 9.86^{\mathrm{m}}\sin 2\lambda_\odot.
\end{aligned} \tag{1.24}$$

(1) Eq. (1.20) with $\bar\lambda_\odot - \lambda_\odot = \ell - v$.
(2) The two results above, each to first order.
(3) $2e = 0.0334$ and $\tan^2(\epsilon/2) = 0.0430$ radians, times 229.2 minutes per radian.

$$\boxed{\mathrm{EoT} \simeq -7.66^{\mathrm{m}}\sin\ell + 9.86^{\mathrm{m}}\sin 2\lambda_\odot.} \tag{1.25}$$

The first term has a period of one year and is set by e alone. The second has a period of half a year and is set by ϵ alone. Neither depends on where the observer is.

Figure 1.4 compares Eq. (1.25) with the exact gap. The two terms add up to within 0.3 minutes of it. The perihelion is in early January, so $\sin\ell$ peaks in early April, while $\sin 2\lambda_\odot$ peaks in early May and early November. The two terms add in November and in February. The equation of time ranges from −14.2 minutes in mid-February to +16.4 minutes in early November, with smaller turning points of +3.7 minutes in mid-May and −6.5 minutes in late July.

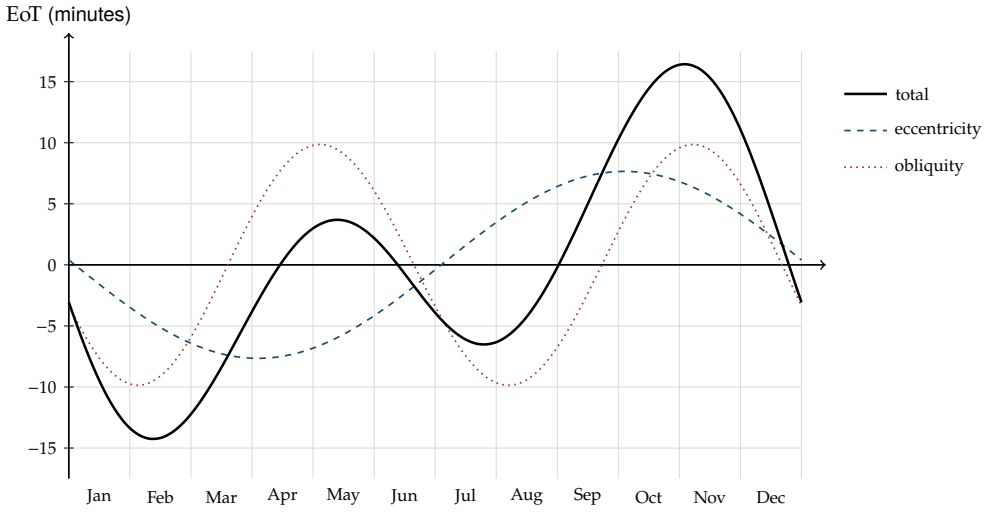


Figure 1.4: The equation of time over one year (solid), with the eccentricity term $-7.66^{\mathrm{m}} \sin \ell$ (dashed) and the obliquity term $9.86^{\mathrm{m}} \sin 2\lambda_\odot$ (dotted). The solid curve is computed from Eq. (1.19) without expanding in e or ϵ.

Worked example 1.2: Apparent noon at IISER TVM

Local mean noon at IISER TVM is at 12:21:27 IST (Section 1.4). The Sun crosses the meridian when AST = 12^{h}, that is when MST = $12^{\mathrm{h}} - \mathrm{EoT}$. On 3 November, with EoT = $+16.4^{\mathrm{m}} = +16^{\mathrm{m}}24^{\mathrm{s}}$, the Sun crosses at 12:05:03 IST. On 11 February, with EoT = -14.2^{m}, it crosses at 12:35:39 IST.

The analemma. Plot the Sun's declination $\delta_\odot$ (Eq. (1.18)) against the equation of time for each day of the year. The curve is a figure eight, the **analemma**. Its height is $2\epsilon = 46.9°$. Its two loops are unequal because the eccentricity term adds to the obliquity term in one half of the year and works against it in the other.

1.8 Precession of the Equinoxes and Reference Epochs

So far the Earth's axis has pointed at a fixed place among the stars, and the equinox $\hat{\gamma}$ has been a fixed zero point.

Over centuries this fails. Hipparchus, comparing his star positions with ones made 150 years earlier, found that the equinox had moved west among the stars. The pole star today, Polaris, was not near the pole in his time.

Spin a top and tilt it. Gravity pulls its centre down and tries to tip it over. It does not fall. Its axis sweeps slowly round a cone instead, sideways to the pull.

The Earth is a spinning top with a bulge at its equator. The Sun and the Moon pull on the bulge and try to turn the equator into the plane of the ecliptic. The axis responds by sweeping round a cone about the pole of the ecliptic. A torque at right angles to the spin turns the axis sideways, which is angular momentum conservation (P1).

A bicycle wheel makes a good gyroscope. Take a wheel of mass 2 kg and radius 0.3 m, spinning 10 times a second, with its axle held horizontal at one end 0.2 m from the wheel. Its angular momentum is $2 \times 0.3^2 \times 2\pi \times 10 = 11.3\,\mathrm{kg\,m^2\,s^{-1}}$, and the weight gives a torque of $2 \times 9.81 \times 0.2 = 3.92\,\mathrm{N\,m}$. The axle swings round at $3.92/11.3 = 0.35$ radians per second, once in 18 seconds.

Picture the Earth's axis drawing a circle of radius 23.44° on the sky, once in about 26 000 years. The same rule makes a spinning top and a gyrocompass precess, and makes the orbit of a satellite around the bulging Earth turn its plane. It gives the rate of the drift of the equinox from the size of the Earth's bulge, and the way it changes every catalogue position.

The gyroscope rule. Let $\hat{p}$ be the unit vector along the spin axis and $\hat{k}$ the north pole of the ecliptic, at angle ϵ from $\hat{p}$. Suppose the angular momentum $\vec{L}$ turns steadily about $\hat{k}$ at rate ω_{prec}, with ϵ fixed. In the ecliptic frame, with z along $\hat{k}$, $\vec{L}(t) = R_z(-\omega_{\mathrm{prec}} t)\,\vec{L}(0)$, where the minus sign anticipates that the turn goes clockwise seen from the north (we find this below). Differentiating at $t = 0$,

$$\begin{aligned}\frac{\mathrm{d}\vec{L}}{\mathrm{d}t} &\overset{(1)}{=} -\omega_{\mathrm{prec}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \vec{L} \\ &\overset{(2)}{=} -\omega_{\mathrm{prec}}\, \hat{k} \times \vec{L}. \end{aligned} \tag{1.26}$$

(1) The derivative of Eq. (1.3) at $\varphi = 0$ is the matrix shown, and $\varphi = -\omega_{\mathrm{prec}} t$.
(2) The matrix times (L_x, L_y, L_z) is $(-L_y, L_x, 0) = \hat{k} \times \vec{L}$.

The size of $\hat{k} \times \vec{L}$ is $L \sin\epsilon$, so

$$\omega_{\mathrm{prec}} = \frac{|\dot{\vec{L}}|}{L \sin\epsilon}. \tag{1.27}$$

The rate change $\dot{\vec{L}}$ is the torque, $\sum \vec{x} \times \vec{F}$ over the Earth's mass (Eq. (6.98)). The torque must be perpendicular to $\vec{L}$ for L and ϵ to stay fixed.

The pull on the bulge. Let a body of mass M, the Sun or the Moon, be at distance a from the Earth's centre in the direction $\hat{s}$. A mass element dm of the Earth at $\vec{x}$ from the centre, with $|\vec{x}| \ll a$, feels the force $dm\, GM(a\hat{s} - \vec{x})/|a\hat{s} - \vec{x}|^3$. To first order in $|\vec{x}|/a$,

$$\begin{aligned}\frac{a\hat{s} - \vec{x}}{|a\hat{s} - \vec{x}|^3} &\overset{(1)}{=} \frac{a\hat{s} - \vec{x}}{a^3}\Big(1 - \frac{2\hat{s}\cdot\vec{x}}{a} + \frac{x^2}{a^2}\Big)^{-3/2} \\ &\overset{(2)}{\simeq} \frac{a\hat{s} - \vec{x}}{a^3}\Big(1 + \frac{3\hat{s}\cdot\vec{x}}{a}\Big) \\ &\overset{(3)}{\simeq} \frac{1}{a^2}\Big[\hat{s} + \frac{3(\hat{s}\cdot\vec{x})\hat{s} - \vec{x}}{a}\Big]. \end{aligned} \tag{1.28}$$

(1) $|a\hat{s} - \vec{x}|^2 = a^2 - 2a\,\hat{s}\cdot\vec{x} + x^2$.
(2) $(1+y)^{-3/2} \simeq 1 - \frac{3}{2}y$, dropping x^2/a^2.
(3) Multiply out and drop the term of order x^2/a^3.

The torque about the Earth's centre is

$$\begin{aligned}\dot{\vec{L}} &\overset{(1)}{=} \frac{GM}{a^2}\Big(\int \vec{x}\, dm\Big) \times \hat{s} + \frac{GM}{a^3}\int \big[3(\hat{s}\cdot\vec{x})\,\vec{x}\times\hat{s} - \vec{x}\times\vec{x}\big] dm \\ &\overset{(2)}{=} \frac{3GM}{a^3}\Big(\int \vec{x}\,\vec{x}^{\mathsf{T}} dm\; \hat{s}\Big) \times \hat{s}. \end{aligned} \tag{1.29}$$

(1) $\vec{x}\times$ the last line of the previous chain, times $GM\,dm$, summed over the Earth.
(2) $\int \vec{x}\, dm = 0$ about the centre of mass, $\vec{x}\times\vec{x} = 0$, and $(\hat{s}\cdot\vec{x})\,\vec{x} = \vec{x}\,\vec{x}^{\mathsf{T}}\hat{s}$.

The matrix $\int \vec{x}\,\vec{x}^{\mathsf{T}} dm$ holds the Earth's shape. In axes fixed in the Earth, with z along $\hat{p}$, it is diagonal, and by symmetry about the axis $\int x^2 dm = \int y^2 dm$. The moments of inertia about the polar axis and about an equatorial axis are $C = \int (x^2 + y^2)\, dm = 2\int x^2 dm$ and $A = \int (y^2 + z^2)\, dm = \int x^2 dm + \int z^2 dm$, so

$$\begin{aligned} \int \vec{x}\,\vec{x}^{\mathsf{T}} dm &= \begin{pmatrix} \int x^2 dm & 0 & 0 \\ 0 & \int x^2 dm & 0 \\ 0 & 0 & \int z^2 dm \end{pmatrix} \\ &\overset{(1)}{=} \Big(\int x^2 dm \Big)\, \mathbb{1} - (C - A)\, \hat{p}\,\hat{p}^{\mathsf{T}}. \end{aligned} \tag{1.30}$$

(1) $\int z^2 dm - \int x^2 dm = A - C$, and $\hat{p}\,\hat{p}^{\mathsf{T}}$ is the matrix with a single 1 in the zz place.

The term with $\mathbb{1}$ gives $\hat{s} \times \hat{s} = 0$. The flattening $C - A$ alone produces the torque,

$$\dot{\vec{L}} = \frac{3GM}{a^3}\, (C - A)\, (\hat{p} \cdot \hat{s})\, (\hat{s} \times \hat{p}). \tag{1.31}$$

A round Earth, $C = A$, feels no torque. Nor does an Earth with the Sun in its equatorial plane, $\hat{p} \cdot \hat{s} = 0$.

Averaged over the orbit. The Sun goes round the ecliptic in a year. That is long compared with a day and short compared with the precession period, so the axis hardly moves in a year and we may average Eq. (1.31) over $\hat{s}$ with $\hat{p}$ held fixed (P6). In the ecliptic frame, $\hat{s} = (\cos\theta, \sin\theta, 0)$ for a circular orbit, and

$$\begin{aligned} \langle (\hat{p} \cdot \hat{s})\, \hat{s} \times \hat{p} \rangle &\overset{(1)}{=} \big(\langle \hat{s}\,\hat{s}^{\mathsf{T}} \rangle\, \hat{p} \big) \times \hat{p} \\ &\overset{(2)}{=} \left[\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \hat{p} \right] \times \hat{p} \\ &\overset{(3)}{=} \tfrac{1}{2} \big(\hat{p} - \cos\epsilon\, \hat{k} \big) \times \hat{p} \\ &\overset{(4)}{=} -\tfrac{1}{2} \cos\epsilon\, \hat{k} \times \hat{p}. \end{aligned} \tag{1.32}$$

(1) $(\hat{p} \cdot \hat{s})\, \hat{s} = \hat{s}\,\hat{s}^{\mathsf{T}} \hat{p}$, and $\hat{p}$ is constant during the average.
(2) $\langle \cos^2\theta \rangle = \langle \sin^2\theta \rangle = \frac{1}{2}$ and $\langle \cos\theta \sin\theta \rangle = 0$ over one turn, and $\hat{s}$ has no z part.
(3) The matrix removes the z part of $\hat{p}$, which is $(\hat{p} \cdot \hat{k})\hat{k} = \cos\epsilon\, \hat{k}$, and halves the rest.
(4) $\hat{p} \times \hat{p} = 0$.

With $\vec{L} = C\omega_\oplus \hat{p}$, which holds to the accuracy of $\omega_{\text{prec}}/\omega_\oplus \sim 10^{-7}$ (P6),

$$\begin{aligned} \dot{\vec{L}} &\overset{(1)}{=} -\frac{3}{2}\, \frac{GM}{a^3}\, (C - A) \cos\epsilon\, \hat{k} \times \hat{p} \\ &\overset{(2)}{=} -\left[\frac{3}{2}\, \frac{GM}{a^3}\, \frac{C - A}{C}\, \frac{\cos\epsilon}{\omega_\oplus} \right] \hat{k} \times \vec{L}. \end{aligned} \tag{1.33}$$

(1) Eq. (1.31), averaged with the result above.
(2) $\hat{p} = \vec{L}/(C\omega_\oplus)$.

This has the form $-\omega_{\text{prec}}\, \hat{k} \times \vec{L}$ of the gyroscope rule. The bracket is positive, so the axis turns clockwise seen from the north ecliptic pole, against the Earth's spin and orbit. The equinox,

where the moving equator meets the ecliptic, therefore moves west. For the Sun, $GM_\odot/a^3 = n^2$ by Kepler's third law, Eq. (6.49), so

$$\boxed{\omega_{\text{prec},\odot} = \frac{3}{2}\,\frac{n^2}{\omega_\oplus}\,\frac{C-A}{C}\,\cos\epsilon.} \tag{1.34}$$

The rate grows with the flattening and falls with the spin. A faster top precesses more slowly.

Numbers for the Sun and the Moon. Satellite orbits give the Earth's flattening as $(C - A)/(M_\oplus R_\oplus^2) = 1.0826 \times 10^{-3}$, and the density profile from seismology gives $C = 0.3307\, M_\oplus R_\oplus^2$, so $(C - A)/C = 0.003274$. With $n/\omega_\oplus = T_{\text{sid}}/T_{\text{yr}} = 1/366.2422$ from Eq. (1.7),

$$\omega_{\text{prec},\odot} = \frac{3}{2} \times \frac{360^\circ}{\text{year}} \times \frac{1}{366.2422} \times 0.003274 \times 0.9175 = 15.9'' \text{ per year}.$$

The Moon gives the same expression with $GM_\odot/a^3$ replaced by $GM_{\text{Moon}}/a_{\text{Moon}}^3$. That is larger by $(M_{\text{Moon}}/M_\odot)(1\,\text{AU}/a_{\text{Moon}})^3 = 2.18$, with $a_{\text{Moon}} = 384\,400\,\text{km}$, so the Moon gives 34.7″ per year. The Moon's orbit is tilted by 5° to the ecliptic, which changes this by about one percent. The total, about 50.6″ per year, is within one percent of the measured value,

$$\boxed{\omega_{\text{prec}} \simeq 50.29'' \text{ per year} = 1^\circ \text{ in } 71.6 \text{ years},} \tag{1.35}$$

one turn in $1\,296\,000''/50.29'' = 25\,770$ years. The measured rate also includes a slow turn of the ecliptic itself, caused by the planets, of about 0.1″ per year.

Epochs. The grid of (α, δ) is tied to the moving equinox $\hat\gamma$, so a catalogue position must state the moment, or **epoch**, whose equator and equinox it uses (Section 2.1.1). The current standard is **J2000.0**, 2000 January 1, 12:00:00 TT.

Precessing a catalogue position. In the frame of the moving equinox, every star turns about the ecliptic pole at ω_{prec} per year, anticlockwise seen from the north, so its ecliptic longitude grows by 50.29″ per year. In equatorial coordinates the ecliptic pole is $\hat k = (0, -\sin\epsilon, \cos\epsilon)$ (Eq. (1.17)). The star's unit vector $\hat r$ then changes as

$$\begin{aligned} \frac{\mathrm{d}\hat r}{\mathrm{d}t} &\overset{(1)}{=} \omega_{\text{prec}}\,\hat k \times \hat r \\ &\overset{(2)}{=} \omega_{\text{prec}} \begin{pmatrix} 0 & -\cos\epsilon & -\sin\epsilon \\ \cos\epsilon & 0 & 0 \\ \sin\epsilon & 0 & 0 \end{pmatrix} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}. \end{aligned} \tag{1.36}$$

(1) A turn about $\hat k$ at rate ω_{prec}, as in the gyroscope rule but anticlockwise.

(2) $\hat k \times \hat r$ written as a matrix acting on $\hat r$, with $\hat k = (0, -\sin\epsilon, \cos\epsilon)$.

The left side is $\dot{\hat r} = \dot\alpha\,\partial_\alpha \hat r + \dot\delta\,\partial_\delta \hat r$. Its third row is $\cos\delta\,\dot\delta = \omega_{\text{prec}} \sin\epsilon \cos\delta\cos\alpha$. Its second row is

$$\cos\delta\cos\alpha\,\dot\alpha - \sin\delta\sin\alpha\,\dot\delta = \omega_{\text{prec}}\cos\epsilon\cos\delta\cos\alpha.$$

Putting $\dot\delta$ from the third row into the second and dividing by $\cos\delta\cos\alpha$,

Result 1.6

$$\alpha(t) \simeq \alpha_0 + \left(\omega_{\text{prec}} \cos\epsilon + \omega_{\text{prec}} \sin\epsilon \, \sin\alpha_0 \tan\delta_0\right) \Delta t, \tag{1.37}$$
$$\delta(t) \simeq \delta_0 + \left(\omega_{\text{prec}} \sin\epsilon \, \cos\alpha_0\right) \Delta t, \tag{1.38}$$

for a star at (α_0, δ_0) at J2000.0 and Δt years later, as long as the change stays small. With $\omega_{\text{prec}} = 50.29''$ per year, $\omega_{\text{prec}} \cos\epsilon = 46.14''$ and $\omega_{\text{prec}} \sin\epsilon = 20.00''$ per year. The tabulated values, $46.12'' = 3.075^{\text{s}}$ and $20.04'' = 1.336^{\text{s}}$ per year, differ in the last digits because of the slow turn of the ecliptic. The first term moves every star equally in α. The second depends on where the star is, and it grows near the celestial poles, where $\tan\delta_0$ is large.

Worked example 1.3: Sirius from J2000.0 to late 2026

Sirius is at $\alpha_0 = 6^{\text{h}}45^{\text{m}}09^{\text{s}}$, $\delta_0 = -16°43'$. Its right ascension grows at $3.075^{\text{s}} + 1.336^{\text{s}} \sin 101.29° \tan(-16.72°) = 2.68^{\text{s}}$ per year, and its declination changes at $20.04'' \cos 101.29° = -3.92''$ per year. Over 26.8 years, α grows by 72^{s} and δ falls by $105''$. The star's own motion across the sky, its proper motion, is a separate correction (Chapter 3).

1.9 Nutation: Mean and Apparent Sidereal Time

The averaged torque of Section 1.8 turns the axis smoothly. The real torque does not stay at its average. It changes as the Sun and the Moon move north and south of the equator, and the Moon's orbit itself turns, once in 18.6 years. The axis wobbles about its smooth cone with small periodic terms, called **nutation**.

The largest term follows the 18.6-year turn of the Moon's orbit. It moves the equinox back and forth along the ecliptic by up to $17.2''$. This shift is the **nutation in longitude** $\Delta\psi$.

Mean and true equinox. The **mean equinox** moves with the smooth precession alone. The **true equinox** also includes nutation. Sidereal time measured from the mean equinox is **Greenwich mean sidereal time** (GMST). Measured from the true equinox it is **Greenwich apparent sidereal time** (GAST).

The equation of the equinoxes. Sidereal time is the hour angle of the equinox. A shift $\Delta\psi$ of the equinox along the ecliptic is a small turn about the ecliptic pole, which is what Eq. (1.37) describes with $\omega_{\text{prec}}\Delta t$ replaced by $\Delta\psi$. For a point on the equator, $\delta = 0$, so only the first term survives, and the shift along the equator is $\Delta\psi \cos\epsilon$. So

Result 1.7

$$\text{GAST} - \text{GMST} = \Delta\psi \cos\epsilon, \tag{1.39}$$

the **equation of the equinoxes**. Here ϵ is the true obliquity. For the largest term, $17.2'' \times 0.917 = 15.8''$, or 1.05^{s}. With the smaller terms the difference stays below 1.2^{s}. GMST is enough for pointing a telescope. GAST is needed for positions better than an arcsecond.

1.10 Modern Time Scales: UT1, TAI, UTC, TT and TDB

Every clock so far has been the turning Earth.

The turning Earth is not a perfect clock. The day grows longer as the tides slow the Earth, and it changes by milliseconds with the winds and with the motion of the Earth's liquid core.

A flywheel with a little friction in its bearing loses speed slowly. A clock that counts its turns falls behind a quartz watch, by more each day.

The tides raised by the Moon are dragged ahead of the Moon by the Earth's spin, and the Moon pulls back on them. Angular momentum (P1) flows from the Earth's spin into the Moon's orbit, and the Moon moves away by about 3.8 cm a year. Records of ancient eclipses show that the day has grown longer by about 1.7 ms per century on average. Atomic clocks keep a steady time, and the time kept by the Earth is corrected against them.

In numbers, if each day lasts 1 ms longer than 86 400 atomic seconds, the Earth's clock falls behind by 0.37 s in a year. A leap second every two or three years keeps civil time with the Earth.

Picture two clocks side by side, one steady and one slowly losing, and a person who now and then holds the steady one back by a second so that the two still agree on when the Sun is overhead. The same split between a clock defined by a motion and a clock defined by an atom is behind every one of the time scales below. Each one fixes a different trade-off between following the Earth and keeping a uniform rate.

1. **UT1 (Universal Time).** The angle through which the Earth has actually turned, read as a time. It is measured with radio telescopes (VLBI) that time the arrival of signals from distant quasars. Because of tidal friction, winds and the core, UT1 does not run at a uniform rate.
2. **TAI (International Atomic Time).** A uniform time kept by a weighted average of several hundred atomic clocks around the world. Its second is the SI second, 9 192 631 770 periods of the radiation from the ground-state hyperfine transition of caesium-133.
3. **UTC (Coordinated Universal Time).** Civil time. UTC runs at the rate of TAI, but whole **leap seconds** are added at the end of June or December to keep $|\mathrm{UTC} - \mathrm{UT1}| < 0.9\,\mathrm{s}$. Since 2017,
$$\mathrm{TAI} - \mathrm{UTC} = 37\,\mathrm{s}.$$
In 2022 the General Conference on Weights and Measures decided to stop adding leap seconds by 2035.
4. **TT (Terrestrial Time).** The uniform time of a clock on the geoid, the surface of mean sea level, free of the Earth's irregular spin. It is defined to run with TAI,
$$\mathrm{TT} = \mathrm{TAI} + 32.184\,\mathrm{s}, \qquad \mathrm{TT} - \mathrm{UTC} = 37\,\mathrm{s} + 32.184\,\mathrm{s} = 69.184\,\mathrm{s}.$$
The 32.184 s keeps TT continuous with the older Ephemeris Time.
5. **TDB (Barycentric Dynamical Time).** The time coordinate for the motion of bodies in the Solar System, referred to its centre of mass, the barycentre. It is used for planetary ephemerides, spacecraft navigation and pulsar timing. It differs from TT because a clock on the Earth moves and sits in the Sun's gravity.

TT against TDB. A clock on the Earth ticks proper time. In a weak field, its rate against the barycentric time t is
$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{1}{c^2}\left(\frac{v^2}{2} + \frac{GM_\odot}{r}\right),$$
with v the Earth's orbital speed and r its distance from the Sun. The Earth's own gravity and spin add a constant, which the definition of TT absorbs. The orbital terms are not constant,

because the orbit is an ellipse. With the vis-viva relation $v^2 = GM_\odot(2/r - 1/a)$, which follows from Eq. (6.33) with $E = \frac{1}{2}\mu v^2 - GM\mu/r$,

$$\frac{v^2}{2} + \frac{GM_\odot}{r} = \frac{2GM_\odot}{r} - \frac{GM_\odot}{2a}.$$

The time average of $1/r$ over an orbit is $1/a$, by the same substitution used below. The average rate is a constant, which the definitions of the time scales take up. The part that changes over the year adds up to

$$\begin{aligned}
\mathrm{TDB} - \mathrm{TT} &\overset{(1)}{=} \frac{2GM_\odot}{c^2} \int \left(\frac{1}{r} - \frac{1}{a}\right) dt \\
&\overset{(2)}{=} \frac{2GM_\odot}{c^2} \int \left(\frac{1}{a(1 - e\cos u)} - \frac{1}{a}\right) \frac{(1 - e\cos u)}{n}\, du \\
&= \frac{2GM_\odot}{c^2\, na} \int e\cos u\, du \\
&\overset{(3)}{=} \frac{2e\sqrt{GM_\odot a}}{c^2} \sin u\,.
\end{aligned} \tag{1.40}$$

(1) The rate above, with its mean removed, integrated over time. The clock on the Earth runs slow when $1/r$ is above its mean, so TT falls behind TDB there.

(2) $r = a(1 - e\cos u)$, Eq. (6.41), and $dt = (1 - e\cos u)\, du/n$ from Eq. (6.55). Here u is the eccentric anomaly of the Earth's orbit. The same steps show that $\int dt/r$ over one orbit is $2\pi/(na)$, so $1/r$ averages to $1/a$.

(3) $GM_\odot/(na) = na^2 = \sqrt{GM_\odot a}$ by Eq. (6.49), and the constant of integration is part of the definition.

With $e = 0.0167$, $GM_\odot = 1.327 \times 10^{20}\ \mathrm{m^3\,s^{-2}}$ and $a = 1.496 \times 10^{11}$ m,

$$\boxed{\mathrm{TDB} \simeq \mathrm{TT} + 0.001657\,\mathrm{s}\ \sin u, \qquad u \simeq \ell + e\sin\ell.} \tag{1.41}$$

The second relation is Kepler's equation, Eq. (6.51), to first order in e, with ℓ the Earth's mean anomaly. The difference is 1.7 ms at most and depends only on the eccentricity and the size of the orbit. The Moon and the planets add terms of a few microseconds. A pulsar timed to a microsecond needs all of them.

1.11 Julian Date and Modified Julian Date

A calendar date is hard to subtract. Months have different lengths, leap years come and go, and the calendar itself was changed in 1582. Astronomers count days instead, with one number that never resets.

Julian Date. The **Julian Date** (JD) is the number of days since Greenwich mean noon on 1 January 4713 BC, in the Julian calendar. The day starts at noon, so that a night of observing has a single date.

Modified Julian Date. The **Modified Julian Date** (MJD) moves the start of the day to midnight and removes the large leading digits,

Result 1.8

$$\mathrm{MJD} \equiv \mathrm{JD} - 2\,400\,000.5. \tag{1.42}$$

MJD = 0.0 is midnight at the start of 17 November 1858.

Standard epochs.

- **J2000.0**, JD = 2 451 545.0, 2000 January 1, 12:00:00 TT.
- **B1950.0**, JD = 2 433 282.423.

From a calendar date. For year Y, month M ($1 \le M \le 12$) and day D, with the time of day as a fraction of D, the following rule (Meeus 1998) gives the Julian Date for a date in the Gregorian calendar. If M is 1 or 2, replace Y by $Y-1$ and M by $M+12$. Then

Result 1.9

$$\mathrm{JD} = \lfloor 365.25(Y+4716) \rfloor + \lfloor 30.6001(M+1) \rfloor + D + 2 - \lfloor Y/100 \rfloor + \lfloor Y/400 \rfloor - 1524.5. \quad (1.43)$$

Each piece has a job. Moving January and February to the end of the previous year puts the leap day at the very end of the year, so it never shifts the months that follow it. The first term counts the days in whole years, one in four of them a leap year. From March, the month lengths run 31, 30, 31, 30, 31 and then repeat, 153 days in five months, or 30.6 a month, so the second term counts the days in the whole months before M. The terms $2 - \lfloor Y/100 \rfloor + \lfloor Y/400 \rfloor$ remove the leap days that the Gregorian rule drops in century years. The constant 1524.5 sets the zero at noon in 4713 BC.

For J2000.0, $Y = 1999$, $M = 13$ and $D = 1.5$. Then $\lfloor 365.25 \times 6715 \rfloor = 2\,452\,653$, $\lfloor 30.6001 \times 14 \rfloor = 428$, and $2 - 19 + 4 = -13$, so

$$\mathrm{JD} = 2\,452\,653 + 428 + 1.5 - 13 - 1524.5 = 2\,451\,545.0,$$

as listed above.

1.11.1 Sidereal Time on Any Date

To point a telescope at a given moment we need LST, and by Eq. (1.14) that needs GMST. The IAU formula gives GMST at 0^{h} UT1 from the number of Julian centuries since J2000.0, $T = (\mathrm{JD} - 2\,451\,545.0)/36\,525$,

$$\mathrm{GMST}(0^{\mathrm{h}}\mathrm{UT1}) = 6^{\mathrm{h}}41^{\mathrm{m}}50.54841^{\mathrm{s}} + 8\,640\,184.812866^{\mathrm{s}}\,T + 0.093104^{\mathrm{s}}\,T^2 - 6.2\times10^{-6\,\mathrm{s}}\,T^3. \quad (1.44)$$

The linear term is the daily gain of the sidereal clock. A Julian century has 36 525 days, and $8\,640\,184.81^{\mathrm{s}}/36\,525 = 236.555^{\mathrm{s}}$ per day, which is the $3^{\mathrm{m}}56.555^{\mathrm{s}}$ found in Section 1.2. At a time t_{UT} later in the same day, the sidereal clock has run faster by the ratio of the two days,

$$\mathrm{GMST}(t_{\mathrm{UT}}) = \mathrm{GMST}(0^{\mathrm{h}}\mathrm{UT1}) + 1.00273790935 \times t_{\mathrm{UT}}, \quad (1.45)$$

where $1.00273790935 = T_{\mathrm{sol}}/T_{\mathrm{sid}} = 366.2422/365.2422$ by Eq. (1.7).

Worked example 1.4: What is on the meridian at IISER TVM?

Take 2026 October 16 at 19:00 IST, which is 13:30 UT. Eq. (1.43) with $Y = 2026$, $M = 10$, $D = 16.0$ gives $\lfloor 365.25 \times 6742 \rfloor = 2\,462\,515$, $\lfloor 30.6001 \times 11 \rfloor = 336$ and $2 - 20 + 5 = -13$, so JD = 2 461 329.5 at 0^{h} UT. Then $T = 0.267885$, and Eq. (1.44), reduced modulo 24^{h}, gives $\mathrm{GMST}(0^{\mathrm{h}}) = 1^{\mathrm{h}}38^{\mathrm{m}}06.6^{\mathrm{s}}$. Adding $1.0027379 \times 13.5^{\mathrm{h}} = 13^{\mathrm{h}}32^{\mathrm{m}}13.1^{\mathrm{s}}$ gives GMST = $15^{\mathrm{h}}10^{\mathrm{m}}19.6^{\mathrm{s}}$. The longitude 77.1382°E is $5^{\mathrm{h}}08^{\mathrm{m}}33.2^{\mathrm{s}}$, so by Eq. (1.14) LST = $20^{\mathrm{h}}18^{\mathrm{m}}52.8^{\mathrm{s}}$. The bright star Deneb, at $\alpha = 20^{\mathrm{h}}41^{\mathrm{m}}$, has $H = -22^{\mathrm{m}}$ and will cross the meridian about 22 minutes later.

1.12 Problem Set: Time Systems and Solar Motion

1.12.1 Problem 1: Local Solar Time and Noon Zenith Angle at IISER TVM

Problem Statement: IISER TVM is at latitude 8.6833°N and longitude 77.1382°E. Indian Standard Time (IST) is the mean solar time of the meridian at 82.5°E, near Prayagraj.

(a) What is the local mean solar time at IISER TVM when it is 12:00 noon IST?

(b) What is the zenith angle of the Sun at noon on the June solstice at IISER TVM?

(c) At what IST does the Sun cross the meridian at IISER TVM on 3 November, when the equation of time is +16.4 minutes?

Essential Ideas:

- The Earth turns 15° per hour relative to the mean Sun, so local mean time differs from standard time by four minutes per degree of longitude, Eq. (1.11). Solar time follows the Sun, and sidereal time follows the stars. A sidereal clock gains about four minutes a day on a solar one (Figure 1.5).

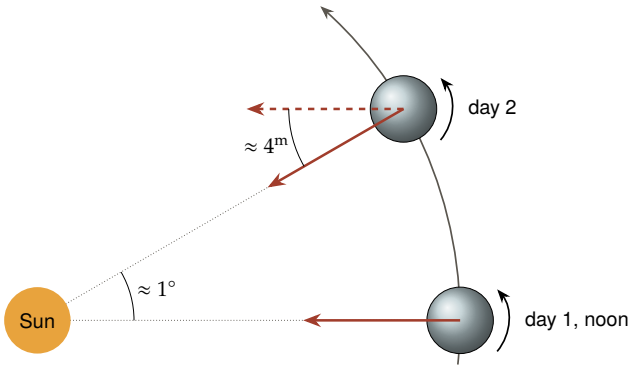


Figure 1.5: The Earth on two successive days, as in Figure 1.1, with the angle moved along the orbit in one day (≈ 1°) and the extra spin needed to face the Sun again (≈ 4^{m}).

- At noon the Sun is on the meridian, and its declination $\delta_\odot$ is the latitude of the point on the Earth where it is overhead. The Sun's rays are parallel, so the noon zenith angle is the angle at the Earth's centre between the observer and that point (Figure 1.6). On the June solstice $\delta_\odot = +\epsilon = +23.44°$ (Table 1.2).
- The Sun crosses the meridian when apparent solar time is 12^{h}, which is when mean solar time is 12^{h} − EoT, Eq. (1.19).

Detailed Solutions:

(a) **Local mean solar time**. IISER TVM is west of the IST meridian by

$$\lambda_{\text{ref}} - \lambda_{\text{obs}} = 82.5° - 77.1382° = 5.3618°.$$

At four minutes per degree this is

$$5.3618° \times 4^{m}/° = 21.4472^{m} = 21^{m}26.83^{s} \approx 21^{m}27^{s}.$$

IISER TVM is west of the reference meridian, so its noon comes later and its local time lags IST,

$$\boxed{\text{local mean solar time} = 12{:}00{:}00 - 0{:}21{:}27 = 11{:}38{:}33.}$$

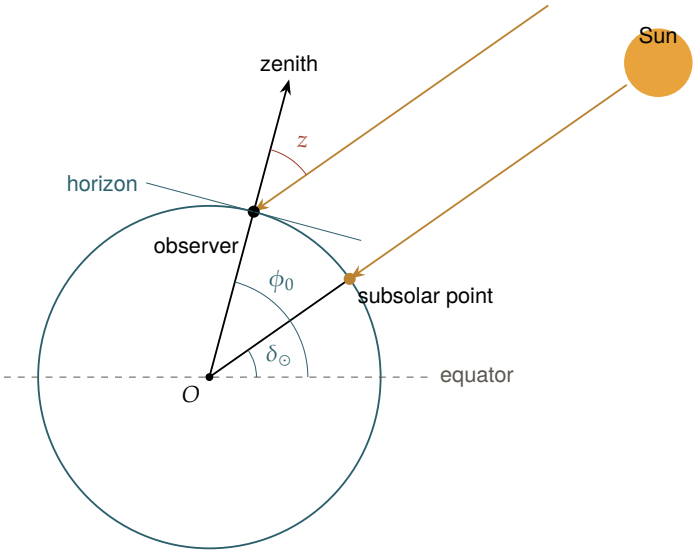


Figure 1.6: A cut through the Earth along the observer's meridian at local noon. The observer is at latitude ϕ_0 and the subsolar point at latitude $\delta_\odot$. Parallel rays from the Sun reach both, and z is the angle at the observer between the zenith and the Sun.

(b) **Zenith angle at solstice noon**. The radii to the observer and to the subsolar point make angles ϕ_0 and $\delta_\odot$ with the equator. The zenith at the observer lies along its radius, and the ray to the Sun is parallel to the radius through the subsolar point. So the angle between them at the observer equals the angle between the two radii at the centre, $|\phi_0 - \delta_\odot|$. (The same result follows from the full rotation between horizon and equator, Eq. (2.32).) With $\phi_0 = 8.6833^\circ$ and $\delta_\odot = 23.44^\circ$,

$$z = |\phi_0 - \delta_\odot| = |8.6833^\circ - 23.44^\circ| = 14.7567^\circ, \qquad \boxed{z \approx 14.76^\circ.}$$

The Sun is north of the zenith, since $\delta_\odot > \phi_0$. Its altitude above the northern horizon is $90^\circ - z = 75.24^\circ$.

(c) **Apparent noon on 3 November**. Mean noon at IISER TVM is at 12:21:27 IST, from part (a). The Sun crosses the meridian when mean solar time is $12^{\mathrm{h}} - 16.4^{\mathrm{m}}$, that is $16^{\mathrm{m}}24^{\mathrm{s}}$ before mean noon,

$$\boxed{12{:}21{:}27 - 0{:}16{:}24 = 12{:}05{:}03 \text{ IST}.}$$

A sundial at IISER TVM in early November reads noon only five minutes after the IST clock does, against twenty-one minutes on average. Anyone setting up a telescope or a sundial from a civil clock must allow for both the longitude and the equation of time.

Chapter 2

Coordinate Systems on the Celestial Sphere

Every direction in the sky is a point on the celestial sphere of Section 1.1. Two angles fix such a point, in the same way that latitude and longitude fix a town on the Earth. The two angles depend on a choice of pole and of a zero point on the equator around that pole. An observer on the ground uses the zenith as the pole. A star catalogue uses the celestial pole. The planets move most simply about the pole of the Earth's orbit, and the Milky Way is simplest about the pole of its own disk.

All four systems describe the same direction. Going from one to another is a turn of the axes, written as a matrix acting on a unit vector. Two physical effects then move a star away from where the geometry puts it. The air bends its light, and the motion of the Earth tilts it.

2.1 The Equatorial Coordinate System

Chapter 1 introduced the celestial poles, the celestial equator and the vernal equinox $\hat{\gamma}$, and built a sidereal clock from them (Section 1.1, Eq. (1.12)).

Words like "north of the equator" or "east of the equinox" cannot yet be computed with. To find the angle between two stars, or to turn to another frame, a direction has to be a vector.

Think of a town on a globe. Its latitude says how far north of the equator it lies. Its longitude says how far east of Greenwich it lies, counted along the equator. An arrow from the centre of the globe to the town is fixed by these two angles. Its height above the plane of the equator is the sine of the latitude. Its shadow on that plane has the cosine of the latitude as its length, and it points in the direction given by the longitude.

The sky uses the same two angles. Declination plays the part of latitude, and right ascension plays the part of longitude, with $\hat{\gamma}$ in place of Greenwich. A direction on the sky is a unit vector whose three components are products of sines and cosines of these two angles.

Vega is at right ascension $18^{\mathrm{h}}36^{\mathrm{m}}56^{\mathrm{s}} = 279.23^\circ$ and declination $+38.78^\circ$. Its arrow rises $\sin 38.78^\circ = 0.626$ above the plane of the equator. Its shadow has length $\cos 38.78^\circ = 0.779$ and points 279.23° round from $\hat{\gamma}$. The two components in the plane are $0.779 \cos 279.23^\circ = 0.125$ and $0.779 \sin 279.23^\circ = -0.769$.

The picture is an arrow from the centre of a ball to a dot on its surface, with the arrow's shadow on the equatorial plane.

The same construction is the spherical polar coordinates of mechanics, with polar angle $90^\circ - \delta$. It gave the line of sight to a binary in Chapter 6 (Section 6.2), and it gives the local directions on the sky used for proper motions in Chapter 3. It is pure geometry, so none of the six principles enters. What it gives is the unit vector on which every later formula acts.

The two angles. The **declination** $\delta \in [-90°, +90°]$ is the angle of a direction north (+) or south (−) of the celestial equator. The great circle through the two poles and a star is the star's **hour circle**. The **right ascension** α is the angle along the celestial equator from $\hat{\gamma}$ eastward to the hour circle. It runs from 0^{h} to 24^{h}, with $1^{\mathrm{h}} = 15°$ (Eq. (1.1)). Both are shown in Figure 2.2.

The unit vector. Put the z axis toward the north celestial pole and the x axis toward $\hat{\gamma}$. The y axis then points to $\alpha = 6^{\mathrm{h}}$, which makes the three axes right-handed. Split the direction $\hat{\mathbf{r}}$ of a star into a part along the pole and a part in the plane of the equator,

$$\begin{aligned} \hat{\mathbf{r}} &\overset{(1)}{=} \sin\delta\,\hat{\mathbf{z}} + \cos\delta\,\big(\cos\alpha\,\hat{\mathbf{x}} + \sin\alpha\,\hat{\mathbf{y}}\big) \\ &\overset{(2)}{=} \cos\delta\cos\alpha\,\hat{\mathbf{x}} + \cos\delta\sin\alpha\,\hat{\mathbf{y}} + \sin\delta\,\hat{\mathbf{z}}. \end{aligned} \tag{2.1}$$

(1) The angle between $\hat{\mathbf{r}}$ and the plane of the equator is δ. So the part along the pole is $\sin\delta$, and the shadow in the plane has length $\cos\delta$ and makes the angle α with $\hat{\mathbf{x}}$.
(2) Multiplying out the bracket.

We write the three components as a column, labelled by the frame they belong to,

$$\boxed{\hat{\mathbf{r}}_{\mathrm{eq}} = \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}.} \tag{2.2}$$

Its length is one for every (α, δ). It does not depend on the time or on where on the Earth the observer stands, because the grid turns with the stars.

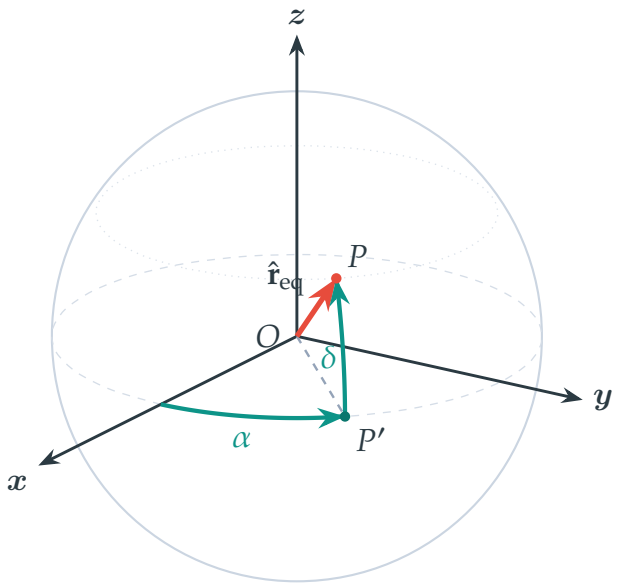


Figure 2.1: The unit vector $\hat{\mathbf{r}}_{\mathrm{eq}}$ toward a point P in the equatorial frame, with its right ascension α measured from the x axis and its declination δ measured up from the equatorial plane.

The values of (α, δ) in a catalogue change only slowly, through the star's own motion across the sky and through precession. The daily rising and setting of the stars comes from the observer's horizon turning under this fixed grid.

The grid of Figure 2.3 is the one that planetarium software draws on the real sky (Figure 2.4). In both, α grows clockwise around the pole for an observer facing north. The reason is that a

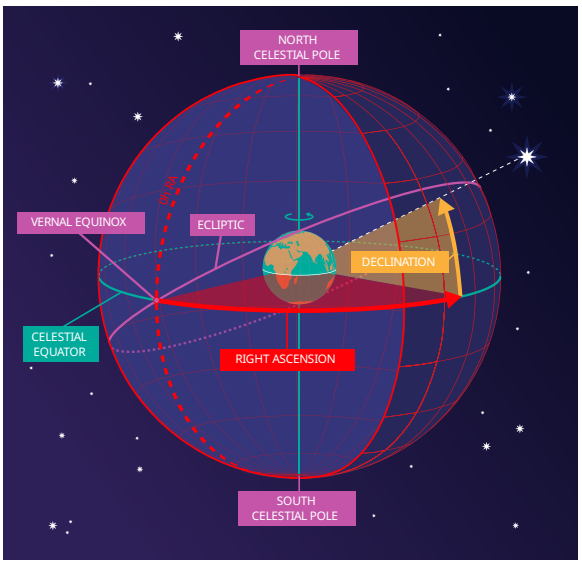


Figure 2.2: The celestial sphere around the Earth, with the north and south celestial poles, the celestial equator, the ecliptic, the vernal equinox, and the right ascension and declination of a star.

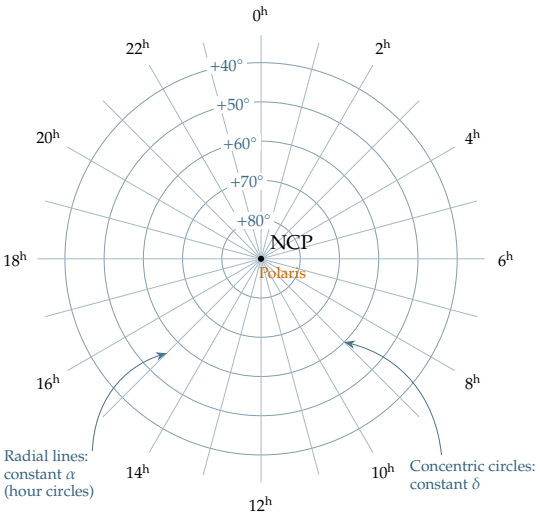


Figure 2.3: The equatorial grid near the north celestial pole, as seen by an observer facing north. Radial lines are hour circles 1^h apart. Circles are parallels of constant δ, $10°$ apart.

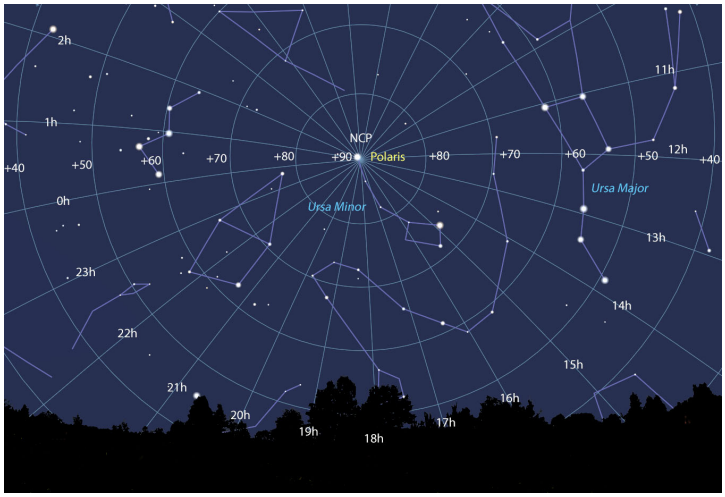


Figure 2.4: An equatorial grid drawn over the sky near the north celestial pole, with Polaris and the constellations Ursa Minor and Ursa Major.

star of larger α reaches the meridian later, so at any moment it lies further east, and east is on the right for someone facing north.

The local sidereal time and the hour angle of Chapter 1 are two more angles on the same sphere. Figure 2.5 marks $\hat{\gamma}$, the observer's meridian and a star S together. The right ascension now on the meridian is the local sidereal time, Eq. (1.12). In that figure the star is still east of the meridian. As the Earth turns, the LST grows while the star's α stays fixed. Figure 2.6 shows the same sky later, after the meridian has passed the star. The arc from the meridian back to the star is then the hour angle H, defined in Section 2.3. In the earlier figure the same arc points the other way and has length $-H$.

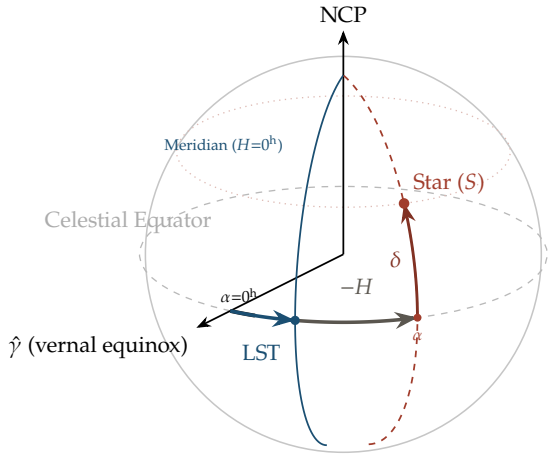


Figure 2.5: The vernal equinox $\hat{\gamma}$, the observer's meridian and a star S at (α, δ) east of the meridian, on one celestial sphere.

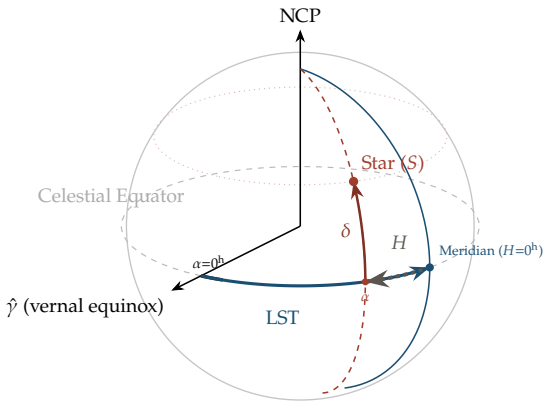


Figure 2.6: The same sky as Figure 2.5 at a later instant. $\hat{\gamma}$ and S are unchanged, and the meridian has moved past the star.

2.1.1 Coordinate Epochs

The equinox $\hat{\gamma}$ that anchors the grid moves. The Earth's axis precesses, and $\hat{\gamma}$ slides west along the ecliptic by about 50.3″ per year, Eq. (1.35). A position (α, δ) therefore means nothing until it states the instant at which the grid was taken. That instant is the **epoch**. The standard today is J2000.0, and older catalogues used B1950.0. Some names carry the epoch. The pulsar PSR J1939+2134 is named after its J2000 position, $\alpha \approx 19^h39^m$ and $\delta \approx +21°34'$.

Moving a position from one epoch to another needs two corrections. One is the precession of the grid, Eqs. (1.37)–(1.38). The other is the star's own motion across the sky, its proper motion $(\mu_\alpha^*, \mu_\delta)$ of Chapter 3. The two are independent, and over a few decades they simply add. To restate a catalogue from B1950.0 to J2000.0 with the stars held fixed, only the precession is needed.

2.2 Turning the Axes

The equatorial frame gives each star a column of three numbers, Eq. (2.2).

The other frames of the sky put their pole or their zero point somewhere else. We need the same star's components in those frames.

Think of a plan of a room drawn with the x axis along one wall. A lamp stands 3 m along that wall from the corner. A friend draws the same room with axes turned 30° anticlockwise. The lamp has not moved. Only the axes have. Its new coordinates are its distances measured along the friend's axes, and each of those is a dot product of the lamp's position with one of the friend's axes.

A star's direction does not change when we change frame either. The idea is that the new components are dot products with the new axes, so the matrix of the change has the new axes as its rows.

The lamp in numbers. The friend's axes are $(\cos 30°, \sin 30°)$ and $(-\sin 30°, \cos 30°)$ in the old frame. The lamp at $(3, 0)$ has new coordinates $3\cos 30° = 2.60$ and $-3\sin 30° = -1.50$. Stacking the two axes as rows gives the same answer as one matrix product,

$$\begin{pmatrix} \cos 30° & \sin 30° \\ -\sin 30° & \cos 30° \end{pmatrix} \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.60 \\ -1.50 \end{pmatrix}.$$

The picture is two sets of axes with one origin, one turned against the other, and a single fixed arrow.

The same matrix is the passive rotation of mechanics and the change of basis of quantum mechanics. Chapter 6 used two such turns to place an orbit in space, Eq. (6.7). It gives the two turns every frame of the sky is built from, and the fact that undoing a turn costs only a transpose.

The rule. Let a new frame have orthonormal axes $\hat{\mathbf{x}}', \hat{\mathbf{y}}', \hat{\mathbf{z}}'$, each written by its components in the old frame. The new components of the direction $\hat{\mathbf{r}}$ are

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} \hat{\mathbf{x}}' \cdot \hat{\mathbf{r}} \\ \hat{\mathbf{y}}' \cdot \hat{\mathbf{r}} \\ \hat{\mathbf{z}}' \cdot \hat{\mathbf{r}} \end{pmatrix} \overset{(2)}{=} \underbrace{\begin{pmatrix} — \, \hat{\mathbf{x}}'^{T} \, — \\ — \, \hat{\mathbf{y}}'^{T} \, — \\ — \, \hat{\mathbf{z}}'^{T} \, — \end{pmatrix}}_{\mathbf{M}} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \tag{2.3}$$

(1) In an orthonormal frame each component of a vector is its dot product with one axis.
(2) Each dot product is the row of the new axis's old components times the column of old components.

The entries of $\mathbf{M}\mathbf{M}^T$ are the dot products of the new axes with each other,

$$\mathbf{M}\mathbf{M}^T = \begin{pmatrix} \hat{\mathbf{x}}' \cdot \hat{\mathbf{x}}' & \hat{\mathbf{x}}' \cdot \hat{\mathbf{y}}' & \hat{\mathbf{x}}' \cdot \hat{\mathbf{z}}' \\ \hat{\mathbf{y}}' \cdot \hat{\mathbf{x}}' & \hat{\mathbf{y}}' \cdot \hat{\mathbf{y}}' & \hat{\mathbf{y}}' \cdot \hat{\mathbf{z}}' \\ \hat{\mathbf{z}}' \cdot \hat{\mathbf{x}}' & \hat{\mathbf{z}}' \cdot \hat{\mathbf{y}}' & \hat{\mathbf{z}}' \cdot \hat{\mathbf{z}}' \end{pmatrix} = \mathbb{1}, \qquad \text{so} \qquad \boxed{\mathbf{M}^{-1} = \mathbf{M}^T.} \tag{2.4}$$

Going back to the old frame multiplies by the transpose, whose columns are the new axes. The determinant of $\mathbf{M}$ is $+1$ when the new axes are right-handed, and then $\mathbf{M}$ is a rotation. It is -1 when they are left-handed. The horizontal frame of Section 2.3 is such a case.

Turning about the z axis. Turn the x and y axes by θ about z, anticlockwise as seen from $+z$. The new axes are $\hat{\mathbf{x}}' = \cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}}$, $\hat{\mathbf{y}}' = -\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}' = \hat{\mathbf{z}}$. Stacked as rows by Eq. (2.3),

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\mathbf{R}_z(\theta)} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \tag{2.5}$$

The same matrix from polar angles. A second route uses angles in the plane of the turn. Take the turn about the x axis, in the y–z plane (Figure 2.7). A point at distance r from the axis and at angle Θ from the old y axis has $(y, z) = (r\cos\Theta, r\sin\Theta)$. Measured from the new y' axis, turned by θ, its angle is $\Theta - \theta$. So

$$\begin{aligned} y' &= r\cos(\Theta - \theta) \overset{(1)}{=} r\cos\Theta\cos\theta + r\sin\Theta\sin\theta \overset{(2)}{=} y\cos\theta + z\sin\theta, \\ z' &= r\sin(\Theta - \theta) \overset{(1)}{=} r\sin\Theta\cos\theta - r\cos\Theta\sin\theta \overset{(2)}{=} -y\sin\theta + z\cos\theta. \end{aligned} \tag{2.6}$$

(1) The angle-difference identities.
(2) $r\cos\Theta = y$ and $r\sin\Theta = z$.

With $x' = x$ this is

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}, \tag{2.7}$$

which is $\mathbf{R}_z(\theta)$ with the axes relabelled $x \to y \to z \to x$, as the rule of rows gives it.

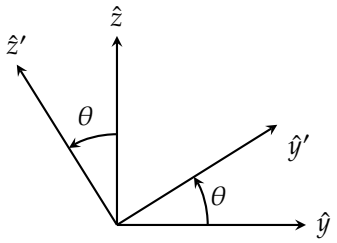


Figure 2.7: The y and z axes and the axes y' and z' turned by θ about the x axis.

What a turn about the pole does to a star. Apply $\mathbf{R}_z(\theta)$ to the equatorial column (2.2),

$$\mathbf{R}_z(\theta)\,\hat{\mathbf{r}}_{\text{eq}} \overset{(1)}{=} \begin{pmatrix} \cos\delta\cos\alpha\cos\theta + \cos\delta\sin\alpha\sin\theta \\ -\cos\delta\cos\alpha\sin\theta + \cos\delta\sin\alpha\cos\theta \\ \sin\delta \end{pmatrix} \overset{(2)}{=} \begin{pmatrix} \cos\delta\cos(\alpha-\theta) \\ \cos\delta\sin(\alpha-\theta) \\ \sin\delta \end{pmatrix}. \tag{2.8}$$

(1) Each row of Eq. (2.5) times the column.
(2) The angle-difference identities.

A turn by θ about the pole subtracts θ from every right ascension and leaves every declination alone.

2.3 Horizontal Coordinates

The equatorial frame gives each star a fixed column, Eq. (2.2), and Section 2.2 gives the matrix for any change of frame.

A star's (α, δ) does not say whether the star is above the horizon now, or where to point. That needs a frame tied to the ground.

Think of pointing out an aeroplane to a friend. You say "thirty degrees up, toward the east-north-east". One angle is the height above the horizon. The other is the compass direction, counted round from north. Neither needs any knowledge of the sky, only of the ground. The answer changes from minute to minute because the plane moves.

For a star it is the ground that moves. The **altitude** and **azimuth** use the zenith as their pole and turn with the Earth. The hour angle measures how far the sky has turned since the star crossed the meridian. The idea is that the horizontal frame is the equatorial frame turned about the pole to the meridian, and then tipped about the east–west line so that the zenith takes the place of the pole.

A star on the celestial equator crossing the meridian of IISER TVM, at latitude $\phi_0 = 8.68°\text{N}$, stands at altitude $90° - 8.68° = 81.32°$, due south. The pole is $8.68°$ above the northern horizon, so the equator, $90°$ from the pole, passes $8.68°$ south of the zenith.

The picture is a dome over flat ground. The zenith is overhead, the horizon is the rim, and the meridian is the arc from the north point through the zenith to the south point.

The east–north–up frame of surveying and of radar is the same frame. The hour angle is the clock of Chapter 1 read for one star. It gives the matrix between the equatorial and horizontal frames, in both directions.

The angles. The **altitude** $h \in [-90°, +90°]$ is the angle of a star above the horizon, $+90°$ at the zenith and $-90°$ at the nadir. The **zenith angle** is its distance from the zenith,

$$z = 90° - h. \tag{2.9}$$

The **azimuth** $A \in [0°, 360°)$ is the angle along the horizon from north (0°) through east (90°), south (180°) and west (270°). The **hour angle** of Eq. (1.15) is the angle along the celestial equator from the meridian westward to the star's hour circle,

$$H = \mathrm{LST} - \alpha, \tag{2.10}$$

with $H > 0$ after the star has crossed the meridian.

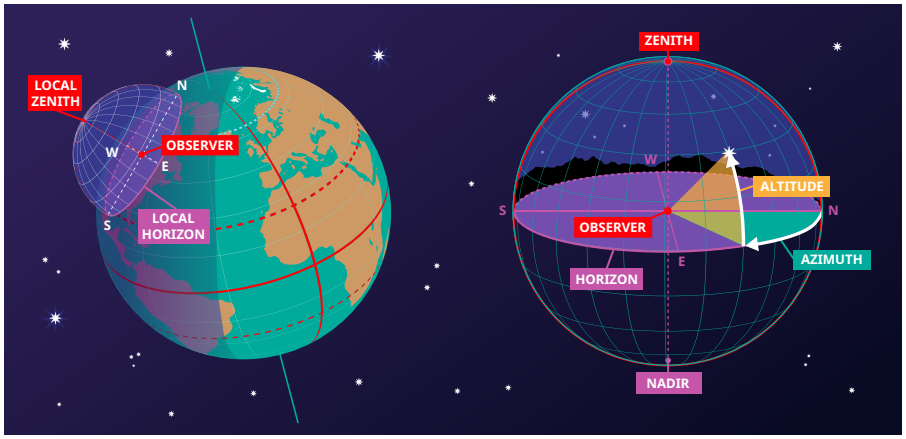


Figure 2.8: Left, an observer on the Earth with the local horizon and zenith. Right, the observer's sky with the horizon, zenith, nadir, the cardinal points, and the altitude and azimuth of a star.

2.3.1 The Meridian Frame

To compare the two systems we first turn the equatorial axes about the pole until the x axis lies in the observer's meridian. The right ascension on the meridian is LST, so the turn is $\mathbf{R}_z(\mathrm{LST})$, and Eq. (2.8) gives

$$\hat{\mathbf{r}}_{\mathrm{mer}} \overset{(1)}{=} \begin{pmatrix} \cos\mathrm{LST} & \sin\mathrm{LST} & 0 \\ -\sin\mathrm{LST} & \cos\mathrm{LST} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix} \tag{2.11}$$

$$\overset{(2)}{=} \begin{pmatrix} \cos\delta\cos(\alpha - \mathrm{LST}) \\ \cos\delta\sin(\alpha - \mathrm{LST}) \\ \sin\delta \end{pmatrix} \tag{2.12}$$

$$\overset{(3)}{=} \begin{pmatrix} \cos\delta\cos H \\ -\cos\delta\sin H \\ \sin\delta \end{pmatrix}. \tag{2.13}$$

(1) Eq. (2.5) with $\theta = \mathrm{LST}$, acting on Eq. (2.2).
(2) Eq. (2.8).
(3) $\alpha - \mathrm{LST} = -H$ by Eq. (2.10), with $\cos(-H) = \cos H$ and $\sin(-H) = -\sin H$.

In this frame the x axis points to where the meridian crosses the celestial equator, the y axis points to the east point of the horizon ($H = -90°$, $\delta = 0$), and the z axis still points to the pole. The right ascension and the sidereal time now enter only through their difference H.

2.3.2 The Local Axes

Next we need the observer's north, east and up directions in the meridian frame.

Up. The zenith lies on the meridian. Its declination is the latitude ϕ_0, because the angle between the celestial equator and the zenith is the angle between the Earth's equator and the local vertical. Setting $H = 0$ and $\delta = \phi_0$ in Eq. (2.13) gives the up direction $\hat{\mathbf{u}}$.

East. The east point of the horizon is 90° from the zenith and 90° from the pole, since it is on the horizon and both the zenith and the pole lie on the meridian. So it lies on the celestial equator, a quarter turn east of the meridian, at $\delta = 0$ and $H = -90°$.

North. North is the horizontal direction that completes the frame, $\hat{\mathbf{n}} = \hat{\mathbf{u}} \times \hat{\mathbf{e}}$. With the two vectors just found,

$$\hat{\mathbf{u}} = \begin{pmatrix} \cos\phi_0 \\ 0 \\ \sin\phi_0 \end{pmatrix}, \qquad \hat{\mathbf{e}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \tag{2.14}$$

$$\hat{\mathbf{n}} = \hat{\mathbf{u}} \times \hat{\mathbf{e}} \overset{(1)}{=} \begin{pmatrix} 0 \cdot 0 - \sin\phi_0 \cdot 1 \\ \sin\phi_0 \cdot 0 - \cos\phi_0 \cdot 0 \\ \cos\phi_0 \cdot 1 - 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} -\sin\phi_0 \\ 0 \\ \cos\phi_0 \end{pmatrix}. \tag{2.15}$$

(1) The components of a cross product, $(u_y e_z - u_z e_y,\ u_z e_x - u_x e_z,\ u_x e_y - u_y e_x)$.

A second route gives the same $\hat{\mathbf{n}}$. The north point is on the meridian, 90° beyond the zenith, past the pole. Putting $H = 0$ and $\delta = 90° + \phi_0$ into Eq. (2.13) gives $(\cos(90° + \phi_0), 0, \sin(90° + \phi_0)) = (-\sin\phi_0, 0, \cos\phi_0)$. For later reference the east vector is

$$\hat{\mathbf{e}} = (0, 1, 0)^T. \tag{2.16}$$

The triple $(\hat{\mathbf{n}}, \hat{\mathbf{e}}, \hat{\mathbf{u}})$ is left-handed, $\hat{\mathbf{n}} \times \hat{\mathbf{e}} = -\hat{\mathbf{u}}$, because the azimuth runs clockwise as seen from above.

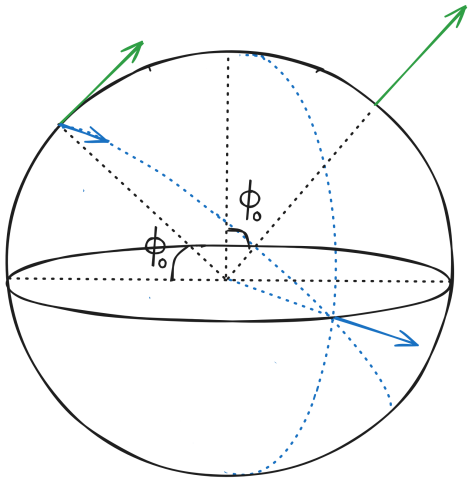


Figure 2.9: The local north (blue) and up (green) directions at an observer at latitude ϕ_0, drawn on the Earth.

2.3.3 The Transformation

The goal is (h, A) in terms of (δ, H). A star at altitude h and azimuth A has, by the same split as Eq. (2.2) with h for δ and A counted from north,

$$\hat{\mathbf{r}} = (\cos h \cos A)\,\hat{\mathbf{n}} + (\cos h \sin A)\,\hat{\mathbf{e}} + (\sin h)\,\hat{\mathbf{u}}. \tag{2.17}$$

In the meridian frame this is a matrix whose columns are the three local axes,

$$\hat{\mathbf{r}}_{\text{mer}} = \begin{pmatrix}\hat{\mathbf{n}} & \hat{\mathbf{e}} & \hat{\mathbf{u}}\end{pmatrix}\hat{\mathbf{r}}_{\text{hor}} = \underbrace{\begin{pmatrix}-\sin\phi_0 & 0 & \cos\phi_0 \\ 0 & 1 & 0 \\ \cos\phi_0 & 0 & \sin\phi_0\end{pmatrix}}_{\mathbf{M}(\phi_0)}\begin{pmatrix}\cos h\cos A \\ \cos h \sin A \\ \sin h\end{pmatrix}. \tag{2.18}$$

To solve for the horizontal column we invert $\mathbf{M}(\phi_0)$,

$$\begin{aligned}\begin{pmatrix}\cos h\cos A \\ \cos h \sin A \\ \sin h\end{pmatrix} &\overset{(1)}{=} \mathbf{M}^T(\phi_0)\,\hat{\mathbf{r}}_{\text{mer}} \\ &\overset{(2)}{=} \begin{pmatrix}-\sin\phi_0 & 0 & \cos\phi_0 \\ 0 & 1 & 0 \\ \cos\phi_0 & 0 & \sin\phi_0\end{pmatrix}\begin{pmatrix}\cos\delta\cos H \\ -\cos\delta\sin H \\ \sin\delta\end{pmatrix} \\ &\overset{(3)}{=} \begin{pmatrix}-\sin\phi_0\cos\delta\cos H + \cos\phi_0\sin\delta \\ -\cos\delta\sin H \\ \cos\phi_0\cos\delta\cos H + \sin\phi_0\sin\delta\end{pmatrix}.\end{aligned} \tag{2.19}$$

(1) The columns of $\mathbf{M}(\phi_0)$ are orthonormal, so $\mathbf{M}^{-1} = \mathbf{M}^T$, Eq. (2.4).
(2) $\mathbf{M}(\phi_0)$ is symmetric, so $\mathbf{M}^T = \mathbf{M}$. The column $\hat{\mathbf{r}}_{\text{mer}}$ is Eq. (2.13).
(3) Each row times the column.

Reading off the three rows,

Result 2.1

$$\sin h = \sin\phi_0\sin\delta + \cos\phi_0\cos\delta\cos H, \tag{2.20}$$
$$\cos h\sin A = -\cos\delta\sin H, \tag{2.21}$$
$$\cos h\cos A = -\sin\phi_0\cos\delta\cos H + \cos\phi_0\sin\delta. \tag{2.22}$$

The star enters only through δ, the time only through H, and the site only through ϕ_0. The site's longitude enters only through the LST inside H. The last row is also $\hat{\mathbf{u}}\cdot\hat{\mathbf{r}} = \cos z$, the dot product of the up vector (2.14) with Eq. (2.13).

The azimuth. Dividing Eq. (2.21) by Eq. (2.22),

$$\tan A = \frac{-\cos\delta\sin H}{-\sin\phi_0\cos\delta\cos H + \cos\phi_0\sin\delta} \overset{(1)}{=} \frac{\sin H}{\sin\phi_0\cos H - \cos\phi_0\tan\delta}. \tag{2.23}$$

(1) Numerator and denominator divided by $-\cos\delta$.

That division flips the sign of both parts, so the second form cannot tell A from $A + 180°$. The quadrant comes from the signs of Eqs. (2.21) and (2.22) themselves. $\sin A$ has the sign of $\cos h\sin A$, and $\cos A$ the sign of $\cos h\cos A$, because $\cos h \geq 0$.

Going back. The product of $\mathbf{M}(\phi_0)$ with itself is

$$\mathbf{M}(\phi_0)\,\mathbf{M}(\phi_0) = \begin{pmatrix} \sin^2\phi_0 + \cos^2\phi_0 & 0 & -\sin\phi_0\cos\phi_0 + \cos\phi_0\sin\phi_0 \\ 0 & 1 & 0 \\ -\cos\phi_0\sin\phi_0 + \sin\phi_0\cos\phi_0 & 0 & \cos^2\phi_0 + \sin^2\phi_0 \end{pmatrix} = \mathbb{1}, \tag{2.24}$$

so the same matrix also takes horizontal to equatorial coordinates. Eq. (2.18) multiplied out reads

$$\boxed{\begin{aligned} \sin\delta &= \sin\phi_0\sin h + \cos\phi_0\cos h\cos A, \\ \cos\delta\cos H &= \cos\phi_0\sin h - \sin\phi_0\cos h\cos A, \\ -\cos\delta\sin H &= \cos h\sin A. \end{aligned}} \tag{2.25}$$

These are Eqs. (2.20)–(2.22) with the pairs (δ, H) and (h, A) exchanged. The reason is that $\mathbf{M}(\phi_0)$ sends the pole $(0, 0, 1)^T$ to the zenith $\hat{\mathbf{u}}$ and the zenith back to the pole, and leaves east alone. It is a mirror, not a rotation. Its determinant is $-\sin^2\phi_0 - \cos^2\phi_0 = -1$, as it must be for a left-handed set of axes. Tipping the pole onto the zenith about the east–west line would be a rotation. Counting the azimuth clockwise reverses one axis on top of that.

The astronomical triangle. The pole, the zenith and the star are the corners of a spherical triangle (Figure 2.10). Its sides are the co-latitude $90° - \phi_0$, the star's distance from the pole $90° - \delta$, and its zenith angle z. Its angle at the pole is H. For a star west of the meridian its angle at the zenith is $360° - A$. Eq. (2.20) is the cosine rule of spherical trigonometry for the side z. The matrix route gives all three relations at once and with their signs. The third angle, at the star, is the parallactic angle q of Section 2.10.2.

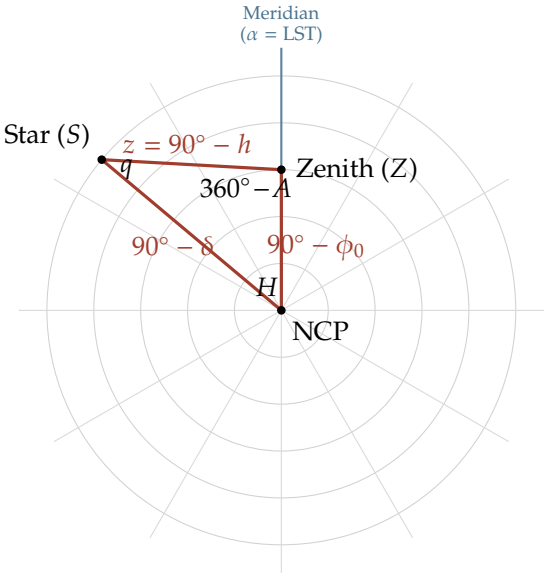


Figure 2.10: The pole–zenith–star triangle for a star west of the meridian, drawn over the equatorial grid near the pole as seen facing north. The sides are $90° - \phi_0$, $90° - \delta$ and z. The angles are H at the pole, $360° - A$ at the zenith and q at the star. The sides are drawn straight, though each is an arc of a great circle.

Worked example, the Sun at an equinox. At an equinox $\delta_\odot = 0$. Three hours after local noon the Sun's hour angle is $H = 3^{\mathrm{h}} = 45^\circ$. At IISER TVM, $\phi_0 = 8.6833^\circ$, Eqs. (2.20)–(2.22) give

$$\begin{aligned} \sin h &= \cos(8.6833^\circ)\cos 45^\circ = 0.6990, & h &= 44.35^\circ, \\ \cos h \sin A &= -\sin 45^\circ = -0.7071, \\ \cos h \cos A &= -\sin(8.6833^\circ)\cos 45^\circ = -0.1068. \end{aligned}$$

Both $\sin A$ and $\cos A$ are negative, so A is in the third quadrant,

$$A = 180^\circ + \arctan(0.7071/0.1068) = 261.41^\circ.$$

The Sun is 8.59° south of due west. At an equinox the Sun runs along the celestial equator, which crosses the horizon at the east and west points and passes south of the zenith at this site. So the Sun stays a little south of the east–west line all day.

2.4 Rising, Setting, and Circumpolar Stars

Eq. (2.20) gives the altitude of a star at any hour angle.

To plan a night one wants the reverse, the times when a star is up at all.

Think of a big wheel at a fair whose lower part dips behind a fence. Seats near the hub may never go behind the fence. Seats on the rim spend part of each turn hidden. How much of each turn a seat is hidden depends on its distance from the hub and on how high the fence is compared with the hub.

Each star moves once a sidereal day on a circle of constant δ around the pole. The horizon cuts that circle, and the height of the pole above the horizon is the latitude. The idea is to set $h = 0$ in Eq. (2.20) and solve for the hour angle.

On the equator, $\phi_0 = 0$, Eq. (2.20) reads $\sin h = \cos\delta\cos H$. Every star is on the horizon at $H = \pm 90^\circ = \pm 6^{\mathrm{h}}$, so every star is up for exactly half a sidereal day.

The picture is a set of parallel circles tilted to the horizon by $90^\circ - \phi_0$ and cut by it (Figure 2.11).

With the Sun's declination in place of the star's, the same formula gives the length of the day, the midnight Sun near the poles and the polar night. It gives the hour angles of rising and setting and the conditions for a star never to set or never to rise.

The rising and setting hour angle. A star rises or sets when its centre crosses the horizon, $h = 0$. In Eq. (2.20),

$$\begin{aligned} 0 &\overset{(1)}{=} \sin\phi_0 \sin\delta + \cos\phi_0 \cos\delta \cos H_0, \\ \cos H_0 &\overset{(2)}{=} -\frac{\sin\phi_0 \sin\delta}{\cos\phi_0 \cos\delta}. \end{aligned} \tag{2.26}$$

(1) Eq. (2.20) with $\sin h = 0$.
(2) Divided by $\cos\phi_0 \cos\delta$, which is positive away from the poles.

$$\boxed{\cos H_0 = -\tan\phi_0 \tan\delta.} \tag{2.27}$$

The equation has two roots. The star sets at $H = +H_0$ and rises at $H = -H_0$, with $H_0 = \arccos\bigl(-\tan\phi_0\tan\delta\bigr)$ between 0° and 180°. By Eq. (2.10) the sidereal times are

$$\mathrm{LST}_{\mathrm{rise}} = \alpha - H_0, \qquad \mathrm{LST}_{\mathrm{set}} = \alpha + H_0. \tag{2.28}$$

The star is up for $2H_0$ of sidereal time. A star north of the equator seen from a northern site has $\tan\phi_0\tan\delta > 0$, so $H_0 > 90^\circ$ and it is up for more than half the day. The formula ignores refraction, which lifts a star on the horizon by about half a degree (Section 2.9), so a star is seen to rise a little before $-H_0$.

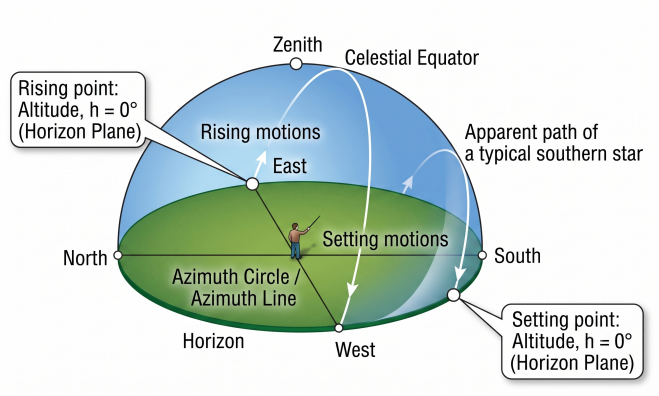


Figure 2.11: Paths of stars across an observer's sky, with a star rising in the east and setting in the west, and the shorter path of a southern star.

Stars that never set or never rise. A root exists only if $|\tan\phi_0 \tan\delta| \le 1$. Take a northern site, $\phi_0 > 0$. If $\tan\phi_0 \tan\delta > 1$ then $\cos H_0 < -1$, there is no crossing, and $h > 0$ all day. The star is **circumpolar**. The condition is

$$\tan\delta \overset{(1)}{>} \frac{1}{\tan\phi_0} \overset{(2)}{=} \tan\bigl(90^\circ - \phi_0\bigr) \quad\Longleftrightarrow\quad \delta > 90^\circ - \phi_0. \tag{2.29}$$

(1) $\tan\phi_0 \tan\delta > 1$ divided by $\tan\phi_0 > 0$.
(2) $1/\tan\phi_0 = \cot\phi_0 = \tan\bigl(90^\circ - \phi_0\bigr)$. The last step uses that tan increases on $(-90^\circ, 90^\circ)$.

In the same way a star with $\tan\phi_0 \tan\delta < -1$, that is $\delta < -(90^\circ - \phi_0)$, never rises. For a southern site the roles of the two poles swap.

A second route checks the condition. The star is lowest at lower transit, $H = 180^\circ$, where Eq. (2.20) gives

$$\sin h_{\text{low}} = \sin\phi_0 \sin\delta - \cos\phi_0 \cos\delta \overset{(1)}{=} -\cos\bigl(\phi_0 + \delta\bigr) \overset{(2)}{=} \sin\bigl(\phi_0 + \delta - 90^\circ\bigr), \tag{2.30}$$

(1) The angle-sum identity for the cosine.
(2) $-\cos x = \sin(x - 90^\circ)$.

so $h_{\text{low}} = \phi_0 + \delta - 90^\circ$. This is above the horizon exactly when $\delta > 90^\circ - \phi_0$, as in Eq. (2.29).

From IISER TVM, $90^\circ - \phi_0 = 81.32^\circ$. Polaris, at $\delta = +89.26^\circ$, is circumpolar. It circles the pole between $h_{\text{low}} = 8.68^\circ + 89.26^\circ - 90^\circ = 7.95^\circ$ and 9.42°, low in the north all night.

2.5 Altitude at Meridian Transit

Eq. (2.20) gives the altitude at any hour angle. A star is highest when it crosses the meridian, at $H = 0$. That is the best moment to observe it, and for the Sun it is local noon. On the meridian the pole, the zenith and the star lie on one great circle. The zenith is the point of that circle at declination ϕ_0, so the zenith angle is just the difference of two declinations, as the distance between two towns on one meridian of the Earth is the difference of their latitudes. Chapter 1 used this for the noon Sun at IISER TVM in its problem set.

The highest altitude. Put $H = 0$ in Eq. (2.20),

$$\begin{aligned} \sin h_{\max} &\overset{(1)}{=} \sin\phi_0 \sin\delta + \cos\phi_0 \cos\delta \\ &\overset{(2)}{=} \cos\big(\phi_0 - \delta\big) \\ &\overset{(3)}{=} \sin\big(90^\circ - |\phi_0 - \delta|\big). \end{aligned} \tag{2.31}$$

(1) $\cos H = 1$.
(2) The angle-difference identity for the cosine.
(3) The cosine is even, so $\cos\big(\phi_0 - \delta\big) = \cos|\phi_0 - \delta|$, and $\cos x = \sin(90^\circ - x)$.

Both $h_{\max}$ and $90^\circ - |\phi_0 - \delta|$ lie between -90° and 90°, so they are equal, and by Eq. (2.9)

$$\boxed{z_{\min} = |\phi_0 - \delta|.} \tag{2.32}$$

The star crosses south of the zenith if $\delta < \phi_0$ and north of it if $\delta > \phi_0$. At local solar noon the Sun is on the meridian, so its zenith angle is $z_{\text{noon}} = |\phi_0 - \delta_\odot|$.

Worked example, the June solstice at IISER TVM. At the June solstice $\delta_\odot = +23.44^\circ$. With $\phi_0 = 8.6833^\circ\text{N}$,

$$z_{\text{noon}} = |8.6833^\circ - 23.44^\circ| = 14.76^\circ.$$

Since $\delta_\odot > \phi_0$, the noon Sun is 14.76° north of the zenith.

2.6 Angular Distance on the Sphere

Each star has a unit vector, Eq. (2.2).

The angle between two stars is not found by subtracting their coordinates. Near the pole, two stars 12^h apart in right ascension can be only a few degrees apart on the sky.

Think of two towns at latitude 60°N on opposite sides of the globe, 180° apart in longitude. Walking west along the parallel covers $180^\circ \times \cos 60^\circ = 90^\circ$ of arc. Going straight over the pole covers only $30^\circ + 30^\circ = 60^\circ$. The shortest way is the great circle, and its angle is the angle between the two arrows from the centre of the Earth.

Two stars are two such arrows. The idea is that the cosine of the angle between them is the dot product of their unit vectors.

For the two towns, with $\delta_1 = \delta_2 = 60^\circ$ and $\alpha_1 - \alpha_2 = 180^\circ$, the formula derived below gives $\cos\theta = \sin^2 60^\circ + \cos^2 60^\circ \cos 180^\circ = 0.75 - 0.25 = 0.5$, so $\theta = 60^\circ$, the route over the pole.

The picture is two arrows from the centre of a ball, with the short arc between their tips.

In navigation this is the great-circle distance, and in spherical trigonometry it is the cosine rule. The scattering angle between two directions in physics is found the same way. It gives the separation of two stars, a form of it that keeps its accuracy for small angles, and the small-angle limit used for proper motions.

The cosine rule. For two unit vectors, $\cos\theta = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2$. With Eq. (2.2),

$$\begin{aligned} \cos\theta &\overset{(1)}{=} (\cos\delta_1\cos\alpha_1)(\cos\delta_2\cos\alpha_2) + (\cos\delta_1\sin\alpha_1)(\cos\delta_2\sin\alpha_2) + \sin\delta_1\sin\delta_2 \\ &\overset{(2)}{=} \cos\delta_1\cos\delta_2\big(\cos\alpha_1\cos\alpha_2 + \sin\alpha_1\sin\alpha_2\big) + \sin\delta_1\sin\delta_2 \\ &\overset{(3)}{=} \sin\delta_1\sin\delta_2 + \cos\delta_1\cos\delta_2\cos(\alpha_1 - \alpha_2). \end{aligned} \tag{2.33}$$

(1) The dot product of the two columns (2.2), component by component.
(2) $\cos\delta_1\cos\delta_2$ taken out of the first two terms.
(3) The angle-difference identity for the cosine.

Result 2.2

$$\cos\theta = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 = \sin\delta_1\sin\delta_2 + \cos\delta_1\cos\delta_2\cos(\alpha_1 - \alpha_2). \tag{2.34}$$

The result depends on the right ascensions only through their difference, as it must, since a turn about the pole changes neither the angle between two stars nor that difference.

Small angles. For close pairs Eq. (2.34) loses accuracy. Then $\cos\theta \approx 1 - \theta^2/2$, and θ comes from the small difference of numbers close to one. A computer stores numbers with a relative error of about 10^{-16}, so $\theta^2/2$ is lost once θ is below about 1.5×10^{-8} rad, which is 3 milliarcseconds. The cure is to compute $1 - \cos\theta = 2\sin^2(\theta/2)$ directly,

$$\begin{aligned} 2\sin^2\frac{\theta}{2} &\overset{(1)}{=} 1 - \sin\delta_1\sin\delta_2 - \cos\delta_1\cos\delta_2\cos(\alpha_2 - \alpha_1) \\ &\overset{(2)}{=} 1 - \sin\delta_1\sin\delta_2 - \cos\delta_1\cos\delta_2 + 2\cos\delta_1\cos\delta_2\sin^2\frac{\alpha_2 - \alpha_1}{2} \\ &\overset{(3)}{=} 1 - \cos(\delta_2 - \delta_1) + 2\cos\delta_1\cos\delta_2\sin^2\frac{\alpha_2 - \alpha_1}{2} \\ &\overset{(4)}{=} 2\sin^2\frac{\delta_2 - \delta_1}{2} + 2\cos\delta_1\cos\delta_2\sin^2\frac{\alpha_2 - \alpha_1}{2}. \end{aligned} \tag{2.35}$$

(1) $1 - \cos\theta = 2\sin^2(\theta/2)$, with $\cos\theta$ from Eq. (2.34).
(2) $\cos x = 1 - 2\sin^2(x/2)$ for $x = \alpha_2 - \alpha_1$.
(3) The angle-difference identity, $\cos\delta_1\cos\delta_2 + \sin\delta_1\sin\delta_2 = \cos(\delta_2 - \delta_1)$.
(4) $1 - \cos x = 2\sin^2(x/2)$ again, for $x = \delta_2 - \delta_1$.

Dividing by two gives the **haversine formula**,

$$\sin^2\left(\frac{\theta}{2}\right) = \sin^2\left(\frac{\delta_2 - \delta_1}{2}\right) + \cos\delta_1\cos\delta_2\sin^2\left(\frac{\alpha_2 - \alpha_1}{2}\right). \tag{2.36}$$

Every term is now small when the stars are close, so nothing is lost.

The flat-sky limit. For small separations each sine equals its argument. Writing $\Delta\delta = \delta_2 - \delta_1$ and $\Delta\alpha = \alpha_2 - \alpha_1$, and keeping the lowest order,

$$\boxed{\theta^2 \approx \Delta\delta^2 + \cos^2\delta\,\Delta\alpha^2.} \tag{2.37}$$

A step in right ascension is shortened on the sky by $\cos\delta$, because the hour circles crowd together toward the pole. This is why the proper motion in right ascension carries the factor $\cos\delta$ in Eq. (3.19).

2.7 The Ecliptic Coordinate System

Section 2.2 gives the matrix for a turn about any axis.

The Sun and the planets move close to the ecliptic, which is tilted to the celestial equator. In equatorial coordinates the Sun's declination swings between $-23.44°$ and $+23.44°$ every year. A frame whose equator is the ecliptic removes that swing.

Think of a model railway laid on a tilted table. Measured from the floor, the train rises and falls as it goes round. Measured from the table top, it stays at height zero. Choosing the table as the reference plane removes the rising and falling.

The ecliptic is the table, tilted by the obliquity $\epsilon = 23.44°$, and the Sun is the train. In ecliptic coordinates its latitude is always zero. The idea is that the ecliptic frame is the equatorial frame turned by ϵ about the line to $\hat{\gamma}$, which lies in both planes.

At the June solstice the Sun is at ecliptic longitude 90°, a quarter of the way round from $\hat{\gamma}$. That is where the tilted table stands highest above the floor, so the Sun's declination is the full tilt, $\delta_\odot = +\epsilon = +23.44°$, at $\alpha = 6^{\mathrm{h}}$.

The picture is two great circles that cross at $\hat{\gamma}$ at the angle ϵ (Figure 2.12).

The ecliptic is the plane of the Earth's orbit, so ϵ is the inclination of the Earth's equator to its orbit. The turn is the same $\mathbf{R}_x$ that tilted an orbital plane by its inclination in Chapter 6, Eq. (6.7). It gives the transformation in both directions and the Sun's declination through the year.

The angles. The **ecliptic longitude** $\lambda \in [0°, 360°)$ is measured eastward along the ecliptic from $\hat{\gamma}$. The **ecliptic latitude** $\beta \in [-90°, +90°]$ is the angle north or south of the ecliptic (Figure 2.13). By the split of Eq. (2.2), a star has the ecliptic column

$$\hat{\mathbf{r}}_{\mathrm{ecl}} = (\cos\beta\cos\lambda,\ \cos\beta\sin\lambda,\ \sin\beta)^T.$$

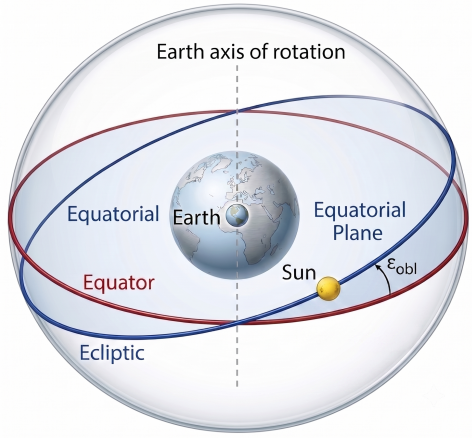


Figure 2.12: The celestial equator and the ecliptic around the Earth, crossing at the obliquity $\epsilon \simeq 23.44°$.

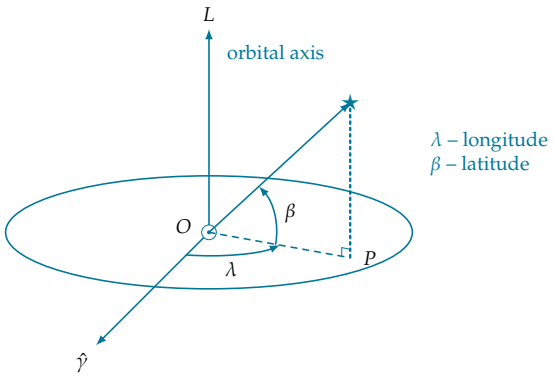


Figure 2.13: Ecliptic longitude λ, measured from $\hat{\gamma}$ in the ecliptic plane, and ecliptic latitude β, measured from that plane toward the orbital axis.

The ecliptic axes in equatorial components. The vernal equinox is on both circles (Section 1.6), so both frames share $\hat{\mathbf{x}}$. The June solstice point is on the ecliptic at $\alpha = 90°$, $\delta = \epsilon$, so its column (2.2) is $(0, \cos\epsilon, \sin\epsilon)^T$. The ecliptic pole is at right angles to both, so it is their cross product, and the third axis completes the set,

$$\hat{\mathbf{z}}_{\mathrm{ecl}} \overset{(1)}{=} \hat{\mathbf{x}} \times \begin{pmatrix} 0 \\ \cos\epsilon \\ \sin\epsilon \end{pmatrix} = \begin{pmatrix} 0\cdot\sin\epsilon - 0\cdot\cos\epsilon \\ 0\cdot 0 - 1\cdot\sin\epsilon \\ 1\cdot\cos\epsilon - 0\cdot 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -\sin\epsilon \\ \cos\epsilon \end{pmatrix},$$

$$\hat{\mathbf{y}}_{\mathrm{ecl}} \overset{(2)}{=} \hat{\mathbf{z}}_{\mathrm{ecl}} \times \hat{\mathbf{x}} = \begin{pmatrix} -\sin\epsilon\cdot 0 - \cos\epsilon\cdot 0 \\ \cos\epsilon\cdot 1 - 0\cdot 0 \\ 0\cdot 0 - (-\sin\epsilon)\cdot 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \cos\epsilon \\ \sin\epsilon \end{pmatrix}. \tag{2.38}$$

(1) The pole of the ecliptic is perpendicular to every direction in the ecliptic, here $\hat{\mathbf{x}}$ and the solstice. The order of the factors puts it on the north side, $\cos\epsilon > 0$.
(2) A right-handed set has $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$.

The ecliptic y axis points to the solstice, $\lambda = 90°$, as it should. The ecliptic pole lies at $\alpha = 18^{\mathrm{h}}$ and $\delta = 90° - \epsilon$.

The transformation. Stacking $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}_{\mathrm{ecl}}$ and $\hat{\mathbf{z}}_{\mathrm{ecl}}$ as rows, by Eq. (2.3), gives exactly $\mathbf{R}_x(\epsilon)$ of Eq. (2.7),

$$\begin{pmatrix} \cos\beta\cos\lambda \\ \cos\beta\sin\lambda \\ \sin\beta \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\epsilon & \sin\epsilon \\ 0 & -\sin\epsilon & \cos\epsilon \end{pmatrix} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix} \tag{2.39}$$
$$\overset{(2)}{=} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha\cos\epsilon + \sin\delta\sin\epsilon \\ -\cos\delta\sin\alpha\sin\epsilon + \sin\delta\cos\epsilon \end{pmatrix}.$$

(1) Eq. (2.3) with the ecliptic axes as rows, acting on Eq. (2.2).
(2) Each row times the column.

Row by row,

Result 2.3

$$\sin\beta = \sin\delta\cos\epsilon - \cos\delta\sin\alpha\sin\epsilon, \tag{2.40}$$
$$\cos\beta\cos\lambda = \cos\delta\cos\alpha, \tag{2.41}$$
$$\cos\beta\sin\lambda = \cos\delta\sin\alpha\cos\epsilon + \sin\delta\sin\epsilon. \tag{2.42}$$

The longitude needs both of the last two rows. Their ratio alone, $\tan\lambda$, cannot tell λ from $\lambda + 180°$.

Going back, and the Sun's declination. The inverse is the transpose, Eq. (2.4), so $\hat{\mathbf{r}}_{\mathrm{eq}} = \mathbf{R}_x^T(\epsilon)\,\hat{\mathbf{r}}_{\mathrm{ecl}}$. For the Sun, $\beta = 0$ and $\hat{\mathbf{r}}_{\mathrm{ecl}} = (\cos\lambda_\odot, \sin\lambda_\odot, 0)^T$, so

$$\begin{pmatrix} \cos\delta_\odot\cos\alpha_\odot \\ \cos\delta_\odot\sin\alpha_\odot \\ \sin\delta_\odot \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\epsilon & -\sin\epsilon \\ 0 & \sin\epsilon & \cos\epsilon \end{pmatrix} \begin{pmatrix} \cos\lambda_\odot \\ \sin\lambda_\odot \\ 0 \end{pmatrix} \overset{(2)}{=} \begin{pmatrix} \cos\lambda_\odot \\ \cos\epsilon\sin\lambda_\odot \\ \sin\epsilon\sin\lambda_\odot \end{pmatrix}. \tag{2.43}$$

(1) The transpose of Eq. (2.7) with $\theta = \epsilon$.
(2) Each row times the column.

$$\boxed{\sin\delta_\odot = \sin\epsilon\,\sin\lambda_\odot.} \tag{2.44}$$

The Sun's declination follows from its longitude alone. It is zero at the equinoxes, $\lambda_\odot = 0°$ and $180°$, and reaches $\pm\epsilon$ at the solstices, which is the table of Section 1.6.

2.8 The Galactic Coordinate System

The ecliptic frame was one turn about one axis, because the two planes share the line to $\hat{\gamma}$.

The disk of the Milky Way is tilted by about 63° to the celestial equator, and its zero point, the direction of the Galactic centre, does not lie on the line where the two planes cross. One turn is not enough.

Think of a camera on a tripod. Any way the camera can point is reached by three turns of the tripod head. The head pans about the vertical, tilts about a horizontal axis, and then the camera rolls about its own lens axis.

The Galactic frame is reached from the equatorial frame by the same three turns. A pan about the celestial pole brings the x axis to the line where the Galactic equator crosses the celestial equator. A tilt about that line carries the z axis to the north Galactic pole. A roll about the new pole brings the x axis to the Galactic centre. The idea is that the Galactic frame is the equatorial frame turned by three known angles.

In numbers the pan is 282.86°, the tilt is 62.87° and the roll is −32.93°. All three come from the pole's position and one longitude, given below.

The picture is the band of the Milky Way as a great circle tilted steeply to the celestial equator and crossing it twice.

Chapter 6 placed an orbit in space by the same turns, the node longitude Ω, the inclination ι and the argument of periapsis, Eq. (6.7). In rigid-body mechanics the three angles are the Euler angles. The matrix found here turns stellar velocities into Galactic ones in Chapter 3, Eq. (3.34).

The angles and the pole. The **Galactic latitude** $b \in [-90°, +90°]$ is the angle north (+) or south (−) of the Galactic plane. The **Galactic longitude** $l \in [0°, 360°)$ is measured along the Galactic equator from the direction of the Galactic centre, toward the direction of the Galaxy's rotation at $l = 90°$. The radio source Sagittarius A*, at the centre, lies within 0.1° of $l = b = 0$. By IAU convention the north Galactic pole and the longitude of the north celestial pole are fixed at the J2000.0 values

$$\begin{aligned} \alpha_{\mathrm{GP}} &= 12^{\mathrm{h}}51^{\mathrm{m}}26.28^{\mathrm{s}} = 192.85948°, \qquad \delta_{\mathrm{GP}} = +27°07'41.7'' = +27.12825°, \\ l_{\mathrm{CP}} &= 122.93192°. \end{aligned} \tag{2.45}$$

Here l_{CP} is the Galactic longitude of the north celestial pole.

The three turns. The Galactic equator is 90° from its pole, so it crosses the celestial equator 90° from the pole's hour circle, at $\alpha = \alpha_{\mathrm{GP}} + 90°$. The angle between the two poles is $90° - \delta_{\mathrm{GP}}$. In the Galactic frame that crossing is 90° from the half circle of longitude l_{CP} that runs through the celestial pole, at $l = l_{\mathrm{CP}} - 90° = 32.93°$, so the last turn is by $-(l_{\mathrm{CP}} - 90°)$. With Eqs. (2.5) and (2.7),

$$\hat{\mathbf{r}}_{\mathrm{gal}} = \mathbf{R}_z(90° - l_{\mathrm{CP}})\, \mathbf{R}_x(90° - \delta_{\mathrm{GP}})\, \mathbf{R}_z(\alpha_{\mathrm{GP}} + 90°)\, \hat{\mathbf{r}}_{\mathrm{eq}}. \tag{2.46}$$

We multiply from the right. The first turn is Eq. (2.8) with $\theta = \alpha_{\mathrm{GP}} + 90°$,

$$\mathbf{R}_z(\alpha_{\mathrm{GP}} + 90°)\, \hat{\mathbf{r}}_{\mathrm{eq}} \overset{(1)}{=} \begin{pmatrix} \cos\delta\cos(\alpha - \alpha_{\mathrm{GP}} - 90°) \\ \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}} - 90°) \\ \sin\delta \end{pmatrix} \overset{(2)}{=} \begin{pmatrix} \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}}) \\ -\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}) \\ \sin\delta \end{pmatrix}. \tag{2.47}$$

(1) Eq. (2.8).
(2) $\cos(x - 90°) = \sin x$ and $\sin(x - 90°) = -\cos x$.

The tilt uses $\cos(90° - \delta_{\mathrm{GP}}) = \sin\delta_{\mathrm{GP}}$ and $\sin(90° - \delta_{\mathrm{GP}}) = \cos\delta_{\mathrm{GP}}$,

$$\begin{aligned} \mathbf{R}_x(90° - \delta_{\mathrm{GP}}) \begin{pmatrix} \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}}) \\ -\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}) \\ \sin\delta \end{pmatrix} &\overset{(1)}{=} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sin\delta_{\mathrm{GP}} & \cos\delta_{\mathrm{GP}} \\ 0 & -\cos\delta_{\mathrm{GP}} & \sin\delta_{\mathrm{GP}} \end{pmatrix} \begin{pmatrix} \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}}) \\ -\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}) \\ \sin\delta \end{pmatrix} \\ &\overset{(2)}{=} \begin{pmatrix} \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}}) \\ \cos\delta_{\mathrm{GP}}\sin\delta - \sin\delta_{\mathrm{GP}}\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}) \\ \sin\delta_{\mathrm{GP}}\sin\delta + \cos\delta_{\mathrm{GP}}\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}) \end{pmatrix}. \end{aligned} \tag{2.48}$$

(1) Eq. (2.7) with $\theta = 90^\circ - \delta_{\mathrm{GP}}$.
(2) Each row times the column.

For the roll it is simpler to move it to the left side. By Eq. (2.4), $\mathbf{R}_z^T(\theta) = \mathbf{R}_z(-\theta)$, so the left side of Eq. (2.46) times $\mathbf{R}_z(l_{\mathrm{CP}} - 90^\circ)$ is

$$\mathbf{R}_z(l_{\mathrm{CP}} - 90^\circ)\begin{pmatrix}\cos b\cos l\\ \cos b\sin l\\ \sin b\end{pmatrix} \overset{(1)}{=} \begin{pmatrix}\cos b\cos(l - l_{\mathrm{CP}} + 90^\circ)\\ \cos b\sin(l - l_{\mathrm{CP}} + 90^\circ)\\ \sin b\end{pmatrix} \overset{(2)}{=} \begin{pmatrix}\cos b\sin(l_{\mathrm{CP}} - l)\\ \cos b\cos(l_{\mathrm{CP}} - l)\\ \sin b\end{pmatrix}. \tag{2.49}$$

(1) Eq. (2.8) with $\theta = l_{\mathrm{CP}} - 90^\circ$.
(2) $\cos(x + 90^\circ) = -\sin x$ and $\sin(x + 90^\circ) = \cos x$, with $x = l - l_{\mathrm{CP}}$.

Setting the last two columns equal, row by row,

Result 2.4

$$\sin b = \sin\delta_{\mathrm{GP}}\sin\delta + \cos\delta_{\mathrm{GP}}\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}), \tag{2.50}$$
$$\cos b\sin(l_{\mathrm{CP}} - l) = \cos\delta\sin(\alpha - \alpha_{\mathrm{GP}}), \tag{2.51}$$
$$\cos b\cos(l_{\mathrm{CP}} - l) = \cos\delta_{\mathrm{GP}}\sin\delta - \sin\delta_{\mathrm{GP}}\cos\delta\cos(\alpha - \alpha_{\mathrm{GP}}). \tag{2.52}$$

The latitude depends only on the star's angle from the Galactic pole, since Eq. (2.50) is the cosine rule (2.34) between the star and the pole. The longitude l needs both of the other two rows to fix its quadrant.

The same formulas as for the horizon. Eqs. (2.50)–(2.52) are Eqs. (2.20)–(2.22) with the Galactic pole in place of the zenith. Replace ϕ_0 by δ_{GP}, the hour angle H by $\alpha_{\mathrm{GP}} - \alpha$, the altitude by b and the azimuth A by $l_{\mathrm{CP}} - l$. The zenith is a pole at declination ϕ_0 on the meridian, and the Galactic pole is a pole at declination δ_{GP} on the hour circle α_{GP}. The minus sign in $l_{\mathrm{CP}} - l$ is there because the azimuth runs clockwise about its pole and l runs anticlockwise.

The numbers. Multiplying out Eq. (2.46) with the values above gives

$$\hat{\mathbf{r}}_{\mathrm{gal}} = \begin{pmatrix}-0.054876 & -0.873437 & -0.483835\\ +0.494109 & -0.444830 & +0.746982\\ -0.867666 & -0.198076 & +0.455984\end{pmatrix}\hat{\mathbf{r}}_{\mathrm{eq}}. \tag{2.53}$$

By the rule of rows (2.3), the first row is the direction of the Galactic centre in equatorial components, the second is the direction $l = 90^\circ$, and the third is the Galactic pole. The third row is $(\cos\delta_{\mathrm{GP}}\cos\alpha_{\mathrm{GP}}, \cos\delta_{\mathrm{GP}}\sin\alpha_{\mathrm{GP}}, \sin\delta_{\mathrm{GP}})$, as Eq. (2.2) requires. The same matrix appears in Eq. (3.34). Two checks follow. The point $\alpha = 266.405^\circ$, $\delta = -28.936^\circ$ comes out at $l = 0.0002^\circ$, $b = 0.0001^\circ$, so it is the zero point of the frame. Sagittarius A*, at $\alpha = 17^{\mathrm{h}}45^{\mathrm{m}}40.04^{\mathrm{s}}$, $\delta = -29^\circ00'28.1''$, comes out at $l = 359.944^\circ$, $b = -0.046^\circ$.

2.9 Atmospheric Refraction and Stellar Aberration

Every formula so far gives the true direction of a star from the centre of the Earth.

The direction in which its light arrives at a telescope is not quite that. The air bends the light, and the motion of the Earth tilts it.

Think of a straw in a glass of water. It looks bent where it enters the water, because light changes direction when it crosses into a medium with a different refractive index. Now think

of running through rain that falls straight down. The drops seem to come at you from in front, and you tilt your umbrella forward. The faster you run, the more you tilt it.

Starlight entering the air bends toward the vertical, so a star looks higher than it is. Starlight met by the moving Earth seems to come from a little ahead of the Earth's motion. The idea is that refraction lifts a star by $(n-1)\tan z$ and aberration tilts it forward by v/c.

With $n-1 = 2.82\times10^{-4}$ for air, a star at $z = 45°$ is lifted by 2.82×10^{-4} rad $= 58''$. The Earth moves round the Sun at $v = 29.78\,\mathrm{km\,s^{-1}}$, so $v/c = 0.99\times10^{-4}$ and the tilt is $20.5''$.

The picture is a ray bending a little at each layer of air, and a stream of slanted raindrops.

Refraction is Snell's law, and it also makes mirages. Aberration is the low-speed limit of the relativistic aberration of light. Both formulas hold only in a range, and that range is set by comparing two lengths (P6). The flat-layer picture of the air works while the height of the air is small compared with the distance over which the ground curves away. These effects are what move a star from its computed place by arcseconds, and near the horizon by half a degree.

2.9.1 Atmospheric Refraction

The goal is the lift $\Delta h = h_{\text{apparent}} - h_{\text{true}}$, the same as the drop in zenith angle.

Snell's law through many layers. Let the air be a stack of flat layers with refractive indices $n_1, n_2, \ldots$ from the top down to the ground, where $n_k = n$, and let space above have index 1. A ray from a star at true zenith angle z enters at the top and reaches the ground at the apparent zenith angle z'. Snell's law at each boundary gives

$$\sin z \overset{(1)}{=} n_1 \sin z_1 \overset{(1)}{=} n_2 \sin z_2 \overset{(1)}{=} \cdots \overset{(1)}{=} n \sin z'. \tag{2.54}$$

(1) Snell's law at each flat boundary. The boundaries are parallel, so the angle leaving one layer is the angle arriving at the next.

All the layers in between cancel. For flat layers the bending depends only on the index at the ground, not on how n varies with height.

The lift. Put $z' = z - \Delta h$ and expand for small Δh,

$$\begin{aligned} \sin z &= n \sin(z - \Delta h) \\ &\overset{(1)}{=} n \sin z \cos \Delta h - n \cos z \sin \Delta h \\ &\overset{(2)}{=} n \sin z - n\,\Delta h \cos z, \end{aligned} \tag{2.55}$$

(1) The angle-difference identity for the sine.
(2) $\cos\Delta h = 1$ and $\sin\Delta h = \Delta h$ to first order in Δh.

so that

$$\Delta h \approx \frac{(n-1)\sin z}{n\cos z} \approx (n-1)\tan z, \tag{2.56}$$

using $n \approx 1$ in the denominator. The index of air is a measured property. For visible light at 10°C and 1010 hPa it is $n - 1 = 2.82\times10^{-4}$. In arcseconds (1 rad = 206 265″),

$$\boxed{\Delta h \approx 58.2''\tan z = 58.2''\cot h \qquad (h > 15°).} \tag{2.57}$$

The lift is zero at the zenith and grows toward the horizon. To first order it does not matter whether z is the true or the apparent zenith angle. Since $n - 1$ is proportional to the density of the air, which goes as P/T, the coefficient $58.2''$ must be scaled by the local pressure and temperature.

When the flat layers fail (P6). The air thins with a scale height of 8.4 km, Eq. (10.4). Call it H_{air} here, to keep it apart from the hour angle. A ray at zenith angle z crosses this lowest scale height while covering a horizontal distance $d = H_{\text{air}} \tan z$. Over a horizontal distance d the curved ground of radius $R_\oplus$ falls away below a flat plane by

$$R_\oplus - \sqrt{R_\oplus^2 - d^2} \overset{(1)}{=} R_\oplus\Big(1 - \sqrt{1 - d^2/R_\oplus^2}\Big) \overset{(2)}{=} \frac{d^2}{2R_\oplus}, \tag{2.58}$$

(1) $R_\oplus$ taken out of the bracket.
(2) $\sqrt{1-x} \approx 1 - x/2$ for $x = d^2/R_\oplus^2 \ll 1$.

The flat picture holds while this drop is small compared with H_{air} itself. The ratio of the two lengths is

$$\frac{d^2/(2R_\oplus)}{H_{\text{air}}} = \frac{H_{\text{air}} \tan^2 z}{2R_\oplus}. \tag{2.59}$$

With $R_\oplus = 6371$ km, $2R_\oplus/H_{\text{air}} = 1517$. The ratio is 0.01 at $\tan z = 3.9$, that is $z = 75.6°$ or $h \approx 14°$, which is why Eq. (2.57) is quoted for $h > 15°$. The ratio reaches one at $\tan z = 39$, or $z = 88.5°$, and there the flat picture fails outright.

Near the horizon. Close to the horizon an empirical fit is used instead. Bennett (1982) fitted the refraction as a function of the apparent altitude, $\Delta h[\text{arcmin}] \approx 1/\tan\big(h_{\text{app}} + 7.31°/(h_{\text{app}} + 4.4°)\big)$. Sæmundsson (1986) turned it round to use the true altitude h, and added the scaling of $n-1$ with pressure P and temperature T,

$$\Delta h[\text{arcmin}] \approx \frac{1.02}{\tan\left(h + \dfrac{10.3°}{h + 5.11°}\right)} \times \left(\frac{P}{1010\,\text{hPa}}\right)\left(\frac{283}{273 + T[°\text{C}]}\right). \tag{2.60}$$

A star whose true position is on the horizon is lifted by 29′. A star that is seen exactly on the horizon is really 34.5′ below it, by Bennett's fit at $h_{\text{app}} = 0$. That is more than the Sun's angular diameter of about 32′, so when the setting Sun appears to touch the horizon it is already geometrically below it. At $h = 45°$ Eq. (2.60) gives 1.01′ = 61″, close to the 58″ of Eq. (2.57).

2.9.2 Stellar Aberration

The goal is the shift of a star's apparent direction caused by the observer's velocity $\vec{v}$.

The general shift. Light from a star in the direction $\hat{\mathbf{r}}$ travels with velocity $-c\,\hat{\mathbf{r}}$ in the frame of the Sun. Since $v \ll c$, Galilean addition of velocities is enough. In the Earth's frame the light moves with velocity $-c\,\hat{\mathbf{r}} - \vec{v}$, so it seems to come from the direction of $c\,\hat{\mathbf{r}} + \vec{v}$. The apparent direction is

$$\begin{aligned} \hat{\mathbf{r}}' = \frac{c\,\hat{\mathbf{r}} + \vec{v}}{|c\,\hat{\mathbf{r}} + \vec{v}|} &\overset{(1)}{=} \frac{\hat{\mathbf{r}} + \vec{v}/c}{\sqrt{1 + 2\,\hat{\mathbf{r}}\cdot\vec{v}/c + v^2/c^2}} \\ &\overset{(2)}{=} \Big(\hat{\mathbf{r}} + \frac{\vec{v}}{c}\Big)\Big(1 - \frac{\hat{\mathbf{r}}\cdot\vec{v}}{c}\Big) \overset{(3)}{=} \hat{\mathbf{r}} + \frac{\vec{v} - (\hat{\mathbf{r}}\cdot\vec{v})\,\hat{\mathbf{r}}}{c}. \end{aligned} \tag{2.61}$$

(1) Divided above and below by c, and $|\hat{\mathbf{r}} + \vec{v}/c|^2$ written out with $\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = 1$.
(2) $(1+x)^{-1/2} \approx 1 - x/2$, keeping first order in v/c.
(3) Multiplied out, dropping the term of order v^2/c^2.

$$\boxed{\hat{\mathbf{r}}' - \hat{\mathbf{r}} = \frac{\vec{v}_\perp}{c},} \tag{2.62}$$

where $\vec{v}_\perp = \vec{v} - (\hat{\mathbf{r}} \cdot \vec{v})\,\hat{\mathbf{r}}$ is the part of the velocity across the line of sight. The star moves toward the direction of motion by the angle $(v/c)\sin\psi$, where ψ is the angle between $\hat{\mathbf{r}}$ and $\vec{v}$. A velocity along the line of sight gives no shift. The relativistic formula differs from this only by a fraction of order $v^2/c^2 \approx 10^{-8}$.

The constant of aberration. The Earth's orbital speed is $v = 2\pi \times 1\,\text{AU}$ per sidereal year, $29.78\,\text{km s}^{-1}$. The largest shift, for $\psi = 90°$, is

$$\boxed{\kappa = \frac{v}{c} = \frac{29.78}{299\,792} \times 206\,265'' = 20.49''.} \tag{2.63}$$

The yearly ellipse. To see the path of a star over a year, work in ecliptic coordinates and take the Earth's orbit as a circle. The Earth is opposite the Sun, so with the Sun at ecliptic longitude $\lambda_\odot$ the Earth is at $-1\,\text{AU}\,(\cos\lambda_\odot, \sin\lambda_\odot, 0)$. Its velocity is the time derivative of that position, with $\lambda_\odot$ growing at 2π per year,

$$\vec{v} = v\,(\sin\lambda_\odot,\ -\cos\lambda_\odot,\ 0)^T. \tag{2.64}$$

The directions of increasing λ and β at the star are Eq. (3.11) with $\alpha \to \lambda$ and $\delta \to \beta$, namely $\hat{\boldsymbol{\lambda}} = (-\sin\lambda, \cos\lambda, 0)^T$ and $\hat{\boldsymbol{\beta}} = (-\sin\beta\cos\lambda, -\sin\beta\sin\lambda, \cos\beta)^T$. Both are perpendicular to $\hat{\mathbf{r}}$, so the components of the shift (2.62) along them are $\vec{v}\cdot\hat{\boldsymbol{\lambda}}/c$ and $\vec{v}\cdot\hat{\boldsymbol{\beta}}/c$,

$$\begin{aligned} \cos\beta\,\Delta\lambda &= \frac{\vec{v}\cdot\hat{\boldsymbol{\lambda}}}{c} \overset{(1)}{=} -\kappa\,(\sin\lambda_\odot\sin\lambda + \cos\lambda_\odot\cos\lambda) \overset{(2)}{=} -\kappa\cos(\lambda_\odot - \lambda), \\ \Delta\beta &= \frac{\vec{v}\cdot\hat{\boldsymbol{\beta}}}{c} \overset{(1)}{=} -\kappa\sin\beta\,(\sin\lambda_\odot\cos\lambda - \cos\lambda_\odot\sin\lambda) \overset{(2)}{=} -\kappa\sin\beta\sin(\lambda_\odot - \lambda). \end{aligned} \tag{2.65}$$

(1) The dot products of the columns, with $v/c = \kappa$.
(2) The angle-difference identities.

As $\lambda_\odot$ runs through the year,

$$\left(\frac{\cos\beta\,\Delta\lambda}{\kappa}\right)^2 + \left(\frac{\Delta\beta}{\kappa\sin\beta}\right)^2 = 1. \tag{2.66}$$

Every star traces an ellipse with semi-axes $20.5''$ along the ecliptic and $20.5''\sin\beta$ across it. At the ecliptic pole it is a circle, and on the ecliptic it closes to a line.

The parallax ellipse of Chapter 3 comes from the same projection with the Earth's position in place of its velocity. A star at distance d is shifted by $-(\text{Earth's position})_\perp/d$, which gives $\cos\beta\,\Delta\lambda = p\sin(\lambda_\odot - \lambda)$ and $\Delta\beta = -p\sin\beta\cos(\lambda_\odot - \lambda)$ with $p = 1\,\text{AU}/d$. The two ellipses have the same shape and the same one-year period, and they are a quarter of a year out of step. They differ in size. The parallax of even the nearest star is below $0.8''$ and shrinks with distance. The aberration ellipse is $20.5''$ for every star at every distance.

2.10 Airmass and the Parallactic Angle

The zenith angle of a star is known from Eq. (2.20), and refraction from Eq. (2.56).

Two practical questions remain. Photometry needs to know how much air the light has crossed. Spectroscopy needs to know in which direction on the sky the air spreads the colours.

Think of crossing a road 10 m wide. Walking straight across covers 10 m. Walking at 60° to the straight line covers $10/\cos 60° = 20$ m. Now think of a prism. It bends blue light more than red, so white light fans out into colours along the direction of bending.

Light from a star at zenith angle z crosses the air on a slant. Air also bends blue light a little more than red, so every star is drawn out into a short spectrum pointing at the zenith. The idea is that the path through flat layers grows as $1/\cos z$, and the parallactic angle gives the direction of the zenith as seen at the star.

A star at $z = 60°$ shines through twice as much air as one at the zenith, as on the road.

The picture is a slanted ray through a stack of flat slabs.

Any slanted ray through flat layers gains the same factor $1/\cos z$. The optical depth of Chapter 4 grows with it, so the dimming of a star in magnitudes grows in proportion to the airmass. Looking slantwise into the Sun's atmosphere reaches higher, cooler layers for the same reason, which is why the edge of the Sun looks darker (Chapter 11). The flat picture holds while the comparison of lengths in Eq. (2.59) is small (P6). It gives the airmass and the parallactic angle, and with them the angle at which to set a spectrograph slit.

2.10.1 Airmass

Let the air be flat layers, with density $\rho(y)$ at height y. A ray at zenith angle z covers a path ds while it rises dy, and the slanted path is the long side of a right triangle whose side along the vertical is dy, so $ds = dy/\cos z$. The **airmass** X is the mass of air along the slanted path divided by that along the vertical,

$$X \equiv \frac{\int \rho \, ds}{\int \rho \, dy} \overset{(1)}{=} \frac{\frac{1}{\cos z}\int \rho(y)\, dy}{\int \rho(y)\, dy} \overset{(2)}{=} \frac{1}{\cos z}. \tag{2.67}$$

(1) $ds = dy/\cos z$, with the same z in every flat layer. Refraction changes z by less than 10^{-3} rad above $h = 15°$, Eq. (2.56).

(2) The integrals cancel, whatever the profile $\rho(y)$.

$$\boxed{X \approx \sec z = \frac{1}{\sin h}.} \tag{2.68}$$

The airmass depends only on the zenith angle, not on how the density falls with height. It diverges at the horizon, while the real path stays finite because the ground curves away. By Eq. (2.59) the curvature matters at the one per cent level from about $z = 75°$. There the empirical fit of Kasten and Young (1989) is used,

$$X = \frac{1}{\cos z + 0.50572\,(96.07995° - z)^{-1.6364}}, \tag{2.69}$$

with z in degrees. It gives 1.994 at $z = 60°$, where $\sec z = 2.000$, and 4.071 at $z = 76°$, where $\sec z = 4.134$, a difference of 1.5%. At the horizon it gives $X = 37.9$ instead of infinity.

2.10.2 The Parallactic Angle

The **parallactic angle** q is the angle at the star between the direction toward the celestial pole and the direction toward the zenith. It is the third angle of the astronomical triangle of Figure 2.10. We measure it from the pole direction through east, which is how position angles on the sky are measured.

The two directions at the star. Work in the meridian frame of Eq. (2.13), with the pole $\hat{\mathbf{p}} = (0,0,1)^T$ and the zenith $\hat{\mathbf{u}}$ of Eq. (2.14). Seen from the star, the great circle toward the pole leaves along the part of $\hat{\mathbf{p}}$ perpendicular to $\hat{\mathbf{r}}$, and the great circle toward the zenith leaves along the part of $\hat{\mathbf{u}}$ perpendicular to $\hat{\mathbf{r}}$,

$$\vec{t}_{\mathrm{P}} = \hat{\mathbf{p}} - \sin\delta\,\hat{\mathbf{r}}, \qquad \vec{t}_{\mathrm{Z}} = \hat{\mathbf{u}} - \sin h\,\hat{\mathbf{r}}, \tag{2.70}$$

since $\hat{\mathbf{p}}\cdot\hat{\mathbf{r}} = \sin\delta$ and $\hat{\mathbf{u}}\cdot\hat{\mathbf{r}} = \sin h$. Their lengths are $|\vec{t}_{\mathrm{P}}|^2 = 1 - 2\sin^2\delta + \sin^2\delta = \cos^2\delta$ and, in the same way, $|\vec{t}_{\mathrm{Z}}| = \cos h$.

The cosine. The angle between the two directions has

$$\begin{aligned}\cos q\,\cos\delta\cos h = \vec{t}_{\mathrm{P}}\cdot\vec{t}_{\mathrm{Z}} &\overset{(1)}{=} \hat{\mathbf{p}}\cdot\hat{\mathbf{u}} - \sin h\,\hat{\mathbf{p}}\cdot\hat{\mathbf{r}} - \sin\delta\,\hat{\mathbf{r}}\cdot\hat{\mathbf{u}} + \sin\delta\sin h\,\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} \\ &\overset{(2)}{=} \sin\phi_0 - \sin h\sin\delta - \sin\delta\sin h + \sin\delta\sin h \\ &= \sin\phi_0 - \sin h\sin\delta.\end{aligned} \tag{2.71}$$

(1) The dot product multiplied out.
(2) $\hat{\mathbf{p}}\cdot\hat{\mathbf{u}} = \sin\phi_0$ from Eq. (2.14), $\hat{\mathbf{p}}\cdot\hat{\mathbf{r}} = \sin\delta$, $\hat{\mathbf{r}}\cdot\hat{\mathbf{u}} = \sin h$ and $\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = 1$.

The sine. The direction a quarter turn east of the pole direction at the star is $\vec{t}_{\mathrm{P}}\times\hat{\mathbf{r}}/\cos\delta$. (On the meridian at the equator, $\vec{t}_{\mathrm{P}} = \hat{\mathbf{p}}$ and $\hat{\mathbf{r}} = \hat{\mathbf{x}}$, and $\hat{\mathbf{p}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}$ points east.) So

$$\begin{aligned}\sin q\,\cos\delta\cos h = \vec{t}_{\mathrm{Z}}\cdot(\vec{t}_{\mathrm{P}}\times\hat{\mathbf{r}}) &\overset{(1)}{=} \hat{\mathbf{u}}\cdot(\hat{\mathbf{p}}\times\hat{\mathbf{r}}) \\ &\overset{(2)}{=} \begin{pmatrix}\cos\phi_0\\0\\\sin\phi_0\end{pmatrix}\cdot\begin{pmatrix}0\cdot\sin\delta - 1\cdot(-\cos\delta\sin H)\\1\cdot\cos\delta\cos H - 0\cdot\sin\delta\\0\end{pmatrix} \\ &\overset{(3)}{=} \cos\phi_0\cos\delta\sin H.\end{aligned} \tag{2.72}$$

(1) The parts of $\vec{t}_{\mathrm{P}}$ and $\vec{t}_{\mathrm{Z}}$ along $\hat{\mathbf{r}}$ drop out, since a triple product with $\hat{\mathbf{r}}$ twice is zero.
(2) $\hat{\mathbf{p}} = (0,0,1)^T$, $\hat{\mathbf{r}}$ from Eq. (2.13) and $\hat{\mathbf{u}}$ from Eq. (2.14).
(3) The dot product.

Dividing both results by $\cos\delta\cos h$,

$$\boxed{\sin q = \frac{\sin H\cos\phi_0}{\cos h}, \qquad \cos q = \frac{\sin\phi_0 - \sin h\sin\delta}{\cos h\cos\delta}.} \tag{2.73}$$

The angle q is zero on the meridian, where the zenith lies straight toward the pole, and has the sign of H. A star in the west has its zenith direction turned toward the east of north. These are the sine and cosine rules of spherical trigonometry for the astronomical triangle.

Why it matters for a slit. The index of air is larger for blue light than for red. Between 400 nm and 700 nm, $n-1$ changes by 7.0×10^{-6} at 15°C and 1013 hPa. By Eq. (2.56), the blue image of a star is lifted more than the red one by $7.0\times10^{-6}\tan z$ radians, or $1.43''\tan z$. At $z = 60°$ that is 2.5″. A spectrograph slit is often about 1″ wide, so a slit set across this spread loses the blue or the red light. Set along the vertical through the star, at position angle q, it takes in all colours.

2.11 Problem Set: Celestial Coordinate Transformations

2.11.1 Problem 1: Horizontal ↔ Equatorial Transformations and Sirius at IISER TVM

Problem Statement: Two coordinate systems turn with the Earth. One is the **horizontal system**, altitude h and azimuth A. The other is the turning **equatorial system**, declination δ and hour angle H.

(a) Show how the two systems are related.

(b) Sirius is at $\alpha = 06^{\mathrm{h}}45^{\mathrm{m}}08.917^{\mathrm{s}}$, $\delta = -16°42'58.02''$. Find its hour angle, altitude and azimuth at LST $= 22^{\mathrm{h}}00^{\mathrm{m}}00^{\mathrm{s}}$ at IISER TVM ($\phi_0 = 8.6833°$N).

(c) Find the sidereal times at which Sirius rises and sets at IISER TVM.

Essential Ideas:

- The local sidereal time is the right ascension on the meridian, Eq. (1.12), and the hour angle is $H = \mathrm{LST} - \alpha$, Eq. (2.10).
- In a frame turned to the meridian the star is, by Eq. (2.13),

$$\hat{\mathbf{r}}_{\mathrm{mer}} = \begin{pmatrix} \cos\delta\cos(\alpha - \mathrm{LST}) \\ \cos\delta\sin(\alpha - \mathrm{LST}) \\ \sin\delta \end{pmatrix} = \begin{pmatrix} \cos\delta\cos H \\ -\cos\delta\sin H \\ \sin\delta \end{pmatrix}. \tag{2.74}$$

  The matrix $\mathbf{M}(\phi_0)$ whose columns are north, east and up, Eq. (2.18), turns it into $(\cos h\cos A, \cos h\sin A, \sin h)^T$.
- Rising and setting happen at $h = 0$, so $\cos H_0 = -\tan\phi_0\tan\delta$, Eq. (2.27).

Detailed Solutions:

(a) **The transformation**. The derivation is in Section 2.3. Since $\mathbf{M}(\phi_0)$ is orthogonal and symmetric, $(\cos h\cos A, \cos h\sin A, \sin h)^T = \mathbf{M}(\phi_0)\,\hat{\mathbf{r}}_{\mathrm{mer}}$, Eq. (2.19), which gives

$$\boxed{\begin{aligned} \sin h &= \cos\phi_0\cos\delta\cos H + \sin\phi_0\sin\delta, \\ \cos h\sin A &= -\cos\delta\sin H, \\ \cos h\cos A &= -\sin\phi_0\cos\delta\cos H + \cos\phi_0\sin\delta. \end{aligned}} \tag{2.75}$$

The azimuth is fixed by the last two rows together. The same matrix takes (h, A) back to (δ, H), Eq. (2.25), because $\mathbf{M}(\phi_0)^2 = \mathbb{1}$.

(b) **Hour angle, altitude and azimuth**. In degrees, $\delta = -16.7161°$ and $\alpha = 15\,(6 + 45/60 + 8.917/3600)° = 101.2872°$. The hour angle is

$$H = \mathrm{LST} - \alpha = 22^{\mathrm{h}}00^{\mathrm{m}}00^{\mathrm{s}} - 06^{\mathrm{h}}45^{\mathrm{m}}08.917^{\mathrm{s}} = 15^{\mathrm{h}}14^{\mathrm{m}}51.08^{\mathrm{s}} = 228.7128°.$$

With $\sin\phi_0 = 0.15097$, $\cos\phi_0 = 0.98854$, $\sin\delta = -0.28763$, $\cos\delta = 0.95774$, $\sin H = -0.75141$ and $\cos H = -0.65983$,

$$\begin{aligned} \sin h &= 0.15097\times(-0.28763) + 0.98854\times 0.95774\times(-0.65983) = -0.6681, \\ \cos h\sin A &= -0.95774\times(-0.75141) = +0.7197, \\ \cos h\cos A &= -0.15097\times 0.95774\times(-0.65983) + 0.98854\times(-0.28763) = -0.1889. \end{aligned}$$

So $h = \arcsin(-0.6681)$. The sine of A is positive and its cosine negative, so A is in the second quadrant, $A = 180° - \arctan(0.7197/0.1889)$.

$$\boxed{H = 228.71°\ (15^{\mathrm{h}}14^{\mathrm{m}}51^{\mathrm{s}}), \qquad h = -41.92°, \qquad A = 104.71°.}$$

Sirius is 42° below the horizon, toward the east-southeast. Its hour angle, $228.71^\circ = -131.29^\circ$, says it is still east of the meridian and has yet to rise. The ratio form (2.23) gives $\arctan(-3.809) = -75.29^\circ$, which is 180° away from the right answer, so the quadrant must come from the two rows separately.

(c) **Rising and setting**. With $\tan\phi_0 = 0.15272$ and $\tan\delta = -0.30032$, Eq. (2.27) gives

$$\cos H_0 = -0.15272 \times (-0.30032) = 0.04587, \qquad H_0 = 87.371^\circ = 5^{\mathrm{h}}49^{\mathrm{m}}29^{\mathrm{s}}.$$

Then $\mathrm{LST}_{\mathrm{rise}} = \alpha - H_0$ and $\mathrm{LST}_{\mathrm{set}} = \alpha + H_0$,

$$\boxed{\mathrm{LST}_{\mathrm{rise}} = 00^{\mathrm{h}}55^{\mathrm{m}}40^{\mathrm{s}}, \qquad \mathrm{LST}_{\mathrm{set}} = 12^{\mathrm{h}}34^{\mathrm{m}}38^{\mathrm{s}}.}$$

Sirius is up for $2H_0 = 11^{\mathrm{h}}39^{\mathrm{m}}$ of sidereal time, a little less than half the day, because it is south of the equator and the site is north of it. At $\mathrm{LST} = 22^{\mathrm{h}}$ it is indeed below the horizon, as part (b) found.

2.11.2 Problem 2: Angular Distance on the Sphere and Ecliptic Coordinates of Sirius and Regulus

Problem Statement: Two stars are at equatorial coordinates (α_1, δ_1) and (α_2, δ_2). What is the angle between them on the celestial sphere? Regulus is at $\alpha = 10^{\mathrm{h}}08^{\mathrm{m}}22.311^{\mathrm{s}}$, $\delta = +11^\circ 58' 01.95''$.

(a) Find the angle between Sirius (Problem 2.11.1) and Regulus.

(b) Find the ecliptic coordinates of Sirius and of Regulus. Use $\epsilon = 23.4393^\circ$.

Essential Ideas:

- Each star is a unit vector, Eq. (2.2), and the angle between two unit vectors has $\cos\theta = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2$, Eq. (2.34).
- The equatorial and ecliptic frames share the x axis toward $\hat{\gamma}$. The ecliptic column is $\mathbf{R}_x(\epsilon)$ times the equatorial column, Eq. (2.39), with the turn of Eq. (2.7) and Figure 2.7.
- The longitude needs both $\cos\beta\cos\lambda$ and $\cos\beta\sin\lambda$ to fix its quadrant.

Detailed Solutions:

(a) **Angular distance**. In degrees, Sirius is at $(\alpha_1, \delta_1) = (101.2872^\circ, -16.7161^\circ)$ from Problem 2.11.1. For Regulus,

$$(\alpha_2, \delta_2) = \left(15\left(10 + \tfrac{8}{60} + \tfrac{22.311}{3600}\right)^\circ,\ 11 + \tfrac{58}{60} + \tfrac{1.95}{3600}\right) = (152.0930^\circ,\ 11.9672^\circ).$$

The two columns (2.2) are

$$\hat{\mathbf{r}}_1 = \begin{pmatrix} -0.18746 \\ 0.93922 \\ -0.28763 \end{pmatrix}, \qquad \hat{\mathbf{r}}_2 = \begin{pmatrix} -0.86450 \\ 0.45787 \\ 0.20735 \end{pmatrix},$$

and their dot product, Eq. (2.34), is

$$\cos\theta = \sin\delta_1\sin\delta_2 + \cos\delta_1\cos\delta_2\cos(\alpha_1 - \alpha_2) = 0.5325,$$

$$\boxed{\theta = 57.83^\circ.}$$

(b) **Ecliptic coordinates**. With $\cos\epsilon = 0.91748$ and $\sin\epsilon = 0.39778$, Eq. (2.39) gives for Sirius

$$\hat{\mathbf{r}}_{\mathrm{ecl}} = \begin{pmatrix} -0.18746 \\ 0.93922 \times 0.91748 + (-0.28763) \times 0.39778 \\ -0.93922 \times 0.39778 + (-0.28763) \times 0.91748 \end{pmatrix} = \begin{pmatrix} -0.18746 \\ 0.74730 \\ -0.63749 \end{pmatrix}.$$

So $\beta = \arcsin(-0.63749)$. For λ, $\cos\lambda < 0$ and $\sin\lambda > 0$, so λ is in the second quadrant, $\lambda = 180^\circ - \arctan(0.74730/0.18746)$. For Regulus,

$$\hat{\mathbf{r}}_{\text{ecl}} = \begin{pmatrix} -0.86450 \\ 0.45787 \times 0.91748 + 0.20735 \times 0.39778 \\ -0.45787 \times 0.39778 + 0.20735 \times 0.91748 \end{pmatrix} = \begin{pmatrix} -0.86450 \\ 0.50256 \\ 0.00811 \end{pmatrix},$$

again with λ in the second quadrant.

$$\boxed{\text{Sirius } (\lambda, \beta) = (104.08^\circ, -39.61^\circ), \qquad \text{Regulus } (\lambda, \beta) = (149.83^\circ, +0.46^\circ).}$$

Taking arctan of the ratio alone gives -75.92° for Sirius and -30.17° for Regulus, each 180° from the true longitude. Regulus lies less than half a degree from the ecliptic. The Moon's orbit is inclined about 5° to the ecliptic, so the Moon often passes in front of Regulus, while Sirius, 40° south of the ecliptic, is never covered by the Moon or a planet.

Chapter 3

Astrometry: Distances, Motions, and the Space Velocity Vector

Chapter 2 gives each star a direction on the sky, the two angles (α, δ). To place the star in space we also need its distance, and to follow it in time we need its velocity. That makes six numbers, three for the position $\vec{r}$ and three for the velocity $\vec{v}$.

Each of the six comes from its own measurement. The direction gives two. The **parallax**, the small yearly shift of the star as the Earth goes round the Sun, gives the distance. The **proper motion**, the slow drift of the star across the sky, gives two components of the velocity. The Doppler shift of the star's spectral lines gives the third, the speed along the line of sight. Six measurements for six unknowns, so the count closes, as the twelve constants of the two-body problem did in Chapter 6. Astrometry is the measurement of these small angles on the sky. With the six numbers we can map where the stars of the Galaxy are and where they are going.

3.1 Trigonometric Parallax and the Parsec

Chapter 2 gives each star a direction, the unit vector $\hat{\mathbf{r}}_{\text{eq}}$ of Eq. (2.2).

A direction alone does not say how far away the star is. A faint star close by and a bright star far away can sit at the same point of the sky.

Hold a thumb up at arm's length and look at it with the left eye, then with the right. The thumb jumps against the far wall, and the wall does not move. The eyes are about 6.5 cm apart and the thumb is about 65 cm away, so the jump is $0.065/0.65 = 0.1$ radian, about 6°. Bring the thumb closer and the jump grows. The jump and the gap between the eyes together give the distance to the thumb.

The Earth's orbit gives the astronomer two eyes. In January and in July the Earth is on opposite sides of the Sun, 2 AU apart. A nearby star then jumps against the far background stars, which hardly move at all. The idea is that a distance is a baseline divided by an angle.

The angles are far smaller than for the thumb. One second of arc is 1/206 265 of a radian, so a star at 206 265 AU shifts by 1″ when the observer moves sideways by 1 AU. Sirius, one of the nearest stars, shifts by only 0.38″.

The picture is a very long and thin triangle. Its short side is the radius of the Earth's orbit, and the star sits at its far corner (Figure 3.1).

A surveyor finds the distance to a far peak from the two ends of a measured line in the same way, and our two eyes judge depth by it. The angle is so small that its tangent can be replaced by the angle itself. That is P6, since the ratio $1\ \text{AU}/d$ is far below one. The triangle gives the parallax, the parsec, and a rule for turning a noisy parallax into a distance.

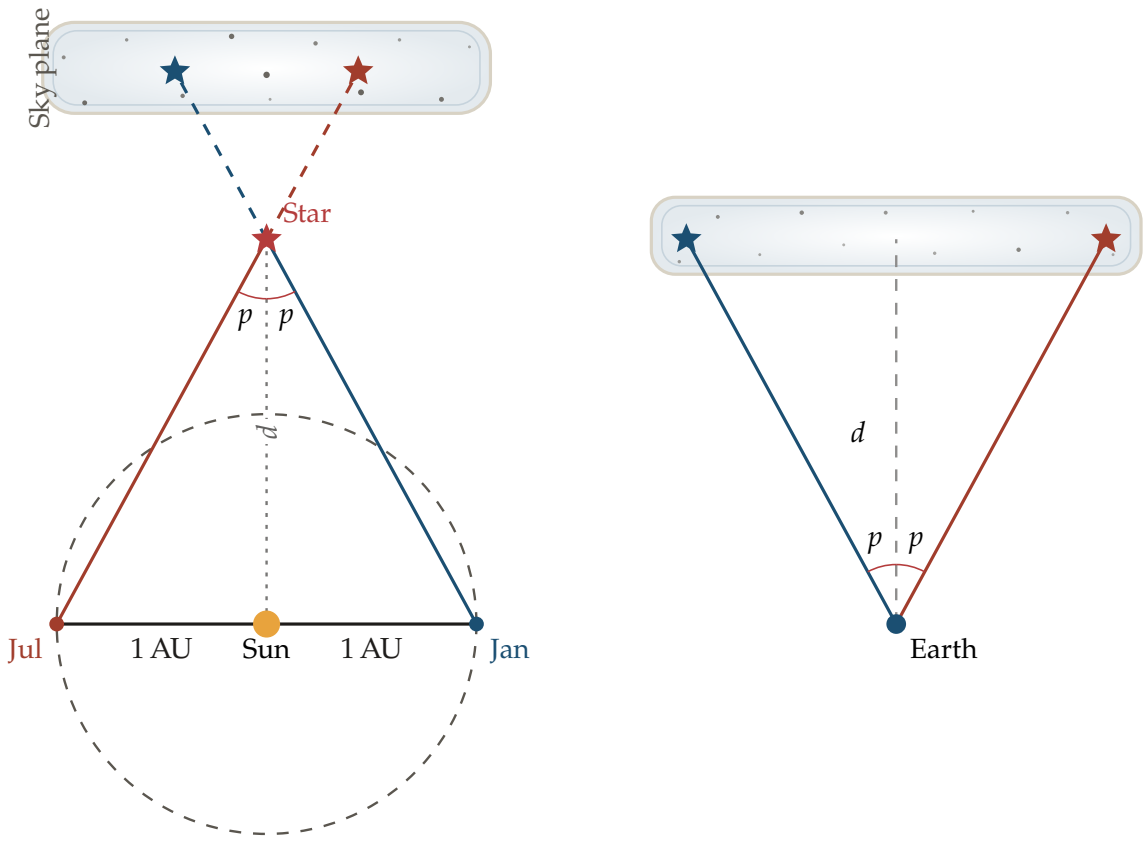


Figure 3.1: Left, the Sun, the Earth in January and in July, a nearby star at distance d, and the star's two apparent positions on the far sky. Right, the same star seen from an Earth held fixed. The baseline is 1 AU and the angle at the star is the parallax p. The angles are drawn far larger than real ones.

The triangle. Take the moment when the line from the Sun to the Earth is at right angles to the line from the Sun to the star. The triangle Sun–Earth–star then has its right angle at the Sun (Figure 3.1, left). The side from the Sun to the Earth is $1\,\mathrm{AU} = 1.495\,978\,707 \times 10^{11}$ m, the side from the Sun to the star is the distance d, and the angle at the star is the **parallax** p. So $\tan p = 1\,\mathrm{AU}/d$, and

$$d = \frac{1\,\mathrm{AU}}{\tan p} \overset{(1)}{=} 1\,\mathrm{AU}\left(\frac{1}{p} - \frac{p}{3} - \ldots\right) \overset{(2)}{=} \frac{1\,\mathrm{AU}}{p}\left(1 - \frac{p^2}{3} - \ldots\right). \tag{3.1}$$

(1) The series $\cot p = 1/p - p/3 - p^3/45 - \ldots$, with p in radians.
(2) Taking $1/p$ out of the bracket.

For $p = 1''$ the correction $p^2/3$ is 7.8×10^{-12}, and every star has $p < 1''$. So, with p in radians,

$$p = \frac{1\,\mathrm{AU}}{d}. \tag{3.2}$$

The same p is the largest shift of the star from the place where it would appear if seen from the Sun. Over a year the star traces a small ellipse whose semi-major axis is p, Eq. (3.17).

The parsec. The **parsec** is the distance at which 1 AU subtends one second of arc. Since $1'' = \pi/648\,000$ radian, Eq. (3.2) gives

$$1\,\mathrm{pc} = \frac{1\,\mathrm{AU}}{\pi/648\,000} = 206\,264.806\,\mathrm{AU} \overset{(1)}{=} 3.085\,677\,58 \times 10^{16}\,\mathrm{m} \overset{(2)}{=} 3.261\,56\,\mathrm{ly}. \tag{3.3}$$

(1) $1\,\mathrm{AU} = 1.495\,978\,707 \times 10^{11}$ m.
(2) $1\,\mathrm{ly} = 9.460\,730\,47 \times 10^{15}$ m, the distance light travels in a Julian year of 365.25 days.

The IAU has defined the parsec in exactly this way, as $648\,000/\pi$ astronomical units, since 2015. For a parallax p'' measured in seconds of arc, the angle in radians is $p''\pi/648\,000$, and Eq. (3.2) becomes

$$d \overset{(1)}{=} \frac{1\,\mathrm{AU}}{p''\,\pi/648\,000} \overset{(2)}{=} \frac{1}{p''}\,\frac{1\,\mathrm{AU}}{\pi/648\,000} \overset{(3)}{=} \frac{1}{p''}\,\mathrm{pc}. \tag{3.4}$$

(1) Eq. (3.2) with the angle converted to radians.
(2) Taking the number $1/p''$ out.
(3) The definition of the parsec above.

Result 3.1

$$d\,[\mathrm{pc}] = \frac{1}{p\,[\mathrm{arcsec}]}. \tag{3.5}$$

A star with half the parallax is twice as far away. The rule uses nothing about the star itself, only the size of the Earth's orbit.

Worked example 3.1: Distance to Sirius

The *Hipparcos* catalogue gives Sirius a parallax of $p = 379.21\,\mathrm{mas} = 0.37921''$. By Eq. (3.5),

$$d = \frac{1}{0.37921}\,\mathrm{pc} = 2.637\,\mathrm{pc} = 8.601\,\mathrm{ly}.$$

Worked example 3.2: Parallax as Measured from Mars

From Mars, the baseline would be the radius of the orbit of Mars, 1.524 AU. The distance d belongs to the star and does not change. By Eq. (3.2) the parallax is proportional to the baseline,

$$\frac{p_{\mathrm{Mars}}}{p_{\mathrm{Earth}}} = \frac{1.524\,\mathrm{AU}/d}{1\,\mathrm{AU}/d} = 1.524, \qquad p_{\mathrm{Mars}} = 1.524 \times 0.37921'' = 0.578''.$$

A longer baseline gives a larger angle, which is easier to measure. The price is time. Mars goes round the Sun in 1.88 years, so the star takes 1.88 years to trace its full ellipse, against one year from the Earth.

From a measured parallax to a distance. A parallax is measured with an error, and the goal is to see what that error does to $d = 1/p$. Write the measured value as $p_{\mathrm{obs}} = p + \varepsilon$, where p is the

true parallax and the error ε has mean $\langle \varepsilon \rangle = 0$ and spread $\langle \varepsilon^2 \rangle = \sigma_p^2$. On average the inverse is

$$\left\langle \frac{1}{p_{\text{obs}}} \right\rangle = \left\langle \frac{1}{p}\,\frac{1}{1+\varepsilon/p} \right\rangle \overset{(1)}{=} \frac{1}{p}\left\langle 1 - \frac{\varepsilon}{p} + \frac{\varepsilon^2}{p^2} - \dots \right\rangle \overset{(2)}{=} \frac{1}{p}\left(1 + \frac{\sigma_p^2}{p^2} + \dots\right). \tag{3.6}$$

(1) The geometric series $1/(1+x) = 1 - x + x^2 - \dots$, valid while $|\varepsilon| < p$.
(2) $\langle \varepsilon \rangle = 0$ and $\langle \varepsilon^2 \rangle = \sigma_p^2$.

Even with errors that scatter evenly up and down, $1/p_{\text{obs}}$ overestimates the distance on average, by the fraction σ_p^2/p^2. At $\sigma_p/p = 0.15$ that is 2%. The series needs σ_p/p small (P6). At $\sigma_p/p = 0.5$, 2.3% of the measured parallaxes come out negative, and $1/p_{\text{obs}}$ means nothing for them.

A second bias comes from where the stars are. A shell of radius d and thickness $\mathrm{d}d$ has volume $4\pi d^2\,\mathrm{d}d$, so in a region of even density there are more stars far away than near. Among all the stars measured at one p_{obs}, more have a smaller true parallax and were scattered up than have a larger one and were scattered down. In terms of the parallax, the number of stars is

$$\mathrm{d}N \propto d^2\,\mathrm{d}d \overset{(1)}{=} \frac{1}{p^2}\,\frac{\mathrm{d}p}{p^2} = \frac{\mathrm{d}p}{p^4}. \tag{3.7}$$

(1) $d = 1/p$, so $|\mathrm{d}d| = \mathrm{d}p/p^2$.

By Bayes' rule, the chance of a true parallax p given the measured one is this count times the chance of the error, $p^{-4}\exp\left[-(p_{\text{obs}}-p)^2/2\sigma_p^2\right]$. Its peak is where the derivative of its logarithm vanishes,

$$0 \overset{(1)}{=} -\frac{4}{p} + \frac{p_{\text{obs}} - p}{\sigma_p^2} \quad\Longrightarrow\quad p^2 - p_{\text{obs}}\,p + 4\sigma_p^2 \overset{(2)}{=} 0$$

$$\Longrightarrow\quad p_{\text{peak}} \overset{(3)}{=} \frac{p_{\text{obs}}}{2}\left(1 + \sqrt{1 - \frac{16\sigma_p^2}{p_{\text{obs}}^2}}\right). \tag{3.8}$$

(1) The derivative of $-4\ln p - (p_{\text{obs}} - p)^2/2\sigma_p^2$ with respect to p.
(2) Multiplied by $-p\,\sigma_p^2$.
(3) The larger root of the quadratic. The smaller root is a minimum, since the chance grows without limit as $p \to 0$.

At $\sigma_p/p_{\text{obs}} = 0.15$ the bracket is 1.8, so $p_{\text{peak}} = 0.9\,p_{\text{obs}}$ and the most likely distance is 11% beyond $1/p_{\text{obs}}$. For $\sigma_p/p_{\text{obs}} > 1/4$ the square root is imaginary and there is no peak at all. The measurement alone then cannot fix the distance, and the answer depends on what is assumed about where the stars are. This is the **Lutz–Kelker bias**. The *Gaia* catalogues handle it by assuming a density of stars that falls off exponentially with distance, which keeps the chance finite as $p \to 0$.

3.2 Three-Dimensional Cartesian Position Vector

The parallax gives the distance d, and Chapter 2 gives the direction $\hat{\mathbf{r}}_{\text{eq}}$. Together they place the star in space.

To describe how the star moves we need more. We need three directions at the star itself, because a motion is seen as a part along the line of sight and a part across the sky.

On the Earth every town has its own up, north and east. They differ from town to town, but in each town they are at right angles to each other. A pilot reports a speed as so much north and so much east, at the place where the plane is.

A point on the sky has the same three directions. Up is outward along the line of sight, north is the way of increasing declination, and east is the way of increasing right ascension. The idea is that the three local directions, written in equatorial components, are the rows of a rotation matrix.

At $\alpha = 0$, $\delta = 0$ the three directions are the x, y and z axes themselves, and the matrix is the identity.

The picture is three arrows at right angles planted at the star, one pointing away from us.

In mechanics these are the unit vectors of spherical polar coordinates, with north along minus the polar unit vector. Chapter 2 built local axes in the same way for the observer's horizon, Eq. (2.16). Only geometry enters, so no principle is used. The local axes give the position vector, the matrix between local and equatorial components, and the yearly parallax ellipse.

The position. The star is at distance d along $\hat{\mathbf{r}}_{\text{eq}}$, Eq. (2.2), so its position relative to the Sun is

$$\boxed{\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = d\,\hat{\mathbf{r}}_{\text{eq}} = \begin{pmatrix} d\cos\delta\cos\alpha \\ d\cos\delta\sin\alpha \\ d\sin\delta \end{pmatrix}.} \tag{3.9}$$

For Sirius, at $\alpha = 101.287°$, $\delta = -16.716°$ and $d = 2.637\,\text{pc}$, this is

$$\vec{r} = (-0.494,\ 2.477,\ -0.758)^T\ \text{pc}.$$

The local directions. The direction of increasing δ at the star is the rate of change of $\hat{\mathbf{r}}$ with δ, and the same for α. Differentiating Eq. (2.2),

$$\frac{\partial\hat{\mathbf{r}}}{\partial\delta} = \begin{pmatrix} -\sin\delta\cos\alpha \\ -\sin\delta\sin\alpha \\ \cos\delta \end{pmatrix}, \qquad \frac{\partial\hat{\mathbf{r}}}{\partial\alpha} = \begin{pmatrix} -\cos\delta\sin\alpha \\ \cos\delta\cos\alpha \\ 0 \end{pmatrix} = \cos\delta\begin{pmatrix} -\sin\alpha \\ \cos\alpha \\ 0 \end{pmatrix}. \tag{3.10}$$

The first has length $\sqrt{\sin^2\delta + \cos^2\delta} = 1$. The second has length $\cos\delta$, because a circle of constant declination has radius $\cos\delta$ on the unit sphere and shrinks to a point at the poles. Dividing it by $\cos\delta$ gives a unit vector. So the three directions at the star are

$$\hat{\mathbf{r}} = \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}, \qquad \hat{\boldsymbol{\alpha}} = \frac{1}{\cos\delta}\frac{\partial\hat{\mathbf{r}}}{\partial\alpha} = \begin{pmatrix} -\sin\alpha \\ \cos\alpha \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{\delta}} = \frac{\partial\hat{\mathbf{r}}}{\partial\delta} = \begin{pmatrix} -\sin\delta\cos\alpha \\ -\sin\delta\sin\alpha \\ \cos\delta \end{pmatrix}. \tag{3.11}$$

They are at right angles to each other at every point,

$$\begin{aligned} \hat{\mathbf{r}}\cdot\hat{\boldsymbol{\alpha}} &= -\cos\delta\cos\alpha\sin\alpha + \cos\delta\sin\alpha\cos\alpha = 0, \\ \hat{\mathbf{r}}\cdot\hat{\boldsymbol{\delta}} &= -\sin\delta\cos\delta\,(\cos^2\alpha + \sin^2\alpha) + \sin\delta\cos\delta = 0, \\ \hat{\boldsymbol{\alpha}}\cdot\hat{\boldsymbol{\delta}} &= \sin\delta\sin\alpha\cos\alpha - \sin\delta\sin\alpha\cos\alpha = 0, \end{aligned} \tag{3.12}$$

and $\hat{\mathbf{r}}\times\hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\delta}}$, so in the order $(\hat{\mathbf{r}}, \hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\delta}})$ they are right-handed, like (x, y, z).

The matrix of the local axes. By the rule of rows, Eq. (2.3), the matrix whose rows are the three local directions turns equatorial components into local ones. We call it $\mathbf{Q}$,

$$\mathbf{Q}(\alpha,\delta)=\begin{pmatrix}\cos\delta\cos\alpha & \cos\delta\sin\alpha & \sin\delta\\ -\sin\alpha & \cos\alpha & 0\\ -\sin\delta\cos\alpha & -\sin\delta\sin\alpha & \cos\delta\end{pmatrix}. \tag{3.13}$$

Its rows are orthonormal and right-handed, so $\mathbf{Q}$ is a rotation and $\mathbf{Q}^{-1}=\mathbf{Q}^T$ by Eq. (2.4). The star itself has local components $\mathbf{Q}\,\hat{\mathbf{r}}_{\text{eq}}=(1,0,0)^T$, straight up.

The same matrix comes out of two turns, as a second check. A pan by α about the pole, $\mathbf{R}_z(\alpha)$ of Eq. (2.5), brings the star to $(\cos\delta,0,\sin\delta)^T$ by Eq. (2.8). A tilt by δ about the new y axis then brings it to the first axis. By the rule of rows the tilt has the rows $(\cos\delta,0,\sin\delta)$, $(0,1,0)$ and $(-\sin\delta,0,\cos\delta)$, so

$$\begin{aligned}\begin{pmatrix}\cos\delta & 0 & \sin\delta\\ 0 & 1 & 0\\ -\sin\delta & 0 & \cos\delta\end{pmatrix}\mathbf{R}_z(\alpha) &\overset{(1)}{=}\begin{pmatrix}\cos\delta & 0 & \sin\delta\\ 0 & 1 & 0\\ -\sin\delta & 0 & \cos\delta\end{pmatrix}\begin{pmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{pmatrix}\\ &\overset{(2)}{=}\begin{pmatrix}\cos\delta\cos\alpha & \cos\delta\sin\alpha & \sin\delta\\ -\sin\alpha & \cos\alpha & 0\\ -\sin\delta\cos\alpha & -\sin\delta\sin\alpha & \cos\delta\end{pmatrix}=\mathbf{Q}(\alpha,\delta).\end{aligned} \tag{3.14}$$

(1) Eq. (2.5) with $\theta=\alpha$.
(2) Each row of the tilt times each column of $\mathbf{R}_z(\alpha)$.

The yearly parallax ellipse. The goal is the path a nearby star traces on the sky over a year, as seen from the moving Earth. We work in ecliptic coordinates (λ,β), Eq. (2.39), and take the Earth's orbit as a circle of radius 1 AU. With the Sun at ecliptic longitude $\lambda_\odot$, the Earth is on the opposite side, at $\vec{r}_\oplus=-1\,\text{AU}\,(\cos\lambda_\odot,\sin\lambda_\odot,0)^T$ from the Sun. The star is at $d\,\hat{\mathbf{r}}$ from the Sun, so from the Earth it is at $d\,\hat{\mathbf{r}}-\vec{r}_\oplus$. Its apparent direction is

$$\hat{\mathbf{r}}'=\frac{d\,\hat{\mathbf{r}}-\vec{r}_\oplus}{|d\,\hat{\mathbf{r}}-\vec{r}_\oplus|}\overset{(1)}{=}\hat{\mathbf{r}}-\frac{\vec{r}_\oplus-(\hat{\mathbf{r}}\cdot\vec{r}_\oplus)\,\hat{\mathbf{r}}}{d}. \tag{3.15}$$

(1) The first-order expansion of Eq. (2.62), with $-\vec{r}_\oplus/d$ in place of $\vec{v}/c$. It holds since $1\,\text{AU}/d\ll 1$ (P6).

The star moves away from the Earth's position, by the part of $\vec{r}_\oplus/d$ across the line of sight. The local directions in the ecliptic frame are Eq. (3.11) with $\alpha\to\lambda$ and $\delta\to\beta$, so $\hat{\boldsymbol{\lambda}}=(-\sin\lambda,\cos\lambda,0)^T$ and $\hat{\boldsymbol{\beta}}=(-\sin\beta\cos\lambda,-\sin\beta\sin\lambda,\cos\beta)^T$. Both are perpendicular to $\hat{\mathbf{r}}$, so the shifts along them are $-\vec{r}_\oplus\cdot\hat{\boldsymbol{\lambda}}/d$ and $-\vec{r}_\oplus\cdot\hat{\boldsymbol{\beta}}/d$. With $p=1\,\text{AU}/d$ from Eq. (3.2),

$$\begin{aligned}\cos\beta\,\Delta\lambda &\overset{(1)}{=} p\,(-\cos\lambda_\odot\sin\lambda+\sin\lambda_\odot\cos\lambda)\overset{(2)}{=}p\sin(\lambda_\odot-\lambda),\\ \Delta\beta &\overset{(1)}{=} -p\sin\beta\,(\cos\lambda_\odot\cos\lambda+\sin\lambda_\odot\sin\lambda)\overset{(2)}{=}-p\sin\beta\cos(\lambda_\odot-\lambda).\end{aligned} \tag{3.16}$$

(1) The dot products of $(\cos\lambda_\odot,\sin\lambda_\odot,0)$ with $\hat{\boldsymbol{\lambda}}$ and with $\hat{\boldsymbol{\beta}}$.
(2) The angle-difference identities.

A step $\Delta\lambda$ in longitude is an arc $\cos\beta\,\Delta\lambda$ on the sky, for the same reason as the $\cos\delta$ in Eq. (3.10). As $\lambda_\odot$ runs through the year,

$$\boxed{\left(\frac{\cos\beta\,\Delta\lambda}{p}\right)^2+\left(\frac{\Delta\beta}{p\sin\beta}\right)^2=1.} \tag{3.17}$$

Every star traces an ellipse with semi-axes p along the ecliptic and $p \sin\beta$ across it. Near the ecliptic pole it is a circle of radius p. On the ecliptic it closes to a line of half-length p. The semi-major axis is p for every star, which is why p is the number a catalogue lists. The aberration ellipse of Chapter 2 has the same shape and a quarter of a year's offset, but it is 20.5″ for every star, so the two are told apart by size and by timing.

3.3 Proper Motion and Transverse Velocity

The parallax and the direction give the star's position $\vec{r} = d\,\hat{\mathbf{r}}$, Eq. (3.9).

Stars move, so $\vec{r}$ changes. A catalogue position slowly goes out of date, as the change of epoch in Section 2.1.1 already showed. We need to read a velocity from the change.

Stand by a straight road and watch two cars, both at 30 m/s. The one 300 m away crosses your view at 0.1 radian, about 6°, per second. The one 3 km away crosses at only 0.6° per second. The angle swept per second is the sideways speed divided by the distance.

A star moving across the line of sight drifts across the sky in the same way. The drift is its **proper motion**. The idea is that the sideways speed is the proper motion times the distance.

A star 1 pc away moving sideways at 4.74 km/s covers 1 AU a year, so by the definition of the parsec it drifts by 1″ a year.

The picture is the velocity arrow of the star split into a part along the line of sight and a part across it. Only the part across shows as a drift on the sky.

Angular speed equals sideways speed over distance in every branch of mechanics. Here it is the parallax again, with the baseline of 1 AU replaced by the distance the star itself moves in a year. It gives the two components of the proper motion, why one of them carries $\cos\delta$, and the constant 4.74.

How the direction changes. As the star moves, its α and δ change with time, and so does $\hat{\mathbf{r}}(\alpha, \delta)$. By the chain rule,

$$\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} = \frac{\partial\hat{\mathbf{r}}}{\partial\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}t} + \frac{\partial\hat{\mathbf{r}}}{\partial\delta}\frac{\mathrm{d}\delta}{\mathrm{d}t} \overset{(1)}{=} \underbrace{\cos\delta\,\frac{\mathrm{d}\alpha}{\mathrm{d}t}}_{\mu_\alpha^*}\,\hat{\boldsymbol{\alpha}} + \underbrace{\frac{\mathrm{d}\delta}{\mathrm{d}t}}_{\mu_\delta}\,\hat{\boldsymbol{\delta}}. \tag{3.18}$$

(1) The partial derivatives (3.10), written with the unit vectors (3.11).

The two components are the **proper motion in right ascension** $\mu_\alpha^* = \cos\delta\,\dot{\alpha}$ and the **proper motion in declination** $\mu_\delta = \dot{\delta}$. Both are true angles on the sky per unit time, usually given in mas/yr. The rate $\dot{\alpha}$ itself is not, because a step in α is an arc of only $\cos\delta$ times its size, as at Eq. (3.10). The two directions are at right angles, so the total proper motion is

$$\mu = \left|\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}\right| = \sqrt{(\mu_\alpha^*)^2 + \mu_\delta^2} = \sqrt{(\cos\delta\,\dot{\alpha})^2 + \mu_\delta^2}. \tag{3.19}$$

The velocity of the star. The goal is the star's velocity $\vec{v} = \dot{\vec{r}}$ in terms of what is measured. Differentiating $\vec{r} = d\,\hat{\mathbf{r}}$,

$$\vec{v} \overset{(1)}{=} \dot{d}\,\hat{\mathbf{r}} + d\,\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \overset{(2)}{=} \underbrace{\dot{d}}_{v_r}\,\hat{\mathbf{r}} + \underbrace{d\,\mu_\alpha^*}_{v_\alpha}\,\hat{\boldsymbol{\alpha}} + \underbrace{d\,\mu_\delta}_{v_\delta}\,\hat{\boldsymbol{\delta}}. \tag{3.20}$$

(1) The product rule on $\vec{r} = d\,\hat{\mathbf{r}}$.

(2) Eq. (3.18).

The **radial velocity** $v_r = \dot{d}$ is the rate at which the distance grows. It does not change the direction, so astrometry cannot see it, and it is measured from the Doppler shift (Section 3.4). The other two components lie across the line of sight. Their size is the **transverse velocity**

$$v_t = \sqrt{v_\alpha^2 + v_\delta^2} = d\,\mu, \tag{3.21}$$

with μ in radians per unit time.

The constant 4.74. Catalogues give μ in seconds of arc per year and d through p in seconds of arc, and we want v_t in km/s.

$$\begin{aligned} v_t &\overset{(1)}{=} \frac{1\,\mathrm{AU}}{p}\,\mu \overset{(2)}{=} \frac{\mu\,[''/\mathrm{yr}]}{p\,['']}\,\frac{1\,\mathrm{AU}}{1\,\mathrm{yr}} \overset{(3)}{=} \frac{\mu\,[''/\mathrm{yr}]}{p\,['']}\,\frac{1.495\,978\,707\times10^{8}\,\mathrm{km}}{3.155\,76\times10^{7}\,\mathrm{s}} \\ &= 4.740\,47\,\frac{\mu\,[''/\mathrm{yr}]}{p\,['']}\ \mathrm{km/s}. \end{aligned} \tag{3.22}$$

(1) Eq. (3.21) with $d = 1\,\mathrm{AU}/p$ from Eq. (3.2).
(2) μ/p is a ratio of two angles, the same in any angular unit, so μ in ″/yr and p in ″ can be used directly.
(3) 1 AU in km, and the Julian year of 365.25 days in seconds.

With $1/p'' = d$ in parsecs from Eq. (3.5),

Result 3.2

$$v_t\,[\mathrm{km/s}] = 4.74047\,\mu\,[\mathrm{arcsec/yr}]\,d\,[\mathrm{pc}] = 4.74047\,\frac{\mu\,[\mathrm{arcsec/yr}]}{p\,[\mathrm{arcsec}]}. \tag{3.23}$$

The number 4.74047 is just 1 AU per year in km/s. The transverse velocity depends only on the ratio of the two angles, the yearly drift over the yearly wobble. The same holds for each component, $v_\alpha = 4.74047\,\mu_\alpha^*/p$ and $v_\delta = 4.74047\,\mu_\delta/p$.

Worked example 3.3: Transverse Velocity of Sirius

Sirius has $\mu_\alpha^* = -546.01$ mas/yr and $\mu_\delta = -1223.07$ mas/yr. By Eq. (3.19),

$$\mu = \sqrt{0.54601^2 + 1.22307^2}\ ''/\mathrm{yr} = 1.3394''/\mathrm{yr}.$$

With $d = 2.637$ pc, Eq. (3.23) gives

$$v_t = 4.74047 \times 1.3394 \times 2.637\ \mathrm{km/s} = 16.74\ \mathrm{km/s}.$$

The components are $v_\alpha = -6.83$ km/s and $v_\delta = -15.29$ km/s, so Sirius drifts mainly south.

3.4 Radial Velocity and the Doppler Effect

Proper motion and parallax give the two components of the velocity across the line of sight, Eq. (3.20).

The third component, $v_r = \dot{d}$, changes the distance but not the direction. A star moving straight toward us shows no drift on the sky at all, and its parallax changes far too slowly to measure.

Picture a friend who walks away from you at 1 m/s and throws you a ball every second, each at 10 m/s. Each ball starts 1 m further away than the one before, so it needs an extra 0.1 s to arrive. The balls reach you every 1.1 seconds, not every second. Walking toward you, the friend would send them every 0.9 seconds.

A star sends wave crests instead of balls. If it moves away, each crest starts a little further off, and the crests arrive further apart. The light is shifted to longer wavelengths. The idea is that the fractional stretch of the wavelength is v_r/c.

Sirius approaches at 5.5 km/s. Its Hα line, at 656.46 nm in the laboratory, arrives shortened by $656.46 \times 5.5/299\,792$ nm $= 0.012$ nm.

The picture is a row of wave crests bunched up in front of a moving source and spread out behind it.

The same stretch lowers the pitch of a passing siren and lets a radar gun time a car. It gives the radial-velocity curves of spectroscopic binaries in Chapter 6 and the line widths and Galactic rotation of the gas in Chapter 8. For stars $v_r/c \sim 10^{-5}$, so the first-order formula is enough (P6). We derive that formula, the exact one, and the correction for the Earth's own motion.

The stretched period. Take a source that emits crests one period $T_0 = \lambda_0/c$ apart in its own frame, and moves straight away from the observer at $v_r \ll c$.

$$T_{\text{obs}} \overset{(1)}{=} T_0 + \frac{v_r T_0}{c} = T_0\left(1 + \frac{v_r}{c}\right),$$
$$z \equiv \frac{\lambda_{\text{obs}} - \lambda_0}{\lambda_0} \overset{(2)}{=} \frac{T_{\text{obs}} - T_0}{T_0} \overset{(3)}{=} \frac{v_r}{c}. \tag{3.24}$$

(1) Between two crests the source moves a further $v_r T_0$ away, so the second crest travels that far more and arrives $v_r T_0/c$ later.
(2) $\lambda = cT$ for both wavelengths.
(3) The first line.

Result 3.3

$$z = \frac{\lambda_{\text{obs}} - \lambda_0}{\lambda_0} = \frac{v_r}{c} \qquad (v_r \ll c). \tag{3.25}$$

A positive v_r is a recession and a shift to the red. A negative v_r is an approach and a shift to the blue. The shift fixes v_r alone, and the motion across the line of sight leaves it unchanged at this order.

The exact formula. For speeds near c, such as relativistic jets, the clock of the moving source runs slow. In the observer's frame its period is $T_0/\sqrt{1 - v^2/c^2}$, where v is its full speed. For motion straight along the line of sight, $v = v_r$, and

$$1 + z = \frac{T_{\text{obs}}}{T_0} \overset{(1)}{=} \frac{1 + v_r/c}{\sqrt{1 - v_r^2/c^2}} \overset{(2)}{=} \frac{1 + v_r/c}{\sqrt{(1 + v_r/c)(1 - v_r/c)}} = \sqrt{\frac{1 + v_r/c}{1 - v_r/c}}. \tag{3.26}$$

(1) The stretch of the first line of the classical chain, applied to the dilated period.
(2) $1 - v_r^2/c^2 = (1 + v_r/c)(1 - v_r/c)$.

Squaring and solving for v_r/c,

$$1 + z = \sqrt{\frac{1 + v_r/c}{1 - v_r/c}} \quad \Longrightarrow \quad \frac{v_r}{c} = \frac{(1+z)^2 - 1}{(1+z)^2 + 1}. \tag{3.27}$$

For small v_r/c the root is $1 + v_r/c + \frac{1}{2}(v_r/c)^2 + \ldots$, so Eq. (3.25) is its first term. For Sirius the second term is 1.7×10^{-10}, far below what a spectrograph can measure. A source moving across the line of sight also shows a shift, $z = 1/\sqrt{1 - v^2/c^2} - 1 \approx v^2/2c^2$, from the slow clock alone. The redshift of distant galaxies is a different effect. It comes from the expansion of space, and Eq. (3.27) does not apply to it.

The Earth's own motion. A spectrograph on the Earth moves with it, at about 29.78 km/s round the Sun. That is $2\pi \times 4.74$ km/s, since the Earth covers 2π AU a year. The shift measures how fast the distance from the observer to the star grows. With the star at $\vec{r}_*$ and the Earth at $\vec{r}_\oplus$,

$$\frac{\mathrm{d}}{\mathrm{d}t}|\vec{r}_* - \vec{r}_\oplus| \overset{(1)}{=} \frac{(\vec{r}_* - \vec{r}_\oplus)\cdot(\vec{v}_* - \vec{v}_\oplus)}{|\vec{r}_* - \vec{r}_\oplus|} \overset{(2)}{=} (\vec{v}_* - \vec{v}_\oplus)\cdot\hat{\mathbf{r}} = v_r - \vec{v}_\oplus\cdot\hat{\mathbf{r}}. \tag{3.28}$$

(1) $\mathrm{d}|\vec{a}|/\mathrm{d}t = \vec{a}\cdot\dot{\vec{a}}/|\vec{a}|$.
(2) The star is so far away that $(\vec{r}_* - \vec{r}_\oplus)/|\vec{r}_* - \vec{r}_\oplus|$ is $\hat{\mathbf{r}}$ to first order in $1\ \mathrm{AU}/d$.

So the velocity relative to the Sun is the measured one plus $\vec{v}_\oplus\cdot\hat{\mathbf{r}}$. With the Earth's velocity of Chapter 2, this term swings once a year between $\pm 29.78\cos\beta$ km/s, where β is the star's ecliptic latitude. It is larger than many stellar radial velocities, so every catalogue value has it removed. The Earth's spin adds up to 0.47 km/s at the equator, which is removed in the same way.

3.5 The Full 3D Space Velocity Vector

Eq. (3.20) gives the star's velocity as three components $(v_r, v_\alpha, v_\delta)$ along its local axes.

Those axes differ from star to star. To compare two stars, or to add the Sun's own motion, we need all velocities in one fixed frame.

A ship's log gives the current as so much ahead and so much to starboard. To draw it on a chart the navigator turns those two numbers through the ship's heading into north and east. The current itself has not changed, only the axes it is written on.

A star's velocity is written on its own local axes, and the fixed frame is the equatorial one. The idea is that the local components turn into equatorial ones with the transpose of $\mathbf{Q}$, Eq. (3.13).

At $\alpha = \delta = 0$ the local axes are the equatorial ones, $\mathbf{Q}$ is the identity, and $(v_x, v_y, v_z) = (v_r, v_\alpha, v_\delta)$.

The picture is the same arrow drawn twice, once on the star's local axes and once on the fixed axes.

It is the change of basis of Chapter 2, applied to a velocity. It gives the velocity in fixed components and the total speed.

Turning to the equatorial frame. Stacking the three local directions of Eq. (3.11) as columns,

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \overset{(1)}{=} v_r\,\hat{\mathbf{r}} + v_\alpha\,\hat{\boldsymbol{\alpha}} + v_\delta\,\hat{\boldsymbol{\delta}} \overset{(2)}{=} \begin{pmatrix} | & | & | \\ \hat{\mathbf{r}} & \hat{\boldsymbol{\alpha}} & \hat{\boldsymbol{\delta}} \\ | & | & | \end{pmatrix} \begin{pmatrix} v_r \\ v_\alpha \\ v_\delta \end{pmatrix} \overset{(3)}{=} \mathbf{Q}^T(\alpha,\delta) \begin{pmatrix} v_r \\ v_\alpha \\ v_\delta \end{pmatrix}. \tag{3.29}$$

(1) Eq. (3.20), with each unit vector in equatorial components.
(2) A matrix times a column is the sum of its columns weighted by the entries.
(3) The rows of $\mathbf{Q}$, Eq. (3.13), are $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\delta}}$, so its columns are those of $\mathbf{Q}^T$.

$$\boxed{\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \begin{pmatrix} \cos\delta\cos\alpha & -\sin\alpha & -\sin\delta\cos\alpha \\ \cos\delta\sin\alpha & \cos\alpha & -\sin\delta\sin\alpha \\ \sin\delta & 0 & \cos\delta \end{pmatrix} \begin{pmatrix} v_r \\ v_\alpha \\ v_\delta \end{pmatrix}.} \tag{3.30}$$

Going back costs only a transpose, $(v_r, v_\alpha, v_\delta)^T = \mathbf{Q}\,(v_x, v_y, v_z)^T$, by Eq. (2.4).

The speed. A rotation keeps lengths, so the speed is the same in both sets of components,

$$v_x^2 + v_y^2 + v_z^2 = \begin{pmatrix} v_r & v_\alpha & v_\delta \end{pmatrix} \mathbf{Q}\,\mathbf{Q}^T \begin{pmatrix} v_r \\ v_\alpha \\ v_\delta \end{pmatrix} \overset{(1)}{=} v_r^2 + v_\alpha^2 + v_\delta^2 \overset{(2)}{=} v_r^2 + v_t^2. \tag{3.31}$$

(1) $\mathbf{Q}\mathbf{Q}^T = \mathbb{1}$, Eq. (2.4).
(2) Eq. (3.21).

$$v_{\text{total}} = \sqrt{v_r^2 + v_t^2} = \sqrt{v_x^2 + v_y^2 + v_z^2}. \tag{3.32}$$

Worked example 3.4: Space Velocity of Sirius

Sirius is at $\alpha = 101.287°$, $\delta = -16.716°$, with $v_r = -5.50\,\text{km/s}$ and, from the previous example, $v_\alpha = -6.83\,\text{km/s}$, $v_\delta = -15.29\,\text{km/s}$. Eq. (3.30) gives

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \begin{pmatrix} -0.18746 & -0.98066 & -0.05630 \\ 0.93922 & -0.19573 & 0.28207 \\ -0.28763 & 0 & 0.95774 \end{pmatrix} \begin{pmatrix} -5.50 \\ -6.83 \\ -15.29 \end{pmatrix} \text{km/s} = \begin{pmatrix} 8.59 \\ -8.14 \\ -13.06 \end{pmatrix} \text{km/s}.$$

The speed is $\sqrt{8.59^2 + 8.14^2 + 13.06^2} = 17.62\,\text{km/s}$, the same as $\sqrt{5.50^2 + 16.74^2}$ from Eq. (3.32).

3.6 Galactic Space Velocity Components (U, V, W)

Eq. (3.30) gives every star's velocity in one fixed frame, the equatorial one.

That frame is set by the Earth's spin axis, which means nothing to the Galaxy. Stars orbit the Galactic centre in the Galactic disk, so their motions are simplest on Galactic axes. The velocities are also measured relative to the Sun, which itself moves.

A wind gauge on a moving train reads the wind relative to the train. To get the wind over the ground we first turn the reading to compass directions and then add the train's own velocity.

The Sun is the train. We turn (v_x, v_y, v_z) to Galactic axes with the matrix of Chapter 2, and then add the Sun's own motion relative to its neighbours. The idea is that a velocity turns with the same matrix as a position, and that velocities relative to moving frames add.

For Sirius the result is 12.96 km/s toward the Galactic centre, 1.89 km/s against the Galactic rotation and 11.79 km/s toward the south Galactic pole. Relative to the Sun's neighbours it moves at 24.1, 10.3 and −4.5 km/s along the same three axes.

The picture is the disk of the Galaxy seen from above, with the Sun on its circle and an arrow for the star's motion drawn from the Sun.

The same two steps, turn the axes and then add the frame's velocity, take any measured velocity to a frame that moves. The rotation is a change of basis (Chapter 2), and the addition is Galilean, since every speed here is far below c. They give the velocity components (U, V, W) and their values relative to the local standard of rest.

The axes. Galactic velocities are given in a right-handed Cartesian frame centred on the Sun,

- U toward the Galactic centre ($l = 0°$, $b = 0°$),
- V along the direction of Galactic rotation ($l = 90°$, $b = 0°$),
- W toward the north Galactic pole ($b = +90°$).

Some authors take U positive toward the anticentre, which makes the frame left-handed. We do not.

Why a velocity turns like a position. Chapter 2 turned positions with a fixed matrix, $\hat{\mathbf{r}}_{\text{gal}} = \mathbf{T}\,\hat{\mathbf{r}}_{\text{eq}}$ with $\mathbf{T}$ the product of three turns, Eq. (2.46). Multiplying by d gives the same for the full position $\vec{r}$. The Galactic pole and centre are fixed by convention at their J2000.0 values, so $\mathbf{T}$ does not change with time, and

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \frac{\mathrm{d}}{\mathrm{d}t}\,\vec{r}_{\text{gal}} \overset{(1)}{=} \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{T}\,\vec{r}_{\text{eq}}\right) \overset{(2)}{=} \mathbf{T}\,\frac{\mathrm{d}\vec{r}_{\text{eq}}}{\mathrm{d}t} = \mathbf{T}\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}. \tag{3.33}$$

(1) Eq. (2.46), multiplied by d.
(2) $\mathbf{T}$ is constant.

The two frames share their origin at the Sun, so no extra velocity enters here.

The rotation. The three turns of Eq. (2.46), multiplied out as in Eq. (2.53), give

$$\begin{aligned}\begin{pmatrix} U \\ V \\ W \end{pmatrix} &= \mathbf{R}_z(90^\circ - l_{\text{CP}})\,\mathbf{R}_x(90^\circ - \delta_{\text{GP}})\,\mathbf{R}_z(\alpha_{\text{GP}} + 90^\circ)\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \\ &\overset{(1)}{=} \begin{pmatrix} -0.054876 & -0.873437 & -0.483835 \\ +0.494109 & -0.444830 & +0.746982 \\ -0.867666 & -0.198076 & +0.455984 \end{pmatrix}\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}.\end{aligned} \tag{3.34}$$

(1) The IAU values of α_{GP}, δ_{GP} and l_{CP}, as in Eq. (2.53).

By the rule of rows, the rows of $\mathbf{T}$ are the directions of U, V and W in equatorial components. Putting in Eq. (3.30) for the equatorial components and Eq. (3.23) for v_α and v_δ takes us from the measured numbers to (U, V, W) in one line,

Result 3.4

$$\begin{pmatrix} U \\ V \\ W \end{pmatrix} = \mathbf{T}\,\mathbf{Q}^T(\alpha, \delta)\begin{pmatrix} v_r \\ 4.74047\,\mu^*_\alpha/p \\ 4.74047\,\mu_\delta/p \end{pmatrix}, \tag{3.35}$$

in km/s for v_r in km/s, μ in ″/yr and p in ″. All six measured numbers $(\alpha, \delta, p, \mu^*_\alpha, \mu_\delta, v_r)$ enter, and nothing else. For Sirius, $\mathbf{T}$ times the column (8.59, −8.14, −13.06) km/s of the last example gives $(U, V, W) = (12.96,\ -1.89,\ -11.79)$ km/s.

The local standard of rest. The **local standard of rest** (LSR) is a frame that moves on a circular orbit round the Galactic centre at the Sun's distance from it, $R_0 \approx 8.2$ kpc, with circular speed $v_0 \approx 220$–240 km/s. The Sun moves relative to it. The measured value (Schönrich et al. 2010) is

$$(U_\odot, V_\odot, W_\odot) = (11.10^{+0.69}_{-0.75},\ 12.24^{+0.47}_{-0.47},\ 7.25^{+0.37}_{-0.36})\ \text{km/s}. \tag{3.36}$$

The velocity of the star relative to the LSR is its velocity relative to the Sun plus that of the Sun relative to the LSR. Both are written on the same Galactic axes, so the components add,

$$(U_{\text{LSR}}, V_{\text{LSR}}, W_{\text{LSR}}) = (U + U_\odot,\ V + V_\odot,\ W + W_\odot). \tag{3.37}$$

For Sirius this is $(24.06,\ 10.35,\ -4.54)$ km/s. Relative to the LSR, the velocity of a disk star is the small departure of its Galactic orbit from a circle.

3.7 The Cosmic Distance Ladder

Parallax gives a distance from geometry alone, Eq. (3.5).

Its angle shrinks as $1/d$, while the error of the measurement does not. Beyond some distance the parallax is lost in the error.

To find how far away a lighthouse is at night, we cannot pace the distance. But if we have measured how bright such a lamp is from close by, its faintness tells us how far it is.

Astronomers do the same with stars whose true brightness can be known from some other property. The nearby ones are calibrated by parallax, and they then give distances far beyond its reach. The idea is that each kind of distance calibrates the next, like the rungs of a ladder.

A distance is good to 10% when $\sigma_p/p = 0.1$. Since $d = 1/p$, the fractional errors of d and p are equal, and with $p = 1/d$ in seconds of arc and parsecs,

$$\frac{\sigma_d}{d} \overset{(1)}{=} \frac{\sigma_p}{p} \overset{(2)}{=} \sigma_p\, d \quad \Longrightarrow \quad d_{10\%} = \frac{0.1}{\sigma_p}. \tag{3.38}$$

(1) $|\mathrm{d}d/d| = |\mathrm{d}p/p|$ for $d = 1/p$.
(2) Eq. (3.5).

Hipparcos measured $\sigma_p \approx 1$ mas for bright stars, which gives 10% distances to 100 pc. *Gaia* reaches $\sigma_p \approx 0.02$ mas for bright stars, which gives 5 kpc, still well short of the far side of the Galaxy.

The picture is a ladder whose bottom rung stands on the Earth's orbit and whose top reaches the distant galaxies.

The ratio σ_p/p decides whether a parallax distance can be trusted, and that is P6. The ladder lists the rungs that take over beyond it.

1. **Spectroscopic parallax.** The absolute magnitude M_V follows from the spectral type and luminosity class on the H–R diagram. The distance then follows from the distance modulus $m - M = 5\log_{10} d\,[\mathrm{pc}] - 5$, Eq. (4.11).
2. **Main-sequence fitting.** The colour–magnitude diagram of a star cluster is shifted in magnitude until its main sequence lies on the zero-age main sequence (ZAMS) of nearby stars. The shift is the distance modulus.
3. **Standard candles, Cepheids and RR Lyrae.** Leavitt's period–luminosity relation for classical Cepheid variables, $M_V \approx -2.81\log_{10}(P/1\,\mathrm{day}) - 1.43$ with P the pulsation period, carries distances out to about 30 Mpc (the Hubble Space Telescope key project).
4. **Type Ia supernovae.** These are thermonuclear explosions of white dwarfs. The Phillips relation between the width of the light curve and the peak luminosity makes them standard, with $M_B \approx -19.3$, and they reach cosmological distances ($z \sim 1$–2).
5. **The Hubble–Lemaître law.** On cosmological scales, galaxies recede at a speed proportional to their distance, because space expands,

$$v_r = H_0\, d, \qquad H_0 \approx 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{3.39}$$

Every rung beyond parallax rests on an apparent brightness, a colour or the shape of a light curve. These are the quantities of photometry, Chapter 4.

3.8 Problem Set: Stellar Astrometry, Kinematics, and Space Velocities

3.8.1 Problem 1: Position, Space Velocity and Doppler Shift of Regulus

Problem Statement: Regulus (α Leonis), the brightest star in Leo, lies at $\alpha = 152.093°$, $\delta = +11.967°$. Its measured parallax is $p = 41.13\,\text{mas}$, its radial velocity $v_r = 4.39\,\text{km/s}$, and its proper motions are $\mu_\alpha^* = -248.73\,\text{mas/yr}$ and $\mu_\delta = +5.59\,\text{mas/yr}$.

(a) What is its distance d in parsecs?

(b) Find its position vector $\vec{r}$ in equatorial components.

(c) Find its velocity $\vec{v}$ relative to the Sun, in equatorial components, and its speed $|\vec{v}|$.

(d) At what wavelength is its Hα line observed ($\lambda_0 = 656.46\,\text{nm}$ in vacuum)?

(e) Find its Galactic velocity (U, V, W) and its velocity relative to the local standard of rest.

Essential Ideas:

- Parallax gives the distance, $d\,[\text{pc}] = 1/p\,['']$, Eq. (3.5).
- Proper motion gives the velocity across the line of sight, $v_\alpha = 4.74047\,\mu_\alpha^* d$ and $v_\delta = 4.74047\,\mu_\delta d$ in km/s, Eq. (3.23). The number 4.74047 is 1 AU per Julian year in km/s.
- The local components turn into equatorial ones with $\mathbf{Q}^T(\alpha, \delta)$, Eq. (3.30), and into Galactic ones with $\mathbf{T}$ after that, Eq. (3.34).
- For $v_r \ll c$ the Doppler shift is $(\lambda_{\text{obs}} - \lambda_0)/\lambda_0 = v_r/c$, Eq. (3.25).

Detailed Solutions:

(a) **Distance**. With $p = 41.13\,\text{mas} = 0.04113''$,

$$\boxed{d = \frac{1}{0.04113}\,\text{pc} = 24.31\,\text{pc}.}$$

(b) **Position vector**. With $\cos\delta = 0.97827$, $\sin\delta = 0.20735$, $\cos\alpha = -0.88371$ and $\sin\alpha = 0.46804$, Eq. (3.9) gives

$$\boxed{\vec{r} = 24.31\,\text{pc}\begin{pmatrix} 0.97827 \times (-0.88371) \\ 0.97827 \times 0.46804 \\ 0.20735 \end{pmatrix} = \begin{pmatrix} -21.02 \\ 11.13 \\ 5.04 \end{pmatrix}\text{pc}.}$$

(c) **Velocity and speed**. The proper motions give the local components

$$v_\alpha = 4.74047 \times (-0.24873) \times 24.31\,\text{km/s} = -28.67\,\text{km/s},$$
$$v_\delta = 4.74047 \times 0.00559 \times 24.31\,\text{km/s} = +0.64\,\text{km/s}.$$

Eq. (3.30) turns them into equatorial components,

$$\vec{v} = \begin{pmatrix} -0.86450 & -0.46804 & 0.18324 \\ 0.45787 & -0.88371 & -0.09705 \\ 0.20735 & 0 & 0.97827 \end{pmatrix}\begin{pmatrix} 4.39 \\ -28.67 \\ 0.64 \end{pmatrix}\text{km/s} = \begin{pmatrix} 9.74 \\ 27.28 \\ 1.54 \end{pmatrix}\text{km/s}.$$

By Eq. (3.32) the speed can be found from either set of components,

$$\boxed{|\vec{v}| = \sqrt{4.39^2 + 28.67^2 + 0.64^2}\,\text{km/s} = 29.01\,\text{km/s}.}$$

(d) **Observed wavelength of** Hα. The star recedes at $v_r = 4.39\,\mathrm{km/s}$, so by Eq. (3.25)

$$\lambda_{\mathrm{obs}} - \lambda_0 = \lambda_0 \frac{v_r}{c} = 656.46\,\mathrm{nm} \times \frac{4.39}{299\,792.458} = 0.0096\,\mathrm{nm},$$

$$\boxed{\lambda_{\mathrm{obs}} = 656.4696\,\mathrm{nm}.}$$

The shift is one part in 68 000 of the wavelength.

(e) **Galactic velocity**. Eq. (3.34) acting on the column of part (c) gives

$$\boxed{(U, V, W) = (-25.11,\ -6.17,\ -13.15)\,\mathrm{km/s},}$$

with the same length, 29.01 km/s, as a rotation must keep. Adding the Sun's motion relative to the LSR,

$$\boxed{(U_{\mathrm{LSR}}, V_{\mathrm{LSR}}, W_{\mathrm{LSR}}) = (-14.01,\ 6.07,\ -5.90)\,\mathrm{km/s}.}$$

Most of the speed relative to the Sun is the U component, directed away from the Galactic centre. Relative to the LSR, Regulus moves outward from the centre at 14 km/s, a little ahead of circular rotation, and slowly toward the south side of the disk.

3.8.2 Problem 2: Closest Approach of a Fast Nearby Star

Problem Statement: A nearby red dwarf has parallax $p = 0.547''$, total proper motion $\mu = 10.39''/\mathrm{yr}$ and radial velocity $v_r = -110\,\mathrm{km/s}$. These are close to the values for Barnard's star. Treat its motion relative to the Sun as a straight line.

(a) Find its distance and its transverse velocity.

(b) Find its speed and the angle between its velocity and the line of sight.

(c) How close will it come to the Sun, and when?

(d) What will its proper motion be at closest approach?

Essential Ideas:

- With no force acting, momentum is conserved (P1) and the star moves on a straight line, $\vec{r}(t) = \vec{r}_0 + \vec{v}\,t$.
- The velocity splits into v_r along the line of sight and $v_t = 4.74047\,\mu/p$ across it, Eq. (3.23), and $|\vec{v}|^2 = v_r^2 + v_t^2$, Eq. (3.32).
- The closest approach is where $|\vec{r}(t)|^2$ has its minimum.

Detailed Solutions:

(a) **Distance and transverse velocity**. By Eqs. (3.5) and (3.23),

$$\boxed{d = \frac{1}{0.547}\,\mathrm{pc} = 1.828\,\mathrm{pc}, \qquad v_t = 4.74047 \times \frac{10.39}{0.547}\,\mathrm{km/s} = 90.0\,\mathrm{km/s}.}$$

(b) **Speed and direction**. By Eq. (3.32),

$$|\vec{v}| = \sqrt{110^2 + 90.0^2}\,\mathrm{km/s} = 142.2\,\mathrm{km/s}.$$

The angle ψ between $\vec{v}$ and $\hat{\mathbf{r}}$ has $\cos\psi = v_r/|\vec{v}| = -0.774$,

$$\boxed{|\vec{v}| = 142.2\,\mathrm{km/s}, \qquad \psi = 140.7^\circ.}$$

The star is coming toward us, at 39.3° from the line that joins it to the Sun.

(c) **Closest approach**. Write $\vec{r}_0 = d\,\hat{\mathbf{r}}$ now. The squared distance and its minimum are

$$|\vec{r}(t)|^2 \overset{(1)}{=} d^2 + 2d\,(\hat{\mathbf{r}}\cdot\vec{v})\,t + |\vec{v}|^2 t^2 \overset{(2)}{=} d^2 + 2d\,v_r\,t + |\vec{v}|^2 t^2,$$

$$t_{\min} \overset{(3)}{=} -\frac{d\,v_r}{|\vec{v}|^2}, \qquad d^2_{\min} \overset{(4)}{=} d^2 - \frac{d^2 v_r^2}{|\vec{v}|^2} = d^2\,\frac{v_t^2}{|\vec{v}|^2}.$$

(1) $|\vec{r}_0 + \vec{v}\,t|^2$ multiplied out.
(2) $\hat{\mathbf{r}}\cdot\vec{v} = v_r$, Eq. (3.20).
(3) The derivative with respect to t set to zero.
(4) $t_{\min}$ put back, then $|\vec{v}|^2 - v_r^2 = v_t^2$.

So $d_{\min} = d\,v_t/|\vec{v}| = 1.828 \times 90.0/142.2$ pc. With $d = 1.828 \times 3.0857 \times 10^{13}$ km $= 5.64 \times 10^{13}$ km, $t_{\min} = 5.64 \times 10^{13} \times 110/142.2^2$ s $= 3.07 \times 10^{11}$ s,

$$\boxed{d_{\min} = 1.16\,\text{pc}, \qquad t_{\min} = 9.7 \times 10^3 \text{ years from now.}}$$

(d) **Proper motion at closest approach**. There the whole velocity is across the line of sight, so $v_t = |\vec{v}|$ and, by Eq. (3.23),

$$\boxed{\mu = \frac{|\vec{v}|}{4.74047\,d_{\min}} = \frac{142.2}{4.74047 \times 1.158}\,''/\text{yr} = 25.9''/\text{yr}.}$$

A second route uses the angular momentum about the Sun. With no force it is conserved (P1), and per unit mass it is $|\vec{r}\times\vec{v}| = d\,v_t = \mu\,d^2$ by Eq. (3.21). So $\mu\,d^2$ is the same now and at closest approach, and $\mu = 10.39\times(1.828/1.158)^2\,''/\text{yr} = 25.9''/\text{yr}$ again. The same law gives the present rate of growth. Differentiating $\mu\,d^2 = \text{const}$ gives $\dot{\mu} = -2\mu\,\dot{d}/d = -2\mu\,v_r/d$, which is $0.0013''/\text{yr}$ per year. The velocity is constant, yet the proper motion grows, because the distance shrinks. This growth is the **secular acceleration**, and for Barnard's star it shifts the position by about half a second of arc in thirty years.

Chapter 4

Stellar Photometry: Flux, Magnitudes, and Radii

Chapter 3 found the distances of nearby stars from their parallax and used magnitudes to reach further. Photometry is the measurement of how much light arrives from a star. Together with the distance, the light gives the power of the star. Its colour gives the temperature of the surface. Power and temperature together give the size of a star that no telescope shows as a disc.

On its way to us the light spreads out with distance, and dust in space dims it and makes it redder. Each effect is undone by one formula, and astronomers record the result on a logarithmic scale of magnitudes.

4.1 Flux and the Inverse-Square Law

Chapter 3 gave the distance d of a nearby star from its parallax p, with $d\,[\mathrm{pc}] = 1/p\,['']$ (Eq. (3.5)).

A distance alone says nothing about the star itself. A star can look bright because it is powerful or because it is close. To separate the two we need a rule for how light thins out with distance.

Stand under a street lamp at night with a book, then walk away until you are twice as far. The page is much dimmer. The lamp sends out the same light every second as before. At twice the distance that light is spread over a sphere with four times the area, so the page gets a quarter of it.

A star is such a lamp, and the Earth holds the page. The power of a star is shared out over a sphere centred on it, so the light per square metre falls as one over the square of the distance.

Suppose the lamp sends out 100 W of light, equally in all directions. At 2 m the sphere has area $4\pi(2\,\mathrm{m})^2 = 50.3\,\mathrm{m}^2$, and each square metre gets $100/50.3 = 1.99\,\mathrm{W\,m^{-2}}$. At 4 m the area is $201\,\mathrm{m}^2$, and each square metre gets $0.497\,\mathrm{W\,m^{-2}}$, a quarter as much.

Picture the star inside a set of nested spheres, like the skins of an onion. The same power passes through every skin, and the outer skins are larger.

This is energy conservation, principle P1 (Section 9.1). Counting what crosses a closed surface in the same way gives the $1/r^2$ of Coulomb's law and of the gravity of a point mass. In Section 9.2 it gives the gravity of a sphere, P2. It also sets how the sound of a loudspeaker fades in open air. Here it gives the flux at any distance from the power of the star, and the power from a measured flux and a known distance.

Luminosity and flux. The **luminosity** L of a star is the total power it sends out, over all wavelengths, in watts. The **flux** F at a point is the power that crosses one square metre facing the star, in $\mathrm{W\,m^{-2}}$. A flux summed over all wavelengths is called **bolometric**.

Spreading over a sphere. To find the flux at distance d, we draw a sphere of radius d about the star and count the power that crosses it (P1). The star is steady, and nothing between it and the sphere absorbs light.

$$\begin{aligned} L &\overset{(1)}{=} \oint F \, dA \\ &\overset{(2)}{=} F(d) \oint dA \\ &\overset{(3)}{=} 4\pi d^2 \, F(d). \end{aligned} \tag{4.1}$$

(1) In a steady state the energy that leaves the star each second crosses the sphere in the same second, because none is stored or absorbed on the way (P1).
(2) The star sends out light equally in all directions, so F is the same at every point of the sphere.
(3) The area of a sphere of radius d.

Result 4.1

$$F = \frac{L}{4\pi d^2} \quad \left[\mathrm{W\,m^{-2}}\right]. \tag{4.2}$$

The flux depends on the star and its distance only through L/d^2. A measured flux and a parallax distance therefore give the luminosity, $L = 4\pi d^2 F$.

The flux at the surface. The sphere can be drawn anywhere outside the star, including at its surface, at radius R. There the flux is $F_{\mathrm{surf}} = L/(4\pi R^2)$. Dividing Eq. (4.2) by it,

$$\boxed{F(d) = F_{\mathrm{surf}} \left(\frac{R}{d}\right)^2}. \tag{4.3}$$

The light thins out by the ratio of the two areas. This form returns in Section 4.5, where it gives the angular size of a star from its flux.

Worked example 4.1: The solar constant

The Sun has $L_\odot = 3.828 \times 10^{26}$ W. The Earth is at 1 AU $= 1.495\,978\,707 \times 10^{11}$ m from it, and Eq. (4.2) gives

$$F_\odot = \frac{3.828 \times 10^{26}\ \mathrm{W}}{4\pi \left(1.495\,978\,707 \times 10^{11}\ \mathrm{m}\right)^2} = 1361\ \mathrm{W\,m^{-2}}.$$

This is the **solar constant**, the bolometric flux of the Sun above the Earth's air. Satellites measure it, and the Sun's luminosity is found by running the same formula backwards.

4.2 The Magnitude Scale

We can now give the light of a star as a flux in $\mathrm{W\,m^{-2}}$.

Fluxes of stars cover an enormous range. The Sun gives $1361\ \mathrm{W\,m^{-2}}$, and the faintest stars the eye can see give about 10^{-13} of that. A flux is also hard to measure in watts. A detector and the air above it change from night to night, so what an astronomer measures well is the ratio of a star's flux to that of a standard star seen through the same instrument.

Our ears face the same problem. They judge loudness roughly by ratios, not by differences. Going from one violin to ten sounds like one step, and going from ten to a hundred sounds like another step of the same size. Sound engineers therefore count in decibels, ten times the logarithm of a ratio of powers. Ten decibels is a factor of 10 in power, and twenty decibels a factor of 100, whatever the level we start from.

The eye also judges brightness roughly by ratios (the Weber–Fechner law). In the 2nd century BCE Hipparchus sorted the stars seen by the naked eye into six classes, from first magnitude for the brightest to sixth for the faintest. In 1856 Norman Pogson noticed that a first-magnitude star gives about 100 times the light of a sixth-magnitude one, and made this exact. A magnitude is a logarithm of a ratio of two fluxes, with five magnitudes set equal to a factor of 100.

Two stars differ by 5 magnitudes when one gives 100 times the flux of the other. One magnitude is then a factor of $100^{1/5} = 2.512$, and 2.5 magnitudes a factor of 10. Five steps of 2.512 make $2.512^5 = 100$.

Picture a ruler whose marks are equally spaced, where each mark means "multiply by 2.512". It runs backwards, with brighter stars at smaller numbers, as in the ranking of Hipparchus.

Decibels, the pH of a solution and the Richter scale of earthquakes are the same device. A magnitude never says how much light a star sends. It says only how the star compares with another, so it is the comparison of two scales, P6 (Section 9.6), in its plainest form. It gives Pogson's formula, the absolute magnitude, and the distance modulus, which turns a difference of magnitudes into a distance.

4.2.1 Apparent Magnitude and Pogson's Formula

Fixing the constant. A magnitude falls as the flux rises and changes by equal steps for equal ratios of flux. So it is a logarithm of the flux with a negative factor, $m = -k \log_{10} F + \text{constant}$. For two stars the constant cancels,

$$m_1 - m_2 = -k \log_{10} \frac{F_1}{F_2}.$$

Pogson's rule sets $m_1 - m_2 = 5$ when star 1 is fainter by a factor of 100,

$$\begin{aligned} 5 &\overset{(1)}{=} -k \log_{10} \frac{1}{100} \\ &= 2k, \end{aligned} \tag{4.4}$$

(1) Pogson's rule with $F_1/F_2 = 1/100$.

so $k = 5/2$. The **apparent magnitudes** of two stars with fluxes F_1 and F_2 then differ by

Result 4.2

$$m_1 - m_2 = -2.5 \log_{10} \frac{F_1}{F_2}. \tag{4.5}$$

Turned round, a difference of magnitudes gives a ratio of fluxes,

$$\frac{F_1}{F_2} = 10^{-0.4(m_1 - m_2)}, \tag{4.6}$$

and one magnitude is a factor of $10^{0.4} = 2.512$.

The zero point. One reference of known flux fixes the whole scale. Let star 2 be a reference that has magnitude zero, with flux F_{ref}. Eq. (4.5) becomes

$$m = -2.5 \log_{10} \frac{F}{F_{\text{ref}}} = -2.5 \log_{10} F + 2.5 \log_{10} F_{\text{ref}}. \tag{4.7}$$

The last term is a constant, different for each filter. In the visible bands F_{ref} is close to the flux of the star Vega, which has $V = 0.03$. For bolometric magnitudes the International Astronomical Union fixed $F_{\text{ref}} = 2.518 \times 10^{-8}\,\text{W}\,\text{m}^{-2}$ in 2015. The Sun then has the apparent bolometric magnitude

$$m_{\text{bol},\odot} = -2.5 \log_{10} \frac{1361}{2.518 \times 10^{-8}} = -26.83.$$

4.2.2 Absolute Magnitude and the Distance Modulus

An apparent magnitude mixes the star's power with its distance, just as the flux does. To compare stars we put them all at one distance. The **absolute magnitude** M of a star is the apparent magnitude it would have at a distance of 10 pc.

Absolute magnitude and luminosity. At one common distance the fluxes of two stars are in the ratio of their luminosities.

$$\begin{aligned} M_1 - M_2 &\overset{(1)}{=} -2.5 \log_{10} \frac{F_1(10\,\text{pc})}{F_2(10\,\text{pc})} \\ &\overset{(2)}{=} -2.5 \log_{10} \frac{L_1/\left(4\pi(10\,\text{pc})^2\right)}{L_2/\left(4\pi(10\,\text{pc})^2\right)} \\ &= -2.5 \log_{10} \frac{L_1}{L_2}. \end{aligned} \tag{4.8}$$

(1) Eq. (4.5) for the two stars placed at 10 pc.
(2) Eq. (4.2) for each star.

Taking the Sun as star 2,

$$\boxed{M - M_\odot = -2.5 \log_{10} \frac{L}{L_\odot}.} \tag{4.9}$$

For bolometric magnitudes, a star that gives the zero-point flux F_{ref} at 10 pc has the luminosity

$$4\pi(10\,\text{pc})^2 F_{\text{ref}} = 3.0128 \times 10^{28}\,\text{W}.$$

Using this star as star 2 instead, the Sun has

$$M_{\text{bol},\odot} = -2.5 \log_{10} \frac{3.828 \times 10^{26}}{3.0128 \times 10^{28}} = +4.74.$$

The distance modulus. Now compare one star with itself, at its true distance d and at 10 pc.

$$\begin{aligned} m - M &\overset{(1)}{=} -2.5 \log_{10} \frac{F(d)}{F(10\,\text{pc})} \\ &\overset{(2)}{=} -2.5 \log_{10} \frac{L/(4\pi d^2)}{L/\left(4\pi(10\,\text{pc})^2\right)} \\ &= -2.5 \log_{10} \left(\frac{10\,\text{pc}}{d}\right)^2 \\ &\overset{(3)}{=} 5 \log_{10} \frac{d}{10\,\text{pc}}. \end{aligned} \tag{4.10}$$

(1) Eq. (4.5), with m the magnitude at d and M the magnitude at 10 pc.
(2) Eq. (4.2) at both distances.
(3) $\log_{10} x^2 = 2\log_{10} x$ and $\log_{10}(1/x) = -\log_{10} x$.

Result 4.3

$$m - M = 5\log_{10}\frac{d}{10\,\mathrm{pc}} = 5\log_{10} d\,[\mathrm{pc}] - 5. \tag{4.11}$$

The luminosity has cancelled. Without dust, $m - M$ depends on the distance alone, and every 5 magnitudes is a factor of 10 in distance. The difference $m - M$ is called the **distance modulus**. With the parallax distance of Eq. (3.5), $\log_{10} d\,[\mathrm{pc}] = -\log_{10} p\,['']$, and

$$m - M = -5\log_{10} p\,[''] - 5. \tag{4.12}$$

A star with $p = 0.1''$ is at 10 pc and has $m = M$.

As a check, the Sun is at 1 AU $= 4.848 \times 10^{-6}$ pc, so Eq. (4.11) gives $m - M = -31.57$ for it. Then $m_{\mathrm{bol},\odot} = 4.74 - 31.57 = -26.83$, the value found above from the zero point.

Filter and bolometric magnitudes. A magnitude through a filter counts only part of the light. The difference $m_{\mathrm{bol}} - m_V$ is the **bolometric correction**. It depends on the temperature, because a hot star sends much of its light out in the ultraviolet and a cool star in the infrared, where the V filter does not look. Eqs. (4.5)–(4.12) hold in any one band and for bolometric magnitudes alike.

4.3 Light from a Hot Surface

We can now find the luminosity of a star from its flux and its distance.

The luminosity does not say whether the star is small and hot or large and cool. For that we need to know how much light a hot surface gives off, and in which colours.

Watch the coil of an electric stove as it heats. At first it gives warmth but no glow. Then it glows dull red, then orange. A blacksmith's iron, hotter still, turns yellow and then almost white. Two things change together. The colour moves from red toward blue, and the heat felt on the face grows very fast.

Stars do the same. In Orion, Betelgeuse is orange-red and Rigel is blue-white, because Rigel's surface is hotter. A dense hot body that absorbs all the light falling on it, a **blackbody**, gives off a spectrum fixed by its temperature alone, and the surface of a star is close to one.

A stove coil at 1000 K is brightest at 2.9 μm, in the infrared, so we see only the red tail of its light. The Sun's surface, at 5772 K, is brightest at 502 nm, in the middle of the visible. Each square metre of the Sun gives off 6.29×10^7 W, which is 1110 times as much as a square metre of the coil, although the Sun is only 5.77 times hotter. The factor is 5.772^4.

Picture a hump-shaped curve of brightness against wavelength. As the temperature rises, the hump grows taller at every wavelength and its peak slides toward the blue.

The shape of the hump comes from counting the quantum states of light and filling them at temperature T, principle P3. It is derived that way in Section 9.3. The same curve describes the glow of a lamp filament, the picture in a thermal camera, and the radiation left over from the early universe at 2.7 K. Here it gives the wavelength of the peak, Wien's law, and the total power of a hot surface, the Stefan–Boltzmann law. The second defines the temperature of a star.

4.3.1 The Planck Function

The **specific intensity** of light is the power carried along a ray, per unit area, per unit solid angle of direction and per unit frequency. How it is defined in full and how it changes inside matter is the subject of Chapter 11. For a blackbody at temperature T the specific intensity is the **Planck function**, derived by counting states in Eq. (9.26),

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/k_BT}-1} \qquad \left[\mathrm{W\,m^{-2}\,sr^{-1}\,Hz^{-1}}\right]. \tag{4.13}$$

Here h is Planck's constant, c the speed of light and k_B Boltzmann's constant. B_ν depends on T and ν only, and it is the same in every direction. The photon energy in units of k_BT,

$$x = \frac{h\nu}{k_BT} = \frac{hc}{\lambda k_BT},$$

appears in every formula below.

Per unit wavelength. Filters are described by wavelength, so we also need the intensity per unit wavelength, B_λ. The power in a narrow band of the spectrum cannot depend on whether we label the band by wavelength or by frequency, so $B_\lambda\,|d\lambda| = B_\nu\,|d\nu|$.

$$\begin{aligned} B_\lambda &\overset{(1)}{=} B_\nu \left|\frac{\mathrm{d}\nu}{\mathrm{d}\lambda}\right| \\ &\overset{(2)}{=} B_\nu\,\frac{c}{\lambda^2} \\ &\overset{(3)}{=} \frac{2hc^3}{c^2\lambda^3}\,\frac{c}{\lambda^2}\,\frac{1}{e^{hc/\lambda k_BT}-1}. \end{aligned} \tag{4.14}$$

(1) Equal power in the same band of the spectrum.
(2) $\nu = c/\lambda$, so $d\nu/d\lambda = -c/\lambda^2$.
(3) Eq. (4.13) with $\nu = c/\lambda$.

Collecting the powers of c and λ,

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_BT}-1} \qquad \left[\mathrm{W\,m^{-2}\,sr^{-1}\,m^{-1}}\right]. \tag{4.15}$$

The factor c/λ^2 stretches the curve differently from B_ν, and this moves the peak, as we see next.

4.3.2 Wien's Displacement Law

To find where B_λ peaks, we write it through x. Since $x = hc/(\lambda k_BT)$ falls steadily as λ grows, a peak in λ is a peak in x.

$$\begin{aligned} B_\lambda &\overset{(1)}{=} 2hc^2\left(\frac{k_BT}{hc}\right)^5 \frac{x^5}{e^x-1} \\ &= \frac{2k_B^5T^5}{h^4c^3}\,\frac{x^5}{e^x-1}. \end{aligned} \tag{4.16}$$

(1) Eq. (4.15) with $1/\lambda = k_BT\,x/(hc)$.

At fixed T only the factor $x^5/(e^x-1)$ changes with wavelength. Its derivative vanishes at the peak,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^5}{e^x-1}\right) &\overset{(1)}{=} \frac{5x^4(e^x-1)-x^5e^x}{(e^x-1)^2}=0,\\ 5(e^x-1) &\overset{(2)}{=} xe^x,\\ x &\overset{(3)}{=} 5\left(1-e^{-x}\right).\end{aligned} \tag{4.17}$$

(1) The quotient rule.
(2) The numerator vanishes, $5x^4(e^x-1)=x^5e^x$. Divide by x^4.
(3) Divide by e^x.

This has no solution in closed form. Starting from $x=5$ and putting each value back into the right-hand side gives 4.96631, then 4.96516, then 4.96512, and the values settle at $x=4.965114$. The peak wavelength $\lambda_{\max}$ is then

$$\begin{aligned}\lambda_{\max}T &\overset{(1)}{=} \frac{hc}{4.965114\,k_B}\\ &= \frac{(6.626\times10^{-34})(2.998\times10^{8})}{4.965114\times1.381\times10^{-23}}\,\mathrm{m\,K}.\end{aligned} \tag{4.18}$$

(1) The definition of x at the peak.

Result 4.4

$$\lambda_{\max}T = 2.898\times10^{-3}\,\mathrm{m\,K} = 2898\,\mu\mathrm{m\,K}. \tag{4.19}$$

The wavelength of the peak is inversely proportional to the temperature and depends on nothing else. The Sun, at 5772 K, peaks at 502 nm.

Per unit frequency the peak lies elsewhere. In terms of x, $B_\nu=(2k_B^3T^3/h^2c^2)\,x^3/(e^x-1)$. The same steps with the power 3 in place of 5 give $x=3(1-e^{-x})$, so $x=2.821$. For the Sun this is $\nu_{\max}=2.821\,k_BT/h=339$ THz, whose wavelength $c/\nu_{\max}=883$ nm is far from 502 nm. Neither peak is wrong. A curve per unit wavelength and a curve per unit frequency put different widths of spectrum under each point, so "the colour at which a star is brightest" depends on which curve is meant. The total power, found next, does not.

4.3.3 The Stefan–Boltzmann Law

The flux leaving a blackbody surface is the power per square metre summed over all outward directions and all frequencies. This is the same counting of energy through a surface as in Section 4.1 (P1), now done at the surface itself.

Summing over directions. Take a small patch of surface. A ray leaving it at the angle ψ to the outward normal sees the patch foreshortened by $\cos\psi$. Per unit area of the patch, it carries the power $B_\nu\cos\psi\,d\Omega\,d\nu$ into the solid angle $d\Omega=\sin\psi\,d\psi\,d\phi$, with ϕ the angle around the

normal. Only outward rays count, $0 \le \psi \le \pi/2$.

$$\begin{aligned} F_{\text{surf}} &\overset{(1)}{=} \int_0^\infty d\nu \int_0^{2\pi} d\phi \int_0^{\pi/2} B_\nu \cos\psi \sin\psi \, d\psi \\ &\overset{(2)}{=} \int_0^\infty B_\nu \, d\nu \times 2\pi \times \left[\frac{\sin^2\psi}{2}\right]_0^{\pi/2} \\ &= \pi \int_0^\infty B_\nu \, d\nu. \end{aligned} \tag{4.20}$$

(1) The power per unit area, summed over the outward half of all directions and over frequency.
(2) B_ν is the same in every direction, so it leaves the angle integrals. The ϕ integral is 2π, and $\int \cos\psi \sin\psi \, d\psi = \frac{1}{2}\sin^2\psi$.

The factor is π, not the 2π of a half-sphere of directions, because rays that leave at a slant cross the surface at a slant. The same steps at one frequency give πB_ν per unit frequency, and πB_λ per unit wavelength.

Summing over frequency. The substitution $\nu = k_B T x/h$ takes all the temperature out of the integral.

$$\begin{aligned} \int_0^\infty B_\nu \, d\nu &\overset{(1)}{=} \frac{2h}{c^2} \int_0^\infty \frac{\nu^3 \, d\nu}{e^{h\nu/k_B T} - 1} \\ &\overset{(2)}{=} \frac{2h}{c^2} \left(\frac{k_B T}{h}\right)^4 \int_0^\infty \frac{x^3 \, dx}{e^x - 1} \\ &\overset{(3)}{=} \frac{2k_B^4 T^4}{h^3 c^2} \frac{\pi^4}{15}. \end{aligned} \tag{4.21}$$

(1) Eq. (4.13).
(2) $\nu = k_B T x/h$ and $d\nu = (k_B T/h)\, dx$, so $\nu^3 d\nu = (k_B T/h)^4 x^3 dx$. The limits stay 0 and ∞.
(3) The integral is $\pi^4/15$, Eq. (4.22) below.

The integral. Multiplying the top and bottom by e^{-x} turns the integrand into a geometric series.

$$\begin{aligned} \int_0^\infty \frac{x^3 \, dx}{e^x - 1} &\overset{(1)}{=} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} \, dx \\ &\overset{(2)}{=} \sum_{n=1}^\infty \int_0^\infty x^3 e^{-nx} \, dx \\ &\overset{(3)}{=} \sum_{n=1}^\infty \frac{1}{n^4} \int_0^\infty y^3 e^{-y} \, dy \\ &\overset{(4)}{=} 6 \sum_{n=1}^\infty \frac{1}{n^4} \\ &\overset{(5)}{=} \frac{6\pi^4}{90} = \frac{\pi^4}{15}. \end{aligned} \tag{4.22}$$

(1) Multiply the top and the bottom by e^{-x}.
(2) $e^{-x}/(1 - e^{-x}) = \sum_{n \ge 1} e^{-nx}$, a geometric series with ratio $e^{-x} < 1$ for $x > 0$. All terms are positive, so the sum and the integral may be swapped.

(3) The substitution $y = nx$.
(4) Three integrations by parts give $\int_0^\infty y^3 e^{-y}\,dy = 3! = 6$.
(5) $\sum_{n\ge1} 1/n^4 = \pi^4/90$, shown next.

The sum of $1/n^4$. On $-\pi \le x \le \pi$ the function x^2 has the Fourier series $x^2 = \pi^2/3 + \sum_{n\ge1} a_n \cos nx$, with $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 \cos nx\,dx = 4(-1)^n/n^2$ from two integrations by parts. Parseval's theorem says the mean square of a function is the sum of the squares of its Fourier terms, $\frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx = 2(\pi^2/3)^2 + \sum_n a_n^2$ for this series.

$$\begin{aligned}
\frac{1}{\pi}\int_{-\pi}^{\pi} x^4\,dx &\overset{(1)}{=} 2\left(\frac{\pi^2}{3}\right)^2 + \sum_{n=1}^{\infty}\frac{16}{n^4},\\
\frac{2\pi^4}{5} &\overset{(2)}{=} \frac{2\pi^4}{9} + 16\sum_{n=1}^{\infty}\frac{1}{n^4},\\
\sum_{n=1}^{\infty}\frac{1}{n^4} &= \frac{1}{16}\left(\frac{2\pi^4}{5} - \frac{2\pi^4}{9}\right) = \frac{1}{16}\,\frac{8\pi^4}{45} = \frac{\pi^4}{90}.
\end{aligned} \tag{4.23}$$

(1) Parseval's theorem with $f = x^2$ and $a_n^2 = 16/n^4$.
(2) $\int_{-\pi}^{\pi} x^4 dx = 2\pi^5/5$.

The law. Putting the frequency sum into Eq. (4.20),

$$F_{\text{surf}} = \pi\,\frac{2k_B^4 T^4}{h^3 c^2}\,\frac{\pi^4}{15} = \sigma_{\text{SB}} T^4, \tag{4.24}$$

where the **Stefan–Boltzmann constant** collects all the other factors,

$$\boxed{\sigma_{\text{SB}} = \frac{2\pi^5 k_B^4}{15 c^2 h^3} = 5.670 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}.} \tag{4.25}$$

The power per square metre grows as the fourth power of the temperature, so doubling T multiplies it by 16. It depends on nothing about the surface except its temperature, because the light of a blackbody is set by T alone.

4.3.4 Effective Temperature

A real star is not a perfect blackbody. Its spectrum has absorption lines, and its light comes from layers at different temperatures (Section 11.7). We still give it one temperature, by asking which blackbody of the same size gives off the same power. The **effective temperature** T_{eff} of a star of luminosity L and radius R is the temperature of a blackbody sphere of radius R that gives off L. The sphere has area $4\pi R^2$, and each square metre gives off $\sigma_{\text{SB}} T_{\text{eff}}^4$ by Eq. (4.24), so

Result 4.5

$$L = 4\pi R^2\,\sigma_{\text{SB}} T_{\text{eff}}^4. \tag{4.26}$$

For the Sun, with $L_\odot = 3.828 \times 10^{26}\,\text{W}$ and $R_\odot = 6.957 \times 10^{8}\,\text{m}$,

$$T_{\text{eff},\odot} = \left(\frac{L_\odot}{4\pi R_\odot^2\,\sigma_{\text{SB}}}\right)^{1/4} = 5772\,\text{K},$$

the value used throughout.

4.4 Colours and the Dimming by Dust

A magnitude through one filter measures one flux, and the Planck function says how the flux of a hot surface is shared among the colours.

A temperature from Wien's law needs the whole spectrum, which takes a long time to measure for a faint star. Also, the light of distant stars passes through dust, which dims it. A dimmed star looks further away than it is, and the distance modulus gives the wrong distance.

The setting Sun shows both problems at once. Its light crosses a long path of air near the horizon. The air removes blue light from the beam more than red, so the Sun looks fainter and redder than at noon. Someone who did not know about the air might think the Sun had cooled. Comparing its brightness through a blue glass and a red glass shows the change, and knowing how the air treats each colour undoes it.

Astronomers measure each star through a blue filter B and a yellow-green filter V. The difference of the two magnitudes, the colour, measures the temperature without needing the distance, because distance dims both filters alike. Dust between the stars removes blue light faster than yellow, like the air at sunset, so a dimmed star also looks redder. A colour and a magnitude, together with the spectral type of the star, separate temperature, dust and distance, which one magnitude alone cannot do.

For blackbodies at 5000 K and 10 000 K, the values of $B - V$ differ by 0.71 magnitudes (Eq. (4.30) below). A dust cloud that dims a star by one magnitude in V makes it 0.32 magnitudes redder in $B - V$ (Eq. (4.42) below).

Picture the spectrum seen through two windows, one blue and one yellow. A hotter star tips the spectrum toward the blue window. Dust tips it back toward the yellow window and dims both.

A colour compares two fluxes. Whether dust scatters light strongly is decided by comparing the size of a grain with the wavelength. The optical depth, which measures the dimming, compares the length of the path with the mean distance between grains. All three are comparisons of two scales, P6 (Section 9.6). The same law of dimming gives the airmass correction for the Earth's air (Section 2.10), and it returns in the transfer of light through a star (Chapter 11). Here it gives the colour as a thermometer, the extinction from the optical depth, and the distance modulus corrected for dust.

4.4.1 Passbands and the Colour Index

Astronomical detectors measure flux through standard filters. The Johnson–Cousins set has passbands centred near U (365 nm), B (440 nm), V (550 nm), R (640 nm) and I (790 nm), shown as shaded strips in Figure 4.1.

A **colour index** is the difference of two magnitudes of the same star, such as $B - V$. Applying Eq. (4.7) in each band,

$$\begin{aligned} B - V &\overset{(1)}{=} -2.5 \log_{10} \frac{F_B}{F_{B,\mathrm{ref}}} + 2.5 \log_{10} \frac{F_V}{F_{V,\mathrm{ref}}} \\ &\overset{(2)}{=} -2.5 \log_{10} \frac{F_B}{F_V} + 2.5 \log_{10} \frac{F_{B,\mathrm{ref}}}{F_{V,\mathrm{ref}}}. \end{aligned} \tag{4.27}$$

(1) Eq. (4.7) for the B and V fluxes, each with its own zero point.
(2) Collecting the logarithms.

The last term is a constant, fixed so that Vega and other stars of type A0 V have $B - V \approx 0$. The distance and the radius of the star cancel from F_B/F_V, because by Eq. (4.3) each flux is its surface value times the same factor $(R/d)^2$.

Colour as a thermometer. For a blackbody, F_B/F_V depends on the temperature alone. We treat each filter as a single wavelength, λ_B and λ_V. The flux at the Earth per unit wavelength is then πB_λ (Eq. (4.20) at one wavelength) times $(R/d)^2$.

$$\begin{aligned}\frac{F_B}{F_V} &\overset{(1)}{=} \frac{\pi B_\lambda(\lambda_B, T)\,(R/d)^2}{\pi B_\lambda(\lambda_V, T)\,(R/d)^2} \\ &\overset{(2)}{=} \left(\frac{\lambda_V}{\lambda_B}\right)^5 \frac{e^{hc/\lambda_V k_B T} - 1}{e^{hc/\lambda_B k_B T} - 1} \\ &\overset{(3)}{\approx} \left(\frac{\lambda_V}{\lambda_B}\right)^5 \exp\left[-\frac{hc}{k_B T}\left(\frac{1}{\lambda_B} - \frac{1}{\lambda_V}\right)\right].\end{aligned} \tag{4.28}$$

(1) Eq. (4.3) with the surface flux πB_λ at each wavelength.
(2) Eq. (4.15). The factors π, $2hc^2$ and $(R/d)^2$ cancel.
(3) Drop the -1 in each bracket, since $e^x \gg 1$. At 440 nm and 10^4 K, $x = 3.3$, and the -1 changes the ratio by a few per cent. For cooler stars it matters even less.

Putting this into Eq. (4.27),

$$\begin{aligned}B - V &\overset{(1)}{=} \frac{2.5}{\ln 10}\,\frac{hc}{k_B}\left(\frac{1}{\lambda_B} - \frac{1}{\lambda_V}\right)\frac{1}{T} - 12.5\log_{10}\frac{\lambda_V}{\lambda_B} + 2.5\log_{10}\frac{F_{B,\mathrm{ref}}}{F_{V,\mathrm{ref}}} \\ &\overset{(2)}{=} \frac{7100\,\mathrm{K}}{T} + \text{constant}.\end{aligned} \tag{4.29}$$

(1) $\log_{10} e^{-y} = -y/\ln 10$ and $\log_{10} a^5 = 5\log_{10} a$.
(2) $hc/k_B = 1.4388 \times 10^{-2}$ m K, $\lambda_B = 440$ nm and $\lambda_V = 550$ nm. The constant collects the terms without T.

$$\boxed{B - V \approx \frac{7100\,\mathrm{K}}{T} + \text{constant}.} \tag{4.30}$$

Hotter stars have smaller $B - V$, and the hottest have negative values, bluer than Vega. Cool stars have large positive values. The colour depends on the temperature, not on the distance or the size of the star. Real filters are broad and real spectra have lines, so in practice $B - V$ is turned into T_{eff} by a relation measured on stars of known temperature. Eq. (4.30) explains its shape.

4.4.2 Extinction from Optical Depth

Interstellar dust grains are mostly smaller than a micrometre. Each grain removes light from a beam, either by absorbing it or by scattering it out of the beam. The two together are called **extinction**. The area of beam that one grain removes is its **extinction cross-section** $\sigma_{\mathrm{ext}}(\lambda)$.

The dimming law. Follow a beam through dust that has n_d grains per unit volume, with s the distance along the beam. In a thin slab of thickness ds, one square metre of beam meets $n_d\,ds$ grains, which together block the fraction $n_d \sigma_{\mathrm{ext}}\,ds$ of it. The grains only remove light from the beam and add none (P1 for the energy in the beam), so

$$dF_\lambda = -n_d\,\sigma_{\mathrm{ext}}\,F_\lambda\,ds. \tag{4.31}$$

Let $F_{\lambda,0}$ be the flux that would arrive with no dust. Dividing by F_λ and integrating along the whole path through the dust,

$$\begin{aligned}\ln\frac{F_\lambda}{F_{\lambda,0}} &\overset{(1)}{=} -\int n_d\,\sigma_{\mathrm{ext}}\,ds \\ &\overset{(2)}{=} -\sigma_{\mathrm{ext}}(\lambda)\,N_d.\end{aligned} \tag{4.32}$$

(1) $\int dF_\lambda/F_\lambda = \ln F_\lambda$, from $F_{\lambda,0}$ where the beam enters the dust to F_λ where it leaves.
(2) All grains are taken to be alike, so σ_{ext} is the same along the path. The **column density** $N_d = \int n_d\, ds$ is the number of grains in a tube of cross-section $1\,\text{m}^2$ along the line of sight.

The **optical depth** is

$$\tau_\lambda = \int n_d\, \sigma_{\text{ext}}\, ds = N_d\, \sigma_{\text{ext}}(\lambda), \tag{4.33}$$

and the flux falls exponentially with it,

$$\boxed{F_\lambda = F_{\lambda,0}\, e^{-\tau_\lambda}.} \tag{4.34}$$

The optical depth counts the grain cross-sections that lie along the line of sight. The mean distance a photon travels between grains is $1/(n_d\sigma_{\text{ext}})$, so τ_λ is the length of the path divided by this distance, a comparison of two lengths (P6). Light passes almost freely when $\tau_\lambda \ll 1$ and hardly at all when $\tau_\lambda \gg 1$.

Extinction in magnitudes. The **extinction** A_λ is the number of magnitudes by which dust dims a star, $A_\lambda = m_\lambda - m_{\lambda,0}$, where $m_{\lambda,0}$ is the magnitude the star would have with no dust.

$$\begin{aligned} A_\lambda &\overset{(1)}{=} -2.5 \log_{10} \frac{F_\lambda}{F_{\lambda,0}} \\ &\overset{(2)}{=} -2.5 \log_{10} e^{-\tau_\lambda} \\ &\overset{(3)}{=} \frac{2.5}{\ln 10}\, \tau_\lambda. \end{aligned} \tag{4.35}$$

(1) Eq. (4.5) between the dimmed flux and the flux with no dust.
(2) Eq. (4.34).
(3) $\log_{10} e^{-y} = -y/\ln 10$.

Result 4.6

$$A_\lambda = 1.086\, \tau_\lambda = 1.086\, N_d\, \sigma_{\text{ext}}(\lambda). \tag{4.36}$$

One magnitude of extinction is close to one optical depth. The extinction is proportional to the column of dust, so it adds up along the line of sight. The magnitude without dust is $m_\lambda - A_\lambda$, and Eq. (4.11) for it becomes

$$m_\lambda = M_\lambda + 5 \log_{10} d\,[\text{pc}] - 5 + A_\lambda. \tag{4.37}$$

4.4.3 Reddening and the Colour Excess

Why dust reddens. The grains have radii a near $0.1\,\mu$m. How they treat light is decided by the ratio of the grain's circumference to the wavelength, $2\pi a/\lambda$ (P6). At 550 nm this ratio is 1.1, so visible light is in the Mie regime, between the λ^{-4} law of particles much smaller than the wavelength and the flat law of particles much larger (Section 8.2.1). Measured across the visible, the extinction by real dust, with its mix of grain sizes, falls roughly as $1/\lambda$ (Figure 4.1). At $2.2\,\mu$m the ratio is 0.29, and the extinction is only about a tenth of that in V. Dust clouds that are dark in visible light can therefore be seen through in the infrared.

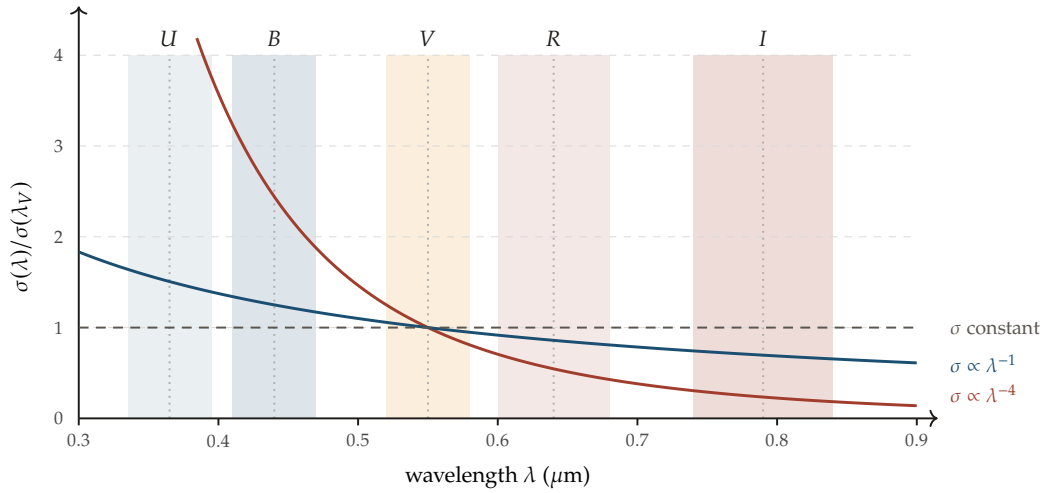


Figure 4.1: Cross-sections that vary as λ^{-4}, as λ^{-1} and not at all, each divided by its value at $\lambda_V = 0.55\,\mu$m, against wavelength from 0.3 to 0.9 μm. The shaded strips mark the U, B, V, R and I passbands, with dotted lines at their central wavelengths.

The colour excess. The measured colour of a star behind dust is

$$\begin{aligned}(B-V)_{\text{obs}} &\overset{(1)}{=} m_B - m_V \\ &\overset{(2)}{=} (m_{B,0} + A_B) - (m_{V,0} + A_V) \\ &\overset{(3)}{=} (B-V)_0 + (A_B - A_V).\end{aligned} \tag{4.38}$$

(1) The definition of the colour index.
(2) $A_\lambda = m_\lambda - m_{\lambda,0}$ in each band.
(3) $(B-V)_0 = m_{B,0} - m_{V,0}$ is the **intrinsic colour**, the colour with no dust.

The **colour excess** is $E(B-V) \equiv A_B - A_V$. By Eq. (4.36),

$$E(B-V) = 1.086\, N_d\, [\sigma_{\text{ext}}(\lambda_B) - \sigma_{\text{ext}}(\lambda_V)] > 0,$$

since σ_{ext} is larger at the shorter wavelength. So

$$(B-V)_{\text{obs}} = (B-V)_0 + E(B-V), \tag{4.39}$$

and the intrinsic colour is recovered as

$$\boxed{(B-V)_0 = (B-V)_{\text{obs}} - E(B-V).} \tag{4.40}$$

In practice it is used the other way round. The intrinsic colour of a star is known from its spectral type, so its measured colour gives $E(B-V)$.

From reddening to dimming. The distance needs the dimming A_V, but what we measure is the reddening $E(B-V)$. Their ratio is

$$\begin{aligned}\frac{A_V}{E(B-V)} &\overset{(1)}{=} \frac{1.086\, N_d\, \sigma_{\text{ext}}(\lambda_V)}{1.086\, N_d\, [\sigma_{\text{ext}}(\lambda_B) - \sigma_{\text{ext}}(\lambda_V)]} \\ &= \frac{\sigma_{\text{ext}}(\lambda_V)}{\sigma_{\text{ext}}(\lambda_B) - \sigma_{\text{ext}}(\lambda_V)} \\ &\overset{(2)}{=} \frac{1}{\lambda_V/\lambda_B - 1} = \frac{1}{550/440 - 1} = 4.0.\end{aligned} \tag{4.41}$$

(1) Eq. (4.36) in the V and B bands.
(2) An extinction cross-section proportional to $1/\lambda$.

The column of dust cancels, so the ratio depends on the kind of grains and not on how many there are. Measured along many lines of sight in the diffuse Milky Way, the ratio is

$$\boxed{A_V = 3.1\,E(B-V).} \tag{4.42}$$

The ratio 3.1 is often called R_V. It is smaller than the 4.0 of a pure $1/\lambda$ law, so between B and V the extinction falls a little faster than $1/\lambda$. A law $\lambda^{-\beta}$ with $(550/440)^{\beta} = 1 + 1/3.1$ fits it, which gives $\beta = 1.25$.

The distance corrected for dust. Eq. (4.37) in the V band gives the distance.

$$\begin{aligned} 5\log_{10} d\,[\mathrm{pc}] - 5 &\overset{(1)}{=} m_V - M_V - A_V \\ &\overset{(2)}{=} m_V - M_V - 3.1\,E(B-V). \end{aligned} \tag{4.43}$$

(1) Eq. (4.37) solved for the distance term.
(2) Eq. (4.42).

Result 4.7

$$(m_V - M_V)_0 \equiv m_V - M_V - 3.1\,E(B-V) = 5\log_{10} d\,[\mathrm{pc}] - 5. \tag{4.44}$$

The corrected distance modulus $(m_V - M_V)_0$ needs three measurements, m_V, $(B-V)_{\mathrm{obs}}$ and the spectral type, which gives both M_V and $(B-V)_0$. Leaving out the dust makes a star seem further away by the factor $10^{0.2A_V}$, which is 2.9 for $A_V = 2.3$ magnitudes (Problem 4.6.1).

4.5 The Radii of Stars

We now have the luminosity of a star, from its flux and its distance, and its effective temperature, from its colour or its spectrum.

Almost no star shows a disc in a telescope. Even the nearest stars look like points, so their sizes cannot be read off the sky with a ruler.

Two electric room heaters glow the same orange. Their surfaces are then at the same temperature, and each square centimetre gives off the same power. If one heater warms the room twice as much as the other, it must have twice the glowing area. Its size follows from its power and its colour, and nobody has to measure it.

A star is such a heater. Its temperature fixes the power per square metre, $\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4$, and its luminosity is the total. The area of the star is the luminosity divided by the power per unit area.

A star as hot as the Sun with 100 times its luminosity has 100 times its area, so 10 times its radius. Aldebaran gives off 439 times the Sun's power, but it is cooler, at 3900 K. Each square metre of it gives $(3900/5772)^4 = 0.208$ of what a square metre of the Sun gives. Its area is therefore $439/0.208 = 2106$ times the Sun's, and its radius is 46 times the Sun's.

Picture the surface of the star tiled with square metres, each giving off $\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4$. Counting the tiles is the same as counting the luminosity.

This is energy conservation (P1) with the power per area that counting states gives (P3). The same relation draws the lines of equal radius on the Hertzsprung–Russell diagram (Section 12.6). Here it gives the radius from L and T_{eff}, the angular size of the star on the sky, and a second route to that angle from the flux alone.

4.5.1 Radius from Luminosity and Temperature

Dividing Eq. (4.26) for a star by the same equation for the Sun,

$$\frac{L}{L_\odot} \overset{(1)}{=} \frac{4\pi R^2 \sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4}{4\pi R_\odot^2 \sigma_{\mathrm{SB}} T_{\mathrm{eff},\odot}^4} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T_{\mathrm{eff}}}{T_{\mathrm{eff},\odot}}\right)^4 . \tag{4.45}$$

(1) Eq. (4.26) for the star and for the Sun.

Result 4.8

$$\frac{R}{R_\odot} = \sqrt{\frac{L}{L_\odot}} \left(\frac{T_{\mathrm{eff},\odot}}{T_{\mathrm{eff}}}\right)^2 . \tag{4.46}$$

The factors 4π and σ_{SB} cancel, so only ratios to the Sun are needed, with $R_\odot = 6.957 \times 10^8$ m and $T_{\mathrm{eff},\odot} = 5772$ K.

With bolometric magnitudes. Luminosities are often given as absolute bolometric magnitudes. Taking the logarithm,

$$\log_{10} \frac{R}{R_\odot} \overset{(1)}{=} \frac{1}{2} \log_{10} \frac{L}{L_\odot} - 2 \log_{10} \frac{T_{\mathrm{eff}}}{T_{\mathrm{eff},\odot}} \overset{(2)}{=} -0.2\,(M_{\mathrm{bol}} - M_{\mathrm{bol},\odot}) - 2 \log_{10} \frac{T_{\mathrm{eff}}}{T_{\mathrm{eff},\odot}}, \tag{4.47}$$

(1) The logarithm of Eq. (4.46).
(2) Eq. (4.9) gives $\log_{10}(L/L_\odot) = -0.4(M_{\mathrm{bol}} - M_{\mathrm{bol},\odot})$.

with $M_{\mathrm{bol},\odot} = +4.74$ from Section 4.2.

4.5.2 Angular Diameter

A sphere of radius R at distance d is seen as a disc of **angular diameter** θ. The lines of sight that just graze its edge touch the sphere, so each meets the radius drawn to the point of contact at a right angle. In the right triangle formed by the observer, the centre of the star and that point, the hypotenuse is d and the side opposite the half-angle $\theta/2$ is R. So

$$\sin \frac{\theta}{2} = \frac{R}{d} \quad \Longrightarrow \quad \theta \approx \frac{2R}{d} \quad \text{(radians)}, \tag{4.48}$$

where the small-angle form holds because $R \ll d$ for every star. In the units used for stars,

$$\theta\,[\mathrm{mas}] \overset{(1)}{=} 2.0626 \times 10^8 \times \frac{2R}{d} \overset{(2)}{=} 2.0626 \times 10^8 \times \frac{2 \times 6.957 \times 10^8\ \mathrm{m}}{3.0857 \times 10^{16}\ \mathrm{m}} \frac{R}{R_\odot} \frac{1\,\mathrm{pc}}{d}. \tag{4.49}$$

(1) One radian is $(180/\pi) \times 3600 \times 1000$ mas $= 2.0626 \times 10^8$ mas.
(2) R in units of $R_\odot$ and d in parsecs.

Result 4.9

$$\theta = 9.30\,\text{mas}\left(\frac{R}{R_\odot}\right)\left(\frac{1\,\text{pc}}{d}\right). \tag{4.50}$$

The Sun seen from 1 pc would be 9.3 mas across.

Worked example 4.2: The angular diameter of Betelgeuse

Betelgeuse (α Orionis) is a red supergiant with $R \approx 764\,R_\odot$ at a distance $d \approx 168$ pc ($p \approx 5.95$ mas). Eq. (4.50) gives

$$\theta = 9.30\,\text{mas} \times \frac{764}{168} = 42.3\,\text{mas}.$$

Infrared interferometers measure 42 to 44 mas. Two telescopes separate angles down to about the wavelength divided by their separation (Eq. (5.15)). At 2.2 μm, 42.3 mas = 2.05×10^{-7} rad needs a separation of $2.2 \times 10^{-6}/2.05 \times 10^{-7} = 11$ m. Most stars are far smaller on the sky than Betelgeuse, and are measured with telescopes hundreds of metres apart (Chapter 5).

4.5.3 Angular Diameter from Flux and Temperature

A second route to the same angle needs no distance at all. The bolometric flux at the Earth is

$$\begin{aligned} F &\overset{(1)}{=} F_{\text{surf}}\left(\frac{R}{d}\right)^2 \\ &\overset{(2)}{=} \sigma_{\text{SB}} T_{\text{eff}}^4 \left(\frac{R}{d}\right)^2 \\ &\overset{(3)}{=} \sigma_{\text{SB}} T_{\text{eff}}^4 \left(\frac{\theta}{2}\right)^2. \end{aligned} \tag{4.51}$$

(1) Eq. (4.3).
(2) $F_{\text{surf}} = L/(4\pi R^2) = \sigma_{\text{SB}} T_{\text{eff}}^4$ by Eq. (4.26).
(3) Eq. (4.48).

$$\boxed{\theta = 2\sqrt{\frac{F}{\sigma_{\text{SB}} T_{\text{eff}}^4}}.} \tag{4.52}$$

The angle depends only on the flux and the temperature, and R and d never enter separately. A star of known flux and colour therefore has a known angular size. Turned round, a measured angular size and a measured flux give T_{eff} without any model of the star. For the Sun, $F = 1361\,\text{W m}^{-2}$ and $\theta = 31.97'$ give $T_{\text{eff}} = 5772$ K (Problem 4.6.3). For Aldebaran the two routes give the same 20.1 mas (Problem 4.6.2).

4.6 Problem Set: Photometry, Colours and Extinction

4.6.1 Problem 1: Distance and Extinction from a Colour

Problem Statement: An A0 V star has intrinsic colour $(B-V)_0 = 0.00$ and absolute magnitude $M_V = +0.60$. It is seen behind an interstellar cloud with apparent magnitudes $m_B = 14.55$ and $m_V = 13.80$. The dust follows $A_V = 3.1\,E(B-V)$.

(a) What is the colour excess $E(B-V)$?

(b) What is the visual extinction A_V?

(c) What is the true distance d?

(d) What distance would we find if we ignored the dust?

Essential Ideas:

- Dust adds the same colour excess to every star behind it, $(B-V)_{\rm obs} = (B-V)_0 + E(B-V)$, Eq. (4.39). The spectral type gives $(B-V)_0$.
- The dimming in V is $A_V = 3.1\,E(B-V)$, Eq. (4.42).
- The corrected distance modulus is $m_V - M_V - A_V = 5\log_{10} d\,[\text{pc}] - 5$, Eq. (4.44).

Detailed Solutions:

(a) **Colour excess**. The measured colour is $(B-V)_{\rm obs} = m_B - m_V = 14.55 - 13.80 = 0.75$. So

$$\boxed{E(B-V) = (B-V)_{\rm obs} - (B-V)_0 = 0.75 - 0.00 = 0.75.}$$

(b) **Visual extinction**. By Eq. (4.42),

$$\boxed{A_V = 3.1 \times 0.75 = 2.325 \text{ magnitudes.}}$$

The cloud lets through $10^{-0.4\times 2.325} = 12\%$ of the V light.

(c) **True distance**. The corrected distance modulus is

$$m_V - M_V - A_V = 13.80 - 0.60 - 2.325 = 10.875,$$

so

$$\boxed{d = 10^{(10.875+5)/5}\,\text{pc} = 10^{3.175}\,\text{pc} = 1496\,\text{pc} \approx 1.50\,\text{kpc}.}$$

(d) **Distance without the dust correction**. With A_V left out,

$$\boxed{d = 10^{(13.80-0.60+5)/5}\,\text{pc} = 10^{3.64}\,\text{pc} = 4365\,\text{pc}.}$$

This is too large by $10^{0.2A_V} = 10^{0.465} = 2.92$. Dust in the plane of the Milky Way makes uncorrected distances too large in this way, which is why every distance from a magnitude needs a colour as well.

4.6.2 Problem 2: Distance, Magnitudes and Radius of Aldebaran

Problem Statement: Aldebaran (α Tauri) has luminosity $L = 439\,L_\odot$, effective temperature $T_{\rm eff} = 3900\,\text{K}$ and parallax $p = 47\,\text{mas}$.

(a) What is the distance to Aldebaran in parsecs and in metres?

(b) What bolometric flux F arrives at the Earth?

(c) With $M_{\text{bol},\odot} = +4.74$ and $m_{\text{bol},\odot} = -26.83$, find the absolute and apparent bolometric magnitudes of Aldebaran.

(d) What is its radius in solar radii?

(e) What is its angular diameter in milliarcseconds?

Essential Ideas:

- Parallax gives the distance, $d\,[\mathrm{pc}] = 1/p\,['']$, Eq. (3.5), and the flux follows from $F = L/(4\pi d^2)$, Eq. (4.2).
- Absolute magnitudes compare luminosities, $M_{\mathrm{bol}} - M_{\mathrm{bol},\odot} = -2.5\log_{10}(L/L_\odot)$, Eq. (4.9). Apparent magnitudes compare fluxes, $m_{\mathrm{bol}} - m_{\mathrm{bol},\odot} = -2.5\log_{10}(F/F_\odot)$, Eq. (4.5). The two are linked by $m - M = 5\log_{10} d\,[\mathrm{pc}] - 5$, Eq. (4.11).
- The radius follows from L and T_{eff}, $R/R_\odot = \sqrt{L/L_\odot}\,(T_{\mathrm{eff},\odot}/T_{\mathrm{eff}})^2$, Eq. (4.46), and the angle from $\theta = 2R/d$, Eq. (4.48), or from $\theta = 2\sqrt{F/\sigma_{\mathrm{SB}}T_{\mathrm{eff}}^4}$, Eq. (4.52).

Detailed Solutions:

(a) **Distance**. With $p = 47\,\mathrm{mas} = 0.047''$,

$$\boxed{d = \frac{1}{0.047}\,\mathrm{pc} = 21.28\,\mathrm{pc} = 6.565\times10^{17}\,\mathrm{m}.}$$

(b) **Bolometric flux at the Earth**. With $L = 439\times3.828\times10^{26}\,\mathrm{W} = 1.680\times10^{29}\,\mathrm{W}$,

$$\boxed{F = \frac{1.680\times10^{29}\,\mathrm{W}}{4\pi\left(6.565\times10^{17}\,\mathrm{m}\right)^2} = 3.10\times10^{-8}\,\mathrm{W\,m^{-2}}.}$$

(c) **Absolute and apparent bolometric magnitudes**. By Eq. (4.9),

$$\boxed{M_{\mathrm{bol}} = 4.74 - 2.5\log_{10}439 = 4.74 - 2.5\times2.6425 = -1.87.}$$

The distance modulus is $5\log_{10}21.28 - 5 = 1.64$, so

$$\boxed{m_{\mathrm{bol}} = M_{\mathrm{bol}} + 1.64 = -1.87 + 1.64 = -0.23.}$$

The flux gives the same answer by a second route,

$$m_{\mathrm{bol}} = -26.83 - 2.5\log_{10}\frac{3.10\times10^{-8}}{1361} = -26.83 + 26.61 = -0.23.$$

The Sun's magnitude in this check must be the bolometric one, -26.83. Its V magnitude, -26.74, would give -0.13.

(d) **Radius**. By Eq. (4.46),

$$\frac{R}{R_\odot} = \sqrt{439}\left(\frac{5772\,\mathrm{K}}{3900\,\mathrm{K}}\right)^2 = 20.95\times2.190, \qquad \boxed{R = 45.9\,R_\odot.}$$

In metres, $R = 45.9\times6.957\times10^8\,\mathrm{m} = 3.19\times10^{10}\,\mathrm{m}$, or $0.213\,\mathrm{AU}$. Placed at the Sun, Aldebaran would reach more than half way to the orbit of Mercury.

(e) **Angular diameter**. By Eq. (4.48),

$$\theta = \frac{2\times3.193\times10^{10}\,\mathrm{m}}{6.565\times10^{17}\,\mathrm{m}} = 9.726\times10^{-8}\,\mathrm{rad},$$

and in milliarcseconds

$$\boxed{\theta = 9.726\times10^{-8}\times2.0626\times10^{8}\,\mathrm{mas} = 20.1\,\mathrm{mas}.}$$

The flux route of Eq. (4.52) needs neither R nor d,

$$\theta = 2\sqrt{\frac{3.10\times10^{-8}}{5.670\times10^{-8}\times3900^4}}\,\mathrm{rad} = 9.72\times10^{-8}\,\mathrm{rad} = 20.1\,\mathrm{mas},$$

the same angle. An angle of 20 mas is easily measured with optical interferometers such as CHARA and the VLTI, so for Aldebaran the radius can be checked directly.

4.6.3 Problem 3: The Sun as a Blackbody

Problem Statement: Above the Earth's air the Sun gives the flux $F_\odot = 1361\,\mathrm{W\,m^{-2}}$, and its disc is 31.97′ across. The Earth is 1 AU $= 1.496 \times 10^{11}$ m from the Sun.

(a) Find the effective temperature of the Sun from these two measurements alone.

(b) At what wavelength does B_λ peak, and at what frequency does B_ν peak? What wavelength belongs to that frequency?

(c) Find the radius and the luminosity of the Sun. Check that they satisfy $L = 4\pi R^2 \sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4$.

Essential Ideas:

- The flux at the Earth is the surface flux diluted by $(\theta/2)^2$, so $\theta = 2\sqrt{F/\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4}$, Eq. (4.52).
- B_λ peaks at $x = hc/(\lambda k_B T) = 4.965$ (Eq. (4.19)), and B_ν at $x = h\nu/k_B T = 2.821$.
- $R = d\,\theta/2$ (Eq. (4.48)) and $L = 4\pi d^2 F$ (Eq. (4.2)).

Detailed Solutions:

(a) **Effective temperature**. In radians, $\theta = 31.97' = 9.301 \times 10^{-3}$. Solving Eq. (4.52) for the temperature,

$$\begin{aligned} T_{\mathrm{eff}} &\overset{(1)}{=} \left(\frac{F_\odot}{\sigma_{\mathrm{SB}}}\right)^{1/4} \left(\frac{2}{\theta}\right)^{1/2} \\ &= \left(\frac{1361}{5.670 \times 10^{-8}}\right)^{1/4} \left(\frac{2}{9.301 \times 10^{-3}}\right)^{1/2} \mathrm{K} \\ &= 393.6 \times 14.66\,\mathrm{K}. \end{aligned}$$

(1) Square Eq. (4.52), $\theta^2/4 = F_\odot/(\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4)$, and take the fourth root.

$$\boxed{T_{\mathrm{eff},\odot} = 5772\,\mathrm{K}.}$$

The distance to the Sun is not needed, only the flux and the angle.

(b) **The two peaks**. By Eq. (4.19),

$$\boxed{\lambda_{\max} = \frac{2.898 \times 10^{-3}\,\mathrm{m\,K}}{5772\,\mathrm{K}} = 502\,\mathrm{nm}.}$$

Per unit frequency,

$$\boxed{\nu_{\max} = \frac{2.821\,k_B T}{h} = \frac{2.821 \times 1.381 \times 10^{-23} \times 5772}{6.626 \times 10^{-34}}\,\mathrm{Hz} = 3.39 \times 10^{14}\,\mathrm{Hz},}$$

whose wavelength is $c/\nu_{\max} = 883$ nm, in the near infrared. The Sun is "brightest in the green" only when its spectrum is counted per unit wavelength.

(c) **Radius and luminosity**. From the angle and the distance,

$$R_\odot = d\,\frac{\theta}{2} = 1.496 \times 10^{11}\,\mathrm{m} \times 4.650 \times 10^{-3} = 6.957 \times 10^{8}\,\mathrm{m},$$

and from the flux and the distance, as in the solar constant example,

$$L_\odot = 4\pi d^2 F_\odot = 4\pi \left(1.496 \times 10^{11}\,\mathrm{m}\right)^2 \times 1361\,\mathrm{W\,m^{-2}} = 3.828 \times 10^{26}\,\mathrm{W}.$$

Then

$$\boxed{4\pi R_\odot^2\,\sigma_{\rm SB}T_{{\rm eff},\odot}^4 = 4\pi\left(6.957\times10^8\right)^2\times5.670\times10^{-8}\times5772^4\,{\rm W} = 3.828\times10^{26}\,{\rm W} = L_\odot.}$$

The check must work, since $T_{\rm eff}$ in part (a) was found from the same flux and angle. It shows that Eqs. (4.2), (4.26) and (4.52) are one statement seen three ways. Any two of the flux, the angle and the temperature give the third.

Chapter 5

Angular Resolution: Baselines, Diffraction, and Pixel Sampling

Chapter 4 measured how much light a star sends us, and Chapter 3 measured where the star is. Both took for granted that a telescope shows each star as a point. It does not. Every telescope spreads a star into a small patch, and two stars closer together than that patch merge into one.

The size of the patch is set by a few lengths compared with one another. The wavelength compared with the size of the mirror, or with the distance between two radio dishes, sets the finest angle the waves can carry. The focal length turns that angle into a distance on the detector, and the size of a pixel decides whether the detector keeps it. From the ground, the size of the patches of still air above the telescope can set a coarser limit than all of these. Each comparison is a ratio of two scales, principle P6 of Chapter 9. Before any of this, the light has to reach the telescope.

5.1 Windows on the Universe

Chapter 4 measures the flux of a star in a band of wavelengths, and takes for granted that the light arrives.

Most of it does not. Above the telescope there are about 10^4 kg of air over every square metre, and at most wavelengths that is enough to absorb every photon.

Ordinary window glass shows the trick. It lets visible light through, so the room is bright, but it stops most of the ultraviolet that causes sunburn. Glass is not simply clear or dark. Light of each colour either finds something in the glass that can take its energy, or passes straight through.

The air is a window of the same kind, with only a few panes open. The idea is that a photon gets through when its energy matches no transition of the molecules in the air, and a radio wave gets through when its frequency is above the plasma frequency of the ionosphere.

A photon of wavelength λ has the energy $hc/\lambda = 1239.84\,\text{eV nm}/\lambda$. Visible light, from 400 to 700 nm, carries 3.10 to 1.77 eV. Ozone absorbs below 300 nm, which is above 4.13 eV, by lifting an electron to a higher level. Water vapour absorbs in bands near 2.7 and 6.3 μm, at 0.46 and 0.20 eV, and carbon dioxide near 4.3 and 15 μm, at 0.29 and 0.083 eV, by making the molecules vibrate. Visible photons fall in the gap between the two kinds of transition.

The picture is a plot of the fraction of light that crosses the air against wavelength. It is zero almost everywhere, with one open stretch around the visible, a few narrow gaps in the infrared, and a wide opening in the radio.

Every material gets its colour in this way, from the photon energies its electrons and molecules can take up. The comparison is principle P6 (Section 9.6), a photon energy set against a

transition energy, or a wave frequency set against the plasma frequency. It gives the bands that reach the ground, and the plasma frequency, which also decides how X-rays can be focused (Section 5.3).

Band by band.

- **Gamma rays and X-rays** ionise the atoms of the air and are absorbed high up. This is what protects life on the ground from them. They can be observed only from orbit (Chandra, NuSTAR).
- **Ultraviolet light** below about 300 nm is absorbed by ozone (O_3) in the stratosphere, and below about 200 nm also by O_2. Both absorb by electronic transitions. So ultraviolet astronomy below the ozone cutoff is done from space.
- **Visible light** (350–800 nm) has too little energy for the electronic transitions of O_2 and O_3, and too much for the vibration bands of H_2O, CO_2 and CH_4. Only a few weak bands remain, such as the oxygen A band at 760 nm. Rayleigh scattering by the air molecules, whose cross-section goes as λ^{-4} (Figure 4.1), removes some of the blue and makes the day sky glow, but it does not close the window. This gap is the **optical window**, and our eyes are tuned to it.
- **Infrared light** passes only through narrow gaps (the J, H, K, L and M bands) between the strong bands of water vapour and carbon dioxide. So infrared observatories sit at high, dry sites (Mauna Kea, the Atacama), or go above most of the water vapour on aircraft (SOFIA, which flew until 2022), on balloons, or in space (JWST).
- **Radio waves** pass from about 10 MHz up to a few hundred GHz. This is the **radio window**. Below about 10 MHz the ionosphere reflects the waves back into space, Eq. (5.2). The same reflection lets short-wave radio signals travel round the globe. At the millimetre and submillimetre end, water vapour absorbs again. This is why ALMA is built at 5000 m in the Atacama desert, above most of the water.

The plasma frequency. The ionosphere is a thin gas of free electrons, n_e of them per cubic metre. We want the refractive index n it has for a wave of frequency ν, and so whether the wave can enter it. The electric field of the wave, $E\cos(2\pi\nu t)$, shakes each electron. Newton's law for the electron, of mass m_e and charge $-e$, is $m_e\ddot{x} = -eE\cos(2\pi\nu t)$. Its steady solution is $x = x_0\cos(2\pi\nu t)$, so $\ddot{x} = -(2\pi\nu)^2 x$, and

$$-m_e(2\pi\nu)^2 x = -eE\cos(2\pi\nu t) \quad\Longrightarrow\quad x \overset{(1)}{=} \frac{eE\cos(2\pi\nu t)}{m_e(2\pi\nu)^2},$$

$$n^2 \overset{(2)}{=} 1 + \frac{n_e(-e)x}{\varepsilon_0 E\cos(2\pi\nu t)} \overset{(3)}{=} 1 - \frac{n_e e^2}{\varepsilon_0 m_e(2\pi\nu)^2} \overset{(4)}{=} 1 - \frac{\nu_p^2}{\nu^2}. \tag{5.1}$$

(1) Divide by $-m_e(2\pi\nu)^2$. The electron moves in step with the field and opposite to the force on a positive charge.
(2) For a non-magnetic medium n^2 is the relative permittivity, $1 + P/(\varepsilon_0 E)$, where the polarisation P is the dipole moment per volume, n_e times $(-e)x$.
(3) The value of x from the first line.
(4) The definition of the **plasma frequency** ν_p below.

$$\boxed{\nu_p = \frac{1}{2\pi}\sqrt{\frac{n_e e^2}{\varepsilon_0 m_e}} = 8.98\,\text{Hz} \times \sqrt{n_e\,[\text{m}^{-3}]}\,.} \tag{5.2}$$

Above ν_p, n^2 is positive and the wave travels on. Below ν_p, n^2 is negative, n is imaginary, and the wave dies away within a short distance and is reflected. At its densest the ionosphere has

$n_e \approx 10^{12}\,\mathrm{m}^{-3}$, which gives $\nu_p \approx 9\,\mathrm{MHz}$. That is the lower edge of the radio window. The result uses only the electron density, so any plasma reflects waves below its own ν_p, whatever it is made of.

Two windows matter most for everyday astronomy, the optical and the radio. They favour two different ways of getting a sharp image. Radio astronomers can combine dishes that stand a continent apart, because the air barely disturbs radio wavefronts over such distances. Optical astronomers are mostly held to a single mirror, whole or made of segments. Its sharpness is set by diffraction, and from the ground it is spoiled further by turbulence in the air, even though the light itself is not absorbed.

5.2 Interferometric Baselines

The radio window lets waves from about 10 MHz to a few hundred GHz reach the ground, and Chapter 2 gives any place on a sphere a unit vector.

A single radio dish is a poor camera. At the 21 cm line of hydrogen a 25 m dish blurs every source over about half a degree, the size of the full Moon.

Close your eyes and listen to a friend calling from one side. You know the direction at once. The sound reaches the nearer ear first, by up to $0.2\,\mathrm{m}/(343\,\mathrm{m\,s^{-1}}) = 0.58\,\mathrm{ms}$ for ears 0.2 m apart, and the brain turns that delay into an angle. A wider head would hear a smaller change of angle as the same change of delay.

Two radio dishes far apart are two ears. A wave from a source off to one side reaches one dish first. The idea is that the finest angle two receivers can tell apart is the wavelength divided by their separation, and it does not depend on the size of each receiver.

Two dishes 1 km apart, observing at 21 cm, resolve $0.21/1000 = 2.1 \times 10^{-4}\,\mathrm{rad} = 43''$. One 25 m dish alone gives about $1.22 \times 0.21/25\,\mathrm{rad} = 35'$, fifty times coarser.

The picture is two dishes on the ground and a flat wavefront coming in at a slant. It touches one dish, and travels on a little further before it touches the other.

Our ears do it, and so does every array of antennas, from radar to the receivers of mobile phones. The delay is a path difference compared with a wavelength, P6 once more. It gives the length of the straight line between two places on the Earth, and the sharpness that line buys. In Very Long Baseline Interferometry (VLBI), and in connected arrays such as the GMRT or ALMA, this straight line between two antennas is the **baseline**.

5.2.1 The Baseline Between Two Stations

The station vectors. Let two stations sit at latitudes ϕ_1, ϕ_2 and longitudes Λ_1, Λ_2 on a sphere of radius $R_E = 6371.0\,\mathrm{km}$, the mean radius of the Earth. The longitude gets a capital letter because λ is the wavelength here. In the frame with its z axis through the North Pole and its x axis through longitude 0°, each station has the unit vector of Eq. (2.2), with latitude in place of declination and longitude in place of right ascension,

$$\hat{\mathbf{r}}_i = \begin{pmatrix} \cos\phi_i \cos\Lambda_i \\ \cos\phi_i \sin\Lambda_i \\ \sin\phi_i \end{pmatrix}, \qquad i \in \{1, 2\}. \tag{5.3}$$

The central angle. The angle γ between the two station vectors, at the centre of the Earth, is the angle between two unit vectors. Chapter 2 found it as their dot product, Eq. (2.34). With $\delta \to \phi$ and $\alpha \to \Lambda$,

$$\cos\gamma = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 = \sin\phi_1 \sin\phi_2 + \cos\phi_1 \cos\phi_2 \cos(\Lambda_2 - \Lambda_1). \tag{5.4}$$

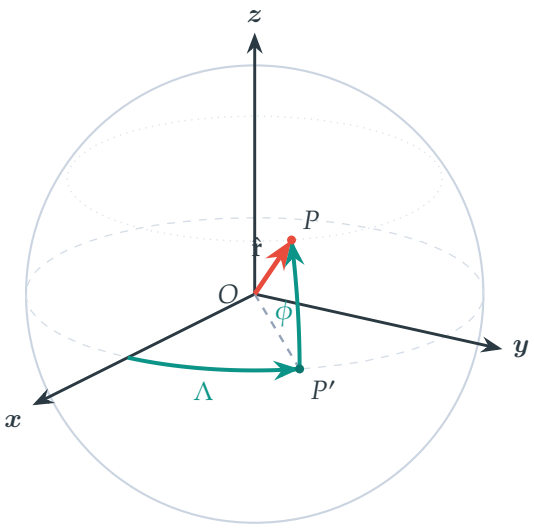


Figure 5.1: A point P on a sphere with centre O, its unit vector, its foot P' in the equatorial plane, its latitude ϕ, and its longitude (marked λ) measured from the x axis.

For stations a few degrees apart, $\cos\gamma$ is close to 1 and loses digits. The haversine form, Eq. (2.36), keeps them,

$$\sin^2\frac{\gamma}{2} = \sin^2\frac{\phi_2-\phi_1}{2} + \cos\phi_1\cos\phi_2\sin^2\frac{\Lambda_2-\Lambda_1}{2}. \tag{5.5}$$

The chord. The baseline B is the straight line through the Earth from one station to the other, the length of $R_E\hat{\mathbf{r}}_1 - R_E\hat{\mathbf{r}}_2$,

$$\begin{aligned} B^2 &= R_E^2\,(\hat{\mathbf{r}}_1-\hat{\mathbf{r}}_2)\cdot(\hat{\mathbf{r}}_1-\hat{\mathbf{r}}_2) \overset{(1)}{=} R_E^2\,(1+1-2\,\hat{\mathbf{r}}_1\cdot\hat{\mathbf{r}}_2) \\ &\overset{(2)}{=} 2R_E^2(1-\cos\gamma) \overset{(3)}{=} 4R_E^2\sin^2\frac{\gamma}{2}. \end{aligned} \tag{5.6}$$

(1) Multiply out. Each unit vector has $\hat{\mathbf{r}}_i\cdot\hat{\mathbf{r}}_i = 1$.
(2) Eq. (5.4).
(3) $1-\cos\gamma = 2\sin^2(\gamma/2)$.

$$\boxed{B = 2R_E\sin\left(\frac{\gamma}{2}\right) = R_E\sqrt{2(1-\cos\gamma)}.} \tag{5.7}$$

The same result follows from the isosceles triangle of Figure 5.2. The line from the centre to the midpoint M of the chord cuts it at a right angle, so each half is $R_E\sin(\gamma/2)$. Together with Eq. (5.5), $\sin(\gamma/2)$ comes straight from the coordinates, and no inverse cosine is needed.

5.2.2 Two Sources Seen by Two Antennas

We now want the finest angle a pair of antennas a distance B apart can tell apart, at wavelength λ. First we find what the pair measures for any sky, then put in two point sources.

One source. A source at angle θ from the normal to the baseline sends a flat wavefront. It reaches A_2 first. To reach A_1 it must travel further by the side of the right-angled triangle in Figure 5.3. For the two sources of the figure,

$$\Delta L_1 = B\sin\theta_1, \tag{5.8}$$
$$\Delta L_2 = B\sin\theta_2. \tag{5.9}$$

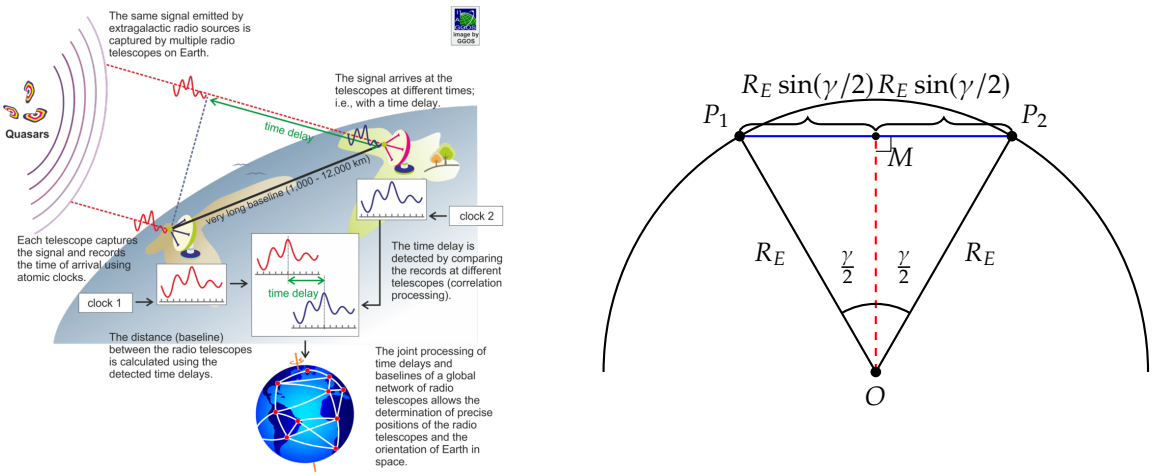


Figure 5.2: Left, the global VLBI network. Right, two stations P_1 and P_2 on a sphere of radius R_E and centre O, the central angle γ, and the chord between them, cut in half at M.

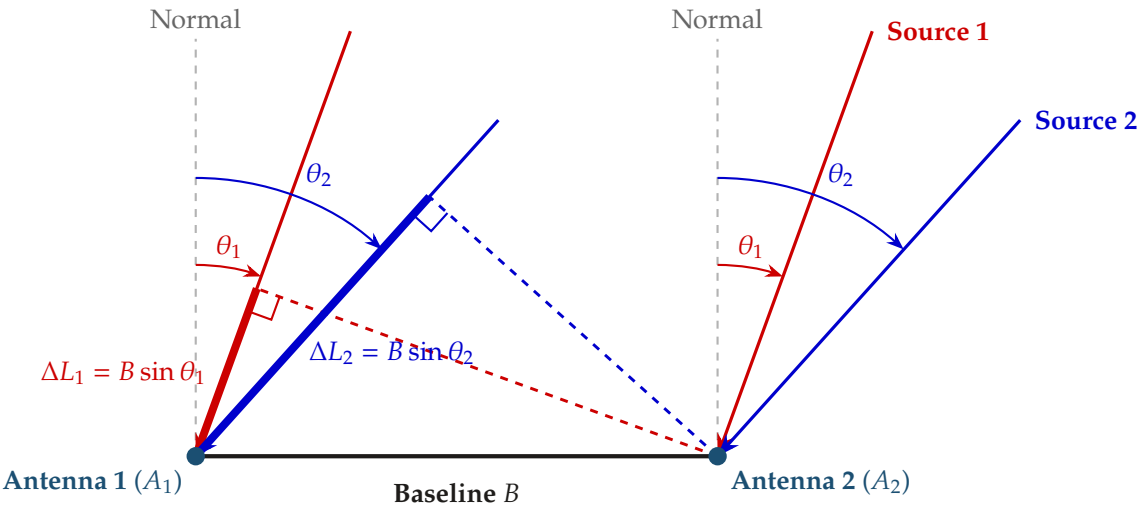


Figure 5.3: Two antennas A_1 and A_2 a baseline B apart, and two distant sources at angles θ_1 and θ_2 from the normal to the baseline. For each source the dashed line is the wavefront through A_2, and the thick segment is the extra path ΔL_1 or ΔL_2 to A_1.

Each antenna turns the wave into a voltage. For one source of frequency $\nu = c/\lambda$, take $v_2(t) = \cos(2\pi\nu t)$ at A_2. At A_1 the same wave arrives later by $\Delta L/c$, so $v_1(t) = \cos(2\pi\nu t - 2\pi\Delta L/\lambda)$. The **correlator** multiplies the two voltages and averages over many periods,

$$\langle v_1 v_2 \rangle \overset{(1)}{=} \frac{1}{2}\left\langle \cos\left(4\pi\nu t - \frac{2\pi\Delta L}{\lambda}\right)\right\rangle + \frac{1}{2}\cos\frac{2\pi\Delta L}{\lambda} \overset{(2)}{=} \frac{1}{2}\cos\frac{2\pi B\sin\theta}{\lambda}. \tag{5.10}$$

(1) The product rule $\cos x \cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$.
(2) The first term swings at twice the wave frequency and averages to zero. Then $\Delta L = B\sin\theta$.

As the Earth turns, θ changes, $B\sin\theta$ runs through many wavelengths, and the output swings between $+\frac{1}{2}$ and $-\frac{1}{2}$. These swings are the **fringes**.

Many sources. Different sources send waves with unrelated phases. In the product of the summed voltages, every cross term between two sources averages to zero, so the outputs of the sources simply add, each weighted by its power. For a sky of brightness $I_{\text{sky}}(\theta)$ along the

baseline, near the normal where $\sin\theta \approx \theta$,

$$\langle v_1 v_2 \rangle \propto \int I_{\text{sky}}(\theta) \cos\!\left(2\pi \frac{B}{\lambda}\,\theta\right) d\theta. \tag{5.11}$$

One pair of antennas measures one Fourier component of the sky, the one with B/λ cycles per radian. Real correlators also form the product with one voltage shifted by a quarter period, which gives the sine component at the same B/λ.

Two equal sources. Put two sources of equal power at θ_1 and θ_2. By Eq. (5.10) the output is

$$\cos\frac{2\pi\Delta L_1}{\lambda} + \cos\frac{2\pi\Delta L_2}{\lambda} \overset{(1)}{=} 2\cos\!\left(\frac{\pi(\Delta L_1 + \Delta L_2)}{\lambda}\right)\cos\!\left(\frac{\pi(\Delta L_2 - \Delta L_1)}{\lambda}\right). \tag{5.12}$$

(1) $\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{y-x}{2}$.

The first factor is the fringe of a single source at the mean position. The second factor changes slowly and sets the size of the fringe. It depends on the difference of the two path lags,

$$\Delta L_2 - \Delta L_1 = B\,(\sin\theta_2 - \sin\theta_1) \overset{(1)}{=} 2B\cos\frac{\theta_1+\theta_2}{2}\,\sin\frac{\theta_2-\theta_1}{2}. \tag{5.13}$$

(1) $\sin y - \sin x = 2\cos\frac{x+y}{2}\sin\frac{y-x}{2}$.

Two close sources have a small separation $\Delta\theta = \theta_2 - \theta_1$, so $\sin(\Delta\theta/2) \approx \Delta\theta/2$. Near the normal the cosine is close to 1 as well,

$$\Delta L_2 - \Delta L_1 \approx B\cos\!\left(\frac{\theta_1+\theta_2}{2}\right)\Delta\theta \approx B\,\Delta\theta. \tag{5.14}$$

Away from the normal the baseline counts only through $B\cos\theta$, its length as seen from the source.

When the pair is resolved. By Eqs. (5.12) and (5.14), the fringe of the pair has the size $|\cos(\pi B\Delta\theta/\lambda)|$ times that of a single source of the same total power. For $\Delta\theta \ll \lambda/B$ it is 1, and the pair cannot be told from a single source. At $B\Delta\theta = \lambda/2$ the two fringe patterns are half a cycle apart and cancel. So the pair shows itself once its extra path $B\Delta\theta$ is a fair part of a wavelength. The angle λ/B is the period of the fringe pattern on the sky, the finest ripple one baseline records. It is quoted as the resolution,

$$B\,\Delta\theta \approx \lambda. \tag{5.15}$$

At exactly $B\Delta\theta = \lambda$ the patterns are a whole cycle apart and agree again. A real array removes this ambiguity with many baselines of different lengths and directions, and the turning Earth changes each projected baseline through the night. With $1\text{ rad} = 648\,000''/\pi = 206\,265''$,

Result 5.1

$$\Delta\theta \approx \frac{\lambda}{B}\ \textit{rad} = 206\,265\,\frac{\lambda}{B}\ \textit{arcsec}. \tag{5.16}$$

Only the ratio of the wavelength to the baseline enters. The size of each dish does not. It sets how much power is collected, and the width λ/D_{dish} of the patch of sky each dish sees, inside which the fine detail λ/B is measured. For the GMRT and the Ooty Radio Telescope, $B = 901$ km and at 1 GHz the resolution is 68.6 mas (Problem 5.7.1).

5.3 Telescope Optics

A radio dish records its own wave, so dishes can stand far apart and be combined later (Section 5.2).

Visible light cannot be recorded wave by wave, and above any telescope the air keeps the wavefront flat only over tens of centimetres (Section 5.6). So an optical telescope collects light with one large mirror, whole or made of segments, and brings it to a point.

A magnifying glass held in sunshine makes a small bright image of the Sun on a sheet of paper. Move the glass away from the paper and the image gets sharp at one distance, the focal length. The image is a disc. Its size is the angle the Sun covers, times the focal length.

A telescope does the same for every star. The idea is that each direction on the sky goes to one point in the focal plane, at a distance from the axis equal to the focal length times the angle.

The Sun covers $0.533^\circ = 9.30 \times 10^{-3}$ rad. A lens of focal length 1 m makes an image 9.3 mm across.

The picture is two bundles of parallel rays from two stars, entering at a small angle to each other and meeting at two points in the focal plane.

The same mapping works in a camera lens and in the eye, whose retina is its focal plane. Only geometry enters, so no principle is used. It gives the reasons for mirrors, the ways a beam is led to a detector, how each band of the spectrum is focused, the size of the secondary mirror, and the angle on the sky covered by one pixel.

5.3.1 Reflectors versus Refractors

The main collecting element of a telescope is either a lens (a *refractor*) or a mirror (a *reflector*). Nearly all research telescopes are reflectors. Every reason comes from the light bouncing off the front surface of the mirror instead of passing through glass.

- The light never enters the mirror, so there is no chromatic aberration and almost no absorption. A lens sends different wavelengths to different foci, and thick glass absorbs the ultraviolet and parts of the infrared.
- Only one surface must be polished to the right shape, against two for a lens, and a mirror can be made much lighter than a lens of the same size.
- A mirror can be supported across its whole back. A lens can be held only at its edge, which sags under gravity for large sizes.
- The back has no optical job, so it can carry actuators that bend the mirror to cancel its sag. This is *active optics*, a slow correction of the whole shape, related to the fast adaptive optics of Section 5.6.

The price of a reflector is that the secondary mirror, or the detector itself, must hang *in front of* the primary, in the incoming beam, and block part of it (Section 5.3.3).

5.3.2 Focus Configurations: Prime, Newtonian, Cassegrain, and Coudé

One primary mirror can send its light to a detector in several standard ways. They differ in what stops the beam before the primary focus.

- **Prime focus.** The detector sits at the primary's own focus, with no second mirror. It is the simplest layout and loses no light to a second reflection, but the detector blocks the beam and is hard to reach on a large telescope.
- **Newtonian.** A flat mirror at 45°, just before the primary focus, turns the beam out through the side of the tube. It suits small telescopes.

- **Cassegrain.** A secondary mirror sends the converging beam back through a hole in the primary, so the instrument sits compactly behind the primary.
- **Coudé.** A chain of extra flat mirrors, often along the axes of the mount, carries the beam to a fixed focus in a separate, stable room. Some light is lost at each reflection, but very large and heavy instruments, notably high-resolution echelle spectrographs, can then stand still while the telescope moves.

Whatever the focus, the mount must turn the telescope to follow the daily motion of the sky. It tracks at the sidereal rate in hour angle, Eq. (2.10), either about one axis parallel to the Earth's (an equatorial mount) or about two axes under computer control (an altitude–azimuth mount).

5.3.3 Obstruction Effects and Observatory Domes

The secondary mirror, or a prime-focus detector, is held by a frame of thin vanes across the tube, the *spider*. The vanes are thin opaque strips inside the opening, so they change the aperture mask $t(\xi, \eta)$ of Eq. (5.29). Each strip adds its own diffraction pattern, a streak at right angles to the vane, on top of the star's Airy pattern (Section 5.4 works this out). These are the four- or six-pointed *diffraction spikes* on bright stars in reflector images. The straight edges of a segmented mirror, like the hexagons of JWST, add more spikes in the same way.

Telescopes stand inside a turning *dome*. It keeps off wind, rain and changes of temperature, and it blocks stray light from the site. Moving air of different temperatures inside an open tube would spoil the image in the same way as the seeing of Section 5.6.

5.3.4 Telescope Technologies Across the Electromagnetic Spectrum

The atmosphere decides from where each band can be observed. How each band is focused into an image is a separate question, because ordinary mirrors stop working as the photon energy rises.

Ultraviolet and infrared. Ultraviolet and infrared telescopes use the same geometry as optical reflectors. Only the reflecting coating changes, chosen to reflect well at the wavelength wanted. They must still go to space (AstroSat-UVIT in the ultraviolet, JWST in the infrared) or to high, dry sites, because the air is the limit, not the optics.

X-rays and grazing incidence. At X-ray energies a mirror at near-normal incidence reflects almost nothing. The photon frequency is far above the plasma frequency of the electrons in the mirror metal, so by Eq. (5.1) the metal has a refractive index slightly below one,

$$n = \sqrt{1 - \frac{\nu_p^2}{\nu^2}} \approx 1 - \frac{\nu_p^2}{2\nu^2}, \tag{5.17}$$

where every electron is treated as free, which holds roughly when the photon energy is well above the binding energies of most of the electrons. The X-rays travel faster in the metal than in vacuum. Light going from glass into air is totally reflected beyond a critical angle, and X-rays going from vacuum into the metal are totally reflected in the same way, provided they skim the surface.

Let the X-ray meet the mirror at the grazing angle α, measured from the surface, so its angle from the normal is $90^\circ - \alpha$. Snell's law gives the angle i from the normal inside the metal,

$$1 \cdot \sin(90^\circ - \alpha) = n \sin i \quad \Longrightarrow \quad \sin i \overset{(1)}{=} \frac{\cos \alpha}{n}. \tag{5.18}$$

(1) $\sin(90^\circ - \alpha) = \cos\alpha$.

No ray enters the metal when this would need $\sin i > 1$, that is when $\cos\alpha > n$. The critical grazing angle α_c has $\cos\alpha_c = n$. For small angles,

$$1 - \frac{\alpha_c^2}{2} \overset{(1)}{=} 1 - \frac{\nu_p^2}{2\nu^2} \quad \Longrightarrow \quad \boxed{\alpha_c = \frac{\nu_p}{\nu}.} \tag{5.19}$$

(1) $\cos\alpha \approx 1 - \alpha^2/2$ on the left, and n from above on the right.

The critical angle is the ratio of the two frequencies, P6 again. The ionosphere reflects radio waves below its ν_p at any angle. A metal reflects X-rays far above its ν_p only at grazing angles below ν_p/ν.

Worked example 5.1: Grazing angle for iridium

The Chandra mirrors are coated with iridium, which has a density of $22.56\,\mathrm{g\,cm^{-3}}$, an atomic mass of 192.22 and 77 electrons per atom. So

$$n_e = \frac{22\,560\,\mathrm{kg\,m^{-3}}}{192.22 \times 1.661 \times 10^{-27}\,\mathrm{kg}} \times 77 = 5.44 \times 10^{30}\,\mathrm{m^{-3}}.$$

By Eq. (5.2), $\nu_p = 2.09 \times 10^{16}\,\mathrm{Hz}$, a photon energy $h\nu_p = 86.6\,\mathrm{eV}$. A 10 keV X-ray has $\nu = 2.42 \times 10^{18}\,\mathrm{Hz}$, so

$$\alpha_c = \frac{86.6\,\mathrm{eV}}{10\,000\,\mathrm{eV}} = 8.7 \times 10^{-3}\,\mathrm{rad} = 0.50^\circ.$$

Harder X-rays need even shallower angles, since $\alpha_c \propto 1/\nu$.

X-ray telescopes are therefore built of long, nearly cylindrical shells that the X-rays skim. The *Wolter type I* design reflects each X-ray twice, off a paraboloid and then a hyperboloid, which keeps images sharp away from the axis. Many shells are nested inside one another to collect more area, as in Chandra and XMM-Newton.

Gamma rays. At still higher energies, in hard X-rays and gamma rays, reflection is too weak at any angle, and no mirror can focus. Imaging uses a *coded mask* instead. A plate with a known pattern of opaque and open cells stands in front of a detector that records where each photon lands. Each source casts a shifted copy of the pattern's shadow, and the sky image is found by computing which set of shifted shadows adds up to the recorded one. AstroSat-CZTI and Swift-BAT work this way.

Radio. At the other end of the spectrum, a radio telescope is a metal dish with a receiver at its prime focus. The surface may be a mesh, as long as its holes are small compared with the wavelength of centimetres to metres. This simplicity makes the baselines of Section 5.2 possible. A radio wave oscillates slowly enough (periods of nanoseconds to microseconds) to be digitised directly. Each dish records its signal against an atomic clock (Chapter 1), and the recordings are multiplied in a computer afterwards, Eq. (5.10). Visible light oscillates at about $5 \times 10^{14}\,\mathrm{Hz}$, far too fast to digitise, so an optical interferometer must bring the beams together with mirrors while the light arrives.

5.3.5 Cassegrain Reflector Telescope Geometry

In a Cassegrain telescope (Figure 5.4), parallel light from a star falls on a concave primary mirror of diameter D and focal length F. Before the light reaches the primary focus, a secondary

mirror at distance d_2 in front of the primary catches the converging cone. It sends the light back through a hole in the primary to a detector a distance a behind the primary. We want the smallest diameter b of the secondary, the place a of the detector, and the focal length of the whole system.

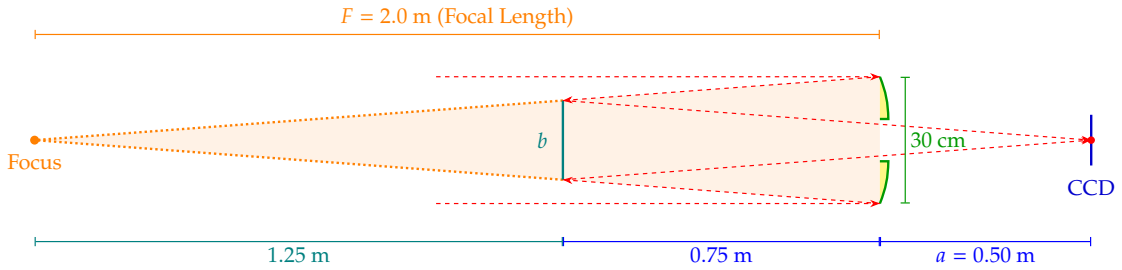


Figure 5.4: A Cassegrain telescope with a 30 cm primary of focal length 2.0 m, a flat secondary of diameter b at 0.75 m from the primary and 1.25 m from the primary focus, and the CCD a distance a behind the primary.

Size of the secondary. The rays from the edge of the primary converge on the primary focus. The cone has diameter D at the primary, a distance F from the focus, and its diameter falls in proportion to the distance from the focus. The secondary is $F - d_2$ from the focus, so by similar triangles

$$\frac{b}{D} = \frac{F - d_2}{F} \qquad \Longrightarrow \qquad \boxed{b = D\left(1 - \frac{d_2}{F}\right).} \tag{5.20}$$

A smaller secondary catches some of the on-axis light only. Placing the secondary closer to the focus makes it smaller, and so it blocks less of the primary.

Place of the focus. The cone aims at the primary focus, a distance $F - d_2$ behind the secondary. A flat secondary forms the image the same distance in front of itself, $F - d_2$ from it. Back from the secondary towards the primary that is d_2 to the primary, and the rest lies behind it,

$$a = (F - d_2) - d_2 = F - 2d_2 \qquad \text{(flat secondary)}. \tag{5.21}$$

A convex secondary, as in a true Cassegrain, sends the image further back, a distance larger than $F - d_2$ from the secondary.

The effective focal length. After the secondary the cone converges from the diameter b over the distance from the secondary to the final focus. Its ratio of length to width is what a single mirror of some focal length F_{eff} and diameter D would give,

$$\frac{F_{\text{eff}}}{D} = \frac{\text{secondary to final focus}}{b} \overset{(1)}{=} \frac{\text{secondary to final focus}}{D\,(F - d_2)/F},$$
$$F_{\text{eff}} = F \times \frac{\text{secondary to final focus}}{\text{secondary to primary focus}}. \tag{5.22}$$

(1) b from Eq. (5.20).

A flat secondary leaves the focal length unchanged, $F_{\text{eff}} = F$. A convex one multiplies it by its magnification, which is how a short tube holds a long focal length. The numbers for the telescope of the figure are in Problem 5.7.2.

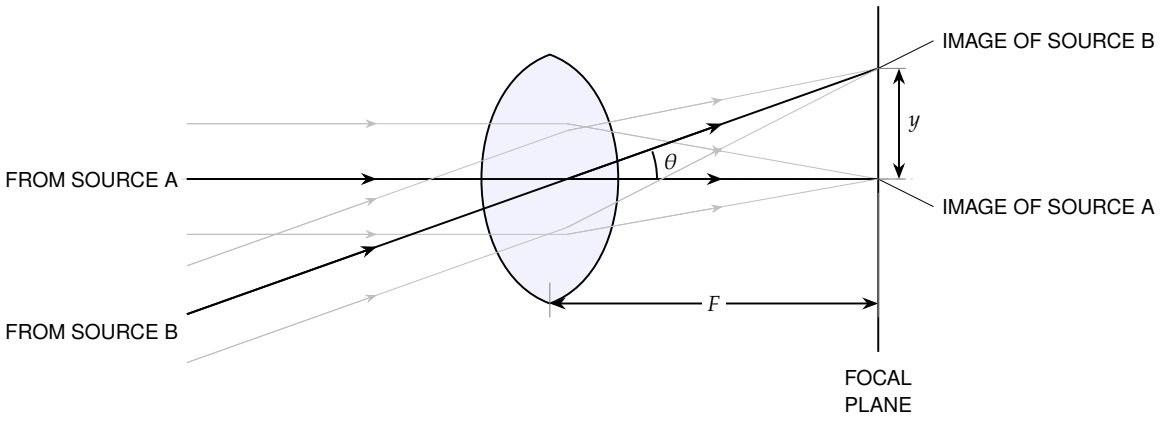


Figure 5.5: Two point sources A and B at the angle θ, a lens of focal length F, and their two images a height y apart in the focal plane. The chief rays through the centre of the lens are dark, and the edge rays are faded.

5.3.6 Optical Ray Diagram and Angular Resolution on the Focal Plane

5.3.7 Plate Scale and Pixel Scale

A detector measures distances in the focal plane, and the astronomer wants angles on the sky. The **plate scale** turns a distance into an angle, and the **pixel scale** is the angle covered by one pixel. Both come from Figure 5.5, where a source at the angle θ from the axis forms its image a height y from the axis.

The exact relation. The chief ray passes through the centre of the lens, or the vertex of the mirror, without bending. It meets the focal plane, a distance F away, at the height

$$y = F\tan\theta \quad\Longrightarrow\quad \theta = \arctan\left(\frac{y}{F}\right). \tag{5.23}$$

With a secondary mirror, F is the effective focal length of Eq. (5.22).

The plate scale. The plate scale is the angle per unit length in the focal plane, the derivative of Eq. (5.23),

$$\frac{\mathrm{d}\theta}{\mathrm{d}y} \overset{(1)}{=} \frac{1}{F}\,\frac{1}{1+(y/F)^2} \overset{(2)}{=} \frac{1}{F}\,\frac{1}{1+\tan^2\theta} \overset{(3)}{=} \frac{\cos^2\theta}{F}. \tag{5.24}$$

(1) $\frac{\mathrm{d}}{\mathrm{d}x}\arctan x = 1/(1+x^2)$, with $x = y/F$.
(2) $y/F = \tan\theta$.
(3) $1+\tan^2\theta = 1/\cos^2\theta$.

So the plate scale is not quite the same everywhere. Away from the axis a millimetre covers slightly fewer seconds of arc. Fields of view rarely exceed a few degrees, and even at $\theta = 1°$, $\cos^2\theta = 0.99970$. In this small-angle limit $\theta \approx y/F$, and the plate scale is a single number,

$$\boxed{\frac{\mathrm{d}\theta}{\mathrm{d}y} = \frac{\theta}{y} = \frac{1}{F},} \tag{5.25}$$

quoted in seconds of arc per millimetre, $206\,265/F$ [mm]. Only wide-field survey cameras need the $\cos^2\theta$ correction.

The pixel scale. A pixel of width y_{pix} covers the angle

$$\theta_{\text{pix}} = \frac{y_{\text{pix}}}{F}, \tag{5.26}$$

which is Eq. (5.25) read at one pixel,

Result 5.2

$$\theta_{\text{pix}} = \left.\frac{y}{F}\right|_{y=y_{\text{pix}}} \text{rad}, \tag{5.27}$$

or, in seconds of arc per pixel,

$$\theta_{\text{pix}} = \frac{206\,265 \times y_{\text{pix}}\,[\text{mm}]}{F\,[\text{mm}]}\ \textit{arcsec per pixel}. \tag{5.28}$$

For pixels 20 μm wide at $F = 2$ m, $\theta_{\text{pix}} = 10^{-5}$ rad $= 2.06''$. A longer focal length spreads the sky over more pixels.

5.4 Diffraction

A telescope maps each direction to a point of the focal plane, Eq. (5.23), and two antennas a distance B apart resolve about λ/B, Eq. (5.16).

Ray optics says a perfect mirror brings a star to a point. Every real telescope, even a perfect one in space such as Hubble or JWST, shows a star as a small disc with faint rings, the **point spread function** (PSF).

Stand beside an open door and you hear people talking in the next room, though you cannot see them. Sound bends round the edge of the door and light does not. The door is about 1 m wide. Sound at 343 Hz has a wavelength of 1 m, the size of the door, so it spreads into every direction. Light has a wavelength two million times smaller, so it spreads by only two millionths of a radian.

The edge of a telescope mirror is the door frame. The idea is that a wave passing an opening of size D spreads into an angle of about λ/D.

For the sound, $\lambda/D = 1$. For light of 500 nm through a 30 cm mirror, $\lambda/D = 1.7 \times 10^{-6}$ rad, and the exact result for a round mirror, $1.22\lambda/D$, is $0.42''$.

The picture is a flat wave meeting a wall with a hole in it. Each point of the hole sends out a small circular wave, and far behind the wall these waves add up into a pattern of bright and dark bands.

The same addition of waves explains the bands from a single slit in the laboratory, the colours from a grating, X-ray diffraction by crystals, the beam of a radio dish, and the pair of antennas of Section 5.2. A photon of momentum h/λ squeezed through the opening gets a sideways momentum of about h/D, by the uncertainty principle, and so the same angle λ/D. Comparing λ with D is P6. It gives the field far behind any opening as one integral, then the single slit, the grating of a spectrograph, the round mirror with its factor 1.22, and the rules for when two stars are resolved.

5.4.1 Diffraction as a Sum of Wavelets

By the **Huygens–Fresnel principle**, every open point of an opening acts as the source of a small spherical wave, a *wavelet*. The waves obey a linear equation, so the field at any point behind the opening is the sum of all the wavelets that reach it, each with the phase of the path it travelled.

The field in the opening. Let the opening lie in the plane $z = 0$, with coordinates (ξ, η). Light from an unresolved star arrives along the z axis as a flat wave of uniform amplitude E_0. The opening is described by its **transmission** $t(\xi, \eta)$, equal to 1 where light passes and 0 where the mirror edge, a baffle or a spider vane blocks it. Just behind the opening the field is

$$E_0\, t(\xi,\eta) = \begin{cases} E_0, & \text{inside the opening,} \\ 0, & \text{where it is blocked.} \end{cases} \tag{5.29}$$

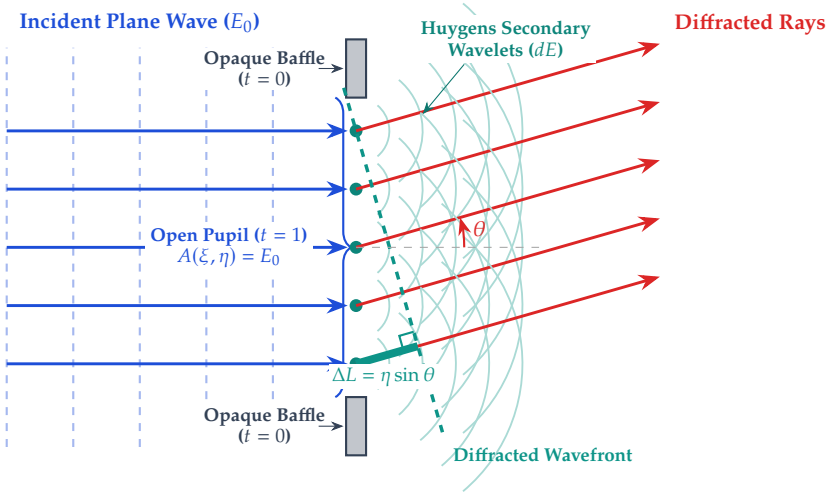


Figure 5.6: A flat wave meeting an opening between two opaque baffles, the wavelets sent out by points of the opening, the rays leaving at the angle θ, and the tilted wavefront. The thick segment marks the extra path of the lowest wavelet.

The path to a far point. Look in the direction of the unit vector

$$\hat{\mathbf{k}} = \begin{pmatrix} \sin\theta\cos\chi \\ \sin\theta\sin\chi \\ \cos\theta \end{pmatrix} \equiv \begin{pmatrix} \theta_x \\ \theta_y \\ \cos\theta \end{pmatrix}, \tag{5.30}$$

where θ is the angle from the axis and χ the angle around it. Its first two components, θ_x and θ_y, are the angles on the sky along the ξ and η directions, for the small angles of astronomy. A point at distance s in this direction is $s\hat{\mathbf{k}}$. The wavelet from the point $\vec{r} = (\xi, \eta, 0)$ of the opening, with $|\vec{r}| = r$, travels to it the distance

$$\begin{aligned} |s\hat{\mathbf{k}} - \vec{r}| = \sqrt{s^2 - 2s\,\vec{r}\cdot\hat{\mathbf{k}} + r^2} &\overset{(1)}{=} s\left(1 - \frac{\vec{r}\cdot\hat{\mathbf{k}}}{s} + \frac{r^2}{2s^2} - \frac{(\vec{r}\cdot\hat{\mathbf{k}})^2}{2s^2} + \dots\right) \\ &\overset{(2)}{=} s - (\xi\theta_x + \eta\theta_y) + \frac{r^2 - (\vec{r}\cdot\hat{\mathbf{k}})^2}{2s} + \dots \end{aligned} \tag{5.31}$$

(1) $\sqrt{1+\epsilon} = 1 + \epsilon/2 - \epsilon^2/8 + \dots$ with $\epsilon = -2\vec{r}\cdot\hat{\mathbf{k}}/s + r^2/s^2$, kept to second order in r/s.
(2) $\vec{r}\cdot\hat{\mathbf{k}} = \xi\theta_x + \eta\theta_y$, since $\vec{r}$ has no z component.

The first term is the same for every wavelet. The second is the path difference that makes the pattern. The last term, at most $(D/2)^2/(2s)$ for an opening of diameter D, changes the phase by at most $\pi D^2/(4\lambda s)$. It can be dropped when

$$s \gg \frac{D^2}{\lambda}, \tag{5.32}$$

a comparison of two lengths (P6). This is the **Fraunhofer** or far-field limit. For a 30 cm mirror at 500 nm, $D^2/\lambda = 180$ km. A telescope brings the far field to its focal plane. A perfect focusing mirror delays each wavelet by just the amount that removes the last term, so the pattern at the focus is the pattern at infinity, with the angle θ mapped to the height $F\theta$.

The integral. A wavelet with extra path $-(\xi\theta_x+\eta\theta_y)$ has the phase factor $\exp\left[-ik(\xi\theta_x+\eta\theta_y)\right]$, with $k = 2\pi/\lambda$. Summing over the opening, and dropping the factor e^{iks}/s and the constant of the Huygens–Fresnel integral, which are the same for every wavelet,

Result 5.3

$$E(\theta_x,\theta_y) = E_0 \iint t(\xi,\eta)\, e^{-ik(\xi\theta_x+\eta\theta_y)}\, d\xi\, d\eta, \qquad I(\theta_x,\theta_y) = |E(\theta_x,\theta_y)|^2. \tag{5.33}$$

The field at the focus is the two-dimensional Fourier transform of the opening, with ξ/λ and η/λ as the frequencies. A wide opening gives a narrow pattern, and a narrow one a wide pattern. Everything below is this one integral for different shapes of t.

5.4.2 Single-Slit Diffraction

The simplest opening is a slit of width a along η and much longer along ξ. The long side makes the pattern narrow along θ_x, so we look only in the plane across the slit, $\theta_x = 0$ and $\theta_y = \sin\theta$, and take the field per unit length of slit.

Pairing the two halves. The dark bands can be found without any integral. Split the slit into a lower half, $-a/2 < \eta < 0$, and an upper half, $0 < \eta < a/2$. Each point η of the lower half has a partner $\eta + a/2$ in the upper half. At the angle θ the wavelets of the two partners differ in path by $(a/2)\sin\theta$. When that is half a wavelength, every pair cancels at once,

$$\frac{a}{2}\sin\theta = \frac{\lambda}{2} \quad\Longrightarrow\quad a\sin\theta = \lambda \quad\Longrightarrow\quad \theta \approx \frac{\lambda}{a}. \tag{5.34}$$

For a slit the number in front of λ/a is exactly 1.

The full pattern. Eq. (5.33) with $t = 1$ for $|\eta| < a/2$ gives

$$\begin{aligned} E(\theta) &= E_0 \int_{-a/2}^{a/2} e^{-ik\eta\sin\theta}\, d\eta \overset{(1)}{=} E_0\, \frac{e^{-ik(a/2)\sin\theta} - e^{ik(a/2)\sin\theta}}{-ik\sin\theta} \\ &\overset{(2)}{=} E_0\, \frac{2\sin(k(a/2)\sin\theta)}{k\sin\theta} \overset{(3)}{=} E_0\, a\, \frac{\sin\beta_a}{\beta_a}, \qquad \beta_a \equiv \frac{\pi a\sin\theta}{\lambda}. \end{aligned} \tag{5.35}$$

(1) The integral of an exponential, between the two edges.
(2) $e^{-ix} - e^{ix} = -2i\sin x$.
(3) $k(a/2)\sin\theta = \pi a\sin\theta/\lambda = \beta_a$, so $k\sin\theta = 2\beta_a/a$.

The intensity, with $I_0 = E_0^2 a^2$ in the straight-through direction, is

$$\boxed{I(\theta) = I_0\left(\frac{\sin\beta_a}{\beta_a}\right)^2 = I_0\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2.} \tag{5.36}$$

The first zero is at $\beta_a = \pi$, so $\sin\theta = \lambda/a$, as the pairing found. The integral $\int_{-\infty}^{\infty}(\sin\beta/\beta)^2 d\beta = \pi$ measures the whole pattern. The part between the first zeros, $(1/\pi)\int_{-\pi}^{\pi}(\sin\beta/\beta)^2 d\beta$, is 0.903, so the central band carries 90.3% of the light.

5.4.3 N-Slit Diffraction Grating: Dispersion and Resolving Power

A diffraction grating is N equal parallel slits of width a, with centres a distance d apart, over a total width $W = Nd$. It spreads light into its colours, and it is the heart of most astronomical spectrographs.

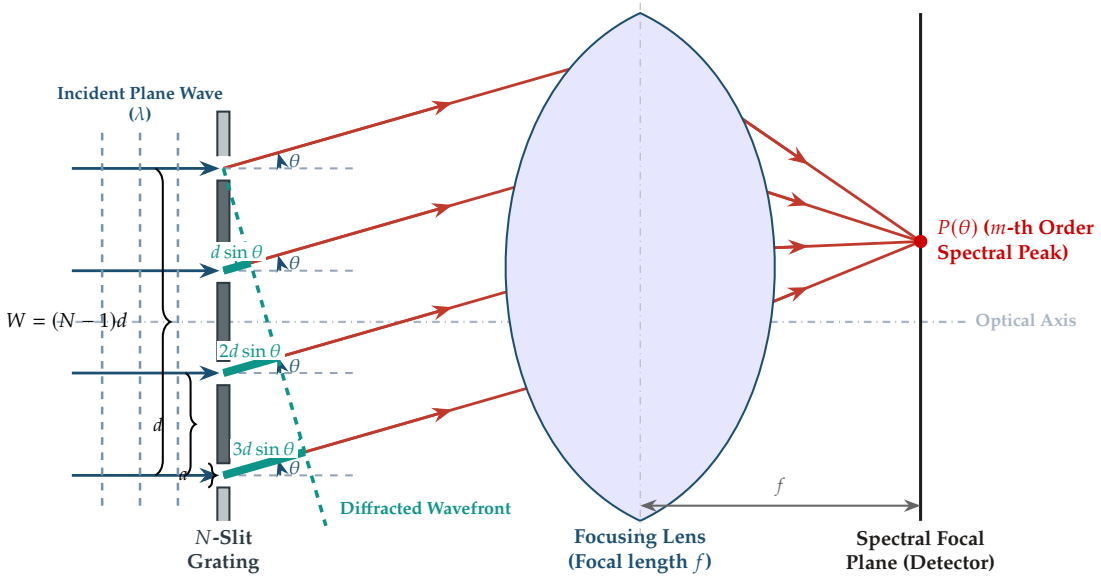


Figure 5.7: A flat wave meeting an N-slit grating, the rays leaving at the angle θ with path lags $d\sin\theta$, $2d\sin\theta$ and $3d\sin\theta$, and a lens of focal length f bringing them to a point $P(\theta)$ in its focal plane.

Summing the slits. The slit numbered $n = 0, \ldots, N-1$ has its centre at $\eta = nd$. Changing the variable to $\eta' = \eta - nd$ turns its integral into that of a slit at the centre, Eq. (5.35), times the phase factor $e^{-iknd\sin\theta}$. Write $\beta_d \equiv \pi d \sin\theta/\lambda$, so that this factor is $e^{-2in\beta_d}$. Then

$$E(\theta) \overset{(1)}{=} E_0 a \,\frac{\sin\beta_a}{\beta_a} \sum_{n=0}^{N-1} e^{-2in\beta_d} \overset{(2)}{=} E_0 a \,\frac{\sin\beta_a}{\beta_a}\,\frac{1-e^{-2iN\beta_d}}{1-e^{-2i\beta_d}}$$
$$\overset{(3)}{=} E_0 a \,\frac{\sin\beta_a}{\beta_a}\,\frac{e^{-iN\beta_d}\left(e^{iN\beta_d}-e^{-iN\beta_d}\right)}{e^{-i\beta_d}\left(e^{i\beta_d}-e^{-i\beta_d}\right)} \overset{(4)}{=} E_0 a \,\frac{\sin\beta_a}{\beta_a}\,\frac{\sin N\beta_d}{\sin\beta_d}\, e^{-i(N-1)\beta_d}. \tag{5.37}$$

(1) Each slit gives Eq. (5.35) times its phase factor $e^{-iknd\sin\theta} = e^{-2in\beta_d}$.
(2) The geometric sum $\sum_{n=0}^{N-1} q^n = (1-q^N)/(1-q)$ with $q = e^{-2i\beta_d}$.
(3) Take $e^{-iN\beta_d}$ out of the top and $e^{-i\beta_d}$ out of the bottom.
(4) $e^{ix} - e^{-ix} = 2i\sin x$ in both, and $e^{-iN\beta_d}/e^{-i\beta_d} = e^{-i(N-1)\beta_d}$.

The phase factor drops out of $|E|^2$, and with $I_0 = E_0^2 a^2$ again,

$$\boxed{I(\theta) = I_0 \underbrace{\left(\frac{\sin\beta_a}{\beta_a}\right)^2}_{\text{one slit}} \underbrace{\left(\frac{\sin N\beta_d}{\sin\beta_d}\right)^2}_{N\text{ slits}}.} \tag{5.38}$$

The width of each slit sets a broad envelope, and their spacing sets sharp peaks inside it.

The principal maxima. When $\beta_d = m\pi$ for a whole number m, all N slits are in phase. The ratio $\sin N\beta_d / \sin\beta_d$ is then $0/0$, and its limit is $\pm N$ (the derivatives of top and bottom give $N\cos N m\pi / \cos m\pi$). So the peak intensity is N^2 times that of one slit. The condition $\beta_d = m\pi$ reads $d\sin\theta = m\lambda$. If the light arrives at the angle θ_i to the normal, the incoming path to slit n also differs, by $nd\sin\theta_i$. Counting θ_i on the opposite side of the normal from θ_m, as is usual for reflection gratings, this lag adds to the lag on the way out, and the peaks of order m are at

$$d\,(\sin\theta_m + \sin\theta_i) = m\lambda \qquad (m \in \mathbb{Z}). \tag{5.39}$$

This is the **grating equation**. For $m \neq 0$ the angle depends on λ, so each colour goes its own way.

The width of a peak. Next to a peak, $\sin N\beta_d$ first vanishes when $N\beta_d$ has moved from $Nm\pi$ to $Nm\pi + \pi$, a step $\Delta\beta_d = \pi/N$. Since $\beta_d = \pi d\sin\theta/\lambda$, a small step in θ gives $\Delta\beta_d = (\pi d\cos\theta_m/\lambda)\Delta\theta$, and

$$\Delta\theta = \frac{\lambda}{Nd\cos\theta_m} = \frac{\lambda}{W\cos\theta_m}. \tag{5.40}$$

Here $W\cos\theta_m$ is the width of the grating seen from the direction θ_m. The peak is as narrow as the diffraction pattern of a single slit as wide as the whole grating.

Resolving power. Two lines at λ and $\lambda + \Delta\lambda$ are just resolved, by the Rayleigh rule of Section 5.4.5, when the peak of one falls on the first zero of the other. In terms of $d\sin\theta$, the peak of $\lambda + \Delta\lambda$ is at $m(\lambda + \Delta\lambda)$, and the first zero beside the peak of λ is at $m\lambda + \lambda/N$. Setting them equal,

$$m(\lambda + \Delta\lambda) = m\lambda + \frac{\lambda}{N} \quad\Longrightarrow\quad m\Delta\lambda = \frac{\lambda}{N}. \tag{5.41}$$

With Eq. (5.39) for m, the **resolving power** is

Result 5.4

$$\mathcal{R} \equiv \frac{\lambda}{\Delta\lambda} = mN = \frac{W\,(\sin\theta_m + \sin\theta_i)}{\lambda}. \tag{5.42}$$

The last form is the difference in path between light from the two ends of the grating, counted in wavelengths. So a high-resolution spectrograph needs a large grating used at a steep angle in a high order. HARPS and ESPRESSO, which reach $\mathcal{R} > 100\,000$ to measure the wobble of stars with planets, use echelle gratings tens of centimetres long.

Worked example 5.2: Resolving the sodium D lines

The two sodium D lines lie at 589.59 and 588.99 nm, so $\Delta\lambda = 0.60$ nm about the mean 589.29 nm. They need

$$\mathcal{R} = \frac{589.29\,\text{nm}}{0.60\,\text{nm}} = 982.$$

In order $m = 2$, Eq. (5.42) asks for $N \geq 982/2 = 491.1$, so at least 492 lit grooves. On a grating with 600 grooves per millimetre that is a beam only $W = 492/600$ mm $= 0.82$ mm wide.

5.4.4 Circular Apertures and the Airy Disk

Telescope mirrors are round, of diameter D. We want the pattern of Eq. (5.33) for a disc, and the angle of its first dark ring.

Why more than one. In a slit every strip parallel to the edges has the same length, so the two halves cancel each other one to one at $\theta = \lambda/a$. In a disc the strips are short near the rim and long, up to D, across the middle. The middle sends more light than the edges. When the path difference between the two edges is λ, the weak wavelets of the narrow edges cannot cancel the strong ones of the wide middle. The angle must open by about 22% more before the whole disc cancels.

Polar coordinates. Take polar coordinates in the opening, $\xi = r\cos\psi$ and $\eta = r\sin\psi$, and on the sky, $\theta_x = \theta\cos\chi$ and $\theta_y = \theta\sin\chi$ for small θ. The phase needs $\xi\theta_x + \eta\theta_y$. Turn the axes of the opening by χ, with the 2×2 upper-left block of $\mathbf{R}_z(\chi)$, Eq. (2.5),

$$\begin{pmatrix}\xi'\\ \eta'\end{pmatrix} = \begin{pmatrix}\cos\chi & \sin\chi\\ -\sin\chi & \cos\chi\end{pmatrix}\begin{pmatrix}r\cos\psi\\ r\sin\psi\end{pmatrix} \overset{(1)}{=} \begin{pmatrix}r\cos(\psi-\chi)\\ r\sin(\psi-\chi)\end{pmatrix},$$

$$\xi\theta_x + \eta\theta_y = \theta\,(\xi\cos\chi + \eta\sin\chi) \overset{(2)}{=} \theta\,\xi' \overset{(3)}{=} \theta\, r\cos(\psi-\chi). \tag{5.43}$$

(1) The angle-difference identities.
(2) The top row of the matrix product.
(3) The first line.

The disc looks the same in the turned axes, so the integral over ψ from 0 to 2π gives the same for every χ. The pattern depends on θ alone, and we may set $\chi = 0$,

$$E(\theta) = E_0 \int_0^{D/2}\int_0^{2\pi} e^{-ikr\theta\cos\psi}\, d\psi\; r\, dr. \tag{5.44}$$

The angle integral and J_0. The integral over ψ defines the Bessel function of order zero,

$$J_0(z) \equiv \frac{1}{2\pi}\int_0^{2\pi} e^{-iz\cos\psi}\, d\psi. \tag{5.45}$$

Its power series follows by expanding the exponential,

$$J_0(z) \overset{(1)}{=} \sum_{n=0}^{\infty}\frac{(-iz)^n}{n!}\,\frac{1}{2\pi}\int_0^{2\pi}\cos^n\psi\, d\psi \overset{(2)}{=} \sum_{j=0}^{\infty}\frac{(-iz)^{2j}}{(2j)!}\,\frac{(2j)!}{4^j(j!)^2}$$
$$\overset{(3)}{=} \sum_{j=0}^{\infty}\frac{(-1)^j}{(j!)^2}\left(\frac{z}{2}\right)^{2j} = 1 - \frac{z^2}{4} + \frac{z^4}{64} - \dots \tag{5.46}$$

(1) The series of e^x, integrated term by term.
(2) Odd powers of $\cos\psi$ average to zero. For $n = 2j$, the average of $\cos^{2j}\psi$ is $(2j)!/(4^j(j!)^2)$, the middle binomial coefficient over 4^j, from $\cos\psi = (e^{i\psi} + e^{-i\psi})/2$.
(3) $(-i)^{2j} = (-1)^j$ and $4^j = 2^{2j}$.

So $E(\theta) = 2\pi E_0 \int_0^{D/2} J_0(kr\theta)\, r\, dr$.

The radial integral and J_1. The Bessel function of order one is

$$J_1(z) \equiv \sum_{j=0}^{\infty}\frac{(-1)^j}{j!\,(j+1)!}\left(\frac{z}{2}\right)^{2j+1} = \frac{z}{2} - \frac{z^3}{16} + \dots \tag{5.47}$$

Differentiating the series term by term gives the identity we need,

$$\frac{\mathrm{d}}{\mathrm{d}z}\big[zJ_1(z)\big] = \frac{\mathrm{d}}{\mathrm{d}z}\sum_{j=0}^{\infty}\frac{(-1)^j\,z^{2j+2}}{2^{2j+1}\,j!\,(j+1)!} \overset{(1)}{=} \sum_{j=0}^{\infty}\frac{(-1)^j\,(2j+2)\,z^{2j+1}}{2^{2j+1}\,j!\,(j+1)!}$$
$$\overset{(2)}{=} z\sum_{j=0}^{\infty}\frac{(-1)^j}{(j!)^2}\left(\frac{z}{2}\right)^{2j} \overset{(3)}{=} zJ_0(z). \tag{5.48}$$

(1) The power rule on each term.
(2) $(2j+2)/(j+1)! = 2/j!$, and $2/2^{2j+1} = 1/2^{2j}$.
(3) Eq. (5.46).

In the same way, differentiating Eq. (5.46) gives $J_0' = -J_1$. Since $zJ_1(z)$ is zero at $z = 0$, Eq. (5.48) integrates to $\int_0^u zJ_0(z)\,dz = u\,J_1(u)$. Now change the variable in the radial integral to $z = kr\theta$. The edge $r = D/2$ becomes

$$\beta_D \equiv k\,\frac{D}{2}\,\theta = \frac{\pi D\theta}{\lambda}, \tag{5.49}$$

the same kind of variable as β_a for the slit, with the radius in place of the half-width. Then

$$E(\theta) \overset{(1)}{=} \frac{2\pi E_0}{(k\theta)^2}\int_0^{\beta_D} zJ_0(z)\,dz \overset{(2)}{=} \frac{2\pi E_0}{(k\theta)^2}\,\beta_D J_1(\beta_D) \overset{(3)}{=} 2\pi E_0\left(\frac{D}{2}\right)^2\frac{J_1(\beta_D)}{\beta_D}$$
$$= E_0 \underbrace{\frac{\pi D^2}{4}}_{\text{area of the mirror}} \frac{2J_1(\beta_D)}{\beta_D}. \tag{5.50}$$

(1) $r = z/(k\theta)$ and $dr = dz/(k\theta)$.
(2) $\int_0^u zJ_0\,dz = uJ_1(u)$, from Eq. (5.48).
(3) $k\theta = 2\beta_D/D$ from Eq. (5.49).

For $\theta \to 0$, $J_1(\beta_D) \approx \beta_D/2$ by Eq. (5.47), so $2J_1(\beta_D)/\beta_D \to 1$. Straight ahead every wavelet is in phase, and the field is E_0 times the area.

The Airy pattern. Squaring, with $I_0 = E_0^2(\pi D^2/4)^2$, gives the **Airy pattern**,

$$\boxed{I(\theta) = I_0\left[\frac{2J_1(\beta_D)}{\beta_D}\right]^2, \qquad \beta_D = \frac{\pi D\sin\theta}{\lambda} \approx \frac{\pi D\theta}{\lambda}.} \tag{5.51}$$

Half the peak intensity is reached at $\beta_D = 1.616$, so the full width at half maximum (FWHM) is $2 \times 1.616/\pi = 1.03\,\lambda/D$.

The first dark ring and the factor 1.22**.** The intensity first vanishes where $J_1(\beta_D) = 0$ for $\beta_D > 0$. The first positive zero of J_1 is $\beta_D = 3.8317$. So the angular radius of the first dark ring is

$$\frac{\pi D\theta}{\lambda} = 3.8317 \quad\Longrightarrow\quad \theta = \frac{3.8317}{\pi}\,\frac{\lambda}{D} = 1.2197\,\frac{\lambda}{D}. \tag{5.52}$$

This is where the factor 1.22 comes from. A second star placed on this ring is the **Rayleigh limit** θ_R of a round telescope,

Result 5.5

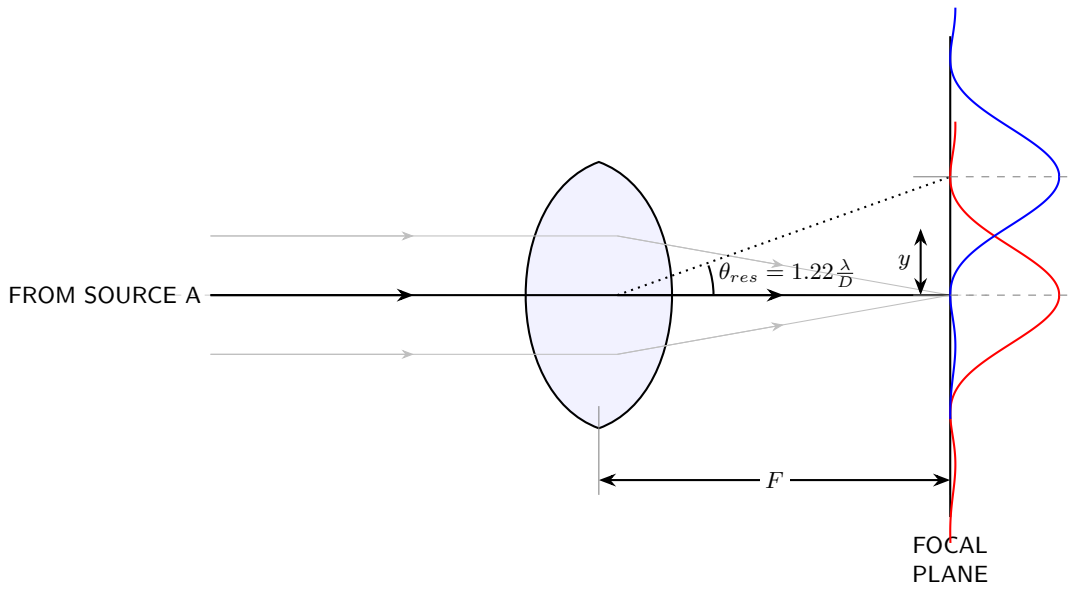


Figure 5.8: Light from source A focused by a lens of focal length F. Beside the focal plane, the Airy profile of source A (red) and that of a second source at the angle $\theta_{\text{res}} = 1.22\lambda/D$ (blue), whose image lies a height y away.

$$\theta_R = 1.22\frac{\lambda}{D}\ rad = 206\,265 \times 1.22\frac{\lambda}{D}\ arcsec. \tag{5.53}$$

Only the ratio λ/D enters. The focal length, the design of the telescope and the brightness of the star do not.

How the light is shared among the rings. The fraction of the light inside the angle θ_0 is the integral of $I(\theta)\,2\pi\theta\,d\theta$ from 0 to θ_0, over the same integral to infinity. With $\theta = \lambda\beta_D/(\pi D)$, both carry the same constant, and only $\int (2J_1/\beta_D)^2\beta_D\,d\beta_D = 4\int J_1^2/\beta_D\,d\beta_D$ remains. An identity does this integral,

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\left[J_0^2 + J_1^2\right] = 2J_0J_0' + 2J_1J_1' \overset{(1)}{=} -2J_0J_1 + 2J_1\left(J_0 - \frac{J_1}{\beta}\right) = -\frac{2J_1^2}{\beta}, \tag{5.54}$$

(1) $J_0' = -J_1$, and $J_1' = J_0 - J_1/\beta$ from expanding Eq. (5.48), $J_1 + \beta J_1' = \beta J_0$.

so $\int_0^{\beta_0} J_1^2/\beta\,d\beta = \frac{1}{2}[1 - J_0^2(\beta_0) - J_1^2(\beta_0)]$, using $J_0(0) = 1$ and $J_1(0) = 0$. Both functions die away at large β, so the whole pattern gives $\frac{1}{2}$, and the fraction inside β_0 is

$$\boxed{\text{fraction inside } \beta_0 = 1 - J_0^2(\beta_0) - J_1^2(\beta_0).} \tag{5.55}$$

At the dark rings $J_1 = 0$, so only J_0^2 is lost. At the first, $J_0(3.8317) = -0.4028$, and the central disc holds $1 - 0.1622 = 83.8\%$ of the light. The second dark ring, at $\beta_D = 7.0156$, encloses 91.0%, so the first bright ring holds 7.2%. The second ring holds 2.8% and the rest goes into fainter rings. The first bright ring peaks at $\beta_D = 5.136$ with only 1.75% of the central intensity.

The total over the whole pattern is a check on the normalisation. It is

$$\int_0^\infty I\,2\pi\theta\,d\theta = I_0\,2\pi\left(\frac{\lambda}{\pi D}\right)^2 \times 4 \times \frac{1}{2} = I_0\,\frac{\lambda^2}{\pi D^2/4} = E_0^2\,\frac{\pi D^2}{4}\,\lambda^2. \tag{5.56}$$

Parseval's theorem applied to Eq. (5.33) gives the same, $E_0^2\lambda^2$ times the open area, for any shape of opening. The light leaving the focus equals the light entering the mirror (P1). It also says the PSF covers an effective solid angle $\lambda^2/(\pi D^2/4)$, the wavelength squared over the collecting area, a rule radio engineers use for the beam of any antenna.

A central obstruction. A Cassegrain secondary of diameter b blocks the middle of the mirror. The open ring is the full disc minus a small disc. The integral of Eq. (5.33) over the ring is the integral over the big disc minus that over the small one, each given by Eq. (5.50),

$$E(\theta) = E_0 \left[\frac{\pi D^2}{4} \frac{2J_1(\beta_D)}{\beta_D} - \frac{\pi b^2}{4} \frac{2J_1(\beta_b)}{\beta_b} \right], \qquad \beta_b = \frac{b}{D}\beta_D. \tag{5.57}$$

The obstruction makes the central disc a little smaller but sends more light into the rings. For $b/D = 0.2$ the first dark ring moves in to $1.17\lambda/D$ and the central disc holds 76.4% of the light that passes, against 83.8% without it. For $b/D = 0.33$ these become $1.10\lambda/D$ and 65.4%. Brighter rings lower the contrast near a bright star, which matters when imaging faint planets beside it.

Spider vanes and spikes. A spider vane is a thin opaque strip of width a across the opening. In the same way, the field is that of the open mirror minus that of the strip. A strip passes what a slit of the same shape would pass, so its pattern is the slit pattern, Eq. (5.36). Across the vane it spreads over about λ/a, wide because the vane is thin, and along the vane it is as narrow as λ over its length. The result is a thin streak of light through the star, at right angles to the vane. Two vanes along one line give the same streak, so a cross of four vanes gives four spikes, and three vanes 120° apart give six.

5.4.5 Resolution Criteria for Point-Source Imaging

Two stars, such as the two of a binary, send light with unrelated phases. Their intensities add, not their fields. For two equal stars a separation $\Delta\theta$ apart, with the profile $I(\theta)$ of one star,

$$I_{\text{tot}}(\theta) = I\left(\theta + \frac{\Delta\theta}{2}\right) + I\left(\theta - \frac{\Delta\theta}{2}\right). \tag{5.58}$$

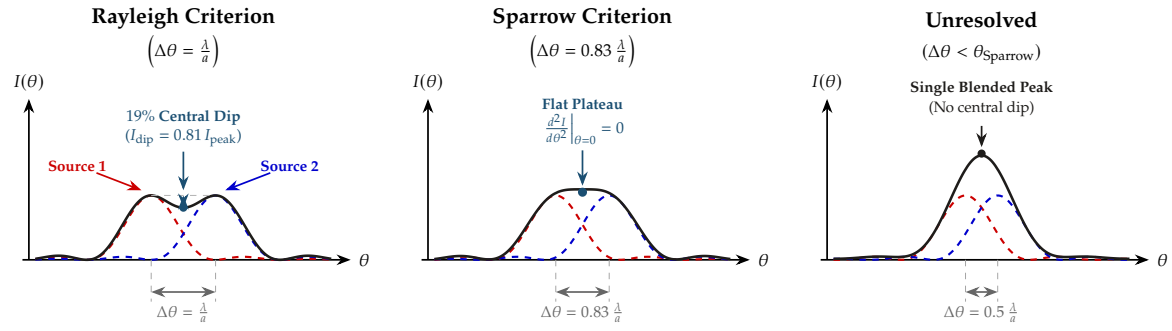


Figure 5.9: The sum of two equal slit patterns at the separations λ/a (Rayleigh, with a 19% dip), $0.83\lambda/a$ (Sparrow, with a flat top) and $0.5\lambda/a$ (a single peak). Dashed curves show the two patterns and solid curves their sum.

The Rayleigh criterion. Rayleigh called two stars resolved when the peak of one falls on the first dark ring of the other, $\Delta\theta = \theta_R$. At the place of each star the sum is then $I_0 + 0 = I_0$. Midway, each pattern is at its half-separation point, and the sum is $2I(\Delta\theta/2)$.

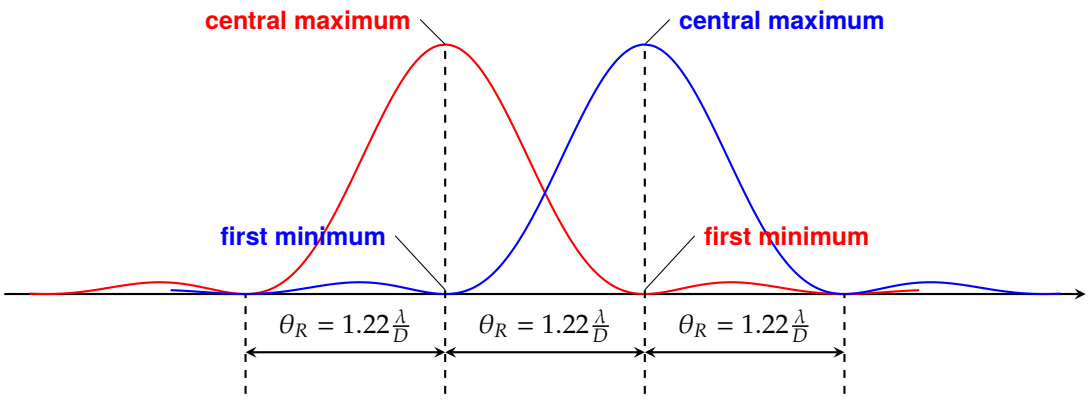


Figure 5.10: Two diffraction patterns whose central maxima are $1.22\lambda/D$ apart, so that each maximum lies on the first minimum of the other.

- For a slit, $\Delta\theta = \lambda/a$ and each pattern is at $\beta_a = \pi/2$. The middle is at $2(\sin(\pi/2)/(\pi/2))^2 = 8/\pi^2 = 0.811$ of the peak, a dip of 19%.
- For a round mirror, $\Delta\theta = 1.22\lambda/D$ and each pattern is at $\beta_D = 3.8317/2 = 1.916$. The middle is at $2\,[2J_1(1.916)/1.916]^2 = 0.735$ of the peak, a dip of 26.5%.

Rayleigh's choice is convenient because the dark ring is easy to compute. A dip of this size is easy to see by eye at the eyepiece.

The Sparrow criterion. Two stars stop showing a dip at all when the sum has a flat top in the middle, with zero second derivative there. By Eq. (5.58),

$$\left.\frac{\mathrm{d}^2 I_{\mathrm{tot}}}{\mathrm{d}\theta^2}\right|_{\theta=0} = I''\left(\frac{\Delta\theta}{2}\right) + I''\left(-\frac{\Delta\theta}{2}\right) \overset{(1)}{=} 2\,I''\left(\frac{\Delta\theta}{2}\right) = 0. \tag{5.59}$$

(1) The profile is even, so its second derivative is even too.

So the **Sparrow limit** is twice the angle at which the profile of one star has its inflection point. The slit profile bends over at $\beta_a = 1.3031$, and the Airy profile at $\beta_D = 1.4878$. This gives

- for a slit, $\Delta\theta = 2 \times 1.3031/\pi\ \lambda/a = 0.83\,\lambda/a$,
- for a round mirror, $\Delta\theta = 2 \times 1.4878/\pi\ \lambda/D = 0.95\,\lambda/D$.

Neither limit is a hard wall. With a high signal-to-noise ratio and a well-measured PSF, two overlapping stars can be separated by fitting, down to the Sparrow limit and below.

Worked example 5.3: Diffraction limit of a 30 cm telescope at 500 nm

For $D = 0.30\,\mathrm{m}$ and $\lambda = 500\,\mathrm{nm}$, Eq. (5.53) gives

$$\theta_R = 1.2197 \times \frac{500 \times 10^{-9}}{0.30}\ \mathrm{rad} = 2.033 \times 10^{-6}\ \mathrm{rad} = 0.42''.$$

5.5 Comparing Pixel Scale to the Diffraction Limit: Nyquist Sampling

A telescope of diameter D blurs a star into an Airy pattern of radius $\theta_R = 1.22\lambda/D$, Eq. (5.53), and each pixel covers the angle θ_{pix}, Eq. (5.26).

A detector keeps one number per pixel. If the pixels are too coarse, fine detail is lost. Worse, detail that is not there can appear.

In old films the wheels of a moving wagon sometimes seem to turn slowly backwards. The camera takes 24 frames a second. If a spoke moves almost one spoke gap between frames, the eye joins each spoke to the one just behind it, and the wheel seems to creep the wrong way.

The pixels of a detector are the frames, and the star image is the wheel. The idea is that a pattern is recorded faithfully only if there are at least two samples in each period of its finest ripple.

A wheel with 12 spokes turning 1.9 times a second has spokes passing at $12 \times 1.9 = 22.8$ a second. At 24 frames a second this is read as $22.8 - 24 = -1.2$ a second, a slow backward turn.

The picture is a smooth curve with dots at equal steps along it, and a fast wiggle that passes through the very same dots as a slow one.

Compact discs sample sound 44 100 times a second because people hear up to about 20 000 Hz. A digital oscilloscope obeys the same rule. Here the pixel is compared with the finest period in the image, P6 once more. It gives the finest ripple a telescope image can contain, the sampling theorem, and the largest pixel that keeps the image whole.

5.5.1 The Finest Detail in the Image

We want the finest ripple in the image of a telescope. Describe the image by its spatial frequencies $\vec{u} = (u_x, u_y)$, in cycles per radian, through the Fourier transform $\tilde{I}(\vec{u}) = \int I(\vec{\theta})\, e^{-2\pi i \vec{u}\cdot\vec{\theta}}\, d^2\theta$, with $\vec{\theta} = (\theta_x, \theta_y)$. Eq. (5.33) gives $I = EE^*$ as a double integral over two points $\vec{r}$ and $\vec{r}'$ of the opening. Then

$$\tilde{I}(\vec{u}) \overset{(1)}{=} E_0^2 \iint t(\vec{r})\, t(\vec{r}') \left[\int e^{-2\pi i \vec{\theta}\cdot\left[(\vec{r}-\vec{r}')/\lambda + \vec{u}\right]}\, d^2\theta \right] d^2r\, d^2r'$$

$$\overset{(2)}{=} E_0^2 \iint t(\vec{r})\, t(\vec{r}')\, \lambda^2\, \delta^2\big(\vec{r} - \vec{r}' + \lambda\vec{u}\big)\, d^2r\, d^2r' \overset{(3)}{=} E_0^2 \lambda^2 \int t(\vec{r})\, t(\vec{r} + \lambda\vec{u})\, d^2r. \tag{5.60}$$

(1) E and E^* from Eq. (5.33), with $k(\xi\theta_x + \eta\theta_y) = 2\pi\vec{r}\cdot\vec{\theta}/\lambda$, and the order of integration swapped.

(2) $\int e^{-2\pi i \vec{\theta}\cdot\vec{q}}\, d^2\theta = \delta^2(\vec{q})$, and $\delta^2(\vec{x}/\lambda) = \lambda^2\delta^2(\vec{x})$ in two dimensions.

(3) The delta function sets $\vec{r}' = \vec{r} + \lambda\vec{u}$.

Since t is 1 or 0, the last integral is the area where the opening overlaps a copy of itself shifted by $\lambda\vec{u}$. Two discs of diameter D stop overlapping once their centres are D apart. So the image of any scene through a round mirror contains no ripple finer than

$$\boxed{u_{\max} = \frac{D}{\lambda} \text{ cycles per radian.}} \tag{5.61}$$

Each pair of points in the mirror a distance λu apart acts as the pair of antennas of Eq. (5.11), with baseline λu. A mirror measures every baseline inside it at once, up to its diameter D. An interferometer measures only the few its antennas give, but out to the full B.

5.5.2 The Nyquist Sampling Criterion

Take one row of pixels along θ, a distance θ_{pix} apart. The recorded image is the true one times a row of spikes at the pixel centres, $I(\theta)\sum_j \delta(\theta - j\theta_{\text{pix}})$. The row of spikes repeats every θ_{pix}, so it is a Fourier series,

$$\sum_{j=-\infty}^{\infty} \delta(\theta - j\theta_{\text{pix}}) = \frac{1}{\theta_{\text{pix}}} \sum_{j'=-\infty}^{\infty} e^{2\pi i j'\theta/\theta_{\text{pix}}}, \tag{5.62}$$

where each coefficient is $(1/\theta_{\text{pix}}) \int_{-\theta_{\text{pix}}/2}^{\theta_{\text{pix}}/2} \delta(\theta)\, e^{-2\pi i j'\theta/\theta_{\text{pix}}} d\theta = 1/\theta_{\text{pix}}$. The transform of the recorded image is

$$\int I(\theta) \sum_j \delta(\theta - j\theta_{\text{pix}})\, e^{-2\pi i u\theta}\, d\theta \overset{(1)}{=} \frac{1}{\theta_{\text{pix}}} \sum_{j'} \int I(\theta)\, e^{-2\pi i (u - j'/\theta_{\text{pix}})\theta}\, d\theta \overset{(2)}{=} \frac{1}{\theta_{\text{pix}}} \sum_{j'} \tilde{I}\left(u - \frac{j'}{\theta_{\text{pix}}}\right). \tag{5.63}$$

(1) Eq. (5.62), and the two exponentials combined.
(2) The definition of $\tilde{I}$, shifted by j'/θ_{pix}.

Sampling makes copies of the true spectrum, spaced $1/\theta_{\text{pix}}$ apart. Each copy runs from $-u_{\max}$ to $+u_{\max}$. They do not overlap if $1/\theta_{\text{pix}} \geq 2u_{\max}$. Then the copy with $j' = 0$ is the true spectrum, untouched, between $\pm 1/(2\theta_{\text{pix}})$, and the image can be rebuilt exactly. Keeping only that copy and transforming back gives

$$I(\theta) = \sum_j I(j\theta_{\text{pix}}) \frac{\sin\big(\pi(\theta - j\theta_{\text{pix}})/\theta_{\text{pix}}\big)}{\pi(\theta - j\theta_{\text{pix}})/\theta_{\text{pix}}}, \tag{5.64}$$

since a flat band of width $1/\theta_{\text{pix}}$ transforms into the slit pattern of Eq. (5.35). If the copies overlap, a ripple at a frequency u just above $1/(2\theta_{\text{pix}})$ is recorded as one at $u - 1/\theta_{\text{pix}}$. At every pixel, $\cos\big(2\pi u j\theta_{\text{pix}}\big) = \cos\big(2\pi(u - 1/\theta_{\text{pix}})j\theta_{\text{pix}}\big)$, because the two phases differ by $2\pi j$, a whole number of turns. This false ripple is **aliasing**, the backward wagon wheel. Real pixels collect light over their whole width rather than at a point. That multiplies the spectrum by a slit pattern, which lowers the fine ripples a little but does not change the condition.

With $u_{\max} = D/\lambda$ from Eq. (5.61), the **Nyquist–Shannon** condition $\theta_{\text{pix}} \leq 1/(2u_{\max})$ becomes

Result 5.6

$$\theta_{\text{pix}} \leq \frac{\lambda}{2D} = \frac{\theta_R}{2.44}. \tag{5.65}$$

The FWHM of the Airy pattern is $1.03\lambda/D$, so this asks for about **two pixels across the FWHM**. The common rule of two pixels across θ_R itself is a little too coarse. A row along the other axis of the detector obeys the same condition.

5.5.3 Configuration Critique: Comparison of Pixel Scale and Diffraction Limit

The 30 cm telescope of Figure 5.4, with a flat secondary, $F = 2\,\text{m}$ and $20\,\mu\text{m}$ pixels, has

$$\theta_{\text{pix}} = 2.06'' \text{ per pixel}, \qquad \theta_R = 0.42'' \text{ at } 500\,\text{nm}.$$

Their ratio is

$$\boxed{\frac{\theta_{\text{pix}}}{\theta_R} = \frac{2.06''}{0.42''} = 4.9.} \tag{5.66}$$

One pixel spans almost five Airy radii. Eq. (5.65) asks for $\theta_{\text{pix}} \leq \lambda/(2D) = 0.17''$, twelve times finer than the pixels give. The same comparison in the focal plane reads more simply. The first dark ring has the radius $1.22\lambda F/D = 4.1\,\mu\text{m}$, and Nyquist sampling would need pixels of $\lambda F/(2D) = 1.7\,\mu\text{m}$. The whole Airy disc falls inside one $20\,\mu\text{m}$ pixel. The image is badly *undersampled*. Its fine detail is aliased, and the position of a star cannot be measured to a fraction of the diffraction blur.

Keeping these pixels, Nyquist sampling at 500 nm would need an effective focal length of $20\,\mu\text{m}/(\lambda/2D) = 24\,\text{m}$, which a convex secondary of magnification 12 could give, Eq. (5.22).

From the ground, the seeing of Section 5.6, about 1″, is the real blur. Two pixels across 1″ need $\theta_{\text{pix}} \leq 0.5''$, so $F_{\text{eff}} \geq 8.3\,\text{m}$. Even then this telescope at $F = 2\,\text{m}$ undersamples by a factor of four.

Sampling regime	Condition	Consequences
Critically sampled	$\theta_{\text{pix}} \approx \lambda/(2D)$	Keeps every spatial frequency without wasting detector area.
Undersampled	$\theta_{\text{pix}} > \lambda/(2D)$	Aliasing, blocky star images, and poorer star positions.
Oversampled	$\theta_{\text{pix}} \ll \lambda/(2D)$	Spreads the light over many pixels, each adding read noise, $\sqrt{n_{\text{pix}}}\,\sigma_{\text{RON}}$ in total (Eq. (7.18)).

Table 5.1: The three sampling regimes of a detector, for a diffraction-limited image.

5.6 Atmospheric Seeing, PSF Profiles, and Adaptive Optics

The optical window lets visible light through the air (Section 5.1), and a telescope resolves $1.22\lambda/D$ when the wavefront reaching it is flat, Eq. (5.53).

From the ground the wavefront is not flat. Transparent air is not still air. On an ordinary night an 8 m telescope shows stars no sharper than a 10 cm one.

Look at a coin at the bottom of a swimming pool while the surface ripples. The coin wobbles and blurs. Each ripple acts as a weak lens and bends the light a little, and the ripples keep moving. Over a hot road in summer the far cars shimmer for the same reason.

The air above a telescope is a pool of warm and cool pockets, each with a slightly different refractive index. The idea is that the sharpness is set by the size r_0 of the patch of air over which the wavefront stays flat to about a radian, and not by the size of the mirror.

At 500 nm with $r_0 = 10\,\text{cm}$ the blur is about $0.98\lambda/r_0 = 1.0''$. An 8 m mirror could reach $1.22\lambda/D = 0.016''$, sixty times sharper, if the air were still.

The picture is a wrinkled wavefront falling on a large mirror. Over any patch of size r_0 it is nearly flat, but neighbouring patches are tilted different ways.

Stars twinkle for the same reason, and so does the heat haze over a road. The ratio D/r_0 decides the regime, P6 again. For $D < r_0$ a telescope is limited by diffraction, and for $D > r_0$ by the air. It gives the seeing angle and how it changes with wavelength, the profile of a star seen through the air, and the way adaptive optics removes the blur.

5.6.1 Atmospheric Seeing and the Fried Parameter

Turbulence mixes air of slightly different temperatures and so of different refractive index. A wavefront that left the star flat reaches the ground with small random extra paths across it. The **Fried parameter** r_0 is the distance across the wavefront over which the phase difference grows to about one radian. Typical values are 10 to 20 cm in the V band at the best mountain sites. Patches closer than r_0 still add in phase as in an undisturbed Airy pattern. Patches farther apart have unrelated phases. So a mirror larger than r_0 acts like about $(D/r_0)^2$ separate small mirrors of size r_0, and the blur is that of diffraction by r_0, not by D.

How r_0 changes with wavelength. Kolmogorov's theory of turbulence, which we quote, gives the mean square difference in extra path between two points a distance r apart as $C\,r^{5/3}$. The constant C measures the strength of the turbulence, and it hardly depends on wavelength,

because the refractive index of air changes little across the optical and infrared. The phase is $k = 2\pi/\lambda$ times the path. Fried wrote the mean square phase difference as $6.88\,(r/r_0)^{5/3}$, with the number chosen so that r_0 is the diameter of a telescope whose diffraction blur matches the seeing. Then

$$6.88\left(\frac{r}{r_0}\right)^{5/3} \overset{(1)}{=} \left(\frac{2\pi}{\lambda}\right)^2 C\,r^{5/3} \quad\Longrightarrow\quad r_0 \overset{(2)}{=} \left(\frac{6.88\,\lambda^2}{4\pi^2 C}\right)^{3/5} \propto \lambda^{6/5}. \tag{5.67}$$

(1) The mean square of k times the path difference.
(2) Cancel $r^{5/3}$ and raise both sides to the power $-3/5$.

For a mirror larger than r_0, the full width at half maximum of the long-exposure star image is, from the same theory,

Result 5.7

$$\theta_{\text{see}} \approx 0.98\frac{\lambda}{r_0} \approx 0.5''\text{–}1.5''. \tag{5.68}$$

It is a width at half maximum, to be compared with the FWHM $1.03\lambda/D$ of the Airy pattern, not with its first dark ring. With $r_0 \propto \lambda^{6/5}$, $\theta_{\text{see}} \propto \lambda^{-1/5}$, so the seeing improves slowly toward the infrared. If $r_0 = 10\,\text{cm}$ at 500 nm, then at 2.2 μm (the K band) $r_0 = 10 \times 4.4^{6/5}\,\text{cm} = 59\,\text{cm}$ and $\theta_{\text{see}} = 0.75''$, against $1.0''$ at 500 nm.

5.6.2 Point Spread Function (PSF) Profiles: The Moffat Function

Seeing turns the Airy pattern into a wider round blob with broad wings. Astronomers fit it with the **Moffat function**,

$$I(\theta) = I_0\left[1 + \left(\frac{\theta}{\theta_{\text{M}}}\right)^2\right]^{-\beta_{\text{M}}}, \tag{5.69}$$

with a core width θ_{M} and a power β_{M}. Fits to real stars give β_{M} between about 2.5 and 4.5. Far from the core the profile falls as the power law $\theta^{-2\beta_{\text{M}}}$, much more slowly than a Gaussian. The width at half maximum follows from setting $I = I_0/2$, at the half width $\theta_{1/2}$,

$$\left(\frac{\theta_{1/2}}{\theta_{\text{M}}}\right)^2 \overset{(1)}{=} 2^{1/\beta_{\text{M}}} - 1,$$

$$\text{FWHM} = 2\theta_{1/2} \overset{(2)}{=} 2\theta_{\text{M}}\sqrt{2^{1/\beta_{\text{M}}} - 1}. \tag{5.70}$$

(1) $[1 + (\theta_{1/2}/\theta_{\text{M}})^2]^{-\beta_{\text{M}}} = \frac{1}{2}$ from Eq. (5.69), raised to the power $-1/\beta_{\text{M}}$, then 1 subtracted.
(2) The square root, and the full width is twice the half width.

As $\beta_{\text{M}} \to \infty$ with $\theta_{\text{M}}^2/\beta_{\text{M}}$ fixed, $(1 + \theta^2/\theta_{\text{M}}^2)^{-\beta_{\text{M}}} \to \exp\!\left(-\beta_{\text{M}}\theta^2/\theta_{\text{M}}^2\right)$, a Gaussian. A finite β_{M} keeps the wings.

5.6.3 Adaptive Optics (AO)

Adaptive optics removes the blur of the air as it happens.

1. A **wavefront sensor**, such as a Shack–Hartmann array of small lenses, measures the tilt of the wavefront over each patch, using a bright natural guide star or a laser guide star. The laser excites sodium atoms in a layer at about 90 km height, at 589 nm, one of the D lines resolved above.

2. A fast computer works out the shape that cancels the measured errors.
3. A **deformable mirror**, with hundreds to thousands of actuators, takes that shape and removes the extra paths.

The numbers follow from r_0. An 8 m mirror with $r_0 = 10$ cm has $(8/0.1)^2 = 6400$ patches, which is why so many actuators are needed, and fewer in the infrared, where r_0 is larger. The wind blows the pattern past at about $10\,\mathrm{m\,s^{-1}}$, so it changes in about $0.31\,r_0/(10\,\mathrm{m\,s^{-1}}) = 3$ ms, and the loop must run at about a kilohertz. With the correction on, 8 to 30 m telescopes recover a core of width close to λ/D from the ground.

5.7 Problem Set: Interferometric Baselines, Optics, and Resolution

5.7.1 Problem 1: GMRT to Ooty Radio Interferometer Baseline and Resolution

Problem Statement: Find the latitude and longitude of the Giant Metrewave Radio Telescope (GMRT) and of the Ooty Radio Telescope from a map.

(a) What is the straight-line distance between them?

(b) If we do very-long-baseline interferometry between these telescopes at 1000 MHz, what is the best resolution we can get?

Essential Ideas:

- Each station is a unit vector, Eq. (5.3). The central angle γ follows from Eq. (5.4), or better from its haversine form, Eq. (5.5).
- The shortest distance in three dimensions is the chord through the Earth, $B = 2R_E \sin(\gamma/2)$, Eq. (5.7).
- The baseline sets the resolution, Eq. (5.16),

$$\theta \simeq \frac{\lambda}{B}. \tag{5.71}$$

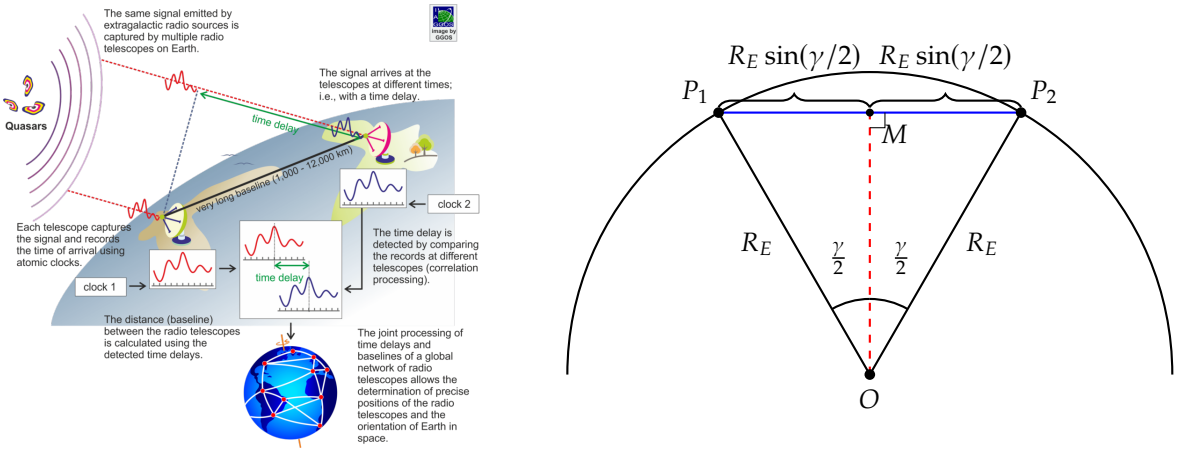


Figure 5.11: Left, the global VLBI network. Right, the chord between two stations P_1 and P_2 on a sphere of radius R_E, with the central angle γ and the midpoint M.

Detailed Solutions:

(a) **Baseline distance**. The coordinates (ϕ, Λ) are about

$$\text{GMRT } (19.09°\text{N},\ 74.05°\text{E}), \qquad \text{Ooty } (11.38°\text{N},\ 76.66°\text{E}).$$

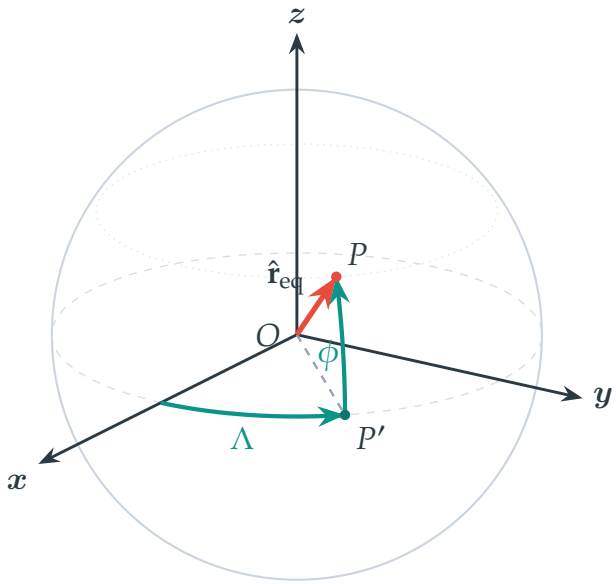


Figure 5.12: A point P on a sphere, its unit vector, its foot P' in the equatorial plane, its latitude ϕ and its longitude (marked λ).

The differences are $\phi_1 - \phi_2 = 7.71^\circ$ and $\Lambda_2 - \Lambda_1 = 2.61^\circ$. With $\cos\phi_1 \cos\phi_2 = 0.94501 \times 0.98034 = 0.92643$, Eq. (5.5) gives

$$\begin{aligned}\sin^2\frac{\gamma}{2} &= \sin^2(3.855^\circ) + 0.92643\,\sin^2(1.305^\circ)\\ &= 0.0045201 + 0.92643 \times 0.00051868 = 0.0050006,\end{aligned}$$

so $\sin(\gamma/2) = 0.070715$ and $\gamma = 8.110^\circ$. By Eq. (5.7) with $R_E = 6371\,\mathrm{km}$,

$$\boxed{B = 2 \times 6371\,\mathrm{km} \times 0.070715 = 901.1\,\mathrm{km}.}$$

The cosine form, Eq. (5.4), gives $\cos\gamma = 0.9899988$. Rounded to 0.9899 it would give $B = 905.5\,\mathrm{km}$, 4 km too long. A cosine near 1 loses the digits that matter, which is why the haversine form is used.

(b) **Angular resolution**. At $\nu = 1000\,\mathrm{MHz} = 10^9\,\mathrm{Hz}$ the wavelength is

$$\lambda = \frac{c}{\nu} = \frac{2.998 \times 10^8\,\mathrm{m\,s^{-1}}}{10^9\,\mathrm{Hz}} = 0.2998\,\mathrm{m}.$$

With $B = 9.011 \times 10^5\,\mathrm{m}$, Eq. (5.71) gives

$$\theta \simeq \frac{0.2998}{9.011 \times 10^5} = 3.327 \times 10^{-7}\,\mathrm{rad}.$$

With $1\,\mathrm{rad} = 2.06265 \times 10^8\,\mathrm{mas}$,

$$\boxed{\theta \simeq 68.6\,\mathrm{mas}.}$$

A single dish would need to be 900 km wide to do as well. The pair records only the sky's ripples at this one baseline, Eq. (5.11), so a real image needs many such pairs.

5.7.2 Problem 2: Cassegrain Reflector Optics, Pixel Scale, and Sampling

Problem Statement: A Cassegrain telescope has a primary mirror of diameter 30 cm and focal length 2 m, and a flat secondary 0.75 m in front of the primary (Figure 5.13).

(a) What are the lengths shown as b and a?

(b) We put a CCD with pixels 20 μm wide at the focal plane. What is the pixel scale?

(c) What is the resolution of this telescope at 500 nm?

(d) Compare the diffraction-limited resolution with the pixel scale. Is this a good configuration?

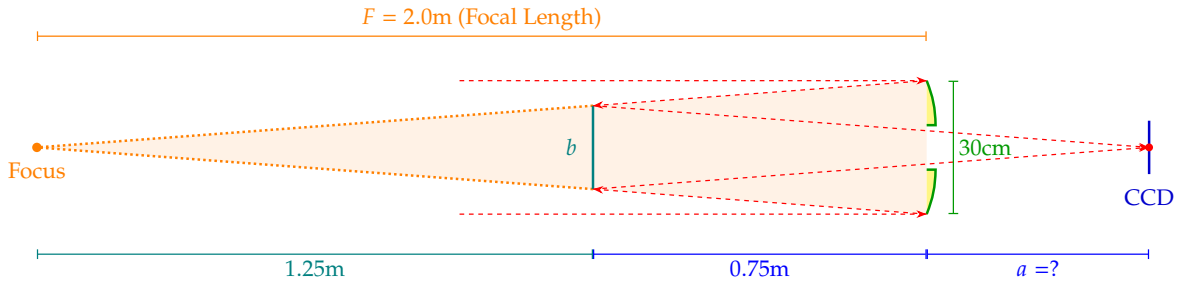


Figure 5.13: The Cassegrain telescope of the problem, with the secondary of diameter b at 0.75 m from the primary and 1.25 m from the primary focus, and the unknown distance a from the primary to the CCD.

Essential Ideas:

- Similar triangles in the converging cone give the size of the secondary, $b = D(1 - d_2/F)$, Eq. (5.20). A flat secondary puts the focus at $a = F - 2d_2$ behind the primary, Eq. (5.21), and leaves the focal length unchanged.
- A pixel of width y_{pix} covers $\theta_{\text{pix}} = y_{\text{pix}}/F$, Eq. (5.27).
- A round mirror resolves $\theta_R = 1.22\lambda/D$, Eq. (5.53), and Nyquist sampling needs $\theta_{\text{pix}} \leq \lambda/(2D)$, Eq. (5.65).

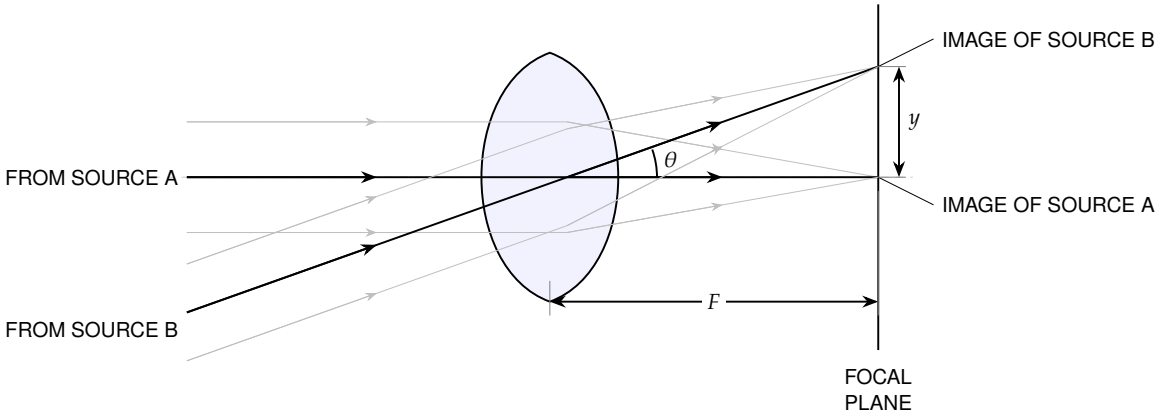


Figure 5.14: Two point sources A and B at the angle θ, a lens of focal length F, and their images a height y apart. The chief rays are dark, and the edge rays faded.

Detailed Solutions:

(a) **The lengths b and a.** With $D = 30\,\text{cm}$, $F = 2.0\,\text{m}$ and $d_2 = 0.75\,\text{m}$, the secondary is $F - d_2 = 1.25\,\text{m}$ from the primary focus. By Eq. (5.20),

$$b = 30\,\text{cm} \times \frac{1.25}{2.0} = 18.75\,\text{cm}.$$

The flat secondary forms the image 1.25 m in front of itself. The primary is 0.75 m away, so by Eq. (5.21)

$$a = 1.25\,\text{m} - 0.75\,\text{m} = 0.50\,\text{m}.$$

$$\boxed{b = 18.75\,\text{cm}, \qquad a = 0.50\,\text{m}.}$$

The secondary blocks $(b/D)^2 = (0.625)^2 = 39\%$ of the mirror's area. A convex secondary closer to the primary focus would block less and lengthen the focal length.

(b) **Pixel scale.** The flat secondary leaves $F = 2\,\text{m}$. With $y_{\text{pix}} = 20 \times 10^{-6}\,\text{m}$, Eq. (5.27) gives

$$\theta_{\text{pix}} = \frac{20 \times 10^{-6}\,\text{m}}{2\,\text{m}} = 1.0 \times 10^{-5}\,\text{rad}.$$

With $1\,\text{rad} = 206\,265''$,

$$\boxed{\theta_{\text{pix}} = 2.06'' \text{ per pixel.}}$$

(c) **Diffraction-limited resolution.**

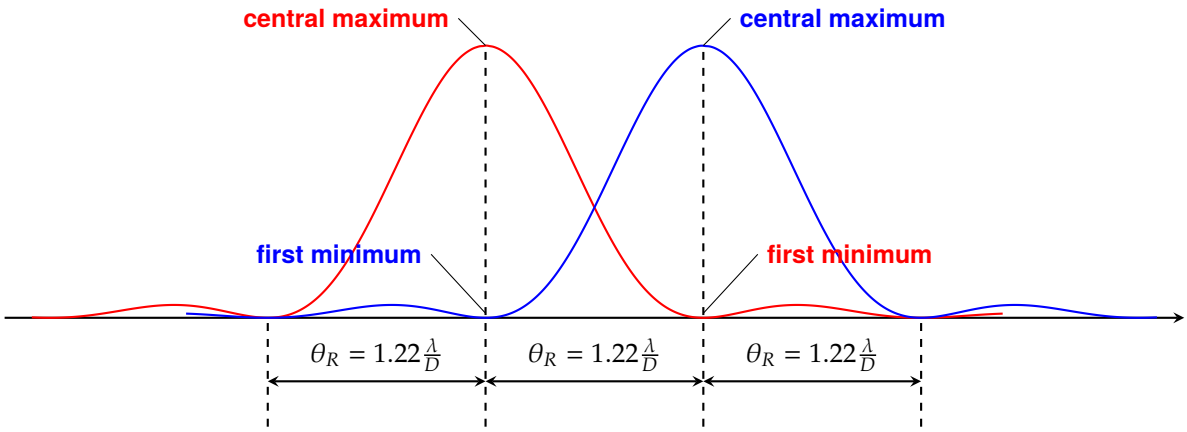


Figure 5.15: Two diffraction patterns whose central maxima are $1.22\lambda/D$ apart.

With $\lambda = 500 \times 10^{-9}\,\text{m}$ and $D = 0.30\,\text{m}$, Eq. (5.53) gives

$$\theta_R = 1.22 \times \frac{500 \times 10^{-9}}{0.30}\,\text{rad} = 2.03 \times 10^{-6}\,\text{rad}, \qquad \boxed{\theta_R = 2.03 \times 10^{-6} \times 206\,265'' = 0.42''.}$$

(d) **Comparison.**

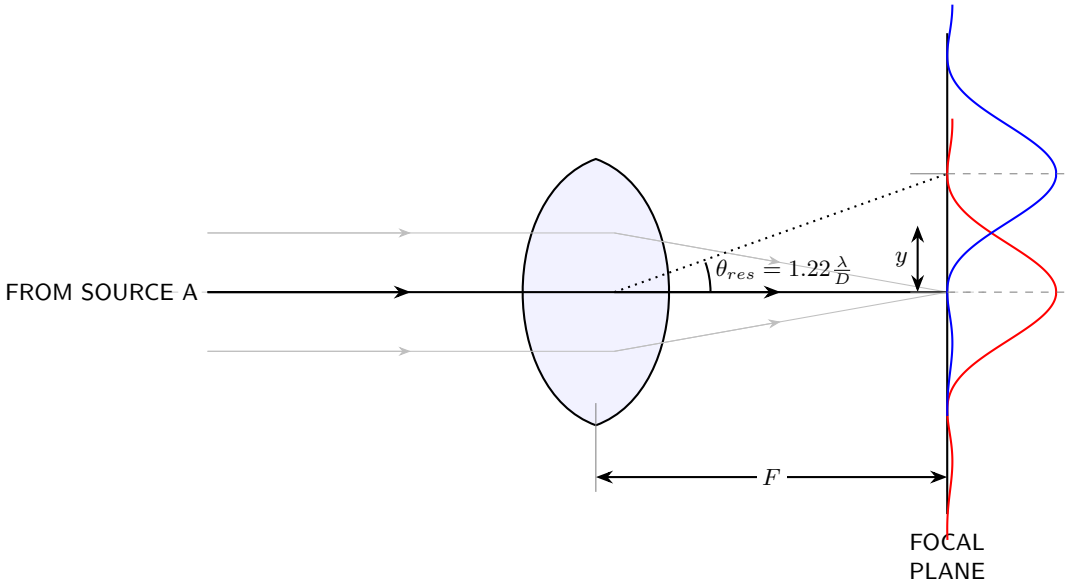


Figure 5.16: The focal-plane Airy profiles of two sources $\theta_{\text{res}} = 1.22\lambda/D$ apart, formed by a lens of focal length F.

The ratio of the pixel scale to the resolution is

$$\frac{\theta_{\text{pix}}}{\theta_R} = \frac{2.06''}{0.42''} = 4.9.$$

Nyquist sampling needs $\theta_{\text{pix}} \le \lambda/(2D)$, Eq. (5.65),

$$\frac{\lambda}{2D} = \frac{500 \times 10^{-9}}{0.60}\ \text{rad} = 8.3 \times 10^{-7}\ \text{rad} = 0.17''.$$

$$\boxed{\theta_{\text{pix}} = 2.06'' \text{ is 12 times larger than the } 0.17'' \text{ needed.}}$$

It is not a good configuration. One pixel covers almost five Airy radii, so the image is badly undersampled, its fine detail is aliased, and star positions are poor. Even the 1″ seeing on the ground would need $\theta_{\text{pix}} \le 0.5''$, four times finer. A convex secondary that lengthens the focal length to about 8 m would match the pixels to the seeing, and 24 m to the diffraction limit (Section 5.5).

Chapter 6

The Two-Body Problem

6.1 Central Force Problem and Conservation Laws

We know how to solve Newton's equations for one body in a given force. A stone thrown across a field, or a mass on a spring, moves in a force that does not care where anything else is.

Two stars are different. Each one pulls the other, so each moves the source of the other's force. Newton's laws then give six coupled second-order equations, and neither star can be followed alone.

Think of two skaters on ice holding the ends of a rope and pulling on it. Seen from the side, both move in complicated loops. But the rope pulls them with equal and opposite forces, so the balance point between them, nearer the heavier skater, never speeds up. And the rope only pulls along itself, so it cannot change how fast the pair turns about that point. Only the length of the rope and the direction it points are left to work out.

A binary star is the pair of skaters, with gravity as the rope. The idea is to split off the motion of the balance point and then use the conserved energy and angular momentum, so that the two-body problem becomes one body moving in a fixed central force.

The skaters in numbers. Let them weigh 60 kg and 40 kg and stand 5 m apart. The balance point is $5 \times 40/100 = 2$ m from the heavier skater and 3 m from the lighter one. Their separation moves like one body of mass $60 \times 40/100 = 24$ kg on the end of the rope.

The picture is a dumbbell whose centre drifts along a straight line while the two weights swing round it.

The same split gives the reduced mass of the electron in the hydrogen atom. It also makes the Earth wobble round a point 4700 km from its centre each month, as the Earth and the Moon turn about their common centre of mass. Conservation of energy and angular momentum is principle P1 (Chapter 9). Treating each star as a point is P2, since outside a sphere its pull is that of a point at the centre. The split and the conserved quantities together fix all twelve constants of the motion.

6.1.1 Central force problem

A central force acts along the line joining two particles and depends only on their separation $r = |\vec{x}_1 - \vec{x}_2|$. Such a force comes from a scalar potential $V(r)$. We write ∇ for the gradient with respect to $\vec{r} = \vec{x}_1 - \vec{x}_2$, so that $\nabla V = V'(r)\,\hat{r}$ with $V' \equiv dV/dr$. The force on particle 1 is $-\nabla V$, and by Newton's third law the force on particle 2 is $+\nabla V$. Newton's equations for the pair are

$$\frac{d^2}{dt^2}\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} -\nabla V(r) \\ \nabla V(r) \end{bmatrix}. \tag{6.1}$$

These are six second-order equations, so the general solution has 12 constants of integration.

6.1.2 Separating centre-of-mass motion

With $M = m_1 + m_2$ and the **reduced mass** $\mu = m_1 m_2/M$, we change to the relative and centre-of-mass coordinates,

$$\begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ m_1/M & m_2/M \end{bmatrix} \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix}, \qquad \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} m_2/M & 1 \\ -m_1/M & 1 \end{bmatrix} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix}.$$

We substitute the second form into (6.1) and multiply both sides from the left by the matrix

$$\begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix},$$

chosen so that it turns the mass matrix diagonal,

$$\begin{aligned} \begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix} \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} m_2/M & 1 \\ -m_1/M & 1 \end{bmatrix} \frac{d^2}{dt^2} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} &= \begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\nabla V \\ \nabla V \end{bmatrix} \\ \begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \mu & m_1 \\ -\mu & m_2 \end{bmatrix} \frac{d^2}{dt^2} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} &\overset{(1)}{=} \begin{bmatrix} -\nabla V \\ 0 \end{bmatrix} \\ \begin{bmatrix} \mu & 0 \\ 0 & M \end{bmatrix} \frac{d^2}{dt^2} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} &\overset{(2)}{=} \begin{bmatrix} -\nabla V \\ 0 \end{bmatrix}. \end{aligned} \tag{6.2}$$

(1) On the left, the mass matrix times the inverse transformation has the entries $m_1 m_2/M = \mu$, m_1, $-\mu$ and m_2. On the right, the first row is $-(m_2 + m_1)\nabla V/M = -\nabla V$ and the second is $-\nabla V + \nabla V = 0$.

(2) The first row of the product is $(\mu m_2 + \mu m_1)/M = \mu$ and $(m_1 m_2 - m_2 m_1)/M = 0$. The second row is $\mu - \mu = 0$ and $m_1 + m_2 = M$.

The mass matrix is now diagonal, and the two rows decouple,

$$\mu\, \ddot{\vec{r}} = -\nabla V(r), \tag{6.3}$$

$$M\, \ddot{\vec{X}} = 0. \tag{6.4}$$

Equation (6.4) integrates at once to $\vec{X}(t) = \vec{V}_{\mathrm{CM}}\, t + \vec{X}_0$. The six constants $\vec{V}_{\mathrm{CM}}, \vec{X}_0$ fix an inertial frame in which the centre of mass is at rest at the origin. All the dynamics is in the *relative* coordinate $\vec{r}$, governed by (6.3). It is a one-body central-force problem with the reduced mass μ.

6.1.3 Conserved quantities

Angular momentum. Define $\vec{L} = \vec{r} \times \vec{p} = \mu\, \vec{r} \times \dot{\vec{r}}$, with $\vec{p} = \mu \dot{\vec{r}}$. Then

$$\begin{aligned} \dot{\vec{L}} &\overset{(1)}{=} \mu \left[\dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}} \right] \\ &\overset{(2)}{=} -\vec{r} \times \nabla V(r) \\ &\overset{(3)}{=} -V'(r)\, r\, \hat{r} \times \hat{r} = 0. \end{aligned} \tag{6.5}$$

(1) The product rule.

(2) $\dot{\vec{r}} \times \dot{\vec{r}} = 0$, and $\mu \ddot{\vec{r}} = -\nabla V$ from (6.3).

(3) $\vec{r} = r\hat{r}$ and $\nabla V = V'(r)\hat{r}$. The force is central.

$\vec{L}$ supplies three constants of motion. They are its magnitude L and two angles fixing its direction, the **inclination** ι and the **longitude of the ascending node** Ω (Section 6.2). Since $\vec{r} \cdot \vec{L} = \mu\, \vec{r} \cdot (\vec{r} \times \dot{\vec{r}}) = 0$, the body stays in the plane through the origin perpendicular to $\vec{L}$.

Energy. $E = K + V$ with $K = \frac{1}{2}\mu\dot{\vec{r}}^2$. The kinetic energy of the centre of mass is a separate constant and drops out. Writing $\dot{r} = d|\vec{r}|/dt$, we have $\hat{r} \cdot \dot{\vec{r}} = \dot{r}$, because $d(\vec{r} \cdot \vec{r})/dt = 2r\dot{r} = 2\vec{r} \cdot \dot{\vec{r}}$. Then

$$\begin{aligned} \dot{E} &\overset{(1)}{=} \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + V'(r)\,\dot{r} \\ &\overset{(2)}{=} \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + \nabla V \cdot \dot{\vec{r}} \\ &= \left(\mu\ddot{\vec{r}} + \nabla V\right) \cdot \dot{\vec{r}} \overset{(3)}{=} 0. \end{aligned} \tag{6.6}$$

(1) The product rule, and the chain rule $dV/dt = V'(r)\,\dot{r}$.
(2) $\nabla V \cdot \dot{\vec{r}} = V'(r)\,\hat{r} \cdot \dot{\vec{r}} = V'(r)\,\dot{r}$.
(3) The equation of motion (6.3).

So E is conserved for *any* central potential $V(r)$, not just gravity. The **total energy and angular momentum** give four constants of integration. There is one more conserved quantity, the Runge–Lenz vector of Section 6.13. It points towards the periapsis and fixes the orientation of the orbit within its plane. The last constant is the time t_0 of a passage through periapsis, which fixes where on the orbit the body is at a given time. The count is

$$\underbrace{\vec{X}_0, \vec{V}_{\mathrm{CM}}}_{6} + \underbrace{a, e, \iota, \Omega, \omega, t_0}_{6} = 12.$$

6.2 Coordinate Systems and Orbital Elements

The angular momentum fixes a plane in space, and the body stays in it.

A telescope does not see that plane face on. It sees the orbit squashed onto the sky, and to compare the two we must know how the orbital plane sits relative to the line of sight.

Take a hoop and tilt it in front of a wall. Seen from the front, the round hoop looks like an ellipse. To say how it is tilted, we name the line on the wall about which it was turned, like a hinge, and the angle of the turn. To say where a mark on the hoop is, we give one more angle measured round the hoop from the hinge.

An orbit is the hoop and the plane of the sky is the wall. The idea is that the orbit's own frame is the sky frame turned twice, first by Ω about the line of sight and then by ι about the hinge, the line of nodes. One more angle, ω, places the ellipse within its plane.

The hoop in numbers. A circle of radius 1 tilted by $\iota = 60^\circ$ about a line through its centre appears as an ellipse with semi-axes 1 and $\cos 60^\circ = 0.5$. Lengths along the hinge stay the same and lengths across it shrink by $\cos\iota$.

The picture is two planes, the orbit and the sky, joined at a hinge along the line of nodes.

The same two turns build the ecliptic and Galactic frames of Chapter 2, and in rigid-body mechanics they are the Euler angles. Only geometry enters here, so none of the six principles is used. The two turns give the matrix from the sky frame to the orbital frame, the orbit as projected on the sky, and the six orbital elements.

The sky frame. Fix an equatorial (reference) frame with the vernal equinox $\hat{\gamma}$ along X and celestial north $\hat{N}$ along Z, and let $\hat{Y}$ complete the right-handed triad. For a source at right ascension α and declination δ, the line of sight from the observer to the source is (Chapter 2, Eq. (2.2))

$$\hat{R} = \cos\delta\cos\alpha\,\hat{\gamma} + \cos\delta\sin\alpha\,\hat{Y} + \sin\delta\,\hat{N}.$$

The local axes in the plane of the sky point along increasing α and δ. They are the normalised derivatives of $\hat{R}$ (Chapter 3, Eq. (3.11)),

$$\hat{\alpha} = \frac{\frac{\partial \hat{R}}{\partial \alpha}}{\left|\frac{\partial \hat{R}}{\partial \alpha}\right|} = -\sin\alpha\,\hat{\gamma} + \cos\alpha\,\hat{Y}, \qquad \hat{\delta} = \frac{\frac{\partial \hat{R}}{\partial \delta}}{\left|\frac{\partial \hat{R}}{\partial \delta}\right|} = -\sin\delta\cos\alpha\,\hat{\gamma} - \sin\delta\sin\alpha\,\hat{Y} + \cos\delta\,\hat{N}.$$

The triad $(\hat{\alpha}, \hat{\delta}, \hat{R})$ is right-handed, since $\hat{\alpha} \times \hat{\delta} = \hat{R}$.

The orbital frame. The orbital plane has its own frame. Its origin is the centre of mass, its X axis lies along the **line of nodes**, where the orbital plane cuts the sky plane, and its Z axis lies along $\vec{L}$. We write $\hat{n}$ for the unit vector towards the **ascending node**, where the body crosses the sky plane moving away from us. The frame is reached by two turns. A turn by Ω about $\hat{R}$ brings the first axis from $\hat{\alpha}$ to $\hat{n}$, and a turn by ι about $\hat{n}$ brings the third axis from $\hat{R}$ to $\hat{L}$. Each turn is a matrix whose rows are the new axes (Chapter 2, Eqs. (2.3), (2.5) and (2.7)), so

$$\begin{pmatrix}\hat{n}\\ \hat{y}\\ \hat{L}\end{pmatrix}_{\text{Orbital Plane}} = \begin{pmatrix}1 & 0 & 0\\ 0 & \cos\iota & \sin\iota\\ 0 & -\sin\iota & \cos\iota\end{pmatrix}\begin{pmatrix}\cos\Omega & \sin\Omega & 0\\ -\sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\hat{\alpha}\\ \hat{\delta}\\ \hat{R}\end{pmatrix}_{\text{Sky plane}}. \tag{6.7}$$

Multiplying out the two matrices,

$$\begin{pmatrix}\hat{n}\\ \hat{y}\\ \hat{L}\end{pmatrix} \overset{(1)}{=} \begin{pmatrix}\cos\Omega & \sin\Omega & 0\\ -\sin\Omega\cos\iota & \cos\Omega\cos\iota & \sin\iota\\ \sin\Omega\sin\iota & -\cos\Omega\sin\iota & \cos\iota\end{pmatrix}\begin{pmatrix}\hat{\alpha}\\ \hat{\delta}\\ \hat{R}\end{pmatrix}. \tag{6.8}$$

(1) Each row of $\mathbf{R}_x(\iota)$ times each column of $\mathbf{R}_z(\Omega)$. The first row of $\mathbf{R}_x$ copies the first row of $\mathbf{R}_z$. The second and third rows mix the second and third rows of $\mathbf{R}_z$ with weights $(\cos\iota, \sin\iota)$ and $(-\sin\iota, \cos\iota)$.

Reading off the rows,

$$\hat{n} = \cos\Omega\,\hat{\alpha} + \sin\Omega\,\hat{\delta}, \tag{6.9}$$

$$\hat{y} = -\sin\Omega\cos\iota\,\hat{\alpha} + \cos\Omega\cos\iota\,\hat{\delta} + \sin\iota\,\hat{R}, \tag{6.10}$$

$$\hat{L} = \sin\Omega\sin\iota\,\hat{\alpha} - \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}. \tag{6.11}$$

The third row gives $\iota = \arccos\left(\hat{L}\cdot\hat{R}\right)$, the angle between the orbital plane and the sky plane. The rows of a rotation are orthonormal, so $(\hat{n}, \hat{y}, \hat{L})$ is a right-handed orthonormal triad. Its first two vectors span the orbital plane and the third is its normal. In the plane we use polar coordinates (r, ϕ), with ϕ measured from $\hat{n}$. The radial and angular unit vectors are $(\hat{n}, \hat{y})$ turned by ϕ,

$$\begin{pmatrix}\hat{r}\\ \hat{\phi}\end{pmatrix} = \begin{pmatrix}\cos\phi & \sin\phi\\ -\sin\phi & \cos\phi\end{pmatrix}\begin{pmatrix}\hat{n}\\ \hat{y}\end{pmatrix}, \qquad \text{that is} \qquad \hat{r} = \cos\phi\,\hat{n} + \sin\phi\,\hat{y}, \quad \hat{\phi} = -\sin\phi\,\hat{n} + \cos\phi\,\hat{y}. \tag{6.12}$$

The projected orbit in the sky plane. The body sits in its plane at the polar angle $\phi = \omega + v$. Here ω is the angle from the node to the periapsis and v, the **true anomaly**, is the angle from the periapsis to the body. Both are found in Section 6.5. The body's orbital components are $(r\cos(\omega+v),\ r\sin(\omega+v),\ 0)$. To get its sky components (X, Y, Z) along $(\hat{\alpha}, \hat{\delta}, \hat{R})$ we undo

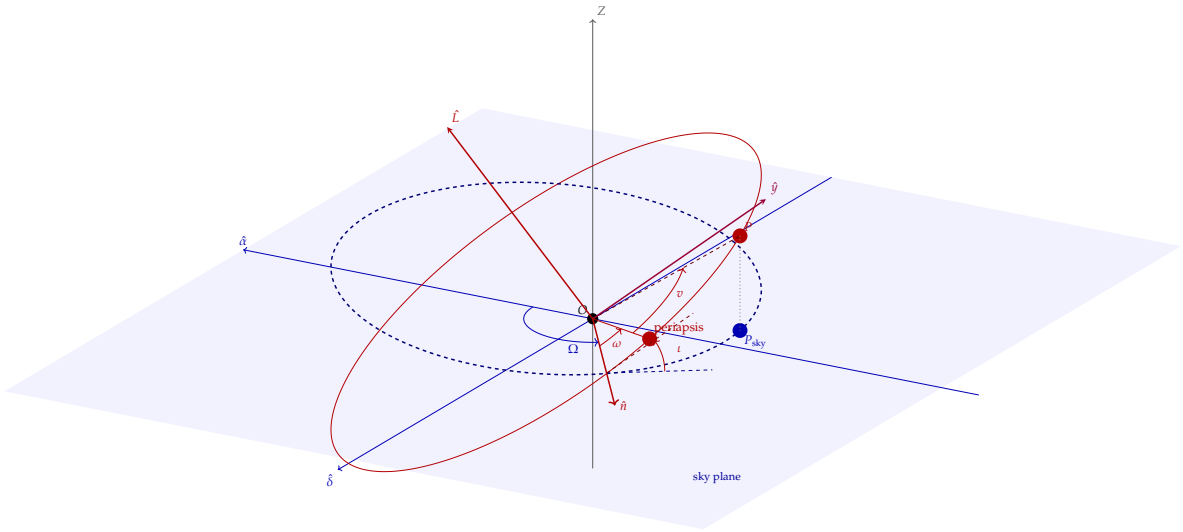


Figure 6.1: An orbit (red) and the plane of the sky (blue), with the sky axes $\hat{\alpha}$, $\hat{\delta}$ and the orbital axes $\hat{n}$, $\hat{y}$, $\hat{L}$. Ω is the angle in the sky plane from $\hat{\alpha}$ to the ascending node. ω is the angle in the orbital plane from the ascending node to the periapsis. ι is the angle between the two planes at the node. v is the angle in the orbital plane from the periapsis to the body P, and $P_{\rm sky}$ is the projection of P on the sky plane.

Eq. (6.7). A rotation is undone by its transpose (Chapter 2, Eq. (2.4)), and a product is undone in reverse order, so

$$\begin{aligned}\begin{pmatrix}X\\Y\\Z\end{pmatrix}_{\text{Sky Plane}} &= \begin{pmatrix}\cos\Omega & -\sin\Omega & 0\\ \sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & \cos\iota & -\sin\iota\\ 0 & \sin\iota & \cos\iota\end{pmatrix}\begin{pmatrix}r\cos(\omega+v)\\ r\sin(\omega+v)\\ 0\end{pmatrix}_{\text{Orbital Plane}}\\ &\overset{(1)}{=}\begin{pmatrix}\cos\Omega & -\sin\Omega & 0\\ \sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}r\cos(\omega+v)\\ r\sin(\omega+v)\cos\iota\\ r\sin(\omega+v)\sin\iota\end{pmatrix}\\ &\overset{(2)}{=} r\begin{pmatrix}\cos(\omega+v)\cos\Omega-\sin(\omega+v)\sin\Omega\cos\iota\\ \cos(\omega+v)\sin\Omega+\sin(\omega+v)\cos\Omega\cos\iota\\ \sin(\omega+v)\sin\iota\end{pmatrix}.\end{aligned} \tag{6.13}$$

(1) $\mathbf{R}_x(\iota)^T$ times the column.
(2) $\mathbf{R}_z(\Omega)^T$ times the column.

X and Y give the position on the sky. Z is the distance along the line of sight, beyond the centre of mass, and it returns in the timing of pulsars (Section 6.18).

Figure 6.2 takes north as the first sky axis and west as the second. This pair is $(\hat{\alpha}, \hat{\delta})$ turned by 90° about $\hat{R}$, so Eq. (6.13) holds unchanged with Ω counted from north. In the same way we may choose the first sky axis along the line of nodes. That sets $\Omega = 0$, and the sky position becomes

$$X = r\cos(\omega + v), \tag{6.14}$$
$$Y = r\sin(\omega + v)\cos\iota. \tag{6.15}$$

Lengths along the line of nodes are unchanged, and every length across it shrinks by the factor $\cos\iota$. A circular orbit therefore appears as an ellipse with axis ratio $\cos\iota$, and it appears as a circle only when $\iota = 0$ (Problem 1).

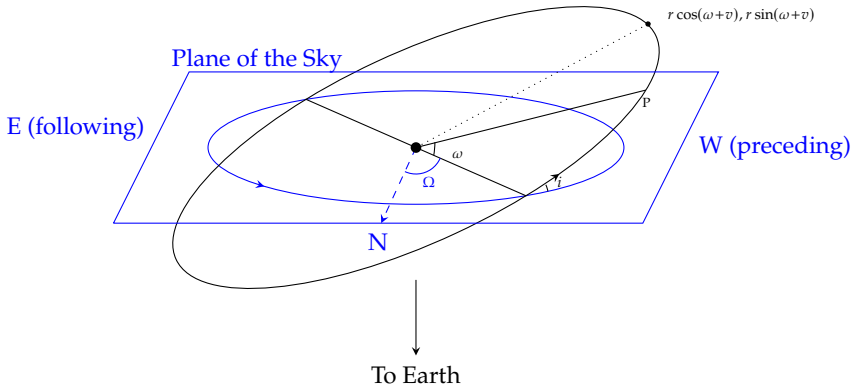


Figure 6.2: An orbit (black) and its projection (blue) on the plane of the sky. N, E and W mark north, east and west on the sky, and the arrow points towards the Earth. Ω is measured in the sky plane from north to the node line, and ω in the orbital plane from the node line to the periapsis. P marks a position $(r\cos(\omega+v), r\sin(\omega+v))$ on the orbit.

The six orbital elements. Six constants appear as the orbit is solved. The angles Ω and ι give the orientation of the plane and come from $\vec{L}$. The semi-major axis a and the eccentricity e give the size and shape and come from E and L. The **argument of periapsis** ω gives the orientation of the ellipse *within* the plane. The time t_0 of periapsis passage fixes where on the ellipse the body is at a given time. Figure 6.1 shows them together.

6.3 Planar Motion and Kepler's Second Law

We have a conserved angular momentum, and with it an orbit that stays in one plane.

Conservation alone does not say how fast the body moves at each point of its path.

A ball slides on a smooth table, tied to a string that passes down through a small hole. Pull the string from below and the ball circles faster as its circle shrinks. The string pulls only towards the hole, so it never pushes the ball sideways. What stays fixed is the distance from the hole times the sideways speed.

A planet does the same. It moves fastest where it is nearest the Sun. The idea is that the area swept by the line from the Sun to the planet grows at a fixed rate, $L/(2\mu)$.

The Earth in numbers. It is 0.983 AU from the Sun at perihelion in January and 1.017 AU at aphelion in July. At both points the velocity is perpendicular to the radius, so the distance times the speed is the same at both. The Earth is therefore $1.017/0.983 = 1.035$ times faster in January, about 3.5 per cent.

The picture is a fan of thin triangles with a common tip at the Sun, long and narrow far away and short and wide nearby, all of the same area.

The same conservation of angular momentum (P1) spins up a skater who pulls in her arms, and spins up a collapsing cloud until it flattens into a disk (Section 10.6). It holds for any central force, not only gravity. It gives Kepler's second law and the single radial equation that the rest of the chapter solves.

Areal velocity. In a time dt the radius vector moves from $\vec{r}$ to $\vec{r} + \dot{\vec{r}}\,dt$ and sweeps a thin triangle. Its area is half the size of the cross product of its two sides,

$$dA = \tfrac{1}{2}\left|\vec{r}\times\dot{\vec{r}}\,dt\right| \overset{(1)}{=} \frac{L}{2\mu}\,dt, \qquad \text{so} \qquad \frac{dA}{dt} = \frac{r^2\dot{\phi}}{2} \overset{(2)}{=} \frac{L}{2\mu} = \text{const.} \tag{6.16}$$

(1) $\vec{L} = \mu\,\vec{r} \times \dot{\vec{r}}$.
(2) In polar coordinates the triangle has sides r and $r\,d\phi$, so $dA = \frac{1}{2}r^2 d\phi$. Then $r^2\dot{\phi} = L/\mu$, as Eq. (6.21) below shows.

This is **Kepler's second law**, equal areas in equal times. It follows only from $\vec{L}$ being conserved, that is, from the force being central. It does not depend on the form of $V(r)$.

Equations of motion in (r, ϕ). With $Z \parallel \vec{L}$ and $X \parallel \hat{n}$, we have $\vec{r} = r\hat{r}$. The unit vectors (6.12) turn with ϕ. Differentiating their matrix,

$$\frac{d}{dt}\begin{pmatrix}\hat{r}\\ \hat{\phi}\end{pmatrix} = \dot{\phi}\begin{pmatrix}-\sin\phi & \cos\phi\\ -\cos\phi & -\sin\phi\end{pmatrix}\begin{pmatrix}\hat{n}\\ \hat{y}\end{pmatrix} = \dot{\phi}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\hat{r}\\ \hat{\phi}\end{pmatrix}, \tag{6.17}$$

so $\dot{\hat{r}} = \dot{\phi}\,\hat{\phi}$ and $\dot{\hat{\phi}} = -\dot{\phi}\,\hat{r}$. The velocity and acceleration follow by the product rule,

$$\begin{aligned}\dot{\vec{r}} &= \dot{r}\,\hat{r} + r\dot{\hat{r}} \overset{(1)}{=} \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi},\\ \ddot{\vec{r}} &= \ddot{r}\,\hat{r} + \dot{r}\dot{\phi}\,\hat{\phi} + \dot{r}\dot{\phi}\,\hat{\phi} + r\ddot{\phi}\,\hat{\phi} + r\dot{\phi}\dot{\hat{\phi}} \overset{(1)}{=} \big(\ddot{r} - r\dot{\phi}^2\big)\hat{r} + \big(2\dot{r}\dot{\phi} + r\ddot{\phi}\big)\hat{\phi}.\end{aligned} \tag{6.18}$$

(1) $\dot{\hat{r}} = \dot{\phi}\,\hat{\phi}$ and $\dot{\hat{\phi}} = -\dot{\phi}\,\hat{r}$, from Eq. (6.17).

The same rates follow from the Christoffel symbols of polar coordinates, $\nabla_\mu \mathbf{e}_\nu = \Gamma^\alpha_{\mu\nu}\mathbf{e}_\alpha$. With $\Gamma^\phi_{\phi r} = 1/r$ and the coordinate basis vector $\mathbf{e}_\phi = r\hat{\phi}$, we get $\dot{\hat{r}} = \dot{\phi}\,\Gamma^\mu_{\phi r}\mathbf{e}_\mu = \dot{\phi}\,\mathbf{e}_\phi/r = \dot{\phi}\,\hat{\phi}$. Equation (6.3), with $\nabla V = V'(r)\hat{r}$, splits into radial and tangential parts,

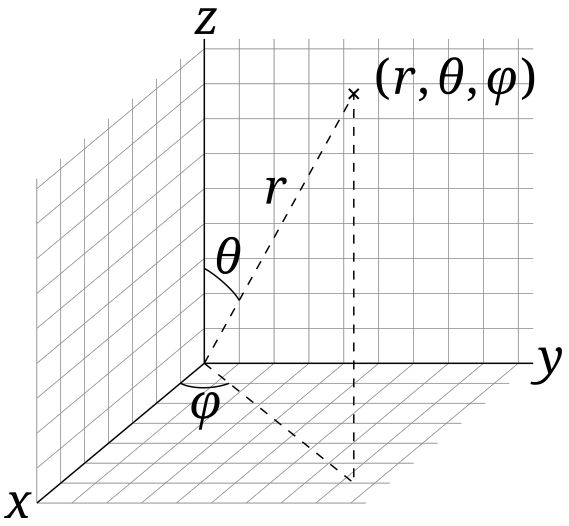


Figure 6.3: Spherical polar coordinates and their unit vectors.

$$\mu\big(\ddot{r} - r\dot{\phi}^2\big) = -V'(r), \tag{6.19}$$

$$\mu\big(2\dot{r}\dot{\phi} + r\ddot{\phi}\big) = 0 \implies \frac{\mathrm{d}(r^2\dot{\phi})}{\mathrm{d}t} = r\big(2\dot{r}\dot{\phi} + r\ddot{\phi}\big) = 0. \tag{6.20}$$

These look like two coupled equations. But Eq. (6.20) is only the conservation of angular momentum, Kepler's second law again, because

$$\vec{L} = \mu\vec{r} \times \dot{\vec{r}} = \mu\, r\hat{r} \times \big(\dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}\big) = \mu r^2\dot{\phi}\,\hat{L}.$$

So

$$\dot{\phi} = \frac{L}{\mu r^2}, \tag{6.21}$$

and Eq. (6.19) becomes, eliminating $\dot{\phi}$,

$$\mu \ddot{r} - \frac{L^2}{\mu r^3} = -V'(r). \tag{6.22}$$

This is one-dimensional motion in r alone. The term $L^2/(\mu r^3)$ is the centrifugal push, and Problem 5 treats it as an effective potential.

6.4 The Orbit Equation

We have one radial equation, (6.22), with time as the variable.

For a closed orbit the first thing we want is its shape, the curve $r(\phi)$. Time only gets in the way.

A hiker's track drawn on a map shows the whole route without saying how fast she walked. Each point of the route is named by where it is, not by when she was there.

The angle ϕ of an orbit grows steadily, since $\dot{\phi} = L/(\mu r^2) > 0$ whenever $L \neq 0$. So ϕ can replace t as the clock. The idea is to change the variable from t to ϕ and to write $\rho = 1/r$. For gravity the equation then becomes linear.

The smallest example is a circle, where ρ is constant and $d^2\rho/d\phi^2 = 0$. For gravity the orbit equation (6.26) below then gives $\rho = GM\mu^2/L^2$, a circle of radius $(L/\mu)^2/GM$. For the Earth, $L/\mu = r\,|\dot{\vec{r}}| = 1.496 \times 10^{11}\,\mathrm{m} \times 29.78\,\mathrm{km\,s^{-1}} = 4.455 \times 10^{15}\,\mathrm{m^2\,s^{-1}}$, and with $GM_\odot = 1.327 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$ the radius is 1.000 AU, as it must be.

The picture is the orbit drawn on paper with no clock marks along it.

The same change of variable, Binet's, gives the bending of light and the turning of Mercury's orbit in general relativity. Here it uses only the conservation of L (P1). It gives an equation for the shape of the orbit in any central potential.

From time to angle. Using Eq. (6.21), $\dfrac{dr}{dt} = \dot{\phi}\dfrac{dr}{d\phi} = \dfrac{L}{\mu r^2}\dfrac{dr}{d\phi}$, and

$$\begin{aligned} \frac{d^2r}{dt^2} &\overset{(1)}{=} \frac{L}{\mu r^2}\frac{d}{d\phi}\left[\frac{L}{\mu r^2}\frac{dr}{d\phi}\right] \\ &\overset{(2)}{=} \frac{L^2}{\mu^2 r^4}\left[\frac{d^2r}{d\phi^2} - \frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right]. \end{aligned} \tag{6.23}$$

(1) $d/dt = \dot{\phi}\, d/d\phi$, applied to dr/dt.

(2) The product rule, with $\dfrac{d}{d\phi}\dfrac{1}{r^2} = -\dfrac{2}{r^3}\dfrac{dr}{d\phi}$.

Substituting into Eq. (6.22) and dividing by L^2/μ,

$$\frac{1}{r^4}\left[\frac{d^2r}{d\phi^2} - \frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right] - \frac{1}{r^3} = -\frac{\mu}{L^2}V'(r). \tag{6.24}$$

Binet's substitution. We set $r = 1/\rho$. Then

$$\frac{dr}{d\phi} = -\frac{1}{\rho^2}\frac{d\rho}{d\phi}, \qquad \frac{d^2r}{d\phi^2} = -\frac{1}{\rho^2}\frac{d^2\rho}{d\phi^2} + \frac{2}{\rho^3}\left(\frac{d\rho}{d\phi}\right)^2,$$

so the left side of Eq. (6.24) is

$$\rho^4\left[-\frac{1}{\rho^2}\frac{d^2\rho}{d\phi^2} + \frac{2}{\rho^3}\left(\frac{d\rho}{d\phi}\right)^2 - 2\rho\cdot\frac{1}{\rho^4}\left(\frac{d\rho}{d\phi}\right)^2\right] - \rho^3 \overset{(1)}{=} -\rho^2\frac{d^2\rho}{d\phi^2} - \rho^3. \tag{6.25}$$

(1) The two $(d\rho/d\phi)^2$ terms cancel.

Setting this equal to $-(\mu/L^2)V'(1/\rho)$ and dividing by $-\rho^2$ gives the **orbit equation**,

$$\boxed{\frac{d^2\rho}{d\phi^2} + \rho = \frac{\mu}{L^2}\frac{1}{\rho^2}V'\left(\frac{1}{\rho}\right).} \tag{6.26}$$

This is an equation for the orbit's shape alone, valid for any central potential.

6.5 Keplerian Orbits

The orbit equation (6.26) holds for any central force.

For gravity we want the actual shape, and we want to know whether the orbit closes on itself.

Hang a weight on a spring. Gravity pulls it down with a constant force, and the weight bobs up and down exactly as it would without gravity. The only change is that it bobs about a lower rest point. A constant force on an oscillator just moves the centre of the oscillation.

For gravity the orbit equation is that spring, with ϕ as the time. The idea is that $\rho = 1/r$ oscillates about a constant value, once per turn of ϕ, so the orbit closes and is a conic section.

The spring in numbers. A mass of 1 kg on a spring of stiffness $100\,\mathrm{N\,m^{-1}}$ hangs 9.8 cm lower than the unstretched spring would put it, and oscillates about that point with the same period, $2\pi\sqrt{1/100}\,\mathrm{s} = 0.63\,\mathrm{s}$, as it would in free space.

The picture is ρ plotted against ϕ, a cosine wave riding on a constant level, with one wave per turn.

Only the $1/r$ potential and the harmonic potential give closed orbits for every bound motion (Bertrand's theorem). In the hydrogen atom the same fact shows up as the equal energies of states with the same principal quantum number. The pull of a spherical star or planet is that of a point at its centre (P2), so the result holds for real stars and planets, not only for points. It gives the conic $r(\phi)$, its axes, and Kepler's first law.

The conic. For Newtonian gravity $V(r) = -GM\mu/r$, so $V'(r) = GM\mu/r^2 = GM\mu\rho^2$. Equation (6.26) becomes

$$\frac{d^2\rho}{d\phi^2} + \rho = \frac{\mu}{L^2}\frac{1}{\rho^2}GM\mu\rho^2 = \frac{GM\mu^2}{L^2}.$$

Shifting $\tilde\rho = \rho - GM\mu^2/L^2$ turns this into the simple-harmonic-oscillator equation $d^2\tilde\rho/d\phi^2 + \tilde\rho = 0$, with solution $\tilde\rho = C\cos(\phi - \omega)$ for constants $C \geq 0$ and ω. In terms of $r = 1/\rho$,

$$r(\phi) = \left(\frac{GM\mu^2}{L^2} + C\cos(\phi-\omega)\right)^{-1} = \frac{L^2/(GM\mu^2)}{1 + \dfrac{L^2C}{GM\mu^2}\cos(\phi-\omega)}.$$

We define the **semi-latus rectum** $\lambda \equiv L^2/(GM\mu^2)$, which is the value of r at $\phi = \omega + 90°$, and the **eccentricity** $e \equiv \lambda C$. Then

$$\boxed{r(\phi) = \frac{\lambda}{1 + e\cos(\phi - \omega)}.} \tag{6.27}$$

This is the polar equation of a conic section with one focus at the origin, semi-latus rectum λ and eccentricity e. Its major axis is turned by ω, the **argument of periapsis**, and the periapsis is at $\phi = \omega$. The angle from the periapsis, $v \equiv \phi - \omega$, is the **true anomaly**.

The axes. For $e < 1$ the orbit is closed. Its nearest point, the periapsis, is at $r_\mathrm{p} = \lambda/(1+e)$, and its farthest, the apoapsis, is at $r_\mathrm{a} = \lambda/(1-e)$. The major axis is their sum, so the **semi-major axis** is

$$a = \frac{r_\mathrm{p} + r_\mathrm{a}}{2} = \frac{\lambda}{2}\,\frac{(1-e)+(1+e)}{1-e^2} = \frac{\lambda}{1-e^2}.$$

The centre lies midway between the two ends, at $a - r_\mathrm{p} = a - a(1-e) = ae$ from the focus. To

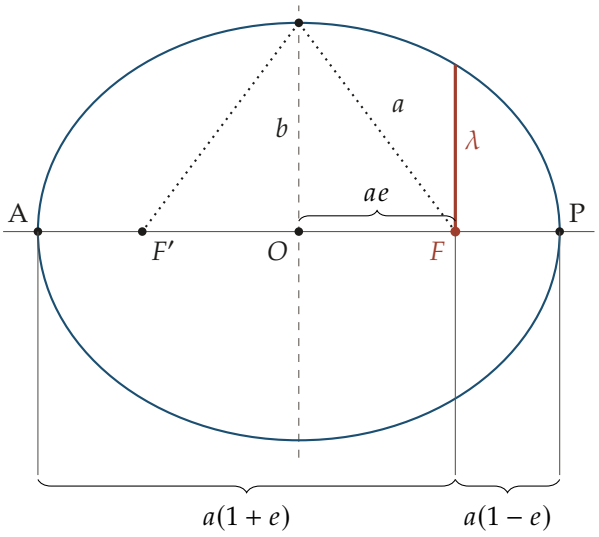


Figure 6.4: An ellipse with semi-axes a and b, centre O and foci F and F', each at distance ae from O. The periapsis P and the apoapsis A lie at distances $a(1-e)$ and $a(1+e)$ from F. The dotted lines join the end of the minor axis to the two foci. The semi-latus rectum λ is drawn at F, perpendicular to the major axis.

see that the conic is the ellipse $x^2/a^2 + y^2/b^2 = 1$ with **semi-minor axis** $b = a\sqrt{1-e^2}$, we put the origin of (x, y) at the centre and the periapsis on the $+x$ axis (taking $\omega = 0$). The focus is then at $(ae, 0)$, and a point at (r, ϕ) about the focus has

$$x = ae + r\cos\phi, \tag{6.28}$$

$$y = r\sin\phi. \tag{6.29}$$

Substituting into the ellipse with $b^2 = a^2(1-e^2)$,

$$\begin{aligned} 1 &= \frac{(ae + r\cos\phi)^2}{a^2} + \frac{r^2\sin^2\phi}{a^2(1-e^2)} \\ 0 &\overset{(1)}{=} r^2(1 - e^2\cos^2\phi) + 2ae(1-e^2)\,r\cos\phi - a^2(1-e^2)^2 \\ r &\overset{(2)}{=} \frac{a(1-e^2)(1-e\cos\phi)}{(1-e\cos\phi)(1+e\cos\phi)} = \frac{a(1-e^2)}{1+e\cos\phi}. \end{aligned} \tag{6.30}$$

(1) Multiply by $a^2(1-e^2)$, expand, and use $\sin^2\phi = 1-\cos^2\phi$ and $(1-e^2)a^2e^2 - a^2(1-e^2) = -a^2(1-e^2)^2$.

(2) The quadratic formula. The discriminant is $4a^2(1-e^2)^2\left[e^2\cos^2\phi + 1 - e^2\cos^2\phi\right] = 4a^2(1-e^2)^2$, and the root with $r > 0$ is kept. Then $1 - e^2\cos^2\phi$ is factored.

This is the conic (6.27) with $\lambda = a(1-e^2)$, so the orbit is this ellipse. The point $(0, b)$ at the end of the minor axis is at distance $\sqrt{b^2 + a^2e^2} = a$ from each focus (Figure 6.4). At $x = ae$ the ellipse has $y = b\sqrt{1-e^2} = b^2/a = \lambda$, which is the semi-latus rectum again.

Eccentricity	Conic section
$e = 0$	circle
$0 < e < 1$	ellipse ($E < 0$, bound)
$e = 1$	parabola ($E = 0$)
$e > 1$	hyperbola ($E > 0$, unbound)

The sign of E in each row is shown in Section 6.6. Bound, closed elliptical orbits ($E < 0$) are **Kepler's first law**. In the centre-of-mass frame, $\vec{X} = 0$, the inverse transformation of Section 6.1.2 gives the two bodies' positions,

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}, \qquad \vec{x}_2 = -\frac{m_1}{M}\vec{r}.$$

6.6 Energy and Angular Momentum in Terms of a, e

We have the shape of the orbit, with constants λ and e, and two conserved numbers, E and L.

The shape constants came out of solving a differential equation. We do not yet know how they depend on E and L.

A child on a swing stops for a moment at the top of each swing. There all of the energy is height, so the height of the swing tells us the energy without any need to know the speed elsewhere.

An orbit has two such turning points of r, the periapsis and the apoapsis, where $\dot{r} = 0$. The idea is to evaluate the conserved E at a turning point. The size a then depends on E alone, and e on E and L.

The swing in numbers. A child of 20 kg who rises 0.5 m has $E = 20 \times 9.8 \times 0.5 = 98$ J more than at rest, and passes the bottom at $\sqrt{2 \times 9.8 \times 0.5} = 3.1\,\mathrm{m\,s^{-1}}$.

The picture is the two ends of the ellipse, where the body moves straight across the radius.

The same turning points are the roots of the effective potential in Problem 5. In the hydrogen atom the energy depends only on the principal quantum number, the quantum form of E depending only on a. The whole calculation is P1. It gives E, L and the speed in terms of a and e, and the inverse.

6.6.1 Energy

Writing the energy in polar coordinates,

$$\begin{aligned} E &= \frac{1}{2}\mu\dot{\vec{r}}^2 - \frac{GM\mu}{r} \\ &\overset{(1)}{=} \frac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{GM\mu}{r} \\ &\overset{(2)}{=} \frac{1}{2}\mu\dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{GM\mu}{r}. \end{aligned} \tag{6.31}$$

(1) $\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}$ with $\hat{r} \perp \hat{\phi}$.
(2) $\dot{\phi} = L/(\mu r^2)$, Eq. (6.21).

Since E is conserved, we evaluate it at periapsis ($\phi = \omega$). There $r = \lambda/(1+e)$, and $\dot{r} = 0$ because r has a minimum,

$$\begin{aligned} E &\overset{(1)}{=} \frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} \\ &\overset{(2)}{=} \frac{GM\mu(1+e)^2}{2\lambda} - \frac{GM\mu(1+e)}{\lambda} \\ &\overset{(3)}{=} -\frac{GM\mu}{2\lambda}(1-e^2) \overset{(4)}{=} -\frac{GM\mu}{2a}. \end{aligned} \tag{6.32}$$

(1) Eq. (6.31) with $\dot{r} = 0$ and $r = \lambda/(1+e)$.
(2) $L^2 = GM\mu^2\lambda$, the definition of λ.
(3) $(1+e)^2 - 2(1+e) = (1+e)(e-1) = -(1-e^2)$.
(4) $a = \lambda/(1-e^2)$.

$$\boxed{E = -\frac{GM\mu}{2a}.} \tag{6.33}$$

The energy depends on the size of the orbit only, not on its shape. Since $\lambda > 0$ always, the sign of E follows e. Bound orbits ($E < 0$) have $e < 1$ and $a > 0$. Unbound orbits ($E \geq 0$) have $e \geq 1$, and hyperbolic orbits formally have $a < 0$.

6.6.2 Angular momentum

At the apoapsis, $r = a + ae = \lambda/(1-e)$, and at the periapsis, $r = a - ae = \lambda/(1+e)$, we have $\dot{r} = 0$. Equating the energy at the two points as a check,

$$\begin{aligned} \frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} &= \frac{L^2(1-e)^2}{2\mu\lambda^2} - \frac{GM\mu(1-e)}{\lambda} \\ \frac{L^2\big[(1+e)^2 - (1-e)^2\big]}{2\mu\lambda^2} &\overset{(1)}{=} \frac{GM\mu\big[(1+e) - (1-e)\big]}{\lambda} \\ \frac{2eL^2}{\mu\lambda^2} &\overset{(2)}{=} \frac{2eGM\mu}{\lambda}. \end{aligned} \tag{6.34}$$

(1) Collect the L^2 terms on the left and the $GM\mu$ terms on the right.
(2) $(1+e)^2 - (1-e)^2 = 4e$.

So $L^2 = GM\mu^2\lambda$, the definition of λ again. The two turning points agree, as they must. With $\lambda = a(1-e^2)$ the angular momentum in terms of a and e is

$$L = \sqrt{GM\mu^2\lambda}, \qquad \text{i.e.} \qquad L^2 = GM\mu^2 a(1-e^2), \tag{6.35}$$

and, with the direction $\hat{L}$ from Eq. (6.11),

$$\vec{L} = \sqrt{GM\mu^2\lambda}\,\big(\sin\Omega\sin\iota\,\hat{\alpha} - \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}\big).$$

6.6.3 Instantaneous kinetic energy

From $E = K + V$ and Eq. (6.33),

$$K = \tfrac{1}{2}\mu\dot{\vec{r}}^2 = E - V = GM\mu\left(\frac{1}{r} - \frac{1}{2a}\right), \qquad \text{or} \qquad |\dot{\vec{r}}|^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right). \tag{6.36}$$

This is the *vis-viva* equation. The speed at any point depends only on the distance and on a. The kinetic energy is shared between the two bodies through $\vec{x}_1 = (m_2/M)\vec{r}$ and $\vec{x}_2 = -(m_1/M)\vec{r}$,

$$K_1 = \tfrac{1}{2}m_1|\dot{\vec{x}}_1|^2 = \tfrac{1}{2}m_1\frac{m_2^2}{M^2}\dot{\vec{r}}^2 = \frac{m_2}{M}\cdot\tfrac{1}{2}\mu\dot{\vec{r}}^2 = \frac{m_2}{M}K, \qquad K_2 = \frac{m_1}{M}K,$$

using $\mu = m_1m_2/M$. The lighter body carries the larger share.

Inverting for a, e. Equations (6.33) and (6.35) give the orbital elements directly from the conserved energy and angular momentum. The first gives $a = -GM\mu/(2E)$. Then

$$e^2 = 1 - \frac{L^2}{GM\mu^2 a} \overset{(1)}{=} 1 + \frac{2EL^2}{(GM)^2\mu^3}, \tag{6.37}$$

(1) $1/a = -2E/(GM\mu)$.

so

$$a = -\frac{GM\mu}{2E}, \qquad e = \sqrt{1 + \frac{2EL^2}{(GM)^2\mu^3}}. \tag{6.38}$$

6.7 True and Eccentric Anomaly; Kepler's Equation

We have the shape $r(\phi)$ and the size of the orbit from its energy.

The shape alone does not say where the body is at a given time, and that is what an observer needs to point a telescope or predict an eclipse.

Draw a clock face on a rubber sheet and stretch the sheet in one direction. The hour marks move, and the angles measured on the stretched sheet are no longer 30° apart. But each mark can still be named by its old angle on the unstretched face, and with that name the arithmetic stays simple.

An ellipse is a circle of radius a squashed by the factor b/a across its major axis. A point on the ellipse at $(a\cos u,\ b\sin u)$ from the centre is named by the angle u on that circle, the **eccentric anomaly**. The idea is that with $r = a(1 - e\cos u)$ the time integral of the orbit becomes elementary.

The ellipse in numbers. For $e = 0.5$ and $u = 90°$, the body is at $r = a(1-0) = a$. Its true anomaly follows from Eq. (6.43) below, $\cos v = (0 - 0.5)/(1 - 0) = -0.5$, so $v = 120°$. Seen from the focus the body is 120° round from periapsis, while on the circle it is 90° round.

The picture is the ellipse inside its circumscribed circle, with a line across the major axis joining the body to the circle and the angle u drawn at the centre.

The same parameter $x = a\cos u$, $y = b\sin u$ describes any ellipse in geometry, and in calculus the same substitution removes the square root of a quadratic from an integral. For a hyperbola cosh and sinh take the place of cos and sin. The integrand comes from E and L (P1). The result is the time $t(u)$, the true anomaly $v(u)$, and the same for hyperbolic orbits.

The time integral. We now want $r(t)$, and hence $\phi(t)$, not just $r(\phi)$. Solving the energy (6.31) for $\dot{r}$,

$$\dot{r} \overset{(1)}{=} \sqrt{\frac{2}{\mu}\left(E - \frac{L^2}{2\mu r^2} + \frac{GM\mu}{r}\right)}$$
$$\overset{(2)}{=} \sqrt{-\frac{GM}{a} - \frac{GM\lambda}{r^2} + \frac{2GM}{r}} \overset{(3)}{=} \sqrt{GM}\,\frac{1}{r}\left(2r - \frac{r^2}{a} - \lambda\right)^{1/2}. \tag{6.39}$$

(1) Eq. (6.31), on the outbound half of the orbit where $\dot r > 0$.
(2) $E = -GM\mu/(2a)$ and $L^2 = GM\mu^2\lambda$, Eqs. (6.33) and (6.35).
(3) Take GM/r^2 out of the square root.

We separate the variables and integrate from the periapsis, which the body passes at the time t_0 with $r = a(1-e)$,

$$\begin{aligned} t - t_0 &= \frac{1}{\sqrt{GM}} \int_{a(1-e)}^{r} \frac{r'\,dr'}{(2r' - r'^2/a - \lambda)^{1/2}} \\ &\overset{(1)}{=} \sqrt{\frac{a}{GM}} \int_{a(1-e)}^{r} \frac{r'dr'}{\sqrt{a^2e^2 - (r'-a)^2}}. \end{aligned} \tag{6.40}$$

(1) Multiply the bracket by a. With $\lambda = a(1-e^2)$, $2ar - r^2 - a^2(1-e^2) = a^2e^2 - (r-a)^2$.

6.7.1 Elliptical orbits: eccentric anomaly

For $a > 0$ we substitute

$$r = a\big(1 - e\cos u\big), \tag{6.41}$$

where u is the **eccentric anomaly**. This choice keeps $r > 0$ throughout, since $-1 \le \cos u \le 1$, and it makes the square root a perfect square,

$$\begin{aligned} t - t_0 &\overset{(1)}{=} \sqrt{\frac{a}{GM}} \int_0^u \frac{a(1 - e\cos u')\, ae\sin u'\, du'}{ae\sin u'} \\ &\overset{(2)}{=} \frac{a^{3/2}}{\sqrt{GM}} \int_0^u (1 - e\cos u')\, du' = \frac{a^{3/2}}{\sqrt{GM}} \big[u - e\sin u\big]. \end{aligned} \tag{6.42}$$

(1) $dr = ae\sin u\, du$, and $a^2e^2 - (r-a)^2 = a^2e^2 - a^2e^2\cos^2 u = a^2e^2\sin^2 u$, whose root is $ae\sin u$ for $0 < u < \pi$. The periapsis $r = a(1-e)$ is $u = 0$.
(2) Cancel $ae\sin u'$.

On the inbound half, $\pi < u < 2\pi$, both $\dot r$ and $\sin u$ change sign, so the same formula holds for all u. The time t_0 of periapsis passage, where $u = 0$, is the sixth and final constant of integration.

Relating the true anomaly v and the eccentric anomaly u. The true anomaly $v = \phi - \omega$ gives $r = \lambda/(1 + e\cos v)$ from Eq. (6.27). Equating this with Eq. (6.41),

$$\begin{aligned} 1 + e\cos v &\overset{(1)}{=} \frac{1 - e^2}{1 - e\cos u} \\ e\cos v &= \frac{1 - e^2 - 1 + e\cos u}{1 - e\cos u} = \frac{e(\cos u - e)}{1 - e\cos u}, \qquad \text{so} \qquad \cos v = \frac{\cos u - e}{1 - e\cos u}. \end{aligned} \tag{6.43}$$

(1) $a(1 - e\cos u) = a(1-e^2)/(1 + e\cos v)$, with $\lambda = a(1-e^2)$.

The sine follows from $\sin^2 v = 1 - \cos^2 v$,

$$\sin^2 v = \frac{(1 - e\cos u)^2 - (\cos u - e)^2}{(1 - e\cos u)^2} \overset{(1)}{=} \frac{(1-e^2)(1 - \cos^2 u)}{(1 - e\cos u)^2}, \tag{6.44}$$

(1) The numerator is $1 - 2e\cos u + e^2\cos^2 u - \cos^2 u + 2e\cos u - e^2 = (1-e^2) - (1-e^2)\cos^2 u$.

and v and u lie in the same half of the orbit, so the positive root is taken where $\sin u > 0$. Together,

$$\sin v = \frac{\sqrt{1-e^2}\sin u}{1 - e\cos u}, \qquad \tan v = \frac{\sqrt{1-e^2}\sin u}{\cos u - e}, \qquad \tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{u}{2}. \tag{6.45}$$

The half-angle form is proved in Problem 4. At $u = 0$ and $u = \pi$ we get $v = u$. The periapsis and the apoapsis are the same points for the two anomalies, as they must be by symmetry. Multiplying $\cos v$ and $\sin v$ by $r = a(1 - e\cos u)$ gives the position in the orbital plane, with x towards the periapsis,

$$r\cos v = a(\cos u - e), \qquad r\sin v = a\sqrt{1-e^2}\,\sin u. \tag{6.46}$$

This is the point $(a\cos u,\ b\sin u)$ of the opening, measured from the focus at $(ae, 0)$.

6.7.2 Hyperbolic trajectories

For $a < 0$ and $e > 1$, the matching substitution is $r = a(1 - e\cosh u) = (-a)(e\cosh u - 1)$. Since $\cosh u \geq 1$ and $a < 0$, it keeps $r > 0$. Multiplying the bracket of the time integral by $-a > 0$ instead of a,

$$\begin{aligned} t - t_0 &\overset{(1)}{=} \sqrt{\frac{-a}{GM}} \int_{-a(e-1)}^{r} \frac{r'\,dr'}{\sqrt{(r'-a)^2 - a^2e^2}} \\ &\overset{(2)}{=} \frac{(-a)^{3/2}}{\sqrt{GM}} \int_0^u (e\cosh u' - 1)\,du' = \frac{(-a)^{3/2}}{\sqrt{GM}}\big[e\sinh u - u\big]. \end{aligned} \tag{6.47}$$

(1) $-a\big(2r - r^2/a - \lambda\big) = (r-a)^2 - a^2e^2$, with $\lambda = a(1-e^2)$. The periapsis is at $r = -a(e-1)$.

(2) $dr = (-a)e\sinh u\,du$, and $(r-a)^2 - a^2e^2 = a^2e^2(\cosh^2 u - 1) = a^2e^2\sinh^2 u$, whose root is $(-a)e\sinh u$ for $u > 0$.

This is the **hyperbolic** anomaly relation. The true anomaly follows as for the ellipse, with $\cosh u$ in place of $\cos u$. Equating $\lambda/(1 + e\cos v)$ with $a(1 - e\cosh u)$ gives $\cos v = (\cosh u - e)/(1 - e\cosh u)$. Then $1 - \cos^2 v = (e^2-1)\sinh^2 u/(e\cosh u - 1)^2$, and the half-angle form follows from

$$\frac{1-\cos v}{1+\cos v} = \frac{(e+1)(\cosh u - 1)}{(e-1)(\cosh u + 1)} = \frac{e+1}{e-1}\tanh^2\frac{u}{2}.$$

The true-anomaly relations are

$$\cos v = \frac{e - \cosh u}{e\cosh u - 1}, \qquad \sin v = \frac{\sqrt{e^2-1}\,\sinh u}{e\cosh u - 1}, \qquad \tan\frac{v}{2} = \sqrt{\frac{e+1}{e-1}}\tanh\frac{u}{2}. \tag{6.48}$$

6.8 Kepler's Third Law

We have the time $t(u)$ along an elliptical orbit, Eq. (6.42).

What we usually measure first for a binary or a planet is the period, the one time scale of the orbit.

A pendulum clock is set by the length of its pendulum alone. A longer pendulum swings more slowly, in a fixed way, whatever the size of its swing.

An orbit is set the same way by its semi-major axis alone. The idea is that one full turn of u is one period, and the eccentricity drops out.

The Earth in numbers. With $a = 1$ AU and $GM_\odot = 1.327 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$, the period $2\pi\sqrt{a^3/GM}$ below is 365.26 days. Mars, at $a = 1.524$ AU, takes $1.524^{3/2} = 1.881$ years.

The picture is two ellipses with the same major axis, one round and one thin, that are run round in the same time.

The same law weighs the Earth from the Moon's orbit, the stars of a binary (Problem 9) and the dark halo of a galaxy (Problem 10). With zero angular momentum it gives the free-fall time of a collapsing cloud (Section 10.5). It combines P1 and P2. It gives the mean motion n and $n^2a^3 = GM$.

The period. In one complete orbit u advances by 2π, while t advances by the orbital period P. We define the **mean motion** $n \equiv 2\pi/P$, the mean angular rate. From Eq. (6.42),

$$\frac{2\pi}{n} = \frac{a^{3/2}}{\sqrt{GM}}\big[2\pi - e\sin(2\pi)\big] = \frac{a^{3/2}}{\sqrt{GM}}\cdot 2\pi \quad\Longrightarrow\quad \boxed{n = \sqrt{\frac{GM}{a^3}}, \qquad n^2a^3 = GM.} \tag{6.49}$$

The period depends on a and the total mass M only. It does not depend on e or on how M is split between the two bodies.

6.9 Kepler's Equation

We have t as a function of u, and the mean motion n.

An observer has the time t and needs the position, so the relation must be run the other way.

Two runners start together on a circular track. One keeps a steady pace. The other speeds up on one side of the track and slows down on the other, and both finish the lap together. The steady runner's position is easy to give at any time, and the other's position can be named by how far ahead or behind the steady one she is.

For an orbit the steady runner is a fictitious body moving round at the mean rate n, like the mean Sun of Chapter 1. Its angle is the **mean anomaly** $\ell = n(t - t_0)$. The idea is that Kepler's equation ties ℓ to the real body's eccentric anomaly u.

The runners in numbers. For $e = 0.5$, a quarter of a period after periapsis the mean anomaly is $\ell = 90°$. Solving $u - 0.5\sin u = \pi/2$ gives $u = 115.8°$, and Eq. (6.43) gives $v = 140.2°$. The real body is 50° ahead of the mean one.

The picture is the steady runner and the real one on the same track, the real one ahead in the first half of the lap and behind in the second.

The same mean body gives the equation of time in Chapter 1. Nothing new enters beyond P1, through Kepler's second law. It gives Kepler's equation for ellipses and hyperbolas.

Mean anomaly. Defining the **mean anomaly** $\ell \equiv n(t - t_0)$,

$$\ell \overset{(1)}{=} n\,\frac{a^{3/2}}{\sqrt{GM}}\,(u - e\sin u) \overset{(2)}{=} u - e\sin u, \tag{6.50}$$

(1) Eq. (6.42).
(2) $n = \sqrt{GM/a^3}$, Eq. (6.49).

so that

$$\boxed{\ell = u - e\sin u} \qquad \text{(elliptical Kepler's equation).} \tag{6.51}$$

This transcendental equation must be solved for $u(\ell)$, that is $u(t)$, numerically (Problem 3) or by a series (Section 6.12). The hyperbolic analogue, with $n \equiv \sqrt{GM/(-a)^3}$, is

$$\ell = e\sinh u - u. \tag{6.52}$$

6.10 Keplerian Parametrization

With Kepler's equation the position follows at any time. We collect the formulas, then add the rates $\dot{u}$, $\dot{r}$ and $\dot{\phi}$ that the perturbation theory and the timing of binaries need.

6.10.1 Elliptical orbits

Collecting Eqs. (6.41), (6.45) and (6.51),

$$r = a(1 - e\cos u), \qquad \phi = v + \omega, \tag{6.53}$$

$$\tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{u}{2}, \qquad \ell = u - e\sin u. \tag{6.54}$$

Differentiating Kepler's equation (6.51) in t gives $n = \dot{u}(1 - e\cos u)$, since $\dot{\ell} = n$. So

$$\dot{u} = \frac{n}{1 - e\cos u}. \tag{6.55}$$

Differentiating $r = a(1 - e\cos u)$,

$$\dot{r} = ae\sin u\,\dot{u} = \frac{an\,e\sin u}{1 - e\cos u}. \tag{6.56}$$

For $\dot{\phi}$ we need dv/du. Differentiating $\cos v$ from Eq. (6.43),

$$\begin{aligned} -\sin v\,\frac{dv}{du} &\overset{(1)}{=} \frac{-\sin u\,(1 - e\cos u) - (\cos u - e)\,e\sin u}{(1 - e\cos u)^2} = -\frac{(1-e^2)\sin u}{(1 - e\cos u)^2}, \\ \frac{dv}{du} &\overset{(2)}{=} \frac{(1-e^2)\sin u}{(1 - e\cos u)^2}\cdot\frac{1 - e\cos u}{\sqrt{1-e^2}\sin u} = \frac{\sqrt{1-e^2}}{1 - e\cos u}. \end{aligned} \tag{6.57}$$

(1) The quotient rule.
(2) Divide by $-\sin v$, with $\sin v$ from Eq. (6.45).

So

$$\dot{\phi} = \dot{v} = \frac{dv}{du}\dot{u} = \frac{n\sqrt{1-e^2}}{(1 - e\cos u)^2}. \tag{6.58}$$

As a check, Eq. (6.21) gives $\dot{\phi} = L/(\mu r^2) = \sqrt{GMa(1-e^2)}/\left[a^2(1 - e\cos u)^2\right]$, which is the same since $\sqrt{GM/a^3} = n$.

6.10.2 Hyperbolic trajectories

In the same way, with $r = -a(e\cosh u - 1)$, $\phi = v + \omega$, $\tan(v/2) = \sqrt{(e+1)/(e-1)}\tanh(u/2)$ and $\ell = e\sinh u - u$,

$$\dot{u} = \frac{n}{e\cosh u - 1}, \quad \dot{r} = -an\,\frac{e\sinh u}{e\cosh u - 1}, \quad \dot{\phi} = \dot{v} = \frac{n\sqrt{e^2-1}}{(e\cosh u - 1)^2}. \tag{6.59}$$

The first is the derivative of $\ell = e\sinh u - u$, the second is $\dot{r} = (-a)e\sinh u\,\dot{u}$, and the third uses $dv/du = \sqrt{e^2-1}/(e\cosh u - 1)$, found from $\cos v$ as for the ellipse.

6.11 Hyperbolic-Trajectory Parameters

A body that passes a planet or a star once, on an unbound orbit with $e > 1$, is described by a few numbers read off the hyperbola. We find the speed far away, the angle of the asymptotes, the closest approach, the miss distance and the angle of the velocity. Problem 12 uses them for a flyby of Jupiter.

Hyperbolic excess speed v_∞. Far away $r \to \infty$ and $V \to 0$, so the energy is all kinetic. From $\frac{1}{2}\mu\dot{\vec{r}}_\infty^2 = E = -GM\mu/(2a)$, with $a < 0$,

$$v_\infty \equiv |\dot{\vec{r}}_\infty| = \sqrt{-\frac{GM}{a}}. \tag{6.60}$$

Asymptote angle θ_∞. As $u \to \infty$ the body runs off to infinity at the true anomaly v_{asy}, with $\cos v_{\text{asy}} = \lim(e - \cosh u)/(e\cosh u - 1) = -1/e$. The asymptote makes the angle $\theta_\infty = \pi - v_{\text{asy}}$ with the axis on the far side from the periapsis, which is half the angle between the two asymptotes,

$$\cos\theta_\infty = -\lim_{u\to\infty}\frac{e - \cosh u}{e\cosh u - 1} = \frac{1}{e}. \tag{6.61}$$

The body is turned by the **deflection angle** $\delta = \pi - 2\theta_\infty$, so $\sin(\delta/2) = \cos\theta_\infty = 1/e$.

Periapsis distance. The smallest r is at $v = 0$,

$$r_{\text{p}} = \min_v \frac{a(1-e^2)}{1 + e\cos v} = \frac{a(1-e^2)}{1+e} = -a(e-1). \tag{6.62}$$

Semi-minor axis and impact parameter. The semi-minor axis $b = -a\sqrt{e^2-1}$ satisfies $b^2 = a^2(e^2-1) = -a\lambda$, since $\lambda = a(1-e^2)$. The impact parameter b' is the distance from the focus to the incoming asymptote, the miss distance had there been no pull. The asymptote passes through the centre of the hyperbola, which is $-a + r_{\text{p}} = -ae$ from the focus, and it makes the angle θ_∞ with the axis, so

$$b' = (-a + r_{\text{p}})\sin\theta_\infty = -ae\sin\theta_\infty \overset{(1)}{=} -ae\sqrt{1 - \frac{1}{e^2}} = -a\sqrt{e^2-1} = b,$$

(1) $\cos\theta_\infty = 1/e$ from Eq. (6.61).

so the impact parameter and the semi-minor axis are the same length.

Flight-path angle ψ, the angle between $\dot{\vec{r}}$ and the in-plane direction $\hat{\phi}$ perpendicular to $\hat{r}$,

$$\sin\psi = \frac{\dot{\vec{r}}\cdot\hat{r}}{|\dot{\vec{r}}|} = \frac{\dot{r}}{\sqrt{\dot{r}^2 + r^2\dot{\phi}^2}}. \tag{6.63}$$

It is zero at the periapsis, where $\dot{r} = 0$.

6.12 Fourier Solution of Kepler's Equation

Kepler's equation gives ℓ from u in one line, but going from ℓ to u takes an iteration (Problem 3).

For work by hand, in perturbation theory or for small e, we want u as an explicit function of ℓ.

A musician who hears a steady beat that is slightly uneven can describe it as the beat plus a few overtones. Anything that repeats can be written as a sum of sines of the repeat frequency and its multiples.

The difference $u - \ell = e\sin u$ repeats every orbit, and it changes sign with ℓ. So it is a sum of $\sin(s\ell)$. The idea is to find the coefficients by changing the variable of the integral from ℓ to u, where Kepler's equation makes it simple.

The series in numbers. For $e = 0.1$, the first term below is $2J_1(0.1)\sin\ell = 0.0999\sin\ell$, close to $e\sin\ell$. The second is $J_2(0.2)\sin 2\ell = 0.0050\sin 2\ell$, close to $\frac{1}{2}e^2\sin 2\ell$. The terms fall off as powers of e.

The picture is the straight line $u = \ell$ with a small wave on top of it, one wave per orbit.

The same Bessel functions give the sidebands of a frequency-modulated radio signal and the diffraction pattern of a circular aperture (Chapter 5). Here they come from pure analysis, with e as the small number (P6). The result is $u(\ell)$ as a series.

The coefficients. Kepler's equation $u - \ell = e \sin u$ is solved as a Fourier series in ℓ,

$$e \sin u = \sum_{s=1}^{\infty} c_s \sin(s\ell).$$

The function $u(\ell) - \ell$ is odd and has period 2π, and it vanishes at $\ell = 0$ and $\ell = \pi$, where $u = \ell$. So

$$\begin{aligned} c_s &\overset{(1)}{=} \frac{2}{\pi} \int_0^{\pi} (u - \ell) \sin(s\ell)\, d\ell \\ &\overset{(2)}{=} \frac{2}{s\pi} \Big[-(u - \ell) \cos(s\ell) \Big]_0^{\pi} + \frac{2}{s\pi} \int_0^{\pi} \cos(s\ell) \left(\frac{du}{d\ell} - 1 \right) d\ell \\ &\overset{(3)}{=} \frac{2}{s\pi} \int_0^{\pi} \cos(s\ell)\, du \\ &\overset{(4)}{=} \frac{2}{s\pi} \int_0^{\pi} \cos(su - se \sin u)\, du \overset{(5)}{=} \frac{2}{s} J_s(se). \end{aligned} \tag{6.64}$$

(1) The Fourier sine coefficient of an odd function of period 2π.
(2) Integration by parts.
(3) The boundary term vanishes since $u = \ell$ at both ends. $\int_0^{\pi} \cos(s\ell)\, d\ell = 0$ for a whole number $s \geq 1$. Then $(du/d\ell)\, d\ell = du$, and u runs from 0 to π as ℓ does.
(4) $\ell = u - e \sin u$, Kepler's equation.
(5) The integral form of the Bessel function of the first kind, $J_s(x) = \frac{1}{\pi} \int_0^{\pi} \cos(s\tau - x \sin \tau)\, d\tau$.

So

$$u(\ell) = \ell + \sum_{s=1}^{\infty} \frac{2}{s} J_s(se) \sin(s\ell). \tag{6.65}$$

This is the classical Bessel-function series solution of Kepler's equation. Since $J_s(se) \sim (se/2)^s / s!$ for small e, the s-th term is of order e^s.

6.13 The Laplace–Runge–Lenz Vector

The energy and the angular momentum fix four of the six elements, a, e, ι and Ω.

They do not say where the periapsis points within the plane. The angle ω came out of solving the orbit equation, not from anything conserved. Yet the ellipse of a $1/r$ potential does not turn, so something must hold its long axis in place.

A marble rolling round the inside of a salad bowl traces a loop that slowly turns, a rosette. In a bowl where the restoring force is exactly proportional to the distance from the centre, the loop closes on itself and its long axis stays put. Energy and angular momentum are the same in both bowls, so in the second something more is conserved.

The Kepler ellipse does not turn either. The idea is that a conserved vector $\vec{A}$ points from the focus to the periapsis, and its direction gives ω.

The Earth in numbers. Its orbit has $e = 0.0167$. In units of $GM\mu^2$ the vector $\vec{A}$ is an arrow of length 0.0167 pointing to the perihelion, which the Earth passes in early January.

The picture is an arrow from the focus to the periapsis that stays fixed while the body goes round.

In the hydrogen atom the same vector, made into an operator, explains why states of equal principal quantum number have equal energy. A small extra force turns $\vec{A}$ slowly, which is the precession of an orbit (Section 6.14). Its conservation is P1 together with the $1/r^2$ law (P2). It gives a fifth conserved quantity, its direction and size, and a count of the new information it carries.

Conservation. Define $\vec{A} \equiv \vec{p} \times \vec{L} - GM\mu^2 \hat{r}$, with $\vec{p} = \mu \dot{\vec{r}}$. For the pure Kepler problem,

$$\begin{aligned} \dot{\vec{A}} &\overset{(1)}{=} \dot{\vec{p}} \times \vec{L} - GM\mu^2 \dot{\hat{r}} \\ &\overset{(2)}{=} -\frac{GM\mu}{r^2} L\big(\hat{r} \times \hat{L}\big) - GM\mu^2 \dot{\phi}\, \hat{\phi} \\ &\overset{(3)}{=} \frac{GM\mu L}{r^2}\, \hat{\phi} - \frac{GM\mu L}{r^2}\, \hat{\phi} = 0. \end{aligned} \tag{6.66}$$

(1) The product rule, with $\dot{\vec{L}} = 0$.
(2) $\dot{\vec{p}} = -GM\mu\hat{r}/r^2$ from Eq. (6.3), $\vec{L} = L\hat{L}$, and $\dot{\hat{r}} = \dot{\phi}\,\hat{\phi}$ from Eq. (6.17).
(3) $(\hat{r}, \hat{\phi}, \hat{L})$ is right-handed, so $\hat{r} \times \hat{L} = -\hat{\phi}$. And $\dot{\phi} = L/(\mu r^2)$ from Eq. (6.21).

So $\vec{A}$ is conserved for the unperturbed problem. It is a constant of the Kepler problem beyond E and $\vec{L}$, and it reflects an extra hidden symmetry of the $1/r$ potential. For bound orbits, $\vec{L}$ and a rescaled $\vec{A}$ together generate the rotation group $SO(4)$ of four dimensions.

Direction and magnitude of $\vec{A}$. The vector $\vec{A}$ lies in the orbital plane, since $\vec{A} \cdot \vec{L} = (\vec{p} \times \vec{L}) \cdot \vec{L} - GM\mu^2 \hat{r} \cdot \vec{L} = 0$. So we may write $\vec{A} = A_r \hat{r} + A_\phi \hat{\phi}$.

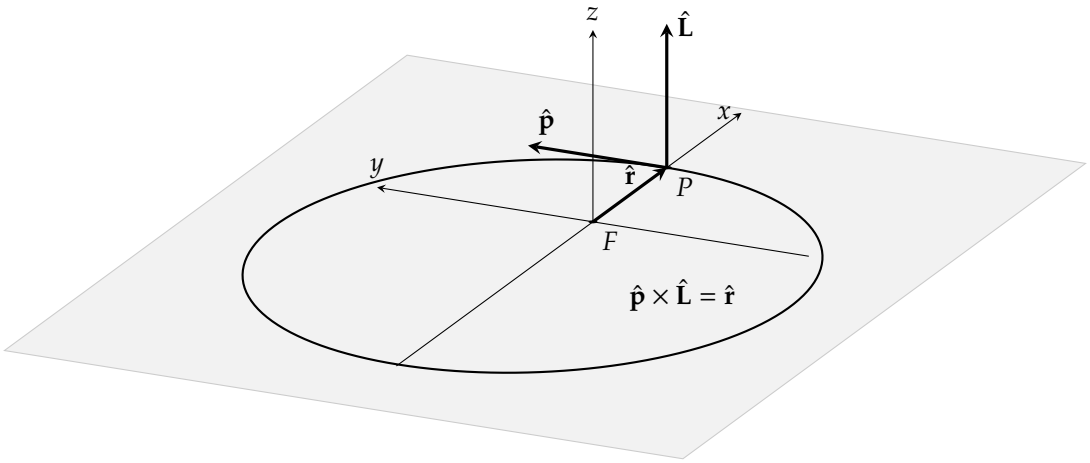


Figure 6.5: An orbit in its plane, with the focus F at the origin and the body at periapsis P. The unit vectors $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$ and $\hat{\mathbf{L}}$ are drawn at P, where $\hat{\mathbf{p}} \times \hat{\mathbf{L}} = \hat{\mathbf{r}}$.

At periapsis the velocity is purely tangential, $\vec{p} \perp \hat{r}$. Then

$$\hat{r} \times \vec{A}\big|_{\text{peri}} = \hat{r} \times (\vec{p} \times \vec{L})\big|_{\text{peri}} = \big[\vec{p}\,(\hat{r} \cdot \vec{L}) - \vec{L}\,(\hat{r} \cdot \vec{p})\big]_{\text{peri}} = 0,$$

because both dot products vanish there. So $\vec{A}$ is parallel to $\hat{r}$ at periapsis (Figure 6.5). To find

which way it points, and how long it is, we dot $\vec{A}$ into $\vec{r}$,

$$\begin{aligned}\vec{A}\cdot\vec{r} &= (\vec{p}\times\vec{L})\cdot\vec{r} - GM\mu^2 r \\ &\overset{(1)}{=} \epsilon_{kij}\, p_i L_j\, r_k - GM\mu^2 r \overset{(2)}{=} \epsilon_{jki}\, r_k p_i\, L_j - GM\mu^2 r \\ &= \underbrace{(\vec{r}\times\vec{p})}_{\vec{L}}\cdot\vec{L} - GM\mu^2 r = L^2 - GM\mu^2 r. \end{aligned} \tag{6.67}$$

(1) $(\vec{p}\times\vec{L})_k = \epsilon_{kij}\, p_i L_j$, summed over repeated indices.
(2) The cyclic symmetry $\epsilon_{kij} = \epsilon_{jki}$, and $(\vec{r}\times\vec{p})_j = \epsilon_{jki}\, r_k p_i$.

At periapsis $r = \lambda/(1+e)$ and $L^2 = GM\mu^2\lambda$, so $\vec{A}\cdot\vec{r} = GM\mu^2\lambda\, e/(1+e) > 0$ for $e > 0$. Hence $\vec{A}$ points from the focus *towards the periapsis*, and the angle between $\vec{A}$ and $\vec{r}$ is the true anomaly v. Writing $\vec{A}\cdot\vec{r} = Ar\cos v$ and using

$$r = \frac{L^2/(GM\mu^2)}{1+e\cos v} \implies GM\mu^2 r = \frac{L^2}{1+e\cos v},$$

we find

$$\begin{aligned} Ar\cos v &= L^2 - GM\mu^2 r \\ A\,\frac{L^2/(GM\mu^2)}{1+e\cos v}\cos v &\overset{(1)}{=} L^2\left(1 - \frac{1}{1+e\cos v}\right) = \frac{L^2 e\cos v}{1+e\cos v}. \end{aligned} \tag{6.68}$$

(1) The conic (6.27) for r on the left, and $GM\mu^2 r = L^2/(1+e\cos v)$ on the right.

Cancelling $L^2\cos v/(1+e\cos v)$ on both sides,

$$\boxed{A = GM\mu^2 e.} \tag{6.69}$$

A second route runs the other way. Eq. (6.67) with $\vec{A}\cdot\vec{r} = Ar\cos v$ is $r = L^2/(GM\mu^2 + A\cos v) = \lambda/\big(1 + (A/GM\mu^2)\cos v\big)$. This is the conic (6.27) again, found from two conserved vectors without solving any differential equation.

The periapsis is at the polar angle ω, so by Eq. (6.12),

$$\hat{A} = \cos\omega\,\hat{n} + \sin\omega\,\hat{y}. \tag{6.70}$$

Equivalently, the **eccentricity vector**

$$\vec{e} \equiv \frac{\vec{A}}{GM\mu^2} = e\hat{A} = e_x\,\hat{n} + e_y\,\hat{y}, \qquad e_x \equiv e\cos\omega, \quad e_y \equiv e\sin\omega, \tag{6.71}$$

is a conserved vector that holds (e,ω) together. Since e, Ω and ι are already fixed by $(E,\vec{L})$, the only *new* information in $\vec{A}$ is ω, the direction of the periapsis from the ascending node.

How many new constants $\vec{A}$ gives

The three components of $\vec{A}$ obey two conditions set by E and $\vec{L}$. One is $\vec{A}\cdot\vec{L} = 0$. The

other fixes its length, since

$$|\vec{A}|^2 = G^2M^2\mu^4e^2 \overset{(1)}{=} G^2M^2\mu^4\left(1+\frac{2EL^2}{G^2M^2\mu^3}\right) = G^2M^2\mu^4 + 2\mu EL^2,$$

where (1) is Eq. (6.38). So $\vec{A}$ supplies one new constant, not three. With E (one), $\vec{L}$ (three), $\vec{A}$ (one) and t_0 (one) the count of six for the relative orbit closes.

6.14 Perturbed Orbits

The Kepler orbit is exact for two point masses alone in space.

Real orbits feel other forces as well. The other planets tug on each one, a flattened planet pulls its satellites off the inverse square, gas drags on low satellites, an engine pushes a spacecraft, and general relativity adds a small extra pull. The orbit is then no longer a fixed ellipse.

A navigator on a ship in a slow current plots, each hour, the course the ship would follow in still water from where it is now and how it is moving now. The current shifts that course a little from one hour to the next. The ship's true track is a chain of these still-water courses, each one touching the track for a moment.

A planet tugged by the others is the ship. At each instant it moves on some Kepler ellipse that fits its present position and velocity, and the elements of that ellipse drift. The idea is to describe the true path by a Kepler ellipse whose elements change, at rates set by the extra force.

Mercury in numbers. Its perihelion turns by about 575″ per century against the stars. About 532″ of this comes from the pull of the other planets and 43″ from general relativity. Mercury makes about 415 orbits in a century, so the ellipse turns by about 1.4″ per orbit, while its size and shape barely change.

The picture is an ellipse that turns slowly, so that the path traced over many orbits is a rosette.

The same method is the variation of constants for a forced oscillator, and in quantum mechanics it is the interaction picture, where the constants of the free motion change slowly under a weak perturbation. The rates come from conservation (P1). That the extra force is small, and that its effects can be ordered by powers of it, is P6. It gives exact rates for E, $\vec{L}$ and $\vec{A}$, and the osculating orbit that turns them into rates of the elements.

Equation of motion. We add a force $\vec{F}$ to the relative equation of motion,

$$\boxed{\mu\ddot{\vec{r}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}.} \tag{6.72}$$

The central Newtonian term defines the Kepler problem. The **perturbing force** $\vec{F}$ need not be central.

The momentum is $\vec{p} = \mu\dot{\vec{r}}$. In the orbital plane, with $\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}$ and $L = \mu r^2\dot{\phi}$,

$$\vec{p} = \mu\dot{r}\,\hat{r} + \frac{L}{r}\,\hat{\phi}, \qquad L = \mu r^2\dot{\phi}. \tag{6.73}$$

Orbital perturbation theory represents the actual trajectory by Keplerian orbital elements that vary with time, and derives their rates directly from Eq. (6.72).

6.14.1 Osculating Kepler orbit

We first fix which Kepler ellipse belongs to the true path at each instant. For $\vec{F} = 0$, the Kepler solution can be written in terms of six constants

$$Q = (Q_1, \ldots, Q_6) = (a, e, \iota, \Omega, \omega, t_0), \tag{6.74}$$

so that

$$\vec{r} = \vec{r}(Q, t), \qquad \vec{p} = \vec{p}(Q, t). \tag{6.75}$$

For fixed Q, these functions satisfy the unperturbed equations of motion.

For the perturbed problem, let

$$Q_i = Q_i(t). \tag{6.76}$$

The actual trajectory is represented in terms of a set of time-dependent osculating orbital elements $Q(t)$ as

$$\vec{r}(t) = \vec{r}_{\text{osc}}\big(Q(t), t\big), \qquad \vec{p}(t) = \vec{p}_{\text{osc}}\big(Q(t), t\big). \tag{6.77}$$

Here, for fixed values of Q, the functions $\vec{r}_{\text{osc}}(Q, t)$ and $\vec{p}_{\text{osc}}(Q, t)$ describe the corresponding Keplerian orbit. The time dependence of $Q(t)$ accounts for the slow evolution of this osculating Keplerian orbit under the perturbing force.

At each instant, the osculating Kepler orbit is chosen to have the same position and momentum as the actual trajectory. Thus,

$$\vec{r}_{\text{osc}}\big(Q(t), t\big) = \vec{r}(t), \qquad \vec{p}_{\text{osc}}\big(Q(t), t\big) = \vec{p}(t). \tag{6.78}$$

This does not make the actual trajectory Keplerian. At each instant it is represented by the Keplerian orbit that touches it at that instant, the **osculating** orbit.

The corresponding accelerations are different,

$$\dot{\vec{p}}_{\text{actual}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}, \tag{6.79}$$

$$\dot{\vec{p}}_{\text{osc}} = -\frac{GM\mu}{r^2}\hat{r}. \tag{6.80}$$

Hence

$$\boxed{\dot{\vec{p}}_{\text{actual}} - \dot{\vec{p}}_{\text{osc}} = \vec{F}.} \tag{6.81}$$

The two trajectories therefore agree in position and momentum at the instant at which the osculating orbit is constructed, but they have different subsequent evolution.

6.14.2 Variation of parameters

The osculating condition becomes six equations for the six rates $\dot{Q}_i$. Differentiating Eq. (6.77) by the chain rule,

$$\dot{\vec{r}} = \left.\frac{\partial \vec{r}}{\partial t}\right|_Q + \sum_{i=1}^{6} \frac{\partial \vec{r}}{\partial Q_i}\dot{Q}_i. \tag{6.82}$$

For fixed Q, the first term is the velocity of the Kepler solution. The osculating condition requires this Kepler velocity to coincide with the actual velocity,

$$\dot{\vec{r}} = \left.\frac{\partial \vec{r}}{\partial t}\right|_Q. \tag{6.83}$$

Therefore

$$\boxed{\sum_{i=1}^{6} \frac{\partial \vec{r}}{\partial Q_i} \dot{Q}_i = 0.} \tag{6.84}$$

This is a vector equation and therefore supplies three scalar conditions.

Now differentiate $\vec{p}(Q(t), t)$,

$$\dot{\vec{p}} = \left.\frac{\partial \vec{p}}{\partial t}\right|_{Q} + \sum_{i=1}^{6} \frac{\partial \vec{p}}{\partial Q_i} \dot{Q}_i. \tag{6.85}$$

For fixed Q, $\vec{p}(Q, t)$ obeys the Kepler equation,

$$\left.\frac{\partial \vec{p}}{\partial t}\right|_{Q} = -\frac{GM\mu}{r^2} \hat{r}. \tag{6.86}$$

The actual momentum obeys

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2} \hat{r} + \vec{F}. \tag{6.87}$$

Subtracting the Kepler contribution gives

$$\boxed{\sum_{i=1}^{6} \frac{\partial \vec{p}}{\partial Q_i} \dot{Q}_i = \vec{F}.} \tag{6.88}$$

Equations (6.84) and (6.88) form six scalar equations for the six quantities $\dot{Q}_i$. This is the formal variation-of-parameters construction of the osculating elements.

Six equations for six unknowns, so the count closes. The perturbing force enters only through the change it makes in the instantaneous position and momentum, and the Kepler orbit is redefined at each instant from that state.

6.14.3 Perturbed Kepler invariants

The Keplerian energy, angular momentum and eccentricity vector are defined from the instantaneous state by

$$E \equiv \frac{p^2}{2\mu} - \frac{GM\mu}{r}, \tag{6.89}$$

$$\vec{L} \equiv \vec{r} \times \vec{p}, \tag{6.90}$$

$$\vec{A} \equiv \vec{p} \times \vec{L} - GM\mu^2 \hat{r}. \tag{6.91}$$

For the unperturbed Kepler problem these are constants. In the perturbed problem they become time-dependent quantities.

Their magnitudes determine the instantaneous Keplerian parameters,

$$E = -\frac{GM\mu}{2a(t)}, \tag{6.92}$$

$$L^2 = GM\mu^2 a(t)(1 - e^2(t)), \tag{6.93}$$

$$A \equiv |\vec{A}| = GM\mu^2 e(t). \tag{6.94}$$

Thus a and e may be regarded as functions of the actual state even when the full trajectory is not Keplerian.

6.14.4 Exact Rate of the Energy

We differentiate Eq. (6.89) along the actual motion,

$$\begin{aligned}\dot{E} &\overset{(1)}{=} \frac{\vec{p}}{\mu}\cdot\dot{\vec{p}} + \frac{GM\mu}{r^2}\dot{r} \\ &\overset{(2)}{=} \dot{\vec{r}}\cdot\left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F}\right) + \frac{GM\mu}{r^2}\dot{r} \\ &\overset{(3)}{=} -\frac{GM\mu}{r^2}\dot{r} + \dot{\vec{r}}\cdot\vec{F} + \frac{GM\mu}{r^2}\dot{r} = \dot{\vec{r}}\cdot\vec{F}.\end{aligned} \tag{6.95}$$

(1) $d(p^2)/dt = 2\vec{p}\cdot\dot{\vec{p}}$ and $d(1/r)/dt = -\dot{r}/r^2$.
(2) $\vec{p}/\mu = \dot{\vec{r}}$, and $\dot{\vec{p}}$ from the equation of motion (6.72).
(3) $\dot{\vec{r}}\cdot\hat{r} = \dot{r}$.

$$\boxed{\dot{E} = \dot{\vec{r}}\cdot\vec{F}.} \tag{6.96}$$

The Kepler energy changes only by the work done by the extra force.

6.14.5 Exact Rate of angular momentum

From $\vec{L} = \vec{r}\times\vec{p}$,

$$\dot{\vec{L}} \overset{(1)}{=} \dot{\vec{r}}\times\vec{p} + \vec{r}\times\dot{\vec{p}} \overset{(2)}{=} \vec{r}\times\left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F}\right) \overset{(3)}{=} \vec{r}\times\vec{F}. \tag{6.97}$$

(1) The product rule.
(2) $\dot{\vec{r}}\times\vec{p} = \mu\,\dot{\vec{r}}\times\dot{\vec{r}} = 0$, and $\dot{\vec{p}}$ from Eq. (6.72).
(3) The central term vanishes because $\vec{r} \parallel \hat{r}$.

$$\boxed{\dot{\vec{L}} = \vec{r}\times\vec{F}.} \tag{6.98}$$

Only the part of $\vec{F}$ perpendicular to $\vec{r}$ exerts a torque.

6.14.6 Exact Rate of the Runge–Lenz vector

We need one kinematic fact first. Since $\hat{r} = \vec{r}/r$,

$$\dot{\hat{r}} = \frac{\dot{\vec{r}}}{r} - \frac{\dot{r}\,\vec{r}}{r^2} \overset{(1)}{=} \frac{r^2\dot{\vec{r}} - (\vec{r}\cdot\dot{\vec{r}})\,\vec{r}}{r^3} \overset{(2)}{=} \frac{(\vec{r}\times\dot{\vec{r}})\times\vec{r}}{r^3} = \frac{\vec{L}\times\vec{r}}{\mu r^3}. \tag{6.99}$$

(1) $r\dot{r} = \vec{r}\cdot\dot{\vec{r}}$.
(2) $(\vec{a}\times\vec{b})\times\vec{c} = \vec{b}\,(\vec{a}\cdot\vec{c}) - \vec{a}\,(\vec{b}\cdot\vec{c})$.

It uses only the definitions, so it holds with the perturbation on. Now we differentiate Eq. (6.91),

$$\begin{aligned}\dot{\vec{A}} &\overset{(1)}{=} \dot{\vec{p}}\times\vec{L} + \vec{p}\times\dot{\vec{L}} - GM\mu^2\,\dot{\hat{r}} \\ &\overset{(2)}{=} -\frac{GM\mu}{r^2}\hat{r}\times\vec{L} + \vec{F}\times\vec{L} + \vec{p}\times(\vec{r}\times\vec{F}) - GM\mu^2\,\dot{\hat{r}} \\ &\overset{(3)}{=} -\frac{GM\mu}{r^2}\hat{r}\times\vec{L} + \vec{F}\times\vec{L} + \vec{p}\times(\vec{r}\times\vec{F}) + \frac{GM\mu}{r^2}\,\hat{r}\times\vec{L} \\ &= \vec{F}\times\vec{L} + \vec{p}\times(\vec{r}\times\vec{F}).\end{aligned} \tag{6.100}$$

(1) The product rule.

(2) $\dot{\vec{p}}$ from Eq. (6.72) and $\dot{\vec{L}} = \vec{r} \times \vec{F}$ from Eq. (6.98).

(3) $-GM\mu^2 \dot{\hat{r}} = -(GM\mu/r^2)\, \vec{L} \times \hat{r} = (GM\mu/r^2)\, \hat{r} \times \vec{L}$, from Eq. (6.99) with $\vec{r} = r\hat{r}$.

The two central terms cancel. Therefore

$$\boxed{\dot{\vec{A}} = \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}).} \tag{6.101}$$

This equation is the starting point for the evolution of the eccentricity and of the direction of the periapsis.

6.15 Planar perturbations

We have exact rates for E, $\vec{L}$ and $\vec{A}$ under any extra force.

To follow an orbit we need the rates of the elements a, e and ω themselves, written with the parts of the force along and across the radius.

Pushing a child on a swing adds energy only if the push is along the motion. A push at the top of the swing, where the child is at rest, does no work at all. When and in which direction we push decides what changes.

An orbit is the same. A push along the motion changes the energy and so the size a. A push along the radius at the periapsis, where $\dot{r} = 0$, does no work, but it still changes the shape and turns the ellipse. The idea is to split $\vec{F}$ into its radial and angular parts and read off $\dot{a}$, $\dot{e}$ and $\dot{\omega}$ from the exact rates. These are the planar **Gauss equations**.

A thruster in numbers. A spacecraft on a circular orbit of radius $a = 2.03 \times 10^4$ km about the Earth fires an engine giving $0.2\,\mathrm{m\,s^{-2}}$ along its motion (Problem 8). Eq. (6.145) below with $e = 0$ gives $\dot{a} = 2\,(F_\phi/\mu)\sqrt{a^3/GM_\oplus} = 1.8\,\mathrm{km\,s^{-1}}$, so the orbit starts to grow by almost 2 km every second.

The picture is the force arrow at one point of the orbit, split into a part along the radius and a part across it.

The same equations follow satellite drag, the push of sunlight on a solar sail, and the slow shrinking of a binary pulsar's orbit as it radiates gravitational waves. They combine the conservation laws (P1) with the smallness of the force (P6). They give $\dot{a}$, $\dot{e}$, $\dot{\omega}$ and $\dot{v}$ for any force in the plane.

Force in the plane. We take a perturbing force confined to the orbital plane,

$$\boxed{\vec{F} = F_r \hat{r} + F_\phi \hat{\phi}.} \tag{6.102}$$

Its torque is $\vec{r} \times \vec{F} = rF_\phi\, \hat{r} \times \hat{\phi} = rF_\phi\, \hat{L}$, along $\hat{L}$. So $\dot{\vec{L}}$ is parallel to $\vec{L} = L\hat{L}$, and only the length of $\vec{L}$ changes. The plane itself stays fixed,

$$\boxed{i = 0, \qquad \dot{\Omega} = 0.} \tag{6.103}$$

In the orbital plane, by Eq. (6.73),

$$\vec{p} = \mu\dot{r}\, \hat{r} + \mu r\dot{\phi}\, \hat{\phi} = \mu\dot{r}\, \hat{r} + \frac{L}{r}\hat{\phi}. \tag{6.104}$$

The equations of motion are the $\hat{r}$ and $\hat{\phi}$ parts of Eq. (6.72), with the acceleration of Section 6.3,

$$\mu(\ddot{r} - r\dot{\phi}^2) = -\frac{GM\mu}{r^2} + F_r, \tag{6.105}$$

$$\mu(r\ddot{\phi} + 2\dot{r}\dot{\phi}) = F_\phi. \tag{6.106}$$

The second equation gives the rate of L,

$$\frac{dL}{dt} = \frac{d}{dt}(\mu r^2\dot{\phi}) = \mu\left(2r\dot{r}\dot{\phi} + r^2\ddot{\phi}\right) = r\,\mu\left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right) = rF_\phi, \tag{6.107}$$

so

$$\boxed{\dot{L} = rF_\phi.} \tag{6.108}$$

The energy equation (6.96) becomes

$$\dot{E} \overset{(1)}{=} \dot{r}F_r + r\dot{\phi}F_\phi \overset{(2)}{=} \dot{r}F_r + \frac{L}{\mu r}F_\phi. \tag{6.109}$$

(1) $\dot{\vec{r}} \cdot \vec{F}$ with $\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}$.
(2) $r\dot{\phi} = L/(\mu r)$.

6.15.1 Keplerian geometry of the instantaneous ellipse

We need the osculating ellipse's r, $\dot{r}$ and $\dot{\phi}$ in terms of its elements. The true anomaly is

$$v \equiv \phi - \omega. \tag{6.110}$$

For the instantaneous Kepler orbit,

$$\boxed{r = \frac{\lambda}{1 + e\cos v},} \tag{6.111}$$

where

$$\boxed{\lambda \equiv a(1 - e^2) = \frac{L^2}{GM\mu^2}.} \tag{6.112}$$

The instantaneous angular velocity is

$$\dot{\phi} = \frac{L}{\mu r^2}. \tag{6.113}$$

For the radial velocity we start from the general case, where the elements depend on time,

$$a(t) = a_0 + \delta a(t), \qquad e(t) = e_0 + \delta e(t), \qquad \omega(t) = \omega_0 + \delta\omega(t), \tag{6.114}$$

and the changes δa, δe and $\delta\omega$ are produced by the perturbing force. Along the actual path, by the chain rule,

$$\dot{r} = \left.\frac{\partial r}{\partial t}\right|_Q + \sum_{i=1}^{6} \frac{\partial r}{\partial Q_i}\dot{Q}_i. \tag{6.115}$$

Since $r = \hat{r} \cdot \vec{r}$ and $\hat{r} \cdot \partial\hat{r}/\partial Q_i = 0$, each $\partial r/\partial Q_i$ is $\hat{r} \cdot \partial\vec{r}/\partial Q_i$. The sum is therefore the radial part of $\sum_i (\partial\vec{r}/\partial Q_i)\dot{Q}_i$, which vanishes by the osculating condition (6.84). So $\dot{r}$ is the radial velocity

of the osculating ellipse with its elements held fixed. On that ellipse only the true anomaly moves, at the rate $\dot{v} = \dot{\phi}$ of Eq. (6.113), giving

$$\boxed{\dot{r} = \frac{dr}{dv}\dot{v} = \frac{\lambda e \sin v}{(1 + e\cos v)^2}\frac{L}{\mu r^2} = \frac{Le\sin v}{\mu\lambda}.} \tag{6.116}$$

The last step uses $1/r^2 = (1 + e\cos v)^2/\lambda^2$. The result is exact for the osculating elements.

Since

$$\frac{\lambda}{r} = 1 + e\cos v, \tag{6.117}$$

we also have

$$\boxed{\frac{\mu r\dot{r}}{L} = \frac{e\sin v}{1 + e\cos v}.} \tag{6.118}$$

We name this ratio,

$$\boxed{\xi \equiv \frac{e\sin v}{1 + e\cos v}.} \tag{6.119}$$

6.15.2 Rate of the semi-major axis

At a given instant the osculating orbit has the same energy as the actual orbit, so its semi-major axis $a(t)$ follows the energy. Its Keplerian energy and its rate are

$$E\Big|_{\text{osculating}} = -\frac{GM\mu}{2a}, \qquad \therefore \qquad \dot{E}\Big|_{\text{osculating}} = \frac{GM\mu}{2a^2}\dot{a}. \tag{6.120}$$

The right side of Eq. (6.109) is evaluated with the osculating r, $\dot{r}$ and L,

$$\dot{E}\Big|_{\text{osculating}} = \dot{r}F_r + \frac{L}{\mu r}F_\phi \overset{(1)}{=} \frac{L}{\mu\lambda}e\sin v\, F_r + \frac{L}{\mu r}F_\phi. \tag{6.121}$$

(1) $\dot{r} = Le\sin v/(\mu\lambda)$, Eq. (6.116).

Every quantity here is the osculating value, so this is exact. Since the perturbing force is small, $F_r, F_\phi = O(F)$, the rates of the elements are $O(F)$ too. To first order the quantities multiplying F_r and F_ϕ may therefore be evaluated on the unperturbed Keplerian orbit, and this is what the orbit averages of Section 6.16 use. Equating with Eq. (6.120),

$$\begin{aligned}\dot{a} &= \frac{2a^2L}{GM\mu^2}\left[\frac{e\sin v}{\lambda}F_r + \frac{1}{r}F_\phi\right] \\ &\overset{(1)}{=} \frac{2a^2\sqrt{GM\lambda}}{GM\mu\lambda}\left[e\sin v\, F_r + (1 + e\cos v)F_\phi\right] \overset{(2)}{=} \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\left[e\sin v\, F_r + (1 + e\cos v)F_\phi\right].\end{aligned} \tag{6.122}$$

(1) $L = \mu\sqrt{GM\lambda}$ and $1/r = (1 + e\cos v)/\lambda$.
(2) $2a^2/(\mu\sqrt{GM\lambda}) = \sqrt{4a^4/(GM\mu^2 a(1-e^2))}$, with $\lambda = a(1-e^2)$.

Here we used

$$L = \mu\sqrt{GM\lambda}, \qquad \lambda = a(1-e^2). \tag{6.123}$$

So

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\left[e\sin v\, F_r + (1 + e\cos v)F_\phi\right].} \tag{6.124}$$

6.15.3 Rate of the eccentricity

Since $A = GM\mu^2 e$, it is enough to find $\dot{A}$ from Eq. (6.101). For the planar force, with $\vec{L} = L\hat{L}$, the two terms are

$$\vec{F} \times \vec{L} = L\left(F_r \hat{r} + F_\phi \hat{\phi}\right) \times \hat{L} \overset{(1)}{=} L\left(-F_r \hat{\phi} + F_\phi \hat{r}\right),$$
$$\vec{p} \times (\vec{r} \times \vec{F}) \overset{(2)}{=} \left(\mu \dot{r} \hat{r} + \frac{L}{r}\hat{\phi}\right) \times \left(r F_\phi \hat{L}\right) \overset{(1)}{=} L F_\phi \hat{r} - \mu r \dot{r} F_\phi \hat{\phi}. \tag{6.125}$$

(1) $\hat{r} \times \hat{L} = -\hat{\phi}$ and $\hat{\phi} \times \hat{L} = \hat{r}$, since $(\hat{r}, \hat{\phi}, \hat{L})$ is right-handed.
(2) $\vec{p}$ from Eq. (6.104), and $\vec{r} \times \vec{F} = r F_\phi \hat{L}$.

Adding them,

$$\dot{\vec{A}} = \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}) = 2L F_\phi \hat{r} - \left(L F_r + \mu r \dot{r} F_\phi\right) \hat{\phi}. \tag{6.126}$$

With $\mu r \dot{r} = L\xi$ from Eq. (6.119),

$$\boxed{\dot{\vec{A}} = L\left[2F_\phi \hat{r} - (F_r + \xi F_\phi)\hat{\phi}\right].} \tag{6.127}$$

For a purely radial force this is $\dot{\vec{A}} = -F_r L\, \hat{\phi}$.

The unit vector $\hat{A}$ points to the periapsis, and the true anomaly is measured from it. Let $\hat{q}$ be $\hat{A}$ turned by 90° towards increasing v. Then $(\hat{r}, \hat{\phi})$ is $(\hat{A}, \hat{q})$ turned by v, and the inverse turn is the transpose,

$$\begin{pmatrix} \hat{r} \\ \hat{\phi} \end{pmatrix} = \begin{pmatrix} \cos v & \sin v \\ -\sin v & \cos v \end{pmatrix} \begin{pmatrix} \hat{A} \\ \hat{q} \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} \hat{A} \\ \hat{q} \end{pmatrix} = \begin{pmatrix} \cos v & -\sin v \\ \sin v & \cos v \end{pmatrix} \begin{pmatrix} \hat{r} \\ \hat{\phi} \end{pmatrix}. \tag{6.128}$$

The first row is

$$\boxed{\hat{A} = \cos v\, \hat{r} - \sin v\, \hat{\phi}.} \tag{6.129}$$

Writing $\vec{A} = A\hat{A}$,

$$\dot{\vec{A}} = \dot{A}\hat{A} + A\dot{\hat{A}}, \tag{6.130}$$
$$\dot{\vec{A}} \cdot \hat{A} \overset{(1)}{=} \dot{A}, \tag{6.131}$$

(1) $\hat{A} \cdot \hat{A} = 1$, so $2\hat{A} \cdot \dot{\hat{A}} = 0$.

and therefore

$$\dot{A} = \dot{\vec{A}} \cdot \hat{A} \overset{(1)}{=} L\left[2F_\phi \cos v + (F_r + \xi F_\phi)\sin v\right], \tag{6.132}$$

(1) Eq. (6.127) dotted into Eq. (6.129), with $\hat{r} \cdot \hat{r} = \hat{\phi} \cdot \hat{\phi} = 1$ and $\hat{r} \cdot \hat{\phi} = 0$.

that is,

$$\boxed{\dot{A} = L\left[F_r \sin v + F_\phi\,(2\cos v + \xi \sin v)\right].} \tag{6.133}$$

Up to this point all relations have been exact. We now write both sides with the osculating elements, $A = GM\mu^2 e$ and $L = \sqrt{GM\mu^2 a(1-e^2)}$,

$$\dot{A}\Big|_{\text{osculating}} = L\left[F_r \sin v + F_\phi\,(2\cos v + \xi \sin v)\right]\Big|_{\text{osculating}}$$
$$GM\mu^2 \dot{e} = \sqrt{GM\mu^2 a(1-e^2)}\left[F_r \sin v + F_\phi\,(2\cos v + \xi \sin v)\right], \tag{6.134}$$

and obtain

$$\boxed{\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r \sin v + F_\phi\,(2\cos v + \xi \sin v)\right].} \tag{6.135}$$

6.15.4 Rate of periapsis

The direction of $\vec{A}$ gives ω. The unit vector perpendicular to $\hat{A}$, towards increasing true anomaly, is the second row of Eq. (6.128),

$$\boxed{\hat{q} = \sin v\,\hat{r} + \cos v\,\hat{\phi}.} \tag{6.136}$$

Differentiating Eq. (6.129),

$$\begin{aligned}\dot{\hat{A}} &= -\dot{v}\sin v\,\hat{r} + \cos v\,\dot{\hat{r}} - \dot{v}\cos v\,\hat{\phi} - \sin v\,\dot{\hat{\phi}} \\ &\overset{(1)}{=} (\dot{\phi} - \dot{v})\left(\sin v\,\hat{r} + \cos v\,\hat{\phi}\right) \overset{(2)}{=} \dot{\omega}\,\hat{q}.\end{aligned} \tag{6.137}$$

(1) $\dot{\hat{r}} = \dot{\phi}\,\hat{\phi}$ and $\dot{\hat{\phi}} = -\dot{\phi}\,\hat{r}$, Eq. (6.17).
(2) $v = \phi - \omega$ gives $\dot{v} = \dot{\phi} - \dot{\omega}$, and Eq. (6.136).

Since

$$\dot{\vec{A}} = \dot{A}\,\hat{A} + A\dot{\hat{A}} = \dot{A}\,\hat{A} + A\dot{\omega}\,\hat{q}, \tag{6.138}$$

projection onto $\hat{q}$ gives

$$A\dot{\omega} = \dot{\vec{A}} \cdot \hat{q}. \tag{6.139}$$

Using Eqs. (6.127) and (6.136),

$$A\dot{\omega}\Big|_{\text{osculating}} = L\left[2F_\phi \sin v - (F_r + \xi F_\phi)\cos v\right]\Big|_{\text{osculating}}. \tag{6.140}$$

With $A = GM\mu^2 e$,

$$\boxed{\dot{\omega} = -\frac{L}{GM\mu^2 e}\left[F_r \cos v + F_\phi\,(\xi\cos v - 2\sin v)\right].} \tag{6.141}$$

Using

$$\frac{L}{GM\mu^2 e} = \frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}}, \tag{6.142}$$

we obtain

$$\boxed{\dot{\omega} = -\frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r \cos v + F_\phi\,(\xi\cos v - 2\sin v)\right].} \tag{6.143}$$

6.15.5 Rate of the true anomaly

Since $v = \phi - \omega$, we have $\dot{v} = \dot{\phi} - \dot{\omega}$. The instantaneous Keplerian value of $\dot{\phi}$ is $L/(\mu r^2)$, Eq. (6.113). With the osculating orbit,

$$\begin{aligned}\dot{v} &= \frac{L}{\mu r^2} - \dot{\omega}\Big|_{\text{osculating}} \\ &\overset{(1)}{=} \frac{\mu\sqrt{GMa(1-e^2)}\,(1+e\cos v)^2}{\mu\, a^2(1-e^2)^2} - \dot{\omega} \overset{(2)}{=} \sqrt{\frac{GM}{a^3}}\frac{(1+e\cos v)^2}{(1-e^2)^{3/2}} - \dot{\omega}.\end{aligned} \tag{6.144}$$

(1) $r = a(1-e^2)/(1+e\cos v)$ and $L = \mu\sqrt{GMa(1-e^2)}$.
(2) $\sqrt{a(1-e^2)}/\left(a^2(1-e^2)^2\right) = a^{-3/2}(1-e^2)^{-3/2}$.

6.15.6 Planar Gauss equations

Collecting the results,

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\left[e\sin v\, F_r + (1+e\cos v)F_\phi\right], \tag{6.145}$$

$$\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r\sin v + F_\phi\,(2\cos v + \xi\sin v)\right], \tag{6.146}$$

$$\dot{\omega} = -\frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r\cos v + F_\phi\,(\xi\cos v - 2\sin v)\right], \tag{6.147}$$

$$\dot{v} = \sqrt{\frac{GM}{a^3}}\frac{(1+e\cos v)^2}{(1-e^2)^{3/2}} - \dot{\omega}, \tag{6.148}$$

where

$$\xi = \frac{e\sin v}{1+e\cos v}. \tag{6.149}$$

For a perturbation confined to the plane,

$$\boxed{i = \dot{\Omega} = 0.} \tag{6.150}$$

These are equations for the *osculating* elements. The quantities a, e and ω are therefore functions of time, even though the actual position and momentum at each instant are those of a Keplerian orbit. Only the force in the plane enters. A push along the motion (F_ϕ at $e = 0$) changes a, while a radial push changes e and ω but, at the apsides, not a.

6.16 Orbit-averaged evolution

The Gauss equations give the rates of the elements at each instant.

Over one orbit most of that change goes forward and then back again. Over many orbits only the net drift matters, and the instantaneous rates hide it.

The sea rises and falls twice a day with the tides. To see the sea level creep up over decades, we average over each day and watch how the daily average changes.

The elements of an orbit wobble within each orbit in the same way. The idea is to average the rates over one unperturbed Kepler orbit. To first order in the force, what is left is the slow, **secular** drift.

The example below in numbers. A small extra radial force χ/r^3 with $\chi\mu/L^2 = 0.01$ turns the periapsis by $-\pi \times 0.01 = -0.031$ rad $= -1.8°$ per orbit, while a and e return to their starting values after each orbit.

The picture is a rate that swings up and down within each orbit around a small offset. The offset is the drift.

The same averaging gives the mean Sun of Chapter 1, the adiabatic invariants of mechanics and the slow precession of the Earth's axis. It works because the orbital period is short compared with the time over which the elements change (P6). It gives the averaging rule and a worked example, checked against an exact solution.

The average. For any quantity X,

$$\boxed{\langle \dot{X} \rangle = \frac{1}{P} \int_0^P \dot{X}\, dt.} \tag{6.151}$$

For an unperturbed Kepler ellipse the elements are fixed, so $\dot{v} = \dot{\phi}$ and

$$L = \mu r^2 \dot{v}, \tag{6.152}$$

and therefore

$$\boxed{dt = \frac{\mu r^2}{L}\, dv.} \tag{6.153}$$

On the true path $\dot{v} = \dot{\phi} - \dot{\omega}$ differs from $\dot{\phi}$ by $\dot{\omega} = O(F)$. Inside an average that is already multiplied by F, the difference between $d\phi$ and dv changes the result only at $O(F^2)$.

The average rate can therefore be written as

$$\langle \dot{X} \rangle = \frac{1}{P} \int_0^{2\pi} \dot{X} \frac{\mu r^2}{L}\, dv. \tag{6.154}$$

A nonzero value of $\langle \dot{X} \rangle$ is a secular change. Terms whose integral over $0 \le v \le 2\pi$ vanishes describe only oscillations within each orbit.

6.16.1 Example: Perturbation by an Inverse-Cube Force $F_r = \chi/r^3$

We take the perturbing force

$$\boxed{\vec{F} = \frac{\chi}{r^3} \hat{r},} \tag{6.155}$$

with a constant χ that is positive for a push outward. Thus

$$F_r = \frac{\chi}{r^3}, \qquad F_\phi = 0. \tag{6.156}$$

We find the three rates directly from E, L and $\vec{A}$, and then compare with the Gauss equations.

Rate of the semimajor axis

From Eq. (6.120),

$$\dot{E} = \frac{GM\mu}{2a^2} \dot{a}. \tag{6.157}$$

The exact rate of change of the energy of the actual orbit is, by Eq. (6.96),

$$\dot{E} = \vec{F} \cdot \dot{\vec{r}}. \tag{6.158}$$

The perturbation is radial and $\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}$, so only the radial velocity does work,

$$\dot{E} = \frac{\chi}{r^3} \dot{r}. \tag{6.159}$$

For the osculating Keplerian orbit, Eq. (6.116) with $\lambda = L^2/(GM\mu^2)$ gives

$$\dot{r} = \frac{Le \sin v}{\mu} \frac{GM\mu^2}{L^2} = \frac{GM\mu}{L} e \sin v. \tag{6.160}$$

Therefore

$$\dot{E} = \frac{\chi GM\mu}{Lr^3} e \sin v. \tag{6.161}$$

Comparing this result with Eq. (6.157),

$$\frac{GM\mu}{2a^2}\dot{a} = \frac{\chi GM\mu}{Lr^3} e \sin v, \tag{6.162}$$

so that

$$\boxed{\dot{a} = \frac{2\chi a^2 e \sin v}{Lr^3}}. \tag{6.163}$$

Rate of the eccentricity

The Runge–Lenz vector is

$$\vec{A} = \vec{p} \times \vec{L} - GM\mu^2 \hat{r}, \tag{6.164}$$

and for the osculating Keplerian orbit

$$A = GM\mu^2 e. \tag{6.165}$$

Since the perturbing force is radial, it exerts no torque,

$$\dot{\vec{L}} = \vec{r} \times \vec{F} = \vec{r} \times \frac{\chi}{r^3}\hat{r} = 0. \tag{6.166}$$

Differentiating $L^2 = GM\mu^2 a(1-e^2)$ gives

$$2L\dot{L} = GM\mu^2 \left[(1-e^2)\dot{a} - 2ae\dot{e}\right], \tag{6.167}$$

and the left side is zero. Therefore

$$\dot{e} = \frac{1-e^2}{2ae}\dot{a}. \tag{6.168}$$

Substituting Eq. (6.163),

$$\dot{e} = \frac{1-e^2}{2ae}\,\frac{2\chi a^2 e \sin v}{Lr^3} = \frac{\chi a(1-e^2)\sin v}{Lr^3}. \tag{6.169}$$

As a check, the Gauss equation (6.146) with $F_\phi = 0$ gives $\dot{e} = \sqrt{a(1-e^2)/(GM\mu^2)}\,\chi \sin v / r^3$. With $L = \mu\sqrt{GMa(1-e^2)}$ the two are the same.

Rate of the argument of periapsis

The direction of $\vec{A}$ gives the periapsis. Differentiating Eq. (6.164),

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} + \vec{p} \times \dot{\vec{L}} - GM\mu^2 \dot{\hat{r}}. \tag{6.170}$$

Since $\dot{\vec{L}} = 0$,

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} - GM\mu^2 \dot{\hat{r}}. \tag{6.171}$$

The momentum equation is

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \frac{\chi}{r^3}\hat{r}. \tag{6.172}$$

Therefore

$$\dot{\vec{A}} = \left(-\frac{GM\mu}{r^2} + \frac{\chi}{r^3}\right) \underbrace{\hat{r} \times \vec{L}}_{=-L\hat{\phi}} \underbrace{- GM\mu^2 \underbrace{\dot{\hat{r}}}_{=\dot{\phi}\,\hat{\phi}}}_{=\frac{GM\mu L}{r^2}\hat{\phi}}. \tag{6.173}$$

The terms with the central gravitational force cancel, leaving

$$\dot{\vec{A}} = -\frac{\chi L}{r^3}\hat{\phi}. \tag{6.174}$$

We use the periapsis frame $(\hat{A}, \hat{q})$ of Eq. (6.128). Here $\hat{A}$ points toward the periapsis, $\hat{A} = \cos\omega\,\hat{n} + \sin\omega\,\hat{y}$ by Eq. (6.70), and $\hat{q}$ is $\hat{A}$ turned by 90° towards increasing true anomaly. Then

$$\vec{A} = A\hat{A}, \tag{6.175}$$

and, since $\dot{\hat{A}} = \dot{\omega}\,\hat{q}$ (Section 6.15),

$$\dot{\vec{A}} = \dot{A}\,\hat{A} + A\dot{\omega}\,\hat{q}. \tag{6.176}$$

The radial and angular unit vectors in the periapsis frame are, from Eq. (6.128),

$$\hat{r} = \cos v\,\hat{A} + \sin v\,\hat{q}, \tag{6.177}$$

and

$$\hat{\phi} = -\sin v\,\hat{A} + \cos v\,\hat{q}. \tag{6.178}$$

Substituting the latter into Eq. (6.174),

$$\dot{\vec{A}} = -\frac{\chi L}{r^3}\left(-\sin v\,\hat{A} + \cos v\,\hat{q}\right) = \frac{\chi L}{r^3}\sin v\,\hat{A} - \frac{\chi L}{r^3}\cos v\,\hat{q}. \tag{6.179}$$

Comparing this with Eq. (6.176),

$$\dot{A} = \frac{\chi L}{r^3}\sin v, \tag{6.180}$$

and

$$A\dot{\omega} = -\frac{\chi L}{r^3}\cos v. \tag{6.181}$$

With $A = GM\mu^2 e$,

$$\boxed{\dot{\omega} = -\frac{\chi L}{GM\mu^2 e\, r^3}\cos v}. \tag{6.182}$$

6.16.2 Semi-major axis

We now average the three rates over one orbit, starting with a. As a check on Eq. (6.163), the Gauss equation (6.145) gives

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\, e\sin v\,\frac{\chi}{r^3} \overset{(1)}{=} \frac{2a^2}{\mu\sqrt{GM\lambda}}\,\frac{\mu\lambda\dot{r}}{L}\,\frac{\chi}{r^3}, \tag{6.183}$$

(1) $\sqrt{4a^3/(GM\mu^2(1-e^2))} = 2a^2/(\mu\sqrt{GM\lambda})$, and $e\sin v = \mu\lambda\dot{r}/L$ from Eq. (6.116).

and with $L = \mu\sqrt{GM\lambda}$,

$$\boxed{\dot{a} = \frac{2\chi a^2}{GM\mu r^3}\dot{r}.} \tag{6.184}$$

For the Keplerian ellipse we use the eccentric anomaly u,

$$r = a(1 - e\cos u), \tag{6.185}$$

and, from Eqs. (6.56) and (6.49),

$$\dot{r} = \frac{ane\sin u}{1 - e\cos u}, \qquad n = \sqrt{\frac{GM}{a^3}}. \tag{6.186}$$

Therefore

$$\boxed{\dot{a} = \frac{2\chi en\sin u}{GM\mu(1 - e\cos u)^4}.} \tag{6.187}$$

Over one unperturbed orbit, from Eq. (6.55),

$$dt = \frac{1 - e\cos u}{n}\,du, \tag{6.188}$$

so

$$\begin{aligned}\Delta a &= \int_0^{2\pi} \dot{a}\,dt = \frac{2\chi e}{GM\mu}\int_0^{2\pi} \frac{\sin u}{(1 - e\cos u)^3}\,du \\ &\overset{(1)}{=} \frac{2\chi e}{GM\mu}\left[-\frac{1}{2e\,(1 - e\cos u)^2}\right]_0^{2\pi} = 0.\end{aligned} \tag{6.189}$$

(1) $\dfrac{d}{du}(1 - e\cos u)^{-2} = -2e\sin u\,(1 - e\cos u)^{-3}$.

The same follows without any integral. Eq. (6.184) is $\dot{a} = -(\chi a^2/GM\mu)\,d(r^{-2})/dt$, and r returns to its starting value after one orbit. Thus

$$\boxed{\langle\dot{a}\rangle = 0} \tag{6.190}$$

to first order in the perturbation.

The osculating semi-major axis oscillates during an orbit but has no secular drift.

6.16.3 Eccentricity

Angular momentum is conserved, since the force is radial, so

$$L^2 = GM\mu^2 a(1 - e^2) \tag{6.191}$$

is constant, and therefore

$$0 = (1 - e^2)\dot{a} - 2ae\dot{e}. \tag{6.192}$$

Hence

$$\boxed{\dot{e} = \frac{1 - e^2}{2ae}\dot{a}.} \tag{6.193}$$

To first order the factor $(1 - e^2)/(2ae)$ is constant over the orbit, so since $\langle\dot{a}\rangle = 0$,

$$\boxed{\langle\dot{e}\rangle = 0.} \tag{6.194}$$

6.16.4 Apsidal precession

For the radial force, Eq. (6.147) with $F_\phi = 0$ and $L/(GM\mu^2 e) = \sqrt{a(1-e^2)/(GM\mu^2)}/e$ reduces to Eq. (6.182),

$$\boxed{\dot\omega = -\frac{\chi L \cos v}{GM\mu^2 e\, r^3}.} \tag{6.195}$$

To first order in χ we evaluate the secular change on the unperturbed Keplerian orbit,

$$r = \frac{\lambda}{1 + e\cos v}, \qquad \lambda = a(1-e^2), \tag{6.196}$$

together with Eq. (6.153),

$$dt = \frac{\mu r^2}{L}\, dv. \tag{6.197}$$

Then

$$\begin{aligned}
\Delta\omega &= \int_0^{2\pi} \dot\omega\, \frac{\mu r^2}{L}\, dv \overset{(1)}{=} -\frac{\chi}{GM\mu e}\int_0^{2\pi} \frac{\cos v}{r}\, dv \\
&\overset{(2)}{=} -\frac{\chi}{GM\mu e}\int_0^{2\pi} \frac{1 + e\cos v}{\lambda}\cos v\, dv \\
&= -\frac{\chi}{GM\mu e\lambda}\left[\int_0^{2\pi}\cos v\, dv + e\int_0^{2\pi}\cos^2 v\, dv\right].
\end{aligned} \tag{6.198}$$

(1) $\dfrac{\chi L}{GM\mu^2 e\, r^3}\cdot\dfrac{\mu r^2}{L} = \dfrac{\chi}{GM\mu e\, r}$.
(2) $1/r = (1 + e\cos v)/\lambda$.

Since

$$\int_0^{2\pi}\cos v\, dv = 0, \qquad \int_0^{2\pi}\cos^2 v\, dv = \pi, \tag{6.199}$$

we obtain

$$\boxed{\Delta\omega = -\frac{\pi\chi}{GM\mu\, a(1-e^2)}.} \tag{6.200}$$

Dividing by the period

$$P = 2\pi\sqrt{\frac{a^3}{GM}}, \tag{6.201}$$

the secular rate is

$$\boxed{\langle\dot\omega\rangle = -\frac{\chi}{2\mu\sqrt{GM}\, a^{5/2}(1-e^2)}.} \tag{6.202}$$

An outward push ($\chi > 0$) turns the periapsis backwards, against the motion, and an extra inward pull ($\chi < 0$) turns it forwards. The size a and the shape e do not drift.

A second route, exact. The same result follows from the exact orbit equation (6.26) for the full radial force

$$F_r = -\frac{GM\mu}{r^2} + \frac{\chi}{r^3}, \tag{6.203}$$

which is $-V'(r)$. With $\rho = 1/r$ the right side of Eq. (6.26) is $-(\mu/L^2)F_r/\rho^2 = GM\mu^2/L^2 - \chi\mu\rho/L^2$, so

$$\frac{d^2\rho}{d\phi^2} + \left(1 + \frac{\chi\mu}{L^2}\right)\rho = \frac{GM\mu^2}{L^2}. \tag{6.204}$$

This is the shifted oscillator of Section 6.5 with a stiffer spring. In ϕ the oscillation of ρ has the angular frequency $\sqrt{1+\chi\mu/L^2}$, so one periapsis follows the next after the angle

$$\Delta\phi_{\text{apsis}} = \frac{2\pi}{\sqrt{1+\chi\mu/L^2}}. \tag{6.205}$$

The exact shift of the periapsis per orbit, relative to 2π, is therefore

$$\Delta\omega_{\text{exact}} = 2\pi\left[\frac{1}{\sqrt{1+\chi\mu/L^2}} - 1\right]. \tag{6.206}$$

For a weak perturbation,

$$\frac{\chi\mu}{L^2} \ll 1, \tag{6.207}$$

so

$$\begin{aligned}\Delta\omega_{\text{exact}} &\overset{(1)}{=} -\frac{\pi\chi\mu}{L^2} + O(\chi^2)\\ &\overset{(2)}{=} -\frac{\pi\chi}{GM\mu\, a(1-e^2)} + O(\chi^2),\end{aligned} \tag{6.208}$$

(1) $(1+x)^{-1/2} = 1 - x/2 + O(x^2)$.
(2) $L^2 = GM\mu^2 a(1-e^2)$, Eq. (6.35).

which agrees with the orbit-averaged result (6.200).

6.17 Nearly circular orbits

The Gauss equation (6.147) for $\dot\omega$ carries a factor $1/e$.

As $e \to 0$ the periapsis loses its meaning, because a circle has no nearest point. Then $\dot\omega$ blows up, although nothing physical does.

Ask which point of a round table is nearest the wall. For a table almost centred in the room the answer swings right round with the smallest nudge, while the nudge itself is tiny. It is better to give how far the table's centre has moved east and north. Those two numbers are always well defined and change smoothly.

For an orbit the two numbers are the components of the eccentricity vector (6.71). The idea is to use $e_x = e\cos\omega$ and $e_y = e\sin\omega$ in place of e and ω. They go smoothly to zero with e, and the factor $1/e$ disappears.

The table in numbers. Take $e = 0.001$ and $\omega = 0$, so $(e_x, e_y) = (0.001, 0)$. A kick that adds 0.001 to e_y turns ω by 45° and raises e only to 0.0014, while neither component changed by more than 0.001.

The picture is the tip of the vector $\vec{e}$ wandering near the origin of a plane. Its angle swings about wildly, but its position does not.

The same trouble is the meaningless longitude at the Earth's pole (Chapter 2), cured the same way by Cartesian components. Here e is the small number and terms are kept to first order in it (P6). The result is r and ϕ to first order in e, and the Gauss equations in e_x and e_y.

The orbit to first order in e. The components of the eccentricity vector along $\hat n$ and $\hat y$ are

$$\boxed{e_x = e\cos\omega, \qquad e_y = e\sin\omega.} \tag{6.209}$$

For small e we define the **mean longitude**

$$\bar{\phi} \equiv \omega + \ell, \tag{6.210}$$

the polar angle of a body moving round at the mean rate. For $e = 0$ it is the polar angle itself. To first order in e, $v = \ell + 2e\sin\ell + O(e^2)$, derived from Kepler's second law in Eq. (1.22), where it gives the equation of time. Then

$$\phi = \omega + v \overset{(1)}{=} \bar{\phi} + 2e\sin\big(\bar{\phi} - \omega\big) + O(e^2) \overset{(2)}{=} \bar{\phi} + 2e_x\sin\bar{\phi} - 2e_y\cos\bar{\phi} + O(e^2). \tag{6.211}$$

(1) $v = \ell + 2e\sin\ell$ and $\ell = \bar{\phi} - \omega$.
(2) $e\sin\big(\bar{\phi} - \omega\big) = e\cos\omega\sin\bar{\phi} - e\sin\omega\cos\bar{\phi}$.

In the same way $r = a(1 - e\cos u) = a(1 - e\cos\ell) + O(e^2)$ and $e\cos\big(\bar{\phi} - \omega\big) = e_x\cos\bar{\phi} + e_y\sin\bar{\phi}$, so

$$\boxed{r = a\left(1 - e_x\cos\bar{\phi} - e_y\sin\bar{\phi}\right) + O(e^2).} \tag{6.212}$$

For fixed elements $\dot{\bar{\phi}} = \dot{\ell} = n$, with

$$n = \sqrt{\frac{GM}{a^3}}, \tag{6.213}$$

so differentiating,

$$\dot{r} = an\left(e_x\sin\bar{\phi} - e_y\cos\bar{\phi}\right) + O(e^2). \tag{6.214}$$

Also,

$$\boxed{\dot{\phi} = n\left(1 + 2e_x\cos\bar{\phi} + 2e_y\sin\bar{\phi}\right) + O(e^2).} \tag{6.215}$$

We name the first-order offset of the polar angle from the mean longitude,

$$\delta\phi \equiv \phi - \bar{\phi} = 2e_x\sin\bar{\phi} - 2e_y\cos\bar{\phi}. \tag{6.216}$$

Then $\dot{r} = an\,\delta\phi/2$, and to first order $\xi = e\sin v/(1 + e\cos v) = e\sin\big(\bar{\phi} - \omega\big) + O(e^2) = \delta\phi/2$.

The variables e_x and e_y are better than ω near a circle because they stay finite and smooth as $e \to 0$.

6.17.1 First-order equations for nearly circular motion

We now write the Gauss equations in a, e_x, e_y to first order in e. For a, Eqs. (6.120) and (6.109) give $\dot{a} = (2a^2/GM\mu)\big[\dot{r}F_r + (L/\mu r)F_\phi\big]$. To first order, $\dot{r} = an\,\delta\phi/2$ and $L/(\mu r) = \sqrt{GMa}\,(1 + e_x\cos\bar{\phi} + e_y\sin\bar{\phi})/a = an(1 + e_x\cos\bar{\phi} + e_y\sin\bar{\phi})$. With $2a^3n/(GM\mu) = \sqrt{4a^3/(GM\mu^2)}$,

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2}}\left[\tfrac{1}{2}\delta\phi\,F_r + \left(1 + e_x\cos\bar{\phi} + e_y\sin\bar{\phi}\right)F_\phi\right] + O(e^2F).} \tag{6.217}$$

The eccentricity vector is, exactly,

$$\vec{A} = GM\mu^2\left(e_x\hat{n} + e_y\hat{y}\right). \tag{6.218}$$

The frame $(\hat{n}, \hat{y})$ is fixed for a planar force, so $\dot{e}_x = \dot{\vec{A}}\cdot\hat{n}/(GM\mu^2)$ and $\dot{e}_y = \dot{\vec{A}}\cdot\hat{y}/(GM\mu^2)$. With $\hat{r}\cdot\hat{n} = \cos\phi$, $\hat{\phi}\cdot\hat{n} = -\sin\phi$, $\hat{r}\cdot\hat{y} = \sin\phi$ and $\hat{\phi}\cdot\hat{y} = \cos\phi$ from Eq. (6.12), Eq. (6.127) gives

$$\begin{aligned}\dot{e}_x &\overset{(1)}{=} \frac{L}{GM\mu^2}\Big[F_r\sin\phi + F_\phi\big(2\cos\phi + \xi\sin\phi\big)\Big] \\ &\overset{(2)}{=} \sqrt{\frac{a}{GM\mu^2}}\Big[F_r\big(\sin\bar{\phi} + \delta\phi\cos\bar{\phi}\big) + F_\phi\big(2\cos\bar{\phi} - 2\delta\phi\sin\bar{\phi} + \tfrac{1}{2}\delta\phi\sin\bar{\phi}\big)\Big] + O(e^2F),\end{aligned} \tag{6.219}$$

(1) $\dot{\vec{A}} \cdot \hat{n} = L\big[2F_\phi \cos\phi + (F_r + \xi F_\phi)\sin\phi\big]$.

(2) $L/(GM\mu^2) = \sqrt{a(1-e^2)/(GM\mu^2)} = \sqrt{a/(GM\mu^2)} + O(e^2)$, $\sin\phi = \sin\bar{\phi} + \delta\phi\cos\bar{\phi}$, $\cos\phi = \cos\bar{\phi} - \delta\phi\sin\bar{\phi}$ and $\xi = \delta\phi/2$, all to first order.

and in the same way for $\dot{e}_y$, with $\dot{\vec{A}} \cdot \hat{y} = L\big[-F_r\cos\phi + F_\phi(2\sin\phi - \xi\cos\phi)\big]$. Collecting terms,

$$\boxed{\dot{e}_x = \sqrt{\frac{a}{GM\mu^2}}\left[\sin\bar{\phi}\,F_r + 2\cos\bar{\phi}\,F_\phi + \tfrac{1}{2}\delta\phi\left(2\cos\bar{\phi}\,F_r - 3\sin\bar{\phi}\,F_\phi\right)\right] + O(e^2F),} \tag{6.220}$$

$$\boxed{\dot{e}_y = \sqrt{\frac{a}{GM\mu^2}}\left[-\cos\bar{\phi}\,F_r + 2\sin\bar{\phi}\,F_\phi + \tfrac{1}{2}\delta\phi\left(2\sin\bar{\phi}\,F_r + 3\cos\bar{\phi}\,F_\phi\right)\right] + O(e^2F).} \tag{6.221}$$

Neither equation contains $1/e$.

For the unperturbed orbit $\dot{\bar{\phi}} = n$ exactly. The force changes this rate only by $O(F)$,

$$\boxed{\dot{\bar{\phi}} = n + O(F).} \tag{6.222}$$

The rates above are already $O(F)$, so dividing them by $\dot{\bar{\phi}}$ instead of n changes them only at $O(F^2)$.

Using $\bar{\phi}$ as the independent variable, $d/d\bar{\phi} = n^{-1}d/dt$, with $\sqrt{a/(GM\mu^2)}/n = a^2/(GM\mu)$ and $\sqrt{4a^3/(GM\mu^2)}/n = 2a^3/(GM\mu)$,

$$\boxed{\frac{da}{d\bar{\phi}} = \frac{2a^3}{GM\mu}\left[\tfrac{1}{2}\delta\phi\,F_r + \left(1 + e_x\cos\bar{\phi} + e_y\sin\bar{\phi}\right)F_\phi\right],} \tag{6.223}$$

$$\boxed{\frac{de_x}{d\bar{\phi}} = \frac{a^2}{GM\mu}\left[\sin\bar{\phi}\,F_r + 2\cos\bar{\phi}\,F_\phi + \tfrac{1}{2}\delta\phi\left(2\cos\bar{\phi}\,F_r - 3\sin\bar{\phi}\,F_\phi\right)\right],} \tag{6.224}$$

$$\boxed{\frac{de_y}{d\bar{\phi}} = \frac{a^2}{GM\mu}\left[-\cos\bar{\phi}\,F_r + 2\sin\bar{\phi}\,F_\phi + \tfrac{1}{2}\delta\phi\left(2\sin\bar{\phi}\,F_r + 3\cos\bar{\phi}\,F_\phi\right)\right].} \tag{6.225}$$

6.18 Rømer delay for a Keplerian binary

We have the position of each star in its orbit, and its distance along the line of sight, the Z of Eq. (6.13).

For a pulsar in a binary we cannot see the position at all. What we measure is the arrival time of each pulse.

A drummer marching to and fro along the line between us and a distant field beats a perfectly steady rhythm. We hear the beats bunch up as he comes towards us and spread out as he goes away, because the sound from the far end has farther to travel.

A pulsar in a binary is the drummer, and its pulses are the beats. The idea is that each pulse arrives late by the extra light-travel distance $\vec{x}_1 \cdot \hat{R}/c$, so the timing measures the line-of-sight coordinate of the orbit.

The Earth in numbers. Light crosses 1 AU in 499 s. A source in the plane of the Earth's orbit has its signals arrive up to 499 s early or late over a year, as the Earth moves towards it and away. Rømer found the speed of light in 1676 from this delay in the eclipses of Jupiter's moon Io.

The picture is the orbit seen edge on, with the pulsar moving nearer to us and farther away.

The same delay, for the Earth's own orbit, is part of correcting every pulse arrival time to the centre of mass of the Solar System, and in a pulsar's timing it reveals planets around the pulsar. The orbit comes from P1 and P2, and the rest is geometry. It gives the delay in terms of the eccentric anomaly, and with it the masses of neutron stars in Chapter 13.

The delay. For a binary of component masses m_1 and m_2, $M = m_1 + m_2$. Let $\vec{r}$ be the relative separation. The primary's position relative to the centre of mass is, by Section 6.1.2,

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}. \tag{6.226}$$

$\hat{R}$ points from the observer towards the source (Section 6.2). A primary that is farther away by $\vec{x}_1 \cdot \hat{R}$ than the centre of mass sends pulses that arrive later by

$$\Delta_R = \frac{\vec{x}_1 \cdot \hat{R}}{c}. \tag{6.227}$$

In the orbital-plane basis,

$$\vec{r} = r\left[\cos\phi\,\hat{n} + \sin\phi\,\hat{y}\right], \tag{6.228}$$

with

$$\phi = \omega + v, \tag{6.229}$$

and, from the third components of Eqs. (6.9) and (6.10),

$$\hat{y} \cdot \hat{R} = \sin\iota, \qquad \hat{n} \cdot \hat{R} = 0, \tag{6.230}$$

so that

$$\Delta_R = \frac{m_2}{Mc}\, r \sin\iota\, \sin(\omega + v). \tag{6.231}$$

This is $(m_2/M)\,Z/c$ with Z from Eq. (6.13).

We define the primary's semi-major axis about the centre of mass,

$$a_1 = \frac{m_2}{M}a, \tag{6.232}$$

and use the eccentric-anomaly relations (6.46),

$$r\cos v = a(\cos u - e), \tag{6.233}$$

$$r\sin v = a\sqrt{1-e^2}\sin u. \tag{6.234}$$

Then

$$\begin{aligned}\Delta_R &= \frac{m_2}{Mc}\sin\iota\left(r\cos v\,\sin\omega + r\sin v\,\cos\omega\right)\\ &\overset{(1)}{=} \frac{a_1\sin\iota}{c}\left[(\cos u - e)\sin\omega + \sqrt{1-e^2}\sin u\cos\omega\right],\end{aligned} \tag{6.235}$$

(1) $\sin(\omega+v) = \sin\omega\cos v + \cos\omega\sin v$ was used in the first line, then Eq. (6.46) and $a_1 = (m_2/M)a$.

so

$$\boxed{\Delta_R = \frac{a_1\sin\iota}{c}\left[(\cos u - e)\sin\omega + \sqrt{1-e^2}\sin u\cos\omega\right].} \tag{6.236}$$

Only the projected size $a_1\sin\iota$ enters, the same combination that a radial-velocity curve gives (Problem 9). Timing alone does not give ι.

6.19 Applied Orbital Mechanics: Binary Stars, Transfer Orbits, and Perturbations

6.19.1 Problem 1: Astrometric Binary White Dwarf System

Problem Statement: Two white dwarfs in a binary system appear on the sky plane as concentric circles of radii 10 μas and 5 μas, respectively, moving anticlockwise. The observed orbital period is 1 day. The system has a parallax of 1 mas.

(a) What is the semi-major axis of the relative orbit?

(b) The stars appear to move with constant angular speed along circular trajectories. What are the orbital eccentricity e and inclination ι?

(c) Find the individual masses m_1 and m_2 and the total mass M.

(d) If the orbit were inclined, what would be the orientation of the orbit in space? Specify the angles Ω and ω.

(e) What is the magnitude of the Runge–Lenz vector for this system?

Essential Ideas:

- **Center of mass**: For a binary system, both stars orbit their mutual center of mass with identical period P. Their individual barycentric semi-major axes satisfy

$$a = a_1 + a_2, \qquad m_1 a_1 = m_2 a_2,$$

  where a is the semi-major axis of the relative orbit.

- **Projected orbit**: A circular orbit tilted with respect to the sky plane projects to an ellipse with axis ratio $\cos\iota$ (Section 6.2). Therefore an observed circle requires $\iota = 0°$. Constant orbital speed then implies a circular Kepler orbit, so $e = 0$.

- **Runge–Lenz vector**: The Runge–Lenz vector points towards the periapsis with magnitude

$$|\vec{A}| = GM\mu^2 e.$$

  For a circular orbit ($e = 0$), $\vec{A} = \vec{0}$.

Detailed Solutions:

(a) **Semi-major axis of the relative orbit.**

The distance d to the binary follows from the parallax of $1\,\text{mas} = 10^{-3}''$ (Chapter 3, Eq. (3.5)),

$$d = \frac{1\,\text{pc}}{10^{-3}} = 10^3\,\text{pc}.$$

An angle of 1″ at 1 pc is 1 AU, so the physical barycentric orbital radii are

$$a_1 = \theta_1 d = (10\,\mu\text{as})(10^3\,\text{pc}) = (10^{-5}'')(10^3\,\text{pc}) = 10^{-2}\,\text{AU} \simeq 1.496 \times 10^9\,\text{m},$$
$$a_2 = \theta_2 d = (5\,\mu\text{as})(10^3\,\text{pc}) = (5 \times 10^{-6}'')(10^3\,\text{pc}) = 0.5 \times 10^{-2}\,\text{AU} \simeq 7.48 \times 10^8\,\text{m}.$$

Therefore the relative separation has the semi-major axis

$$\boxed{a = a_1 + a_2 = 1.5 \times 10^{-2}\,\text{AU} = 2.244 \times 10^9\,\text{m}}.$$

(b) **Eccentricity and inclination.**

Since the observed trajectories are circles, the projection factor satisfies $\cos\iota = 1$, so $\boxed{\iota = 0°}$. Since the orbital speed is constant, the orbit is circular, $\boxed{e = 0}$.

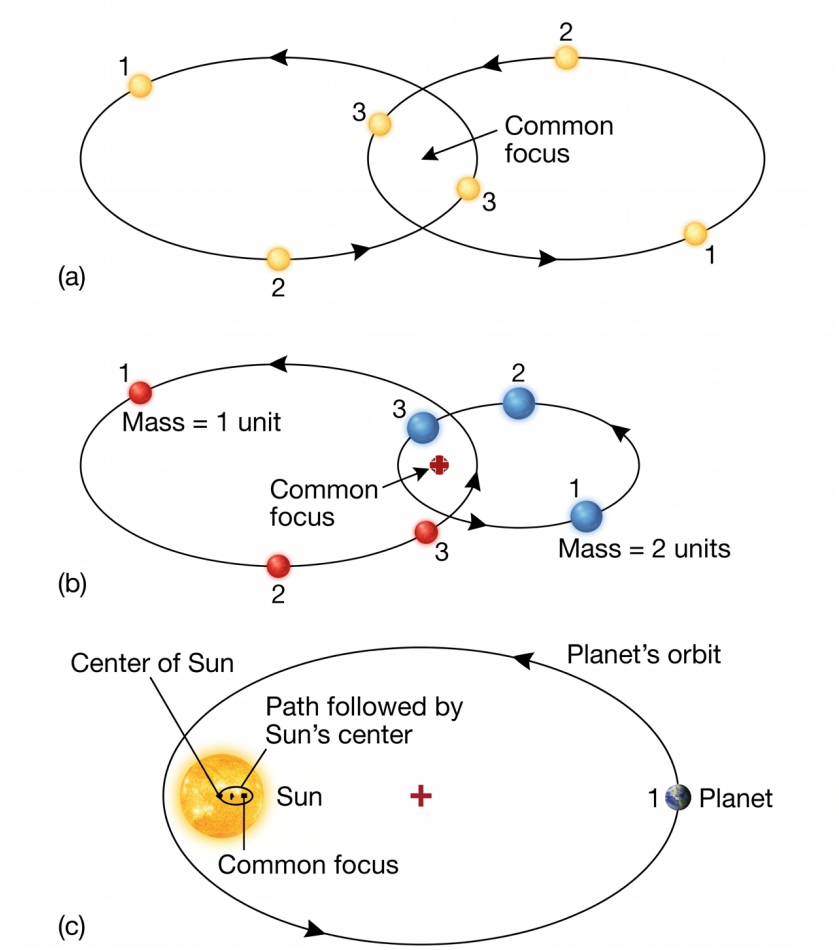


Figure 6.6: Two stars and their common center of mass, with the position vectors $\vec{r}_1 = (m_2/M)\vec{r}$ and $\vec{r}_2 = -(m_1/M)\vec{r}$ along the relative separation $\vec{r}$.

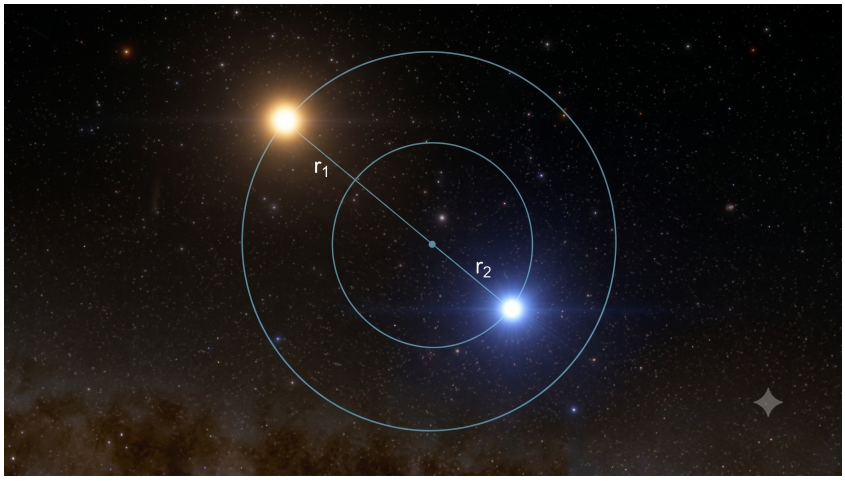


Figure 6.7: Two stars on circular orbits about their common centre of mass, with the barycentric radii r_1 and r_2 marked.

(c) **Masses.**

The mass ratio is the inverse of the ratio of the barycentric semi-major axes,

$$\frac{m_1}{m_2} = \frac{a_2}{a_1} = \frac{5}{10} = \frac{1}{2} \implies m_2 = 2m_1.$$

From Kepler's third law, Eq. (6.49),

$$P^2 = \frac{4\pi^2 a^3}{G(m_1 + m_2)} = \frac{4\pi^2 a^3}{3Gm_1}.$$

With $P = 1\,\text{day} = 86400\,\text{s}$ and $a = 2.244 \times 10^9\,\text{m}$,

$$M = m_1 + m_2 = \frac{4\pi^2 a^3}{GP^2} \simeq 8.95 \times 10^{29}\,\text{kg} \simeq 0.450\,M_\odot.$$

Thus

$$\boxed{m_1 \simeq 0.150\,M_\odot}, \qquad \boxed{m_2 \simeq 0.300\,M_\odot}.$$

The same follows in solar units, $M/M_\odot = (a/\text{AU})^3/(P/\text{yr})^2 = (1.5{\times}10^{-2})^3{\times}365.25^2 = 0.450$. Both masses are well below the Chandrasekhar mass and are typical of the low-mass white dwarfs found in close double-degenerate binaries. Their separation, $3.2\,R_\odot$, is small enough for a one-day orbit and far larger than either white dwarf. The orbit spans $10\,\mu$as on the sky, near the limit of what Gaia can measure.

(d) **Orientation angles Ω and ω.**

For a circular face-on orbit ($\iota = 0°$, $e = 0$), the line of nodes is undefined, since the orbital and sky planes do not intersect, and the periapsis is undefined, since the distance is constant. Hence neither Ω nor ω is defined. If the same circular orbit were inclined, the node line would exist and Ω would be defined, but ω would still be undefined because a circle has no periapsis.

(e) **Runge–Lenz vector.**

Because the orbit is circular ($e = 0$), $\vec{A} = \vec{0}$. A circle has no distinguished periapsis direction, which is why the Runge–Lenz vector vanishes.

6.19.2 Problem 2: Bounds on the Symmetric Mass Ratio

Problem Statement: Prove that the symmetric mass ratio $\eta = \frac{m_1 m_2}{M^2}$ (where $M = m_1 + m_2$) satisfies $\eta \in (0, 0.25]$.

Solution: Since $m_1, m_2 > 0$, $\eta > 0$. For the upper bound,

$$\begin{aligned}(m_1 - m_2)^2 \geq 0 &\implies m_1^2 - 2m_1 m_2 + m_2^2 \geq 0 \\ &\implies (m_1 + m_2)^2 \geq 4m_1 m_2 \\ &\implies \eta = \frac{m_1 m_2}{(m_1 + m_2)^2} \leq \frac{1}{4}.\end{aligned}$$

Equality holds if and only if $m_1 = m_2$. Hence $\boxed{\eta \in (0, 0.25]}$. Since $\eta = \mu/M$, the reduced mass is at most a quarter of the total mass.

6.19.3 Problem 3: Numerical Solution of Kepler's Equation

Problem Statement: Solve Kepler's equation $\ell = u - e\sin u$ using Newton's method and plot $u(\ell)$ for $e = 0, 0.1, 0.3, 0.5, 0.7, 0.9$.

Newton's Iteration Scheme: For fixed mean anomaly ℓ and eccentricity e, let $g(u) = u - e\sin u - \ell$. Then $g'(u) = 1 - e\cos u$, and Newton's method gives the recurrence

$$u_{k+1} = u_k - \frac{u_k - e\sin u_k - \ell}{1 - e\cos u_k}. \tag{6.237}$$

The derivative $1 - e\cos u \geq 1 - e > 0$ never vanishes for $e < 1$, so each step is defined. Starting from $u_0 = \ell$, the error is already $e\sin\ell$, and Newton's method then roughly squares it at each step. The listing below implements it (the code calls the mean anomaly `l`).

```
import matplotlib.pyplot as plt
import numpy as np

def solver(l_vals, e, tol=1e-12, max_iter=100):
    u_vals = np.zeros_like(l_vals)
    for i, l in enumerate(l_vals):
        u = l
        for _ in range(max_iter):
            f = u - e * np.sin(u) - l
            fp = 1 - e * np.cos(u)
            u = u - f / fp
            if abs(f / fp) < tol:
                break
        u_vals[i] = u
    return u_vals

l_grid = np.linspace(0, 2 * np.pi, 500)
eccentricities = [0, 0.1, 0.3, 0.5, 0.7, 0.9]

plt.figure(figsize=(8, 6))
for e in eccentricities:
    u_grid = solver(l_grid, e)
    plt.plot(l_grid, u_grid, label=f"e = {e}")
plt.xlabel(r"Mean anomaly $l$ (rad)")
plt.ylabel(r"Eccentric anomaly $u$ (rad)")
plt.title(r"Kepler equation: $u(l)$")
plt.grid(True)
plt.legend()
plt.show()
```

6.19.4 Problem 4: Relation Between True and Eccentric Anomalies

Problem Statement: Prove that the eccentric anomaly u and true anomaly v satisfy

$$\tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{u}{2}. \tag{6.238}$$

Proof: Equating the two expressions for the orbital radius, Eqs. (6.41) and (6.27),

$$r = a(1 - e\cos u) = \frac{a(1-e^2)}{1 + e\cos v} \implies 1 + e\cos v = \frac{1-e^2}{1 - e\cos u}.$$

Solving for $\cos v$ as in Eq. (6.43),

$$\cos v = \frac{\cos u - e}{1 - e\cos u}.$$

Then

$$1 - \cos v = \frac{1 - e\cos u - \cos u + e}{1 - e\cos u} = \frac{(1+e)(1-\cos u)}{1 - e\cos u},$$
$$1 + \cos v = \frac{1 - e\cos u + \cos u - e}{1 - e\cos u} = \frac{(1-e)(1+\cos u)}{1 - e\cos u}.$$

Dividing the two equations, and using $\tan^2(x/2) = (1-\cos x)/(1+\cos x)$,

$$\frac{1-\cos v}{1+\cos v} = \frac{1+e}{1-e}\frac{1-\cos u}{1+\cos u} \implies \tan^2\frac{v}{2} = \frac{1+e}{1-e}\tan^2\frac{u}{2}.$$

The half-angles $u/2$ and $v/2$ lie in the same quadrant, since u and v are on the same side of the periapsis, so taking the positive root gives

$$\boxed{\tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{u}{2}}. \tag{6.239}$$

6.19.5 Problem 5: Effective Potential and Hohmann Transfer Orbits

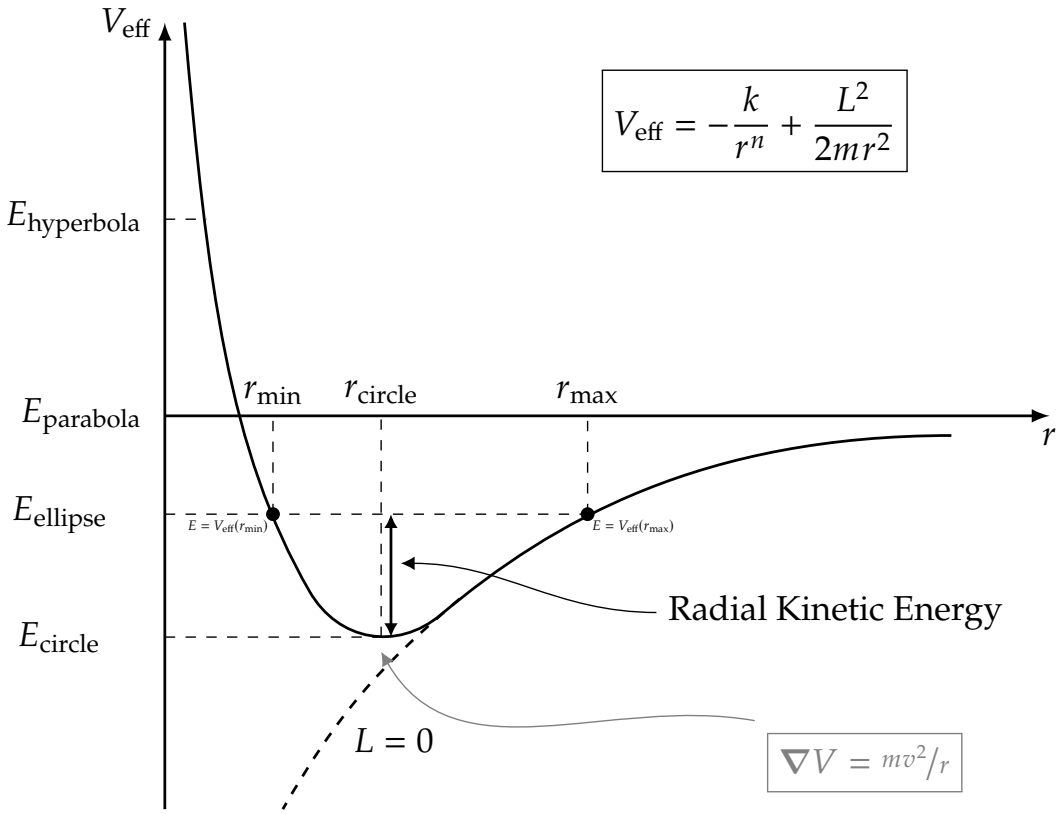


Figure 6.8: The effective potential $V_{\text{eff}}(r) = -k/r^n + L^2/(2mr^2)$ against r, with the energy levels of parabolic, circular, elliptical and hyperbolic orbits, the turning points r_{min} and r_{max} at one bound energy, and the curve for $L = 0$.

Problem Statement: The relative-orbit energy (6.31), divided by μ as in Problem 10, reads

$$\mathcal{E} \equiv \frac{E}{\mu} = \frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r), \qquad V_{\text{eff}}(r) \equiv \frac{h^2}{2r^2} - \frac{GM}{r}, \qquad h \equiv \frac{L}{\mu}.$$

This is one-dimensional motion in the *effective potential* $V_{\text{eff}}(r)$ per unit mass. It is Newtonian gravity's case of the general power-law form $V_{\text{eff}} = -k/r^n + L^2/(2mr^2)$ of Figure 6.8, with $n = 1$, $k = GM$ and $m = 1$. Since $\frac{1}{2}\dot{r}^2 \geq 0$ always, the radii reachable at fixed $\mathcal{E}$ are exactly those with $V_{\text{eff}}(r) \leq \mathcal{E}$.

(a) Find the circular-orbit radius r_{circle}, where V_{eff} is stationary, and the corresponding energy $\mathcal{E}_{\text{circle}} \equiv V_{\text{eff}}(r_{\text{circle}})$, in terms of h and GM.

(b) For $\mathcal{E}_{\text{circle}} < \mathcal{E} < 0$, solve $V_{\text{eff}}(r) = \mathcal{E}$ for its two roots $r_{\min}, r_{\max}$, and show they are exactly the periapsis and apoapsis $a(1 \mp e)$ already implied by Eqs. (6.33) and (6.35).

(c) Show that $\mathcal{E} = \mathcal{E}_{\text{circle}}$, $\mathcal{E} = 0$, and $\mathcal{E} > 0$ correspond to the circular, parabolic, and hyperbolic rows of the table of conic sections in Section 6.5.

(d) An object orbits the Sun ($GM_\odot = 1.327 \times 10^{20}\,\text{m}^3\,\text{s}^{-2}$) with $h = 4.089 \times 10^{15}\,\text{m}^2\,\text{s}^{-1}$ and $\mathcal{E} = -4.423 \times 10^{8}\,\text{J}\,\text{kg}^{-1}$. Find $r_{\min}$, $r_{\max}$, and e.

Detailed Solutions:

(a) Setting $V'_{\text{eff}}(r) = -h^2/r^3 + GM/r^2 = 0$,

$$\boxed{r_{\text{circle}} = \frac{h^2}{GM}}, \qquad \mathcal{E}_{\text{circle}} = \frac{h^2}{2r_{\text{circle}}^2} - \frac{GM}{r_{\text{circle}}} = \frac{GM}{2r_{\text{circle}}} - \frac{GM}{r_{\text{circle}}} \implies \boxed{\mathcal{E}_{\text{circle}} = -\frac{G^2M^2}{2h^2}.}$$

(b) Multiplying $V_{\text{eff}}(r) = \mathcal{E}$ through by $2r^2$ gives the quadratic $2\mathcal{E}r^2 + 2GMr - h^2 = 0$, with roots

$$r = \frac{-GM \pm \sqrt{G^2M^2 + 2\mathcal{E}h^2}}{2\mathcal{E}}.$$

Substituting $\mathcal{E} = -GM/(2a)$ and $h^2 = GMa(1 - e^2)$, which are Eqs. (6.33) and (6.35) divided by μ and μ^2, gives $2\mathcal{E}h^2 = -G^2M^2(1 - e^2)$. The discriminant becomes $G^2M^2e^2$, and with $1/(2\mathcal{E}) = -a/GM$,

$$\boxed{r_{\min} = a(1 - e), \qquad r_{\max} = a(1 + e),}$$

the periapsis and apoapsis distances. The turning points of the effective potential and the geometry of the conic are two descriptions of the same orbit.

(c) From $\mathcal{E} = -GM/(2a)$. $\mathcal{E} = \mathcal{E}_{\text{circle}}$ forces $r_{\min} = r_{\max}$, so $e = 0$ (circular). $\mathcal{E} = 0$ forces $a \to \infty$, so $e = 1$ (parabolic). $\mathcal{E} > 0$ forces $a < 0$, so $e > 1$ (hyperbolic). This is the classification in the table of Section 6.5.

(d) With $GM = 1.327 \times 10^{20}\,\text{m}^3\,\text{s}^{-2}$, $r_{\text{circle}} = h^2/GM = 1.260 \times 10^{11}\,\text{m}$. From $\mathcal{E} = -GM/(2a)$, $a = -GM/(2\mathcal{E}) = 1.500 \times 10^{11}\,\text{m}$. Then $h^2 = GMa(1 - e^2)$ gives $1 - e^2 = h^2/(GMa) = 0.8400$, so

$$\boxed{e \simeq 0.400, \qquad r_{\min} = a(1 - e) \simeq 9.00 \times 10^{10}\,\text{m}, \qquad r_{\max} = a(1 + e) \simeq 2.10 \times 10^{11}\,\text{m}.}$$

6.19.6 Problem 6: Hohmann Transfer Impulses and Energy Budget

Problem Statement: A satellite of mass $m = 1000\,\text{kg}$ transfers from an initial circular orbit of period $P_1 = 6\,\text{h}$ to a geostationary orbit ($P_{\text{GEO}} \approx 86164\,\text{s}$), using $GM_\oplus = 3.986 \times 10^{14}\,\text{m}^3\,\text{s}^{-2}$.

(a) Find the intermediate transfer eccentricity e.

(b) Find the impulsive velocity boosts $\Delta v_1, \Delta v_2$ and the required impulses $m\Delta v_1$, $m\Delta v_2$.

(c) Find the total energy and angular momentum supplied.

Solution:

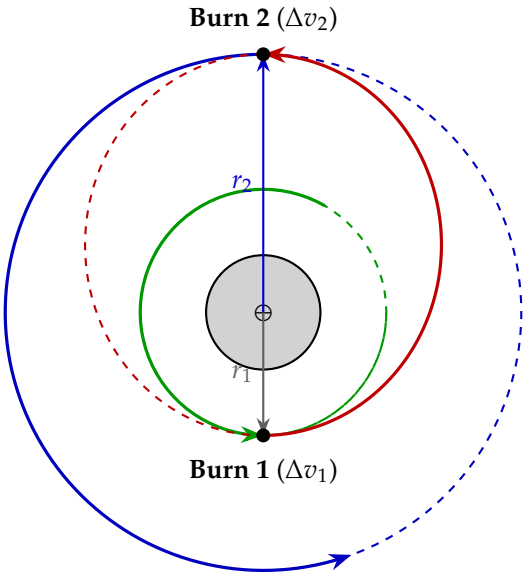


Figure 6.9: A Hohmann transfer ellipse between an initial circular orbit of radius r_1 and a geostationary orbit of radius r_2.

(a) By Kepler's third law (6.49), the initial and final radii are

$$r_1 = \left(\frac{GM_\oplus P_1^2}{4\pi^2}\right)^{1/3} \simeq 1.676 \times 10^7\,\text{m}, \qquad r_2 = \left(\frac{GM_\oplus P_{\text{GEO}}^2}{4\pi^2}\right)^{1/3} \simeq 4.216 \times 10^7\,\text{m}.$$

The transfer ellipse touches both circles, with periapsis $r_1 = a_t(1-e)$ and apoapsis $r_2 = a_t(1+e)$. So $a_t = (r_1 + r_2)/2$, and the eccentricity is

$$\boxed{e = \frac{r_2 - r_1}{r_2 + r_1} \simeq 0.431}.$$

(b) The circular speeds are $v_{c1} = \sqrt{GM_\oplus/r_1} \simeq 4.876\,\text{km/s}$ and $v_{c2} = \sqrt{GM_\oplus/r_2} \simeq 3.075\,\text{km/s}$. At perigee of the transfer orbit, the vis-viva equation (6.36) gives

$$v_p = \sqrt{GM_\oplus(2/r_1 - 1/a_t)} \simeq 5.833\,\text{km/s}.$$

Thus

$$\Delta v_1 = v_p - v_{c1} \simeq 0.957\,\text{km/s} \implies \boxed{m\Delta v_1 \simeq 9.57 \times 10^5\,\text{N s}}.$$

At apogee, $v_a = \sqrt{GM_\oplus(2/r_2 - 1/a_t)} \simeq 2.319\,\text{km/s}$. Thus

$$\Delta v_2 = v_{c2} - v_a \simeq 0.755\,\text{km/s} \implies \boxed{m\Delta v_2 \simeq 7.55 \times 10^5\,\text{N s}}.$$

(c) The energy of a circular orbit is $-GM_\oplus m/(2r)$ by Eq. (6.33), and its angular momentum is $m\sqrt{GM_\oplus r}$ by Eq. (6.35) with $e = 0$. The total energy and angular momentum supplied are

$$\boxed{\Delta E = -\frac{GM_\oplus m}{2r_2} + \frac{GM_\oplus m}{2r_1} \simeq 7.16 \times 10^9\,\text{J}},$$

$$\boxed{\Delta L = m\sqrt{GM_\oplus r_2} - m\sqrt{GM_\oplus r_1} \simeq 4.79 \times 10^{13}\,\text{kg m}^2\,\text{s}^{-1}}.$$

6.19.7 Problem 7: Determining Orbital Elements from Initial State Vectors

Problem Statement: Given the initial relative position $\vec{r}_0$ and velocity $\dot{\vec{r}}_0$ at $t = 0$, derive the orbital elements $(a, e, \Omega, \iota, \omega, t_0)$ of a bound orbit.

Solution:

(a) **Conserved quantities.** The specific energy $\mathcal{E} = E/\mu = \dot{\vec{r}}_0^{\,2}/2 - GM/r_0$, the specific angular momentum $\vec{h} = \vec{L}/\mu = \vec{r}_0 \times \dot{\vec{r}}_0$, and the eccentricity vector $\vec{e} = \vec{A}/(GM\mu^2) = \dot{\vec{r}}_0 \times \vec{h}/GM - \hat{r}_0$, Eq. (6.71).

(b) **Orbital elements.** From the vis-viva equation (6.36) and Eqs. (6.69), (6.49) and (6.11),

$$a = \left(\frac{2}{r_0} - \frac{\dot{\vec{r}}_0^{\,2}}{GM} \right)^{-1}, \qquad e = |\vec{e}|, \qquad n = \sqrt{\frac{GM}{a^3}}, \qquad \iota = \arccos\left(\frac{\vec{h} \cdot \hat{R}}{h} \right).$$

The node vector $\vec{N} = \hat{R} \times \vec{h}$ points to the ascending node, since $\hat{n}$ is perpendicular to both $\hat{R}$ and $\hat{L}$. Then

$$\Omega = \operatorname{atan2}(\vec{N} \cdot \hat{\delta},\ \vec{N} \cdot \hat{\alpha}), \qquad \omega = \operatorname{atan2}\left(\frac{\hat{h} \cdot (\vec{N} \times \vec{e})}{Ne}, \frac{\vec{N} \cdot \vec{e}}{Ne} \right),$$

from Eqs. (6.9) and (6.70).

(c) **Time of periapsis.** The initial true anomaly has $\cos v_0 = \vec{e} \cdot \hat{r}_0 / e$, with $\sin v_0$ of the same sign as $\dot{r}_0 = \hat{r}_0 \cdot \dot{\vec{r}}_0$. The eccentric anomaly u_0 follows from Eq. (6.45), $\tan(u_0/2) = \sqrt{(1-e)/(1+e)}\,\tan(v_0/2)$. Kepler's equation at $t = 0$ gives $\ell_0 = -n t_0 = u_0 - e \sin u_0$, so the time of periapsis passage is

$$t_0 = -\frac{u_0 - e \sin u_0}{n}.$$

6.19.8 Problem 8: Continuous Tangential Thrust Orbital Evolution

Problem Statement: A 500 kg satellite initially moves in a circular Earth orbit of period $P = 8\,\mathrm{h}$. At $t = 0$, a thruster of force $F_\mathrm{T} = 100\,\mathrm{N}$ is switched on, always directed along the instantaneous velocity $\dot{\vec{r}}$.

(a) Find the initial orbital radius $a(0)$.

(b) Derive the rate of change of the semi-major axis $\dot{a}$.

(c) Derive the rate of change of the eccentricity $\dot{e}$ from the exact rate of the Laplace–Runge–Lenz vector $\vec{A}$.

(d) Can the coupled evolution equations for (a, e) be solved analytically?

Essential Ideas:

- **Thrust acceleration and initial radius**: The satellite's mass is negligible next to the Earth's, so $\mu = m = 500\,\mathrm{kg}$. The thrust acceleration is $F_\mathrm{T}/\mu = 100\,\mathrm{N}/500\,\mathrm{kg} = 0.2\,\mathrm{m\,s^{-2}}$. For an initially circular orbit ($e = 0$) of period $P = 8\,\mathrm{h} = 28{,}800\,\mathrm{s}$ around the Earth ($GM_\oplus \approx 3.986 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$), Kepler's third law gives

$$a(0) = \left(\frac{GM_\oplus P^2}{4\pi^2} \right)^{1/3} \simeq 2.03 \times 10^7\,\mathrm{m} \approx 20{,}300\,\mathrm{km}.$$

- **Perturbation force vector**: The thrust is tangential to the path, $\vec{F} = F_{\mathrm{T}}\,\dot{\vec{r}}/|\dot{\vec{r}}|$. In polar coordinates the velocity is $\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi} = \dot{r}\,\hat{r} + \frac{L}{\mu r}\,\hat{\phi}$, so $\vec{F} = F_r\hat{r} + F_\phi\hat{\phi}$ has the components

$$F_r = F_{\mathrm{T}}\frac{\dot{r}}{|\dot{\vec{r}}|}, \qquad F_\phi = \frac{F_{\mathrm{T}}L}{\mu r|\dot{\vec{r}}|}.$$

  With $\xi \equiv \mu r\dot{r}/L = e\sin v/(1+e\cos v)$ (Eq. (6.119)), the radial component is $F_r = \xi F_\phi$.

Detailed Solutions:

(a) **Initial semi-major axis**.

$$\boxed{a(0) = \left(\frac{GM_\oplus P^2}{4\pi^2}\right)^{1/3} = \left(\frac{3.986\times10^{14}\times(28800)^2}{4\pi^2}\right)^{1/3} \simeq 2.03\times10^7\ \mathrm{m}}.$$

(b) **Evolution of the semi-major axis $\dot{a}$**. The work done per unit time by the tangential thrust is, by Eq. (6.96),

$$\dot{E} = \vec{F}\cdot\dot{\vec{r}} = F_{\mathrm{T}}|\dot{\vec{r}}|.$$

The energy $E = -GM_\oplus\mu/(2a)$ changes at the rate $\dot{E} = (GM_\oplus\mu/2a^2)\dot{a}$, so

$$\frac{GM_\oplus\mu}{2a^2}\dot{a} = F_{\mathrm{T}}|\dot{\vec{r}}| \implies \dot{a} = \frac{2a^2}{GM_\oplus}\frac{F_{\mathrm{T}}}{\mu}\,|\dot{\vec{r}}|.$$

The vis-viva equation (6.36) with $r = a(1-e^2)/(1+e\cos v)$ gives

$$|\dot{\vec{r}}|^2 = GM_\oplus\left(\frac{2}{r}-\frac{1}{a}\right) = GM_\oplus\,\frac{2(1+e\cos v)-(1-e^2)}{a(1-e^2)} = \frac{GM_\oplus}{a}\,\frac{1+2e\cos v+e^2}{1-e^2},$$

so the exact rate of change of the semi-major axis is

$$\boxed{\frac{da}{dt} = \frac{2F_{\mathrm{T}}}{\mu}\sqrt{\frac{a^3}{GM_\oplus}}\sqrt{\frac{1+2e\cos v+e^2}{1-e^2}}}. \tag{6.240}$$

(c) **Evolution of the eccentricity $\dot{e}$ via the Laplace–Runge–Lenz vector**. The Laplace–Runge–Lenz vector $\vec{A} = \vec{p}\times\vec{L} - GM\mu^2\hat{r}$ has magnitude $A = GM_\oplus\mu^2 e$ and points to the periapsis, $\hat{A} = \cos v\,\hat{r} - \sin v\,\hat{\phi}$ (Eq. (6.129)).

Under a general planar perturbing force $\vec{F} = F_r\hat{r} + F_\phi\hat{\phi}$, the exact rate of change of $\vec{A}$ is Eq. (6.127),

$$\dot{\vec{A}} = \vec{F}\times\vec{L} + \vec{p}\times(\vec{r}\times\vec{F}) = L\left[2F_\phi\hat{r} - (F_r+\xi F_\phi)\hat{\phi}\right].$$

Since $\hat{A}\cdot\hat{A} = 1$ gives $\hat{A}\cdot\dot{\hat{A}} = 0$, the rate $\dot{A}$ is the projection of $\dot{\vec{A}}$ on $\hat{A}$,

$$\begin{aligned}\dot{A} &= \dot{\vec{A}}\cdot\hat{A} = L\left[2F_\phi\hat{r} - (F_r+\xi F_\phi)\hat{\phi}\right]\cdot(\cos v\,\hat{r} - \sin v\,\hat{\phi})\\ &= L\left[2F_\phi\cos v + (F_r+\xi F_\phi)\sin v\right]\\ &= L\left[F_r\sin v + F_\phi(2\cos v + \xi\sin v)\right].\end{aligned}$$

Evaluating this on the instantaneous osculating orbit with $A = GM_\oplus\mu^2 e$ and $L = \mu\sqrt{GM_\oplus a(1-e^2)}$ gives Gauss's equation for the eccentricity, Eq. (6.135),

$$\dot{e} = \sqrt{\frac{a(1-e^2)}{GM_\oplus\mu^2}}\left[F_r\sin v + F_\phi(2\cos v + \xi\sin v)\right]. \tag{6.241}$$

Now we substitute the components for tangential thrust, $F_r = \xi F_\phi$ and $F_\phi = \dfrac{L}{r}\dfrac{F_\mathrm{T}}{\mu|\dot{\vec{r}}|}$,

$$\begin{aligned} F_r \sin v + F_\phi(2\cos v + \xi \sin v) &= \frac{F_\mathrm{T} L}{\mu r|\dot{\vec{r}}|}\left[\xi \sin v + 2\cos v + \xi \sin v\right] \\ &= \frac{2F_\mathrm{T} L}{\mu r|\dot{\vec{r}}|}\left(\cos v + \xi \sin v\right). \end{aligned}$$

Using $\xi = e\sin v/(1 + e\cos v)$,

$$\cos v + \xi \sin v = \cos v + \frac{e\sin^2 v}{1 + e\cos v} = \frac{\cos v(1 + e\cos v) + e\sin^2 v}{1 + e\cos v} = \frac{\cos v + e}{1 + e\cos v}.$$

With $\dfrac{L}{r} = \dfrac{\mu\sqrt{GM_\oplus a(1-e^2)}}{a(1-e^2)/(1+e\cos v)} = \dfrac{\mu\sqrt{GM_\oplus}\,(1 + e\cos v)}{\sqrt{a(1-e^2)}}$,

$$\begin{aligned} \dot{e} &= \sqrt{\frac{a(1-e^2)}{GM_\oplus \mu^2}} \cdot \frac{2F_\mathrm{T}}{\mu|\dot{\vec{r}}|}\left[\frac{\mu\sqrt{GM_\oplus}(1 + e\cos v)}{\sqrt{a(1-e^2)}}\right]\left[\frac{\cos v + e}{1 + e\cos v}\right] \\ &= \frac{2F_\mathrm{T}}{\mu}\frac{\cos v + e}{|\dot{\vec{r}}|}. \end{aligned}$$

Writing $|\dot{\vec{r}}|$ in terms of the elements with the vis-viva equation,

$$\boxed{\frac{de}{dt} = \frac{2F_\mathrm{T}}{\mu}\frac{\cos v + e}{|\dot{\vec{r}}|} = \frac{2F_\mathrm{T}}{\mu}\sqrt{\frac{a(1-e^2)}{GM_\oplus}}\,\frac{\cos v + e}{\sqrt{1 + 2e\cos v + e^2}}}. \tag{6.242}$$

Differentiating $e^2 = 1 + 2\mathcal{E}h^2/(GM_\oplus)^2$ (Eq. (6.38) with $\mathcal{E} = E/\mu$ and $h = L/\mu$) would give $\dot{e}$ divided by e, an undefined $0/0$ at $e = 0$. The rate of the Runge–Lenz vector avoids this and gives the finite initial rate $\dot{e}\big|_{e=0} = (2F_\mathrm{T}/\mu)\sqrt{a/GM_\oplus}\cos v$.

(d) **Analytical solvability**. Equations (6.240) and (6.242) do not form a closed system for a and e, because both rates depend on the orbital phase $v(t)$. The phase itself evolves by Eq. (6.148),

$$\dot{v} = \frac{L}{\mu r^2} - \dot{\omega},$$

where the periapsis rate $\dot{\omega}$ is also driven by the thrust. The thrust changes the orbit throughout each revolution, so the coupled system $(a(t), e(t), v(t))$ has no closed-form solution in elementary functions. In mission design it is solved by numerical integration, or by *orbit averaging* over each orbital period (Section 6.16),

$$\langle \dot{a} \rangle = \frac{1}{2\pi}\int_0^{2\pi} \dot{a}\, d\ell = \frac{1}{P}\int_0^{2\pi} \dot{a}\,\frac{\mu r^2}{L}\, dv.$$

6.19.9 Problem 9: Double-Lined Spectroscopic Binary — Radial Velocities and Masses

Problem Statement: In a *double-lined spectroscopic binary* (SB2), we see absorption lines from *both* stars in the combined spectrum. Each line is Doppler-shifted by that star's own orbital motion around the center of mass. A Doppler shift only measures the *radial velocity* $v_r \equiv \dot{\vec{x}} \cdot \hat{R}$, the part of a star's velocity $\dot{\vec{x}}$ along the line of sight $\hat{R}$. Motion across the sky, perpendicular to $\hat{R}$, does not stretch or compress the wavelength, so it leaves no Doppler signal at all. That

sideways motion only shows up as proper motion (Chapter 3), not in the spectrum. Tracking each star's wavelength shift separately gives two radial-velocity curves, $v_{r,1}(t)$ and $v_{r,2}(t)$. Both share the same period P and eccentricity e, but usually have different semi-amplitudes K_1 and K_2 (Figures 6.10 and 6.11). A particular system is measured to have $P = 4.20\,\mathrm{d}$, $e = 0.15$, $K_1 = 88\,\mathrm{km\,s^{-1}}$, and $K_2 = 132\,\mathrm{km\,s^{-1}}$.

(a) Without knowing the orbital inclination ι, what can already be said about the mass ratio m_1/m_2?

(b) Derive $(m_1 + m_2)\sin^3\iota$ in terms of P, K_1, K_2, and e, and evaluate it numerically.

(c) Hence find $m_1 \sin^3\iota$ and $m_2 \sin^3\iota$ individually.

(d) This same system is also observed to be eclipsing, so $\iota \simeq 90°$. What are the actual masses m_1 and m_2?

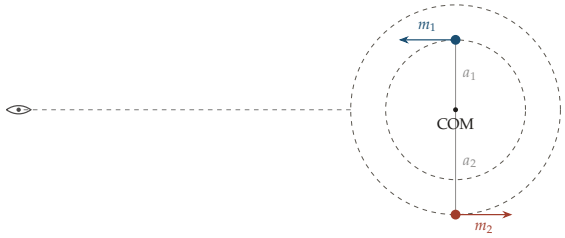


Figure 6.10: A binary at quadrature, with m_1, m_2, and the COM in a line. m_1 moves away from and m_2 towards the observer, along the line of sight $\hat{R}$.

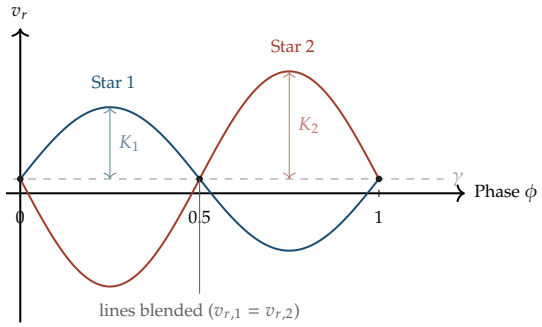


Figure 6.11: Radial-velocity curves of both stars for a circular orbit. Two sine curves in antiphase about γ, with semi-amplitudes $K_1 \neq K_2$.

Essential Ideas:

- Start from $\vec{x}_1 = (m_2/M)\vec{r}$, $\vec{x}_2 = -(m_1/M)\vec{r}$ (Section 6.1.2), shifted by the centre-of-mass motion $\vec{V}_{\mathrm{CM}}t$. Differentiate and project onto $\hat{R}$,

$$v_{r,1} - \gamma = \frac{m_2}{M}\, v_{r,\mathrm{rel}}, \qquad v_{r,2} - \gamma = -\frac{m_1}{M}\, v_{r,\mathrm{rel}}, \qquad \gamma \equiv \vec{V}_{\mathrm{CM}} \cdot \hat{R}.$$

  Here $v_{r,\mathrm{rel}} \equiv v_{r,1} - v_{r,2}$, and γ is constant (Eq. (6.4)). This holds at every phase. So

$$\frac{K_1}{K_2} = \frac{m_2}{m_1},$$

  whatever ι and e are.

- A Keplerian orbit's line-of-sight speed has semi-amplitude $K = 2\pi a \sin\iota/(P\sqrt{1-e^2})$ (Problem 11). Apply this to the relative orbit ($a = a_1 + a_2$). Then $v_{r,\mathrm{rel}} = v_{r,1} - v_{r,2}$ has amplitude $K_1 + K_2$.

Detailed Solutions:

(a) **Mass ratio**.

$$\boxed{\frac{m_1}{m_2} = \frac{K_2}{K_1} = \frac{132}{88} = 1.50.}$$

(b) **Total mass**. Apply $K = 2\pi a \sin\iota/(P\sqrt{1-e^2})$ to the relative orbit, whose amplitude is K_1+K_2, so $a\sin\iota = (K_1+K_2)P\sqrt{1-e^2}/(2\pi)$. Kepler's third law (6.49) is $P^2 = 4\pi^2 a^3/[G(m_1+$

$m_2)$], so

$$(m_1+m_2)\sin^3\iota = \frac{4\pi^2(a\sin\iota)^3}{GP^2} = \frac{4\pi^2}{GP^2}\,\frac{(K_1+K_2)^3P^3(1-e^2)^{3/2}}{8\pi^3}$$
$$= \frac{P(K_1+K_2)^3(1-e^2)^{3/2}}{2\pi G}.$$

With $P = 3.629\times10^5\,\mathrm{s}$, $K_1+K_2 = 2.20\times10^5\,\mathrm{m\,s^{-1}}$ and $(1-e^2)^{3/2} = 0.966$,

$$\boxed{(m_1+m_2)\sin^3\iota \simeq 8.91\times10^{30}\,\mathrm{kg} \simeq 4.48\,M_\odot.}$$

(c) **Individual masses**. Use $m_1 = \dfrac{K_2}{K_1+K_2}(m_1+m_2)$ and $m_2 = \dfrac{K_1}{K_1+K_2}(m_1+m_2)$, the same centre-of-mass weighting as in (a).

$$\boxed{m_1\sin^3\iota \simeq 2.69\,M_\odot, \qquad m_2\sin^3\iota \simeq 1.79\,M_\odot.}$$

(d) **Eclipsing case**. $\iota \simeq 90°$, so $\sin^3\iota \simeq 1$ and the factors in (c) drop out.

$$\boxed{m_1 \simeq 2.69\,M_\odot, \qquad m_2 \simeq 1.79\,M_\odot.}$$

Double-lined eclipsing binaries are valuable for exactly this reason. Both masses, and from the light curve both radii, come out with almost no model assumptions.

6.19.10 Problem 10: Dynamical Mass of a Host Halo from a Satellite's Orbit

Problem Statement: A satellite of mass m_s, for example a dwarf galaxy or globular cluster, orbits a much bigger dark-matter halo of mass M_h. We measure its present distance r from the Galactic center and its full relative velocity $\dot{\vec{r}}$ (from proper motion plus radial velocity). We also estimate its orbital eccentricity e independently, for example from similar orbits in cosmological simulations. As in Section 6.1.2, let $M \equiv m_s + M_h$ be the total mass and $\mu \equiv m_s M_h/M$ the reduced mass. If the satellite sits well outside the halo's inner mass, we can treat the halo as a point mass (P2). Then the satellite follows an ordinary Keplerian orbit, with energy $E = -GM\mu/(2a)$ (Eq. (6.33)) and angular momentum $\vec{L} = \mu\,\vec{r}\times\dot{\vec{r}}$, $L^2 = GM\mu^2 a(1-e^2)$ (Eq. (6.35)).

(a) Show that μ cancels out of both the vis-viva equation and the angular-momentum relation, leaving two equations in the directly observable quantities r, $|\dot{\vec{r}}|$, and $h \equiv |\vec{r}\times\dot{\vec{r}}| = L/\mu$. Combine them to show that the total mass M satisfies

$$\frac{G^2(1-e^2)}{h^2}M^2 - \frac{2G}{r}M + \dot{\vec{r}}^{\,2} = 0,$$

and solve for M in terms of $r, |\dot{\vec{r}}|, h, e$.

(b) Suppose instead the satellite happens to be caught exactly at apocenter, where the velocity is purely tangential ($\dot{\vec{r}} \perp \vec{r}$) and $r = r_{\mathrm{apo}} = a(1+e)$. Show that the mass estimate then becomes the much simpler

$$M = \frac{v_{\mathrm{apo}}^2\, r_{\mathrm{apo}}}{G(1-e)},$$

where $v_{\mathrm{apo}} = |\dot{\vec{r}}|$ at apocenter.

(c) A dwarf satellite is observed at apocenter with $r_{\mathrm{apo}} = 250\,\mathrm{kpc}$ and $v_{\mathrm{apo}} = 95\,\mathrm{km\,s^{-1}}$, and its orbit is modeled to have $e = 0.6$. Evaluate M, in kg and in $M_\odot$, and argue that it is an excellent estimate of M_h alone.

(d) Eccentricity is only one possible choice of the "extra" datum used here. What property of the governing equations makes some third quantity beyond r and $|\dot{\vec{r}}|$ necessary at all, and what else could play that role?

(e) This method returns only the total mass $M = m_s + M_h$, never m_s and M_h separately. Explain why no amount of additional *orbital* data — more precise r, $\dot{\vec{r}}$, or e, or even more satellites on more orbits — could ever split M into m_s and M_h individually, and say what kind of information could.

Essential Ideas:

- μ cancels. In the vis-viva equation $\dot{\vec{r}}^2 = GM(2/r - 1/a)$, Eq. (6.36), the orbit's shape and motion depend on m_s and M_h only through M. μ drops out once E and L are turned into relations among r, $|\dot{\vec{r}}|$ and $h \equiv |\vec{r} \times \dot{\vec{r}}|$.
- There are three unknowns (a, e, M) but only two equations relating r, $|\dot{\vec{r}}|$ and h. So one more independent piece of data is needed, and e is the usual choice.
- At apocenter $\dot{r} = 0$, so the velocity is purely tangential and $r_{\text{apo}} = a(1+e)$ directly, without needing h.
- Test-particle limit. Since $m_s \ll M_h$, $M = m_s + M_h$ is almost exactly M_h.
- **Degeneracy, not just ignorance**. μ is the one combination of m_s and M_h that is *not* their sum, and it cancels completely from both relations. So every orbital observable depends on the masses only through M, and any split of M gives the identical orbit. This is different from Problem 9's mass *ratio*, where two tracers gave one comparison. Here there is only one tracer, so only one orbit.

Detailed Solutions:

(a) **General mass estimator**. Divide $E = \frac{1}{2}\mu\dot{\vec{r}}^2 - \dfrac{GM\mu}{r} = -\dfrac{GM\mu}{2a}$ by μ to recover vis-viva. Divide $L^2 = GM\mu^2 a(1-e^2)$ by μ^2 and write $h \equiv L/\mu = |\vec{r} \times \dot{\vec{r}}|$, which needs no mass.

$$\dot{\vec{r}}^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right), \qquad h^2 = GMa(1-e^2) \implies a = \frac{h^2}{GM(1-e^2)}.$$

Substituting $1/a$ into the first,

$$\dot{\vec{r}}^2 = \frac{2GM}{r} - \frac{G^2M^2(1-e^2)}{h^2} \implies \boxed{\frac{G^2(1-e^2)}{h^2}M^2 - \frac{2G}{r}M + \dot{\vec{r}}^2 = 0,}$$

a quadratic in M, giving

$$\boxed{M = \frac{h^2}{G(1-e^2)}\left[\frac{1}{r} \pm \sqrt{\frac{1}{r^2} - \frac{(1-e^2)\dot{\vec{r}}^2}{h^2}}\right].}$$

The product of the two roots is $h^2\dot{\vec{r}}^2/[G^2(1-e^2)] > 0$ and their sum is positive, so both roots are positive and both are possible orbits. With $\lambda = h^2/(GM)$ the conic gives $e\cos v = \lambda/r - 1$. The + root, the larger M, has $r > \lambda$ and puts the satellite on the far half of its orbit. The − root puts it on the near half. The sign of $\dot{r}$ does not decide between them, so one more fact about the phase is needed. At apocenter, part (b), it is the + root.

(b) **Apocenter simplification**. Here $\dot{r} = 0$, and $r = r_{\text{apo}} = a(1+e)$, so $1/a = (1+e)/r_{\text{apo}}$.

$$v_{\text{apo}}^2 = GM\left(\frac{2}{r_{\text{apo}}} - \frac{1+e}{r_{\text{apo}}}\right) = \frac{GM(1-e)}{r_{\text{apo}}} \implies \boxed{M = \frac{v_{\text{apo}}^2\, r_{\text{apo}}}{G(1-e)}.}$$

This matches the + root of part (a) at apocenter, where $h = r_{\rm apo} v_{\rm apo}$ and the square root becomes $e/r_{\rm apo}$.

(c) **Numerical evaluation**. Use $r_{\rm apo} = 7.714 \times 10^{21}$ m, $v_{\rm apo} = 9.5 \times 10^{4}\,\mathrm{m\,s^{-1}}$, $e = 0.6$.

$$M = \frac{(9.5\times10^4)^2(7.714\times10^{21})}{(6.674\times10^{-11})(1-0.6)} \implies \boxed{M \simeq 2.61\times10^{42}\,\mathrm{kg} \simeq 1.31\times10^{12}\,M_\odot,}$$

which is typical for a Milky-Way-sized halo. Dwarf-satellite masses (10^4–$10^8\,M_\odot$) are tiny next to this, at most about 10^{-4} of it, so $M \simeq M_h$ to four figures or better.

(d) **Why a third datum is unavoidable**. r and $|\dot{\vec{r}}|$ give two independent numbers, but the orbit has three unknowns (a, e, M), related by only two equations, vis-viva and $h^2 = GMa(1-e^2)$. So one more independent piece of data is needed to close the system. It could be e, as used here, or h itself, or the time elapsed since a known pericenter passage, found from Kepler's equation (Section 6.9). This last option is the basis of the Local Group timing argument for the Milky Way–M31 mass.

(e) **The barycenter location is the missing parameter**. To specify $\vec{x}_s(t)$ and $\vec{x}_h(t)$ separately, we need the six orbital elements $(a, e, \iota, \Omega, \omega, t_0)$ of the *relative* orbit $\vec{r} \equiv \vec{x}_s - \vec{x}_h$, the total mass M, and one more number. The six elements describe only the shape, orientation, and timing of $\vec{r}(t)$. They say nothing about how the total mass M is split between the two bodies, and it is that split we are still missing. The extra number is the mass fraction M_h/M, which is the same as the barycenter's location on the line joining the two bodies. Parts (a)–(d) give only the relative orbit and M. The relative orbit cannot see M_h/M, and this is an exact symmetry, not just a practical limit. Newton's third law gives $\ddot{\vec{x}}_s = -GM_h\hat{r}/r^2$ and $\ddot{\vec{x}}_h = Gm_s\hat{r}/r^2$, so

$$\ddot{\vec{r}} = \ddot{\vec{x}}_s - \ddot{\vec{x}}_h = -\frac{GM}{r^2}\hat{r},$$

which depends on m_s and M_h only through M. So $\vec{r}(t)$ stays exactly the same under $m_s \to m_s + \delta$, $M_h \to M_h - \delta$ at fixed M. No relative-orbit measurement, however precise, can pin down M_h/M.

The individual position $\vec{x}_s = \vec{X} + (M_h/M)\vec{r}$ does depend on M_h/M. And $M_h/M = |\vec{x}_s - \vec{X}|/r$ is a purely geometric ratio, so it needs no dynamics to measure. This is the missing parameter. It is the same idea as Problem 1, where $m_1/m_2 = a_2/a_1$ came from each star's distance to the visible barycenter.

For a satellite and its halo, the problem is one of scale, not principle. Since $m_s/M_h \sim 10^{-4}$–10^{-8}, the barycenter sits only that tiny fraction of r from the halo's center, far too close to measure astrometrically (unlike Problem 1's pair of comparable masses). Without it, we estimate m_s some other way, for example from its luminosity via a mass-to-light ratio, or from its own internal velocity dispersion via the virial theorem ($M \sim \sigma_v^2 R/G$, Section 10.2). Then $M_h = M - m_s$.

6.19.11 Problem 11: The Line-of-Sight Direction, in Matrices and Without Them

Problem Statement: Problem 9 projected an orbital velocity onto the line of sight $\hat{R}$, through the semi-amplitude K and the systemic velocity $\gamma = \vec{V}_{\rm CM} \cdot \hat{R}$, without ever pinning down what $\hat{R}$ *is* in terms of the orbital elements (ι, Ω, ω).

(a) Starting from the rotation relating the orbital frame $(\hat{n}, \hat{y}, \hat{L})$ to the observer frame $(\hat{\alpha}, \hat{\delta}, \hat{R})$, Eq. (6.7), invert it to express $\hat{R}$ in terms of $\hat{n}, \hat{y}, \hat{L}$. Show that the result does not depend on Ω at all,

$$\hat{R} = \sin\iota\,\hat{y} + \cos\iota\,\hat{L}.$$

(b) Rederive the same identity without any matrix algebra. Use only two facts. First, the node line $\hat{n}$ is, by construction, the intersection of the orbital plane and the sky plane. Second, the inclination is the angle between the two planes' normals, $\hat{L}$ and $\hat{R}$.

(c) A body follows a Keplerian orbit of elements (a, e, ι, ω) in the plane spanned by $\hat{n}, \hat{y}$, at true anomaly v (so its polar angle from $\hat{n}$ is $\phi = \omega + v$). Using $\hat{R} = \sin\iota\,\hat{y} + \cos\iota\,\hat{L}$ together with the fact that the velocity lies entirely in the orbital plane, show that the line-of-sight (radial) velocity $v_r = \dot{\vec{r}} \cdot \hat{R}$ takes the form

$$v_r = \gamma + K\big[\cos(\omega + v) + e\cos\omega\big],$$

and identify K in terms of G, M, a, e, ι.

(d) Using Kepler's third law, show that this K is the semi-amplitude

$$K = \frac{2\pi a \sin\iota}{P\sqrt{1-e^2}}$$

quoted without proof in Problem 9.

Essential Ideas:

- A rotation matrix is orthogonal, so $\mathbf{R}^{-1} = \mathbf{R}^T$ (Chapter 2, Eq. (2.4)). Undoing it costs no more than transposing and reversing the product order.
- Simple geometric facts about an orthonormal triad, such as which vectors are perpendicular, or what a named angle is the cosine of, can rebuild a rotation result with less algebra than the matrix itself.
- Angular momentum conservation confines $\vec{r}$ and $\dot{\vec{r}}$ to the orbital plane, so $\vec{r} \cdot \hat{L} = \dot{\vec{r}} \cdot \hat{L} = 0$. This drops $\hat{L}$ from $v_r = \dot{\vec{r}} \cdot \hat{R}$ in part (c).

Detailed Solutions:

(a) **Matrix version**. Both 3×3 factors in Eq. (6.7) are rotation matrices, so they are orthogonal, $\mathbf{R}^{-1} = \mathbf{R}^T$. Inverting the product reverses the order and transposes each factor.

$$\begin{pmatrix}\hat{\alpha}\\ \hat{\delta}\\ \hat{R}\end{pmatrix} = \begin{pmatrix}\cos\Omega & \sin\Omega & 0\\ -\sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}^T \begin{pmatrix}1 & 0 & 0\\ 0 & \cos\iota & \sin\iota\\ 0 & -\sin\iota & \cos\iota\end{pmatrix}^T \begin{pmatrix}\hat{n}\\ \hat{y}\\ \hat{L}\end{pmatrix}$$

$$= \begin{pmatrix}\cos\Omega & -\sin\Omega & 0\\ \sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & \cos\iota & -\sin\iota\\ 0 & \sin\iota & \cos\iota\end{pmatrix}\begin{pmatrix}\hat{n}\\ \hat{y}\\ \hat{L}\end{pmatrix}.$$

The third row of the product is the third row of the first matrix, $(0, 0, 1)$, times the second matrix. That picks out the second matrix's third row, $(0, \sin\iota, \cos\iota)$, so

$$\hat{R} = \underbrace{(0)}_{\hat{n}\text{-coefficient}}\,\hat{n} + \sin\iota\,\hat{y} + \cos\iota\,\hat{L} \implies \boxed{\hat{R} = \sin\iota\,\hat{y} + \cos\iota\,\hat{L},}$$

since Ω enters only the first two rows of the first matrix, which never touch the row producing $\hat{R}$.

(b) **Geometric version**. $\hat{n}$ is the intersection of the orbital and sky planes. So it is perpendicular to both planes' normals, $\hat{n} \cdot \hat{L} = 0$ and $\hat{n} \cdot \hat{R} = 0$. Since $\{\hat{n}, \hat{y}, \hat{L}\}$ is orthonormal, any unit vector with zero $\hat{n}$-component is a combination of $\hat{y}$ and $\hat{L}$ alone.

$$\hat{R} = (\hat{R} \cdot \hat{y})\,\hat{y} + (\hat{R} \cdot \hat{L})\,\hat{L}.$$

By definition, ι is the angle between the planes' normals, so $\hat{R}\cdot\hat{L}=\cos\iota$. Since $\hat{R}\cdot\hat{n}=0$ and $\hat{R}$ is a unit vector, $(\hat{R}\cdot\hat{y})^2=\sin^2\iota$. The orientation convention, that $\hat{y}$ leads $\hat{n}$ by 90° toward $\hat{R}$, fixes the sign.

$$\boxed{\hat{R}=\sin\iota\,\hat{y}+\cos\iota\,\hat{L},}$$

the same as part (a), with no matrix multiplied.

(c) **Radial velocity**. Angular momentum conservation confines the orbit to the plane spanned by $\hat{n},\hat{y}$, so $\dot{\vec{r}}\cdot\hat{L}=0$, and

$$v_r=\dot{\vec{r}}\cdot\hat{R}=\sin\iota\,(\dot{\vec{r}}\cdot\hat{y}).$$

With $\vec{r}=r\cos\phi\,\hat{n}+r\sin\phi\,\hat{y}$ and the planar-polar velocity $\dot{\vec{r}}=\dot{r}\,\hat{r}+r\dot{\phi}\,\hat{\phi}$,

$$\dot{\vec{r}}\cdot\hat{y}=\dot{r}\sin\phi+r\dot{\phi}\cos\phi.$$

Substitute $\dot{r}=\dfrac{Le\sin v}{\mu\lambda}$ (Eq. (6.116)) and $r\dot{\phi}=\dfrac{L}{\mu r}=\dfrac{L(1+e\cos v)}{\mu\lambda}$ (Eq. (6.113)), with $\phi=\omega+v$.

$$\dot{\vec{r}}\cdot\hat{y}=\frac{L}{\mu\lambda}\Big[e\sin v\sin(\omega+v)+(1+e\cos v)\cos(\omega+v)\Big]=\frac{L}{\mu\lambda}\Big[\cos(\omega+v)+e\cos\omega\Big],$$

using $\sin v\sin(\omega+v)+\cos v\cos(\omega+v)=\cos\omega$. Since $L=\mu\sqrt{GM\lambda}$ (Eq. (6.35)) and $\lambda=a(1-e^2)$, $L/(\mu\lambda)=\sqrt{GM/[a(1-e^2)]}$. Add back γ, Problem 9's line-of-sight velocity of the centre of mass, as the baseline.

$$v_r=\sin\iota\sqrt{\frac{GM}{a(1-e^2)}}\Big[\cos(\omega+v)+e\cos\omega\Big]+\gamma.$$

Matching to the stated form,

$$\boxed{K=\sin\iota\sqrt{\frac{GM}{a(1-e^2)}}.}$$

(d) **Consistency with Problem 9**. Kepler's third law gives $GM=4\pi^2a^3/P^2$, so

$$K=\sin\iota\sqrt{\frac{4\pi^2a^3/P^2}{a(1-e^2)}}=\sin\iota\sqrt{\frac{4\pi^2a^2}{P^2(1-e^2)}}\;\Longrightarrow\;\boxed{K=\frac{2\pi a\sin\iota}{P\sqrt{1-e^2}},}$$

the radial-velocity semi-amplitude of a spectroscopic binary, Problem 9's K.

6.19.12 Problem 12: The Gravitational Slingshot

Problem Statement: A spacecraft flies by Jupiter. Jupiter's own speed around the Sun is $V_p\simeq 13.1\,\mathrm{km\,s^{-1}}$. Relative to Jupiter, the spacecraft moves on a hyperbolic orbit with $e=1.2$ and $v_\infty=10\,\mathrm{km\,s^{-1}}$ (Eq. (6.60)). Find the change in the spacecraft's speed around the Sun, given that its speed relative to Jupiter cannot change.

Solution. In the solar frame, we drop the Jupiter–Sun terms that do not involve $\vec{x}_{\rm sc}$, and write the Lagrangian of the spacecraft,

$$\mathcal{L}=\frac{1}{2}m_{\rm sc}\dot{\vec{x}}_{\rm sc}^2+\frac{GM_\odot m_{\rm sc}}{|\vec{x}_{\rm sc}|}+\frac{Gm_{\rm sc}M_J}{|\vec{x}_{\rm sc}-\vec{X}_J(t)|}. \tag{6.243}$$

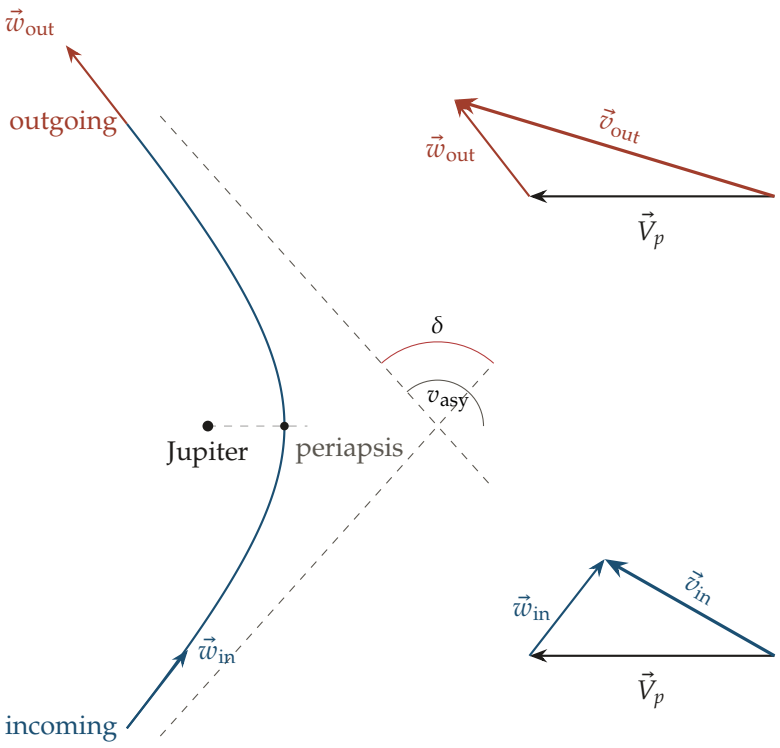


Figure 6.12: The gravitational slingshot. Left, Jupiter's frame, with the incoming and outgoing branches of the hyperbola. The dashed lines are the asymptotes, crossing at the hyperbola's center, with the angle δ between them and the angle $v_{\rm asy}$ from the periapsis direction. $\vec{w}_{\rm in}$ and $\vec{w}_{\rm out}$ are the velocities along the asymptotes. Right, $\vec{w}_{\rm in}$ and $\vec{w}_{\rm out}$ added to Jupiter's heliocentric velocity $\vec{V}_p$.

Let $\vec{r} \equiv \vec{x}_{\rm sc} - \vec{X}_J(t)$. Over the brief encounter, $\vec{X}_J(t) \approx \vec{X}_{J,0} + \vec{V}_p t$ and $|\vec{x}_{\rm sc}| \approx X_J(t)$, dropping the small term $O(r/X_J)$. Also $\dot{\vec{x}}_{\rm sc} = \vec{V}_p + \dot{\vec{r}}$, so

$$\mathcal{L} \approx \underbrace{\tfrac{1}{2} m_{\rm sc} V_p^2 + \frac{G M_\odot m_{\rm sc}}{X_J(t)}}_{g(t)} + \underbrace{m_{\rm sc} \vec{V}_p \cdot \dot{\vec{r}}}_{\frac{d}{dt}(m_{\rm sc}\vec{V}_p \cdot \vec{r})} + \frac{1}{2} m_{\rm sc} \dot{\vec{r}}^{\,2} + \frac{G m_{\rm sc} M_J}{r} \tag{6.244}$$

$$\Longrightarrow \quad \mathcal{L}(\vec{r}, \dot{\vec{r}}) = \frac{1}{2} m_{\rm sc} \dot{\vec{r}}^{\,2} + \frac{G m_{\rm sc} M_J}{r}, \tag{6.245}$$

since a function of time alone and a total time derivative do not change the equations of motion. This $\mathcal{L}$ has no explicit t, so by Noether's theorem the energy is conserved,

$$E = \dot{\vec{r}} \cdot \frac{\partial \mathcal{L}}{\partial \dot{\vec{r}}} - \mathcal{L} = \frac{1}{2} m_{\rm sc} \dot{\vec{r}}^{\,2} - \frac{G m_{\rm sc} M_J}{r} = \text{const} \tag{6.246}$$

$$\Longrightarrow \quad v_{\infty,\rm in} = v_{\infty,\rm out} \equiv v_\infty = \sqrt{-G M_J / a}, \tag{6.247}$$

the Jupiter-relative speed at $r \to \infty$, Eq. (6.60). The solar-frame velocity is $\vec{V}_p + \vec{w}$, where $\vec{w} \equiv \dot{\vec{r}}$ far from Jupiter. Its size $|\vec{w}| = v_\infty$ is fixed, but its direction is free (Figure 6.12).

Put the x-axis along the periapsis direction. The orbit is $r(v) = \lambda/(1 + e\cos v)$, and $r \to \infty$ at $v = \pm v_{\rm asy}$, where

$$\cos v_{\rm asy} = -\frac{1}{e}, \tag{6.248}$$

as in Section 6.11. So $\hat{w}_{\text{out}}$ sits at the angle v_{asy} from the axis. At periapsis $\dot{r} = 0$, so the velocity is purely tangential, perpendicular to the axis. The trajectory is a mirror image about the periapsis, so the total turn δ splits evenly on each side. This puts $\hat{w}_{\text{out}}$ at $\delta/2$ past that perpendicular.

$$v_{\text{asy}} = \frac{\pi}{2} + \frac{\delta}{2} \quad \Longrightarrow \quad \sin\frac{\delta}{2} = \sin\left(v_{\text{asy}} - \frac{\pi}{2}\right) = -\cos v_{\text{asy}} = \frac{1}{e}, \tag{6.249}$$

the deflection angle $\delta = \pi - 2\theta_\infty$ of Eq. (6.61). For $e = 1.2$ this gives $\delta \simeq 112.9°$. As $e \to 1^+$, $\delta \to \pi$, and as $e \to \infty$, $\delta \to 0$. By the same symmetry, $\hat{w}_{\text{in}}$ sits at $v_{\text{asy}} - \delta = \pi/2 - \delta/2$ from the axis.

Take $\vec{V}_p = -V_p\hat{x}$, so Jupiter moves parallel to the periapsis direction, to the left. $\hat{w}_{\text{in}}$ sits at $\pi - v_{\text{asy}}$ and $\hat{w}_{\text{out}}$ at v_{asy} from the axis, so their angles with $\vec{V}_p$ are v_{asy} and $\pi - v_{\text{asy}}$. By the law of cosines,

$$v_{\text{in}}^2 = V_p^2 + v_\infty^2 + 2V_p v_\infty \cos v_{\text{asy}} = V_p^2 + v_\infty^2 - \frac{2V_p v_\infty}{e} \tag{6.250}$$

$$= 13.1^2 + 10^2 - \frac{2(13.1)(10)}{1.2} \implies v_{\text{in}} \simeq 7.3\,\text{km}\,\text{s}^{-1}, \tag{6.251}$$

$$v_{\text{out}}^2 = V_p^2 + v_\infty^2 + 2V_p v_\infty \cos(\pi - v_{\text{asy}}) = V_p^2 + v_\infty^2 + \frac{2V_p v_\infty}{e} \tag{6.252}$$

$$= 13.1^2 + 10^2 + \frac{2(13.1)(10)}{1.2} \implies v_{\text{out}} \simeq 22.1\,\text{km}\,\text{s}^{-1}. \tag{6.253}$$

The spacecraft's speed around the Sun grows by $\boxed{v_{\text{out}} - v_{\text{in}} \simeq 14.8\,\text{km}\,\text{s}^{-1}}$.

By momentum conservation, $m_{\text{sc}}\Delta\vec{v}_{\text{sc}} = -M_J\Delta\vec{V}_p$, so for a spacecraft of about a tonne

$$\Delta V_p = \frac{m_{\text{sc}}}{M_J}\,\Delta v_{\text{sc}} < 10^{-20}\,\text{km}\,\text{s}^{-1}, \tag{6.254}$$

undetectable on Jupiter.

Chapter 7

Signal, Noise, and Detection in CCD Photometry

Chapter 4 measured the light of a star as a flux and wrote it as a magnitude. Chapter 5 focused that light into an image on a detector. Most optical and infrared images today are taken with a CCD (charge-coupled device). It is a thin slab of silicon divided into a grid of pixels. Light frees electrons in the silicon, each pixel holds its electrons during the exposure, and at the end the electrons in every pixel are counted.

A count of electrons is a whole number, and it changes from one exposure to the next even when the star does not. How much it changes decides whether a faint star is seen at all, how well its magnitude is known, and whether one long exposure is better than several short ones. All three answers come from one equation for the signal-to-noise ratio, built from the statistics of counting.

7.1 Counting Photoelectrons

Chapter 4 gave the light of a star as a flux F in $\mathrm{W\,m^{-2}}$, and Chapter 5 gave the size of its image on the detector.

A CCD does not report watts. It reports a number of electrons in each pixel, and part of that number did not come from the star at all.

Picture a grid of buckets on a lawn, with a garden sprinkler running over one spot during a light drizzle. After ten minutes the buckets near the sprinkler hold most of its water. Every bucket also holds the same depth of drizzle, and a little water that has condensed on its inside. To read a bucket we pour it into a measuring jug and read the scale, which adds a small error of its own. To find what the sprinkler gave, we add up the buckets around it and take away what the drizzle and the condensation would have put there.

The pixels are the buckets. The star is the sprinkler, the faint glow of the night sky is the drizzle, electrons freed by heat in the silicon are the condensation, and the electronics that measure each pixel's charge are the jug. The signal is the number of electrons that the star alone put into the pixels around its image.

In numbers, a star that frees 50 electrons per second in its image gives $50 \times 60 = 3000$ electrons in a one-minute exposure. If its image covers 20 pixels and the sky frees 2 electrons per second in each pixel, the sky adds $2 \times 20 \times 60 = 2400$ electrons under the star.

Picture the image as a small hill of electrons standing on a wide, flat plain of sky electrons. The signal is the volume of the hill above the plain.

A Geiger counter measures a radioactive sample in the same way, on top of the counts from

cosmic rays and the walls of the room. So does a rain gauge in a garden that is also watered. Here we name the rates and times in which the noise of a count is written.

Photons become electrons. A photon that enters a pixel frees one electron with a probability $\eta(\lambda)$, the **quantum efficiency** of the detector. For modern astronomical CCDs η is about 0.5 to 0.85. So

$$\text{number of electrons} = \eta \times \text{number of photons that reach the pixel},$$

and 1000 photons give about 800 electrons when $\eta = 0.8$. The factor η multiplies every rate that comes from light, so we fold it into those rates and count only electrons from here on. Every rate below is a rate of detected electrons.

The rates. The light of the star is spread over several pixels by the image blur. We therefore take R_{src} to be the rate at which the star frees electrons in all the pixels of the measuring aperture together. The sky is the same in every pixel, so R_{sky} is a rate per pixel. Heat frees **dark electrons** at the rate R_{dark} in every pixel, light or no light, and they look exactly like electrons freed by photons. The aperture contains n_{pix} pixels. Detector manuals often quote the dark rate per hour, and dividing by 3600 turns it into a rate per second. All the symbols are collected in Table 7.1.

Symbol	Meaning	Units
R_{src}	Rate of electrons from the star, summed over the aperture	$\text{e}^- \text{s}^{-1}$
R_{sky}	Rate of electrons from the sky, per pixel	$\text{e}^- \text{s}^{-1} \text{pix}^{-1}$
R_{dark}	Rate of dark electrons freed by heat, per pixel	$\text{e}^- \text{s}^{-1} \text{pix}^{-1}$
n_{pix}	Number of pixels in the aperture, proportional to θ^2_{seeing}	pixels
t_{exp}	Length of one exposure	s
N_{exp}	Number of exposures added together	—
t	Total time, $t = N_{\text{exp}} t_{\text{exp}}$	s
S	Signal, the electrons from the star in the total time, $S = R_{\text{src}} t$	e^-
$\bar{n}$	Mean of a count of electrons	e^-
σ_{RON}	Readout noise of one pixel in one reading (standard deviation)	e^-
σ_x	Standard deviation added by the process x	e^-
SNR	Signal-to-noise ratio, S/σ_{tot}	—
m, δm	Apparent magnitude and its error	mag

Table 7.1: Symbols, meanings and units for the CCD noise budget.

The signal. Over a total time t the star gives

$$S = R_{\text{src}}\, t \tag{7.1}$$

electrons. In practice t is often split into N_{exp} exposures of length t_{exp} each, which are added afterwards,

$$t = N_{\text{exp}}\, t_{\text{exp}}, \qquad S = R_{\text{src}}\, N_{\text{exp}}\, t_{\text{exp}}. \tag{7.2}$$

The difference between t and N_{exp} matters. Electrons pile up in proportion to the total time only. The electronics add their error once at every reading, so that part grows with the number of exposures.

7.2 Shot Noise and the Poisson Distribution

We can now write the signal as a mean number of electrons, $S = R_{\text{src}} t$.

A real exposure does not give the mean. It gives one count, and a second exposure of the same steady star gives a different count. To say how far a count can stray from its mean we need the probability of every possible count.

Stand by a quiet road at night and count the cars that pass in each minute. The traffic is steady, about ten cars a minute on average. Yet one minute brings seven and the next thirteen. Nothing about the road changed. The cars simply come one at a time, each at its own moment, without arranging themselves to arrive evenly.

Photons from a steady star arrive in the same way, one at a time and independently. So the number of electrons in a fixed exposure scatters about its mean. The scatter is called **shot noise**. It comes from counting, not from any fault of the detector, and its size is the square root of the mean count.

In numbers, a mean of 10 cars a minute gives a scatter of about $\sqrt{10} \approx 3.2$, so most minutes bring between 7 and 13 cars. A mean of 100 gives a scatter of 10, which is 10 per cent of the mean. A mean of 10 000 gives 100, only 1 per cent. The scatter grows with the mean, but more slowly, so it becomes a smaller fraction of it.

Picture a histogram of many repeated counts. It is a hump centred on the mean count, with a width equal to its square root. For a larger mean the hump moves right and gets wider, but it gets narrower compared with its distance from zero.

The same counting statistics give the shot noise of an electric current, first studied by Schottky in vacuum tubes, and the scatter in the clicks of a Geiger counter. The total count is a sum of many independent small pieces, and its variance grows in proportion to the number of pieces. This is the mathematics of the random walk, principle P5 (Section 9.5), where the square of the distance grows in proportion to the number of steps. Here it gives the Poisson distribution of a count, its mean and variance, and the Gaussian shape it takes when the count is large.

Three assumptions. The derivation rests on two physical facts about the arrivals and one mild mathematical condition.

- **Independence.** One arrival says nothing about when the next one comes. There is no memory and no bunching.
- **Constant rate.** The source does not change its brightness during the exposure, so the chance of an arrival is the same in every interval of the same length.
- **Smoothness.** The probability $f(\Delta t)$ of an arrival in an interval of length Δt is a smooth function near $\Delta t = 0$, so it has a Taylor series there.

One short interval. An interval of zero length holds nothing, so $f(0) = 0$. For a short interval,

$$\begin{aligned} f(\Delta t) &\overset{(1)}{=} f(0) + f'(0)\,\Delta t + O(\Delta t^2) \\ &\overset{(2)}{=} f'(0)\,\Delta t + O(\Delta t^2). \end{aligned} \tag{7.3}$$

(1) Taylor series about $\Delta t = 0$, allowed by the smoothness condition.
(2) $f(0) = 0$.

So for any short interval the chance of an arrival is proportional to its length. This is true of any probability that starts from zero, not just of light. Two arrivals in the same interval need two independent rare events at once, so their probability is of order Δt^2.

Many short intervals. Divide the exposure t into N equal intervals of length $\Delta t = t/N$. In each one, an arrival happens with probability

$$p = f'(0)\,\frac{t}{N} \equiv \frac{\bar{n}}{N}, \qquad \bar{n} \equiv f'(0)\,t,$$

and two or more arrivals happen with a probability of order $1/N^2$, which we drop as $N \to \infty$. Each interval then holds either no arrival or one. The intervals are independent, so the probability of a particular pattern with arrivals in n chosen intervals is $p^n(1-p)^{N-n}$, and there are $\binom{N}{n}$ ways to choose the n intervals. The count n therefore follows the binomial distribution. Letting $N \to \infty$ with $\bar{n}$ fixed,

$$\begin{aligned} P(n) &\overset{(1)}{=} \lim_{N\to\infty} \frac{N!}{n!\,(N-n)!} \left(\frac{\bar{n}}{N}\right)^n \left(1-\frac{\bar{n}}{N}\right)^{N-n} \\ &\overset{(2)}{=} \frac{\bar{n}^{\,n}}{n!} \lim_{N\to\infty} \left[\frac{N!}{(N-n)!\,N^n}\right] \left(1-\frac{\bar{n}}{N}\right)^{N} \left(1-\frac{\bar{n}}{N}\right)^{-n} \\ &\overset{(3)}{=} \frac{\bar{n}^{\,n}}{n!} \cdot 1 \cdot e^{-\bar{n}} \cdot 1. \end{aligned} \tag{7.4}$$

(1) The binomial distribution with $p = \bar{n}/N$.
(2) Regroup, taking out the factors that do not depend on N.
(3) $N!/[(N-n)!\,N^n] = \prod_{j=0}^{n-1}(1-j/N)$ is a product of a fixed number n of factors that each tend to 1. The logarithm of the middle factor is $N\ln(1-\bar{n}/N) = N(-\bar{n}/N - \bar{n}^2/2N^2 - \dots) \to -\bar{n}$. The last factor has a fixed power n and tends to 1.

This is the **Poisson distribution**,

$$\boxed{P(n) = \frac{\bar{n}^{\,n}\,e^{-\bar{n}}}{n!}.} \tag{7.5}$$

It depends on the rate and the time only through $\bar{n}$. Its probabilities add up to one (Problem 7.7.1).

Mean. The mean count is

$$\begin{aligned} \langle n \rangle &\overset{(1)}{=} \sum_{n=0}^{\infty} n\,\frac{\bar{n}^{\,n} e^{-\bar{n}}}{n!} \\ &\overset{(2)}{=} e^{-\bar{n}} \sum_{n=1}^{\infty} \frac{\bar{n}^{\,n}}{(n-1)!} \\ &\overset{(3)}{=} \bar{n}\,e^{-\bar{n}} \sum_{j=0}^{\infty} \frac{\bar{n}^{\,j}}{j!} \\ &\overset{(4)}{=} \bar{n}. \end{aligned} \tag{7.6}$$

(1) The definition of the mean, with Eq. (7.5).
(2) The $n = 0$ term is zero, and $n/n! = 1/(n-1)!$.
(3) Put $j = n-1$ and take one factor $\bar{n}$ out of the sum.
(4) The sum is the series of $e^{\bar{n}}$.

So $\bar{n}$ is the mean count, and $f'(0) = \bar{n}/t$ is the mean number of arrivals per second. For the star it is R_{src}, and for the sky and the dark current it is $n_{\mathrm{pix}}R_{\mathrm{sky}}$ and $n_{\mathrm{pix}}R_{\mathrm{dark}}$.

Variance. The variance is $\sigma^2 \equiv \left\langle (n - \langle n \rangle)^2 \right\rangle$. The factor $n(n-1)$ cancels cleanly against $n!$, so we first find the average of $n(n-1)$,

$$\begin{aligned} \langle n(n-1) \rangle &\overset{(1)}{=} e^{-\bar{n}} \sum_{n=2}^{\infty} \frac{\bar{n}^{\,n}}{(n-2)!} \\ &\overset{(2)}{=} \bar{n}^2 \, e^{-\bar{n}} \sum_{j=0}^{\infty} \frac{\bar{n}^{\,j}}{j!} = \bar{n}^2. \end{aligned} \tag{7.7}$$

(1) The terms $n = 0$ and $n = 1$ are zero, and $n(n-1)/n! = 1/(n-2)!$.
(2) Put $j = n - 2$, take $\bar{n}^2$ out, and sum the series of $e^{\bar{n}}$.

Then

$$\begin{aligned} \sigma^2 &\overset{(1)}{=} \left\langle n^2 \right\rangle - \langle n \rangle^2 \\ &\overset{(2)}{=} \langle n(n-1) \rangle + \langle n \rangle - \langle n \rangle^2 \\ &\overset{(3)}{=} \bar{n}^2 + \bar{n} - \bar{n}^2 = \bar{n}. \end{aligned} \tag{7.8}$$

(1) Expand the square. $\langle n \rangle$ is a constant, so $\langle 2n \, \langle n \rangle \rangle = 2 \, \langle n \rangle^2$.
(2) $n^2 = n(n-1) + n$.
(3) The two averages just found, Eq. (7.6) and the one above.

The shot noise, the standard deviation of the count, is therefore

$$\boxed{\sigma = \sqrt{\bar{n}}.} \tag{7.9}$$

The noise grows as the square root of the mean, so the fractional noise $\sigma/\bar{n} = 1/\sqrt{\bar{n}}$ falls as the count grows. A computer algebra check of the three sums gives 1, $\bar{n}$ and $\bar{n}$ exactly. Figure 7.1 shows the distribution for three means.

Why the variance grows as the count. The same variance follows from a second route, which shows where the square root comes from. We need one general fact first. Let a quantity $X = \sum_i X_i$ be a sum of independent parts, and write $\delta X_i = X_i - \langle X_i \rangle$ for the deviation of each part from its mean. Then

$$\begin{aligned} \sigma_X^2 &\overset{(1)}{=} \left\langle \Big(\sum_i \delta X_i \Big)^2 \right\rangle \\ &\overset{(2)}{=} \sum_i \sum_j \left\langle \delta X_i \, \delta X_j \right\rangle \\ &\overset{(3)}{=} \sum_i \left\langle \delta X_i^2 \right\rangle + \sum_{i \neq j} \langle \delta X_i \rangle \left\langle \delta X_j \right\rangle \\ &\overset{(4)}{=} \sum_i \sigma_i^2. \end{aligned} \tag{7.10}$$

(1) $X - \langle X \rangle = \sum_i \delta X_i$, because the mean of a sum is the sum of the means.
(2) Expand the square of the sum into a double sum.
(3) Split off the terms with $i = j$. For $i \neq j$ the parts are independent, so the average of the product is the product of the averages.
(4) $\langle \delta X_i \rangle = 0$ by its definition, and $\left\langle \delta X_i^2 \right\rangle = \sigma_i^2$.

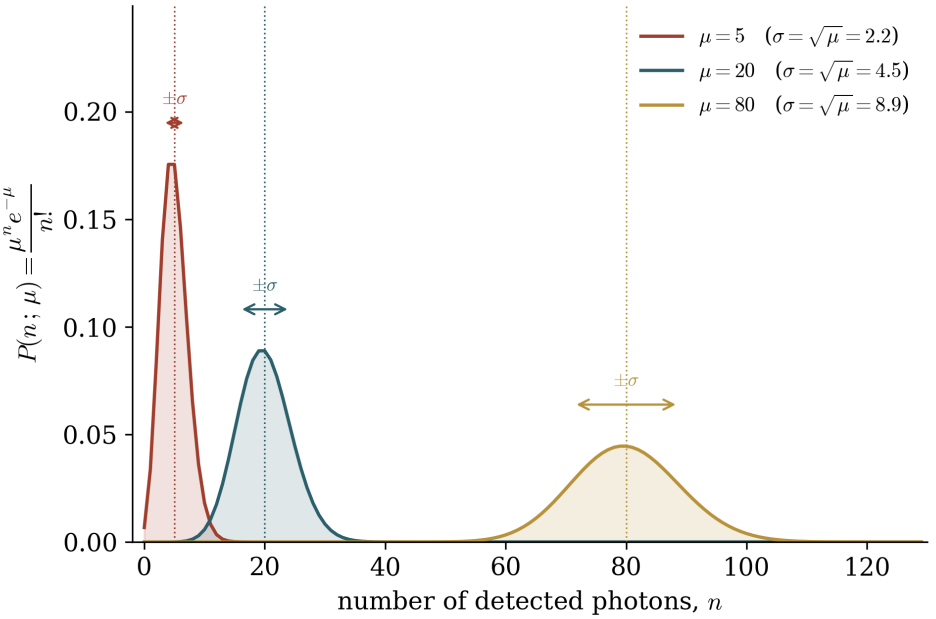


Figure 7.1: The Poisson distribution for mean counts of 5, 20 and 80 (labelled μ in the plot). A dotted line marks each mean, and a double arrow spans one standard deviation on each side of it.

The variances of independent parts add, whatever the distribution of each part. Standard deviations therefore add **in quadrature**, $\sigma_X = \sqrt{\sum_i \sigma_i^2}$. The cross terms die for the same reason as in the random walk, where they gave $R_{\text{rms}} = \sqrt{N}\, s$ (Eq. (9.36)).

Now let X_i be the number of arrivals in the i-th short interval. It is 1 with probability p and 0 otherwise, so $\langle X_i \rangle = p$ and $\langle X_i^2 \rangle = p$, which gives $\sigma_i^2 = p - p^2$. The count is $n = \sum_i X_i$, and

$$\begin{aligned} \sigma^2 &\overset{(1)}{=} \sum_{i=1}^{N} p\,(1-p) \\ &\overset{(2)}{=} \bar{n}\left(1 - \frac{\bar{n}}{N}\right) \\ &\to \bar{n} \quad (N \to \infty). \end{aligned} \tag{7.11}$$

(1) Eq. (7.10), with the same variance $p(1-p)$ for every interval.
(2) $Np = \bar{n}$.

This agrees with Eq. (7.8). Each interval adds the same small variance, so the variance grows in proportion to the number of intervals, and so in proportion to the mean count. The noise grows only as its square root.

Large counts look Gaussian. For a large mean the Poisson hump becomes the Gaussian bell curve. To see this we write $n = \bar{n} + x$ and use Stirling's formula, $\ln n! \approx n \ln n - n + \frac{1}{2}\ln(2\pi n)$,

for large n.

$$\begin{aligned}
\ln P(n) &\overset{(1)}{=} n\ln\bar{n} - \bar{n} - n\ln n + n - \tfrac{1}{2}\ln(2\pi n) \\
&\overset{(2)}{=} -(\bar{n}+x)\ln\left(1+\frac{x}{\bar{n}}\right) + x - \tfrac{1}{2}\ln(2\pi n) \\
&\overset{(3)}{=} -(\bar{n}+x)\left(\frac{x}{\bar{n}} - \frac{x^2}{2\bar{n}^2} + \ldots\right) + x - \tfrac{1}{2}\ln(2\pi\bar{n}) + \ldots \\
&\overset{(4)}{=} -\frac{x^2}{2\bar{n}} - \tfrac{1}{2}\ln(2\pi\bar{n}) + O\left(\frac{x^3}{\bar{n}^2}, \frac{x}{\bar{n}}\right).
\end{aligned} \tag{7.12}$$

(1) The logarithm of Eq. (7.5), with Stirling's formula for $\ln n!$.
(2) $n\ln\bar{n} - n\ln n = -n\ln(n/\bar{n})$ with $n = \bar{n} + x$, and $-\bar{n} + n = x$.
(3) Taylor series $\ln(1+y) = y - y^2/2 + \ldots$, and $\ln(2\pi n) = \ln(2\pi\bar{n}) + \ln(1 + x/\bar{n})$.
(4) Multiply out. The terms in x cancel, and $x^2/2\bar{n} - x^2/\bar{n} = -x^2/2\bar{n}$.

Dropping the small terms,

$$\boxed{P(n) \approx \frac{1}{\sqrt{2\pi\bar{n}}}\exp\left[-\frac{(n-\bar{n})^2}{2\bar{n}}\right].} \tag{7.13}$$

This is a Gaussian with mean $\bar{n}$ and variance $\bar{n}$, as it must be. The dropped term $x^3/\bar{n}^2$ is $1/\sqrt{\bar{n}}$ at one standard deviation, $x = \sqrt{\bar{n}}$. So the Gaussian is good when $\bar{n} \gg 1$ and within a few standard deviations of the mean. The Poisson distribution is defined only at whole numbers and the Gaussian is smooth, but when $\sqrt{\bar{n}} \gg 1$ the steps of one electron are tiny compared with the width, and the difference cannot be seen (Figure 7.7 in Problem 7.7.1). A computer algebra expansion of $\ln P(n)$ to second order in x gives the same leading terms.

7.3 The Noise Budget and the Signal-to-Noise Equation

We now know that a count with mean $\bar{n}$ has a standard deviation $\sqrt{\bar{n}}$.

The number we read from a CCD is not one clean count. It is the star's electrons on top of sky and dark electrons, measured by electronics that add an error of their own. We need the noise of the star's share alone, after the rest has been taken away.

Weigh yourself on a bathroom scale while holding a cat, then weigh yourself alone, and subtract. Each reading of the scale wanders by about 0.2 kg. Your own weight cancels in the difference, but the wandering of the second reading does not. The cat's weight carries both.

In an image, the star's electrons sit on top of the sky and dark electrons. We subtract the mean sky and dark level, but their scatter stays in the result, and the readout adds more. The idea is that the variances of independent sources of noise add.

In numbers, the cat's weight has an error of $\sqrt{0.2^2 + 0.2^2} = 0.28$ kg, not 0.4 kg. Two errors that are independent partly cancel as often as they add.

Picture each source of noise as a side of a box, all at right angles to each other. The total noise is the long diagonal of the box, as in Pythagoras' theorem.

Every laboratory error budget adds independent errors in this way, and so does an engineer adding noise powers in a circuit. It is the same rule that made the random walk grow as $\sqrt{N}$ (P5, Eq. (9.36)), and it was proved for any independent parts in Eq. (7.10). Applied to the four sources of electrons in an aperture, it gives the signal-to-noise equation of a CCD, which every exposure time calculator evaluates.

Signal-to-noise ratio. Let σ_{tot} be the standard deviation of the measured signal S, after the mean sky and dark levels have been subtracted. The measurement is reported as $S \pm \sigma_{\text{tot}}$, and its **signal-to-noise ratio** is

$$\text{SNR} \equiv \frac{S}{\sigma_{\text{tot}}}. \tag{7.14}$$

Its inverse $1/\text{SNR} = \sigma_{\text{tot}}/S$ is the fractional error of the flux. Our goal is σ_{tot} in terms of the rates and times of Table 7.1.

Shot noise of the star. The electrons from the star in time t have the mean $S = R_{\text{src}} t$ (Eq. (7.1)). The rate R_{src} is already summed over the aperture, so no factor n_{pix} appears. Eq. (7.9) gives

$$\sigma_{\text{src}} = \sqrt{S} = \sqrt{R_{\text{src}}\, t}. \tag{7.15}$$

Shot noise of the sky. The sky gives R_{sky} per pixel, so the aperture collects a mean of $n_{\text{pix}} R_{\text{sky}} t$ sky electrons. We assume that the mean sky level is measured over a large empty area nearby, so that its own error is small. Subtracting it removes the mean, but the Poisson scatter of the sky electrons inside the aperture remains,

$$\sigma_{\text{sky}} = \sqrt{n_{\text{pix}}\, R_{\text{sky}}\, t}. \tag{7.16}$$

The aperture must hold the blurred image, so its area grows as the square of the seeing, $n_{\text{pix}} \propto \theta^2_{\text{seeing}}$ (Eq. (5.68)). Good seeing means fewer pixels and less sky noise.

Shot noise of the dark current. Heat in the silicon frees electrons at random moments, independently and at a steady rate, just as photons arrive. So the dark electrons are also Poisson counts, with mean $n_{\text{pix}} R_{\text{dark}} t$ in the aperture,

$$\sigma_{\text{dark}} = \sqrt{n_{\text{pix}}\, R_{\text{dark}}\, t}. \tag{7.17}$$

Cooling the detector lowers R_{dark}, which is why astronomical CCDs are kept cold.

Readout noise. The readout noise is not a count of anything. Dark current adds extra charge to a pixel before it is measured. Readout noise is an error in the measurement itself, after the charge is fixed. The charge of each pixel is turned into a voltage by an output transistor and then into a number by an analogue-to-digital converter. Both have their own electrical noise, the thermal (Johnson–Nyquist) noise of any resistor and a slowly wandering "$1/f$" noise. Each reading of a pixel therefore gets a random error with a Gaussian distribution and a standard deviation σ_{RON}, whatever charge the pixel holds and however long the exposure was. Every reading is independent, and the aperture is read n_{pix} times in each of N_{exp} exposures, so

$$\begin{aligned} \sigma^2_{\text{read}} &\overset{(1)}{=} \sum_{\text{readings}} \sigma^2_{\text{RON}} \\ &= n_{\text{pix}}\, N_{\text{exp}}\, \sigma^2_{\text{RON}}. \end{aligned}$$

(1) Eq. (7.10), with the same variance for every reading.

$$\boxed{\sigma_{\text{read}} = \sqrt{n_{\text{pix}}\, N_{\text{exp}}}\ \sigma_{\text{RON}}.} \tag{7.18}$$

The readout variance grows with N_{exp}, not with t. Splitting a total time into N_{exp} exposures pays the readout variance N_{exp} times.

Adding the four. The star, the sky, the heat in the silicon and the readout electronics are independent of each other, so Eq. (7.10) adds their variances,

$$\sigma_{\text{tot}}^2 \overset{(1)}{=} \sigma_{\text{src}}^2 + \sigma_{\text{sky}}^2 + \sigma_{\text{dark}}^2 + \sigma_{\text{read}}^2 \tag{7.19}$$
$$\overset{(2)}{=} R_{\text{src}}\, t + n_{\text{pix}} R_{\text{sky}}\, t + n_{\text{pix}} R_{\text{dark}}\, t + n_{\text{pix}} N_{\text{exp}}\, \sigma_{\text{RON}}^2.$$

(1) Eq. (7.10) for four independent parts.
(2) Eqs. (7.15)–(7.18).

Putting this and $S = R_{\text{src}} t$ into Eq. (7.14) gives the **CCD signal-to-noise equation**,

Result 7.1

$$\text{SNR} = \frac{R_{\text{src}}\, t}{\sqrt{R_{\text{src}}\, t \;+\; n_{\text{pix}} R_{\text{sky}}\, t \;+\; n_{\text{pix}} R_{\text{dark}}\, t \;+\; n_{\text{pix}} N_{\text{exp}}\, \sigma_{\text{RON}}^2}}, \qquad t = N_{\text{exp}} t_{\text{exp}}. \tag{7.20}$$

The signal and the first three variances grow in proportion to t. The last variance grows with N_{exp}. So the signal-to-noise ratio depends on how the time is split as well as on the total time. Figure 7.2 shows the share of each variance in three typical observations.

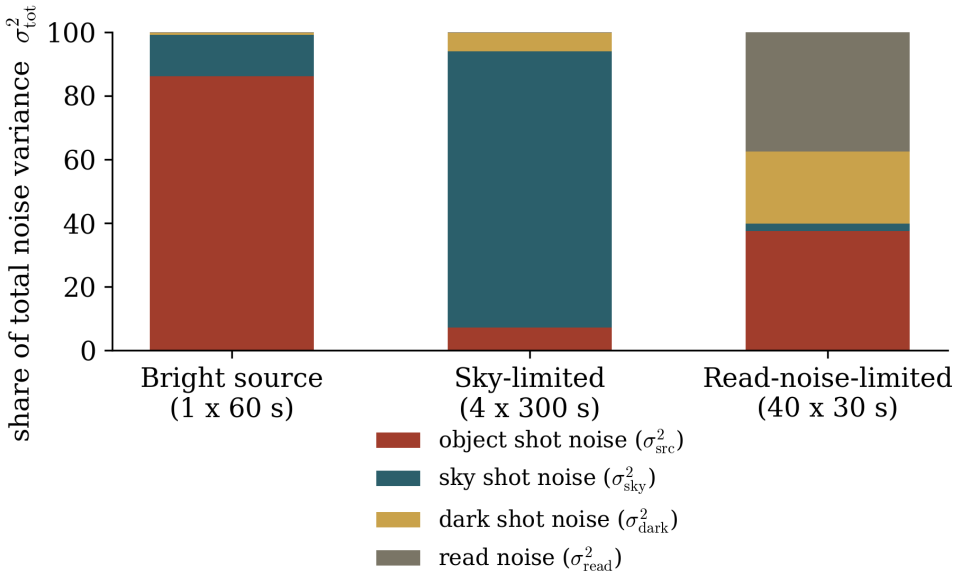


Figure 7.2: Share of each term in the total variance σ_{tot}^2, in per cent, for three observations. A bright source in one 60 s exposure, a sky-limited source in four 300 s exposures, and a faint source in forty 30 s exposures.

Dark current and read noise from two dark frames. The detector's own two terms can be measured with the shutter closed. Such a **dark frame** has no star and no sky. Its variance over an area of n_{pix} pixels, for an exposure of length t, is the last two terms of Eq. (7.19) with $N_{\text{exp}} = 1$,

$$\sigma^2(t) = n_{\text{pix}} R_{\text{dark}}\, t + n_{\text{pix}} \sigma_{\text{RON}}^2.$$

This is a straight line in t. Its slope is the dark variance per unit time and its intercept is the read variance. Two unknowns need two measurements, and two dark frames of lengths t_1 and

t_2 with variances σ_1^2 and σ_2^2 give

$$\begin{pmatrix} t_1 & 1 \\ t_2 & 1 \end{pmatrix} \begin{pmatrix} n_{\mathrm{pix}} R_{\mathrm{dark}} \\ n_{\mathrm{pix}} \sigma_{\mathrm{RON}}^2 \end{pmatrix} = \begin{pmatrix} \sigma_1^2 \\ \sigma_2^2 \end{pmatrix}.$$

The determinant of the matrix is $t_1 - t_2$. Two frames of the same length give the same equation twice and cannot separate the two terms. For $t_1 \neq t_2$,

$$\begin{aligned} \begin{pmatrix} n_{\mathrm{pix}} R_{\mathrm{dark}} \\ n_{\mathrm{pix}} \sigma_{\mathrm{RON}}^2 \end{pmatrix} &\overset{(1)}{=} \frac{1}{t_1 - t_2} \begin{pmatrix} 1 & -1 \\ -t_2 & t_1 \end{pmatrix} \begin{pmatrix} \sigma_1^2 \\ \sigma_2^2 \end{pmatrix} \\ &= \frac{1}{t_2 - t_1} \begin{pmatrix} \sigma_2^2 - \sigma_1^2 \\ t_2\sigma_1^2 - t_1\sigma_2^2 \end{pmatrix}. \end{aligned} \tag{7.21}$$

(1) The inverse of a 2×2 matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

The dark variance per unit time is the rise in variance divided by the rise in time, the slope of the line in Figure 7.3. The read variance is where the line meets $t = 0$. A third frame of another length tests whether the points really lie on a straight line. Problem 7.7.4 puts in numbers.

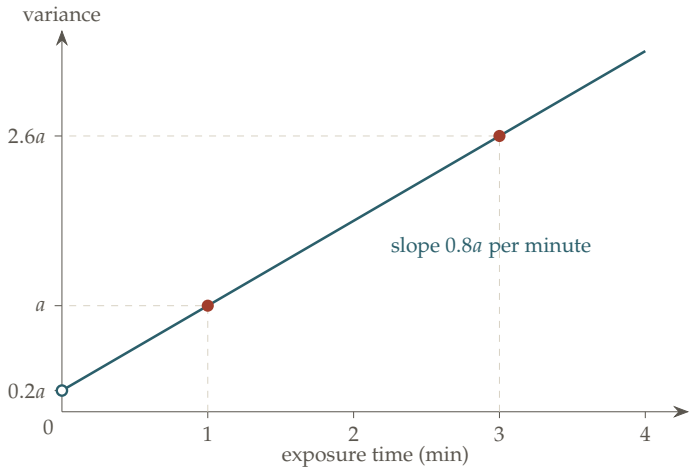


Figure 7.3: Variance of a dark frame against exposure time. The filled points are frames of 1 and 3 minutes with variances a and $2.6a$. The straight line through them meets the vertical axis at $0.2a$ (open circle).

Defects that are not noise. Some flaws of a CCD frame are not random scatter and must be found and removed before Eq. (7.19) means anything.

- **Hot and cold pixels.** Single pixels with a far higher dark rate (hot) or a far lower quantum efficiency (cold). They are found in calibration frames and flagged or filled in from their neighbours.
- **Saturation and bleeding.** A pixel holds only a limited charge, its **full well** (Section 7.4.1). Near it the response stops being linear, and beyond it extra charge spills into neighbouring pixels along a column, leaving a bright *bleed trail*.
- **Cosmic rays.** Fast charged particles leave a sharp spike of charge in one frame only. Comparing several exposures (Section 7.4.2) or the shape of the spike, which is sharper than any star, removes them.

- **Airglow and city lights.** Light emitted by the upper atmosphere and light scattered from towns raise R_{sky} itself, and with it σ_{sky} in Eq. (7.16), above the level of a dark site on a moonless night.

Dithering, moving the telescope by a few arcseconds between exposures, deals with several of these at once. A given point of the sky then falls on a different pixel in each exposure. A bad pixel or a bleed trail stays fixed on the detector and a cosmic ray appears in one frame only, so when the frames are lined up on the sky and combined they can be rejected.

7.4 Three Regimes of the Signal-to-Noise Equation

Eq. (7.20) gives the signal-to-noise ratio of any exposure.

As it stands it is hard to plan with. Should we take one long exposure or many short ones? Does good seeing matter? Does a larger telescope help a faint star more than a bright one? The answer depends on which of the four terms under the square root is largest.

Think of a taxi fare. There is a fixed charge for each ride and a charge for each minute. On one long ride the minutes dominate and the fixed charge hardly matters. Split the same trip into many short rides and the fixed charges pile up. Whether splitting matters depends on which part of the fare is larger.

In the signal-to-noise equation the star, the sky and the dark current charge their variance per second, and the readout charges its variance per exposure. When one term is much larger than the others, the equation simplifies and tells us directly what to do. Which term wins is decided by comparing two numbers, principle P6 (Section 9.6).

In numbers, take a taxi that charges 50 rupees per ride and 10 rupees per minute. One 20-minute ride costs 250 rupees, while ten rides of 2 minutes cost 700. For a CCD with $\sigma_{\text{RON}} = 5\,\text{e}^-$, an aperture of 20 pixels and a sky of $2\,\text{e}^-\,\text{s}^{-1}$ per pixel, each reading adds a variance of $20 \times 25 = 500\,\text{e}^{-2}$ and the sky adds $40\,\text{e}^{-2}$ per second. A 300 s exposure collects a sky variance of $12\,000\,\text{e}^{-2}$, so its one reading hardly matters. A 10 s exposure collects only $400\,\text{e}^{-2}$ of sky variance, less than the readout.

Picture the four variances as bars, three that grow with time and one that grows with the number of exposures. The tallest bar sets how the signal-to-noise ratio responds to any change.

Ratios of this kind choose the regime everywhere in physics, from the Reynolds number of a flow to the optical depth τ of a dust cloud (Chapter 4). Here they give the shortest useful exposure and the three limits of the equation, each with its own way of observing.

How long one exposure should be. For a faint star, R_{src} is small next to the sky and dark rates. Then the variance in Eq. (7.19) is

$$\begin{aligned}\sigma_{\text{tot}}^2 &\overset{(1)}{\approx} n_{\text{pix}}(R_{\text{sky}} + R_{\text{dark}})\,t + n_{\text{pix}} N_{\text{exp}}\,\sigma_{\text{RON}}^2 \\ &\overset{(2)}{=} n_{\text{pix}} N_{\text{exp}} \left[(R_{\text{sky}} + R_{\text{dark}})\,t_{\text{exp}} + \sigma_{\text{RON}}^2\right] \\ &= n_{\text{pix}}(R_{\text{sky}} + R_{\text{dark}})\,t \left[1 + \frac{\sigma_{\text{RON}}^2}{(R_{\text{sky}} + R_{\text{dark}})\,t_{\text{exp}}}\right].\end{aligned} \tag{7.22}$$

(1) Drop the star's own shot noise, $R_{\text{src}} t$.
(2) $t = N_{\text{exp}} t_{\text{exp}}$.

The last factor compares the read variance of one pixel with the sky and dark variance that the pixel collects in one exposure. Splitting the time costs little when that ratio is small,

$$\boxed{t_{\mathrm{exp}} \gg \frac{\sigma_{\mathrm{RON}}^2}{R_{\mathrm{sky}} + R_{\mathrm{dark}}}.} \tag{7.23}$$

For the numbers above the right side is $25/2 = 12.5\,\mathrm{s}$. Exposures of 300 s lower the signal-to-noise ratio by a factor $(1 + 12.5/300)^{-1/2} = 0.98$, only 2 per cent. The bound depends only on the detector and the sky, not on the star.

7.4.1 Regime 1: Bright Source, Limited by Its Own Shot Noise

For a bright star the first term under the root is much larger than the others, $R_{\mathrm{src}} t \gg n_{\mathrm{pix}}(R_{\mathrm{sky}} + R_{\mathrm{dark}})t + n_{\mathrm{pix}} N_{\mathrm{exp}} \sigma_{\mathrm{RON}}^2$. Then

$$\begin{aligned} \mathrm{SNR} &\overset{(1)}{\approx} \frac{R_{\mathrm{src}}\, t}{\sqrt{R_{\mathrm{src}}\, t}} \\ &= \sqrt{R_{\mathrm{src}}\, t}, \end{aligned}$$

(1) Keep only the star's own term in Eq. (7.20).

$$\boxed{\mathrm{SNR} \approx \sqrt{R_{\mathrm{src}}\, t} \propto \sqrt{t}.} \tag{7.24}$$

The signal-to-noise ratio is the square root of the number of electrons from the star. It does not depend on the sky, the detector or the way the time is split. Reaching a given ratio takes $t = \mathrm{SNR}^2 / R_{\mathrm{src}}$, so ten times the precision costs a hundred times the time.

Here the difficulty is not to collect enough light but to avoid too much. A pixel saturates at its full well, typically about 10^5 electrons, and its response stops being linear before that. A bright star is therefore observed with shorter exposures, which cost little here, or with the telescope slightly out of focus, which spreads the light over more pixels.

7.4.2 Regime 2: Faint Source on a Bright Sky

For a faint star seen through a broad optical filter, the sky term is the largest, $n_{\mathrm{pix}} R_{\mathrm{sky}} t \gg R_{\mathrm{src}} t,\ n_{\mathrm{pix}} R_{\mathrm{dark}} t,\ n_{\mathrm{pix}} N_{\mathrm{exp}} \sigma_{\mathrm{RON}}^2$. Then

$$\begin{aligned} \mathrm{SNR} &\overset{(1)}{\approx} \frac{R_{\mathrm{src}}\, t}{\sqrt{n_{\mathrm{pix}} R_{\mathrm{sky}}\, t}} \\ &= \frac{R_{\mathrm{src}}}{\sqrt{n_{\mathrm{pix}} R_{\mathrm{sky}}}}\, \sqrt{t}, \end{aligned}$$

(1) Keep only the sky term under the root in Eq. (7.20).

$$\boxed{\mathrm{SNR} \approx \frac{R_{\mathrm{src}} \sqrt{t}}{\sqrt{n_{\mathrm{pix}} R_{\mathrm{sky}}}} \propto \frac{\sqrt{t}}{\sqrt{n_{\mathrm{pix}}}}.} \tag{7.25}$$

The ratio still grows as $\sqrt{t}$, but now in proportion to R_{src}, not its square root. A star four times fainter has a quarter of the signal-to-noise ratio and needs sixteen times the time.

Seeing. Solving Eq. (7.25) for the time needed to reach a given ratio,

$$t \overset{(1)}{=} \mathrm{SNR}^2 \, \frac{n_{\mathrm{pix}} R_{\mathrm{sky}}}{R_{\mathrm{src}}^2} \overset{(2)}{\propto} \theta_{\mathrm{seeing}}^2 . \tag{7.26}$$

(1) Square Eq. (7.25) and solve for t.
(2) The aperture must hold the seeing disc, so $n_{\mathrm{pix}} \propto \theta_{\mathrm{seeing}}^2$, with the star and the sky fixed.

Seeing of 0.6″ instead of 1.2″ needs a quarter of the time for the same result, or doubles the signal-to-noise ratio in the same time. On a bright sky, good seeing is worth as much as a larger telescope.

Splitting is free. Here the signal-to-noise ratio depends on t and not on N_{exp}, as long as each exposure satisfies Eq. (7.23). Splitting 3000 s into ten exposures of 300 s costs almost nothing. It allows cosmic rays to be rejected and the pointing to be dithered, and it keeps bright stars in the field below saturation.

7.4.3 Regime 3: Faint Source Limited by Readout Noise

When the background is very low, as through a narrow filter, in a spectrograph that spreads the light thinly, or in very short exposures, the readout term is the largest, $n_{\mathrm{pix}} N_{\mathrm{exp}} \sigma_{\mathrm{RON}}^2 \gg (R_{\mathrm{src}} + n_{\mathrm{pix}} R_{\mathrm{sky}} + n_{\mathrm{pix}} R_{\mathrm{dark}})\, t$. Then

$$\mathrm{SNR} \overset{(1)}{\approx} \frac{R_{\mathrm{src}}\, t}{\sqrt{n_{\mathrm{pix}} N_{\mathrm{exp}}}\, \sigma_{\mathrm{RON}}},$$

(1) Keep only the readout term under the root in Eq. (7.20).

$$\boxed{\mathrm{SNR} \approx \frac{R_{\mathrm{src}}\, t}{\sqrt{n_{\mathrm{pix}} N_{\mathrm{exp}}}\, \sigma_{\mathrm{RON}}} \propto \frac{t}{\sqrt{n_{\mathrm{pix}} N_{\mathrm{exp}}}}.} \tag{7.27}$$

The noise no longer depends on the light at all, so the ratio grows in proportion to the signal. At a fixed number of exposures it grows as t, faster than in the other two regimes. At a fixed total time it falls as $1/\sqrt{N_{\mathrm{exp}}}$. A star twice as far away gives a quarter of the electrons and a quarter of the signal-to-noise ratio (Problem 7.7.3).

Binning. The charges of a block of 2×2 pixels can be added on the chip before they are read. The block is then read once, so the number of readings in the aperture falls by 4. The read variance $n_{\mathrm{pix}} N_{\mathrm{exp}} \sigma_{\mathrm{RON}}^2$ falls by 4, while the shot-noise terms stay the same, because the electrons are the same. In this regime the signal-to-noise ratio rises by $\sqrt{4} = 2$. Binning coarsens the pixel scale, so it stops being useful once about two binned pixels span the seeing disc, the sampling limit of Section 5.5.

Long exposures. At a fixed total time, fewer and longer exposures give fewer readings. The useful length is limited mainly by cosmic rays, which pile up in a long exposure. For CCDs on the ground the usual limit is about 30 to 45 minutes.

Figure 7.4 shows both cases at a fixed total time.

The regimes side by side. The term that dominates the variance decides how the ratio scales and how to observe. Table 7.2 lists the three cases.

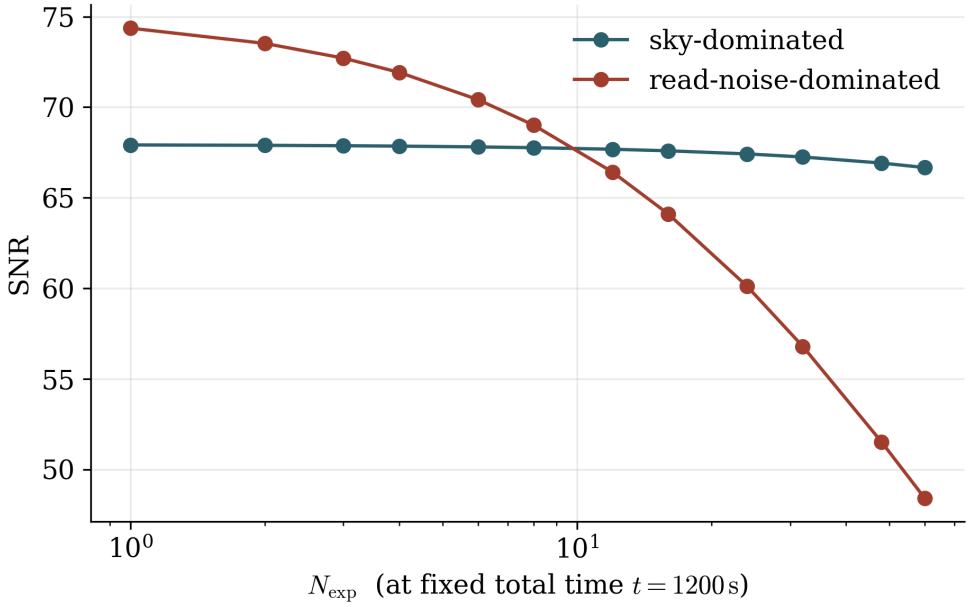


Figure 7.4: Signal-to-noise ratio against the number of exposures N_{exp}, at a fixed total time $t = 1200\,\mathrm{s}$, for a sky-dominated case and a read-noise-dominated case.

Regime	Largest variance	Scaling	How to observe
Bright source	$\sigma^2_{\mathrm{src}} = R_{\mathrm{src}} t$	$\mathrm{SNR} \propto \sqrt{t}$	Keep below the full well. Use short exposures or defocus.
Bright sky	$\sigma^2_{\mathrm{sky}} = n_{\mathrm{pix}} R_{\mathrm{sky}} t$	$\mathrm{SNR} \propto \dfrac{\sqrt{t}}{\sqrt{n_{\mathrm{pix}}}}$	Use the smallest aperture good seeing allows. Split freely into exposures that satisfy Eq. (7.23).
Readout	$\sigma^2_{\mathrm{read}} = n_{\mathrm{pix}} N_{\mathrm{exp}} \sigma^2_{\mathrm{RON}}$	$\mathrm{SNR} \propto \dfrac{t}{\sqrt{n_{\mathrm{pix}} N_{\mathrm{exp}}}}$	Bin on the chip up to the sampling limit. Use the longest exposures cosmic rays allow.

Table 7.2: The three regimes of the signal-to-noise equation, with the largest variance, the scaling of SNR, and the way of observing in each.

7.5 From Signal-to-Noise to Magnitudes and Detections

The signal-to-noise equation tells us how well the electrons of a star are counted.

Astronomers want two other things. They want the error of a magnitude, the unit in which brightness is reported. And for a faint smudge in an image they want to know whether it is a star at all, or just the noise happening to be high in one place.

Toss a coin ten times and get eight heads. It is odd, but a fair coin does it once in about 18 tries. Toss it a hundred times and get eighty heads, and the coin is surely bent. A fair coin does that about once in two billion tries. The fraction of heads is the same. What changed is how far the result lies from the mean, measured in standard deviations.

A smudge in an image is a count of electrons above the mean sky. If the excess is a few times the noise, chance can produce it. If it is many times the noise, chance almost never does. The idea is that significance is the excess in units of the standard deviation, and the Gaussian shape

of the noise turns it into a probability.

In numbers, ten tosses have a mean of 5 heads and a standard deviation of $\sqrt{10 \times \frac{1}{4}} = 1.58$, so eight heads is 1.9 standard deviations high, with a probability of 5.5 per cent. A hundred tosses have a mean of 50 and a standard deviation of 5, so eighty heads is 6 standard deviations high, with a probability of 5.6×10^{-10}.

Picture the bell curve of the noise. A detection threshold is a line drawn far out on its right side, and the false alarms are the thin sliver of the curve beyond the line.

The same reasoning sets the confidence of a medical test, the "five sigma" that particle physicists demand before announcing a new particle, and quality control in a factory. Here it gives the error of a magnitude and the threshold for believing a detection.

7.5.1 The Error of a Magnitude

A magnitude is $m = -2.5 \log_{10} F + \text{constant}$ (Eq. (4.7)). With the same telescope, filter and exposure, the signal S is proportional to the flux F, so $m = -2.5 \log_{10} S + \text{constant}$. A signal that is high by one standard deviation makes the star look brighter by

$$\begin{aligned} \delta m &\equiv m(S) - m(S + \sigma_{\text{tot}}) \\ &\overset{(1)}{=} 2.5 \log_{10} \frac{S + \sigma_{\text{tot}}}{S} \\ &\overset{(2)}{=} 2.5 \log_{10} \left(1 + \frac{1}{\text{SNR}}\right). \end{aligned} \tag{7.28}$$

(1) The constant cancels, and $-\log_{10} a + \log_{10} b = \log_{10}(b/a)$.
(2) Eq. (7.14).

For a large signal-to-noise ratio the logarithm is nearly linear,

$$\begin{aligned} \delta m &\overset{(1)}{=} \frac{2.5}{\ln 10} \ln\left(1 + \frac{1}{\text{SNR}}\right) \\ &\overset{(2)}{\approx} \frac{2.5}{\ln 10} \cdot \frac{1}{\text{SNR}}, \end{aligned} \tag{7.29}$$

(1) $\log_{10} y = \ln y / \ln 10$.
(2) $\ln(1 + y) \approx y$ for $y \ll 1$.

$$\boxed{\delta m \approx \frac{1.0857}{\text{SNR}}.} \tag{7.30}$$

The factor $2.5/\ln 10 = 1.0857$ is the one that turned an optical depth into magnitudes of extinction in Chapter 4 (Eq. (4.36)). Both times a natural logarithm is turned into magnitudes. A signal-to-noise ratio of 10 gives an error of about 0.1 magnitude, and 100 gives 0.01.

The error is not symmetric. A signal low by one standard deviation makes the star look fainter by $-2.5 \log_{10}(1 - 1/\text{SNR})$, which is larger. At SNR = 3 the two shifts are 0.31 and 0.44 magnitude, and Eq. (7.30) gives 0.36, between them. At SNR = 10 they are 0.103 and 0.114, and the linear value 0.109 differs from each by about 5 per cent. Figure 7.5 compares the brighter-side shift with the linear form. Below SNR ≈ 5 the linear form is a poor guide, and both sides should be given. Table 7.3 lists what typical signal-to-noise ratios mean in practice.

7.5.2 Detection and the False-Alarm Probability

At a signal-to-noise ratio of 2 to 5 the question is whether there is a source at all.

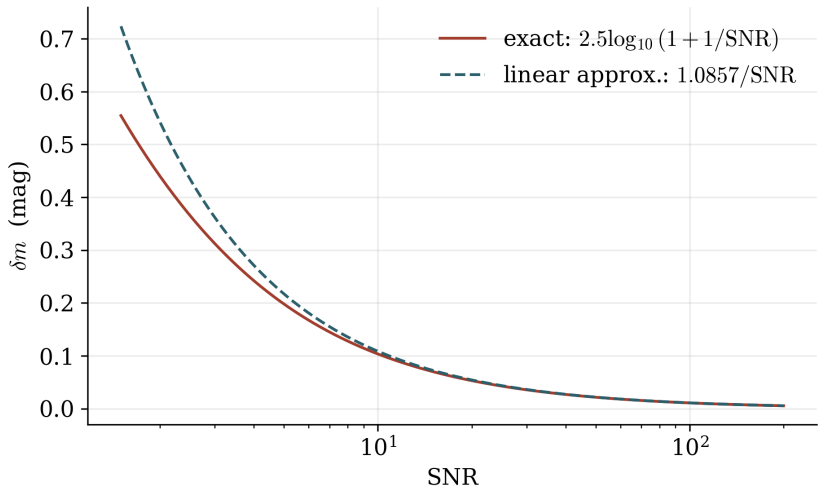


Figure 7.5: Magnitude error δm against SNR. The solid curve is $2.5\log_{10}(1 + 1/\mathrm{SNR})$ and the dashed curve is $1.0857/\mathrm{SNR}$.

SNR	1/SNR	**Meaning**
2–3	33–50%	Barely detected. In a search over a whole image, often a false alarm.
5	20%	The usual threshold for claiming a detection (5σ).
10	10%	A routine measurement, about 0.1 mag.
100	1%	Precise photometry, about 0.01 mag, as needed for planetary transits and variable stars.

Table 7.3: Typical signal-to-noise ratios, the fractional error of each, and what each means for an observation.

The noise is Gaussian. A large Poisson count has the Gaussian shape of Eq. (7.13), and the readout noise is Gaussian from the start. The total noise is a sum of these independent parts. A sum of independent Gaussian quantities is again Gaussian, and by the central limit theorem a sum of many independent parts of any kind is close to Gaussian. So the measured excess, when no source is present, scatters about zero with a Gaussian distribution of standard deviation σ_{tot}. Measuring the excess in units of σ_{tot}, as $k\sigma_{\mathrm{tot}}$, turns every case into the same standard Gaussian.

Probability of a false alarm. With no source present, the probability that the noise alone rises by at least $k\sigma_{\mathrm{tot}}$ is the area of the standard Gaussian beyond k,

$$\boxed{P_{\mathrm{tail}}(k) = \int_k^\infty \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}\, dx.} \tag{7.31}$$

The integral has no closed form, but for large k it is easy to estimate.

$$\begin{aligned} \int_k^\infty e^{-x^2/2}\, dx &\overset{(1)}{=} \int_k^\infty \frac{1}{x}\, d\left(-e^{-x^2/2}\right) \\ &\overset{(2)}{=} \frac{e^{-k^2/2}}{k} - \int_k^\infty \frac{e^{-x^2/2}}{x^2}\, dx \\ &\overset{(3)}{\approx} \frac{e^{-k^2/2}}{k}. \end{aligned} \tag{7.32}$$

(1) $x\,e^{-x^2/2}\,dx = d\big(-e^{-x^2/2}\big)$.
(2) Integration by parts. The boundary term at infinity is zero.
(3) The remaining integral is smaller than the first term by a factor of order $1/k^2$.

$$P_{\text{tail}}(k) \approx \frac{e^{-k^2/2}}{k\sqrt{2\pi}} \qquad (k \gg 1). \tag{7.33}$$

At $k = 5$ this gives 2.97×10^{-7}, against the exact 2.87×10^{-7}. The factor $e^{-k^2/2}$ is why the probability falls so steeply. Going from 5σ to 6σ multiplies it by about $e^{-5.5} \times 5/6$, or $1/294$, against the exact $1/290$. Table 7.4 gives exact values, and Figure 7.6 plots them.

k	$P_{\text{tail}}(k)$	about one in
2	2.28×10^{-2}	44
3	1.35×10^{-3}	741
5	2.87×10^{-7}	3.49×10^{6}
10	7.62×10^{-24}	1.3×10^{23}

Table 7.4: Probability that Gaussian noise alone exceeds k standard deviations in a single trial.

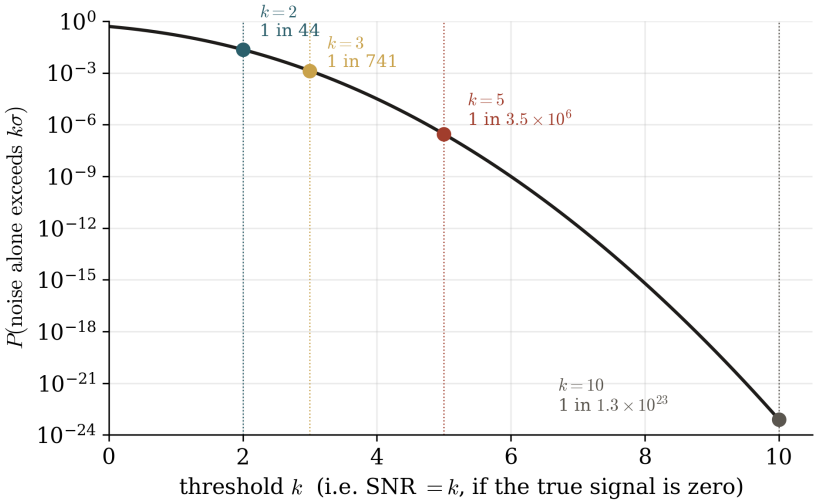


Figure 7.6: Single-trial probability $P_{\text{tail}}(k)$ that Gaussian noise with zero mean exceeds $k\sigma$, on a logarithmic scale, with the values at $k = 2$, 3, 5 and 10 marked.

7.5.3 Searching a Whole Image

The probability $P_{\text{tail}}(k)$ applies to one place chosen in advance, such as the position of a known star. A search for new objects, such as supernovae or asteroids, tests every place in the image at once. Let the image hold n_{trial} independent places, each of which could hold a source. The probability that at least one of them shows a false excess of $k\sigma$ is

$$\begin{aligned} P_{\text{any}}(k) &\overset{(1)}{=} 1 - \big[1 - P_{\text{tail}}(k)\big]^{n_{\text{trial}}} \\ &\overset{(2)}{\approx} n_{\text{trial}}\, P_{\text{tail}}(k). \end{aligned} \tag{7.34}$$

(1) The places are independent, so the probability that none of them exceeds $k\sigma$ is the product of n_{trial} equal factors.

(2) $(1-y)^n \approx 1 - ny$ when $ny \ll 1$.

The product $n_{\text{trial}} P_{\text{tail}}(k)$ is also exactly the mean number of false alarms in the image, whether or not it is small. When it is small it is the chance of even one.

Take a 4000×4000 CCD and count every pixel as a place, $n_{\text{trial}} = 1.6 \times 10^7$. Neighbouring pixels share the light of one star, so the true number of independent places is smaller, and this count gives an upper estimate.

- At $k = 3$ the mean number of false alarms is $1.6 \times 10^7 \times 1.35 \times 10^{-3} \approx 2.2 \times 10^4$. The real objects would be lost among them.
- At $k = 5$ it is $1.6 \times 10^7 \times 2.87 \times 10^{-7} \approx 4.6$ in the whole image.
- At $k = 5.5$ it is about 0.3.

One false alarm per image is expected at $k = 5.3$. This is why large surveys require 5σ or more before they claim a new object.

7.6 Noise Budgets Beyond the CCD

The variance sum of Eq. (7.19) used nothing particular to a CCD. Any measurement made of independent parts has a budget of the same kind, and the part that dominates decides what to improve. Timing the radio pulses of a pulsar is a second example. The measured quantity is the arrival time of a pulse instead of a number of electrons, and its total error is again a sum of independent variances.

- **Radiometer noise.** The noise of the receiver and its electronics, like the read and dark noise of a CCD.
- **Sky noise.** Radio emission from everything else in the beam, the radio form of σ_{sky}.
- **Source noise.** Changes in the pulsar itself, with no CCD counterpart. Single pulses differ in shape (*pulse jitter*), and the spin of the neutron star wanders slightly (*spin noise*).
- **Dispersion noise.** Free electrons in the ionised gas between the stars (Chapter 8) delay a pulse by an amount that depends on the radio frequency. The column of electrons along the line of sight changes as the pulsar and the gas move, so the delay changes too. A CCD image has nothing like it, because the atmosphere and the dust only dim and scatter its light (Section 5.1).
- **Radio-frequency interference.** Man-made radio signals from television, mobile phones, radar and power lines, which must be found and cut out, much as cosmic-ray hits are removed from a CCD frame.

As for a CCD, the term that dominates a given observation decides how it should be planned and reduced.

7.7 Problem Set: Poisson Statistics and CCD Signal-to-Noise

7.7.1 Problem 1: Normalisation, Moments, and the Gaussian Limit of the Poisson Distribution

Problem Statement: A count n has the Poisson distribution $P(n) = \bar{n}^{\,n} e^{-\bar{n}}/n!$ for $n = 0, 1, 2, \ldots$

(a) Prove that it is normalised, $\sum_{n=0}^{\infty} P(n) = 1$.

(b) Prove that its mean and its variance are both equal to $\bar{n}$.

(c) Plot the distribution for $\bar{n} = 0.1$, 1, 10, 100 and 1000, and show that it approaches a Gaussian when $\bar{n}$ is large.

Essential Ideas:

- The exponential series, $\sum_{n=0}^{\infty} \bar{n}^{\,n}/n! = e^{\bar{n}}$, does every sum.
- The mean and variance are derived in Eqs. (7.6) and (7.8). A second route differentiates the exponential series with respect to $\bar{n}$.
- For $\bar{n} \gg 1$, Stirling's formula turns $P(n)$ into a Gaussian of mean $\bar{n}$ and variance $\bar{n}$, Eq. (7.13).

Detailed Solutions:

(a) **Normalisation**.

$$\sum_{n=0}^{\infty} P(n) \overset{(1)}{=} e^{-\bar{n}} \sum_{n=0}^{\infty} \frac{\bar{n}^{\,n}}{n!} \overset{(2)}{=} e^{-\bar{n}}\, e^{\bar{n}}$$

(1) $e^{-\bar{n}}$ does not depend on n.
(2) The exponential series.

$$\boxed{\sum_{n=0}^{\infty} P(n) = 1.}$$

(b) **Mean and variance**. The direct sums are Eqs. (7.6) and (7.8). As a second route, write $\bar{n}\,\frac{\mathrm{d}}{\mathrm{d}\bar{n}}$ for the operator that multiplies the n-th term of the series by n, since $\bar{n}\,\frac{\mathrm{d}}{\mathrm{d}\bar{n}}\bar{n}^{\,n} = n\,\bar{n}^{\,n}$. Applying it once and twice to the exponential series,

$$\sum_{n=0}^{\infty} n\,\frac{\bar{n}^{\,n}}{n!} \overset{(1)}{=} \bar{n}\,\frac{\mathrm{d}}{\mathrm{d}\bar{n}}\, e^{\bar{n}} = \bar{n}\, e^{\bar{n}},$$

$$\sum_{n=0}^{\infty} n^2\,\frac{\bar{n}^{\,n}}{n!} \overset{(2)}{=} \bar{n}\,\frac{\mathrm{d}}{\mathrm{d}\bar{n}}\left(\bar{n}\, e^{\bar{n}}\right) = \bar{n}\,(1+\bar{n})\, e^{\bar{n}}.$$

(1) Differentiate the series term by term and multiply by $\bar{n}$.
(2) Do it again to the first line.

Multiplying by $e^{-\bar{n}}$ gives $\langle n\rangle = \bar{n}$ and $\langle n^2\rangle = \bar{n} + \bar{n}^2$. Then

$$\boxed{\langle n\rangle = \bar{n}, \qquad \sigma^2 = \langle n^2\rangle - \langle n\rangle^2 = \bar{n}.}$$

(c) **Plot and the Gaussian limit**. The distribution can be drawn directly from its definition.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import poisson, norm

lambdas = [0.1, 1, 10, 100, 1000]
fig, axes = plt.subplots(1, len(lambdas), figsize=(18, 3.2))

for ax, lam in zip(axes, lambdas):
    k_max = int(lam + 6*np.sqrt(lam) + 5)
    k = np.arange(0, k_max + 1)
    ax.bar(k, poisson.pmf(k, lam), color="C0", width=1.0, label="
        Poisson")
    if lam >= 10:
```

```
        x = np.linspace(0, k_max, 400)
        ax.plot(x, norm.pdf(x, loc=lam, scale=np.sqrt(lam)), "C3--",
            label="Gaussian")
    ax.set_title(rf"$\lambda={lam}$")
    ax.set_xlabel("$k$")
axes[0].set_ylabel("$P[X=k]$")
axes[-1].legend()
plt.tight_layout()
plt.show()
```

(In the code, `lam` is $\bar{n}$ and `k` is n.) Figure 7.7 shows the five distributions with the Gaussian of the same mean and variance, Eq. (7.13).

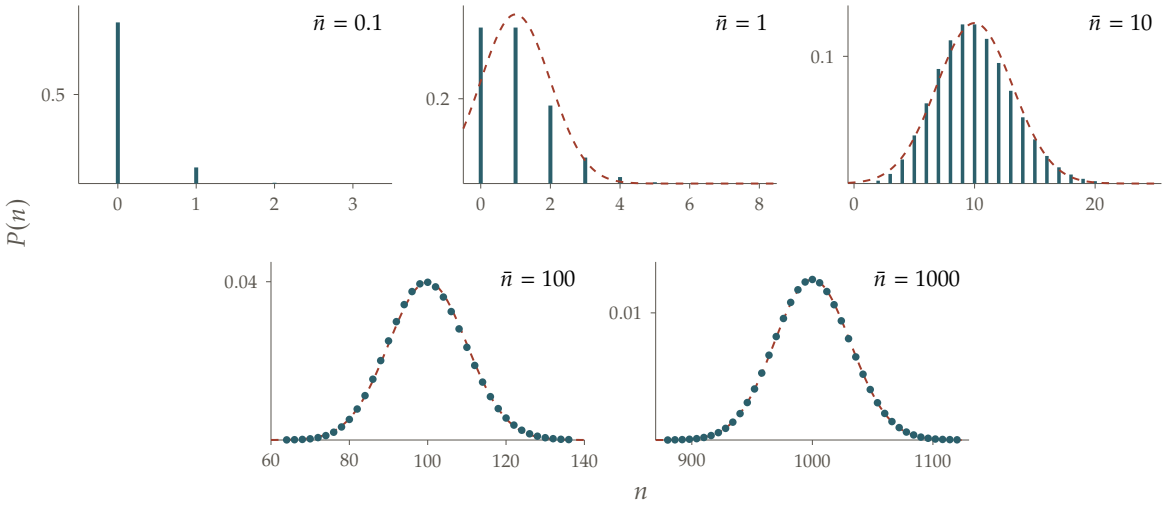


Figure 7.7: Poisson probabilities $P(n)$ for $\bar{n} = 0.1$, 1, 10, 100 and 1000 (bars, or dots at every second and every sixth n in the last two panels), with the Gaussian of mean $\bar{n}$ and variance $\bar{n}$ (dashed) for $\bar{n} \geq 1$.

For $\bar{n} = 0.1$ the count is 0 with probability $e^{-0.1} = 0.905$ and 1 with probability $0.1\,e^{-0.1} = 0.090$, and larger counts almost never happen. No bell curve fits this. For $\bar{n} = 1$ the counts 0 and 1 are equally likely, and the Gaussian puts probability at negative n, which cannot occur. For $\bar{n} = 10$ the shape is close to a bell but lopsided, with a longer tail to the right. For $\bar{n} = 100$ and 1000 the points lie on the Gaussian. The peak height checks this, since Eq. (7.13) gives $1/\sqrt{2\pi\bar{n}} = 0.012616$ at $\bar{n} = 1000$, against the Poisson value 0.012615. The lopsidedness comes from the dropped term of order $(n-\bar{n})^3/\bar{n}^2$, which is $1/\sqrt{\bar{n}}$ at one standard deviation, so

$$\boxed{P(n) \to \frac{1}{\sqrt{2\pi\bar{n}}} \exp\left[-\frac{(n-\bar{n})^2}{2\bar{n}}\right] \quad \text{for } \bar{n} \gg 1.}$$

This is why the noise of large counts may be treated as Gaussian, as in Section 7.5.2.

7.7.2 Problem 2: Signal-to-Noise with Exposure Time and Adding Images

Problem Statement: Two images I_1 and I_2 of the same bright object are taken with the same instrument. Both are dominated by the shot noise of the source. I_1 has an exposure time of 1 min and SNR = 100.

(a) What is the SNR of I_2 if its exposure time is 5 min?

(b) If the two images are added pixel by pixel, what is the SNR of the sum?

Essential Ideas:

- With the source's own shot noise dominant, SNR $= \sqrt{R_{\mathrm{src}} t} \propto \sqrt{t}$, Eq. (7.24).
- Variances of independent images add, Eq. (7.10), and the variance of a Poisson count equals its mean, Eq. (7.8).

Detailed Solutions:

(a) **Longer exposure**. Only the time differs between the two images.

$$\begin{aligned} \mathrm{SNR}_2 &\overset{(1)}{=} \sqrt{R_{\mathrm{src}} t_2} \\ &\overset{(2)}{=} \mathrm{SNR}_1 \sqrt{\frac{t_2}{t_1}} = 100\sqrt{5} \end{aligned}$$

(1) Eq. (7.24).
(2) Divide by $\mathrm{SNR}_1 = \sqrt{R_{\mathrm{src}} t_1}$.

$$\boxed{\mathrm{SNR}_2 = 223.6.}$$

(b) **Sum of the two images**. The signals add, $S_{1+2} = S_1 + S_2 = R_{\mathrm{src}}(t_1 + t_2)$. The two images are independent, so their variances add, and each variance equals the signal.

$$\begin{aligned} \mathrm{SNR}_{1+2} &\overset{(1)}{=} \frac{S_1 + S_2}{\sqrt{\sigma_1^2 + \sigma_2^2}} \\ &\overset{(2)}{=} \frac{S_1 + S_2}{\sqrt{S_1 + S_2}} = \sqrt{R_{\mathrm{src}}(t_1 + t_2)} \\ &\overset{(3)}{=} \mathrm{SNR}_1 \sqrt{1 + \frac{t_2}{t_1}} = 100\sqrt{6} \end{aligned}$$

(1) Eq. (7.14), with the variance of the sum from Eq. (7.10).
(2) $\sigma_i^2 = S_i$ for a Poisson count, Eq. (7.8).
(3) Divide by $\mathrm{SNR}_1 = \sqrt{R_{\mathrm{src}} t_1}$.

$$\boxed{\mathrm{SNR}_{1+2} = 244.9.}$$

As a check by counting electrons, $\mathrm{SNR}_1 = 100$ means $S_1 = 100^2 = 10^4$ electrons, so $S_2 = 5 \times 10^4$ and the sum has 6×10^4 electrons, with $\sqrt{6 \times 10^4} = 244.9$. Adding the images gives the same result as one exposure of 6 min. That stops being exactly true when readout noise matters, because the sum has been read twice and the single exposure only once.

7.7.3 Problem 3: A Faint Star Twice as Far Away

Problem Statement: The image of a faint star has SNR = 5, and its noise is dominated by instrumental noise.

(a) What would the SNR be if the star were twice as far away, with everything else unchanged?

(b) What would it be if the noise had instead been dominated by the star's own shot noise?

Essential Ideas:

- Instrumental noise, the dark and readout terms, does not depend on the star, so SNR $\propto R_{\mathrm{src}}$, Eq. (7.27).

- With the star's own shot noise dominant, SNR $\propto \sqrt{R_{\rm src}}$, Eq. (7.24).
- The flux, and so $R_{\rm src}$, falls as $1/d^2$, Eq. (4.2).

Detailed Solutions:

(a) **Instrumental noise**. The variance $n_{\rm pix}R_{\rm dark}t + n_{\rm pix}N_{\rm exp}\sigma_{\rm RON}^2$ contains no light from the star, so it is the same at either distance. Only the signal changes.

$$\frac{\mathrm{SNR}(2d)}{\mathrm{SNR}(d)} \overset{(1)}{=} \frac{R_{\rm src}(2d)}{R_{\rm src}(d)} \overset{(2)}{=} \left(\frac{d}{2d}\right)^2 = \frac{1}{4}$$

(1) SNR $= R_{\rm src}t/\sigma_{\rm tot}$ with t and $\sigma_{\rm tot}$ unchanged.
(2) $R_{\rm src} \propto F \propto 1/d^2$.

$$\boxed{\mathrm{SNR}(2d) = \frac{5}{4} = 1.25.}$$

The star is no longer detected. An excess of 1.25σ is produced by noise alone about once in ten trials.

(b) **Shot-noise limit**. Now SNR $= \sqrt{R_{\rm src}t} \propto \sqrt{R_{\rm src}} \propto 1/d$, so

$$\boxed{\mathrm{SNR}(2d) = \frac{5}{2} = 2.5.}$$

A star limited by its own light loses less when it is moved away, because its noise falls with its signal. The same light loss costs a faint star on a bright sky or on a noisy detector twice as much.

7.7.4 Problem 4: Dark Current and Readout Noise from Two Dark Frames

Problem Statement: Two dark frames are taken with the same telescope and CCD. The first has an exposure time of 1 min and a noise variance a. The second has an exposure time of 3 min and a noise variance $2.6a$.

(a) Find the noise variance of the dark current per unit time.

(b) Find the noise variance of the readout noise.

Essential Ideas:

- A dark frame has no star and no sky. Its variance is $\sigma^2(t) = n_{\rm pix}R_{\rm dark}t + n_{\rm pix}\sigma_{\rm RON}^2$, a straight line in t. For a single pixel, $n_{\rm pix} = 1$.
- Two frames of different lengths give two equations for the two unknowns, solved in Eq. (7.21).

Detailed Solutions: The two frames give

$$\begin{pmatrix} 1\,\mathrm{min} & 1 \\ 3\,\mathrm{min} & 1 \end{pmatrix} \begin{pmatrix} n_{\rm pix}R_{\rm dark} \\ n_{\rm pix}\sigma_{\rm RON}^2 \end{pmatrix} = \begin{pmatrix} a \\ 2.6a \end{pmatrix}.$$

(a) **Dark variance per unit time**. By Eq. (7.21),

$$\boxed{n_{\rm pix}R_{\rm dark} = \frac{\sigma_2^2 - \sigma_1^2}{t_2 - t_1} = \frac{2.6a - a}{3\,\mathrm{min} - 1\,\mathrm{min}} = 0.8a \text{ per minute.}}$$

(b) **Read variance**. By Eq. (7.21),

$$\boxed{n_{\mathrm{pix}}\sigma_{\mathrm{RON}}^2 = \frac{t_2\sigma_1^2 - t_1\sigma_2^2}{t_2 - t_1} = \frac{3a - 2.6a}{2} = 0.2a.}$$

As a check, the first frame gives $0.8a \times 1 + 0.2a = a$. In the 1-minute frame the dark current gives 80 per cent of the variance, and in the 3-minute frame 92 per cent. The read variance is the same $0.2a$ in every frame, whatever its length. A 5-minute frame should show $0.8a \times 5 + 0.2a = 4.2a$, and taking one is how the straight line of Figure 7.3 would be tested.

Chapter 8

The Interstellar Medium

Chapters 3 and 4 measured stars by their light alone, as if the light crossed empty space. It does not. Between the stars lies a thin gas with a little dust in it, the **interstellar medium**. Every photon we collect has passed through it, and the medium has dimmed it, reddened it and delayed it on the way.

The same gas is the raw material of stars. Stars form out of it, and when they die they return much of their mass to it, enriched with heavier elements.

8.1 Composition, Density and Temperature

The earlier chapters treated a star as a point of light, and read its position, distance, flux and colour from that light.

They took the space between the star and the telescope to be empty. It is not, and a star seen through a dusty part of the Galaxy can be dimmed by many magnitudes.

Think of a light mist on a cool morning. Across a room nobody notices it. A hill a kilometre away is hidden by it. Each metre of mist holds very little water, but a kilometre of it holds a thousand times as much.

The interstellar medium is a mist of this kind, far thinner than any mist on Earth but thousands of parsecs deep. What decides its effect on light is the total amount of matter along the line of sight, not how dense it is at any point.

Toward the centre of the Galaxy, 8 kpc away, gas with one atom per cubic centimetre puts 2.5×10^{22} atoms into a column of cross-section 1 cm^2. Air at sea level holds 2.5×10^{19} molecules per cubic centimetre, so a column of air only 10 m long holds as many.

Picture a long thin tube from the star to the telescope. What matters is how much matter the tube holds.

That amount is the column density of Chapter 4. The optical depth compares the length of the tube with the mean distance between grains, a comparison of two scales (P6, Section 9.6). The same count sets the strength of absorption lines and the brightness of the 21 cm line of hydrogen. The temperature of the gas is set by a balance of rates (P4) and the widths of its lines by counting states (P3). Here we list what the medium is made of, how dense and hot it is, and why it matters.

What it is. The **interstellar medium** is the matter between the stars of a galaxy. It is gas, dust, fast charged particles called cosmic rays, and a magnetic field. It holds of order ten to twenty per cent of the ordinary matter of the Milky Way. Its bulk properties are in Table 8.1.

Property	Typical value
Number density n	10^{-4}–$10^{6}\,\mathrm{cm}^{-3}$
Temperature T	10–$10^{6}\,\mathrm{K}$
Composition (by mass)	70% H, 28% He, $\lesssim$ 2% heavier elements
Phase split (by mass)	99% gas, 1% dust
Magnetic field strength	$\sim 10^{-10}\,\mathrm{T}$ ($\sim 1\,\mu\mathrm{G}$)

Table 8.1: Bulk properties of the interstellar medium.

How thin it is. The best laboratory vacuum, called extreme high vacuum, still holds about 10^5 molecules per cubic centimetre. Much of the interstellar medium has fewer than one atom per cubic centimetre, so it is emptier than any vacuum made on Earth. In such thin gas an excited atom can wait a long time before a collision disturbs it. Transitions too slow to be seen in laboratory gas, the so-called forbidden lines, then become the main way the gas radiates (Section 8.4).

What it is made of. The gas holds ions such as H^+, He^+ and He^{2+}, free electrons, neutral atoms, and molecules such as H_2, CO, CN, H_2O and NH_3. The dust is grains from about 1 nm to 1 μm across, made of silicates, carbon in several forms, and metals such as iron and nickel, often coated with ices of water and ammonia. Dust is only one per cent of the mass. Per kilogram, small grains block far more light than gas does (Eq. (8.13)), so dust causes almost all of the dimming (Section 8.2). Its surfaces are also where molecular hydrogen forms (Section 8.4.2).

Why it matters. There are three reasons. Every photon used in the earlier chapters is scattered, absorbed, delayed and dispersed by the medium, so the corrections for dust in Chapter 4 rest on understanding it. It holds conditions that no laboratory can make, gas thinner than any vacuum and colder than 20 K, with chemistry far from equilibrium. And it is where stars are born and where their matter returns (Chapter 10).

8.2 How the Medium Acts on Light

Chapter 4 found that dust dims a beam by the factor $e^{-\tau_\lambda}$, with optical depth $\tau_\lambda = N_d\,\sigma_{\mathrm{ext}}(\lambda)$ (Eq. (4.33)), and that across the visible the cross-section σ_{ext} falls roughly as $1/\lambda$.

That cross-section was taken from measurement. Why it depends on wavelength at all, why the sky is blue while clouds are white, and why free electrons scatter every colour alike was left open.

Hold a weight on a soft spring and move your hand up and down. Move it slowly and the weight follows the hand. Move it at the rate the weight bounces by itself and it swings wildly. Shake fast and the weight hardly moves. What decides is how fast you shake compared with the spring's own rate.

Light shakes the electrons in an atom, a molecule or a grain, and each is held in place as if by a spring. A shaken charge radiates, and the light it sends off in new directions is the scattered light. One shaken spring gives the blue sky, the spectral lines and the scattering by free electrons.

For a molecule of air, visible light shakes far below the natural rate. The scattered power then grows as the fourth power of the frequency, so blue light of 450 nm is scattered $(700/450)^4 = 5.9$ times more strongly than red light of 700 nm.

The picture is a curve of cross-section against frequency. It climbs steeply from zero, spikes at the natural rate, and levels off far above it.

The ratio of the driving frequency to the natural one picks the regime, and for a grain the ratio of its size to the wavelength does the same. Both are comparisons of two scales (P6). The same shaken charge gave the colours of materials and the reflection of radio waves by the ionosphere in Chapter 5, and it gives the refractive index of glass. Here it gives the cross-section of one oscillator, its regimes, the scattering by grains, and the delay of radio waves by free electrons.

Extinction. Light is taken out of a beam in two ways. It can be absorbed, its energy turned into heat or internal excitation of the absorber. Or it can be scattered, sent off in a new direction. The total loss is the **extinction**,

$$\text{extinction} = \text{absorption} + \text{scattering}. \tag{8.1}$$

Absorption happens either at sharp frequencies, as lines of atoms and molecules, or over broad bands, in grains. Scattering is sorted by whether the photon keeps its energy.

8.2.1 One Oscillator, Three Regimes

The driven electron. Model a bound electron as a charge $-e$ of mass m_e held by a spring of natural angular frequency ω_0. Left alone it also loses energy, at the rate Γ, so its energy dies away as $e^{-\Gamma t}$. A light wave with electric field $E_0 \cos \omega t$ along x drives it. Newton's law is

$$m_e \ddot{x} = -m_e \omega_0^2 x - m_e \Gamma \dot{x} - e E_0 \cos \omega t. \tag{8.2}$$

To find how far it moves, write x as the real part of $\tilde{x}\, e^{-i\omega t}$ and the force as the real part of $-e E_0\, e^{-i\omega t}$.

$$\begin{aligned} -\omega^2 \tilde{x} &\overset{(1)}{=} -\omega_0^2 \tilde{x} + i\omega\Gamma\, \tilde{x} - \frac{e}{m_e} E_0 \\ \Longrightarrow \qquad \tilde{x} &\overset{(2)}{=} -\frac{e}{m_e}\, \frac{E_0}{\omega_0^2 - \omega^2 - i\Gamma\omega}. \end{aligned} \tag{8.3}$$

(1) Each time derivative becomes a factor $-i\omega$, so $\ddot{x} \to -\omega^2 \tilde{x}$ and $\dot{x} \to -i\omega\tilde{x}$. Divide by m_e.
(2) Collect the terms in $\tilde{x}$ and divide.

The cross-section. The moving electron is a dipole $-ex$. The scattering cross-section is the power it radiates divided by the power per unit area of the wave that drives it. Larmor's formula gives the radiated power of a dipole $p_0 \cos \omega t$, averaged over a cycle, as $p_0^2 \omega^4 / (12\pi\varepsilon_0 c^3)$. The wave carries $c\varepsilon_0 E_0^2/2$ per unit area, averaged over a cycle.

$$\begin{aligned} \sigma(\omega) &\overset{(1)}{=} \frac{e^2 |\tilde{x}|^2 \omega^4 / (12\pi\varepsilon_0 c^3)}{c\varepsilon_0 E_0^2/2} \\ &\overset{(2)}{=} \frac{e^4}{6\pi\varepsilon_0^2 m_e^2 c^4}\, \frac{\omega^4}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2} \\ &\overset{(3)}{=} \frac{8\pi}{3} \left(\frac{e^2}{4\pi\varepsilon_0 m_e c^2} \right)^2 \frac{\omega^4}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2}. \end{aligned} \tag{8.4}$$

(1) Larmor's formula with dipole amplitude $e|\tilde{x}|$, over the flux of the wave.
(2) $|\tilde{x}|^2$ from Eq. (8.3). The factor E_0^2 cancels.
(3) $6\pi\varepsilon_0^2 = \frac{3}{8\pi}(4\pi\varepsilon_0)^2$.

The constant in front is the **Thomson cross-section** σ_T, written with the **classical electron radius** $r_e = e^2/(4\pi\varepsilon_0 m_e c^2) = 2.818 \times 10^{-15}$ m. So

$$\boxed{\sigma(\omega) = \sigma_T \frac{\omega^4}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2}.} \tag{8.5}$$

The answer depends on the frequency only through ω/ω_0 and Γ/ω_0. The damping is weak, $\Gamma \ll \omega_0$, so three regimes appear.

Slow shaking, Rayleigh scattering. For $\omega \ll \omega_0$ the denominator is ω_0^4 to leading order, and

$$\sigma \approx \sigma_T \left(\frac{\omega}{\omega_0}\right)^4 \propto \frac{1}{\lambda^4}. \tag{8.6}$$

The electrons of air molecules have natural frequencies in the ultraviolet, so visible light is in this regime. Blue light is scattered most, which makes the daytime sky blue and the setting Sun red. This is **Rayleigh scattering**.

Fast shaking, Thomson scattering. For $\omega \gg \omega_0$ the spring no longer matters, and $\sigma \to \sigma_T$. A free electron, with no spring at all ($\omega_0 = 0$), has $\sigma = \sigma_T$ at every frequency. This is the electron of Chapter 5 (Eq. (5.1)), now asked how much it radiates rather than how it bends a wave.

$$\sigma_\mathrm{T} = \frac{8\pi}{3}\left(\frac{e^2}{4\pi\varepsilon_0 m_e c^2}\right)^2 = \frac{8\pi}{3} r_e^2 = 6.652 \times 10^{-29}\,\mathrm{m}^2. \tag{8.7}$$

It does not depend on the wavelength, so free electrons scatter all colours alike. **Thomson scattering** is the main opacity of hot ionised gas, in the interiors and envelopes of hot stars (Chapter 11), in accretion disks, and in the early universe before the cosmic microwave background was released. It holds while the photon energy is small compared with $m_e c^2$.

Shaking at resonance, a line. Near ω_0, write $\omega = \omega_0 + \delta$ with $|\delta| \ll \omega_0$.

$$\begin{aligned} \sigma &\overset{(1)}{=} \sigma_T \frac{\omega_0^4}{(2\omega_0\delta)^2 + \Gamma^2\omega_0^2} \\ &\overset{(2)}{=} \sigma_T \frac{\omega_0^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4}. \end{aligned} \tag{8.8}$$

(1) $\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \approx -2\omega_0\delta$, and $\omega \approx \omega_0$ elsewhere.
(2) Divide the top and bottom by $4\omega_0^2$.

The cross-section has a sharp peak of full width Γ at half its height, at the frequency where the oscillator absorbs. At the peak it is $\sigma_T(\omega_0/\Gamma)^2$, far larger than σ_T, which is why a thin gas makes strong lines. This peak is an absorption line, and its shape is found again by a second route in Section 8.3.

8.2.2 Scattering by Grains

A grain holds many electrons. How they radiate together depends on the size a of the grain compared with the wavelength. The ratio that decides is $2\pi a/\lambda$ (P6).

Grains much smaller than the wavelength. When $a \ll \lambda$ the field of the wave is the same across the whole grain. A sphere of material with relative permittivity ε_r in a uniform field becomes a dipole

$$\tilde{p} = 4\pi\varepsilon_0 a^3 \frac{\varepsilon_r - 1}{\varepsilon_r + 2} E_0,$$

a standard result of electrostatics. Here ε_r is complex when the material absorbs. All the electrons move in step, so the grain radiates as one dipole. Its scattering cross-section is

$$\begin{aligned} \sigma_{\text{sca}} &\overset{(1)}{=} \frac{|\tilde{p}|^2 \omega^4 / (12\pi\varepsilon_0 c^3)}{c\varepsilon_0 E_0^2/2} \\ &\overset{(2)}{=} \frac{16\pi^2 a^6\, \omega^4}{6\pi c^4} \left|\frac{\varepsilon_r - 1}{\varepsilon_r + 2}\right|^2 \\ &\overset{(3)}{=} \frac{8\pi}{3} \left(\frac{2\pi}{\lambda}\right)^4 a^6 \left|\frac{\varepsilon_r - 1}{\varepsilon_r + 2}\right|^2 . \end{aligned} \tag{8.9}$$

(1) Larmor's formula for the dipole $\tilde{p}$, over the flux of the wave, as for the electron.
(2) $|\tilde{p}|^2/E_0^2 = 16\pi^2\varepsilon_0^2 a^6 |(\varepsilon_r - 1)/(\varepsilon_r + 2)|^2$, and ε_0^2 cancels.
(3) $\omega/c = 2\pi/\lambda$.

$$\boxed{\sigma_{\text{sca}} \propto \frac{a^6}{\lambda^4}.} \tag{8.10}$$

The λ^{-4} is the same as for one molecule. The a^6 comes from radiating in step. The number of electrons grows as a^3, the total dipole grows with it, and the power goes as the square. Blue light scattered by small grains makes the blue reflection nebulae around young stars, such as the Pleiades.

The grain also absorbs. The field does work on the dipole at the rate $\frac{1}{2}\omega E_0 \operatorname{Im} \tilde{p}$, averaged over a cycle, so

$$\begin{aligned} \sigma_{\text{abs}} &\overset{(1)}{=} \frac{\frac{1}{2}\omega E_0 \operatorname{Im} \tilde{p}}{c\varepsilon_0 E_0^2/2} \\ &\overset{(2)}{=} \frac{2\pi}{\lambda} 4\pi a^3 \operatorname{Im} \frac{\varepsilon_r - 1}{\varepsilon_r + 2} \;\propto\; \frac{a^3}{\lambda}. \end{aligned} \tag{8.11}$$

(1) Power taken from the wave, over its flux.
(2) The dipole $\tilde{p}$ above, and $\omega/c = 2\pi/\lambda$.

For small grains absorption beats scattering, since $\sigma_{\text{sca}}/\sigma_{\text{abs}} \propto (a/\lambda)^3$.

Grains near the wavelength, Mie scattering. When $2\pi a/\lambda$ is near one, the field changes across the grain and its parts radiate out of step. The exact answer, found by Mie, expands the fields inside and outside the sphere in vector spherical harmonics. Over this range the cross-section of a grain grows roughly as its area times a/λ, so $\sigma_{\text{ext}} \propto a^3/\lambda$, the same form as Eq. (8.11). Interstellar grains have $a \approx 0.1\,\mu$m, so $2\pi a/\lambda = 1.1$ at 550 nm. Visible light is in the **Mie** regime, and the mix of grain sizes gives the measured extinction that falls roughly as $1/\lambda$ (Figure 4.1).

Grains much larger than the wavelength. When $a \gg \lambda$ the grain casts a sharp shadow and blocks its own area πa^2. It also bends light out of the beam by diffraction at its edge. By Babinet's principle a disc diffracts as much power as a hole of the same size lets through,

another πa^2. So $\sigma_{\text{ext}} \to 2\pi a^2$, the same for every colour. Water droplets in clouds and fog are this large, which is why clouds are white or grey. Large grains in the shells around old stars behave the same way.

Tyndall scattering. Scattering by fine particles suspended in a gas or a liquid, in the range from Rayleigh to Mie, is called **Tyndall scattering**. It makes a sunbeam visible in a dusty room or a smoky forest. Dusty interstellar clouds lit by nearby stars shine in the same way.

8.2.3 Scattering That Changes the Colour

In all the cases above the photon leaves with the frequency it came with. This is **elastic** scattering. In **inelastic** scattering the photon trades energy with the scatterer and leaves with a new frequency.

Raman scattering. The photon excites or de-excites a vibration or rotation of a molecule or atom. It leaves with lower frequency if it gave energy (Stokes) and with higher frequency if it took energy from an already excited molecule (anti-Stokes). In the laboratory this identifies chemical bonds. In symbiotic stars and some planetary nebulae, ultraviolet photons of O VI at 103.2 and 103.8 nm are Raman-scattered by neutral hydrogen into broad bands at 682.5 and 708.2 nm.

Brillouin scattering. Light scatters off sound waves in a dense fluid or solid. The sound wave acts as a moving grating, so the scattered light is Doppler shifted by $\Delta\nu = \pm 2\nu_0 (c_s/c)\sin(\psi/2)$, times the refractive index of the medium, where c_s is the sound speed and ψ the scattering angle.

Compton and inverse Compton scattering. When the photon energy approaches $m_e c^2$, the electron recoils and the photon loses energy. Its wavelength grows by $\Delta\lambda = \lambda_C(1 - \cos\psi)$ (derived in Eq. (15.29)), with the Compton wavelength $\lambda_C = h/(m_e c) = 2.426 \times 10^{-12}$ m. In **inverse Compton** scattering the electron moves near the speed of light, with Lorentz factor $\gamma_e \gg 1$, and gives energy to a low-energy photon of the radio, infrared or microwave background. The photon leaves with about γ_e^2 times its energy, in X-rays or gamma rays. This lights up pulsar wind nebulae, supernova remnants and the jets of active galaxies.

Table 8.2 collects the processes.

Process	Scatterer	Dependence	Photon energy	Where seen
Rayleigh	Atoms, molecules, small grains ($a \ll \lambda$)	$\sigma \propto a^6 \lambda^{-4}$	kept	Blue sky, Pleiades reflection nebula
Mie	Dust grains ($a \sim \lambda$)	$\sigma \propto a^3 \lambda^{-1}$	kept	Interstellar extinction and reddening
Geometric	Large particles ($a \gg \lambda$)	$\sigma \to 2\pi a^2$, no colour	kept	White clouds, dust shells of old stars
Tyndall	Suspended fine particles	between Rayleigh and Mie	kept	Sunbeams, dusty clouds lit by stars
Raman	Molecular and atomic levels	at set energy steps	changed	Symbiotic stars, Raman O VI bands
Brillouin	Sound waves in dense media	shift $\propto c_s/c$	changed	Laboratory solids and liquids
Thomson	Free electrons ($h\nu \ll m_e c^2$)	σ_{T}, no colour	kept	Hot stars, accretion disks, early universe
Inverse Compton	Fast electrons ($\gamma_e \gg 1$)	$\nu \to\sim \gamma_e^2 \nu$	raised	Jets of active galaxies, pulsar wind nebulae

Table 8.2: Scattering processes, the scatterer, how the cross-section or the shift depends on size and wavelength, whether the photon energy changes, and where each is seen.

8.2.4 Dust Extinction and Reddening

Chapter 4 turned the cross-section into magnitudes. The extinction is $A_\lambda = 1.086\,\tau_\lambda = 1.086\,N_d\,\sigma_{\text{ext}}(\lambda)$ (Eq. (4.36)). Because σ_{ext} is larger in B than in V, dust makes a star redder by the colour excess $E(B-V) = A_B - A_V$, and in the Milky Way $A_V = 3.1\,E(B-V)$ (Eq. (4.42)). The

intrinsic colour and the distance modulus corrected for dust follow from Eqs. (4.40) and (4.44). What is new here is what the grains themselves tell us.

Extinction per kilogram of dust. To see why one per cent of the mass does almost all the dimming, write the extinction in terms of the mass of dust along the line of sight. Let each grain be a sphere of radius a and density ρ_{grain}, and let Σ_d be the mass of dust in a column of cross-section $1\,\text{m}^2$.

$$\begin{aligned} A_\lambda &\overset{(1)}{=} 1.086\, N_d\, \sigma_{\text{ext}} \\ &\overset{(2)}{=} 1.086\, \Sigma_d\, \frac{\sigma_{\text{ext}}}{\frac{4}{3}\pi a^3 \rho_{\text{grain}}} \\ &\overset{(3)}{\approx} 1.086\, \Sigma_d\, \frac{3}{4 a\, \rho_{\text{grain}}}. \end{aligned} \tag{8.12}$$

(1) Eq. (4.36).
(2) N_d grains each of mass $\frac{4}{3}\pi a^3 \rho_{\text{grain}}$ make up Σ_d.
(3) In the Mie regime the cross-section of a grain is close to its area, $\sigma_{\text{ext}} \approx \pi a^2$.

$$\boxed{A_\lambda \approx 1.086\, \frac{3\,\Sigma_d}{4\, a\, \rho_{\text{grain}}}.} \tag{8.13}$$

Per kilogram, a grain blocks light in proportion to $1/(a\rho_{\text{grain}})$. Breaking a kilogram of matter into grains of $0.1\,\mu$m gives it an enormous total area. The same kilogram as gas atoms scatters almost nothing at visible wavelengths, because each atom is in the Rayleigh regime, with a cross-section below σ_T by Eq. (8.6).

A check against the measured dust. Take grains of radius $0.1\,\mu$m and density $3000\,\text{kg}\,\text{m}^{-3}$, and one per cent of the mass in dust. With hydrogen making up 70 per cent of the mass, each hydrogen atom comes with $0.01\, m_{\text{H}}/0.70 = 2.4 \times 10^{-29}\,\text{kg}$ of dust. Each grain has mass $1.26{\times}10^{-17}\,\text{kg}$, so there are $1.9{\times}10^{-12}$ grains per hydrogen atom. Each blocks $\pi a^2 = 3.1{\times}10^{-14}\,\text{m}^2$, so per hydrogen atom in the column

$$\frac{A_V}{N_{\text{H}}} \approx 1.086 \times 1.9 \times 10^{-12} \times 3.1 \times 10^{-14}\,\text{m}^2 = 6.5 \times 10^{-26}\,\text{mag}\,\text{m}^2,$$

where N_{H} is the column density of hydrogen atoms. Ultraviolet absorption lines of hydrogen toward reddened stars give $N_{\text{H}}/E(B-V) = 5.8 \times 10^{25}\,\text{m}^{-2}$ per magnitude, and with $A_V = 3.1\,E(B-V)$ that is $A_V/N_{\text{H}} = 5.3 \times 10^{-26}\,\text{mag}\,\text{m}^2$. The simple estimate is right to within 25 per cent. In the units of the field, one magnitude of A_V needs a column of 1.9×10^{21} hydrogen atoms per cm^2.

The extinction curve. Measured toward many stars, the extinction rises from the infrared through the visible into the ultraviolet. At the long-wavelength end it follows the rough $1/\lambda$ of Mie scattering. Near 217.5 nm there is a broad bump, carried by grains that contain carbon, in the form of graphite or of large molecules called polycyclic aromatic hydrocarbons. The curve differs a little from one line of sight to another, as the mix of grains does.

Dark clouds seen in the infrared. At $2.2\,\mu$m, $2\pi a/\lambda = 0.29$ for the same grains. They are close to the Rayleigh regime and block only about a tenth as much as in V. A cloud such as Barnard 68, black against the stars in visible light, is nearly transparent in the near infrared, and the stars behind it reappear, reddened.

8.2.5 Free Electrons Delay Radio Waves

Free electrons do not only scatter light. They also slow it, by an amount that depends on its frequency. In Chapter 5 the free electrons of a plasma gave the refractive index $n^2 = 1 - \nu_p^2/\nu^2$ (Eq. (5.1)), with the plasma frequency ν_p of Eq. (5.2). In interstellar gas with $n_e = 3 \times 10^4\,\mathrm{m}^{-3}$ ($0.03\,\mathrm{cm}^{-3}$), $\nu_p = 1.6\,\mathrm{kHz}$, far below any radio frequency used. The wave passes, but a pulse does not travel at c.

The speed of a pulse. A pulse moves at the group velocity $d\omega/dk$, with k the wavenumber and $\omega = 2\pi\nu$.

$$\begin{aligned} \omega^2 &\overset{(1)}{=} \frac{c^2k^2}{n^2} \\ &\overset{(2)}{=} \omega_p^2 + c^2k^2, \\ \frac{\mathrm{d}\omega}{\mathrm{d}k} &\overset{(3)}{=} \frac{c^2k}{\omega} \overset{(4)}{=} c\sqrt{1 - \frac{\nu_p^2}{\nu^2}}. \end{aligned} \tag{8.14}$$

(1) The phase velocity is $\omega/k = c/n$.
(2) $n^2 = 1 - \omega_p^2/\omega^2$, with $\omega_p = 2\pi\nu_p$, multiplied by ω^2 and rearranged.
(3) Differentiate the second line, $2\omega\, d\omega = 2c^2k\, dk$.
(4) $ck/\omega = n$.

The delay. A pulse that crosses a path of length d to the source arrives after

$$\begin{aligned} t &\overset{(1)}{=} \int_0^d \frac{ds}{c}\left(1 - \frac{\nu_p^2}{\nu^2}\right)^{-1/2} \\ &\overset{(2)}{\approx} \frac{d}{c} + \frac{1}{2c\nu^2}\int_0^d \nu_p^2\, ds \\ &\overset{(3)}{=} \frac{d}{c} + \frac{e^2}{8\pi^2\varepsilon_0 m_e c}\,\frac{1}{\nu^2}\int_0^d n_e\, ds. \end{aligned} \tag{8.15}$$

(1) Time is $\int ds$ over the group velocity.
(2) $(1-y)^{-1/2} \approx 1 + y/2$ for $y = \nu_p^2/\nu^2 \ll 1$.
(3) $\nu_p^2 = n_e e^2/(4\pi^2\varepsilon_0 m_e)$ from Eq. (5.2).

The column of electrons $\int n_e\, ds$ is called the **dispersion measure**, and is usually quoted in $\mathrm{pc\,cm}^{-3}$. The extra delay is

$$\boxed{\Delta t = \frac{e^2}{8\pi^2\varepsilon_0 m_e c}\,\frac{\int n_e\, ds}{\nu^2} = 4.15\,\mathrm{ms} \times \frac{\int n_e\, ds}{1\,\mathrm{pc\,cm}^{-3}}\left(\frac{\nu}{1\,\mathrm{GHz}}\right)^{-2}.} \tag{8.16}$$

Low frequencies arrive later. A pulsar 1 kpc away through gas with 0.03 electrons per cm^3 has a dispersion measure of $30\,\mathrm{pc\,cm}^{-3}$. Its pulses arrive 0.78 s late at 400 MHz, and 0.58 s later than at 800 MHz. Timing a pulse at two frequencies measures the column of electrons, and with a model of the electron density it gives the distance. The column changes as the pulsar and the gas move, which is the dispersion noise of pulsar timing (Section 7.6).

8.3 Spectral Lines and Their Widths

The oscillator of Section 8.2.1 absorbs in a narrow peak near its natural frequency, and Chapter 3 showed that motion along the line of sight shifts a line (Eq. (3.25)).

A real line has a width and a shape, and both carry information. The width tells the temperature of the gas, how it churns, and how dense it is. To read them we need the shape each cause produces.

Strike a bell and let it ring. It rings for a long time, and the note is pure. Touch it with a hand while it rings and the sound stops quickly, and what you hear is a dull clunk with no clear pitch. A short ring does not have a sharp pitch. Now load many identical bells on a merry-go-round and strike them all. Those coming toward you sound higher and those going away sound lower, and together they make a blurred chord.

An atom in an excited state is a bell. It radiates for about its lifetime, so its line has a width set by how long it rings. Collisions that interrupt it shorten the ring and widen the line. The atoms of a gas move at random, so the lines of different atoms are Doppler shifted by different amounts and blur together.

The Lyman α line of hydrogen, at 121.6 nm, decays at the rate $6.27 \times 10^8\ \mathrm{s}^{-1}$. Its width from the lifetime alone is 1.0×10^8 Hz. In gas at 100 K the random motions blur it over 1.8×10^{10} Hz, 176 times more.

Picture the line as a hill with a rounded top and long, low slopes far out on each side. The top is made by the motions, and the far slopes by the lifetime.

The width from the lifetime is the uncertainty principle of quantum mechanics. The width from motion is the Maxwell distribution, which is counting states (P3, Section 9.3). Which of the two dominates, at the centre and in the far wings, is a comparison of two scales (P6). The same three causes shape the lines of stars, where pressure matters more, and the lines of lasers. Here we find the shape of a line from each cause and the shape when they act together.

Formation. An atom or ion has only electronic transitions, from one arrangement of its electrons to another. A molecule can also rotate and vibrate, so its spectrum is much richer. An excited atom or molecule that drops to a lower level by itself sends out a photon, and many such photons make an **emission line**. A cool gas in front of a hotter source of continuous light takes photons out of it at the same frequencies, and makes an **absorption line**. The rates of these processes, and when each kind of line appears, are worked out in Chapter 11 (Section 11.6).

The line profile. Let ν_0 be the frequency at the centre of the line. The **line profile** $\phi(\nu)$ is the fraction of the line's photons per unit frequency, normalised so that $\int \phi(\nu)\, d\nu = 1$. Its width is measured as the full width at half the maximum.

8.3.1 Natural Broadening

The width from the uncertainty principle. Let the upper level decay at the rate Γ, so that it lives for a time $\Delta t = 1/\Gamma$. By the time–energy uncertainty relation, a state that lasts for Δt has an energy spread of about

$$\Delta E\, \Delta t \sim \hbar. \tag{8.17}$$

With $\Delta E = h\, \Delta\nu$ this is a spread in frequency,

$$\Delta\nu \sim \frac{1}{2\pi\, \Delta t} = \frac{\Gamma}{2\pi}. \tag{8.18}$$

Strongly allowed lines decay fast and are broad. Forbidden lines decay slowly and are narrow. If the lower level also decays, as for a line between two excited levels, Γ is the sum of the two rates.

The shape, from a decaying wave. The uncertainty relation gives only the width. To get the shape, treat the radiating atom as an oscillator that is excited at $t = 0$ and then rings down. Its energy decays as $e^{-\Gamma t}$, so the field it sends out decays at half that rate,

$$E(t) = E_0\, e^{-\Gamma t/2}\, e^{-i\omega_0 t} \quad (t \geq 0), \qquad E(t) = 0 \quad (t < 0).$$

The spectrum is the squared size of its Fourier transform.

$$\begin{aligned} \tilde{E}(\omega) &\overset{(1)}{=} \int_0^\infty E_0\, e^{-\Gamma t/2}\, e^{i(\omega-\omega_0)t}\, dt \\ &\overset{(2)}{=} \frac{E_0}{\Gamma/2 - i(\omega-\omega_0)}, \\ |\tilde{E}(\omega)|^2 &\overset{(3)}{=} \frac{E_0^2}{(\omega-\omega_0)^2 + \Gamma^2/4}. \end{aligned} \tag{8.19}$$

(1) The Fourier transform, $\int E(t)\, e^{i\omega t}\, dt$, over the times when the wave exists.
(2) $\int_0^\infty e^{-st} dt = 1/s$ with $s = \Gamma/2 - i(\omega - \omega_0)$, whose real part is positive.
(3) $|x + iy|^2 = x^2 + y^2$.

In terms of $\nu = \omega/2\pi$, and normalised to unit area, this is the **Lorentzian profile**,

$$\boxed{\phi_L(\nu) = \frac{\Gamma/4\pi^2}{(\nu-\nu_0)^2 + (\Gamma/4\pi)^2}.} \tag{8.20}$$

It falls to half its peak at $\nu - \nu_0 = \pm\Gamma/4\pi$, so its full width at half maximum is $\Gamma/2\pi$, the uncertainty estimate (8.18) exactly. The driven oscillator gave the same shape and width near resonance, Eq. (8.8), a second route to the same result. Far from the centre ϕ_L falls only as $(\nu - \nu_0)^{-2}$. These slow wings are the mark of an exponential decay. For Lyman α, $\Gamma = 6.27 \times 10^8\ \mathrm{s}^{-1}$ gives a width of 1.0×10^8 Hz, or 4.9×10^{-6} nm in wavelength.

8.3.2 Pressure Broadening

A collision with a neighbour disturbs the levels of an emitting atom and resets the phase of its wave at random. The wave before and after a collision no longer add in step. A collision cuts the ring short, like the hand on the bell.

The shape. Let collisions strike one atom at random times, at the mean rate $1/t_{\mathrm{col}} = n\,\sigma_{\mathrm{col}}\,\bar{v}$, where n is the density of colliders, σ_{col} the cross-section for a collision and $\bar{v}$ the mean relative speed. The chance of no collision in a time t is $e^{-t/t_{\mathrm{col}}}$, the zero-count probability of the Poisson distribution (Section 7.2). Averaged over many atoms, the part of the wave that is still in step with its start after a time t has decayed by $e^{-\Gamma t/2}$ from the lifetime and by $e^{-t/t_{\mathrm{col}}}$ from collisions. The spectrum of a steady random signal is the Fourier transform of this average, by the Wiener–Khinchin theorem. The transform is that of the previous chain with $\Gamma/2$ replaced by $\Gamma/2 + 1/t_{\mathrm{col}}$. So pressure broadening also gives a Lorentzian, Eq. (8.20) with

$$\boxed{\Gamma \;\to\; \Gamma + \frac{2}{t_{\mathrm{col}}}, \qquad \text{full width} = \frac{1}{2\pi}\left(\Gamma + \frac{2}{t_{\mathrm{col}}}\right).} \tag{8.21}$$

Denser and hotter gas has more frequent collisions and broader lines.

Where it matters. Take a collision cross-section of order $10^{-19}\,\mathrm{m}^2$, a few times the area of an atom. In interstellar gas with $n = 10^6\,\mathrm{m}^{-3}$ and $\bar{v} \approx 2\,\mathrm{km\,s}^{-1}$, $1/t_{\mathrm{col}} \approx 2\times10^{-10}\,\mathrm{s}^{-1}$, far below $\Gamma \approx 10^8\,\mathrm{s}^{-1}$ of a strong line. Pressure broadening does not matter there. In the Sun's photosphere, with $n \approx 10^{23}\,\mathrm{m}^{-3}$ and $\bar{v} \approx 10\,\mathrm{km\,s}^{-1}$, $1/t_{\mathrm{col}} \approx 10^8\,\mathrm{s}^{-1}$, comparable with Γ. Lines of dense dwarf stars are broader than those of thin giant atmospheres for this reason, which is how a spectrum tells a dwarf from a giant of the same temperature.

8.3.3 Doppler Broadening

The velocities of the atoms. An atom moving away from us at line-of-sight velocity v_r has its line shifted by Eq. (3.25). In gas at temperature T, the probability of an energy $\frac{1}{2}mv_r^2$ in this one direction carries the Boltzmann weight $e^{-mv_r^2/2k_BT}$, where m is the mass of the atom (P3). Normalised, this is the one-dimensional Maxwell distribution,

$$f(v_r)\,dv_r = \sqrt{\frac{m}{2\pi k_B T}}\,\exp\!\left(-\frac{mv_r^2}{2k_BT}\right)dv_r, \qquad \sigma_v \equiv \sqrt{\langle v_r^2\rangle} = \sqrt{\frac{k_BT}{m}}. \tag{8.22}$$

It is one component of the Maxwell distribution derived by counting states in Section 9.3, which also gives $\langle v_r^2\rangle = k_BT/m$ and the ideal gas (Eq. (9.23)).

The shape. Each atom gives a line much narrower than the spread of shifts, so the profile is the velocity distribution written in frequency. A photon from an atom with velocity v_r arrives at $\nu = \nu_0/(1+v_r/c) \approx \nu_0(1-v_r/c)$, so $v_r = -c(\nu-\nu_0)/\nu_0$ and $|dv_r| = (c/\nu_0)\,d\nu$.

$$\begin{aligned}
\phi_D(\nu)\,d\nu &\overset{(1)}{=} f(v_r)\,|dv_r| \\
&\overset{(2)}{=} \sqrt{\frac{m}{2\pi k_BT}}\,\exp\!\left[-\frac{mc^2(\nu-\nu_0)^2}{2k_BT\,\nu_0^2}\right]\frac{c}{\nu_0}\,d\nu \\
&\overset{(3)}{=} \frac{1}{\sqrt{2\pi}\,\Delta\nu_D}\,\exp\!\left[-\frac{(\nu-\nu_0)^2}{2\,\Delta\nu_D^2}\right]d\nu.
\end{aligned} \tag{8.23}$$

(1) The fraction of photons in $d\nu$ is the fraction of atoms with velocities in the matching dv_r.
(2) Eq. (8.22) with $v_r = -c(\nu-\nu_0)/\nu_0$.
(3) Define $\Delta\nu_D = \nu_0\sigma_v/c$. Then $mc^2/(k_BT\nu_0^2) = 1/\Delta\nu_D^2$ and $\sqrt{m/2\pi k_BT}\;c/\nu_0 = 1/(\sqrt{2\pi}\Delta\nu_D)$.

This is the **Gaussian profile**,

$$\boxed{\phi_D(\nu) = \frac{1}{\sqrt{2\pi}\,\Delta\nu_D}\,\exp\!\left[-\frac{(\nu-\nu_0)^2}{2\,\Delta\nu_D^2}\right],} \tag{8.24}$$

with the **Doppler width** $\Delta\nu_D$ as its standard deviation,

$$\frac{\Delta\nu_D}{\nu_0} = \frac{\sigma_v}{c} = \sqrt{\frac{k_BT}{mc^2}}. \tag{8.25}$$

It falls to half its peak where $(\nu-\nu_0)^2/2\Delta\nu_D^2 = \ln 2$, so its full width at half maximum is $2\sqrt{2\ln 2}\,\Delta\nu_D = 2.355\,\Delta\nu_D$. Its wings fall as $e^{-(\nu-\nu_0)^2/2\Delta\nu_D^2}$, much faster than the Lorentzian, because fast atoms are exponentially rare. The width grows as $\sqrt{T/m}$. Hydrogen at 100 K has $\sigma_v = 0.91\,\mathrm{km\,s}^{-1}$. The H$\alpha$ line at 656.28 nm from gas at 10^4 K has a full width of $21\,\mathrm{km\,s}^{-1}$, or 0.047 nm.

Turbulence and bulk motion. The gas also moves in eddies and streams. If the line-of-sight velocity of an atom is a thermal part plus an independent turbulent part with spread σ_{turb}, the variances add (Eq. (7.10)),

$$\sigma_v^2 = \frac{k_B T}{m} + \sigma_{\text{turb}}^2.$$

When the turbulent part is also Gaussian, the profile stays Gaussian with this larger σ_v. The thermal part depends on the mass of the atom and the turbulent part does not. Comparing the widths of a light and a heavy atom from the same gas therefore separates the temperature from the turbulence. A steady motion of the whole cloud, such as Galactic rotation, shifts the centre ν_0 and does not widen the line.

8.3.4 The Voigt Profile

A real line has all the causes at once. Each atom has a Lorentzian line from its lifetime and collisions, centred on its own Doppler-shifted frequency. The atoms have the Gaussian spread of those centres.

The convolution. Adding up the atoms, weighted by Eq. (8.24),

$$\boxed{\phi_V(\nu) = \int_{-\infty}^{\infty} \phi_D(\nu')\, \phi_L(\nu - \nu' + \nu_0)\, d\nu',} \tag{8.26}$$

where ν' is the centre of one group of atoms and $\phi_L(\nu - \nu' + \nu_0)$ is the Lorentzian (8.20) moved to that centre. This is the **Voigt profile**. It has no closed form, but both of its limits do.

The core. In interstellar gas the Lorentzian half-width $\Gamma/4\pi$ is far smaller than $\Delta\nu_D$. Near the centre, ϕ_L is then a narrow spike of unit area, and

$$\phi_V(\nu) \approx \phi_D(\nu) \int \phi_L \, d\nu' = \phi_D(\nu).$$

The core of the line is Gaussian.

The wings. Far from the centre, $|\nu - \nu_0| \gg \Delta\nu_D$. The Gaussian is non-zero only for ν' within a few $\Delta\nu_D$ of ν_0, and across that small range $\phi_L(\nu - \nu' + \nu_0)$ hardly changes.

$$\begin{aligned} \phi_V(\nu) &\overset{(1)}{\approx} \phi_L(\nu) \int \phi_D(\nu')\, d\nu' \\ &\overset{(2)}{=} \phi_L(\nu) \overset{(3)}{\approx} \frac{\Gamma/4\pi^2}{(\nu - \nu_0)^2}. \end{aligned} \tag{8.27}$$

(1) Take ϕ_L out of the integral at $\nu' = \nu_0$.
(2) ϕ_D has unit area.
(3) Eq. (8.20) with $|\nu - \nu_0| \gg \Gamma/4\pi$.

The wings are Lorentzian. The Gaussian falls exponentially and the Lorentzian only as a power, so however small Γ is, the Lorentzian wins far enough out. For Lyman α at 100 K, the two are equal at $4.0\,\Delta\nu_D$ from the centre. Inside that the line is Gaussian, and outside it Lorentzian (Figure 8.1). When the column of gas is so large that the core absorbs all the light, the line can only grow in its wings. The wings of Lyman α in the spectra of distant quasars give the columns of the largest hydrogen clouds in this way.

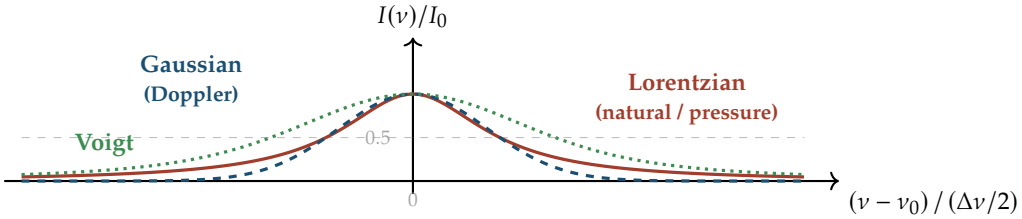


Figure 8.1: A Lorentzian profile (solid), a Gaussian profile (dashed) with the same peak intensity I_0 and the same half-width $\Delta\nu/2$, and their convolution, the Voigt profile (dotted), scaled to the same peak, against $(\nu - \nu_0)/(\Delta\nu/2)$ from -4.5 to 4.5. The Voigt profile has a half-width of 1.64 in these units. The dashed horizontal line marks half the peak.

8.4 Hydrogen in the ISM

Section 8.3 gave the width of a line from the lifetime of its upper level and from the motions of the gas.

Hydrogen is most of the interstellar medium, but its strong lines are hard to use. Almost every atom sits in its ground state, and the first line up from there, Lyman α, needs 10.2 eV. At 100 K the Boltzmann factor for that is e^{-1184}, about 10^{-514}. Cold hydrogen gives no Lyman α of its own, and is seen in it only as absorption against a hot ultraviolet source behind it.

Hold two small bar magnets close together. Lined up north to south they attract and sit comfortably. Turned so that like poles face, they push apart. The two arrangements have different energies, and flipping one magnet over moves between them. For tiny magnets the energy difference is tiny.

The electron and the proton in a hydrogen atom are such magnets, because both spin. Flipping the electron's spin relative to the proton's changes the energy by a few millionths of an electronvolt. The photon from that flip has a wavelength of 21 cm. Even the coldest gas has enough energy to keep both arrangements populated, so every hydrogen atom can send out this line.

The energy 5.9×10^{-6} eV corresponds to the temperature $hc/(\lambda k_B) = 0.068$ K. In gas at 100 K that is negligible, so the upper level, which has three states, holds three times as many atoms as the lower one, which has one. Three quarters of all the atoms are in the upper level.

Picture a tilted magnet that very rarely flips back by itself, but is knocked between its two settings every few hundred years by passing atoms.

The populations follow the Boltzmann weight (P3) only because collisions move atoms between the levels much faster than the atoms decay. That comparison of two rates is P6, and the balance of the rates into and out of each level is P4. The same level balance sets the cooling of the gas (Section 8.6) and decides which lines of nebulae can be seen. Here we find the energy of the line, its decay rate, the populations of its levels, and what it measures, and then turn to ionised and molecular hydrogen.

Three forms. Hydrogen in the interstellar medium is neutral and atomic, written HI, ionised, written HII (the same as H^+), or molecular, H_2. Ionisation needs photons above 13.6 eV, with wavelengths below 91.2 nm. Only hot, massive stars give many of them, so HII is found around such stars. Molecular hydrogen forms on the surfaces of dust grains (Section 8.4.2).

8.4.1 The 21 cm Line

The energy of the spin flip. The electron and proton have magnetic moments $\vec{\mu}_e = -g_e \mu_B \vec{S}_e/\hbar$ and $\vec{\mu}_p = g_p \mu_N \vec{S}_p/\hbar$, with spins $\vec{S}_e$ and $\vec{S}_p$, the Bohr and nuclear magnetons $\mu_B = e\hbar/2m_e$ and $\mu_N = e\hbar/2m_p$, and $g_e \approx 2$, $g_p = 5.586$. The proton's field, averaged over the spherical ground state of the electron, leaves only its contact part at the proton, $(2\mu_0/3)\,\vec{\mu}_p\,\delta^3(\vec{r})$, a standard result of magnetostatics. The electron density there is $|\psi(0)|^2 = 1/(\pi a_0^3)$, with a_0 the Bohr radius.

$$\begin{aligned} E_{\text{mag}} &\overset{(1)}{=} -\left\langle \vec{\mu}_e \cdot \vec{B}_p \right\rangle \\ &\overset{(2)}{=} -\frac{2\mu_0}{3}\,\vec{\mu}_e \cdot \vec{\mu}_p\,|\psi(0)|^2 \\ &\overset{(3)}{=} \frac{2\mu_0}{3}\,\frac{g_e g_p \mu_B \mu_N}{\pi a_0^3}\,\frac{\vec{S}_e \cdot \vec{S}_p}{\hbar^2}. \end{aligned} \tag{8.28}$$

(1) The energy of a magnetic moment in a field, averaged over where the electron is.
(2) Only the contact part of the proton's field survives the average over the spherical cloud, and it acts where the electron overlaps the proton.
(3) The two moments in terms of the spins, and $|\psi(0)|^2 = 1/(\pi a_0^3)$.

The total spin $\vec{S} = \vec{S}_e + \vec{S}_p$ has $S^2 = \hbar^2 s(s+1)$, with $s = 1$ (spins parallel, three states) or $s = 0$ (spins opposed, one state). Each spin has $S_e^2 = S_p^2 = \tfrac{3}{4}\hbar^2$.

$$\begin{aligned} \frac{\vec{S}_e \cdot \vec{S}_p}{\hbar^2} &\overset{(1)}{=} \frac{S^2 - S_e^2 - S_p^2}{2\hbar^2} \\ &\overset{(2)}{=} \tfrac{1}{2}\big(s(s+1) - \tfrac{3}{2}\big) = \begin{cases} +\tfrac{1}{4}, & s = 1, \\ -\tfrac{3}{4}, & s = 0. \end{cases} \end{aligned} \tag{8.29}$$

(1) Square $\vec{S} = \vec{S}_e + \vec{S}_p$ and solve for the cross term.
(2) $S^2 = \hbar^2 s(s+1)$ and $S_e^2 = S_p^2 = \tfrac{3}{4}\hbar^2$.

The two values differ by one. With $g_e = 2$ the gap between the levels is

$$\boxed{E_{21} = \frac{4\mu_0\, g_p\, \mu_B \mu_N}{3\pi a_0^3} = 5.88 \times 10^{-6}\,\text{eV}, \qquad \nu_0 = \frac{E_{21}}{h} = 1421\,\text{MHz}.} \tag{8.30}$$

The measured frequency is 1420.406 MHz, at a wavelength of 21.1 cm. Using $g_e = 2.0023$ and the reduced mass of the electron and proton brings the formula to within 0.01 per cent of it. Written in terms of the ionisation energy of hydrogen, 13.6 eV, the gap is

$$\frac{E_{21}}{13.6\,\text{eV}} = \frac{8}{3}\, g_p\, \alpha^2\, \frac{m_e}{m_p} = 4.3 \times 10^{-7},$$

with α the fine-structure constant. It is small because magnetic energies carry α^2 against electric ones, and because the proton's moment is smaller than the electron's by the ratio of their masses.

How fast it decays. An oscillating electric dipole of matrix element d decays at the rate $\omega^3 |d|^2/(3\pi\varepsilon_0 \hbar c^3)$. A magnetic dipole radiates the same way with $d \to \mu/c$, so

$$A_{21} = \frac{\mu_0\, \omega^3\, |\langle 1|\vec{\mu}|2\rangle|^2}{3\pi\hbar c^3},$$

the spontaneous rate in the notation of Section 11.6, with level 1 the lower and level 2 the upper. The electron's moment is 658 times the proton's, so only it counts. Take the upper state with total spin projection zero, $(\uparrow\downarrow + \downarrow\uparrow)/\sqrt{2}$, with the electron's spin written first. The z part of $-2\mu_B \vec{S}_e/\hbar$ flips the sign of the second term and turns it into the lower state $(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2}$ times $-\mu_B$, so the matrix element is μ_B in size. The other two upper states give the same. Then

$$\boxed{A_{21} = \frac{\mu_0\,\omega_0^3\,\mu_B^2}{3\pi\hbar c^3} = 2.87\times10^{-15}\,\mathrm{s}^{-1}, \qquad \frac{1}{A_{21}} = 1.1\times10^{7}\ \text{years}.} \tag{8.31}$$

The rate is tiny for two reasons. The frequency is low, and the rate goes as ω^3. And the transition is magnetic, which costs a factor of about α^2 against an electric one. Lyman α decays 2×10^{23} times faster. A transition this slow is called **forbidden**. By Eq. (8.18) its natural width is $A_{21}/2\pi = 4.6\times10^{-16}$ Hz. The thermal width at 100 K, from Eq. (8.25), is 10 kHz in full. The width of the 21 cm line measures only the motions of the gas.

Level populations, a balance of rates. Collisions knock atoms between the two levels. Let one atom in the upper level be knocked down at the rate $1/t_{\mathrm{col}}$. By detailed balance (Eq. (9.34), with g_1 and g_2 the numbers of states in the lower and upper levels) an atom in the lower level is knocked up at the rate $(g_2/g_1)\,e^{-E_{21}/k_BT}/t_{\mathrm{col}}$. The upper level also decays at the rate A_{21}. Background radio light can also lift atoms, but for the 21 cm line it does so far more slowly than collisions do, so we leave it out. In a steady state the rates into and out of the upper level balance (P4). With n_1 and n_2 the densities of atoms in the two levels,

$$\begin{aligned} n_2\left(A_{21} + \frac{1}{t_{\mathrm{col}}}\right) &\overset{(1)}{=} n_1\,\frac{g_2}{g_1}\,\frac{e^{-E_{21}/k_BT}}{t_{\mathrm{col}}} \\ \Longrightarrow \qquad \frac{n_2}{n_1} &\overset{(2)}{=} \frac{g_2}{g_1}\,e^{-E_{21}/k_BT}\,\frac{1}{1+A_{21}t_{\mathrm{col}}}. \end{aligned} \tag{8.32}$$

(1) Out of the upper level by decay and by collisions, into it by collisions from below.
(2) Divide by $n_1(A_{21} + 1/t_{\mathrm{col}})$ and multiply the top and bottom by t_{col}.

$$\boxed{\frac{n_2}{n_1} = \frac{g_2}{g_1}\,e^{-E_{21}/k_BT}\,\frac{1}{1+A_{21}t_{\mathrm{col}}}.} \tag{8.33}$$

One number decides, $A_{21}t_{\mathrm{col}}$, the ratio of the decay rate to the collision rate (P6). When collisions are much faster, $A_{21}t_{\mathrm{col}} \ll 1$, the populations take the Boltzmann ratio at the gas temperature (P3). When decay is much faster, $A_{21}t_{\mathrm{col}} \gg 1$, the upper level is nearly empty and every upward collision ends in a photon. The changeover is at the **critical density**, where $A_{21}t_{\mathrm{col}} = 1$, so $n = A_{21}/(\sigma_{\mathrm{col}}\bar{v})$.

The populations of the 21 cm levels. For hydrogen at 100 K the mean relative speed of two atoms is $\sqrt{16k_BT/(\pi m_{\mathrm{H}})} = 2.0\,\mathrm{km\,s^{-1}}$. With $n = 10^6\,\mathrm{m^{-3}}$ ($1\,\mathrm{cm^{-3}}$) and a cross-section of order $10^{-19}\,\mathrm{m^2}$ for the collisions that exchange the spins, $1/t_{\mathrm{col}} = 2\times10^{-10}\,\mathrm{s^{-1}}$. An atom is knocked between the levels about once in 150 years, against once in 11 million years by decay. So $A_{21}t_{\mathrm{col}} = 1.4\times10^{-5}$, the populations follow the gas temperature, and with $g_2/g_1 = 3$ and $E_{21}/k_B = 0.068$ K,

$$\boxed{\frac{n_2}{n_1} = 3\,e^{-0.068\,\mathrm{K}/T} \approx 3, \qquad n_2 \approx \tfrac{3}{4}\,n_{\mathrm{H}}.} \tag{8.34}$$

Three quarters of the atoms are in the upper level whatever the temperature of the cloud, because every cloud is far hotter than 0.068 K.

Why the line is seen. Each atom in the upper level sends out a photon at the rate A_{21} whatever the collisions do, because they keep the upper level filled. The power per atom is tiny, $\frac{3}{4}A_{21}h\nu_0 = 2.0 \times 10^{-39}$ W, but a line of sight through the Galaxy crosses 10^{25} or more atoms per square metre, and together they make the line easy to detect with a radio telescope. A gas cell in a laboratory holds far too few atoms to show it in spontaneous emission. Instead the line is measured in the laboratory by stimulated emission in the hydrogen maser, to better than one part in 10^{12}. Because the populations do not depend on the temperature, the brightness of the line counts the atoms in the column (Problem 8.7.2).

What it maps. The rest frequency is known so well that the Doppler shift of the line, Eq. (3.25), gives the line-of-sight velocity of each cloud of HI along a sight line. Combined with the rotation of the Galaxy, the velocity places each cloud at a distance. The 21 cm line is the main tool for mapping neutral gas in the Milky Way and for measuring the rotation of other galaxies far beyond their stars.

8.4.2 Molecular Hydrogen

How it forms. Two hydrogen atoms that meet in the gas cannot stay together. Binding releases 4.5 eV, and the pair has no fast way to get rid of it, since H_2 has no electric dipole to radiate with. The atoms fly apart again within one vibration. On a grain the energy has somewhere to go. An atom that lands on the grain sticks to its surface and wanders over it, which makes it far more likely to meet a second atom. When they bind, the grain takes up the released energy, and the molecule leaves. In the thin gas, reactions without grains are far too slow to compete.

Where it survives. Ultraviolet starlight breaks H_2 apart. The molecule survives only deep in clouds, where dust absorbs that light (Section 8.2.4).

Why it is hard to see. H_2 is made of two identical atoms, so it has no permanent electric dipole moment and no allowed dipole transitions between its rotation and vibration levels. Its lowest rotational line comes from a level at $E/k_B \approx 510$ K, and is a weak quadrupole line. Cold molecular clouds, at 10 to 20 K, hardly excite it at all. Astronomers trace H_2 instead by molecules that do have a dipole. The most used is CO, whose first rotational level lies at $E/k_B = 5.5$ K and is easily excited at 10 K. The amount of H_2 follows from the strength of the CO line and a ratio calibrated by other means. Many other molecules are also found, among them OH, H_2O, CO_2, NH, CH and CH_2.

8.5 Interstellar Clouds and Nebulae

The medium holds hydrogen in three forms, with dust that dims the light, and each form has its own way of being seen.

The density and temperature change smoothly over many powers of ten, so any division into kinds of cloud is rough.

Weather clouds are named by how they look, puffy or layered or wispy, and not by what they are made of, which is always water. The names are still useful, because how a cloud looks tells something about how it formed.

Interstellar clouds are named the same way. The classes overlap, and the same cloud can be called by different names when seen in different ways. Each class is defined by what dominates its light, the 21 cm line, CO, the dust or the glow of hot gas.

How dark a cloud looks is set by its extinction. A diffuse molecular cloud with A_V between 1 and 5 lets through 40 to 1 per cent of the visible light behind it. A dense globule with $A_V \approx 10$ lets through one part in 10^4.

Picture the same smooth spread of densities, cut into named ranges.

Whether a cloud is dark or bright is the optical depth compared with one (P6). Whether it is seen in a line is the balance of rates in its levels (P4). Here we list the kinds of clouds and nebulae, and find how fast a planetary nebula expands.

Clouds. Table 8.3 gives the main kinds. In **HI clouds** the hydrogen is atomic. In **diffuse and translucent molecular clouds**, with $1 < A_V < 5$, the hydrogen is still mostly atomic, with H_2 where the optical depth is higher. They are irregular in shape. **Giant molecular clouds** are large, clumpy complexes of gas and dust, and the main sites of star formation. The Horsehead Nebula is part of the giant molecular cloud complex in Orion. **Bok globules** are small, nearly round, dense clouds with $A_V \approx 10$ and $T \approx 10\,\mathrm{K}$, where stars also form. They appear to be dense cores whose surrounding gas was stripped away by the light and winds of hot massive stars.

Cloud type	T (K)	Size	Mass ($M_\odot$)	Tracer
HI clouds	~ 100	tens of pc	—	21 cm line
Diffuse molecular	15–50	several pc	3–100	CO, UV absorption
Giant molecular clouds	~ 10–20	~ 50 pc	10^5–10^6	CO
Bok globules	~ 10	below 1 pc	1–1000	Dark silhouette

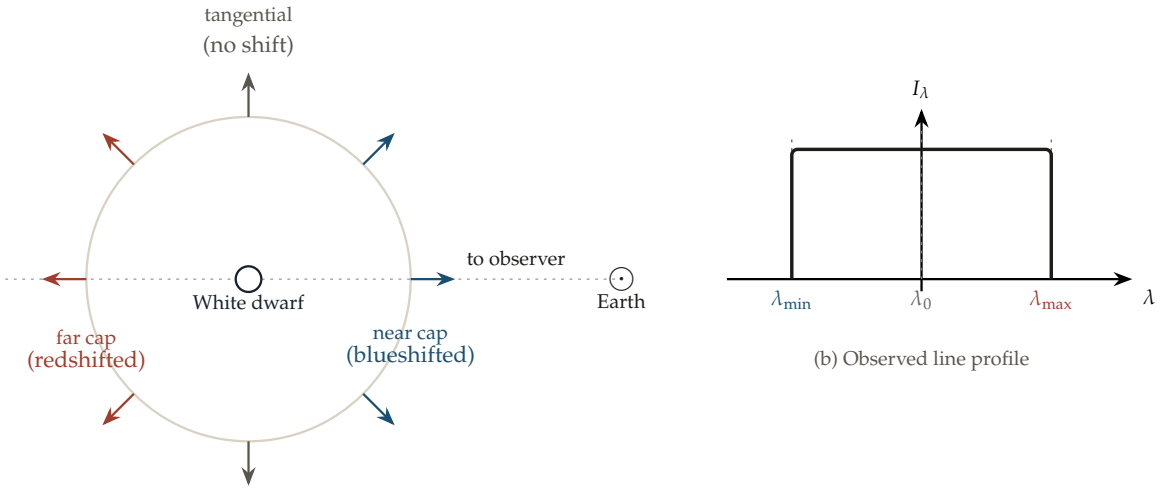
Table 8.3: Approximate temperature, size, mass and tracer of the kinds of interstellar cloud. The ranges overlap between kinds.

Nebulae. *Nebula*, Latin for cloud, is an old name for anything in the sky that looks extended and diffuse rather than point-like. Some nebulae turned out to be distant galaxies, such as Andromeda. The rest are clouds in our own Galaxy, of three kinds.

- **Reflection nebulae** shine by starlight scattered from dust, such as the blue nebula around the Pleiades.
- **Dark nebulae** are clouds whose extinction is high enough to show as silhouettes against the stars behind them. Giant molecular clouds such as the Horsehead, Bok globules and diffuse molecular clouds can all appear this way. Barnard 68 is a well-known example.
- **Emission nebulae** make their own light, mostly in lines. **HII regions** are gas ionised by nearby hot stars, such as the Monkey Head Nebula, NGC 2174. **Planetary nebulae** are gas thrown off by a star late in its life, such as the Ring Nebula. **Supernova remnants** are gas thrown out by a stellar explosion, such as the Crab Nebula, left by the supernova of 1054, which is lit from inside by a pulsar.

8.5.1 Expanding Planetary Nebulae and the Doppler-Split Line

A planetary nebula is a shell of gas thrown off by a star near the end of its life. The hot core left behind, a white dwarf, ionises the shell and makes it glow in emission lines. The shell expands outward, roughly as a sphere centred on the white dwarf, at a speed v_{exp} of a few tens of $\mathrm{km\,s^{-1}}$. Every point of the shell moves outward at v_{exp}, but in different directions relative to us. The near side comes toward us, the far side goes away, and the rim moves across the line of sight (Figure 8.2). We want the line profile of the whole shell and the speed it gives.



(a) Expanding shell, as seen from Earth

(b) Observed line profile

Figure 8.2: (a) An expanding spherical shell around a white dwarf, with Earth to the right. Arrows show the local expansion velocity, blue on the near cap, red on the far cap and grey near the rim. (b) A flat-topped emission-line profile between $\lambda_{\min}$ and $\lambda_{\max}$, with the rest wavelength λ_0 marked between them.

The velocity of one point. Let θ be the angle at the white dwarf between a point of the shell and the direction pointing away from us. The point moves along its radius at speed $v_{\exp}$, and the part of that velocity along the line of sight, away from us, is

$$v_r = v_{\exp} \cos\theta. \tag{8.35}$$

The far cap ($\theta = 0$) has $v_r = +v_{\exp}$, the near cap ($\theta = \pi$) has $v_r = -v_{\exp}$, and the rim ($\theta = \pi/2$) has $v_r = 0$.

How much of the shell has each velocity. Take a thin shell of radius R that glows evenly and is transparent to its own line, all of it inside the telescope's aperture. The band of the shell between θ and $\theta + d\theta$ is a ring of area $2\pi R^2 \sin\theta\, d\theta$. The fraction of the shell in it is

$$\begin{aligned} \frac{dA}{A} &\overset{(1)}{=} \frac{2\pi R^2 \sin\theta\, d\theta}{4\pi R^2} \\ &\overset{(2)}{=} \frac{1}{2}\,|d(\cos\theta)| \\ &\overset{(3)}{=} \frac{|dv_r|}{2v_{\exp}}. \end{aligned} \tag{8.36}$$

(1) The area of the ring over the area of the sphere.
(2) $d(\cos\theta) = -\sin\theta\, d\theta$.
(3) $v_r = v_{\exp}\cos\theta$.

Each equal step of v_r between $-v_{\exp}$ and $+v_{\exp}$ holds an equal share of the shell. So a single sharp line becomes a flat-topped line, as in panel (b) of Figure 8.2. Its edges come from the two caps. By Eq. (3.25),

$$\lambda_{\min} = \lambda_0\left(1 - \frac{v_{\exp}}{c}\right), \qquad \lambda_{\max} = \lambda_0\left(1 + \frac{v_{\exp}}{c}\right).$$

A slit placed across the centre of the nebula sees only the two caps, and the line splits into two lines at these same wavelengths.

The expansion speed. Subtracting the two edges removes the unknown λ_0 from the difference,

$$\lambda_{\max} - \lambda_{\min} = \frac{2\lambda_0\, v_{\exp}}{c} \qquad \Longrightarrow \qquad v_{\exp} = \frac{c}{2}\,\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_0}. \tag{8.37}$$

The speed comes from the two edges of one line, without the distance of the nebula. Their mean is λ_0 shifted by any motion of the nebula as a whole, which cancels from the difference. The thermal width of Section 8.3 rounds the edges. For hydrogen at 10^4 K it is 21 km s^{-1} in full, as large as the expansion itself, so the lines of heavier ions such as O^{2+}, with $\sigma_v = 2.3$ km s^{-1}, give sharper edges.

8.6 Heating and Cooling of the ISM

Section 8.4 balanced the rates into and out of an energy level, Eq. (8.33).

That fixes how atoms share out among levels at a given temperature. It does not say what the temperature is. Gas in the medium is heated by starlight and fast particles and cooled by the light it sends out, and its temperature is wherever the two meet.

Put a pan of water on a flame with no lid. The flame heats it. Water evaporates from the surface and cools it. Evaporation rises steeply with temperature. Turn the flame up or down and the water settles at almost the same temperature, a little below boiling, because a small rise in temperature is enough to carry away a lot more heat.

Interstellar gas is cooled in the same way. Collisions lift atoms and molecules into excited levels, and the photons they then send out escape and carry the energy away. The number of collisions with enough energy grows as e^{-E_{21}/k_BT}, steeply, so the temperature settles near E_{21}/k_B of the main cooling line, almost whatever the heating.

The main coolant of atomic clouds is the ion C^+, with a line at 157.7 μm. Its upper level lies at $E_{21}/k_B = 91$ K, and HI clouds sit near 100 K. In molecular clouds CO cools, with levels at 5.5 and 16.6 K, and these clouds sit at 10 to 20 K.

Picture a flat line for the heating and a steep rising curve for the cooling, crossing at one temperature.

The temperature is a steady state of rates, heat in equal to heat out (P4, Section 9.4). The steep rise comes from the Boltzmann factor (P3). The photons escape only if the cloud is transparent to them, a comparison of two lengths (P6). The same balance sets the temperature of a planet in sunlight, and of a star's atmosphere (Chapter 11). Here we list the sources of heating and cooling, and find the temperature where they balance.

Heating. Several processes put energy into the gas.

- **Cosmic rays**, charged particles of 10 to 10^{14} MeV accelerated in stellar flares and supernova shocks, ionise atoms and molecules. The freed electron shares its energy with the gas by collisions. Cosmic rays reach deep into dense clouds where starlight cannot.
- **Ultraviolet photons** from massive stars ionise atoms, and the freed electrons heat the gas.
- **Photoelectrons from grains**, knocked out of dust grains by ultraviolet photons, are the main heating in much of the diffuse gas.
- **X-rays** from stars ionise more deeply than ultraviolet light.
- **Dust** absorbs continuous starlight and warms.
- **Shocks** from stellar winds and supernovae turn the energy of bulk motion into heat.

Cooling. Collisions excite molecules and the fine-structure levels of atoms and ions, which then decay by sending out infrared and millimetre photons. Grains are much smaller than these wavelengths, so the photons are in the Rayleigh regime, where $\sigma \propto \lambda^{-4}$ is small (Table 8.2). They escape the cloud and carry the energy with them. Dust also cools by its own thermal emission in the far infrared.

The balance. Let each gas particle receive heat at the rate $\dot{E}_{\mathrm{heat}}$, and let a coolant with density n_c have a line of energy E_{21} that escapes. In thin gas, below the critical density ($A_{21} t_{\mathrm{col}} \gg 1$), nearly every coolant particle is in its lower level and every upward collision ends in an escaping photon. With $1/t_{\mathrm{col}} = n \sigma_{\mathrm{col}} \bar{v}$ from Section 8.3, heat in equals heat out (P4),

$$\begin{aligned} n\,\dot{E}_{\mathrm{heat}} &\overset{(1)}{=} n_2\,A_{21}\,E_{21} \\ &\overset{(2)}{=} n_c\,\frac{g_2}{g_1}\,\frac{e^{-E_{21}/k_B T}}{t_{\mathrm{col}}}\,E_{21} \\ &\overset{(3)}{=} n_c\,n\,\sigma_{\mathrm{col}}\bar{v}\,\frac{g_2}{g_1}\,E_{21}\,e^{-E_{21}/k_B T}. \end{aligned} \tag{8.38}$$

(1) Heating per unit volume equals the energy carried off per unit volume by escaping photons.
(2) Eq. (8.33) with $A_{21} t_{\mathrm{col}} \gg 1$, so $n_2 A_{21} = n_1 (g_2/g_1) e^{-E_{21}/k_B T}/t_{\mathrm{col}}$, and $n_1 \approx n_c$.
(3) $1/t_{\mathrm{col}} = n \sigma_{\mathrm{col}} \bar{v}$.

Dividing by n and taking the logarithm,

$$\boxed{k_B T = \frac{E_{21}}{\ln\!\left(n_c\,\sigma_{\mathrm{col}}\,\bar{v}\,(g_2/g_1)\,E_{21}/\dot{E}_{\mathrm{heat}}\right)}}. \tag{8.39}$$

The heating and the density enter only through a logarithm. Changing the heating tenfold changes the logarithm by 2.3, so over a wide range of conditions the gas sits within a factor of a few of E_{21}/k_B of its main coolant. That is why atomic clouds, cooled by C^+, are near 100 K, and molecular clouds, cooled by CO, are near 10 to 20 K. Above the critical density the cooling per particle stops growing with n, but the Boltzmann factor still pins the temperature in the same way.

Why it matters for stars. The temperature fixes the pressure of the gas, $P = n k_B T$ (Eq. (9.23)), and the pressure is what resists gravity. A cloud that can radiate away the heat of its own compression stays cold while it shrinks, and its pressure cannot stop the collapse. Efficient cooling is therefore a condition for star formation. Whether gravity or pressure wins is worked out in Chapter 10.

8.7 Problem Set: The Interstellar Medium

8.7.1 Problem 1: Planetary Nebula Emission and Expansion

Problem Statement: The Hα line from a spherical planetary nebula, observed with the whole nebula in the aperture, spreads from a minimum wavelength of 656.24 nm to a maximum of 656.32 nm. (a) Find the rest wavelength the observation implies. (b) Estimate how fast the nebula expands, in km s^{-1}. (c) Convert the speed to AU yr^{-1}. (d) The gas is at about 10^4 K. Is the thermal width of Hα small compared with the spread of the line?

Essential Ideas:

- The near cap gives the shortest wavelength and the far cap the longest, and the profile is flat between them (Section 8.5.1).
- $v_{\rm exp} = (c/2)(\lambda_{\rm max} - \lambda_{\rm min})/\lambda_0$, Eq. (8.37).
- The thermal full width is $2.355\,\lambda_0\sigma_v/c$ with $\sigma_v = \sqrt{k_B T/m_{\rm H}}$, Eqs. (8.22) and (8.25).

Detailed Solutions:

(a) **Rest wavelength**. The caps move at equal and opposite speeds, so λ_0 is the mean of the edges.

$$\boxed{\lambda_0 = \frac{656.32 + 656.24}{2}\,\text{nm} = 656.28\,\text{nm}.}$$

This is the laboratory wavelength of Hα, so the nebula as a whole is not moving toward or away from us.

(b) **Expansion speed**.

$$\boxed{v_{\rm exp} = \frac{2.998\times10^5\,\text{km s}^{-1}}{2}\times\frac{656.32-656.24}{656.28} \simeq 18.3\,\text{km s}^{-1}.}$$

(c) **In astronomical units per year**. With $1\,\text{yr} = 3.156\times10^7\,\text{s}$ and $1\,\text{AU} = 1.496\times10^8\,\text{km}$,

$$\boxed{v_{\rm exp} = 18.27\,\frac{\text{km}}{\text{s}}\times\frac{3.156\times10^7\,\text{s yr}^{-1}}{1.496\times10^8\,\text{km AU}^{-1}} \simeq 3.85\,\text{AU yr}^{-1}.}$$

This is a typical speed for the shell of a planetary nebula, whose observed values are 10 to $30\,\text{km s}^{-1}$.

(d) **Thermal width**. $\sigma_v = \sqrt{1.381\times10^{-23}\times10^4/1.674\times10^{-27}}\,\text{m s}^{-1} = 9.1\,\text{km s}^{-1}$, so

$$\boxed{\text{thermal full width} = 2.355\times656.28\,\text{nm}\times\frac{9.1}{2.998\times10^5} \simeq 0.047\,\text{nm},}$$

more than half the spread of 0.08 nm. The thermal width is not small. The edges of the Hα line are blurred, and the measured edges should be read where the profile has fallen to half its height. Lines of heavier ions, such as [O III] at 500.7 nm, have four times smaller thermal widths and give the expansion speed more cleanly.

8.7.2 Problem 2: Counting Hydrogen with the 21 cm Line

Problem Statement: A radio telescope sees 21 cm emission from a cloud of HI with no bright source behind it. Radio astronomers express the intensity I_ν as a brightness temperature T_B, the temperature a blackbody would need to give the same I_ν. (a) Show that at this frequency $I_\nu = 2\nu^2 k_B T_B/c^2$. (b) Assume the cloud is transparent to its own line. Show that the column density of hydrogen is

$$N_{\rm H} = \frac{32\pi k_B \nu_0^2}{3hc^3 A_{21}}\int T_B\,dv,$$

where v is the line-of-sight velocity across the line. (c) Evaluate the constant in cm^{-2} per K km s^{-1}. (d) The line is Gaussian with peak $T_B = 50\,\text{K}$ and full width $10\,\text{km s}^{-1}$. Find $N_{\rm H}$ and the visual extinction expected from the dust that comes with it.

Essential Ideas:

- The Planck function at $h\nu \ll k_B T$ (Eq. (4.13)).

- Three quarters of the atoms are in the upper level and each sends out photons at the rate A_{21}, evenly in all directions, Eqs. (8.31) and (8.34).
- The dust per hydrogen atom, $A_V/N_{\rm H} = 5.3\times10^{-22}\,\mathrm{mag\,cm^2}$ (Section 8.2.4).

Detailed Solutions:

(a) **Brightness temperature**. For $\nu_0 = 1420\,\mathrm{MHz}$, $h\nu_0/k_B = 0.068\,\mathrm{K}$, far below any T_B of interest. With $x = h\nu/k_BT_B \ll 1$, $e^x - 1 \approx x$ in the Planck function, and

$$\boxed{I_\nu = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/k_BT_B}-1} \approx \frac{2\nu^2 k_B T_B}{c^2}.}$$

(b) **Column density**. A column of cross-section $1\,\mathrm{m^2}$ holds $\frac{3}{4}N_{\rm H}$ atoms in the upper level. Together they send out the power $\frac{3}{4}N_{\rm H}A_{21}h\nu_0$, spread evenly over 4π steradians. Seen face on through a transparent cloud, the intensity summed over the line is therefore

$$\int I_\nu\,d\nu = \frac{h\nu_0 A_{21}}{4\pi}\,\frac{3}{4}N_{\rm H}.$$

By (a), and with $d\nu = (\nu_0/c)\,dv$ from Eq. (3.25),

$$\int I_\nu\,d\nu = \frac{2\nu_0^2 k_B}{c^2}\,\frac{\nu_0}{c}\int T_B\,dv.$$

Setting the two equal and solving,

$$\boxed{N_{\rm H} = \frac{32\pi k_B \nu_0^2}{3hc^3A_{21}}\int T_B\,dv.}$$

The temperature of the gas has dropped out, because the populations of Eq. (8.34) do not depend on it. Absorption by the cloud of its own line, left out here, matters only when the cloud is not transparent, and is treated with the transfer equation of Chapter 11.

(c) **The constant**. With $\nu_0 = 1.4204\times10^9\,\mathrm{Hz}$ and $A_{21} = 2.87\times10^{-15}\,\mathrm{s^{-1}}$, in SI units the constant is $1.82\times10^{19}\,\mathrm{m^{-2}}$ per $\mathrm{K\,m\,s^{-1}}$. Multiplying by 10^3 for $\mathrm{km\,s^{-1}}$ and by 10^{-4} for $\mathrm{cm^{-2}}$,

$$\boxed{N_{\rm H} = 1.82\times10^{18}\,\mathrm{cm^{-2}}\times\frac{\int T_B\,dv}{1\,\mathrm{K\,km\,s^{-1}}}.}$$

(d) **The cloud**. The area under a Gaussian is its peak times its full width times $\sqrt{\pi/4\ln 2} = 1.064$, so $\int T_B\,dv = 1.064\times50\times10 = 532\,\mathrm{K\,km\,s^{-1}}$, and

$$\boxed{N_{\rm H} \simeq 9.7\times10^{20}\,\mathrm{cm^{-2}}, \qquad A_V \simeq 9.7\times10^{20}\times5.3\times10^{-22} \simeq 0.5\,\mathrm{mag}.}$$

A cloud that glows this brightly in the 21 cm line dims the stars behind it by about half a magnitude. Comparing the two along many lines of sight is how the ratio of dust to gas is measured.

8.7.3 Problem 3: The Two Widths of Lyman Alpha

Problem Statement: The Lyman α line of hydrogen is at 121.567 nm, and its upper level decays at $\Gamma = 6.27\times10^8\,\mathrm{s^{-1}}$. The lower level is the ground state, which does not decay. Take interstellar gas at 100 K. (a) Find the natural full width, in Hz and in nm. (b) Find the Doppler full width in Hz. (c) Which dominates the core of the line, and by how much? (d) How far from the centre, in units of $\Delta\nu_D$, does the Lorentzian wing exceed the Gaussian?

Essential Ideas:

- The natural full width is $\Gamma/2\pi$, Eq. (8.20).
- The Doppler full width is $2.355\,\Delta\nu_D$, with $\Delta\nu_D = \nu_0\sqrt{k_BT/m_{\rm H}}/c$, Eq. (8.25).
- The Voigt profile is Gaussian in the core and Lorentzian in the wings (Eq. (8.26)).

Detailed Solutions:

(a) **Natural width**. In wavelength, $\Delta\lambda = \lambda_0^2\Delta\nu/c$.

$$\boxed{\frac{\Gamma}{2\pi} = \frac{6.27\times10^8}{2\pi}\,\text{Hz} \simeq 1.0\times10^8\,\text{Hz}, \qquad \Delta\lambda \simeq 4.9\times10^{-6}\,\text{nm}.}$$

(b) **Doppler width**. $\nu_0 = c/\lambda_0 = 2.466\times10^{15}$ Hz and $\sigma_v = 0.908\,\text{km s}^{-1}$, so $\Delta\nu_D = \nu_0\sigma_v/c = 7.47\times10^9$ Hz.

$$\boxed{2.355\,\Delta\nu_D \simeq 1.76\times10^{10}\,\text{Hz}.}$$

(c) **The core**. The ratio of the two widths is

$$\boxed{\frac{1.76\times10^{10}}{1.0\times10^8} \simeq 176,}$$

so the core is Gaussian and set by the temperature. Even at 100 K the motions of the atoms blur the line far more than its lifetime does.

(d) **The wings**. Write $y = (\nu-\nu_0)/\Delta\nu_D$. The Gaussian is $e^{-y^2/2}/(\sqrt{2\pi}\Delta\nu_D)$ and the Lorentzian wing is $(\Gamma/4\pi^2)/(y\Delta\nu_D)^2$. They are equal when

$$\frac{e^{-y^2/2}}{\sqrt{2\pi}} = \frac{\Gamma}{4\pi^2\Delta\nu_D}\,\frac{1}{y^2} = \frac{2.12\times10^{-3}}{y^2},$$

which a numerical solution puts at

$$\boxed{y \simeq 4.0.}$$

Beyond four Doppler widths the line is Lorentzian. A cloud thick enough to absorb all the light out to there shows broad Lorentzian wings, and their width gives its column of hydrogen without any knowledge of its temperature.

Chapter 9

Six Principles

Astrophysics can look like a long list of formulas. The Jeans mass, the Eddington luminosity, the free-fall time, the Chandrasekhar mass and the Planck spectrum are usually met one at a time, each with its own derivation. The list grows with every new object.

The list does not have to be remembered. Almost every formula about stars and gas comes from six general principles, applied with the classical and quantum physics the reader already knows. Each result is then a short calculation that can be redone whenever the formula itself is forgotten.

	Principle	Statement
P1	Conservation	The mass, momentum, energy and angular momentum in a region change only by what flows across its boundary and by the forces that act on it.
P2	Gravity of a sphere	Inside a spherical body, a parcel at radius r is pulled only by the mass inside r, as if it sat at the centre.
P3	Counting states	Phase space comes in cells of size h^3. Pressure, energy density and populations follow from counting the states and how full each one is.
P4	Balance of rates	In a steady state every gain is matched by a loss. In equilibrium every process is matched by its own reverse.
P5	Random walk	After N steps in random directions a particle is about $\sqrt{N}$ steps from its start, so anything that scatters often moves down its own gradient.
P6	Compare two scales	The ratio of two times or two lengths decides which physics matters. A ratio near one marks a transition.

Table 9.1: The six principles.

None of the six is new physics. Each is a familiar law put in the form that applies directly to gas, light and gravity. Each has one general form, written with as few symbols as possible, and every result built on them names the principles it combines.

9.1 Conservation (P1)

The Kepler problem of Chapter 6 was solved by two conserved quantities. Conservation of the energy and of the angular momentum turned six second-order equations into an orbit that could be written down (Section 6.1.2 onwards).

A star is not two bodies. It holds about 10^{57} particles, and nobody can follow each of them.

Think of a water tank in a house. Water comes in through one pipe and leaves through another. To know how much water the tank holds a minute from now, nobody follows single molecules. One watches the two pipes. The water inside changes by what flows in minus what flows out, and nothing that happens inside the tank can change that.

A star can be treated the same way. Take any fixed region inside it. The mass in that region changes only by gas flowing across its boundary. Its momentum changes by the momentum that gas carries across the boundary and by the forces acting on the region. This is conservation written for a fixed region instead of a single particle, principle P1.

The tank in numbers. It holds 200 L, water flows in at 3 L/min and out at 5 L/min. The volume changes at $3 - 5 = -2$ L/min, so the tank is empty after 100 minutes. Only the two flows were needed.

The picture to keep is a box with arrows crossing its faces. Arrows pointing in add to what is inside, arrows pointing out take away.

The same bookkeeping is the continuity equation for charge in electromagnetism, the probability current in quantum mechanics, and the conservation of energy and angular momentum in Chapter 6. For stars it gives hydrostatic equilibrium (Chapter 10), the change of light along a ray (Chapter 11) and the energy equation of a star (Chapter 12). For a gas it gives two equations, one for mass and one for momentum.

Mass. Let a gas have density $\rho(\vec{r}, t)$ and velocity $\vec{v}(\vec{r}, t)$. Take a fixed volume V with surface S, and let $d\vec{A}$ be an outward-pointing element of S. Gas crossing $d\vec{A}$ carries mass out at the rate $\rho\,\vec{v}\cdot d\vec{A}$. The mass inside therefore changes as

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\, dV &\overset{(1)}{=} -\oint_S \rho\,\vec{v}\cdot d\vec{A} \\ &\overset{(2)}{=} -\int_V \nabla\cdot(\rho\vec{v})\, dV. \end{aligned} \tag{9.1}$$

(1) Mass inside changes only by the flow across the surface (P1).
(2) The divergence theorem.

The volume is fixed, so the time derivative on the left acts only on ρ and can be taken inside the integral. Both sides are then integrals over the same V,

$$\int_V \left[\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right] dV = 0.$$

This holds for every choice of V, however small, so the integrand itself vanishes,

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0.} \tag{9.2}$$

This is the **continuity equation**. It holds for any fluid and says nothing about forces.

Momentum. The i-th component of the momentum inside V is $\int_V \rho v_i\, dV$. It changes in three ways. Gas crossing the surface carries momentum ρv_i at the rate $\rho v_i\,\vec{v}\cdot d\vec{A}$. The pressure P of the surrounding gas pushes inward on each surface element with the force $-P\,d\vec{A}$. Gravity pulls on every element of mass with the acceleration $\vec{g}$. Writing the surface element as dA_j and

summing over repeated indices,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho v_i \, dV \overset{(1)}{=} -\oint_S \rho v_i v_j \, dA_j - \oint_S P\,\delta_{ij}\, dA_j + \int_V \rho g_i \, dV$$
$$\overset{(2)}{=} \int_V \Big[-\partial_j\big(\rho v_i v_j + P\delta_{ij}\big) + \rho g_i \Big] dV. \tag{9.3}$$

(1) The three ways listed above (P1). The pressure force is written with δ_{ij} so that both surface terms have the same form.
(2) The divergence theorem, applied to each surface term.

The quantity being differentiated, $\rho v_i v_j + P\delta_{ij}$, is the flux of the i-th component of momentum in direction j. It is a 3×3 matrix, the **momentum flux**,

$$\Pi = \begin{pmatrix} \rho v_x^2 + P & \rho v_x v_y & \rho v_x v_z \\ \rho v_y v_x & \rho v_y^2 + P & \rho v_y v_z \\ \rho v_z v_x & \rho v_z v_y & \rho v_z^2 + P \end{pmatrix} = \rho\, \vec{v}\, \vec{v}^{\,\mathsf{T}} + P\, \mathbb{1}. \tag{9.4}$$

The first part is momentum carried by the bulk flow. The second part, P on the diagonal, is momentum carried across the surface by the random motion of the particles, which is what pressure is (Section 9.3). Removing the integrals as for the mass, and using Π,

$$\frac{\partial(\rho v_i)}{\partial t} + \partial_j \Pi_{ij} = \rho g_i. \tag{9.5}$$

The left side can be simplified with the continuity equation. Writing $\Pi_{ij} = \rho v_i v_j + P\delta_{ij}$,

$$\frac{\partial(\rho v_i)}{\partial t} + \partial_j(\rho v_i v_j) \overset{(1)}{=} v_i \left[\frac{\partial \rho}{\partial t} + \partial_j(\rho v_j)\right] + \rho\left[\frac{\partial v_i}{\partial t} + v_j\, \partial_j v_i\right]$$
$$\overset{(2)}{=} \rho\left[\frac{\partial v_i}{\partial t} + v_j\, \partial_j v_i\right]. \tag{9.6}$$

(1) The product rule, applied to ρv_i and to $\rho v_j \cdot v_i$.
(2) The first bracket is the continuity equation (9.2), so it vanishes.

With $\partial_j(P\delta_{ij}) = \partial_i P$, Eq. (9.5) becomes the **Euler equation**,

$$\boxed{\rho\left(\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right) = -\nabla P + \rho\, \vec{g}.} \tag{9.7}$$

The left side is mass times the acceleration of a parcel that moves with the gas. The right side is the force on it. Pressure acts only through its gradient, since a uniform pressure pushes equally on all sides of a parcel. The gravity $\vec{g}$ comes from P2 and the pressure from P3.

Counting unknowns. The unknowns are ρ, the three components of $\vec{v}$, and P, five in all. The equations are continuity and the three components of Euler, four in all. One relation is missing. It is the equation of state, which gives P from the density and the temperature (P3). The temperature is then a sixth unknown, and the sixth equation is conservation of energy, written for a star in Section 12.2. This is the same count of unknowns against equations that fixed the twelve constants of the two-body problem.

Angular momentum. When gravity is the only force and it points to a fixed centre, $\vec{g} \parallel \hat{r}$, each parcel keeps its angular momentum per unit mass $\vec{r} \times \vec{v}$. Following a parcel,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{r} \times \vec{v}\right) = \vec{v} \times \vec{v} + \vec{r} \times \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \vec{r} \times \vec{g} = 0,$$

exactly as for the single body of Eq. (6.16). With pressure present, neighbouring parcels can trade angular momentum through pressure forces, but the internal forces come in equal and opposite pairs, so the total angular momentum of an isolated cloud does not change. This is what turns a collapsing cloud into a disk (Section 10.6).

9.2 Gravity of a Sphere (P2)

Newton's law of gravity between two points, GMm/r^2, ran the whole of Chapter 6.

A star or a gas cloud is not a point. Every bit of it pulls on every other bit, and adding all those pulls one by one looks hopeless.

Imagine a well dug straight down to the centre of the Earth, and suppose for now that the Earth has the same density everywhere. At the surface you have your usual weight. Halfway down you weigh half as much. The rock above your head forms a shell around you, and the pulls from its different sides cancel exactly. Only the ball of rock below you counts, and it is smaller.

Inside a spherical star the same is true. A parcel of gas at radius r feels only the mass inside r, and feels it as if that mass sat at the centre. Everything outside r, however it is arranged in shells, pulls with zero net force. That is principle P2.

The well in numbers. For the uniform Earth the mass inside radius r is $M(r/R)^3$, so the pull is

$$g(r) = \frac{G\,M(r/R)^3}{r^2} = \frac{GM}{R^2}\,\frac{r}{R},$$

which falls linearly to zero at the centre. Halfway down, at $r = R/2$, it is $9.8/2 = 4.9\,\mathrm{m\,s^{-2}}$.

The picture is Figure 9.1. Draw a sphere of radius r through the parcel. The mass inside it sets the pull, and the shell outside it does nothing.

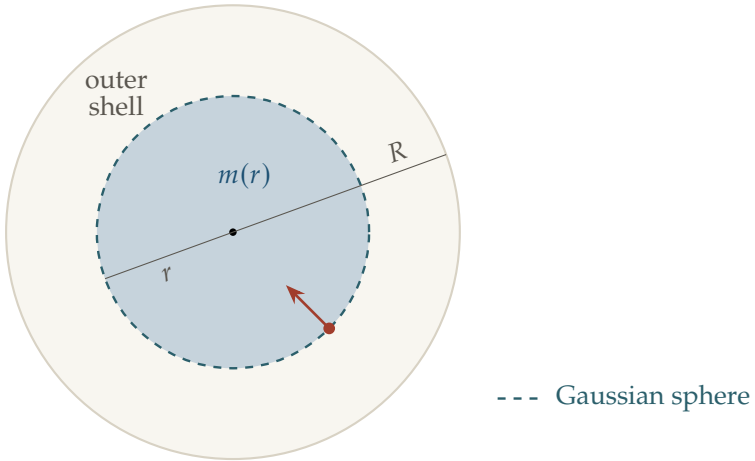


Figure 9.1: A spherical star of radius R. The dashed sphere of radius r passes through a parcel (dot), which is pulled toward the centre (arrow). The shaded ball holds the mass $m(r)$. The outer shell lies between r and R.

The same law with charge in place of mass is Gauss's law of electrostatics. In astrophysics it fixes the rotation speed of stars in a galaxy by the mass inside their orbit, it gives the expansion

of the universe from a single shell of matter, and in general relativity it survives as Birkhoff's theorem. It gives the gravity at every radius from one number, the mass inside.

Gauss's law for gravity. A point mass m at the origin produces $\vec{g} = -Gm\,\hat{r}/r^2$. Through a sphere of radius r around it, with $d\vec{A} = r^2\,d\Omega\,\hat{r}$,

$$\oint \vec{g}\cdot d\vec{A} \overset{(1)}{=} \oint \left(-\frac{Gm}{r^2}\right) r^2\,d\Omega \overset{(2)}{=} -4\pi G m. \tag{9.8}$$

(1) $\hat{r}\cdot\hat{r} = 1$.
(2) The r^2 cancels, and the solid angle of a whole sphere is 4π.

The radius dropped out, so the answer is the same for any closed surface around the mass. Each patch of a surface subtends some solid angle at the mass, and the flux through the patch depends only on that solid angle. A mass outside the surface gives zero, because every line from it enters the surface once and leaves once, with equal and opposite contributions. Adding the fields of many masses therefore gives

$$\boxed{\oint_S \vec{g}\cdot d\vec{A} = -4\pi G\, m_{\text{inside}}.} \tag{9.9}$$

The field of a sphere. For a spherically symmetric body, $\vec{g}$ can only point along $\hat{r}$ and can only depend on r. Any other direction would single out a direction in space, and a sphere has none. So $\vec{g} = g(r)\,\hat{r}$. Define the mass inside radius r,

$$m(r) \equiv \int_0^r 4\pi r'^2 \rho(r')\,dr', \qquad \frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2\rho. \tag{9.10}$$

Applying Eq. (9.9) to a sphere of radius r,

$$4\pi r^2\, g(r) \overset{(1)}{=} -4\pi G\, m(r)$$
$$\Longrightarrow \qquad \vec{g}(r) = -\frac{G\,m(r)}{r^2}\,\hat{r}. \tag{9.11}$$

(1) $\vec{g}\cdot d\vec{A} = g(r)\,dA$ everywhere on the sphere, and the area is $4\pi r^2$.

$$\boxed{\vec{g}(r) = -\frac{G\,m(r)}{r^2}\,\hat{r}} \tag{9.12}$$

The pull at radius r depends only on $m(r)$. How the mass outside r is arranged does not enter, as long as it is spherical.

Poisson's equation. The divergence theorem turns Eq. (9.9) into $\int_V (\nabla\cdot\vec{g} + 4\pi G\rho)\,dV = 0$ for every volume, so $\nabla\cdot\vec{g} = -4\pi G\rho$. With $\vec{g} = -\nabla\Phi$, where Φ is the gravitational potential,

$$\nabla^2\Phi = 4\pi G\rho. \tag{9.13}$$

This local form holds for any shape, spherical or not.

Gravitational energy. Build the sphere shell by shell, bringing each shell in from far away. When the ball already built has radius r and mass $m(r)$, the next shell of mass dm lands at radius r. By Eq. (9.12) the ball pulls it as a point mass $m(r)$ at the centre, and the shells not yet built do not exist. Bringing dm from infinity to r therefore releases the energy $G\,m(r)\,dm/r$. Adding all the shells gives the **gravitational energy**

$$\boxed{U = -\int_0^M \frac{G\,m}{r}\,dm.} \tag{9.14}$$

U is negative, because energy is released in building the body, and it is more negative when the same mass sits at smaller radii. Its value for a uniform sphere is found where it is first used, in Section 10.2.

9.3 Counting States (P3)

The Planck spectrum was quoted in Chapter 4 (Eq. (4.13)), and the Maxwell distribution of velocities in Chapter 8 (Eq. (8.22)). Each was taken as given.

A star needs more than that. Its interior holds hot gas, a bath of light, and in some stars electrons packed so tightly that quantum mechanics stops them from coming closer. All three push. Quoting a separate formula for each hides the fact that they are the same calculation.

Think of a theatre with numbered seats. To know how many people sit in the front rows, two things are needed. One is how many seats there are. The other is how likely each seat is to be taken. Some audiences fill seats at random, some crowd the cheap seats, and some follow a strict rule of one person per seat.

Particles fill quantum states in the same way. Quantum mechanics fixes how many states there are in a given range of position and momentum. Temperature, and whether particles may share a state, fix how full each state is. Counting both gives the pressure of any gas, whether of atoms, photons or electrons. That is principle P3.

The smallest example is the air in a room. At 293 K and normal pressure it has $n = 2.50 \times 10^{25}$ molecules per cubic metre, and

$$P = nk_BT = (2.50 \times 10^{25})(1.381 \times 10^{-23})(293)\,\text{Pa} = 1.01 \times 10^5\,\text{Pa},$$

the pressure we live in. The light in the same room, at the same temperature, pushes with $aT^4/3 = 1.9 \times 10^{-6}$ Pa (Eq. (9.25) below), about 10^{-11} of the air. In the core of the Sun the share of light rises to about 5×10^{-4} (Problem 9.7.2), and in the most massive stars light carries most of the pressure.

The picture is position and momentum together, as one six-dimensional space cut into cells of volume h^3. Each cell holds one quantum state for each spin direction. Particles sit in cells, and the pressure is the momentum they carry across a surface.

The same counting gives the Maxwell, Planck and Fermi–Dirac distributions. It gives the ideal-gas law, blackbody radiation (Chapter 4), the degeneracy pressure that holds up white dwarfs (Chapters 12 and 13), the ionisation of hydrogen, and the radiation left over from the early universe. Here it gives one formula for the pressure that works for every kind of particle.

How many states. Put one particle in a cube of side b and volume $V = b^3$, with periodic boundary conditions. Its wavefunction $e^{i\vec{p}\cdot\vec{r}/\hbar}$ must repeat after a distance b in each direction, so $p_x b/\hbar = 2\pi$ times an integer, and the same for y and z,

$$p_x = \frac{h}{b}\,q_x, \qquad p_y = \frac{h}{b}\,q_y, \qquad p_z = \frac{h}{b}\,q_z, \qquad q_x, q_y, q_z \in \mathbb{Z}. \tag{9.15}$$

The allowed momenta are the points of a grid with spacing h/b. The number of grid points in a small box of momentum space is then

$$dq_x\,dq_y\,dq_z \overset{(1)}{=} \frac{b}{h}\,dp_x\,\frac{b}{h}\,dp_y\,\frac{b}{h}\,dp_z \overset{(2)}{=} \frac{V\,d^3p}{h^3}. \tag{9.16}$$

(1) One grid point per interval h/b in each direction.
(2) $b^3 = V$ and $d^3p = dp_x\,dp_y\,dp_z$.

$$\boxed{\text{number of states in } d^3x\,d^3p = \frac{d^3x\,d^3p}{h^3} \quad \text{(per spin state)}} \tag{9.17}$$

This does not depend on what the particle is. Every particle gets one state per h^3 of phase space for each spin direction.

How full. Let $f(p)$ be the mean number of particles in one state of momentum p. Statistical mechanics gives it in the three cases needed.

- Particles far apart compared with their quantum wavelength (an ordinary gas) occupy states rarely, with $f \propto e^{-p^2/2mk_BT}$.
- Photons can pile into one state without limit, and $f = 1/(e^{pc/k_BT} - 1)$.
- Electrons allow at most one particle per state, $f \le 1$. At low temperature every state up to some momentum is full and every state above it is empty. This case is worked out in Section 12.3.

Combining f with Eq. (9.17), the number of particles per unit volume with momentum in d^3p is

$$dn = (\text{spin states})\,\frac{f(p)\,d^3p}{h^3}, \qquad n = \int dn. \tag{9.18}$$

Pressure as momentum flux. A particle with momentum $\vec{p}$ moves with velocity $\vec{v}(p)$ in the same direction. Across a surface it carries the momentum component p_i at the rate v_j per unit area in direction j, so the particles together carry the momentum flux $\int p_i v_j\,dn$. This is the random-motion part of the momentum flux Π of Eq. (9.4), seen from a frame in which the gas is at rest. When f depends only on the size of $\vec{p}$, as in all three cases above,

$$\begin{pmatrix} \int p_xv_x\,dn & \int p_xv_y\,dn & \int p_xv_z\,dn \\ \int p_yv_x\,dn & \int p_yv_y\,dn & \int p_yv_z\,dn \\ \int p_zv_x\,dn & \int p_zv_y\,dn & \int p_zv_z\,dn \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} P & 0 & 0 \\ 0 & P & 0 \\ 0 & 0 & P \end{pmatrix}. \tag{9.19}$$

(1) Off the diagonal, the integrand p_xv_y changes sign when $p_y \to -p_y$ while dn does not, so the integral vanishes. On the diagonal the three integrals are equal because no direction is special. Their common value is the pressure.

Adding the three diagonal entries gives $3P = \int \vec{p}\cdot\vec{v}\,dn$, and since $\vec{v} \parallel \vec{p}$,

$$\boxed{P = \frac{1}{3}\int p\,v(p)\,dn.} \tag{9.20}$$

The factor $\frac{1}{3}$ is the average of $\cos^2\theta$ over all directions. Only the distribution of momenta enters. The temperature, the quantum rules and the kind of particle enter only through dn.

Pressure and energy density. The kinetic energy per unit volume is $u = \int (\text{kinetic energy})\, dn$. In two limits the integrand of Eq. (9.20) is a fixed multiple of the kinetic energy.

- Slow particles have $v = p/m$ and kinetic energy $p^2/2m$, so $pv = p^2/m = 2\times$ kinetic energy, and $P = \frac{2}{3}u$.
- Particles at the speed of light (photons, or electrons with $pc \gg mc^2$) have $v = c$ and energy pc, so $pv = pc =$ energy, and $P = \frac{1}{3}u$.

Writing both as one formula with an exponent γ,

$$\boxed{P = (\gamma - 1)\, u, \qquad \gamma = \tfrac{5}{3} \text{ (slow particles)}, \qquad \gamma = \tfrac{4}{3} \text{ (particles at speed } c\text{).}} \tag{9.21}$$

This depends only on how a particle's energy grows with its momentum, not on f. It holds for an ordinary gas, for light and for degenerate electrons alike. The difference between $\gamma = \frac{5}{3}$ and $\gamma = \frac{4}{3}$ decides whether a star is firmly bound or barely bound (Section 10.2).

The ideal gas. For an ordinary gas, $f \propto e^{-p^2/2mk_BT}$. Normalising dn so that $\int dn = n$ and writing $d^3p = 4\pi p^2 dp$, Eq. (9.20) gives

$$\begin{aligned} P &\overset{(1)}{=} \frac{1}{3}\, \frac{n \int_0^\infty (p^2/m)\, e^{-p^2/2mk_BT}\, 4\pi p^2\, dp}{\int_0^\infty e^{-p^2/2mk_BT}\, 4\pi p^2\, dp} \\ &\overset{(2)}{=} \frac{1}{3}\, n\, \frac{3mk_BT}{m} \\ &= nk_BT. \end{aligned} \tag{9.22}$$

(1) Eq. (9.20) with $v = p/m$. Dividing by the same integral without p^2/m normalises dn to n.
(2) The Gaussian moments $\int_0^\infty p^4 e^{-p^2/2mk_BT} dp = 3mk_BT \int_0^\infty p^2 e^{-p^2/2mk_BT} dp$, which follows from one integration by parts.

$$\boxed{P = nk_BT, \qquad u = \tfrac{3}{2}nk_BT.} \tag{9.23}$$

The second form follows from $u = \frac{3}{2}P$ (Eq. (9.21) with $\gamma = \frac{5}{3}$). For N particles the total kinetic energy is $K = \frac{3}{2}Nk_BT$. The one-dimensional distribution of Eq. (8.22) is one component of this one.

Light. Photons have two polarisations, energy pc and $f = 1/(e^{pc/k_BT} - 1)$. Their energy density is

$$\begin{aligned} u &\overset{(1)}{=} \int_0^\infty pc\, \frac{2}{h^3}\, \frac{4\pi p^2\, dp}{e^{pc/k_BT} - 1} \\ &\overset{(2)}{=} \frac{8\pi (k_BT)^4}{h^3c^3} \int_0^\infty \frac{x^3\, dx}{e^x - 1} \\ &\overset{(3)}{=} \frac{8\pi^5 k_B^4}{15\, h^3c^3}\, T^4. \end{aligned} \tag{9.24}$$

(1) Eq. (9.18) with two spin states, times the energy pc of each photon.
(2) The substitution $x = pc/k_BT$.
(3) $\int_0^\infty x^3 dx/(e^x - 1) = \pi^4/15$, the same integral that gave the Stefan–Boltzmann constant in Eq. (4.25).

The constant in front is the **radiation constant**. Comparing with Eq. (4.25), it is $a = 4\sigma_{SB}/c = 7.566 \times 10^{-16}\,\mathrm{J\,m^{-3}\,K^{-4}}$. With Eq. (9.21) at $\gamma = \frac{4}{3}$,

$$\boxed{u = aT^4, \qquad P = \tfrac{1}{3}aT^4.} \tag{9.25}$$

The pressure of light depends on the temperature alone, not on any density.

The Planck spectrum. Keeping the photon energy $h\nu = pc$ instead of integrating over it gives the energy per unit volume and per unit frequency. With $p = h\nu/c$ and $dp = h\,d\nu/c$ in the first line above,

$$u_\nu\,d\nu = \frac{8\pi h\nu^3}{c^3}\,\frac{d\nu}{e^{h\nu/k_BT} - 1}.$$

This light moves at c and is spread evenly over all 4π steradians of direction, so the energy crossing a unit area per unit time, per unit frequency and per steradian is $c\,u_\nu/4\pi$,

$$\boxed{B_\nu(T) = \frac{c}{4\pi}\,u_\nu = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/k_BT} - 1}.} \tag{9.26}$$

This is the Planck function used in Chapter 4, now derived by counting states. The quantity it describes, the intensity of light in one direction, is developed in Chapter 11.

The mechanism does not matter. Nothing in this derivation asked how the light was made. It used only the number of photon states, Eq. (9.17), and how full each state is at temperature T. So light that has come into thermal equilibrium with matter at temperature T has the Planck spectrum, whatever produced it. Electrons braking in collisions with ions, atoms jumping between energy levels and gamma rays from nuclear fusion all end in the same $B_\nu(T)$, once the light has been absorbed and re-emitted often enough to share the temperature of the matter. This is detailed balance at work (P4). In equilibrium each process that makes a photon of frequency ν is matched by its own reverse, which absorbs one, and that balance fixes the ratio of emission to absorption at $B_\nu(T)$ for every process separately. This is Kirchhoff's law (Section 11.3).

The Sun shows this. Fusion in its core releases gamma rays with energies near a million electron volts, yet the light that leaves its surface is close to a blackbody at 5772 K. On the way out the energy is absorbed and re-emitted about 5×10^{23} times (Problem 11.8.3), and at every depth it takes on the local temperature, so nothing of the fusion spectrum survives. The microwave background is another case. It was made by many processes in the hot early universe and is today the most perfect blackbody known (Section 16.3).

What the mechanism does decide is how fast equilibrium is reached, and whether it is reached at all. The matter must be thick enough that light is absorbed many times before it escapes, an optical depth much larger than one (Section 11.2). Thin matter shows its mechanism instead. A nebula glows in the spectral lines of its own atoms, and the fast electrons of a radio galaxy give a synchrotron power law (Chapter 15). Neither is a blackbody, because in neither does the light come into equilibrium with the matter.

9.4 Balance of Rates (P4)

The temperature of interstellar gas was set by matching heating to cooling (Section 8.6).

That balance says where the gas ends up. It does not say how long it takes to get there, or how the many processes inside the gas share the work.

Think of two rooms joined by a door, with people walking through it both ways. After a while the number in each room stops changing. People have not stopped walking. As many walk one way as the other. If one room is more pleasant, people leave it more slowly, and it ends up holding more of them.

Atoms in a gas move between energy levels in the same way, kicked up and down by collisions. In a steady state the rate into each level equals the rate out. In full equilibrium each process is balanced by its own reverse. That is principle P4.

The rooms in numbers. Ninety people. Each person in room A leaves once a minute on average, and each person in room B leaves twice a minute. In the steady state the flows through the door are equal, $1 \times N_A = 2 \times N_B$, so $N_A = 60$ and $N_B = 30$. Room A is the pleasant one. The rooms reach this split within about 20 s (Eq. (9.33) below).

The picture is two boxes with flows between them. The boxes stop changing when the flows match, not when they stop.

The same principle is the law of mass action in chemistry. In astrophysics it gives Kirchhoff's law of radiation and the Einstein coefficients (Chapter 11), the ionisation balance of hydrogen, and the moment in the early universe when a reaction becomes too slow to keep up with the expansion. It gives the steady state, the time to reach it, and a link between the rates of a process and its reverse.

Steady state. Let N be a number of particles, supplied at a constant rate S per unit time and each lost at the rate $1/\tau$, so that it survives for a time τ on average. Then

$$\frac{\mathrm{d}N}{\mathrm{d}t} = S - \frac{N}{\tau}. \tag{9.27}$$

Starting from $N = 0$ at $t = 0$,

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\left(N\, e^{t/\tau}\right) \overset{(1)}{=} S\, e^{t/\tau} \\ \Longrightarrow \quad &N(t) \overset{(2)}{=} S\tau\left(1 - e^{-t/\tau}\right). \end{aligned} \tag{9.28}$$

(1) Multiply the equation by $e^{t/\tau}$. The left side becomes a total derivative.
(2) Integrate from 0 to t and divide by $e^{t/\tau}$.

$$\boxed{N \to S\,\tau \quad \text{after a few } \tau.} \tag{9.29}$$

The steady amount is the supply rate times the lifetime. How the loss happens does not matter, only its rate.

Detailed balance. Take two states A and B, with energies E_A and E_B. A particle in A jumps to B at the rate k_{AB}, and one in B jumps back at k_{BA}. The populations change as one matrix equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} -k_{AB} & k_{BA} \\ k_{AB} & -k_{BA} \end{pmatrix}\begin{pmatrix} N_A \\ N_B \end{pmatrix}. \tag{9.30}$$

The solutions are set by the eigenvalues λ of the matrix,

$$\begin{aligned} 0 &\overset{(1)}{=} (-k_{AB} - \lambda)(-k_{BA} - \lambda) - k_{AB}k_{BA} \\ &\overset{(2)}{=} \lambda\,(\lambda + k_{AB} + k_{BA}). \end{aligned} \tag{9.31}$$

(1) The determinant of the matrix minus λ times the unit matrix.

(2) Multiply out. The products $k_{AB}k_{BA}$ cancel.

The eigenvalue $\lambda = 0$ belongs to the state that does not change, the equilibrium. Its eigenvector satisfies $-k_{AB}N_A + k_{BA}N_B = 0$, so

$$\frac{N_B}{N_A} = \frac{k_{AB}}{k_{BA}}. \tag{9.32}$$

The other eigenvalue, $-(k_{AB} + k_{BA})$, is the rate at which any departure from equilibrium dies away,

$$\text{relaxation time} = \frac{1}{k_{AB} + k_{BA}}. \tag{9.33}$$

For the two rooms, $1/(1 + 2)$ minutes, which is 20 s.

Now add P3. In thermal equilibrium at temperature T, the two populations also follow the Boltzmann factor, $N_B/N_A = e^{-(E_B-E_A)/k_BT}$. Both statements hold at once, so

$$\boxed{\frac{k_{AB}}{k_{BA}} = e^{-(E_B-E_A)/k_BT}.} \tag{9.34}$$

The ratio of an upward rate to the downward rate is fixed by the temperature alone, whatever the physics of the collisions. The rates themselves belong to single collisions, which do not know whether the populations are in equilibrium. So the relation still holds when the populations are far from equilibrium, as long as the colliding particles have a Maxwell distribution at temperature T. That is what makes it useful. (When a level contains several states, each side is multiplied by its number of states.)

9.5 Random Walk (P5)

In the earlier chapters light travelled in straight lines, from a star to a telescope.

Inside a star it cannot. A photon is absorbed after a short distance and re-emitted in a new direction, again and again. Treating its path as a straight line gives an escape time that is wrong by many orders of magnitude.

Think of a bottle of perfume opened in the corner of a still room. Each molecule flies a tiny distance, hits an air molecule and goes off in a random direction. Each one moves at hundreds of metres per second, yet in perfectly still air the scent would take days to cross the room (Problem 9.7.4). We smell it within minutes only because air currents carry it.

Anything that scatters many times moves this way, by a random walk. After N steps it is not N steps from its start but only about $\sqrt{N}$ steps. That is principle P5.

In numbers, 100 steps of 1 m in random directions leave a walker about 10 m from the start, and 10 000 steps about 100 m. Figure 9.2 shows 500 such walks of 100 steps. Over 100 000 walks the root-mean-square distance comes out as 9.98 steps.

The same walk is diffusion, Brownian motion, the conduction of heat and the friction inside a gas. In astrophysics it is the slow escape of light from the Sun (Chapter 11), of neutrinos from a collapsing stellar core, and of cosmic rays from the Galaxy's magnetic field. It gives the $\sqrt{N}$ law and the flux that flows down any gradient.

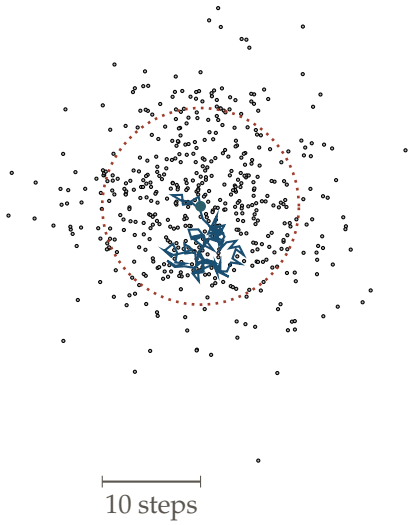


Figure 9.2: End points of 500 random walks of 100 unit steps each in the plane (grey dots), all starting at the centre (dark dot). One of the walks is drawn in full (blue). The dotted circle has a radius of 10 steps.

The $\sqrt{N}$ law. Let a particle take N steps $\vec{s}_1, \ldots, \vec{s}_N$, each of length s and each in a random direction independent of the others. Its displacement is $\vec{R} = \sum_i \vec{s}_i$. Averaging over many walks,

$$\begin{aligned} \langle R^2 \rangle &\overset{(1)}{=} \sum_{i=1}^{N} \sum_{j=1}^{N} \langle \vec{s}_i \cdot \vec{s}_j \rangle \\ &\overset{(2)}{=} \sum_{i=1}^{N} \langle s_i^2 \rangle + \sum_{i \neq j} \langle \vec{s}_i \cdot \vec{s}_j \rangle \\ &\overset{(3)}{=} N s^2 . \end{aligned} \tag{9.35}$$

(1) $R^2 = \vec{R} \cdot \vec{R}$, with each $\vec{R}$ written as its sum of steps.
(2) Split the double sum into the terms with $i = j$ and the rest.
(3) Each step has length s. Two different steps point in independent random directions, so their dot product is as often negative as positive and averages to zero. This is the same step that makes the variances of independent noise sources add (Eq. (7.10)).

$$\boxed{R_{\text{rms}} = \sqrt{N}\, s.} \tag{9.36}$$

Real free paths vary in length. For the usual exponential spread of free paths with mean s, the average of s_i^2 is $2s^2$, which changes only the factor in front, not the $\sqrt{N}$.

To cover a distance D the walk needs $N = (D/s)^2$ steps. At speed v each step takes s/v, so the time is

$$t = N \frac{s}{v} = \frac{D^2}{s\, v}. \tag{9.37}$$

The time grows as D^2. Twice the distance takes four times as long. Compared with flying straight across, the walk is slower by D/s, the number of free paths across the distance.

Flux down a gradient. Let the number density $n(x)$ of the walkers change along x, and look at the particles crossing the plane at x. A particle crossing it at an angle θ to the x-axis last

scattered, on average, one free path back along its flight, at $x - s\cos\theta$. It therefore brings the density found there. Particles fly in all directions equally, a fraction $d\Omega/4\pi$ of them within a solid angle $d\Omega$, so the net number crossing per unit area per unit time is

$$\begin{aligned} J &\overset{(1)}{=} \int \frac{d\Omega}{4\pi}\, v\cos\theta\; n(x - s\cos\theta) \\ &\overset{(2)}{=} \int \frac{d\Omega}{4\pi}\, v\cos\theta \left[n(x) - s\cos\theta\, \frac{\mathrm{d}n}{\mathrm{d}x} \right] \\ &\overset{(3)}{=} -v\, s\, \frac{\mathrm{d}n}{\mathrm{d}x} \int \frac{d\Omega}{4\pi} \cos^2\theta \\ &\overset{(4)}{=} -\frac{1}{3}\, v\, s\, \frac{\mathrm{d}n}{\mathrm{d}x}. \end{aligned} \tag{9.38}$$

(1) Particles moving at angle θ cross the plane at the rate $v\cos\theta$ per unit density, positive for those moving toward $+x$.
(2) A Taylor expansion of n over one free path.
(3) $\int \cos\theta\, d\Omega = 0$, so the uniform part $n(x)$ carries no net flux.
(4) The average of $\cos^2\theta$ over all directions is $\frac{1}{3}$, as in the pressure integral (9.20).

In three dimensions the same holds along each axis,

$$\boxed{\vec{J} = -\frac{1}{3}\, v\, s\, \nabla n.} \tag{9.39}$$

Particles flow from where there are many to where there are few. No force is needed, only random motion and a difference in numbers. The flow is larger when the steps are longer. For photons inside a star, with the energy density aT^4 in place of n, $v = c$ and the mean free path in place of s, this becomes the flow of energy out of the star (Section 11.4).

9.6 Compare Two Scales (P6)

Chapter 5 decided whether two stars are resolved by comparing their separation with λ/D (Eq. (5.53)). Chapter 8 chose between Rayleigh, Mie and geometric scattering by comparing the size of a grain with the wavelength (Table 8.2).

Each time, a single ratio decided which physics mattered. A star has many competing effects at once, and most of them are negligible in any given place. Knowing which ones can be dropped is most of the work.

Think of crossing a road. If you can cross before the car reaches you, you are safe, however fast the car is. If the car arrives first, you are not. Only the ratio of the two times matters, and when it is close to one, small changes decide.

In a gas cloud, sound tries to smooth out a squeeze while gravity tries to deepen it. Whichever acts faster wins. Every regime in astrophysics is chosen in this way. That is principle P6.

The smallest example uses only gravity and density. The constant G has units $\mathrm{m^3\,kg^{-1}\,s^{-2}}$ and a density ρ has units $\mathrm{kg\,m^{-3}}$. Their product $G\rho$ has units $\mathrm{s^{-2}}$, so $(G\rho)^{-1/2}$ is the only time that can be built from them. For the Earth's mean density, $5514\,\mathrm{kg\,m^{-3}}$, it is 27 minutes. Any process driven by gravity alone must run on a time of this order.

A ratio of exactly this kind appears in fluid mechanics as the Reynolds number and the Mach number, in kinetic theory as the Knudsen number, and in the chapters before this one as the optical depth τ (Chapters 4 and 8). It gives a quick way to find the right ratio, by its units, and a rule for reading it.

A time from units, checked against Kepler. The orbit of a satellite that skims the surface of a uniform sphere of density ρ and radius a gives the same time. Kepler's third law, Eq. (6.49), sets the orbital angular speed to $\sqrt{GM/a^3}$, so the period is

$$\begin{aligned} \text{period} &\overset{(1)}{=} 2\pi\sqrt{\frac{a^3}{GM}} \\ &\overset{(2)}{=} 2\pi\sqrt{\frac{a^3}{G\,\frac{4}{3}\pi\rho\,a^3}} \\ &= \sqrt{\frac{3\pi}{G\rho}}. \end{aligned} \tag{9.40}$$

(1) Eq. (6.49), one full turn at the angular speed $\sqrt{GM/a^3}$.
(2) The mass of a uniform sphere, $M = \frac{4}{3}\pi\rho a^3$.

The radius has dropped out. The period depends on the density alone, and it is $(G\rho)^{-1/2}$ times the number $\sqrt{3\pi} \approx 3.1$. For the Earth it is 84 minutes, the period of a satellite skimming the surface. Real satellites, 400 km up, take about 92 minutes. The units had already said that only the number in front could be missing.

Reading a ratio. A ratio far from one means that one of the two effects can be dropped, and the physics simplifies. A ratio near one means that both matter, and something changes there, often an instability. The ratios that decide the regime in stars and gas are listed in Table 9.2, with the place where each is used.

Ratio	Small	Large	Where
size / mean free path of light	transparent	opaque	Ch. 11
sound-crossing time / free-fall time	stable	collapses	Sec. 10.4
free-fall time / Kelvin–Helmholtz time	slow evolution	–	Sec. 10.5
particle spacing / quantum wavelength	degenerate	classical	Sec. 12.3
pressure of light / pressure of gas	gas holds the star	light holds the star	Ch. 12
luminosity / Eddington luminosity	bound	blown apart	Sec. 11.5
GM/Rc^2	Newtonian gravity	general relativity	Ch. 13

Table 9.2: Ratios that decide the regime, and where each is used.

9.7 Problem Set: Six Principles

9.7.1 Problem 1: Weight in a Well

Problem Statement: Model the Earth as a uniform sphere of radius $R = 6371$ km with surface gravity $9.81\ \mathrm{m\,s^{-2}}$.

(a) Find the gravity at a depth d below the surface, and at a height h above it.

(b) At what height above the surface is gravity as weak as it is 1000 km below the surface?

Essential Ideas:

- Inside, only the mass within radius r pulls (Eq. (9.12)), and for uniform density it is $M(r/R)^3$.
- Outside, the whole mass pulls as a point at the centre.

Detailed Solutions:

(a) **Inside and outside**. With $g_0 = GM/R^2$,

$$\boxed{g(R-d) = g_0\left(1 - \frac{d}{R}\right), \qquad g(R+h) = \frac{g_0}{(1+h/R)^2}.}$$

(b) **Equal gravity**. At $d = 1000\,\text{km}$, $g = g_0(1 - 1000/6371) = 0.843\,g_0$. Setting $(1+h/R)^{-2} = 0.843$ gives $1 + h/R = 1.089$.

$$\boxed{h \simeq 0.089\,R \simeq 570\,\text{km}.}$$

Gravity falls faster going up than going down. Going down, the loss of mass below is partly made up by coming closer to the centre. Going up, nothing makes up for the greater distance.

9.7.2 Problem 2: Light Against Gas

Problem Statement: Compare the pressure of light with the pressure of gas.

(a) In a room at 293 K with $n = 2.50 \times 10^{25}\,\text{m}^{-3}$.

(b) At the centre of the Sun, where $\rho = 1.5 \times 10^5\,\text{kg m}^{-3}$, $T = 1.57 \times 10^7\,\text{K}$, and the ionised gas has a mean mass per particle of $0.62\,m_\text{H}$.

(c) How does the ratio scale with T and n?

Essential Ideas:

- Gas pressure nk_BT (Eq. (9.23)), light pressure $aT^4/3$ (Eq. (9.25)).
- The ratio is a P6 ratio, $aT^3/(3nk_B)$.

Detailed Solutions:

(a) **Room**. $P_\text{gas} = 1.01 \times 10^5\,\text{Pa}$ and $P_\text{light} = 7.566 \times 10^{-16} \times 293^4/3 = 1.86 \times 10^{-6}\,\text{Pa}$.

$$\boxed{P_\text{light}/P_\text{gas} \simeq 1.8 \times 10^{-11}.}$$

(b) **Centre of the Sun**. $n = \rho/(0.62\,m_\text{H}) = 1.45 \times 10^{32}\,\text{m}^{-3}$, so $P_\text{gas} = nk_BT = 3.1 \times 10^{16}\,\text{Pa}$. $P_\text{light} = aT^4/3 = 1.5 \times 10^{13}\,\text{Pa}$.

$$\boxed{P_\text{light}/P_\text{gas} \simeq 5 \times 10^{-4}.}$$

(c) **Scaling**. $P_\text{light}/P_\text{gas} = aT^3/(3nk_B)$ grows as T^3/n. Hot, thin gas is held up by light. This is why light matters most in the most massive stars, which are hotter inside but not denser.

9.7.3 Problem 3: A Two-Level Atom

Problem Statement: Collisions move atoms between a ground level A and an excited level B that lies 0.010 eV higher. Both levels are single states.

(a) In gas at 100 K, what is N_B/N_A in equilibrium?

(b) If the downward rate is $k_{BA} = 10^{-3}\,\text{s}^{-1}$, what is the upward rate, and how fast is equilibrium reached?

Essential Ideas:

- Detailed balance, Eq. (9.34), fixes the ratio of the two rates.
- The relaxation time is $1/(k_{AB} + k_{BA})$, Eq. (9.33).

Detailed Solutions:

(a) **Populations**. The gap in temperature units is $0.010\,\mathrm{eV}/k_B = 116\,\mathrm{K}$.

$$\boxed{N_B/N_A = e^{-116/100} = 0.31.}$$

(b) **Rates**. $k_{AB} = 0.31\,k_{BA} = 3.1 \times 10^{-4}\,\mathrm{s}^{-1}$, and the relaxation time is $1/(1.31 \times 10^{-3}\,\mathrm{s}^{-1})$.

$$\boxed{k_{AB} = 3.1 \times 10^{-4}\,\mathrm{s}^{-1}, \qquad \text{relaxation time} \simeq 760\,\mathrm{s}.}$$

9.7.4 Problem 4: Perfume in Still Air

Problem Statement: Air molecules move at about $v = 400\,\mathrm{m\,s}^{-1}$ with a mean free path $s = 6.8 \times 10^{-8}\,\mathrm{m}$.

(a) How long does a perfume molecule take to cross a 3 m room by diffusion alone?

(b) How long would it take flying straight?

Essential Ideas:

- The random-walk time $D^2/(sv)$, Eq. (9.37).
- The ratio of the two times is D/s, a P6 ratio.

Detailed Solutions:

(a) **Diffusion**. $t = (3)^2/(6.8 \times 10^{-8} \times 400)\,\mathrm{s} = 3.3 \times 10^5\,\mathrm{s}$.

$$\boxed{t \simeq 4\ \text{days}.}$$

(b) **Straight flight**. $3/400\,\mathrm{s} = 7.5\,\mathrm{ms}$. The ratio of the two times is $D/s = 4.4 \times 10^7$. In a real room, air currents carry the scent across in minutes, so the bulk motion of the air beats diffusion by far. Inside stars the same competition between carrying energy by motion and by diffusion decides where a star convects (Section 12.2).

9.7.5 Problem 5: Gravity's Clock

Problem Statement: Evaluate $(G\rho)^{-1/2}$ for the Sun (mean density $1410\,\mathrm{kg\,m}^{-3}$) and for a white dwarf (mean density $10^9\,\mathrm{kg\,m}^{-3}$). What does each number say about how quickly the object can respond to a disturbance?

Essential Ideas:

- $(G\rho)^{-1/2}$ is the only time made from G and ρ (Section 9.6).

Detailed Solutions:

$$\boxed{\text{Sun: } (G\rho)^{-1/2} = 3.3 \times 10^3\,\mathrm{s} \simeq 54\,\mathrm{min}, \qquad \text{white dwarf: } 3.9\,\mathrm{s}.}$$

A star responds to a change in its own balance of forces within times of this order. The Sun readjusts within an hour, a white dwarf within seconds. Both are far shorter than the times over which their heat and fuel change, which is why both can be treated as being in balance at every moment (Section 10.5).

Chapter 10

Self-Gravitating Gas

A cold cloud of molecular gas in the Galaxy is tens of parsecs across and about 10 K (Table 8.3). Some such clouds sit unchanged for millions of years. Others break up into dense cores that shrink, heat up and become stars. Three things decide which. Gravity pulls every part of the cloud toward every other part. Pressure pushes back. Rotation, small at first, resists being squeezed sideways.

Pressure is not one thing. In a star it is a sum,

$$P = P_{\text{gas}} + P_{\text{light}} + P_{\text{degenerate}} + \cdots ,$$

and each term is one case of counting states (P3, Section 9.3). Gas pressure holds up the Sun over most of its interior. Light carries a growing share in more massive stars (Problem 9.7.2). Degenerate electrons hold up white dwarfs. In a planet the pressure comes instead from compressed liquid and solid, each with its own equation of state.

Two questions run through what follows. When does gravity win, and why does a body held together by gravity get hotter when it loses energy?

10.1 Hydrostatic Equilibrium

Chapter 9 gave the Euler equation for any moving gas (P1, Eq. (9.7)) and the pull of a sphere on its own parts (P2, Eq. (9.12)).

A star that shines for billions of years is not speeding up or slowing down anywhere inside. The full Euler equation carries terms for motion that such a star does not need.

Think of a pile of books. The bottom book carries the weight of every book above it. The one in the middle carries half the pile. A diver feels the same thing in water. Every ten metres of depth adds the weight of ten metres of water above each square metre, about one atmosphere.

Inside a star each layer holds up all the layers above it, so the pressure must grow toward the centre. At every radius the pressure difference across a thin layer balances the weight of that layer. This is momentum conservation (P1) for gas at rest, with the gravity of a sphere (P2).

The diver in numbers. Water has $\rho = 1000\,\text{kg}\,\text{m}^{-3}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, so the pressure grows by $\rho g = 9800\,\text{Pa}$ per metre, and one atmosphere, $1.01 \times 10^5\,\text{Pa}$, every 10.3 m.

The picture is a thin slab of gas with more pressure pushing up on its bottom face than down on its top face. The difference is exactly its weight.

The same balance sets how fast the air thins with height, how thick a disk of gas around a star is, and how much mass a galaxy cluster must have to hold its hot gas (Problem 10.7.2). It gives the pressure gradient at every radius of a star.

The equation. Set $\vec{v} = 0$ in the Euler equation (9.7) and use the gravity of a sphere (9.12). Only the radial component survives,

$$0 \overset{(1)}{=} -\nabla P + \rho\,\vec{g} \overset{(2)}{=} -\frac{\mathrm{d}P}{\mathrm{d}r}\,\hat{r} - \rho\,\frac{G\,m(r)}{r^2}\,\hat{r}. \tag{10.1}$$

(1) Eq. (9.7) with $\vec{v} = 0$ everywhere and at all times.
(2) Spherical symmetry, so P depends only on r, and $\vec{g}$ from Eq. (9.12).

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{G\,m(r)\,\rho}{r^2}.} \tag{10.2}$$

This is **hydrostatic equilibrium**. It holds for any star at rest, whatever its equation of state and whatever carries its energy. It does not say what P is made of.

How large is the central pressure? Replace the derivative by the drop of the pressure from the centre to the surface, $-P_c/R$, and put the whole mass at a typical radius R. Then $P_c \sim GM^2/R^4$. For the Sun this is 1.1×10^{15} Pa. The detailed models of Chapter 12 give 2.5×10^{16} Pa, larger because the Sun's mass is concentrated toward its centre. The estimate gets the size right to within a factor of a few without any knowledge of what the Sun is made of.

An atmosphere. Near the surface of a planet, $m(r)$ and r hardly change, so Gm/r^2 is the constant surface gravity g. If the air is an ideal gas at one temperature T, with mean mass μm_{H} per particle, Eq. (9.23) gives $\rho = P\,\mu m_{\mathrm{H}}/(k_B T)$. With height z in place of r,

$$\frac{\mathrm{d}P}{\mathrm{d}z} \overset{(1)}{=} -\rho g \overset{(2)}{=} -\frac{\mu m_{\mathrm{H}}\,g}{k_B T}\,P, \tag{10.3}$$

(1) Eq. (10.2) with $Gm/r^2 = g$ held constant.
(2) The ideal-gas law solved for ρ.

so the pressure falls exponentially,

$$P(z) = P(0)\,e^{-z/H}, \qquad H = \frac{k_B T}{\mu m_{\mathrm{H}}\,g}. \tag{10.4}$$

The **scale height** H is the height over which the pressure falls by a factor e. For the Earth's air ($T = 288$ K, mean molecular mass 28.97 atomic mass units) it is 8.4 km. A hotter or lighter gas reaches higher.

Planets, and why their interiors are layered. A planet obeys Eq. (10.2) just as a star does, but its pressure comes from rock, ice and liquid rather than from hot gas and light. Gravity does not change these materials directly. It compresses them. Going down, the pressure grows as Eq. (10.2) demands, and usually the temperature grows with it, so each depth sits at a different point (P, T) of the material's phase diagram. Where that point crosses a phase boundary, the stable form of the material changes. Rock turns into denser mineral structures, and in the giant planets hydrogen is squeezed into a dense, metallic state. A layered interior is hydrostatic equilibrium plus a phase diagram.

10.2 The Virial Theorem

Hydrostatic equilibrium gives the pressure at every radius, but only once the density at every radius is known. That needs the equation of state and the way energy flows, which is the work of Chapter 12.

Often the profile is not wanted at all. The questions are simpler. How hot is a star, on average? How much energy does it hold? What happens to it when it loses some?

Think of a satellite in a circular orbit. The lower its orbit, the faster it moves. A satellite that skims the air feels a little drag, loses energy, and sinks to a lower orbit, where it moves faster. Losing energy has sped it up. The station at 400 km moves at $7.67\,\mathrm{km\,s^{-1}}$. Dropped to 300 km it would move at $7.73\,\mathrm{km\,s^{-1}}$.

The particles of a star are like satellites in the star's own gravity. Averaged over the whole star, their kinetic energy is tied to the depth of the gravitational well in the same way. Since the kinetic energy of a gas measures its temperature, the depth of the well sets the temperature. This is the virial theorem. It is hydrostatic equilibrium (P1 and P2) added up over the whole star, read with P3.

The satellite in symbols. Gravity supplies the centripetal force, $mv^2/r = GMm/r^2$. So the kinetic energy is $K = \frac{1}{2}mv^2 = GMm/(2r)$ and the potential energy is $U = -GMm/r$. Then $2K + U = 0$, and the total energy is $E = K + U = -K$, the result $E = -GM\mu/2a$ of Eq. (6.33) for a circular orbit. Lowering E raises K.

The picture is a bowl. The deeper the bowl, the faster everything in it has to move to stay in it.

The same theorem is Clausius's virial of a gas. Zwicky applied it to the galaxies of the Coma cluster and found that they move far too fast for the mass that shines, the first evidence of dark matter. It turns the speeds of stars into the mass of a galaxy, and the temperature of the hot gas in a cluster into the cluster's mass. Here it gives the theorem for a gas sphere by two routes, the temperature of a star, and how firmly a star is bound.

Route A, from hydrostatic equilibrium. Multiply Eq. (10.2) by $4\pi r^3$ and integrate from the centre to the surface R. On the left,

$$\int_0^R 4\pi r^3 \frac{\mathrm{d}P}{\mathrm{d}r}\,dr \overset{(1)}{=} \Big[4\pi r^3 P\Big]_0^R - 3\int_0^R 4\pi r^2 P\,dr \overset{(2)}{=} -3\int P\,dV. \tag{10.5}$$

(1) Integration by parts.
(2) At the centre $r^3 = 0$, and at the surface $P = 0$. The volume element is $dV = 4\pi r^2 dr$.

On the right,

$$\int_0^R 4\pi r^3 \left(-\frac{Gm\rho}{r^2}\right) dr \overset{(1)}{=} -\int_0^M \frac{Gm}{r}\,dm \overset{(2)}{=} U. \tag{10.6}$$

(1) $4\pi r^2 \rho\,dr = dm$, Eq. (9.10).
(2) The gravitational energy, Eq. (9.14).

Setting the two equal,

$$\boxed{3\int P\,dV + U = 0.} \tag{10.7}$$

Only gravity and the pressure enter, so this holds for any equation of state. It is hydrostatic equilibrium weighted by r and summed over the star. The detail at each radius is lost, and the overall balance is kept.

Route B, for a gas that moves. Route A assumed the gas is at rest. To see what happens when it is not, follow the second moment of the mass, $I \equiv \int r^2\, dm$, where each dm moves with the gas. Differentiating twice,

$$\begin{aligned}\frac{1}{2}\frac{\mathrm{d}^2 I}{\mathrm{d}t^2} &\overset{(1)}{=} \int v^2\, dm + \int \vec{r}\cdot\frac{\mathrm{d}\vec{v}}{\mathrm{d}t}\, dm \\ &\overset{(2)}{=} 2K_{\text{bulk}} + \int \vec{r}\cdot\left(-\nabla P + \rho\,\vec{g}\right) dV \\ &\overset{(3)}{=} 2K_{\text{bulk}} + 3\int P\, dV + U. \end{aligned} \tag{10.8}$$

(1) $\mathrm{d}(r^2)/\mathrm{d}t = 2\vec{r}\cdot\vec{v}$, then once more. The mass elements do not change (P1).

(2) $K_{\text{bulk}} = \frac{1}{2}\int v^2\, dm$ is the kinetic energy of bulk motion. The Euler equation (9.7) gives $\rho\ \mathrm{d}\vec{v}/\mathrm{d}t$.

(3) $-\int \vec{r}\cdot\nabla P\, dV = 3\int P\, dV$ by integration by parts, as in Route A. For a sphere, $\vec{r}\cdot\vec{g} = -Gm/r$ by Eq. (9.12), so the gravity term is U as in Route A. (For any shape the same result follows by writing U as a double sum over pairs of mass elements.)

For a gas of slow particles, $\gamma = \frac{5}{3}$ in Eq. (9.21), so $\int P\, dV = \frac{2}{3}K_{\text{th}}$, where K_{th} is the kinetic energy of the random thermal motion. With the total kinetic energy $K = K_{\text{bulk}} + K_{\text{th}}$,

$$\boxed{\frac{1}{2}\frac{\mathrm{d}^2 I}{\mathrm{d}t^2} = 2K + U.} \tag{10.9}$$

A cloud that neither grows nor shrinks on average has $\langle \ddot{I} \rangle = 0$, and $2K + U = 0$, which is Route A again. If $2K < |U|$, then $\ddot{I} < 0$. The cloud's size accelerates inward and it collapses. If $2K > |U|$, it disperses.

Reading it, gravity sets the temperature. Let the gas be ideal with mean mass μm_{H} per particle, where μ is the **mean molecular weight**, so that the number density is $n = \rho/(\mu m_{\text{H}})$. Then

$$\int P\, dV = \int n k_B T\, dV = \frac{k_B}{\mu m_{\text{H}}}\int T\, dm \equiv \frac{M k_B \bar{T}}{\mu m_{\text{H}}},$$

which defines $\bar{T}$, the temperature averaged over the mass. With Eq. (10.7),

$$\boxed{k_B \bar{T} = \frac{\mu m_{\text{H}}\,|U|}{3M}.} \tag{10.10}$$

The average temperature of a star is set by the depth of its gravitational well per unit mass, and by nothing else. The source of its energy does not appear.

A uniform sphere. To put numbers in, take a sphere of uniform density. The mass inside r is $m = M(r/R)^3$, so $dm = 3Mr^2dr/R^3$, and Eq. (9.14) gives

$$\begin{aligned} U &\overset{(1)}{=} -\int_0^R \frac{G\,M(r/R)^3}{r}\,\frac{3Mr^2}{R^3}\,dr \\ &\overset{(2)}{=} -\frac{3GM^2}{R^6}\int_0^R r^4\,dr \\ &= -\frac{3}{5}\,\frac{GM^2}{R}. \end{aligned} \tag{10.11}$$

(1) Eq. (9.14) with $m = M(r/R)^3$ and $dm = 3Mr^2dr/R^3$.
(2) Collect the powers of r and R.

Then Eq. (10.10) becomes $k_B\bar{T} = GM\mu m_{\mathrm{H}}/(5R)$. For the Sun, with $\mu = 0.6$ for its ionised gas, this is $\bar{T} = 2.8 \times 10^6$ K. The real Sun is denser toward the middle, so its well is deeper and its gas hotter. Its centre is at 1.57×10^7 K. Without knowing anything about nuclear fusion, the virial theorem says that the inside of the Sun is at millions of kelvin. That is hot enough for hydrogen nuclei to fuse (Section 12.3), which is why stars shine by fusion at all.

How firmly a star is bound. Let the gas have $P = (\gamma - 1)u$, Eq. (9.21). Its internal energy is $\int u\,dV = \int P\,dV/(\gamma - 1)$, and the total energy is

$$\begin{aligned} E &\overset{(1)}{=} \frac{1}{\gamma - 1}\int P\,dV + U \\ &\overset{(2)}{=} -\frac{U}{3(\gamma - 1)} + U \\ &= \frac{3\gamma - 4}{3(\gamma - 1)}\,U. \end{aligned} \tag{10.12}$$

(1) Internal energy plus gravitational energy, for a star at rest.
(2) The virial theorem (10.7), $\int P\,dV = -U/3$.

$$\boxed{E = \frac{3\gamma - 4}{3(\gamma - 1)}\,U.} \tag{10.13}$$

For slow particles, $\gamma = \frac{5}{3}$ and $E = U/2 < 0$. The star is bound, and taking it apart costs half its gravitational energy. As $\gamma \to \frac{4}{3}$, $E \to 0$. The star is barely bound, and a small push can make it collapse or fly apart. For $\gamma < \frac{4}{3}$ no static balance exists at all. This single line explains three later results. Stars held up mainly by light are fragile (Section 12.5). White dwarfs whose electrons move near the speed of light have a maximum mass (Section 13.2). And a collapsing protostar collapses a second time while its molecules are being broken apart (Section 10.6).

10.3 Negative Heat Capacity

By the virial theorem a star of slow particles has $E = -K$, Eq. (10.13) with $\gamma = \frac{5}{3}$.

Everyday thermodynamics says that something that loses heat gets colder. Its heat capacity, the energy needed to raise its temperature by one kelvin, is positive. A star seems to break this rule.

Think of a cup of tea left in a room. It cools until it has the temperature of the room, and then its energy wobbles a little around a fixed value as the room's air gives and takes small amounts of heat. The room decides the temperature. The rule that heat capacity is positive is a

statement about objects that sit in a room like this, with a large surrounding body that sets the temperature.

A star has no room around it. It is isolated in empty space, held together by its own gravity, and it loses energy only by radiating into space, which never sends heat back. Nothing sets its temperature from outside. Its temperature follows from its energy through the virial theorem, and that relation runs backwards. The proof that heat capacity is positive does not apply to it.

The Sun in numbers. With $\mu = 0.6$ the Sun holds $N = M/(\mu m_{\mathrm{H}}) = 2.0 \times 10^{57}$ particles, and $E = -K = -\frac{3}{2} N k_B \bar{T}$. If it shone at its present luminosity for a million years with no fusion, it would lose $\Delta E = -1.2 \times 10^{40}\,\mathrm{J}$, and its mean temperature would rise by $3 \times 10^5\,\mathrm{K}$.

The picture is a curve of temperature against energy that slopes the wrong way. Moving left along it, toward lower energy, the temperature climbs.

The same behaviour appears in the core collapse of globular star clusters and in black holes, whose temperature rises as they lose mass to Hawking radiation. It gives the sign of the heat capacity of a star, the reason the usual rule fails, and what a negative heat capacity does for a star.

The heat capacity of a star. Along a sequence of stars in virial balance, each with a slightly different energy, $E = -\frac{3}{2} N k_B \bar{T}$. So

$$\boxed{C \equiv \frac{\mathrm{d}E}{\mathrm{d}\bar{T}} = -\frac{3}{2} N k_B < 0.} \tag{10.14}$$

Losing energy by radiation, $dE < 0$, raises the temperature, $d\bar{T} = dE/C > 0$. Of the gravitational energy released by contraction, half goes into heating the gas and half is radiated away, since $\Delta K = -\Delta U/2$ and $\Delta E = \Delta U/2$.

Why the usual rule says $C \geq 0$. The usual rule belongs to the canonical ensemble, a system in contact with a heat bath at temperature T. With $\beta = 1/k_B T$ its partition function is $Z(\beta) = \sum_s e^{-\beta E_s}$, summed over its states s. The mean energy and its spread follow from Z,

$$\begin{aligned} \langle E \rangle &\overset{(1)}{=} -\frac{\partial \ln Z}{\partial \beta}, \\ \left\langle (\Delta E)^2 \right\rangle = \left\langle E^2 \right\rangle - \langle E \rangle^2 &\overset{(2)}{=} \frac{\partial^2 \ln Z}{\partial \beta^2} \overset{(3)}{=} -\frac{\partial \langle E \rangle}{\partial \beta} \overset{(4)}{=} k_B T^2 \, \frac{\partial \langle E \rangle}{\partial T}. \end{aligned} \tag{10.15}$$

(1) Differentiate $\ln Z$ once. Each state enters with weight $e^{-\beta E_s}/Z$.
(2) Differentiate once more. The second derivative of $\ln Z$ is the mean of E^2 minus the square of the mean of E.
(3) The first line.
(4) $\beta = 1/k_B T$, so $\partial/\partial\beta = -k_B T^2 \, \partial/\partial T$.

$$\boxed{C_{\mathrm{can}} = \frac{\left\langle (\Delta E)^2 \right\rangle}{k_B T^2} \geq 0.} \tag{10.16}$$

A spread in energy cannot be negative, so in contact with a bath the heat capacity cannot be negative either. This is a statement about the canonical ensemble only.

The isolated system. An isolated star has a fixed energy and belongs to the microcanonical ensemble, where the temperature is defined from the entropy $S(E)$ by $1/T = \mathrm{d}S/\mathrm{d}E$. Comparing

neighbouring equilibria with slightly different E,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}E}\left(\frac{1}{T}\right) &\overset{(1)}{=} \frac{\mathrm{d}^2 S}{\mathrm{d}E^2} \\ \Longrightarrow \quad -\frac{1}{T^2}\,\frac{\mathrm{d}T}{\mathrm{d}E} &= \frac{\mathrm{d}^2 S}{\mathrm{d}E^2} \\ \Longrightarrow \quad C_{\text{micro}} = \frac{\mathrm{d}E}{\mathrm{d}T} &\overset{(2)}{=} -\frac{1}{T^2\ \mathrm{d}^2 S/\mathrm{d}E^2}. \end{aligned} \tag{10.17}$$

(1) Differentiate the definition of T once more.
(2) Invert $\mathrm{d}T/\mathrm{d}E$.

Nothing here forbids $C < 0$. It only requires $\mathrm{d}^2 S/\mathrm{d}E^2 > 0$, an entropy that curves upward. For short-range forces, a large system can be split into parts whose energies add, and then the two ensembles agree and $S(E)$ curves downward. Gravity is long-range. The energy of a star is not the sum of the energies of its halves, because the halves attract each other. So the two ensembles need not agree, and a self-gravitating body can have $C < 0$ as an isolated system even though no system with $C < 0$ can sit in steady contact with a bath.

That last point is itself physical. Put a star in a bath slightly hotter than itself. Heat flows in, and with $C < 0$ the star cools, so even more heat flows in. Put it in a slightly cooler bath and it heats as it loses heat. Either way it runs away from the bath's temperature. A self-gravitating body cannot come into steady equilibrium with a heat bath.

What it does for a star. Suppose the nuclear burning in a core runs a little too fast. The extra energy expands the core, and by Eq. (10.14) the core cools, which slows the burning. If the burning runs too slow, the core contracts and heats until the burning catches up. Negative heat capacity makes a star its own thermostat (P4 on the energy, Section 9.4). When fuel runs out in the core, the same rule makes the core contract and heat until the next fuel ignites. Where the pressure stops depending on temperature, in degenerate matter, the thermostat breaks, and burning can run away (Section 12.3).

10.4 The Jeans Criterion

The virial theorem says when a whole cloud is in balance, $2K + U = 0$, and that it collapses when $2K < |U|$.

It does not say which parts of a large cloud collapse, or how big the pieces are.

Think of traffic on a busy road. Now and then cars bunch up a little. Drivers in a bunch slow down, the cars behind catch up, and the bunch grows. Drivers also react and spread out again. A short bunch is smoothed out before it can grow. A long one grows faster than drivers can spread it, and becomes a jam. Only bunches longer than some length grow.

A cloud behaves the same way. A region that is slightly denser than its surroundings pulls in more gas. Its higher pressure pushes back, and that push travels at the speed of sound. A small region is smoothed out by sound before gravity can act. A large region collapses before sound can cross it. The dividing size, where the sound-crossing time equals the free-fall time, is the Jeans length. This is a comparison of two scales (P6), made exact with conservation (P1) and the gravity of the gas (P2).

A dense core in numbers. Molecular gas at 10 K with $n = 10^{10}\,\mathrm{m}^{-3}$ ($10^4\,\mathrm{cm}^{-3}$) and $\mu = 2.33$ has the sound speed $c_s = \sqrt{k_B T/(\mu m_{\mathrm{H}})} = 188\,\mathrm{m\,s}^{-1}$ and the free-fall time $t_{\mathrm{ff}} = 3.4 \times 10^5$ years (Section 10.5). In that time sound covers $c_s t_{\mathrm{ff}} = 0.065\,\mathrm{pc}$. Regions much larger than this cannot be held up.

The picture is Figure 10.1. Short waves oscillate as ordinary sound. Long waves do not oscillate at all. They grow.

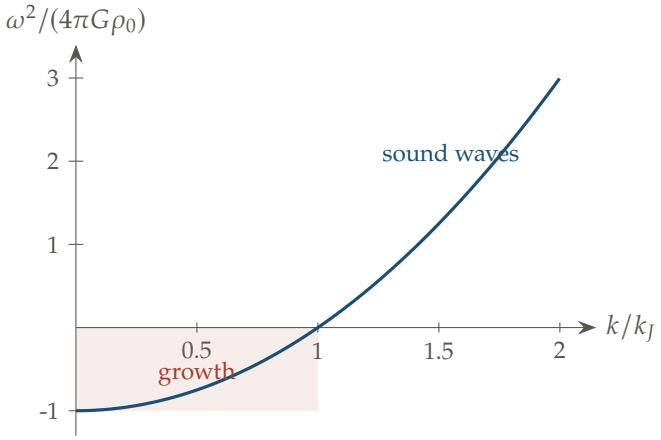


Figure 10.1: The squared frequency ω^2 of a small wave in a uniform self-gravitating gas against its wavenumber k, in units of $4\pi G\rho_0$ and of k_J. Below $k = k_J$ (shaded) $\omega^2 < 0$.

The same instability, with the expansion of the universe added, grows the galaxies and clusters out of the nearly smooth early universe. In a rotating disk it becomes the Toomre criterion for spiral arms and clumps. It gives the critical mass and length, by energy and by waves.

By energy. Take a uniform sphere of mass M, radius R and temperature T. Its thermal energy is $K = \frac{3}{2}Nk_BT = \frac{3}{2}Mk_BT/(\mu m_\mathrm{H})$ (Eq. (9.23)), and its gravitational energy is Eq. (10.11). The marginal case of Eq. (10.9), $2K = |U|$, gives

$$3\,\frac{Mk_BT}{\mu m_\mathrm{H}} = \frac{3}{5}\,\frac{GM^2}{R}$$
$$\Longrightarrow \qquad R = \frac{GM\mu m_\mathrm{H}}{5k_BT}. \tag{10.18}$$

Writing $M = \frac{4}{3}\pi R^3\rho$ and eliminating R or M gives the **Jeans mass** and **Jeans length**,

$$\boxed{M_J = \left(\frac{5k_BT}{G\mu m_\mathrm{H}}\right)^{3/2}\left(\frac{3}{4\pi\rho}\right)^{1/2}, \qquad R_J = \left(\frac{15\,k_BT}{4\pi G\mu m_\mathrm{H}\rho}\right)^{1/2}.} \tag{10.19}$$

A cloud with $M > M_J$ cannot be held up by its thermal pressure. For the dense core above, $M_J = 5.4\,M_\odot$ and $R_J = 0.13\,\mathrm{pc}$. Hot, thin gas is stable. Cold, dense gas is not, which is why stars form in cold molecular clouds rather than in the warm diffuse gas.

By waves. Take an infinite gas at rest with uniform density ρ_0 and one temperature, so that a small change of density ρ_1 changes the pressure by $c_s^2\rho_1$. Disturb it slightly, $\rho = \rho_0 + \rho_1$, $\vec{v} = \vec{v}_1$, $\Phi = \Phi_0 + \Phi_1$. To first order in the small quantities, continuity (9.2), Euler (9.7) and Poisson (9.13) read

$$\frac{\partial\rho_1}{\partial t} + \rho_0\nabla\cdot\vec{v}_1 = 0, \qquad \rho_0\frac{\partial\vec{v}_1}{\partial t} = -c_s^2\nabla\rho_1 - \rho_0\nabla\Phi_1, \qquad \nabla^2\Phi_1 = 4\pi G\rho_1. \tag{10.20}$$

(Strictly, a uniform ρ_0 cannot be at rest, because Poisson's equation then needs a Φ_0 that grows without bound. Jeans set this aside, and the result is still right for regions much smaller

than the cloud.) For a wave $\propto e^{i(kx-\omega t)}$ along x, with $\vec{v}_1 = v_1\hat{x}$, each derivative becomes a factor, $\partial_t \to -i\omega$ and $\partial_x \to ik$. The three equations become one matrix equation for the three amplitudes,

$$\begin{pmatrix} -i\omega & i\rho_0 k & 0 \\ ic_s^2 k & -i\omega\rho_0 & i\rho_0 k \\ 4\pi G & 0 & k^2 \end{pmatrix} \begin{pmatrix} \rho_1 \\ v_1 \\ \Phi_1 \end{pmatrix} = 0. \tag{10.21}$$

A wave exists only if the determinant vanishes,

$$\begin{aligned} 0 &\overset{(1)}{=} -i\omega\big(-i\omega\rho_0 k^2\big) - i\rho_0 k\big(ic_s^2 k^3 - i\rho_0 k \cdot 4\pi G\big) \\ &\overset{(2)}{=} \rho_0 k^2\big(c_s^2 k^2 - 4\pi G\rho_0 - \omega^2\big). \end{aligned} \tag{10.22}$$

(1) Expand along the first row.
(2) Multiply out, using $i^2 = -1$, and collect.

$$\boxed{\omega^2 = c_s^2 k^2 - 4\pi G\rho_0.} \tag{10.23}$$

Short waves, with large k, have $\omega^2 > 0$ and oscillate as sound. Long waves with k below $k_J = \sqrt{4\pi G\rho_0}/c_s$ have $\omega^2 < 0$, so ω is imaginary and the disturbance grows exponentially instead of oscillating. The critical wavelength is

$$\lambda_J = \frac{2\pi}{k_J} = c_s\sqrt{\frac{\pi}{G\rho_0}}. \tag{10.24}$$

It scales with T and ρ exactly as R_J does, and differs only in the number in front, because one route treats a finite sphere and the other an infinite medium. Both say $\lambda_J \approx c_s\, t_{\rm ff}$. With $t_{\rm ff}$ from Eq. (10.28) the ratio is $\sqrt{32/3} = 3.3$.

Why stars form in groups. A giant molecular cloud holds 10^5 to $10^6\,M_\odot$ (Table 8.3), many thousands of Jeans masses. It does not collapse into one star. While it collapses it stays thin enough to radiate away the heat of compression, so its temperature stays near 10 K. By Eq. (10.19), $M_J \propto T^{3/2}\rho^{-1/2}$ then falls as the density rises, so parts that were stable become unstable on their own. The cloud breaks into clumps, the clumps into cores, and each core into stars. This is why most stars are born in clusters and in pairs, although many clusters later drift apart.

10.5 Free Fall and the Three Clocks

The Jeans criterion says when a region collapses.

It does not say how fast.

Imagine a hole drilled straight through a uniform Earth. A ball dropped into it reaches the centre after 21 minutes, whatever the hole's length. The time depends only on the Earth's density, as the units of $G\rho$ demand (Section 9.6).

A cloud whose pressure can no longer resist gravity falls in on itself in the same way. Each shell falls toward the centre pulled by the mass inside it. The time to reach the centre depends only on the density, so every shell of a uniform cloud arrives at the same moment. This is conservation of energy (P1) for each shell, with the gravity of a sphere (P2).

A uniform cloud with the Earth's mean density would collapse in 15 minutes. The Sun's mean density gives 30 minutes. A dense core of a molecular cloud takes 3.4×10^5 years.

The picture is Figure 10.2. The radius falls slowly at first and then plunges to zero.

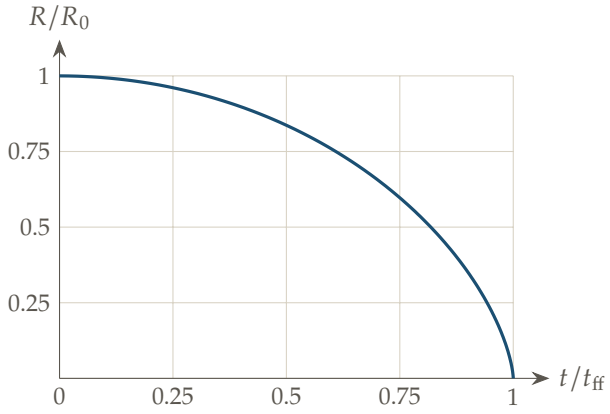


Figure 10.2: The radius of a shell in a uniform, pressure-free sphere that starts at rest, against time in units of the free-fall time.

The same collapse is a Kepler orbit with no angular momentum (Chapter 6). In cosmology it is the collapse of a uniform overdense sphere into a galaxy halo, and in a massive star it sets how fast the core falls in at the start of a supernova. It gives the free-fall time and the three clocks of a star.

The fall of one shell. Take a uniform cloud of density ρ_0 at rest, with no pressure. Follow a shell that starts at radius R_0, with mass $M = \frac{4}{3}\pi R_0^3 \rho_0$ inside it. While the shells do not cross, M stays the same, and by Eq. (9.12)

$$\frac{\mathrm{d}^2 R}{\mathrm{d}t^2} = -\frac{GM}{R^2}. \tag{10.25}$$

Multiplying by $\mathrm{d}R/\mathrm{d}t$ and integrating from rest at R_0 gives the energy of the shell per unit mass,

$$\begin{aligned} &\frac{1}{2}\left(\frac{\mathrm{d}R}{\mathrm{d}t}\right)^2 - \frac{GM}{R} = -\frac{GM}{R_0} \\ \Longrightarrow \quad &\left(\frac{\mathrm{d}R}{\mathrm{d}t}\right)^2 = \frac{2GM}{R_0}\left(\frac{R_0}{R} - 1\right). \end{aligned} \tag{10.26}$$

The substitution $R = R_0 \cos^2\xi$, with ξ running from 0 at the start to $\pi/2$ at the centre, turns the bracket into $\tan^2\xi$,

$$\begin{aligned} &\left(-2R_0 \cos\xi \sin\xi \,\frac{\mathrm{d}\xi}{\mathrm{d}t}\right)^2 \overset{(1)}{=} \frac{2GM}{R_0}\,\frac{\sin^2\xi}{\cos^2\xi} \\ \Longrightarrow \quad &\cos^2\xi \,\frac{\mathrm{d}\xi}{\mathrm{d}t} \overset{(2)}{=} \sqrt{\frac{GM}{2R_0^3}} \\ \Longrightarrow \quad &\xi + \sin\xi\cos\xi \overset{(3)}{=} \sqrt{\frac{2GM}{R_0^3}}\; t. \end{aligned} \tag{10.27}$$

(1) $\mathrm{d}R/\mathrm{d}t = -2R_0 \cos\xi \sin\xi \;\mathrm{d}\xi/\mathrm{d}t$, and $R_0/R - 1 = 1/\cos^2\xi - 1 = \tan^2\xi$.
(2) Divide both sides by $4R_0^2 \sin^2\xi$ and take the positive root, since ξ grows.
(3) $\int_0^{\xi} 2\cos^2\xi'\, d\xi' = \xi + \sin\xi\cos\xi$, with $\xi = 0$ at $t = 0$.

The shell reaches the centre at $\xi = \pi/2$. With $M = \frac{4}{3}\pi R_0^3 \rho_0$, $\sqrt{2GM/R_0^3} = \sqrt{8\pi G\rho_0/3}$, and

$$\boxed{t_{\rm ff} = \frac{\pi/2}{\sqrt{8\pi G\rho_0/3}} = \sqrt{\frac{3\pi}{32\,G\rho_0}}.} \tag{10.28}$$

R_0 has dropped out. Every shell of a uniform cloud reaches the centre at the same time, so shells never cross, which is what the calculation assumed. The time is $(G\rho_0)^{-1/2}$ times a number, as P6 said it must be.

The same result as a Kepler orbit. A shell falling from rest is a body on a Kepler orbit of zero angular momentum. As the angular momentum shrinks to zero at fixed energy, the ellipse of Chapter 6 flattens into a straight segment from R_0 to the centre, with semi-major axis $R_0/2$. The fall from one end to the other is half an orbital period. By Eq. (6.49),

$$t_{\rm ff} = \frac{1}{2}\cdot 2\pi\sqrt{\frac{(R_0/2)^3}{GM}} = \pi\sqrt{\frac{R_0^3}{8GM}} = \sqrt{\frac{3\pi}{32\,G\rho_0}},$$

the same as Eq. (10.28).

The three clocks of a star. A star changes on three very different times.

- The **free-fall** or dynamical time, Eq. (10.28). It is how fast the star restores hydrostatic equilibrium after a disturbance. For the Sun's mean density, 1410 kg m^{-3}, it is 30 minutes.
- The **Kelvin–Helmholtz** time, $t_{\rm KH} = |U|/L$. It is how long the star could shine on its gravitational energy alone, by contracting (Section 10.3). With $|U| \sim GM^2/R$, the Sun gives $t_{\rm KH} \approx 3\times10^7$ years.
- The **nuclear** time. Fusing hydrogen into helium releases 0.7% of the rest mass, and about a tenth of the Sun's hydrogen is burned on the main sequence, so $t_{\rm nuc} \approx 0.007{\times}0.1{\times}Mc^2/L \approx 10^{10}$ years.

Figure 10.3 places them on one axis. Each is separated from the next by many orders of magnitude.

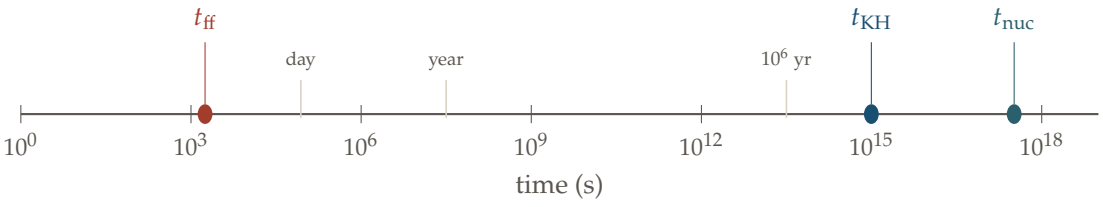


Figure 10.3: The free-fall, Kelvin–Helmholtz and nuclear times of the Sun on a logarithmic axis, with a day, a year and a million years for comparison.

This ordering is a P6 statement, and it is why a star can be described as a sequence of hydrostatic models. A star restores its balance within an hour, while its heat content changes over tens of millions of years and its composition over billions. At every moment it is in hydrostatic equilibrium, to great accuracy, and it moves slowly from one equilibrium to the next as it loses energy or burns fuel. A collapsing cloud is different. During free fall, $t_{\rm ff}$ is the only clock, the gas is not in balance at any moment, and no equilibrium description holds.

10.6 Rotation and the End of Collapse

Free fall assumed a cloud with no pressure, no rotation, and gas that radiates away every bit of heat.

None of these lasts. Real clouds turn a little, and as they become denser they become opaque to their own cooling radiation.

Think of a figure skater spinning with arms held out. When the skater pulls the arms in, the spin becomes much faster. Nothing pushed the skater. The angular momentum stayed the same while the size shrank.

A collapsing cloud does the same (P1 for angular momentum, Section 9.1). Its spin grows as it shrinks, and the outward pull of rotation grows faster than gravity. Along the axis of rotation nothing resists, so the cloud keeps falling there. Across the axis the spin stops it. The cloud flattens into a disk.

In numbers, the ratio of the rotational pull to gravity grows as $1/R$ (below). A core that shrinks from 0.1 pc to the size of the Sun raises that ratio by 4×10^6. A rotation far too slow to notice at the start becomes the main obstacle at the end.

The picture is a sphere that flattens into a spinning disk with a dense centre.

The same spin-up gives the disks in which planets form, the accretion disks around black holes, the disks of spiral galaxies and the fast spin of young neutron stars. It gives the rotation ratio, the end of the collapse, and the three meanings of equilibrium.

The rotation ratio. Let a parcel at radius R circle the axis with speed v_{rot}, so its angular momentum per unit mass is Rv_{rot}, which is conserved. The ratio of the rotational pull to gravity is

$$\begin{aligned} \frac{v_{\mathrm{rot}}^2/R}{GM/R^2} &\overset{(1)}{=} \frac{v_{\mathrm{rot}}^2 R}{GM} \\ &\overset{(2)}{=} \frac{(Rv_{\mathrm{rot}})^2}{GM}\,\frac{1}{R}. \end{aligned} \tag{10.29}$$

(1) Rotational acceleration v_{rot}^2/R divided by the gravity of the mass M inside.
(2) Group the conserved angular momentum per unit mass.

With Rv_{rot} and M fixed, the ratio grows as $1/R$. Shrinking R by some factor raises the ratio by the same factor.

The end of the isothermal collapse. While the gas can radiate away the heat of compression, its temperature stays nearly constant and the collapse goes on. Once the centre becomes dense enough to be opaque to its own cooling radiation, the heat stays in. The pressure now rises faster than the density, close to an adiabat with $\gamma = \frac{7}{5}$ for molecular hydrogen, above $\frac{4}{3}$. By Eq. (10.13) a static balance is again possible. The centre stops falling and forms the **first hydrostatic core**, a few hundredths of a solar mass, which grows by catching the gas still falling in.

The core heats as it grows. Near 2000 K the hydrogen molecules start to break apart. Breaking them takes energy that would otherwise raise the pressure, so during this phase the gas behaves as if $\gamma < \frac{4}{3}$. By Eq. (10.13) no static balance exists, and the core collapses a second time, until the molecules are gone and the gas, now atomic and soon ionised, can stop it again. That second core is the protostar.

Three kinds of equilibrium. The word equilibrium has been used in three senses, and they differ.

- **Hydrostatic equilibrium** is mechanical. The forces on each parcel balance, Eq. (10.2). A star is in it to great accuracy.
- **Global thermodynamic equilibrium** means no net flow of heat anywhere, and in Newtonian physics a single temperature throughout. A star is never in it. Its temperature falls from the centre outward, and that is why energy flows out. (In general relativity the condition is a little different, Section 13.3.)
- **Local thermodynamic equilibrium** means that each small parcel is close to equilibrium inside itself, with one temperature for its gas and its light, even though the temperature changes from parcel to parcel. This holds deep inside stars, and it is what lets the gas pressure and the light of a parcel be written with the formulas of P3. Chapters 11 and 12 rely on it.

A star can be in excellent hydrostatic equilibrium while far from global thermodynamic equilibrium, and for most of its interior local thermodynamic equilibrium sits between the two.

10.7 Problem Set: Self-Gravitating Gas

10.7.1 Problem 1: Scale Heights of Planetary Atmospheres

Problem Statement: Find the scale height of (a) the Earth's atmosphere ($T = 288\,\mathrm{K}$, mean molecular mass $28.97\,\mathrm{u}$, $g = 9.81\,\mathrm{m\,s^{-2}}$) and (b) the atmosphere of Mars ($T = 210\,\mathrm{K}$, mostly CO_2 with mean molecular mass $43.3\,\mathrm{u}$, $g = 3.71\,\mathrm{m\,s^{-2}}$). (c) Why is the scale height of Mars larger, although its air is heavier and colder?

Essential Ideas:

- Hydrostatic equilibrium with an ideal gas at one temperature gives $H = k_B T/(\mu m_{\mathrm{H}} g)$, Eq. (10.4).

Detailed Solutions:

(a) **Earth**. $H = (1.381 \times 10^{-23} \times 288)/(28.97 \times 1.661 \times 10^{-27} \times 9.81)\,\mathrm{m}$.

$$\boxed{H_\oplus \simeq 8.4\,\mathrm{km}.}$$

(b) **Mars**.

$$\boxed{H_{\mathrm{Mars}} \simeq 10.9\,\mathrm{km}.}$$

(c) **Comparison**. $H \propto T/(\mu g)$. The heavier, colder air of Mars lowers H by a factor $(210/288)(28.97/43.3) = 0.49$, but its weaker gravity raises it by $9.81/3.71 = 2.6$, which wins.

10.7.2 Problem 2: The Temperature of a Galaxy Cluster

Problem Statement: The galaxies of a rich cluster move with a line-of-sight velocity dispersion $\sigma = 1000\,\mathrm{km\,s^{-1}}$. The cluster also holds hot ionised gas ($\mu = 0.6$) in hydrostatic equilibrium in the same gravitational well. Estimate the temperature of that gas. In which band of light does it shine?

Essential Ideas:

- The galaxies and the gas feel the same well, so by the virial theorem their particles have the same kinetic energy per unit mass on average.

- For galaxies, the kinetic energy per unit mass is $\frac{3}{2}\sigma^2$ (three directions). For gas, it is $\frac{3}{2}k_BT/(\mu m_{\rm H})$.

Detailed Solutions: Setting the two equal, $k_BT = \mu m_{\rm H}\sigma^2$, so

$$\boxed{T = \frac{0.6 \times 1.673 \times 10^{-27} \times (10^6)^2}{1.381 \times 10^{-23}}\,\mathrm{K} \simeq 7 \times 10^7\,\mathrm{K}.}$$

Gas at this temperature has $k_BT \approx 6\,\mathrm{keV}$ and shines in X-rays, as clusters are observed to. Measuring this temperature and the density profile of the gas, and using Eq. (10.2), gives the mass of the cluster, which comes out far larger than the mass in stars and gas. This is the X-ray version of Zwicky's argument for dark matter.

10.7.3 Problem 3: Jeans Mass of a Dense Core

Problem Statement: For a molecular core with $\mu = 2.33$, find the Jeans mass (a) at 10 K and $n = 10^{10}\,\mathrm{m}^{-3}$, (b) at 20 K and the same density, and (c) at 10 K and $n = 10^{12}\,\mathrm{m}^{-3}$. What does (c) say about a core that collapses while staying at 10 K?

Essential Ideas:

- Eq. (10.19), with $M_J \propto T^{3/2}\rho^{-1/2}$.

Detailed Solutions:

(a) **Reference core**. $\rho = n\mu m_{\rm H} = 3.9 \times 10^{-17}\,\mathrm{kg\,m}^{-3}$ in Eq. (10.19).

$$\boxed{M_J \simeq 5.4\,M_\odot.}$$

(b) **Twice as warm**. M_J grows by $2^{3/2} = 2.8$.

$$\boxed{M_J \simeq 15\,M_\odot.}$$

(c) **A hundred times denser**. M_J falls by $\sqrt{100} = 10$.

$$\boxed{M_J \simeq 0.54\,M_\odot.}$$

A core that collapses while staying cold has a Jeans mass that keeps falling, so its parts become unstable on their own and it fragments.

10.7.4 Problem 4: The Age of the Sun Before Fusion Was Known

Problem Statement: In the nineteenth century, Kelvin and Helmholtz assumed the Sun shines by slowly contracting. Estimate how long it could have shone at its present luminosity $L_\odot = 3.83 \times 10^{26}\,\mathrm{W}$. Compare with the age of the Earth from geology and radioactivity, about 4.5×10^9 years.

Essential Ideas:

- $t_{\rm KH} = |U|/L$ with $|U| \sim GM^2/R$.
- By the virial theorem only half of the released energy is radiated, which changes only the factor in front.

Detailed Solutions:

$$\boxed{t_{\rm KH} \sim \frac{GM_\odot^2}{R_\odot L_\odot} = \frac{6.674 \times 10^{-11} \times (1.989 \times 10^{30})^2}{6.957 \times 10^8 \times 3.83 \times 10^{26}}\,\mathrm{s} \simeq 3 \times 10^7\ \text{years}.}$$

This is about a hundred times shorter than the age of the Earth. Gravity alone cannot have powered the Sun for that long, which pointed to a much larger store of energy, later found in the nucleus.

10.7.5 Problem 5: A Star That Heats as It Cools

Problem Statement: A young star of one solar mass, made of ionised gas with $\mu = 0.6$, contracts toward the main sequence. It shines with $L = L_\odot$ for 10^6 years without any fusion. By how much does its mean temperature change? Does it rise or fall?

Essential Ideas:

- The heat capacity of a star in virial balance is $C = -\frac{3}{2}Nk_B$, Eq. (10.14).
- The energy lost is $L\times$ time.

Detailed Solutions: $N = M/(\mu m_{\rm H}) = 2.0 \times 10^{57}$, so $C = -4.1 \times 10^{34}\,{\rm J\,K^{-1}}$. The energy lost is $\Delta E = -3.83 \times 10^{26} \times 3.16 \times 10^{13}\,{\rm J} = -1.2 \times 10^{40}\,{\rm J}$.

$$\boxed{\Delta \bar{T} = \frac{\Delta E}{C} \simeq +3 \times 10^5\,{\rm K}.}$$

The temperature rises. The star loses energy and gets hotter, and it keeps contracting and heating until its centre is hot enough for hydrogen to fuse.

Chapter 11

Radiation in Matter

Almost everything known about stars arrives as light. That light was made deep inside a star, fought its way out through layers of gas, crossed interstellar space (Chapter 8) and finally reached a telescope. At each stage matter took some light away and added some of its own.

One equation describes all of this. It is energy conservation along a single ray (P1), with gains and losses balanced as in P4. Where the matter is thin, the equation gives emission and absorption lines. Where it is thick, it gives a blackbody and a slow, random-walk leak of energy (P5). This one equation, in its two limits, covers most of what astronomers measure.

11.1 Specific Intensity

Chapter 4 measured light by its flux, the energy crossing a unit area per unit time, which falls as the inverse square of distance (Eq. (4.2)).

Inside a star or a cloud, light comes from every direction at once. A single flux mixes all of them together. To follow light through matter we need the light travelling in each direction separately.

Look at a white wall in an evenly lit room, first from one metre away and then from five. From farther away the wall covers less of your view, but each patch of it looks just as bright. The brightness of a surface does not fade with distance. Only its size shrinks.

The surface of the Sun is the same. Seen from Mars it covers a smaller patch of sky than seen from Earth, but each square degree of that patch is equally bright. The quantity that stays fixed along a ray is the energy per unit area, per unit time, per unit frequency and per unit solid angle, the specific intensity. Its constancy is energy conservation (P1) for a narrow bundle of rays.

The Sun in numbers. From Earth its angular radius is $0.266°$. From Mars, 1.52 times farther, it is $0.175°$. The flux falls by $1.52^2 = 2.3$ because the solid angle falls by that factor, while the intensity of each patch is the same.

The picture is a narrow cone of rays leaving a patch of surface. Farther along, the cone is wider, but it carries the same energy in the same solid angle.

The same constancy makes the surface brightness of a galaxy independent of its distance in a static space, which is what makes it a test of the expansion of the universe. It is why gravitational lensing, which bends rays but does not create or destroy light, leaves surface brightness unchanged. Radio astronomers state it as a brightness temperature. Here it gives the intensity and the three quantities built from it.

Definition. Light of frequency between ν and $\nu + d\nu$ crosses an area dA in time dt, travelling within a solid angle $d\Omega$ about a direction that makes an angle θ with the normal of dA. The

energy it carries defines the **specific intensity** I_ν,

$$dE = I_\nu \cos\theta \, dA \, dt \, d\nu \, d\Omega. \tag{11.1}$$

The factor $\cos\theta$ is the area dA as seen from the direction of travel.

Constant along a ray. Take two small areas dA_1 and dA_2, a distance d apart in empty space, each facing the other. The light leaving dA_1 toward dA_2 is the light arriving at dA_2 from dA_1. Seen from dA_1, the area dA_2 subtends $d\Omega_1 = dA_2/d^2$, and seen from dA_2, dA_1 subtends $d\Omega_2 = dA_1/d^2$. Equating the two energies (P1),

$$\begin{aligned} & I_{\nu,1}\, dA_1\, d\Omega_1 = I_{\nu,2}\, dA_2\, d\Omega_2 \\ \Longrightarrow \quad & I_{\nu,1} \frac{dA_1\, dA_2}{d^2} \overset{(1)}{=} I_{\nu,2} \frac{dA_2\, dA_1}{d^2} \\ \Longrightarrow \quad & I_{\nu,1} = I_{\nu,2}. \end{aligned} \tag{11.2}$$

(1) The two solid angles written with the areas they subtend.

$$\boxed{I_\nu \text{ is constant along a ray in empty space.}} \tag{11.3}$$

The inverse-square fall of the flux is entirely the shrinking of the solid angle that a source covers.

Flux, energy density and pressure. Three quantities used throughout are moments of I_ν over direction,

$$F_\nu = \int I_\nu \cos\theta \, d\Omega, \qquad u_\nu = \frac{1}{c}\int I_\nu \, d\Omega, \qquad P_\nu = \frac{1}{c}\int I_\nu \cos^2\theta \, d\Omega. \tag{11.4}$$

The flux F_ν is the net energy crossing a unit area. Light at speed c carrying intensity I_ν through $d\Omega$ fills space with energy density $I_\nu \, d\Omega/c$, which gives u_ν. Each photon carries momentum equal to its energy over c, and its component $\cos\theta$ across the surface, so the momentum flux, which is the pressure, is P_ν. For light that is the same in every direction,

$$F_\nu = 0, \qquad u_\nu = \frac{4\pi}{c} I_\nu, \qquad P_\nu = \frac{u_\nu}{3},$$

the same $\frac{1}{3}$ as in the pressure integral (9.20), from the average of $\cos^2\theta$. With $I_\nu = B_\nu$ this is the relation $B_\nu = c\, u_\nu/4\pi$ of Eq. (9.26).

A surface that sends intensity B_ν outward into every direction of the outer half-space gives the flux $F_\nu = \pi B_\nu$ (Eq. (4.20)), and integrated over frequency the surface flux $\sigma_{\rm SB}T^4$ of a blackbody, Eq. (4.24). In the language of intensity, the factor π is the first moment of Eq. (11.4) taken over the outward half of the sphere.

11.2 The Transfer Equation

Chapters 4 and 8 described extinction, in which light passing through dust is dimmed by $e^{-\tau}$ (Eq. (4.36)).

That is only half the story. Gas and dust also emit. A hot cloud glows, and a star's atmosphere both absorbs the light from below and adds its own.

Think of a foggy night street. A distant lamp fades as you look at it through more fog, but the fog itself glows with the light it has scattered. Through a little fog you see the lamp. Through enough fog you see only the glowing fog, and nothing about the lamp behind it.

Light travelling through matter loses a fixed fraction of itself per unit length by absorption and gains whatever the matter emits. Its intensity along the ray is a balance of the two, energy conservation (P1) with gains against losses (P4). When the matter is thick enough, the light forgets where it came from and takes on the glow of the matter.

A slab in numbers. Light of intensity I_0 enters a uniform slab whose own glow is S. If the slab has optical depth $\tau = 1$, a fraction $e^{-1} = 0.37$ of the entering light survives and the slab adds $0.63\,S$. At $\tau = 5$ only 0.7% survives and the slab adds 99.3% of S.

The picture is Figure 11.1. Whatever enters, the light leaving a thick slab approaches the slab's own glow.

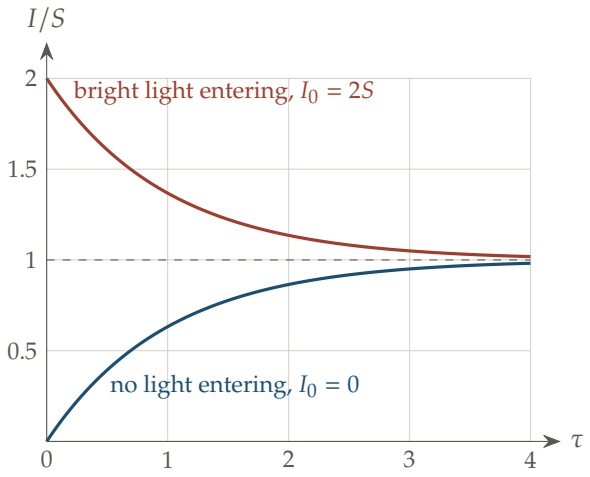


Figure 11.1: The intensity leaving a uniform slab against its optical depth τ, in units of the source function S, for entering intensity $I_0 = 2S$ (upper curve) and $I_0 = 0$ (lower curve).

The same equation governs radio emission from cold gas, X-rays from hot clusters, the Lyman-α absorption of distant quasars and the cosmic microwave background. It gives the equation and its solution.

The equation. Along a path element ds, matter of density ρ removes the fraction $\kappa_\nu \rho\, ds$ of the intensity, where κ_ν is the **opacity**, the absorbing area per unit mass. It also adds its own emission. Let j_ν be the energy it emits per unit volume, time, frequency and solid angle. Then (P1, P4)

$$\boxed{\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\kappa_\nu \rho\, I_\nu + j_\nu.} \tag{11.5}$$

Scattering removes light from a ray just as absorption does, so it counts in κ_ν, and the light it sends into the ray counts in j_ν.

Optical depth and source function. Two combinations make the equation simpler. The **optical depth** along the ray, $d\tau_\nu = \kappa_\nu \rho\, ds$, is the path length counted in mean free paths $1/(\kappa_\nu \rho)$, the same τ as in Chapter 4. The **source function** $S_\nu = j_\nu/(\kappa_\nu \rho)$ is the ratio of emission to absorption. Dividing Eq. (11.5) by $\kappa_\nu \rho$,

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}\tau_\nu} = -I_\nu + S_\nu. \tag{11.6}$$

The intensity always moves toward the source function. If $I_\nu > S_\nu$ it falls, and if $I_\nu < S_\nu$ it rises.

Formal solution. Multiply Eq. (11.6) by e^{τ_ν} and integrate from 0, where the ray enters with $I_\nu(0)$, to τ_ν,

$$\frac{\mathrm{d}}{\mathrm{d}\tau_\nu}\left(I_\nu\, e^{\tau_\nu}\right) \overset{(1)}{=} S_\nu\, e^{\tau_\nu}$$

$$\Longrightarrow \quad I_\nu(\tau_\nu) \overset{(2)}{=} I_\nu(0)\, e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau')\, e^{-(\tau_\nu-\tau')}\, d\tau'. \tag{11.7}$$

(1) The product rule. The terms in I_ν combine into a total derivative.
(2) Integrate, then divide by e^{τ_ν}.

The entering light is dimmed by $e^{-\tau_\nu}$. Each layer adds its emission, dimmed by the matter between it and the end of the ray. For a uniform slab, S_ν is constant and

$$\boxed{I_\nu = I_\nu(0)\, e^{-\tau_\nu} + S_\nu\left(1 - e^{-\tau_\nu}\right).} \tag{11.8}$$

Two limits follow, chosen by the ratio of the size of the slab to the mean free path (P6).

- **Optically thin**, $\tau_\nu \ll 1$. Then $I_\nu \approx I_\nu(0) + S_\nu\tau_\nu$. The slab adds emission in proportion to the amount of matter. Where κ_ν is large, at the frequency of a line, the emission is large, so a thin glowing gas shows emission lines.
- **Optically thick**, $\tau_\nu \gg 1$. Then $I_\nu \to S_\nu$, whatever entered. We see only the last mean free path or so of the matter.

With no emission, $S_\nu = 0$, this is the extinction law of Eq. (4.34).

11.3 Kirchhoff's Law and Local Thermodynamic Equilibrium

The Planck spectrum was derived by counting the states of light (Eq. (9.26)), and Chapter 4 treated stars as blackbodies.

A star is not a closed box at one temperature. Its temperature falls outward and light escapes. It is not obvious why its light should look like a blackbody, or why its spectrum should have dark lines.

Look into a furnace. A poker left in the coals glows with exactly the colour of the coals around it, and its outline nearly disappears. Everything in the furnace glows alike, whatever it is made of. A material that absorbs well also glows well, in just the right proportion, because otherwise the poker would end up hotter or colder than the coals.

The same rule holds for each small parcel of a star. Matter that absorbs strongly at some frequency also emits strongly there, and the ratio of the two is fixed by its temperature. This is detailed balance (P4) for absorption and emission, with the counted spectrum of light (P3).

Two layers in numbers. Suppose the continuum at 500 nm reaches us from a layer at 6000 K, and the centre of a strong line from a higher layer at 4500 K. The intensity at line centre is then $B_\nu(4500\,\mathrm{K})/B_\nu(6000\,\mathrm{K}) = 0.20$ of the continuum. The line is dark.

The picture is two depths in an atmosphere. At most frequencies we see down to a deep, hot layer. At the frequency of a line the gas is more opaque, so we see only a high, cooler layer.

The same law gives the dark Fraunhofer lines of the Sun, the bright lines of glowing nebulae, the darkening of the Sun's edge, and the nearly perfect blackbody of the microwave background. It gives the source function of matter in equilibrium and the way lines form.

Kirchhoff's law. Inside a closed cavity in full thermodynamic equilibrium at temperature T, the light is the same everywhere and in every direction, and by P3 it is Planck light, $I_\nu = B_\nu(T)$. Nothing changes along a ray, so Eq. (11.5) gives

$$0 \overset{(1)}{=} -\kappa_\nu \rho\, B_\nu(T) + j_\nu$$
$$\Longrightarrow \qquad S_\nu \overset{(2)}{=} \frac{j_\nu}{\kappa_\nu \rho} = B_\nu(T). \tag{11.9}$$

(1) $\mathrm{d}I_\nu/\mathrm{d}s = 0$ with $I_\nu = B_\nu$. Every process is balanced by its reverse (P4).
(2) The definition of S_ν.

$$\boxed{j_\nu = \kappa_\nu \rho\, B_\nu(T).} \tag{11.10}$$

Emission and absorption are tied together by the Planck function of the matter's temperature.

Local thermodynamic equilibrium. The emission and absorption of a parcel depend on the parcel itself, on its atoms and how they are excited, not on the light passing through. Where collisions keep each parcel close to equilibrium inside itself, the local thermodynamic equilibrium of Section 10.6, its atoms are excited as in equilibrium at the local temperature. Eq. (11.10) then holds point by point, even though the light passing through is not Planck light,

$$S_\nu = B_\nu\big(T(\vec{r})\big) \qquad \text{(local thermodynamic equilibrium).} \tag{11.11}$$

A thick layer at one temperature then emits Planck light, as a blackbody does, without being a closed box.

Absorption lines. In an atmosphere the temperature falls outward. By Eq. (11.8) we see light from about one mean free path deep, $\tau_\nu \approx 1$ along our line of sight (more exactly $\tau_\nu \approx \frac{2}{3}$, Section 11.7). At the frequency of a line, κ_ν is large, so $\tau_\nu = 1$ is reached higher up, where the gas is cooler and B_ν is smaller. The line is darker than the continuum on either side. If the temperature rose outward, as it does in the solar chromosphere, the same argument would give emission lines. The edge of the Sun looks darker than its centre for the same reason. Looking at the edge, the line of sight runs slantwise and reaches $\tau = 1$ in higher, cooler layers.

11.4 Diffusion of Light

Deep inside a star the optical depth to the surface is enormous, and by the thick limit of Eq. (11.8) the light at each point is nearly Planck light at the local temperature.

If the light were exactly Planck light, the same in every direction, it would carry no net flux (Section 11.1). But the Sun does shine. A small departure from equal light in all directions must carry its whole luminosity outward.

Think again of perfume in still air (Problem 9.7.4). Each molecule wanders at random, but a little more perfume drifts away from the bottle than toward it, because there is more perfume near the bottle. A net flow appears with no push in any direction, only a gradient.

Photons in a star do the same. Each is absorbed and re-emitted after a short free path, in a random direction. There is slightly more light where the gas is hotter, so a little more light drifts outward than inward. This is a random walk (P5), with the energy density of light aT^4 in place of the number of perfume molecules.

The Sun in numbers. Near its centre a photon's free path is of order a millimetre to a centimetre. To random-walk out across a radius of 7×10^8 m takes $R^2/(sc)$, Eq. (9.37), which is 5×10^3 years for $s = 1$ cm and 5×10^4 years for $s = 1$ mm. A straight flight would take 2.3 seconds.

The picture is the random walk of Figure 9.2, with photons in place of walkers, drifting slowly down the slope of temperature.

The same diffusion carries heat through the interiors of stars, carries neutrinos out of a collapsing stellar core, and smooths the early universe on small scales before atoms form. It gives the flux of light deep inside a star, by two routes.

Route A, the random walk. The diffusion flux of Eq. (9.39) applies to anything that walks at random. For light, the walkers are photons and the quantity that flows is energy. With energy density aT^4 (Eq. (9.25)) in place of n, speed c, and step length $1/(\kappa\rho)$,

$$F = -\frac{1}{3}\, c\, \frac{1}{\kappa\rho}\, \frac{\mathrm{d}(aT^4)}{\mathrm{d}r} = -\frac{c}{3\kappa\rho}\, \frac{\mathrm{d}(aT^4)}{\mathrm{d}r}. \tag{11.12}$$

Route B, the momentum of light. Multiply the transfer equation (11.5) by $\cos\theta/c$ and integrate over all directions. Along a radial coordinate, $\mathrm{d}/\mathrm{d}s = \cos\theta\ \mathrm{d}/\mathrm{d}r$ for the part of the change that matters deep inside a star, where curvature over one free path is negligible. Then

$$\frac{\mathrm{d}}{\mathrm{d}r}\,\frac{1}{c}\int I_\nu \cos^2\theta\, d\Omega \overset{(1)}{=} -\frac{\kappa_\nu\rho}{c}\int I_\nu \cos\theta\, d\Omega + \frac{j_\nu}{c}\int \cos\theta\, d\Omega$$
$$\Longrightarrow \qquad \frac{\mathrm{d}P_\nu}{\mathrm{d}r} \overset{(2)}{=} -\frac{\kappa_\nu\rho}{c}\, F_\nu. \tag{11.13}$$

(1) Eq. (11.5) with $\mathrm{d}/\mathrm{d}s = \cos\theta\ \mathrm{d}/\mathrm{d}r$, times $\cos\theta/c$, integrated over $d\Omega$.
(2) The moments of Eq. (11.4). Emission is the same in all directions, so $\int \cos\theta\, d\Omega = 0$ removes it.

This is momentum conservation (P1) for light. The gradient of the pressure of light equals the momentum per unit volume and time that the light gives to the matter. Deep inside, $P = aT^4/3$, and with κ a suitable average over frequency,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{aT^4}{3}\right) = -\frac{\kappa\rho}{c}\, F, \tag{11.14}$$

which is Eq. (11.12) again. (The right average weights $1/\kappa_\nu$ by $\partial B_\nu/\partial T$, so the frequencies where matter is most transparent carry most of the flux. It is called the Rosseland mean.)

The transport equation. Write $F = L(r)/(4\pi r^2)$, where $L(r)$ is the luminosity through the sphere of radius r, and $\mathrm{d}(aT^4)/\mathrm{d}r = 4aT^3\ \mathrm{d}T/\mathrm{d}r$. Then

$$\boxed{\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\,\kappa\rho\,L(r)}{16\pi a c\, r^2 T^3}.} \tag{11.15}$$

The temperature gradient needed to carry a luminosity grows with the opacity. Opaque matter needs a steep fall of temperature to push the same energy through. This is the radiative transport equation of a star (Section 12.2).

11.5 The Eddington Luminosity

The light flowing out of a star pushes on the matter it passes through, with the force found in Route B above.

The brighter a star, the harder its light pushes. At some brightness the push must match gravity.

Think of standing in a strong wind. A breeze does nothing, a gale pushes you over. Somewhere between, the push of the wind equals the grip your weight gives you on the ground.

The matter at the surface of a star is held down by gravity and pushed out by the momentum of the light streaming past it (P1 for light, P2 for gravity). Both forces fall off as $1/r^2$, so their ratio depends only on the luminosity and the mass. Above a certain luminosity, the Eddington luminosity, no star can hold on to its outer layers.

The Sun in numbers. For ionised hydrogen, where light is scattered by free electrons, the limit is 1.3×10^{31} W, about 3.3×10^4 times the Sun's luminosity. The Sun is far below it.

The picture is a parcel of gas with gravity pulling it inward and a stream of photons pushing it outward. The ratio of the two arrows does not depend on the radius.

The same limit caps the luminosity of the most massive stars, and it caps how fast black holes can swallow gas, since infalling gas radiates and the light pushes back. A quasar shining at 10^{39} W must hold at least 10^8 solar masses. It gives the maximum luminosity of a body of mass M.

The limit. From Route B, light of flux F gives the matter a force per unit volume $\kappa\rho F/c$, so a force per unit mass $\kappa F/c$. Gravity pulls with GM/r^2 per unit mass. With $F = L/(4\pi r^2)$ the two balance when

$$\frac{\kappa}{c}\frac{L}{4\pi r^2} = \frac{GM}{r^2}$$
$$\Longrightarrow \qquad L = \frac{4\pi GMc}{\kappa}. \tag{11.16}$$

$$\boxed{L_{\mathrm{Edd}} = \frac{4\pi GMc}{\kappa}.} \tag{11.17}$$

The radius has dropped out. In hot ionised gas the opacity is scattering by free electrons, with the Thomson cross-section σ_T of Eq. (8.7) per electron. For pure hydrogen there is one electron per mass m_{H}, so $\kappa = \sigma_T/m_{\mathrm{H}} = 0.040\,\mathrm{m^2\,kg^{-1}}$ and

$$L_{\mathrm{Edd}} = 1.3 \times 10^{31}\,\mathrm{W}\left(\frac{M}{M_\odot}\right) = 3.3 \times 10^4\,L_\odot\left(\frac{M}{M_\odot}\right).$$

Any larger opacity lowers the limit.

11.6 Einstein Coefficients

Kirchhoff's law fixed the ratio of emission to absorption for matter in local thermodynamic equilibrium (Eq. (11.10)).

It did not say what an atom actually does, or what happens when collisions are too rare to keep its levels in equilibrium, as in thin interstellar gas.

Think of a ball resting at the bottom of a step. A kick can lift it onto the step, and from the step it can roll back down by itself. A kick can also knock it down sooner than it would have rolled. If the ball spends time on and off the step in the proportions the temperature demands, the three kinds of move must be related.

An atom with two levels makes three kinds of transition. Light can lift it, it can drop by itself, and light can push it down sooner (stimulated emission). Detailed balance (P4) with Planck light and Boltzmann populations (P3) fixes how the three rates are related. Since the rates belong to the atom alone, the relations hold everywhere, in equilibrium or not.

The 21 cm line in numbers. The upper level of hydrogen's ground state drops by itself at the rate $2.9 \times 10^{-15}\,\mathrm{s}^{-1}$ (Eq. (8.31)), once in 11 million years. In interstellar gas, collisions change the level far more often than that, so the populations follow the gas temperature (Section 8.4).

The picture is two levels with three arrows between them, one up and two down.

The same relations give the strength of every spectral line, the gain of lasers and of the masers seen in star-forming clouds, and the condition for a line to form out of equilibrium. They give the three coefficients and the relations between them.

Three rates. Let level 1 (lower) and level 2 (upper) be single states separated by $h\nu$. In light of energy density u_ν at the line frequency,

- absorption lifts an atom from 1 to 2 at the rate $B_{12}\,u_\nu$ per atom,
- spontaneous emission drops it from 2 to 1 at the rate A_{21} per atom,
- stimulated emission drops it from 2 to 1 at the rate $B_{21}\,u_\nu$ per atom.

These are the **Einstein coefficients**. They belong to the atom.

Relations. In thermodynamic equilibrium at temperature T, the upward and downward flows balance (P4), the populations follow the Boltzmann factor, $N_2/N_1 = e^{-h\nu/k_BT}$, and the light is Planck light (P3),

$$N_1 B_{12}\,u_\nu \overset{(1)}{=} N_2\big(A_{21} + B_{21}\,u_\nu\big)$$
$$\Longrightarrow \qquad u_\nu \overset{(2)}{=} \frac{A_{21}}{B_{12}\,e^{h\nu/k_BT} - B_{21}}. \tag{11.18}$$

(1) Detailed balance between level 1 and level 2.
(2) Solve for u_ν and use $N_1/N_2 = e^{h\nu/k_BT}$.

This must equal the Planck energy density, $u_\nu = (8\pi h\nu^3/c^3)/(e^{h\nu/k_BT} - 1)$ from Section 9.3, at every temperature. That is possible only if

$$\boxed{B_{12} = B_{21}, \qquad A_{21} = \frac{8\pi h\nu^3}{c^3}\,B_{21}.} \tag{11.19}$$

One coefficient fixes the other two. The relations hold for the atom in any surroundings, since the coefficients do not depend on whether the gas is in equilibrium. Stimulated emission is required by them. Without it no choice of coefficients could reproduce the Planck spectrum. (For levels with several states, each B is multiplied by the number of states of its starting level.)

When local equilibrium fails. The populations follow the gas temperature when collisions move atoms between the levels much faster than spontaneous emission does. The ratio of the collision rate to A_{21} decides (P6). In dense stellar atmospheres collisions win, and local thermodynamic equilibrium holds. In thin nebulae, light and spontaneous decay set the populations instead, and lines must be computed from the rates themselves. If some process fills the upper level faster than the lower one, $N_2 > N_1$, the net absorption $(N_1 - N_2)B_{12}u_\nu$ turns negative and the gas amplifies light passing through it. Such masers are seen in water and hydroxyl molecules in star-forming clouds.

11.7 The Grey Atmosphere

The effective temperature was defined in Chapter 4 by $L = 4\pi R^2 \sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4$ (Eq. (4.26)), as if the star had one surface at one temperature.

A star has no sharp surface. Its temperature falls continuously outward, and the light we see comes from a range of depths. Where is the effective temperature actually found?

Think of looking into a pool of murky water. You see down to some depth where the water stops being clear, and no farther. The bottom you see is not a surface in the water but the depth at which the murk has added up to enough.

A star's photosphere is such a depth. It lies at a fixed optical depth, and the temperature there is the effective temperature. The calculation uses momentum conservation for light (P1), the steady flow of energy through the atmosphere (P4), and Planck light at the local temperature (P3).

In numbers, the result below puts the effective temperature at an optical depth of $\frac{2}{3}$, and the topmost gas, at $\tau = 0$, at $2^{-1/4} = 0.84$ of it. For the Sun that is 4850 K at the top against 5772 K at the photosphere.

The picture is Figure 11.2, a temperature that rises slowly with depth through the layers we see.

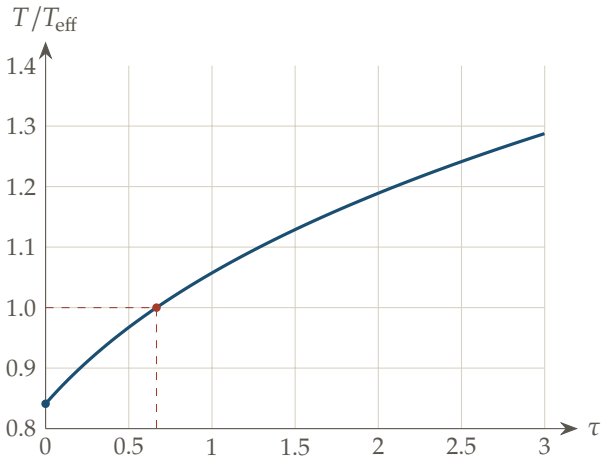


Figure 11.2: The temperature of a grey atmosphere against optical depth, in units of the effective temperature. The point at $\tau = \frac{2}{3}$ marks $T = T_{\text{eff}}$, and the point at $\tau = 0$ marks $T = 2^{-1/4} T_{\text{eff}}$.

The same calculation, with the opacity's dependence on frequency added, is the starting point of every model stellar atmosphere, and of planetary atmospheres warmed from below. It gives the temperature through the atmosphere.

Assumptions. Take the opacity the same at all frequencies (grey), so frequency can be integrated out. Measure the optical depth τ inward from the top. The atmosphere is thin compared with the star, so it is treated as flat, and no energy is made in it, so the flux F is the same at every depth (P1, P4).

Temperature against depth. The momentum equation of Route B, with $d\tau = -\kappa\rho\, dr$, reads $\mathrm{d}P_{\text{light}}/\mathrm{d}\tau = F/c$. With F constant,

$$P_{\text{light}}(\tau) = \frac{F}{c}\,\tau + P_{\text{light}}(0). \tag{11.20}$$

Two approximations close it. Deep down the light is nearly the same in all directions, so $P_{\text{light}} = u/3$, and Eddington took this to hold at all depths. At the top, light only leaves and none comes in. If the outgoing light is spread evenly over the outer half of directions with

intensity I^+, then $F = \pi I^+$ and $u = 2\pi I^+/c$ from Eq. (11.4), so $u(0) = 2F/c$. Then

$$\begin{aligned} \frac{u}{3} &\overset{(1)}{=} \frac{F}{c}\tau + \frac{2F}{3c} \\ \Longrightarrow \quad aT^4 &\overset{(2)}{=} \frac{3F}{c}\left(\tau + \frac{2}{3}\right) \\ &\overset{(3)}{=} \frac{3}{4}aT_{\text{eff}}^4\left(\tau + \frac{2}{3}\right). \end{aligned} \tag{11.21}$$

(1) $P_{\text{light}} = u/3$ at all depths and $P_{\text{light}}(0) = u(0)/3 = 2F/(3c)$.
(2) Local thermodynamic equilibrium, $u = aT^4$.
(3) $F = \sigma_{\text{SB}}T_{\text{eff}}^4$ and $\sigma_{\text{SB}} = ac/4$ (Section 9.3).

$$\boxed{T^4 = \frac{3}{4}T_{\text{eff}}^4\left(\tau + \frac{2}{3}\right).} \tag{11.22}$$

The temperature equals T_{eff} at $\tau = \frac{2}{3}$. That depth is the **photosphere**, the layer the effective temperature describes. Above it the temperature levels off at $2^{-1/4}T_{\text{eff}}$, and below it the temperature keeps rising, as $\tau^{1/4}$ deep down, where the diffusion of Section 11.4 takes over.

11.8 Problem Set: Radiation in Matter

11.8.1 Problem 1: The Sun Seen From Mars

Problem Statement: Mars is 1.52 times farther from the Sun than the Earth. Compare (a) the flux of sunlight and (b) the specific intensity of the solar disk seen from the two planets. (c) Why does a camera on Mars need a longer exposure for a picture of the whole landscape, but not for a picture of the Sun's disk filling the same number of pixels?

Essential Ideas:

- The flux falls as $1/d^2$ (Eq. (4.2)).
- The intensity is constant along a ray (Eq. (11.3)).

Detailed Solutions:

(a) **Flux**. $F_{\text{Mars}}/F_\oplus = 1/1.52^2$.

$$\boxed{F_{\text{Mars}} = 0.43\,F_\oplus.}$$

(b) **Intensity**. Unchanged.

$$\boxed{I_{\text{Mars}} = I_\oplus.}$$

(c) **Exposures**. A landscape is lit by the total flux, which is lower on Mars. A pixel looking at the Sun's disk collects intensity times the solid angle of one pixel, and both are the same.

11.8.2 Problem 2: How Opaque Is a Dark Cloud?

Problem Statement: A dark cloud dims background starlight in the visual band by $A_V = 10$ magnitudes, and the cloud's own glow in that band is negligible. (a) What is its optical depth? (b) What fraction of the starlight gets through? (c) At 2.2 μm the extinction is about 0.11 of the visual value. What fraction gets through there?

Essential Ideas:

- With $S_\nu = 0$, Eq. (11.8) gives $I = I_0 e^{-\tau}$, and $A = 2.5\log_{10}(I_0/I) = 1.086\,\tau$ (Eq. (4.36)).

Detailed Solutions:

(a) **Optical depth**. $\tau_V = 10/1.086$.

$$\boxed{\tau_V \simeq 9.2.}$$

(b) **Visual transmission**. $e^{-9.2}$.

$$\boxed{I/I_0 \simeq 1.0 \times 10^{-4}.}$$

(c) **Infrared**. $\tau = 0.11 \times 9.2 = 1.0$, so $e^{-1.0} = 0.36$.

$$\boxed{I/I_0 \simeq 0.36.}$$

The cloud is opaque in visible light and nearly transparent in the near infrared, which is why young stars inside such clouds are found with infrared telescopes.

11.8.3 Problem 3: How Long Light Takes to Leave the Sun

Problem Statement: Take the mean free path of a photon inside the Sun to be $s = 1\,\text{mm}$ throughout. Estimate the time for energy made at the centre to random-walk to the surface, $R_\odot = 6.96 \times 10^8\,\text{m}$. How many free paths is that?

Essential Ideas:

- The random-walk time $R^2/(sc)$, Eq. (9.37), and $N = (R/s)^2$ steps.

Detailed Solutions:

$$\boxed{t = \frac{(6.96 \times 10^8)^2}{10^{-3} \times 3.00 \times 10^8}\,\text{s} \simeq 1.6 \times 10^{12}\,\text{s} \simeq 5 \times 10^4\ \text{years}, \qquad N \simeq 5 \times 10^{23}.}$$

The light we see today left the core tens of thousands of years ago, as energy. The individual photons were absorbed and re-emitted along the way, about 5×10^{23} times.

11.8.4 Problem 4: Eddington Limits

Problem Statement: Using electron scattering in ionised hydrogen, $\kappa = 0.040\,\text{m}^2\,\text{kg}^{-1}$, find the Eddington luminosity of (a) a star of $100\,M_\odot$ and (b) a black hole of $10^8\,M_\odot$. (c) A quasar emits $10^{40}\,\text{W}$. What is the least mass that can power it?

Essential Ideas:

- $L_{\text{Edd}} = 4\pi GMc/\kappa = 1.3 \times 10^{31}\,\text{W}\,(M/M_\odot)$, Eq. (11.17).

Detailed Solutions:

(a) **Massive star**.

$$\boxed{L_{\text{Edd}} \simeq 1.3 \times 10^{33}\,\text{W} \simeq 3 \times 10^6\,L_\odot.}$$

The most massive stars shine at a good fraction of this, so they lose mass in strong winds.

(b) **Black hole**.

$$\boxed{L_{\text{Edd}} \simeq 1.3 \times 10^{39}\,\text{W}.}$$

(c) **Quasar**. $M \geq L\kappa/(4\pi Gc) = 10^{40}/(1.3 \times 10^{31})\,M_\odot$.

$$\boxed{M \gtrsim 8 \times 10^8\,M_\odot.}$$

11.8.5 Problem 5: The Surface of a Grey Star

Problem Statement: In the grey atmosphere of Eq. (11.22) for a star with $T_{\rm eff} = 5772\,\mathrm{K}$, find the temperature (a) at the top, (b) at $\tau = \frac{2}{3}$, and (c) at $\tau = 10$.

Essential Ideas:

- $T = T_{\rm eff}\,[\frac{3}{4}(\tau + \frac{2}{3})]^{1/4}$.

Detailed Solutions:

(a) **Top**. $T = 2^{-1/4} T_{\rm eff}$.

$$\boxed{T(0) \simeq 4850\,\mathrm{K}.}$$

(b) **Photosphere**.

$$\boxed{T(\tfrac{2}{3}) = T_{\rm eff} = 5772\,\mathrm{K}.}$$

(c) **Deeper**. $[\frac{3}{4} \times 10.67]^{1/4} = 1.68$.

$$\boxed{T(10) \simeq 9700\,\mathrm{K}.}$$

Chapter 12

The Equations of a Star

A star is a ball of gas in hydrostatic equilibrium (Section 10.1), heated inside by nuclear burning and losing energy from its surface as light (Chapter 11). Its whole structure, the density, pressure, temperature and luminosity at every depth, follows from four equations. Each equation is one of the principles at work. Three properties of the gas, its pressure, its opacity and its rate of burning, close the four equations, and each of those is again one of the principles.

The same four equations, with their boundary conditions, are what stellar-evolution codes solve. Understanding them is understanding how a star is computed.

12.1 The Unknowns

Chapter 10 described a star at each radius r, with $m(r)$ the mass inside.

Inside a real star the gas does not stay at one radius. As the star contracts or expands, a given parcel of gas moves, and the radius stops being a good label for it. What follows the gas is the mass inside it.

Think of the rings of a tree. As the tree grows, every ring moves outward, but the ring laid down in a given year stays the same ring. Counting rings from the centre labels the wood, whatever the size of the trunk.

Inside a star the mass coordinate m plays the part of the ring count. The shell of gas with mass m inside it keeps that label while the star contracts, expands or burns. Its composition changes only by its own burning, so everything that happens to a parcel happens at fixed m. The four equations are therefore written with m, from 0 at the centre to M at the surface, as the independent variable.

In numbers, the Sun's radius has grown by about 10% since it began to burn hydrogen, but the shell holding the inner half of its mass is still the shell with $m = M_\odot/2$.

The picture is a set of nested shells labelled by the mass they enclose, not by their radius.

Fluid dynamics calls this the Lagrangian description, following the fluid, as opposed to the Eulerian description at fixed points of space. It gives four unknown functions of m. They are the radius $r(m)$, the pressure $P(m)$, the temperature $T(m)$ and the luminosity $L(m)$ flowing out through the shell.

12.2 The Four Equations

Hydrostatic equilibrium (Eq. (10.2)) and the flow of light (Eq. (11.15)) have been derived already.

They are two of the four equations, and both are written with r. A complete model needs the other two, how the mass relates to the radius and how the luminosity builds up, and needs all

four with m as the variable.

Think of a household budget kept month by month. One line records the money coming in. Another records the money going out. The balance at the end of each month follows from the balance before plus the difference. Nothing else is needed, and nothing can be left out.

A star keeps four such accounts, shell by shell, for space, momentum, energy and the flow of heat. Each one says how a quantity changes from one shell to the next. Mass and energy are conservation (P1). Momentum is conservation with the gravity of a sphere (P1 and P2). The flow of heat is a random walk of light (P5), or, where that is too slow, the bulk motion of the gas itself.

In numbers, the Sun's luminosity of 3.83×10^{26} W is built up almost entirely in the inner quarter of its radius, where the burning happens, and then carried unchanged through the outer three quarters.

The picture is a stack of shells, with each equation stepping one quantity from a shell to the next.

The same four accounts, with the same structure, describe the interior of a planet (with no burning), the core of a white dwarf (with no burning and degenerate gas) and the envelope of an accreting neutron star. They give the four equations of stellar structure.

Mass. Inverting the definition of $m(r)$, Eq. (9.10),

$$\boxed{\frac{\mathrm{d}r}{\mathrm{d}m} = \frac{1}{4\pi r^2 \rho}.} \tag{12.1}$$

Momentum. Hydrostatic equilibrium (10.2), divided by $\mathrm{d}m/\mathrm{d}r = 4\pi r^2 \rho$,

$$\begin{aligned} \frac{\mathrm{d}P}{\mathrm{d}m} &\overset{(1)}{=} \frac{\mathrm{d}P}{\mathrm{d}r}\,\frac{\mathrm{d}r}{\mathrm{d}m} \\ &\overset{(2)}{=} -\frac{Gm\rho}{r^2}\,\frac{1}{4\pi r^2 \rho} \\ &= -\frac{Gm}{4\pi r^4}. \end{aligned} \tag{12.2}$$

(1) The chain rule.
(2) Eq. (10.2) and Eq. (12.1).

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}m} = -\frac{Gm}{4\pi r^4}.} \tag{12.3}$$

The density has dropped out. The pressure falls outward at a rate set by gravity alone.

Energy. Take the shell between m and $m + dm$. Nuclear burning releases energy in it at the rate $\varepsilon\, dm$, where ε is the energy released per unit mass and time. Light carries $L(m)$ in through its inner face and $L(m + dm)$ out through its outer face. Whatever is left heats the shell. With s the entropy per unit mass, the heat added to the shell per unit time is $T\,(\mathrm{d}s/\mathrm{d}t)\,dm$, so by conservation of energy (P1)

$$\begin{aligned} T\,\frac{\mathrm{d}s}{\mathrm{d}t}\,dm &\overset{(1)}{=} \varepsilon\, dm - \big[L(m + dm) - L(m)\big] \\ &\overset{(2)}{=} \left(\varepsilon - \frac{\mathrm{d}L}{\mathrm{d}m}\right) dm. \end{aligned} \tag{12.4}$$

(1) Heat gained by the shell equals energy released inside it minus the net energy carried out.
(2) To first order, $L(m + dm) - L(m) = (\mathrm{d}L/\mathrm{d}m)\, dm$.

$$\boxed{\frac{\mathrm{d}L}{\mathrm{d}m} = \varepsilon - T\,\frac{\mathrm{d}s}{\mathrm{d}t}.} \tag{12.5}$$

In a star that is not changing, $\mathrm{d}s/\mathrm{d}t = 0$ and the luminosity grows outward exactly by the burning in each shell. In a star that contracts, as before hydrogen ignites, the entropy of each shell falls and $-T\ \mathrm{d}s/\mathrm{d}t > 0$ supplies luminosity from gravity. This term is the Kelvin–Helmholtz contraction of Section 10.5, written for each shell. (Neutrinos that escape directly add a loss term to ε.)

Heat flow by light. The radiative transport equation (11.15), divided by $4\pi r^2\rho$,

$$\boxed{\frac{\mathrm{d}T}{\mathrm{d}m} = -\frac{3\kappa L}{64\pi^2 a c\, r^4 T^3} \qquad \text{(radiative).}} \tag{12.6}$$

Heat flow by convection. Where light alone would need too steep a fall of temperature, the gas starts to move and carries heat itself. To find where, follow a blob of gas lifted a little, from pressure P to the lower pressure $P + dP$ above it ($dP < 0$). It rises too fast to exchange heat with its surroundings but slowly enough to match their pressure, since sound crosses it quickly (P6). So it expands along an adiabat, $P \propto \rho^\gamma$, and for an ideal gas its temperature changes as

$$\left(\frac{dT}{T}\right)_{\text{blob}} = \frac{\gamma - 1}{\gamma}\,\frac{dP}{P}.$$

At the same pressure, a hotter blob is less dense than its surroundings (ideal gas, Eq. (9.23)) and keeps rising. The blob ends up hotter than its surroundings if its temperature falls less than theirs,

$$\begin{aligned} \frac{\gamma - 1}{\gamma}\,\frac{dP}{P} &> \left(\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P}\right)_{\text{star}} \frac{dP}{P} \\ \Longrightarrow \qquad \left(\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P}\right)_{\text{star}} &\overset{(1)}{>} \frac{\gamma - 1}{\gamma}. \end{aligned} \tag{12.7}$$

(1) Divide by dP/P, which is negative, so the inequality turns over.

$$\boxed{\text{convection where} \quad \left(\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P}\right)_{\text{radiative}} > \frac{\gamma - 1}{\gamma} \left(= \tfrac{2}{5} \text{ for } \gamma = \tfrac{5}{3}\right).} \tag{12.8}$$

This is the **Schwarzschild criterion**. The left side is computed from Eq. (12.6) and Eq. (12.3). Two things make it large, and so make a region convect. A high opacity needs a steep temperature fall to push the light through, as in the cool outer third of the Sun. A large luminosity from a small mass needs one too, as in the cores of massive stars, whose steeply temperature-sensitive burning (Section 12.3) packs the energy release into a small centre. Once convection starts it is so efficient that it holds the temperature gradient very close to the adiabatic value, so in a convective region

$$\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P} \approx \frac{\gamma - 1}{\gamma} \qquad \text{(convective)} \tag{12.9}$$

replaces Eq. (12.6). Codes use mixing-length theory, a model of how far a blob rises before it mixes, to compute the small excess over the adiabatic gradient. The same criterion sets the fall of temperature with height in the Earth's lower atmosphere (Problem 12.7.5).

The system, and the count. Collect the four unknowns in one vector. The four equations then form one first-order system in m,

$$\frac{\mathrm{d}}{\mathrm{d}m}\begin{pmatrix} r \\ P \\ L \\ T \end{pmatrix} = \begin{pmatrix} 1/(4\pi r^2\rho) \\ -Gm/(4\pi r^4) \\ \varepsilon - T\ \mathrm{d}s/\mathrm{d}t \\ -3\kappa L/(64\pi^2 ac\, r^4 T^3) \text{ or Eq. (12.9)} \end{pmatrix}. \tag{12.10}$$

Four unknown functions, four first-order equations. The right side needs three properties of the gas, its density $\rho(P,T)$, its opacity $\kappa(\rho,T)$ and its burning rate $\varepsilon(\rho,T)$, each depending also on the composition at that m. They are worked out in Section 12.3. Four first-order equations need four boundary conditions, found in Section 12.4.

12.3 Pressure, Opacity and Burning

The pressure of an ideal gas and of light were found by counting states (Eqs. (9.23) and (9.25)). The flow of light needs an opacity (Section 11.4), and the energy equation needs a burning rate.

The system (12.10) cannot be solved without these three. They depend on the composition and on the state of the gas, and in white dwarfs the pressure takes a form not yet met.

Think of a crowded lift. When there is room, people move about freely, and the more energetic they are, the harder they push on the walls. When the lift is packed, nobody can move whatever their mood, and the push on the walls comes only from being squeezed. The push no longer depends on how energetic people feel.

Electrons in very dense gas are the packed lift. Each quantum state holds at most one electron, so once the low states are full the electrons are forced into states of high momentum, and they push however cold the gas is. Counting states (P3) gives this degeneracy pressure, the opacity follows from counting absorbers (P3) along a random walk of light (P5), and the burning rate from counting fast particles (P3) that tunnel through the electric barrier of the nucleus.

In numbers, the Sun's centre has a Fermi energy about half its thermal energy k_BT, so it is only mildly degenerate. The centre of a white dwarf, at $10^9\,\mathrm{kg\,m^{-3}}$ and 10^7 K, has a Fermi energy 190 times its thermal energy (Problem 12.7.2).

The picture is Figure 12.1, the plane of density and temperature divided into the regions where gas, light and degenerate electrons supply most of the pressure.

The same counting gives the interiors of giant planets, the electron pressure of white dwarfs and neutron-star crusts, and, with nuclear physics, the burning in the Big Bang. It gives the composition, the degenerate pressure, the opacity and the burning rate.

Composition and mean molecular weight. Let X, Y and Z be the mass fractions of hydrogen, helium and everything heavier, with $X + Y + Z = 1$. In a fully ionised gas, a mass m_H of hydrogen gives two particles (a proton and an electron), a mass $4m_\mathrm{H}$ of helium gives three (a nucleus and two electrons), and a heavy atom of mass Am_H gives about $A/2$ particles (it has about $A/2$ electrons, and the one nucleus can be neglected). The number of particles per mass m_H is therefore

$$\boxed{\frac{1}{\mu} = 2X + \frac{3}{4}Y + \frac{1}{2}Z, \qquad \frac{1}{\mu_e} = X + \frac{1}{2}Y + \frac{1}{2}Z = \frac{1+X}{2},} \tag{12.11}$$

where μ is the mean molecular weight of Section 10.2, and μ_e counts the mass per electron, so that $n_e = \rho/(\mu_e m_\mathrm{H})$. For the Sun's composition, $X = 0.70$, $Y = 0.28$, $Z = 0.02$, this gives $\mu = 0.617$ and $\mu_e = 1.18$. Pure helium gives $\mu = \frac{4}{3}$ and $\mu_e = 2$, as do carbon and oxygen to good accuracy.

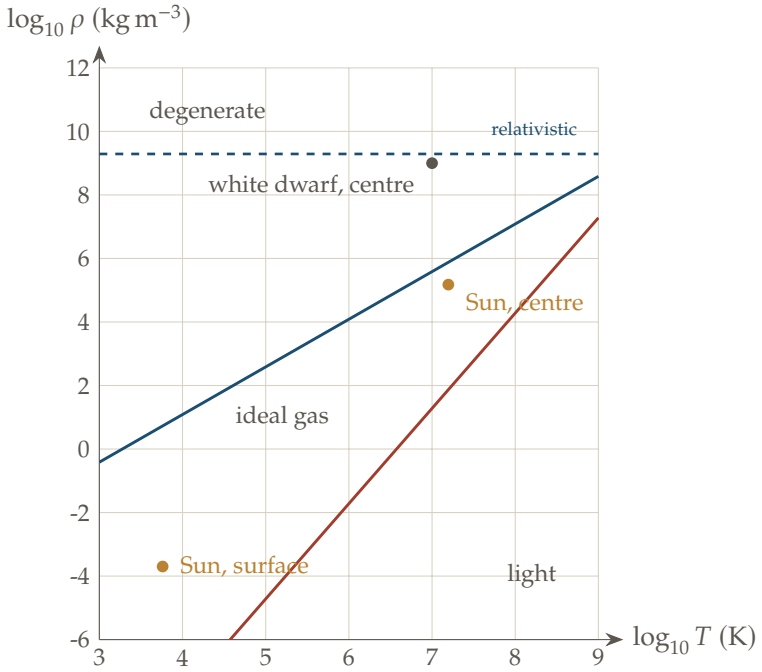


Figure 12.1: Density against temperature, on logarithmic axes. The red line marks equal pressure of light and of an ideal gas ($\mu = 0.62$). The blue line marks a Fermi energy equal to $k_B T$ ($\mu_e = 2$). The dashed line marks the density at which the electrons' Fermi momentum reaches $m_e c$. Points mark the centre and surface of the Sun and the centre of a white dwarf.

Degenerate electrons. At high density and low temperature every electron state up to some momentum p_F, the **Fermi momentum**, is full and every state above it is empty. With two spin states, Eq. (9.18) gives the electron density

$$n_e = \int_0^{p_F} \frac{2}{h^3}\, 4\pi p^2\, dp = \frac{8\pi}{3h^3}\, p_F^3, \qquad p_F = h\left(\frac{3n_e}{8\pi}\right)^{1/3}. \tag{12.12}$$

The pressure integral (9.20) in the two limits of Section 9.3 gives

$$\begin{aligned} P_{\text{slow}} &\overset{(1)}{=} \frac{1}{3}\int_0^{p_F} p\,\frac{p}{m_e}\,\frac{2}{h^3}4\pi p^2\, dp \overset{(2)}{=} \frac{8\pi p_F^5}{15\, m_e h^3} \overset{(3)}{=} \frac{h^2}{20\, m_e}\left(\frac{3}{\pi}\right)^{2/3} n_e^{5/3}, \\ P_{\text{fast}} &\overset{(4)}{=} \frac{1}{3}\int_0^{p_F} p\, c\,\frac{2}{h^3}4\pi p^2\, dp \overset{(5)}{=} \frac{2\pi c\, p_F^4}{3h^3} \overset{(6)}{=} \frac{hc}{8}\left(\frac{3}{\pi}\right)^{1/3} n_e^{4/3}. \end{aligned} \tag{12.13}$$

(1) Slow electrons, $v = p/m_e$.
(2) $\int_0^{p_F} p^4\, dp = p_F^5/5$.
(3) p_F from Eq. (12.12).
(4) Electrons at the speed of light, $v = c$.
(5) $\int_0^{p_F} p^3\, dp = p_F^4/4$.
(6) p_F from Eq. (12.12).

With $n_e = \rho/(\mu_e m_{\text{H}})$ both are power laws of the density alone, $P \propto \rho^{5/3}$ for slow electrons and $P \propto \rho^{4/3}$ for fast ones, the two values of γ from Eq. (9.21). The temperature does not appear. Heating degenerate gas does not raise its pressure, so it does not expand and cool, and the thermostat of Section 10.3 fails. Burning that starts in degenerate gas runs away, as in the helium flash of low-mass giants and in the thermonuclear explosion of a white dwarf. The ions

stay classical, because at the same momentum their much larger mass gives them a far smaller kinetic energy.

When degeneracy matters. Degeneracy matters when the Fermi energy $p_F^2/2m_e$ is comparable with k_BT, or equivalently when the spacing between electrons is comparable with their quantum wavelength (P6). Setting $p_F^2/2m_e = k_BT$ gives the blue line of Figure 12.1, $\rho \propto T^{3/2}$. The electrons move near the speed of light when p_F approaches $m_e c$, which by Eq. (12.12) with $\mu_e = 2$ happens at $\rho = 2.0 \times 10^9\,\mathrm{kg\,m^{-3}}$, the dashed line. The pressure of light exceeds that of the gas where $aT^4/3 > \rho k_BT/(\mu m_\mathrm{H})$, below the red line, at low density and high temperature.

Opacity. In hot, fully ionised gas the electrons scatter light with the Thomson cross-section, so $\kappa = \sigma_T/(\mu_e m_\mathrm{H}) = 0.020(1+X)\,\mathrm{m^2\,kg^{-1}}$, independent of density and temperature. At lower temperatures, electrons bound to ions or passing close to them absorb light. Their cross-section falls with frequency roughly as ν^{-3}, the light that carries the flux has $h\nu \sim k_BT$ (the Rosseland weighting of Section 11.4), and the number of absorbers per unit mass grows with the density. Together these give Kramers' law, $\kappa \propto \rho\, T^{-7/2}$, whose coefficient comes from atomic physics (the free–free part is derived in Eq. (15.18)). Kramers opacity is what makes the outer layers of the Sun opaque enough to convect.

The burning rate and the Gamow peak. Two protons at the Sun's centre, at $k_BT = 1.35\,\mathrm{keV}$, approach each other against an electric barrier of about 1 MeV. Classically they never touch. Quantum mechanically they tunnel, with a probability $e^{-\sqrt{E_G/E}}$ at relative energy E, where the **Gamow energy** $E_G = 2m_r c^2(\pi\alpha Z_1 Z_2)^2$ depends on their charges Z_1, Z_2 and their reduced mass m_r, and α is the fine-structure constant. For two protons, $E_G = 493\,\mathrm{keV}$. The number of pairs at energy E falls as the Maxwell tail e^{-E/k_BT} (P3). The reaction rate is dominated by the energies where the product of the two is largest (Figure 12.2). Setting the derivative of its exponent to zero,

$$0 \overset{(1)}{=} \frac{\mathrm{d}}{\mathrm{d}E}\left(-\frac{E}{k_BT} - \sqrt{\frac{E_G}{E}}\right) \overset{(2)}{=} -\frac{1}{k_BT} + \frac{1}{2}\frac{E_G^{1/2}}{E^{3/2}}$$

$$\Longrightarrow \qquad E_0 \overset{(3)}{=} \left(\frac{E_G\,(k_BT)^2}{4}\right)^{1/3}. \tag{12.14}$$

(1) The maximum of $e^{-E/k_BT}e^{-\sqrt{E_G/E}}$ is where its exponent is largest.
(2) $\mathrm{d}/\mathrm{d}E\ E^{-1/2} = -\frac{1}{2}E^{-3/2}$.
(3) Solve for E.

$$\boxed{E_0 = \left(\frac{E_G\,(k_BT)^2}{4}\right)^{1/3}}. \tag{12.15}$$

For the Sun's centre, $E_0 = 6.1\,\mathrm{keV} = 4.5\,k_BT$. Only the rare protons far out on the Maxwell tail react, which is why a temperature of millions of kelvin, set by gravity through the virial theorem, is enough.

How steeply burning depends on temperature. Near its peak the product is close to a Gaussian of width proportional to $(E_0 k_BT)^{1/2}$, and its height is e^{-3E_0/k_BT} (the two terms of the exponent at E_0 add to $3E_0/k_BT$). The rate per pair, averaged over the Maxwell distribution with its normalisation $(k_BT)^{-3/2}$, is therefore

$$\text{rate} \propto (k_BT)^{-3/2}\,(E_0 k_BT)^{1/2}\,e^{-3E_0/k_BT} \propto T^{-2/3}\,e^{-3E_0/k_BT},$$

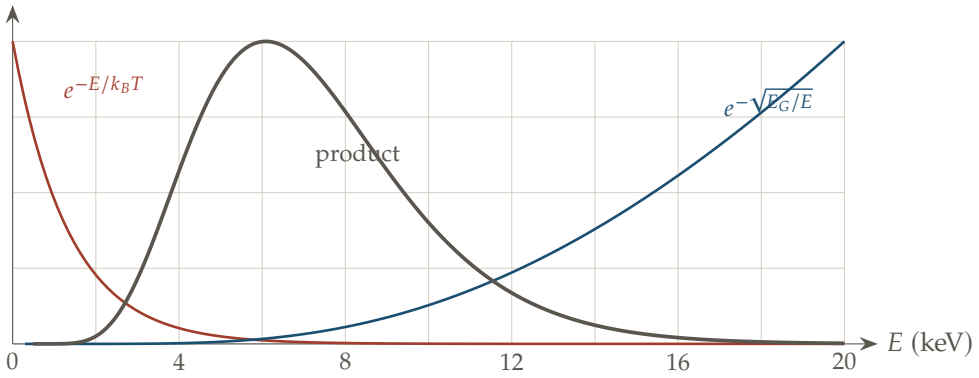


Figure 12.2: Two protons at the Sun's centre ($k_BT = 1.35\,\mathrm{keV}$, $E_G = 493\,\mathrm{keV}$). The Maxwell tail e^{-E/k_BT}, the tunnelling probability $e^{-\sqrt{E_G/E}}$, and their product, each scaled to a maximum of one within the range shown.

using $E_0 \propto T^{2/3}$. Its logarithmic slope is

$$\frac{\mathrm{d}\ln(\text{rate})}{\mathrm{d}\ln T} \overset{(1)}{=} -\frac{2}{3} - \frac{\mathrm{d}}{\mathrm{d}\ln T}\left(\frac{3E_0}{k_BT}\right) \overset{(2)}{=} -\frac{2}{3} + \frac{E_0}{k_BT}. \tag{12.16}$$

(1) The logarithm of the rate above.
(2) $3E_0/k_BT \propto T^{2/3}/T = T^{-1/3}$, so its logarithmic derivative is $-\frac{1}{3} \times 3E_0/k_BT$.

For the proton–proton chain at the Sun's centre this is 3.8, so its rate goes as about T^4. For the CNO cycle, whose slowest step is a proton meeting nitrogen ($Z_2 = 7$, $E_G = 45\,\mathrm{MeV}$), it is about 20. A one percent rise of temperature raises the CNO rate by twenty percent. This is why the thermostat of Section 10.3 holds the temperature of a burning core so tightly, why stars more massive than the Sun, whose hotter cores run on CNO, have convective cores, and why the temperature at which a fuel ignites is so sharply defined. Each fusion of four protons into helium releases 26.7 MeV, 0.7% of their rest mass, the figure used for the nuclear time in Section 10.5.

12.4 Boundary Conditions and How Stars Are Computed

The system (12.10) has four unknowns and four first-order equations, now closed by the properties of the gas.

Four first-order equations need four conditions to fix a solution, and here they are split between the two ends. Two are known at the centre and two at the surface, and neither end alone fixes the star.

Think of building a bridge from both banks. Each team starts from its own bank with a guess for the height and slope of its half. The halves meet in the middle only if the guesses were right, and the gap between them says how to correct the guesses.

A star is computed the same way. Two conditions hold at the centre, two at the surface, and the solution must satisfy all four. Integrating from both ends and closing the gap in the middle, or adjusting the whole profile at once until every equation and every condition holds, gives the star. The method is conservation, applied numerically.

In numbers, a modern stellar model divides the star into about a thousand shells, so it solves for four thousand numbers at once, and repeats this for each of thousands of time steps over the star's life.

The picture is the bridge, two halves built toward each other until they meet.

The same two-ended problem appears in the interior of planets, in the structure of accretion disks and in many boundary-value problems of physics. It gives the boundary conditions, two ways to solve the system, and how a star is evolved in time.

Four boundary conditions. At the centre there is no radius and no luminosity yet,

$$r = 0, \qquad L = 0 \qquad \text{at } m = 0. \tag{12.17}$$

At the surface the pressure has dropped to almost nothing and the temperature is that of the photosphere. With the grey atmosphere of Section 11.7,

$$P \approx 0 \ \text{(more exactly, the pressure at } \tau = \tfrac{2}{3}), \qquad L = 4\pi R^2 \sigma_{\mathrm{SB}} T^4 \qquad \text{at } m = M. \tag{12.18}$$

The second surface condition says that the temperature at $m = M$ is the effective temperature. Two conditions at each end make this a **two-point boundary value problem**.

Shooting. Guess the central pressure P_c and temperature T_c and integrate outward from the centre. Guess the radius R and luminosity L and integrate inward from the surface. Stop both at a fitting mass m_f in between. The four differences between the two solutions there, Δr, ΔP, ΔL and ΔT, are functions of the four guesses. Newton's method corrects all four guesses at once,

$$\begin{pmatrix} \delta P_c \\ \delta T_c \\ \delta R \\ \delta L \end{pmatrix} = - \left[\frac{\partial(\Delta r, \Delta P, \Delta L, \Delta T)}{\partial(P_c, T_c, R, L)} \right]^{-1} \begin{pmatrix} \Delta r \\ \Delta P \\ \Delta L \\ \Delta T \end{pmatrix}, \tag{12.19}$$

where the 4×4 matrix of derivatives is found by repeating the integrations with each guess changed slightly. Near the centre the integration starts from the series $r \approx (3m/4\pi\rho_c)^{1/3}$, since the equations themselves are singular at $r = 0$.

Relaxation. Divide the star into K shells and write the four unknowns at every shell boundary, $4K$ numbers in all. Replace each differential equation by its difference between neighbouring boundaries, $y_{k+1} - y_k = \Delta m \, f\left(\frac{1}{2}(y_k + y_{k+1})\right)$ for the vector y and right side f of Eq. (12.10). This gives $4(K-1)$ equations, and the four boundary conditions make $4K$. Starting from a guessed profile, Newton's method corrects all $4K$ numbers together. Each equation involves only two neighbouring shells, so the matrix of derivatives is zero except for small blocks along its diagonal, and it can be solved in a number of steps proportional to K. This is the Henyey method, used by modern codes such as MESA.

Evolution in time. Burning changes the composition at each m, slowly, at rates set by ε. A star is evolved by alternating two steps. Advance the composition of every shell over a time step, then solve the four equations again for a new model in equilibrium. The ordering of the three clocks of Section 10.5 is what allows this, since the star regains hydrostatic equilibrium far faster than its composition changes. Given its mass and the composition at every m, the four equations with their boundary conditions fix the structure of a star. This is the Vogt–Russell statement, true for ordinary stars, though some sets of conditions admit more than one solution.

12.5 Polytropes and the Isothermal Sphere

The full system (12.10) needs a computer.

Insight often comes from a model simple enough to solve by hand, where the effect of each ingredient can be seen. Hydrostatic equilibrium and the mass equation alone involve only P, ρ and r. They become solvable if the pressure depends on the density alone.

Think of a soft pillow and a firm one, each stacked with books. The firmness of the filling, how much its push grows when squeezed, decides how much the pillow flattens under a given weight. Nothing else about the pillow matters.

In a star the same is true when $P = \mathcal{K}\rho^{\gamma}$ for some constant $\mathcal{K}$. Then the stiffness γ alone decides how the star's mass is spread out, and the mass equation with hydrostatic equilibrium (P1 and P2), closed by a pressure from counting states (P3), give one equation for the density. This happens for degenerate gas, where Eq. (12.13) has exactly this form, and for gas that is convective throughout, where $P \propto \rho^{\gamma}$ along the adiabat.

In numbers, the model with $\gamma = \frac{4}{3}$ scaled to the Sun's mass and radius gives a central density $7.64\times10^{4}\,\mathrm{kg\,m^{-3}}$, a central pressure $1.25\times10^{16}\,\mathrm{Pa}$ and a central temperature $1.22\times10^{7}\,\mathrm{K}$, against $1.5\times10^{5}\,\mathrm{kg\,m^{-3}}$, $2.5\times10^{16}\,\mathrm{Pa}$ and $1.57\times10^{7}\,\mathrm{K}$ in detailed solar models.

The picture is Figure 12.3. A stiffer gas spreads the mass more evenly. A softer one concentrates it at the centre.

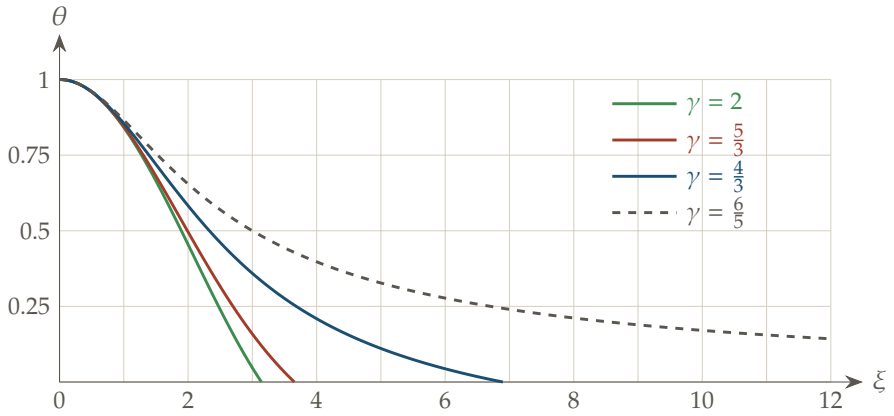


Figure 12.3: Solutions $\theta(\xi)$ of the Lane–Emden equation for $\gamma = 2$, $\frac{5}{3}$, $\frac{4}{3}$ and $\frac{6}{5}$. The density is $\rho_c\theta^{1/(\gamma-1)}$.

The same models describe white dwarfs and the cores of giant planets, fully convective low-mass stars, and, in the limit $\gamma \to 1$, the isothermal sphere that models globular clusters and the dark matter halos of galaxies. They give one equation for the density, its solutions, the mass–radius relation, and two classic applications.

The Lane–Emden equation. Divide hydrostatic equilibrium (10.2) by ρ, multiply by r^2 and differentiate. With $\mathrm{d}m/\mathrm{d}r = 4\pi r^2\rho$,

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{r^2}{\rho}\frac{\mathrm{d}P}{\mathrm{d}r}\right) = -4\pi G\rho. \tag{12.20}$$

Write the density as $\rho = \rho_c\,\theta^{1/(\gamma-1)}$, with ρ_c the central density and $\theta = 1$ at the centre. Then $P = \mathcal{K}\rho_c^{\gamma}\theta^{\gamma/(\gamma-1)}$, and

$$\frac{1}{\rho}\frac{\mathrm{d}P}{\mathrm{d}r} \overset{(1)}{=} \frac{1}{\rho_c\theta^{1/(\gamma-1)}}\,\mathcal{K}\rho_c^{\gamma}\,\frac{\gamma}{\gamma-1}\,\theta^{1/(\gamma-1)}\,\frac{\mathrm{d}\theta}{\mathrm{d}r} = \frac{\gamma\,\mathcal{K}\rho_c^{\gamma-1}}{\gamma-1}\,\frac{\mathrm{d}\theta}{\mathrm{d}r}. \tag{12.21}$$

(1) The chain rule on $\theta^{\gamma/(\gamma-1)}$, whose exponent minus one is $1/(\gamma-1)$.

Putting this into Eq. (12.20) and measuring radius in the unit

$$r_0^2 = \frac{\gamma\,\mathcal{K}\,\rho_c^{\gamma-2}}{4\pi G(\gamma-1)}, \qquad r = r_0\,\xi, \tag{12.22}$$

all the constants collect into r_0 and

$$\boxed{\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right) = -\theta^{1/(\gamma-1)}, \qquad \theta(0) = 1, \quad \theta'(0) = 0.} \tag{12.23}$$

This is the **Lane–Emden equation**. The second condition says that the density is smooth at the centre. The surface of the star is the first zero ξ_1 of θ, where density and pressure vanish.

Solutions. Three values of γ give solutions in closed form, which can be checked by putting them into Eq. (12.23).

- $\gamma \to \infty$, an incompressible star of uniform density, $\theta = 1 - \xi^2/6$, $\xi_1 = \sqrt{6}$.
- $\gamma = 2$, $\theta = \sin\xi/\xi$, $\xi_1 = \pi$.
- $\gamma = \frac{6}{5}$, $\theta = (1 + \xi^2/3)^{-1/2}$, which never reaches zero. The radius is infinite, yet the mass is finite.

The two cases that matter most are solved numerically. For $\gamma = \frac{5}{3}$, $\xi_1 = 3.6538$ and the central density is 5.99 times the mean density. For $\gamma = \frac{4}{3}$, $\xi_1 = 6.8968$ and the central density is 54.18 times the mean.

Mass and radius. The radius is $R = r_0\xi_1$. The mass follows from Eq. (12.23) itself,

$$M \overset{(1)}{=} \int_0^R 4\pi r^2\rho\,dr \overset{(2)}{=} 4\pi r_0^3\rho_c\int_0^{\xi_1}\xi^2\theta^{1/(\gamma-1)}\,d\xi \overset{(3)}{=} 4\pi r_0^3\rho_c\Big[-\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\Big]_{\xi_1}. \tag{12.24}$$

(1) The mass equation.
(2) $r = r_0\xi$ and $\rho = \rho_c\theta^{1/(\gamma-1)}$.
(3) Eq. (12.23) says $\xi^2\theta^{1/(\gamma-1)} = -\,\mathrm{d}/\mathrm{d}\xi\,(\xi^2\,\mathrm{d}\theta/\mathrm{d}\xi\,)$, and the term vanishes at $\xi = 0$.

For a fixed $\mathcal{K}$, Eq. (12.22) gives $r_0 \propto \rho_c^{(\gamma-2)/2}$, so $R \propto \rho_c^{(\gamma-2)/2}$ and $M \propto r_0^3\rho_c \propto \rho_c^{(3\gamma-4)/2}$. Eliminating ρ_c,

$$\boxed{M \propto R^{(3\gamma-4)/(\gamma-2)}.} \tag{12.25}$$

For $\gamma = \frac{5}{3}$, $M \propto R^{-3}$. A heavier star of this kind is smaller, as white dwarfs are (Section 13.2). For $\gamma = \frac{4}{3}$ the exponent vanishes. Only one mass is possible, whatever the radius. The factor $3\gamma - 4$ is the one that made a star barely bound in Eq. (10.13).

The isothermal sphere. As $\gamma \to 1$ the gas has one temperature and $P = c_s^2 \rho$. Writing $\rho = \rho_c e^{-\psi}$ and $r_0^2 = c_s^2/(4\pi G \rho_c)$, Eq. (12.20) becomes

$$\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2 \frac{\mathrm{d}\psi}{\mathrm{d}\xi}\right) = e^{-\psi}, \qquad \psi(0) = 0, \quad \psi'(0) = 0. \tag{12.26}$$

The density never reaches zero, so an isothermal sphere has no surface. Far out it approaches a power law. Trying $\rho = A/r^2$ in hydrostatic equilibrium, with $m = 4\pi A r$,

$$\begin{aligned} c_s^2 \frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{A}{r^2}\right) &= -\frac{G\,(4\pi A r)}{r^2}\,\frac{A}{r^2} \\ \implies \quad -\frac{2c_s^2 A}{r^3} &= -\frac{4\pi G A^2}{r^3} \quad \implies \quad A = \frac{c_s^2}{2\pi G}. \end{aligned} \tag{12.27}$$

So $\rho = c_s^2/(2\pi G r^2)$, which is $\rho/\rho_c = 2/\xi^2$. The numerical solution approaches this law (Figure 12.4). At $\xi = 10^4$ its density is 0.991 of $2\rho_c/\xi^2$. The mass inside r grows as $m = 2c_s^2 r/G$, so a body circling at radius r has the speed

$$v_{\mathrm{circ}}^2 = \frac{Gm}{r} = 2c_s^2, \tag{12.28}$$

the same at every radius. Rotation curves of spiral galaxies are observed to be flat far beyond their visible stars. An isothermal halo of unseen matter is the simplest model that gives this, which is part of the case for dark matter.

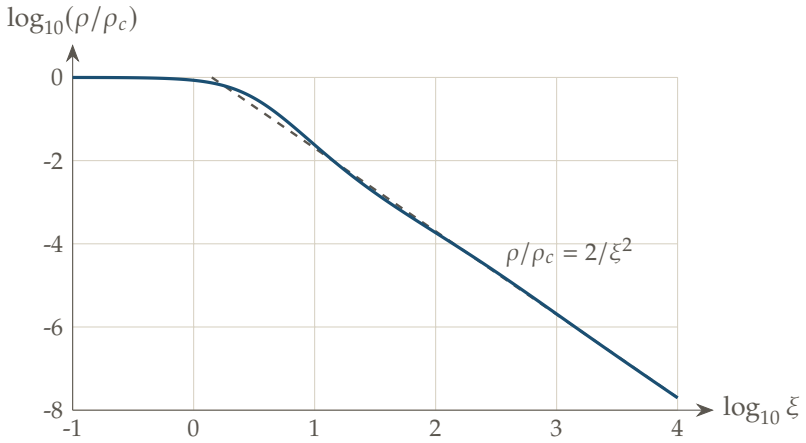


Figure 12.4: The density of the isothermal sphere against radius, on logarithmic axes (solid), and the power law $\rho/\rho_c = 2/\xi^2$ (dashed).

Eddington's standard model, and the most massive stars. Divide the momentum equation for light (Route B of Section 11.4, $\mathrm{d}P_{\mathrm{light}}/\mathrm{d}r = -\kappa\rho L/(4\pi r^2 c)$) by hydrostatic equilibrium (10.2),

$$\frac{\mathrm{d}P_{\mathrm{light}}}{\mathrm{d}P} = \frac{\kappa L(r)}{4\pi c G\, m(r)}. \tag{12.29}$$

Eddington assumed that $\kappa L(r)/m(r)$ is about the same throughout the star. Then P_{light} is a fixed fraction of the total pressure everywhere. Call the gas fraction β, so $P_{\mathrm{gas}} = \beta P$ and $P_{\mathrm{light}} = (1-\beta)P$, with

$$1 - \beta = \frac{\kappa L}{4\pi c G M} = \frac{L}{L_{\mathrm{Edd}}}, \tag{12.30}$$

by Eq. (11.17). Eliminating T between $P_{\text{gas}} = \rho k_B T/(\mu m_{\text{H}})$ and $P_{\text{light}} = aT^4/3$,

$$T \overset{(1)}{=} \frac{\mu m_{\text{H}}\, \beta P}{k_B \rho}$$

$$\Longrightarrow \quad (1-\beta)P \overset{(2)}{=} \frac{a}{3}\left(\frac{\mu m_{\text{H}}\, \beta P}{k_B \rho}\right)^4$$

$$\Longrightarrow \quad P = \left[\frac{3(1-\beta)}{a\,\beta^4}\left(\frac{k_B}{\mu m_{\text{H}}}\right)^4\right]^{1/3} \rho^{4/3}. \tag{12.31}$$

(1) The ideal-gas law for the gas share of the pressure.
(2) The pressure of light, Eq. (9.25).

So the star is a polytrope with $\gamma = \frac{4}{3}$, whose single mass is fixed by $\mathcal{K}$, and so by β. Turned around, the mass fixes β through Eddington's quartic equation. The script gives $\beta = 0.9995$ for the Sun, 0.961 at $10\,M_\odot$, 0.558 at $100\,M_\odot$ and 0.204 at $1000\,M_\odot$. In massive stars light carries a growing share of the pressure, Eq. (12.30) puts them close to the Eddington luminosity, and the effective γ of their gas and light together falls toward $\frac{4}{3}$, where Eq. (10.13) leaves them barely bound. Stars much above about a hundred solar masses blow off their outer layers or become unstable, which sets the upper end of the masses of stars. The profiles of the solar model are drawn in Figure 12.5.

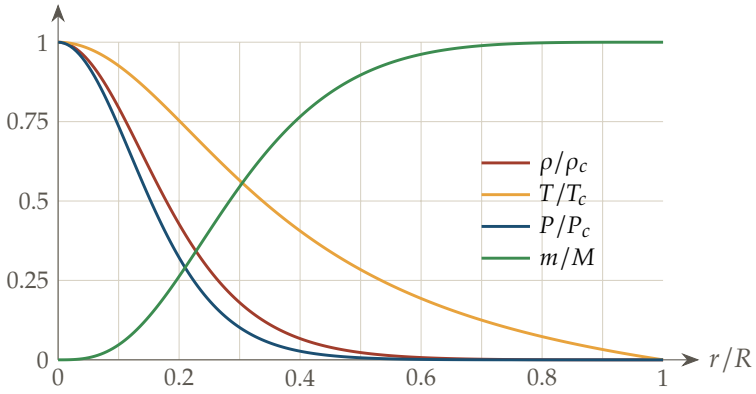


Figure 12.5: Eddington's standard model ($\gamma = \frac{4}{3}$) scaled to the Sun. Density, temperature, pressure and enclosed mass against radius, each relative to its central or total value.

12.6 Scaling Laws and the HR Diagram

Polytropes gave the shape of a star. The full equations are needed for its luminosity.

Solving them for each mass is the work of a code. Yet the way luminosity and lifetime depend on mass can be read from the equations without solving them.

Think of model ships. A model built at half size, with everything in proportion, behaves in a way that can be predicted from the scale alone, without testing each new size from scratch.

Stars of different mass are close to scaled copies of each other. If every quantity in the four equations is replaced by its typical size, each equation becomes a relation between powers of M and R. This is the comparison of scales (P6) applied to the equations themselves.

In numbers, the result $L \propto M^3$ for constant opacity, and the observed $L \propto M^{3.5}$ near the Sun, give a lifetime that falls as $M^{-2.5}$. A star of ten solar masses lives about 3×10^7 years, against 10^{10} for the Sun (Problem 12.7.4).

The picture is the HR diagram of Figure 12.6, where stars of the same radius lie along straight lines.

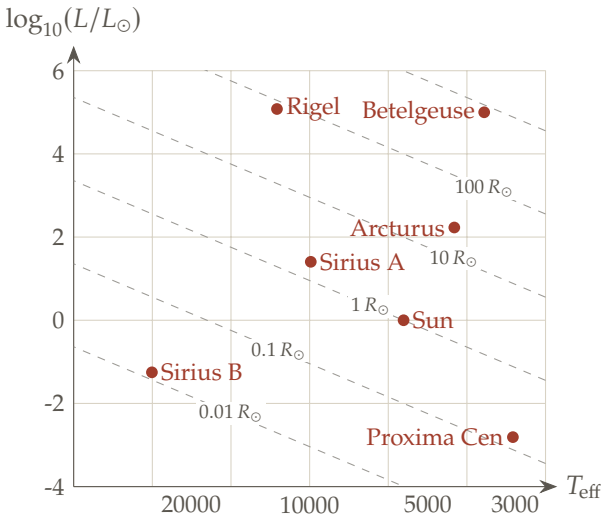


Figure 12.6: Luminosity against effective temperature, with temperature increasing to the left. Dashed lines join points of equal radius, labelled in solar radii. Dots mark seven well-known stars.

The same scaling gives the relation between the speeds of stars in galaxies and the galaxies' luminosities, and the scaling relations of galaxy clusters. It gives the mass–luminosity relation, the smallest mass of a star, and the layout of the HR diagram.

Homology. Replace $\mathrm{d}/\mathrm{d}m$ by $1/M$, r by R and every other quantity by its typical size. The momentum equation (12.3) and the ideal gas give

$$P \sim \frac{GM^2}{R^4}, \qquad T \sim \frac{\mu m_\mathrm{H} P}{k_B \rho} \sim \frac{\mu m_\mathrm{H}}{k_B} \frac{GM^2/R^4}{M/R^3} = \frac{\mu m_\mathrm{H}\, GM}{k_B R}, \tag{12.32}$$

which is the virial temperature again. The radiative equation (12.6) gives

$$\begin{aligned} L &\overset{(1)}{=} \frac{64\pi^2 ac\, R^4 T^4}{3\kappa M} \\ &\overset{(2)}{=} \frac{64\pi^2 ac\, R^4}{3\kappa M} \left(\frac{\mu m_\mathrm{H}\, GM}{k_B R}\right)^4 \\ &\propto \frac{\mu^4 M^3}{\kappa}. \end{aligned} \tag{12.33}$$

(1) $\mathrm{d}T/\mathrm{d}m \sim -T/M$ in Eq. (12.6), solved for L.
(2) The temperature above.

$$\boxed{L \propto \frac{\mu^4 M^3}{\kappa}}. \tag{12.34}$$

The radius has dropped out, and so has the burning. A star's luminosity is set by how fast light can leak out of it, not by how fast it burns. The burning adjusts, through the thermostat, to supply what leaks out. With electron-scattering opacity, as in massive stars, $L \propto M^3$. Kramers opacity, which grows at the lower temperatures of smaller stars, steepens this, and near the Sun $L \propto M^{3.5}$ is observed. The fuel scales as M, so the lifetime scales as $M/L \propto M^{-2.5}$.

The smallest stars. A contracting ball of gas heats up (Section 10.3), but as its density rises its electrons become degenerate and start to push without heating. Add the electrons' Fermi energy to the virial balance of a uniform sphere of ionised hydrogen, with $N = M/m_{\mathrm{H}}$ protons and as many electrons. The protons carry the thermal energy, $\frac{3}{2}Nk_B\bar{T}$, and the electrons their degenerate energy, $\frac{3}{5}NE_F$ with $E_F = (\hbar^2/2m_e)(9\pi N/4R^3)^{2/3}$. The virial theorem, $2\times$ kinetic $= |U| = \frac{3}{5}GM^2/R$, gives

$$k_B\bar{T} = \frac{GMm_{\mathrm{H}}}{5R} - \frac{2}{5}E_F = \frac{\mathcal{A}}{R} - \frac{\mathcal{B}}{R^2}, \tag{12.35}$$

with $\mathcal{A} = GMm_{\mathrm{H}}/5$ and $\mathcal{B} = (\hbar^2/5m_e)(9\pi/4)^{2/3}N^{2/3}$. As R shrinks the temperature first rises and then falls, with a maximum at $R = 2\mathcal{B}/\mathcal{A}$,

$$k_B\bar{T}_{\max} = \frac{\mathcal{A}^2}{4\mathcal{B}} \propto M^{4/3}. \tag{12.36}$$

Hydrogen burns fast enough to stop the contraction only above a few million kelvin. Setting $\bar{T}_{\max} = 3\times10^6\,\mathrm{K}$ gives $M \approx 0.16\,M_\odot$. The centre of a star is about twice as hot as its mean, which lowers this to near the $0.08\,M_\odot$ found by detailed models. Lighter bodies never ignite hydrogen. They are brown dwarfs, and they cool forever at a radius near the peak, about 1.5 times Jupiter's.

The HR diagram. The Stefan–Boltzmann law, $L = 4\pi R^2\sigma_{\mathrm{SB}}T_{\mathrm{eff}}^4$ (Eq. (4.26)), gives

$$\log L = 2\log R + 4\log T_{\mathrm{eff}} + \text{constant},$$

so in a diagram of $\log L$ against $\log T_{\mathrm{eff}}$, stars of equal radius lie on straight lines (Figure 12.6). Stars burning hydrogen in their cores form the main sequence, running from cool dim dwarfs like Proxima Centauri to hot bright stars like Sirius A, ordered by mass through Eq. (12.34). Giants and supergiants like Arcturus and Betelgeuse lie above it at large radius, and white dwarfs like Sirius B below it at about a hundredth of the Sun's radius.

A star leaves the main sequence when its core runs out of hydrogen. The core, now helium, has no burning to support it. It loses energy and, by Section 10.3, contracts and heats. Hydrogen burns on in a shell around it, the shell's output rises as the core shrinks and heats, and the extra luminosity swells the envelope to a hundred times its former radius. The star moves to the upper right of the diagram and becomes a red giant. When the core reaches about $10^8\,\mathrm{K}$, helium ignites.

12.7 Problem Set: The Equations of a Star

12.7.1 Problem 1: Mean Molecular Weights

Problem Statement: Find μ and μ_e for fully ionised gas of (a) solar composition, $X = 0.70$, $Y = 0.28$, $Z = 0.02$, (b) pure helium, and (c) pure carbon.

Essential Ideas:

- Count particles and electrons per mass m_{H}, Eq. (12.11).
- Carbon, $A = 12$, has a nucleus and six electrons, seven particles.

Detailed Solutions:

(a) **Solar**. $1/\mu = 1.40 + 0.21 + 0.01 = 1.62$ and $1/\mu_e = 0.85$.

$$\boxed{\mu = 0.62, \qquad \mu_e = 1.18.}$$

(b) **Helium**. $1/\mu = \frac{3}{4}$ and $1/\mu_e = \frac{1}{2}$.

$$\boxed{\mu = \tfrac{4}{3}, \qquad \mu_e = 2.}$$

(c) **Carbon**. Seven particles and six electrons per $12\,m_{\mathrm{H}}$.

$$\boxed{\mu = \tfrac{12}{7} = 1.71, \qquad \mu_e = 2.}$$

12.7.2 Problem 2: Is It Degenerate?

Problem Statement: Compare the electrons' Fermi energy with $k_B T$ (a) at the Sun's centre ($\rho = 1.5 \times 10^5\,\mathrm{kg\,m^{-3}}$, $T = 1.57 \times 10^7\,\mathrm{K}$, $\mu_e = 1.18$) and (b) at the centre of a white dwarf ($\rho = 10^9\,\mathrm{kg\,m^{-3}}$, $T = 10^7\,\mathrm{K}$, $\mu_e = 2$). (c) In case (b), are the electrons relativistic?

Essential Ideas:

- $n_e = \rho/(\mu_e m_{\mathrm{H}})$, p_F from Eq. (12.12), $E_F = p_F^2/2m_e$ for slow electrons.

Detailed Solutions:

(a) **Sun**.

$$\boxed{E_F/k_B T \simeq 0.5.}$$

Mildly degenerate. The ideal-gas law is accurate to some percent there.

(b) **White dwarf**.

$$\boxed{E_F/k_B T \simeq 190.}$$

Strongly degenerate. The temperature barely matters for the pressure.

(c) **Relativistic?** $p_F/m_e c = 0.80$. The electrons move at a large fraction of c, and neither limit of Eq. (12.13) is exact.

12.7.3 Problem 3: The Gamow Peak

Problem Statement: At the Sun's centre, $k_B T = 1.35\,\mathrm{keV}$. (a) Find the Gamow peak E_0 for two protons ($E_G = 493\,\mathrm{keV}$) and the temperature exponent of the rate. (b) Repeat for a proton and a nitrogen nucleus ($E_G = 45\,\mathrm{MeV}$).

Essential Ideas:

- $E_0 = (E_G (k_B T)^2/4)^{1/3}$, Eq. (12.15), and the exponent $E_0/k_B T - \frac{2}{3}$.

Detailed Solutions:

(a) **Two protons**. $E_0 = (493 \times 1.35^2/4)^{1/3}\,\mathrm{keV}$.

$$\boxed{E_0 \simeq 6.1\,\mathrm{keV} = 4.5\,k_B T, \qquad \text{rate} \propto T^{3.8}.}$$

(b) **Proton and nitrogen**.

$$\boxed{E_0 \simeq 27\,\mathrm{keV} = 20\,k_B T, \qquad \text{rate} \propto T^{19.6}.}$$

12.7.4 Problem 4: Lifetimes

Problem Statement: Take $L \propto M^{3.5}$ and a main-sequence lifetime of 10^{10} years for the Sun. Estimate the lifetimes of stars of 0.5, 10 and $50\,M_\odot$.

Essential Ideas:

- Lifetime $\propto M/L \propto M^{-2.5}$.

Detailed Solutions:

$$\boxed{0.5\,M_\odot:\ 6\times10^{10}\ \text{yr}, \qquad 10\,M_\odot:\ 3\times10^{7}\ \text{yr}, \qquad 50\,M_\odot:\ 6\times10^{5}\ \text{yr}.}$$

Every star of half a solar mass ever formed is still on the main sequence, since the universe is only 1.4×10^{10} years old. Massive stars live and die while the cloud they formed in is still nearby.

12.7.5 Problem 5: The Temperature Fall of the Earth's Air

Problem Statement: The Earth's lower atmosphere is stirred by convection. Use the convective gradient, Eq. (12.9), with $\gamma = \frac{7}{5}$ for air (mean molecular mass 28.97 u) and hydrostatic equilibrium near the ground to find how fast the temperature falls with height.

Essential Ideas:

- $\mathrm{d}\ln T/\mathrm{d}\ln P = (\gamma-1)/\gamma$, and $\mathrm{d}P/\mathrm{d}z = -\rho g$ with $\rho = P\mu m_{\mathrm{H}}/(k_B T)$.

Detailed Solutions:

$$\frac{\mathrm{d}T}{\mathrm{d}z} \overset{(1)}{=} \frac{T}{P}\,\frac{\gamma-1}{\gamma}\,\frac{\mathrm{d}P}{\mathrm{d}z} \overset{(2)}{=} -\frac{\gamma-1}{\gamma}\,\frac{\mu m_{\mathrm{H}}\,g}{k_B}.$$

(1) The convective gradient.
(2) Hydrostatic equilibrium with the ideal gas, $\mathrm{d}P/\mathrm{d}z = -P\mu m_{\mathrm{H}} g/(k_B T)$.

$$\boxed{\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{2}{7}\times\frac{28.97\times1.661\times10^{-27}\times9.81}{1.381\times10^{-23}}\,\mathrm{K\,m^{-1}} \simeq -9.8\,\mathrm{K\,km^{-1}}.}$$

This is the dry adiabatic lapse rate of meteorology. The same criterion that decides where the Sun convects decides how cold mountain tops are.

Chapter 13

Degenerate and Compact Stars

When a star has burned what it can, its core is left without a source of heat. It contracts, and by Section 10.3 it heats, until something other than heat holds it up. Two things can. The degeneracy pressure of electrons holds up white dwarfs, and that of neutrons holds up neutron stars. When neither suffices, nothing does, and the core becomes a black hole.

The word relativistic is used here in two separate senses, and keeping them apart is most of the chapter. Relativistic particles, electrons moving near the speed of light, soften the pressure to $\gamma = \frac{4}{3}$, and that alone, in plain Newtonian gravity, sets the maximum mass of a white dwarf. Relativistic gravity changes the structure equations themselves, and it matters only when a star is compact enough. General relativity itself is not recapped. The relativistic forms below are quoted from it.

13.1 Compactness

The equations of Chapter 12 used Newtonian gravity throughout.

They cannot hold for every star. A body compact enough must be described by general relativity, and the question is where the line falls.

Think of a ball thrown up from the surface of a small asteroid. A gentle throw sends it away for good. On the Earth the same throw brings it straight back. How hard gravity grips a body at the surface depends on how much mass is packed into how small a radius.

The grip of gravity, measured against the speed of light, is the ratio $GM/(Rc^2)$, the **compactness**. It compares the escape energy per unit mass at the surface with c^2. Every correction that general relativity makes to the structure of a star is of this size, so it decides whether Newtonian gravity is good enough (P6).

In numbers, the Sun has $GM/(Rc^2) = 2.1 \times 10^{-6}$, the white dwarf Sirius B 2.6×10^{-4}, and a neutron star of $1.4\,M_\odot$ and 12 km radius 0.17.

The picture is a scale running from 0 to $\frac{1}{2}$. The Sun sits at the bottom, white dwarfs a little above, neutron stars a third of the way up, and a black hole at the top.

The same ratio sets the gravitational redshift of light leaving a surface, the bending of starlight, and the share of rest mass released when matter falls onto a compact object. It gives the size of every relativistic correction.

The redshift of Sirius B, $z = 2.6 \times 10^{-4}$, shifts its spectral lines as a recession at $77\,\mathrm{km\,s^{-1}}$ would, close to what is measured. Why so dense a star can exist at all is worked out in Section 13.2, and Newtonian gravity is enough for it.

Object	$GM/(Rc^2)$	surface redshift z	gravity treated by
Sun	2.1×10^{-6}	2.1×10^{-6}	Newton
Sirius B (white dwarf)	2.6×10^{-4}	2.6×10^{-4}	Newton
Neutron star, 1.4 $M_\odot$, 12 km	0.17	0.24	general relativity
Black hole	0.5	–	general relativity

Table 13.1: Compactness and surface redshift, $z = (1 - 2GM/Rc^2)^{-1/2} - 1$.

13.2 White Dwarfs

The pressure of degenerate electrons is a power of the density alone, $\rho^{5/3}$ for slow electrons and $\rho^{4/3}$ for fast ones (Eq. (12.13)). A star held up by it is therefore a polytrope (Section 12.5).

A white dwarf of the Sun's mass is the size of the Earth. As its mass grows, its electrons are squeezed harder and move faster. The polytrope of slow electrons cannot describe the heaviest ones.

Think of a crowd in a hall, pressed toward the front by a slope. Each person's push on the person behind grows the harder they are squeezed, at first. If everyone is already pushing as hard as they can, adding more people down the slope only crushes the front, and no new push appears to stop it.

Electrons near the speed of light behave like that crowd. Their momentum can keep growing, but their speed cannot exceed c, so their pressure grows more slowly with density, $\gamma \to \frac{4}{3}$. By the virial theorem (Section 10.2), a star with $\gamma = \frac{4}{3}$ is barely bound, and by the mass–radius relation (12.25) only one mass fits. Above that mass, degenerate electrons cannot hold a star up. Counting states (P3) and the virial balance of a sphere (P1, P2) give that mass.

In numbers, the limit is 1.456 $M_\odot$ for the carbon and oxygen of white dwarfs, with two atomic mass units per electron, $\mu_e = 2$.

The picture is Figure 13.1, the radius of white dwarfs shrinking as their mass grows, down to zero at the limit.

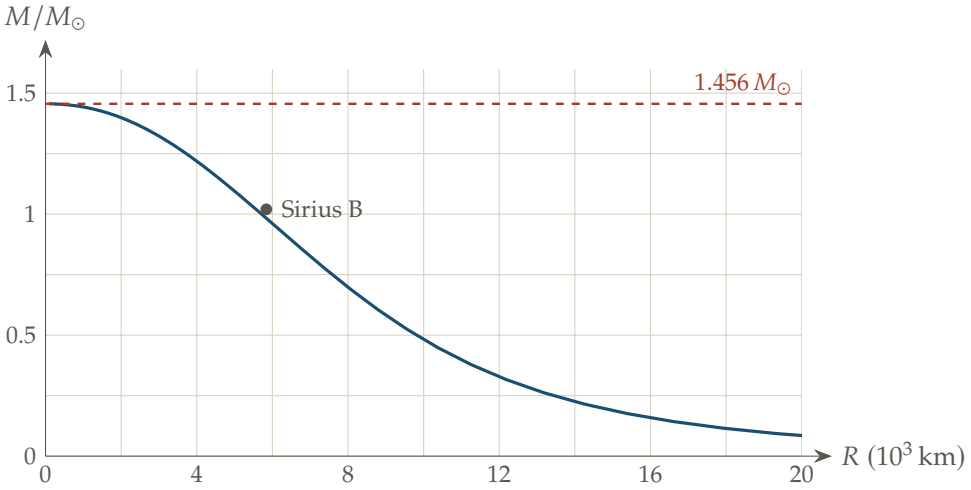


Figure 13.1: The mass of white dwarfs against their radius, computed from the exact pressure of degenerate electrons with $\mu_e = 2$. The dashed line marks 1.456 $M_\odot$. The dot marks the observed mass and radius of Sirius B.

The same limit sets the mass of the iron core of a massive star at the moment it collapses

into a neutron star, and the mass at which a white dwarf gaining matter from a companion explodes as a type Ia supernova. It gives the mass–radius relation of white dwarfs and the Chandrasekhar mass, by two routes.

Slow electrons. A light white dwarf has slow electrons, so $\gamma = \frac{5}{3}$ and, by Eq. (12.25),

$$R \propto M^{-1/3}. \tag{13.1}$$

Adding mass makes it smaller. The degenerate pressure depends on the density alone, so a heavier star must be denser to hold itself up.

Fast electrons, the Chandrasekhar mass. For fast electrons, $P = \mathcal{K}\rho^{4/3}$ with $\mathcal{K} = (hc/8)(3/\pi)^{1/3}/(\mu_e m_{\mathrm{u}})^{4/3}$ from Eq. (12.13) and $n_e = \rho/(\mu_e m_{\mathrm{u}})$, where m_{u} is the atomic mass unit. With $\gamma = \frac{4}{3}$, Eq. (12.22) gives $r_0^2 = \mathcal{K}\rho_c^{-2/3}/(\pi G)$, so $r_0^3\rho_c = (\mathcal{K}/\pi G)^{3/2}$ does not depend on ρ_c, and the mass formula of Section 12.5 gives one mass only,

$$\begin{aligned} M_{\mathrm{Ch}} &\overset{(1)}{=} 4\pi\Big[-\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\Big]_{\xi_1}\left(\frac{\mathcal{K}}{\pi G}\right)^{3/2} \\ &\overset{(2)}{=} 4\pi \times 2.0182 \times \left(\frac{hc}{8\pi G}\right)^{3/2}\left(\frac{3}{\pi}\right)^{1/2}\frac{1}{(\mu_e m_{\mathrm{u}})^2}. \end{aligned} \tag{13.2}$$

(1) The mass of a polytrope, $4\pi r_0^3\rho_c[-\xi^2\theta']_{\xi_1}$, with $r_0^3\rho_c = (\mathcal{K}/\pi G)^{3/2}$.

(2) The Lane–Emden value for $\gamma = \frac{4}{3}$ and $\mathcal{K}$ written out.

$$\boxed{M_{\mathrm{Ch}} = \frac{5.83}{\mu_e^2}\,M_\odot = 1.456\,M_\odot \quad (\mu_e = 2).} \tag{13.3}$$

The mass contains h, c, G and the mass per electron, and nothing about the star's history. Quantum mechanics (h), relativity (c) and gravity (G) together fix a mass of the order of the Sun's.

The same mass by an energy argument. Landau found the limit by comparing two energies of one electron in a star of N electrons and radius R. Its share of gravitational energy is about $-GM\mu_e m_{\mathrm{u}}/R$, with $M = N\mu_e m_{\mathrm{u}}$. A fast electron confined to a spacing $R/N^{1/3}$ has, by the uncertainty principle, a momentum of about $\hbar N^{1/3}/R$ and so a kinetic energy of about $\hbar c N^{1/3}/R$. Both scale as $1/R$, so shrinking the star changes their ratio not at all (P6). If gravity wins at one radius, it wins at every radius, and the star collapses. The two are equal when

$$\begin{aligned} &\frac{\hbar c\, N^{1/3}}{R} = \frac{G\,N(\mu_e m_{\mathrm{u}})^2}{R} \\ \Longrightarrow\quad &N = \left(\frac{\hbar c}{G(\mu_e m_{\mathrm{u}})^2}\right)^{3/2}, \qquad M = N\mu_e m_{\mathrm{u}} = \left(\frac{\hbar c}{G}\right)^{3/2}\frac{1}{(\mu_e m_{\mathrm{u}})^2}. \end{aligned} \tag{13.4}$$

This is $0.47\,M_\odot$ for $\mu_e = 2$. The estimate has the right combination of constants and falls short of Eq. (13.3) only by a number, 3.1, that the full polytrope supplies.

The full curve. Real white dwarfs have electrons that are neither slow nor at the speed of light. Figure 13.1 uses the exact pressure of degenerate electrons at any speed, integrated in Newtonian hydrostatic equilibrium. Light white dwarfs follow $R \propto M^{-1/3}$. As the mass rises toward $1.456\,M_\odot$ the radius falls to zero. Sirius B, $1.02\,M_\odot$, lies on the curve at 5566 km, against 5840 km measured, close enough given that the model ignores temperature and composition details. Throughout, $GM/(Rc^2) < 10^{-3}$, so Newtonian gravity is enough. The limit comes from the particles being relativistic, not gravity.

13.3 The Relativistic Structure Equations

The four structure equations of Section 12.2 assumed Newtonian gravity.

A neutron star has $GM/(Rc^2) \approx 0.2$. Corrections of that size cannot be dropped, and each equation must be replaced by its general-relativistic form.

Think of weighing a box of hot gas. Heat the gas and the box gets heavier, because energy has mass. Squeeze the gas and its pressure adds to the weight as well. In everyday life the extra weight is far too small to notice.

In a neutron star it is not small. The pressure that holds the star up is itself a source of gravity, so pushing harder also pulls harder. General relativity, with conservation of energy and momentum (P1) in curved space-time, gives the equations. They are quoted here from general relativity, not derived again.

In numbers, the pressure at the centre of a neutron star is between about a tenth and a half of its energy density ρc^2. At the centre of the Sun the same ratio is about 2×10^{-6}.

The picture is the Newtonian structure equations of Section 12.2, each with a few correction factors, every one of them close to one when $GM/(rc^2)$ and $P/(\rho c^2)$ are small.

The same equations, in the form given by Thorne (1977), are used for neutron stars, for the hottest proto-neutron stars just after a supernova, and for any star compact enough. They give the full relativistic set, side by side with the Newtonian one.

The metric. A static spherical star has the metric

$$ds^2 = -e^{2\Phi/c^2}c^2dt^2 + \left(1 - \frac{2Gm(r)}{rc^2}\right)^{-1} dr^2 + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right), \tag{13.5}$$

where $\Phi(r)$ becomes the Newtonian potential of Section 9.2 far from the star, and $m(r)$ is the mass inside r.

	Newtonian	**General relativity**
Mass	$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2\rho$	$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2\rho$, ρc^2 = total energy density
Momentum	$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm\rho}{r^2}$	$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{G\left(\rho + \frac{P}{c^2}\right)\left(m + \frac{4\pi r^3 P}{c^2}\right)}{r^2\left(1 - \frac{2Gm}{rc^2}\right)}$
Potential	$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{Gm}{r^2}$	$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{c^2}{\rho c^2 + P}\frac{\mathrm{d}P}{\mathrm{d}r}$
Energy	$\frac{\mathrm{d}L}{\mathrm{d}r} = 4\pi r^2\rho\,\varepsilon$	$\frac{\mathrm{d}}{\mathrm{d}r}\left(L\,e^{2\Phi/c^2}\right) = \frac{4\pi r^2\rho\,\varepsilon\,e^{2\Phi/c^2}}{\sqrt{1 - 2Gm/rc^2}}$
Transport	$L = -\frac{16\pi ac\,r^2T^3}{3\kappa\rho}\frac{\mathrm{d}T}{\mathrm{d}r}$	$L = -\frac{16\pi ac\,r^2T^3}{3\kappa\rho}\sqrt{1 - \frac{2Gm}{rc^2}}\,e^{-\Phi/c^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(T\,e^{\Phi/c^2}\right)$

Table 13.2: The structure equations of a static star in Newtonian gravity and in general relativity (energy equation in a star that does not change).

Reading the corrections. The momentum equation is the Tolman–Oppenheimer–Volkoff (TOV) equation. Each of its three new factors is one physical effect.

- $\rho + P/c^2$ in place of ρ. Pressure has inertia. A parcel under pressure is harder to hold up.

- $m + 4\pi r^3 P/c^2$ in place of m. Pressure gravitates. The pressure that holds the star up adds to the pull it must resist.
- $(1 - 2Gm/rc^2)^{-1}$. Space inside the star is curved, and the radial distance is longer than dr.

All three make gravity stronger. Raising the pressure to support more mass also raises gravity, so beyond some mass no equation of state can win. This is why neutron stars, unlike white dwarfs, have a maximum mass set by gravity itself, whatever the particles.

The energy equation conserves the luminosity as seen from far away, $L\,e^{2\Phi/c^2}$, where one factor e^{Φ/c^2} is the redshift of each photon's energy and the other is the slowing of clocks deep in the potential. The square root turns dr into a proper volume.

Equilibrium in a gravitational field. With no heat flow, $L = 0$, the transport equation gives not a uniform T but

$$\boxed{T\,e^{\Phi/c^2} = \text{constant.}} \tag{13.6}$$

This is the Tolman–Ehrenfest relation. In a gravitational field, a body in full thermal equilibrium is hotter at the bottom, by the same factor by which light climbing out is redshifted. Heat flows only when $T\,e^{\Phi/c^2}$ changes from place to place. The statement in Section 10.6 that global equilibrium means one temperature throughout is the Newtonian limit of this. For the Sun the difference is two parts in a million. Across a neutron star it is about twenty percent.

Mass and binding. In general relativity $m(r)$ counts the total energy inside r, including the negative gravitational binding energy. The mass M that governs an orbit around the star is smaller than the rest mass of all its particles, by about $GM/(Rc^2)$ of it. For a neutron star the difference is about a tenth of the rest mass, 3×10^{46} J, released when the core collapses, almost all of it as neutrinos (Problem 13.6.3).

13.4 Computing a Neutron Star

Table 13.2 gives the equations, and the core of a collapsed star gives the material, neutrons packed to nuclear density.

How are the mass and radius of a neutron star actually found, and what does general relativity do to them?

Think of stacking sand into a cone. Pour more and the cone grows wider and taller. With dry sand the slope holds, and the cone can grow as large as the sand allows. With wet sand the weight of the top crushes the bottom past some height. Then the pile only spreads, and adding sand does not make it taller.

A neutron star is built in the same spirit, one central density at a time. Each central density gives one star, with its own mass and radius. Going to higher central density gives heavier stars, up to a point beyond which the extra gravity of the extra pressure wins and the mass falls. That point is the maximum mass. Finding it is the TOV equation (P1 and P2 in general relativity) solved with the pressure of degenerate neutrons (P3).

In numbers, an ideal gas of neutrons in general relativity reaches at most $0.71\,M_\odot$, at a radius of 9.3 km. The same gas in Newtonian gravity reaches about $5.69\,M_\odot$, eight times more.

The picture is Figure 13.2, two mass–radius curves for the same neutron gas, one turning over at a small mass.

The same procedure, with the best equations of state from nuclear physics, gives the mass–radius curves that are tested against measured neutron-star masses and radii. It gives the procedure and what it shows.

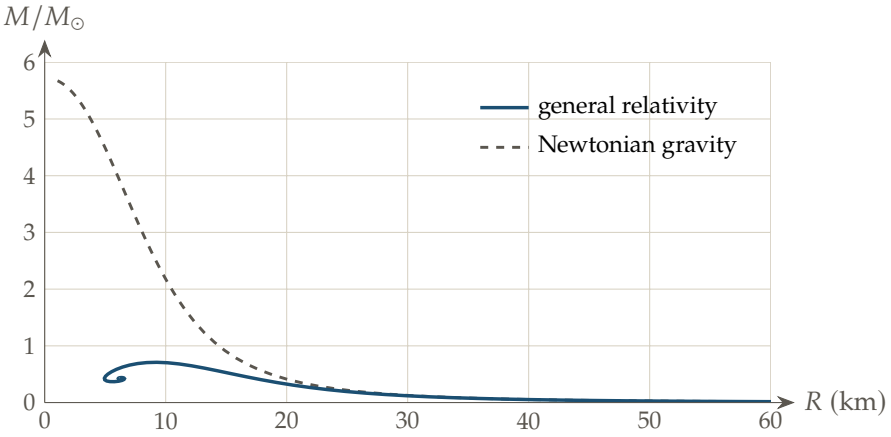


Figure 13.2: Mass against radius for stars of an ideal degenerate neutron gas, computed with the TOV equation (solid) and with Newtonian hydrostatic equilibrium (dashed). Each point is one central density.

The procedure. A cold neutron star has no burning and no heat flow that matter for its structure, so only the mass and momentum equations of Table 13.2 are needed, with an equation of state $P(\rho)$.

1. Choose a central density ρ_c, so $P(0) = P(\rho_c)$ and $m(0) = 0$.
2. Integrate $\mathrm{d}m/\mathrm{d}r$ and the TOV equation outward until $P = 0$. That radius is R and $m(R)$ is M.
3. Repeat for a range of ρ_c. Each gives one point (R, M).

Both boundary conditions sit at the centre, unlike the two-ended problem of Section 12.4, so a single outward integration finishes each star. Along the curve, stars with $\mathrm{d}M/\mathrm{d}\rho_c > 0$ are stable. Past the maximum, where $\mathrm{d}M/\mathrm{d}\rho_c < 0$, a small squeeze grows instead of rebounding.

What the curves show. Figure 13.2 uses the exact pressure of an ideal degenerate neutron gas, and its total energy density as the source of gravity. At low mass, where the star is large and $GM/(Rc^2)$ small, the two curves agree. As the mass grows they part. The Newtonian curve climbs toward the Chandrasekhar-like limit for neutrons, about $5.7\,M_\odot$, the $\mu_e = 1$ value of Eq. (13.3). The relativistic curve turns over at $0.71\,M_\odot$, the Oppenheimer–Volkoff limit, where $GM/(Rc^2) = 0.11$. General relativity has cut the maximum mass by a factor of eight. Pressure that gravitates defeats itself.

Real neutron stars. Pulsars in binaries are weighed with the timing methods of Chapter 6 (Section 6.18), and some are found at about $2\,M_\odot$, far above $0.71\,M_\odot$. Neutrons are not an ideal gas. Nuclear forces repel them strongly at the highest densities and stiffen the pressure, and the best equations of state give maximum masses of about 2 to $2.3\,M_\odot$. The maximum mass itself, and its derivation in general relativity, are taken here from the theory of compact objects, not derived again.

13.5 Black Holes

Above the maximum mass of a neutron star, no pressure can stop a collapse. By the reading of the TOV equation, every increase of pressure adds to the pull it fights.

The collapse continues until the matter has passed inside its own Schwarzschild radius, $r_s = 2GM/c^2$, 3 km for the Sun's mass and 30 km for ten solar masses, where the compactness reaches $\frac{1}{2}$ and nothing, not even light, comes back out. Stars that end their lives with cores well above the neutron-star limit leave black holes. Their properties belong to general relativity and are not repeated here.

13.6 Problem Set: Degenerate and Compact Stars

13.6.1 Problem 1: Gravitational Redshifts

Problem Statement: Light leaving the surface of a static star is redshifted by $1 + z = (1 - 2GM/Rc^2)^{-1/2}$. Find z for (a) the Sun, (b) Sirius B ($1.02\,M_\odot$, 5840 km), expressed as a Doppler velocity cz, and (c) a neutron star of $1.4\,M_\odot$ and 12 km.

Essential Ideas:

- For small compactness, $z \approx GM/(Rc^2)$.

Detailed Solutions:

(a) **Sun**.

$$\boxed{z \simeq 2.1 \times 10^{-6}, \qquad cz \simeq 0.64\,\mathrm{km\,s^{-1}}.}$$

(b) **Sirius B**.

$$\boxed{z \simeq 2.6 \times 10^{-4}, \qquad cz \simeq 77\,\mathrm{km\,s^{-1}}.}$$

The measured shift of Sirius B's lines, about $80\,\mathrm{km\,s^{-1}}$, was one of the first confirmations that white dwarfs are as dense as the theory requires.

(c) **Neutron star**. $2GM/Rc^2 = 0.345$, so $1 + z = (0.655)^{-1/2}$.

$$\boxed{z \simeq 0.24.}$$

13.6.2 Problem 2: The Chandrasekhar Mass for Other Compositions

Problem Statement: Using Eq. (13.3), find the limiting mass of a degenerate core made of (a) helium, carbon or oxygen, (b) iron (^{56}Fe, 26 electrons), and (c) pure hydrogen, if such a white dwarf existed.

Essential Ideas:

- $M_{\mathrm{Ch}} = 5.83\,M_\odot/\mu_e^2$, where μ_e is the mass per electron in atomic mass units.

Detailed Solutions:

(a) **Helium, carbon, oxygen**. $\mu_e = 2$.

$$\boxed{M_{\mathrm{Ch}} \simeq 1.46\,M_\odot.}$$

(b) **Iron**. $\mu_e = 56/26 = 2.15$.

$$\boxed{M_{\mathrm{Ch}} \simeq 1.26\,M_\odot.}$$

An iron core grows in a massive star until it nears this mass, and then collapses.

(c) **Hydrogen**. $\mu_e = 1$.

$$\boxed{M_{\mathrm{Ch}} \simeq 5.8\,M_\odot.}$$

13.6.3 Problem 3: The Energy of a Supernova

Problem Statement: A neutron star of $1.4\,M_\odot$ and radius 12 km forms from the collapse of a core. (a) Estimate the energy released, using the gravitational energy of a uniform sphere. (b) What fraction of the rest energy is this? (c) Compare with all the light the Sun gives off over 10^{10} years.

Essential Ideas:

- $|U| = \frac{3}{5}GM^2/R$ (Eq. (10.11)). The core started far larger, so almost all of $|U|$ is released.

Detailed Solutions:

(a) **Energy**.

$$\boxed{|U| \simeq \frac{3}{5}\,\frac{6.674\times10^{-11}\times(2.785\times10^{30})^2}{1.2\times10^{4}}\,\mathrm{J} \simeq 3\times10^{46}\,\mathrm{J}.}$$

(b) **Fraction**. $Mc^2 = 2.5\times10^{47}$ J.

$$\boxed{|U|/Mc^2 \simeq 0.10.}$$

(c) **Against the Sun**. $L_\odot \times 10^{10}\,\mathrm{yr} = 1.2\times10^{44}$ J. A core collapse releases about two hundred times the Sun's entire lifetime output, within seconds. About 99% leaves as neutrinos, which diffuse out of the hot core (P5) in about ten seconds, as was seen from supernova 1987A.

13.6.4 Problem 4: Hotter at the Bottom

Problem Statement: (a) In thermal equilibrium, by how much does the temperature at the bottom of a 10 km column of the Earth's atmosphere exceed that at the top, according to Eq. (13.6)? (b) Across a neutron star, with Φ/c^2 changing by about 0.2, how large is the effect?

Essential Ideas:

- For a small change of potential, $\Delta T/T \approx -\Delta\Phi/c^2$, with $\Delta\Phi = g\,\Delta h$ near a planet's surface.

Detailed Solutions:

(a) **Earth**. $\Delta\Phi/c^2 = 9.81\times10^4/(9.0\times10^{16})$.

$$\boxed{\Delta T/T \simeq 1.1\times10^{-12}.}$$

Utterly negligible. The real temperature difference in the air comes from convection (Problem 12.7.5).

(b) **Neutron star**.

$$\boxed{\Delta T/T \sim 0.2.}$$

Models of cooling neutron stars must use $T\,e^{\Phi/c^2}$, not T, as the quantity that evens out.

Chapter 14

Galaxies and the Motions of Stars

A galaxy holds some 10^{11} stars with gas and dust between them, all bound by their own gravity. Galaxies in turn gather into groups and into clusters of hundreds or thousands. Seen from far enough away, each star in a galaxy and each galaxy in a cluster is a point that moves in the pull of all the others.

No new physics is needed to describe them. The gravity of a sphere (P2) and conservation of momentum (P1) turn the speed of an orbit into the mass inside it. The virial theorem of Chapter 10 turns random speeds into a mass in the same way. The random walk (P5) decides whether the stars of a system ever trade energy among themselves. Applied to galaxies, these simple tools give a result nobody expected. Most of the mass sends out no light at all.

14.1 Our Galaxy

Chapter 2 set up Galactic coordinates on the band of the Milky Way (Section 2.8). Chapter 8 found the thin gas between the stars and the 21 cm line that traces it (Section 8.4.1).

Both say where things lie on the sky. Neither says how the Galaxy is built in depth, or how much it weighs. We sit inside a dusty disk and cannot step outside to look at it.

Throw a ball straight up. It climbs until gravity has taken away its speed, and then it falls back. Thrown at the same speed on the Moon, it climbs six times higher, because the Moon pulls six times more weakly. Without any scale, the height reached at a given speed weighs the body underneath.

The stars of the Galactic disk do the same. Each one bobs up and down through the plane as it circles the centre, at a few tens of kilometres per second. The disk pulls it back, and how far it rises before turning measures that pull. The pull of a wide flat layer is set by its mass per unit area. So the thickness of the disk weighs the disk.

The ball in symbols. Thrown at speed v against the pull g, it rises to $v^2/2g$. That is 5.1 m for $10\,\mathrm{m\,s^{-1}}$ on the Earth and 31 m on the Moon, where $g = 1.62\,\mathrm{m\,s^{-2}}$. Near the Sun the disk pulls with $g \approx 6 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ (below), so a star that crosses the plane at $20\,\mathrm{km\,s^{-1}}$ rises about 100 pc before it turns back.

The picture is the disk seen edge on, a thin bright layer with stars swinging through it like weights on springs. The faster ones reach further out.

The same balance is hydrostatic equilibrium (Section 10.1) for a gas made of stars. It is momentum conservation (P1) with the gravity of a flat layer from Gauss's law (P2). It sets the thickness of the gas around a young star, and it is how Oort first weighed the disk near the Sun. It gives the parts of the Galaxy, the mass at its centre, and the mass per unit area of its disk.

The parts. The Galaxy has four parts, listed in Table 14.1. The **disk** holds most of the stars, nearly all the gas and dust, and the spiral arms where new stars form. The **bulge**, with a bar, fills the central few kiloparsecs with old stars. The **stellar halo** is a sparse round cloud of old stars poor in metals, with about 150 globular clusters. The **dark halo** is known only from its pull (Section 14.2), and it holds most of the mass. The Sun lies in the disk at $R_0 = 8.2$ kpc from the centre, about 20 pc north of the plane, and it circles at $v_0 \approx 230\,\mathrm{km\,s^{-1}}$, the speed of the local standard of rest of Section 3.6.

Part	Size	Mass ($M_\odot$)	Contents
Disk	scale length 2.6 kpc, thickness 0.3 and 0.9 kpc	stars 4×10^{10}, gas 1×10^{10}	young and old stars, gas, dust, spiral arms
Bulge and bar	central 2–3 kpc	1.5×10^{10}	old stars
Stellar halo	out to 100 kpc and more	1×10^{9}	old metal-poor stars, globular clusters
Dark halo	out to 200–300 kpc	1×10^{12}	unseen
Central black hole	inside 120 AU	4×10^{6}	Sagittarius A*

Table 14.1: The parts of the Milky Way, with rounded sizes and masses. The disk has a thin and a thick component.

The picture took a long time to find. Herschel and later Kapteyn counted stars in every direction and placed the Sun near the middle of a small flattened system. They did not know about dust. Extinction (Chapter 4) hides everything beyond a few kiloparsecs in the plane, so in every direction the counts saw the same nearby patch. Shapley used the globular clusters instead. They lie mostly out of the plane, above the dust, and they form a round swarm whose centre is far away in Sagittarius. That centre is the centre of the Galaxy. The 21 cm line passes through dust and maps the gas over the whole disk. Infrared light also passes through, and shows the bulge.

The mass at the centre. The radio source Sagittarius A* sits at the zero point of Galactic coordinates (Section 2.8). Stars close to it move on Kepler orbits that infrared cameras have followed for decades. The star S2 has a period of 16.05 years and a semi-major axis of 0.1255″ on the sky. To weigh what it orbits, we use Kepler's third law $n^2a^3 = GM$ (Eq. (6.49)) with the mean motion $n = 2\pi/P$ for a period P, and divide by the same law for the Earth round the Sun,

$$\frac{M}{M_\odot} \overset{(1)}{=} \frac{(2\pi/P)^2\,a^3}{(2\pi/1\,\mathrm{yr})^2\,(1\,\mathrm{AU})^3} = \left(\frac{a}{1\,\mathrm{AU}}\right)^3\left(\frac{P}{1\,\mathrm{yr}}\right)^{-2}$$
$$\overset{(2)}{=} \frac{(0.1255\times8180)^3}{16.05^2} = 4.2\times10^6. \tag{14.1}$$

(1) Eq. (6.49) for S2, divided by the same equation for the Earth, $GM_\odot = (2\pi/1\,\mathrm{yr})^2(1\,\mathrm{AU})^3$.
(2) An angle of 1″ at 1 pc spans 1 AU (Eq. (3.2)), so 0.1255″ at 8180 pc is 1027 AU.

S2 comes within 120 AU of the centre. Four million solar masses inside so small a region, giving out very little light, can only be a black hole. The same orbit fixes R_0 as well. The Doppler shift of S2 gives its speed in kilometres per second and its motion on the sky gives the same speed in seconds of arc per year, and their ratio is the distance, 8.18 kpc.

The pull of a flat layer. Near the Sun the disk is about 0.3 kpc thick and changes over a few kiloparsecs along the plane, so locally it is a wide flat layer. Let Σ be its mass per unit area and let z be the height above its middle. By symmetry the pull points toward the plane and

depends only on $|z|$. Take a box of face area A with faces at $+z$ and $-z$, both outside most of the layer, so that it holds the mass ΣA. Gauss's law (9.9) gives

$$\begin{aligned} -4\pi G\,\Sigma A &\overset{(1)}{=} \oint \vec{g}\cdot d\vec{A} \\ &\overset{(2)}{=} -g(z)\,A - g(z)\,A \\ \Longrightarrow \qquad g(z) &= 2\pi G\Sigma. \end{aligned} \tag{14.2}$$

(1) Eq. (9.9) with the mass ΣA inside the box (P2).

(2) On the top face $\vec{g}$ points down and $d\vec{A}$ points up, and on the bottom face the reverse. On the sides $\vec{g}$ lies along the face and gives nothing.

$$\boxed{g = 2\pi G\Sigma} \tag{14.3}$$

Above the layer the pull does not weaken with height, just as the field of a charged plane does not. For $\Sigma = 70\,M_\odot\,\mathrm{pc}^{-2}$ it is $6.1\times10^{-11}\,\mathrm{m\,s}^{-2}$.

A gas of stars above the layer. Now follow one population of stars, say old dwarfs of one type, with mass density $\rho(z)$, and let most of them lie above most of the layer's mass. Let $\sigma_z^2 = \langle v_z^2\rangle$ be the mean square of their vertical speed at height z. The momentum flux (9.4) counted the momentum that particles carry across a surface as they move through it. No collisions entered that count, so it holds for stars. Its zz entry is $\rho\langle v_z^2\rangle = \rho\sigma_z^2$ when there is no mean vertical flow. In a steady flat layer only derivatives along z survive, and the z component of Eq. (9.5) reads

$$\begin{aligned} 0 &\overset{(1)}{=} \frac{\partial(\rho\langle v_z\rangle)}{\partial t} + \frac{\mathrm{d}}{\mathrm{d}z}\Pi_{zz} - \rho\, g_z \\ &\overset{(2)}{=} \frac{\mathrm{d}}{\mathrm{d}z}\left(\rho\sigma_z^2\right) + 2\pi G\Sigma\,\rho. \end{aligned} \tag{14.4}$$

(1) Eq. (9.5), z component, with no change along x or y (P1).

(2) No mean vertical motion and no change in time, $\Pi_{zz} = \rho\sigma_z^2$, and $g_z = -2\pi G\Sigma$ above the layer by Eq. (14.3).

If σ_z is the same at every height, this is $\mathrm{d}\rho/\mathrm{d}z = -\rho/h_z$, and

$$\boxed{\rho(z) \propto e^{-z/h_z}, \qquad h_z = \frac{\sigma_z^2}{2\pi G\Sigma}.} \tag{14.5}$$

This is the scale height of an atmosphere, Eq. (10.4), with $k_BT/(\mu m_\mathrm{H})$ replaced by σ_z^2 and the surface gravity by $2\pi G\Sigma$. Stars with larger vertical speeds make a thicker layer, and a heavier disk makes a thinner one. For $\sigma_z = 20\,\mathrm{km\,s}^{-1}$ and $\Sigma = 70\,M_\odot\,\mathrm{pc}^{-2}$, $h_z = 210\,\mathrm{pc}$, close to the 300 pc of the old thin disk.

Weighing the disk. Turned around, $\Sigma = \sigma_z^2/(2\pi G h_z)$. Counting stars of one type at increasing heights gives h_z, and their spectra give σ_z. Kuijken and Gilmore did this for about five hundred K dwarfs toward the south Galactic pole and found $\Sigma \approx 70\,M_\odot\,\mathrm{pc}^{-2}$ within 1.1 kpc of the plane. The stars and gas that can be seen near the Sun add up to about $50\,M_\odot\,\mathrm{pc}^{-2}$. The difference is about what a round dark halo puts into a slab of that thickness (Problem 14.6.1). Near the Sun the dark matter is spread through the halo, not packed into the disk.

14.2 Rotation Curves and Dark Matter

Chapter 6 weighed the Sun from the orbits of its planets. The gravity of a sphere (P2, Eq. (9.12)) says that an extended body pulls as if the mass inside the orbit sat at the centre.

In a galaxy the mass is spread out over many kiloparsecs, and we do not know how. What we see is light, and light need not follow mass.

Think of the planets. Mercury moves at $47\,\mathrm{km\,s^{-1}}$, the Earth at 30, Neptune at only 5.4. Further out, the planets move slower, because all the mass that holds them is in the Sun at the middle. If Neptune moved as fast as the Earth, the mass inside its orbit would have to be thirty times the mass inside the Earth's orbit, and that extra mass would have to be somewhere between the two.

The stars and gas clouds of a spiral galaxy circle its centre in the same way. Their speed at each radius gives the mass inside that radius, whether that mass shines or not. Beyond the edge of the light, the speeds should fall as Neptune's do. They stay flat. The orbital speed weighs everything inside the orbit, seen or unseen.

The planets in symbols. A circular orbit at radius r about a mass M has $v^2 = GM/r$, so $v \propto r^{-1/2}$, and Neptune at 30.07 AU moves at $29.78/\sqrt{30.07} = 5.43\,\mathrm{km\,s^{-1}}$. The Sun moves round the Galaxy at $v_0 = 230\,\mathrm{km\,s^{-1}}$ at $R_0 = 8.2\,\mathrm{kpc}$, so the mass inside the Sun's orbit is $v_0^2 R_0/G = 1.0 \times 10^{11}\,M_\odot$. One lap takes $2\pi R_0/v_0 = 220$ million years.

The picture is Figure 14.2. The speed rises steeply near the centre, then stays level far beyond the stars.

The same weighing finds the mass of the Earth from the Moon, of Jupiter from its moons, and of the black hole from S2. Here it is P2 with momentum conservation (P1) for a circle, and the flat curve is the isothermal sphere of Section 12.5. In Chapter 16 the same unseen matter decides how galaxies form at all. It gives the circular speed in any round galaxy, how to measure it from inside our own, a model with a halo, and the mass-to-light ratio.

The circular speed. To find the speed of a circular orbit at radius r in any spherical distribution of mass, we equate the acceleration of the orbit with the pull of the mass inside,

$$-\frac{v_{\mathrm{circ}}^2}{r}\,\hat{r} \overset{(1)}{=} \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} \overset{(2)}{=} \vec{g}(r) \overset{(3)}{=} -\frac{G\,m(r)}{r^2}\,\hat{r}. \tag{14.6}$$

(1) Motion on a circle at constant speed v_{circ}.
(2) A star moves under gravity alone (P1 for its momentum).
(3) The gravity of a sphere, Eq. (9.12) (P2).

$$\boxed{v_{\mathrm{circ}}^2 = \frac{G\,m(r)}{r}.} \tag{14.7}$$

The speed at r depends only on the mass inside r, however it is arranged in radius, and the mass outside drops out. That holds exactly for a round halo. For a flat disk the mass outside r also pulls, so for a disk the relation is only a good estimate. The graph of v_{circ} against r is the **rotation curve**.

Reading a rotation curve. Turned around, $m(r) = v_{\mathrm{circ}}^2 r/G$. Each shape of the curve stands for one way of spreading the mass.

Mass	$m(r)$	v_{circ}
all at the centre	constant	$\propto r^{-1/2}$
uniform sphere	$\propto r^3$	$\propto r$
flat curve	$\propto r$	constant

For a flat curve the density follows from $\mathrm{d}m/\mathrm{d}r = 4\pi r^2\rho$,

$$\rho \overset{(1)}{=} \frac{1}{4\pi r^2}\frac{\mathrm{d}m}{\mathrm{d}r} \overset{(2)}{=} \frac{1}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{v_{\text{circ}}^2\, r}{G}\right) = \frac{v_{\text{circ}}^2}{4\pi G\, r^2}. \tag{14.8}$$

(1) Eq. (9.10).
(2) Eq. (14.7) with v_{circ} the same at every radius.

$$\boxed{\rho = \frac{v_{\text{circ}}^2}{4\pi G\, r^2} \qquad \text{(flat rotation curve).}} \tag{14.9}$$

The isothermal sphere of Section 12.5 gives the same law by another route. Far from its centre it has $\rho = c_s^2/(2\pi G r^2)$ and $v_{\text{circ}}^2 = 2c_s^2$, and putting $c_s^2 = v_{\text{circ}}^2/2$ into the first gives Eq. (14.9). A flat rotation curve is what a halo of particles with one velocity spread at every radius produces.

Measuring the rotation from inside. We see our own Galaxy from a point on one of its orbits. To get v_{circ} at other radii, we measure the Doppler shift of the 21 cm line at Galactic longitude l in the plane, and work out what circular motion predicts for it.

Put the origin at the Galactic centre, with $\hat{x}$ from the Sun toward the centre ($l = 0$), $\hat{y}$ toward $l = 90°$, the direction of rotation at the Sun, and $\hat{z}$ toward the north Galactic pole (Figure 14.1). These are the axes of (U, V, W) of Section 3.6, moved to the centre. The Sun is at $\vec{r}_0 = (-R_0,\, 0)$ in the plane. A cloud at distance d along the direction of sight $\hat{\mathbf{s}} = (\cos l,\, \sin l)$ is at $\vec{r} = \vec{r}_0 + d\,\hat{\mathbf{s}}$, at distance $r = |\vec{r}|$ from the centre. Seen from the north pole the Galaxy turns clockwise. A body on a circular orbit at $\vec{r}$ with angular speed $\omega(r) = v_{\text{circ}}(r)/r$ therefore has the velocity

$$\vec{v} = \omega(r)\,\mathbf{J}\,\vec{r}, \qquad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{14.10}$$

a turn of $\vec{r}$ by $-90°$ scaled by ω. At the Sun, $\mathbf{J}\vec{r}_0 = (0,\, R_0)$, so $\vec{v}_0 = \omega_0 R_0\,\hat{y} = v_0\,\hat{y}$, as it should be, with $\omega_0 = v_0/R_0$. The Doppler shift, Eq. (3.25), measures the velocity of the cloud relative to the Sun along $\hat{\mathbf{s}}$, after the Sun's own small motion relative to the local standard of rest has been removed,

$$\begin{aligned} v_r &= \hat{\mathbf{s}}\cdot\left(\omega\,\mathbf{J}\,\vec{r} - \omega_0\,\mathbf{J}\,\vec{r}_0\right) \\ &\overset{(1)}{=} \omega\,\hat{\mathbf{s}}\cdot\mathbf{J}\vec{r}_0 + \omega\, d\,\hat{\mathbf{s}}\cdot\mathbf{J}\hat{\mathbf{s}} - \omega_0\,\hat{\mathbf{s}}\cdot\mathbf{J}\vec{r}_0 \\ &\overset{(2)}{=} (\omega - \omega_0)\,\hat{\mathbf{s}}\cdot\mathbf{J}\vec{r}_0 \\ &\overset{(3)}{=} (\omega - \omega_0)\begin{pmatrix}\cos l & \sin l\end{pmatrix}\begin{pmatrix}0 \\ R_0\end{pmatrix} = (\omega - \omega_0)\,R_0\sin l. \end{aligned} \tag{14.11}$$

(1) $\vec{r} = \vec{r}_0 + d\,\hat{\mathbf{s}}$.
(2) $\hat{\mathbf{s}}\cdot\mathbf{J}\hat{\mathbf{s}} = \cos l\sin l - \sin l\cos l = 0$. A turn by 90° makes any vector perpendicular to the vector it came from.
(3) $\mathbf{J}\vec{r}_0 = (0,\, R_0)$.

$$\boxed{v_r = \left(\frac{v_{\rm circ}(r)}{r} - \frac{v_0}{R_0} \right) R_0 \sin l.} \tag{14.12}$$

The distance d enters only through r. Two checks. Toward the centre or the anticentre, $\sin l = 0$ and $v_r = 0$, since every orbit there moves across the line of sight. A cloud on the Sun's own circle, $r = R_0$, has $v_r = 0$ too, since it keeps its distance from the Sun.

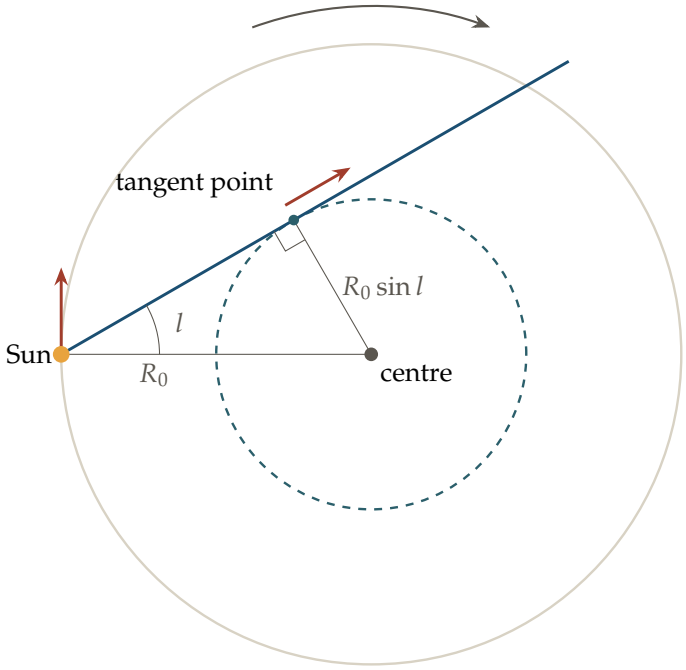


Figure 14.1: The Galactic disk seen from the north Galactic pole, to scale, with $R_0 = 8.2\,\text{kpc}$ and a line of sight at $l = 30°$. The grey circle is the Sun's orbit and the dashed circle has radius $R_0 \sin l$. The red arrows are the velocities of circular orbits at the Sun and at the tangent point. The arc at the top shows the sense of rotation.

The tangent point. Along a line of sight with $0 < l < 90°$, the distance from the centre is smallest where the line touches a circle about the centre, at $r = R_0 \sin l$ (Figure 14.1). If the angular speed $\omega = v_{\rm circ}/r$ falls outward, as it does for a flat or slowly rising curve, it is largest there, and so is v_r. Setting $r = R_0 \sin l$ in Eq. (14.12),

$$v_{r,\max} = \left(\frac{v_{\rm circ}(R_0 \sin l)}{R_0 \sin l} - \frac{v_0}{R_0} \right) R_0 \sin l = v_{\rm circ}(R_0 \sin l) - v_0 \sin l, \tag{14.13}$$

so

$$\boxed{v_{\rm circ}(R_0 \sin l) = v_{r,\max} + v_0 \sin l.} \tag{14.14}$$

The highest velocity in the 21 cm line at each longitude gives the rotation speed at one radius, with no distance needed except R_0. Sweeping l from 0 to 90° maps $v_{\rm circ}$ from the centre out to R_0. Outside the Sun's circle the method fails, because r grows all along the line. There the distances of the gas or of stars must be found separately, from Cepheids or from *Gaia* parallaxes.

Flat rotation curves. In our Galaxy, $v_{\rm circ}$ stays between about 200 and 240 km s^{-1} from 5 kpc out to 25 kpc. For another spiral seen at an angle, the Doppler shift across its disk gives the curve directly. The light of a disk falls off exponentially with radius, over a few kiloparsecs, so a few such lengths out nearly all the light is inside. If the light traced the mass, $m(r)$ would stop

growing there, and v_{circ} would fall as $r^{-1/2}$, like the speeds of the planets. It does not. In the 21 cm line, whose gas reaches much further than the stars, the rotation curves of most spirals stay flat as far out as they can be followed, often several times the radius of the visible disk. By Eq. (14.7) the mass keeps growing as r there, and by Eq. (14.9) the unseen matter has $\rho \propto r^{-2}$.

A model galaxy. Masses add, so the squares of the speeds add,

$$v_{\text{circ}}^2 = \frac{G\big(m_{\text{stars}} + m_{\text{halo}}\big)}{r} = v_{\text{stars}}^2 + v_{\text{halo}}^2. \tag{14.15}$$

For the halo take a density that is level in a core of radius r_c and falls as r^{-2} outside, $\rho = \rho_c/(1 + r^2/r_c^2)$, with ρ_c the central density. Its mass inside r is

$$\begin{aligned} m_{\text{halo}}(r) &\overset{(1)}{=} \int_0^r \frac{4\pi r'^2 \rho_c}{1 + r'^2/r_c^2}\, dr' \\ &\overset{(2)}{=} 4\pi\rho_c r_c^2 \int_0^r \left(1 - \frac{1}{1 + r'^2/r_c^2}\right) dr' \\ &\overset{(3)}{=} 4\pi\rho_c r_c^2 \left(r - r_c \arctan\frac{r}{r_c}\right). \end{aligned} \tag{14.16}$$

(1) Eq. (9.10).
(2) $r'^2/(1 + r'^2/r_c^2) = r_c^2\big[1 - 1/(1 + r'^2/r_c^2)\big]$.
(3) $\int_0^r dr'/(1 + r'^2/r_c^2) = r_c \arctan(r/r_c)$.

By Eq. (14.7),

$$\boxed{v_{\text{halo}}^2 = 4\pi G\rho_c r_c^2 \left[1 - \frac{r_c}{r}\arctan\frac{r}{r_c}\right].} \tag{14.17}$$

Near the centre, $\arctan x \approx x - x^3/3$ gives $v_{\text{halo}}^2 \approx \frac{4}{3}\pi G\rho_c r^2$, the uniform sphere with $v \propto r$. Far out, arctan $\to \pi/2$ and $v_{\text{halo}}^2 \to 4\pi G\rho_c r_c^2$, the flat curve. Figure 14.2 adds this halo to a round model of the stars. The stars set the curve inside a few kiloparsecs and the halo outside, and between them the total is nearly level. The halo's mass grows as r without end, so a real halo must be cut off somewhere. The motions of satellite galaxies show that the Milky Way's halo reaches 200 to 300 kpc and holds about $10^{12}\, M_\odot$.

The mass-to-light ratio. A galaxy's mass divided by its luminosity, in units of $M_\odot/L_\odot$, is its **mass-to-light ratio** M/L. The Sun has $M/L = 1$. A population of old stars has a few, because the mass is mostly in faint low-mass stars and the light comes from the few bright ones. With $L \propto M^{3.5}$ near the Sun (Section 12.6), a star of $0.5\, M_\odot$ alone has $M/L = 0.5^{-2.5} = 5.7$. Inside the visible disk of a spiral, the mass from the rotation curve gives M/L of a few, like its stars. Out to the last point of a flat curve it gives ten to several tens, and for clusters of galaxies about three hundred (Section 14.4). The ratio grows with the radius inside which the mass is measured, because the unseen matter takes over further out. For the Milky Way, a curve flat at 230 km s^{-1} out to 50 kpc would put $6 \times 10^{11}\, M_\odot$ inside that radius, against $6 \times 10^{10}\, M_\odot$ in stars and gas.

What the unseen mass is not. It is not cold gas, since the 21 cm line counts the neutral hydrogen atom by atom (Eq. (8.34)), and hot gas would shine in X-rays. It is not mostly faint stars or dark compact bodies. Such bodies, crossing in front of stars of the Large Magellanic Cloud, would now and then brighten them by bending their light (Section 14.4), and surveys in the 1990s found far too few such events for bodies between about 10^{-7} and $10\, M_\odot$ to make up the halo. The amount of ordinary matter of every kind is fixed by the helium and deuterium

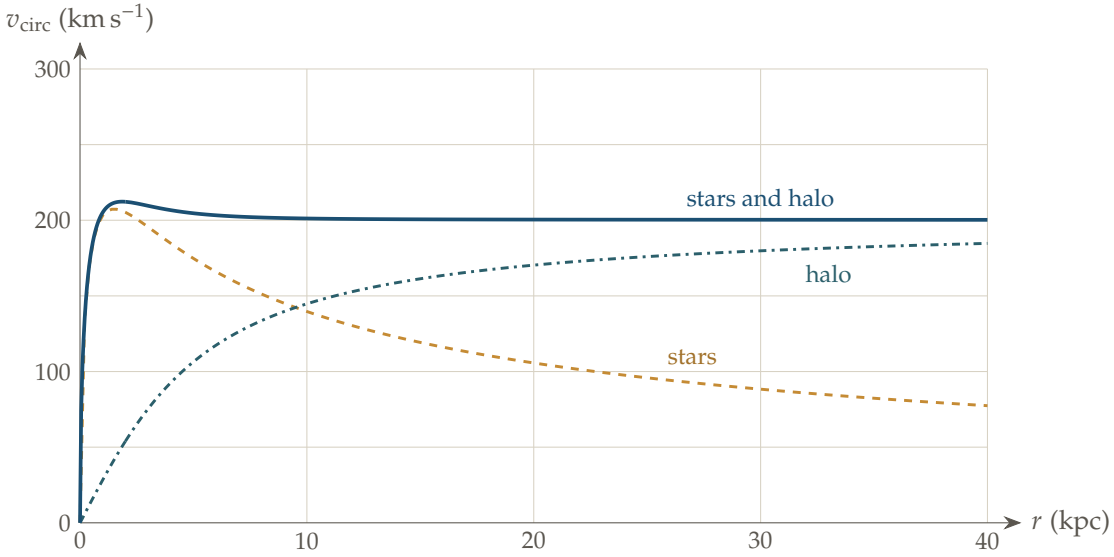


Figure 14.2: The circular speed of a model galaxy against radius. Dashed, stars with enclosed mass $6 \times 10^{10}\, M_\odot\, r^2/(r + 1.5\,\text{kpc})^2$. Dash-dotted, the halo of Eq. (14.17) with $r_c = 4\,\text{kpc}$ and $4\pi G\rho_c r_c^2 = (200\,\text{km s}^{-1})^2$. Solid, both together.

made in the first minutes of the universe (Chapter 16), and it falls well short of the mass that galaxies need. The halo is mostly a kind of matter that neither sends out nor absorbs light, and it is felt only through its gravity.

14.3 Stars as a Gas That Rarely Collides

The atoms of a gas collide often, and that is what gives them a Maxwell distribution of speeds (P3) and lets a gas be treated as a fluid. The disk of Section 14.1 was already treated as a gas of stars.

Stars almost never touch. Near the Sun they are about a parsec apart and a hundred million times smaller than that. A star could cross the Galaxy many times over without hitting another one. If stars never collide, it is not clear that a gas of stars ever forgets how it started, or settles into any kind of equilibrium at all.

Think of a ship crossing the ocean with its rudder tied. Each wave pushes it a little to one side or the other, in no fixed pattern. One wave does nothing worth noticing. But the pushes do not cancel exactly. After many waves the sideways drift has built up, and it grows as the square root of the number of waves. Wait long enough and the ship is heading somewhere else altogether.

A star crossing a galaxy is the ship. Every star it passes tugs it a little, from far away, in a random direction. The tugs add as a random walk in velocity (P5), so the sum of their squares grows steadily. When that sum reaches the square of the star's own speed, the star has forgotten its original orbit. The time this takes is the **relaxation time**, and comparing it with the age of the system decides what kind of system it is (P6).

The ship in numbers. If each wave pushes it sideways by $1\,\text{cm s}^{-1}$ and it sails at $5\,\text{m s}^{-1}$, the drift equals the forward speed after $(500)^2 = 250\,000$ waves. A galaxy of 10^{11} stars needs 5×10^8 crossings to relax (below), each taking some 5×10^7 years. That is 2×10^{16} years, a million times its age. A globular cluster of 10^5 stars needs only about a thousand crossings of under a million years each, so it relaxes many times over in its life.

The picture is Figure 14.3. A star passes a field star on a nearly straight line and gets a small

sideways kick toward it. The pulls before and after closest approach have the same sideways part and opposite forward parts.

The same walk in velocity is Brownian motion, and with charges in place of masses it is the way electrons scatter off ions in a plasma (Chapter 15). Here it combines Gauss's law for one kick (P2), the random walk for many (P5) and a comparison of two times (P6). It gives the kick of one passing star, the relaxation time, and what relaxes and what does not.

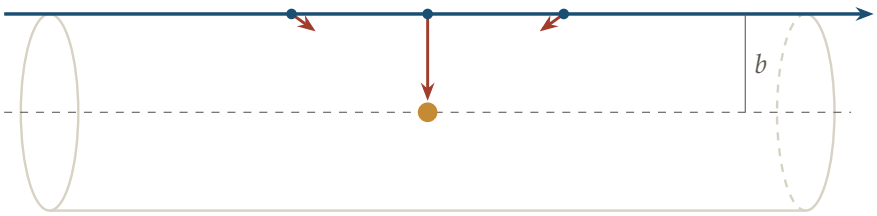


Figure 14.3: A star moving left to right along a straight path (blue) passes a field star (dot on the dashed axis) at impact parameter b. Red arrows are the pull of the field star at three points of the path. The grey cylinder of radius b about the dashed axis contains the path.

One passing star. Let a star move at speed v past a field star of mass $m_\star$, along a straight line that misses it by the **impact parameter** b. We treat the path as straight and the field star as fixed. Both hold while the kick is small compared with v. Put the closest approach at $t = 0$, so the star is at distance $\sqrt{b^2 + v^2t^2}$. The part of the pull along the path changes sign at $t = 0$ and adds nothing. The part across it, toward the line of closest approach, gives the kick,

$$\begin{aligned}\Delta v &\overset{(1)}{=} \int_{-\infty}^{\infty} \frac{Gm_\star}{b^2+v^2t^2}\,\frac{b}{\sqrt{b^2+v^2t^2}}\,dt \\ &\overset{(2)}{=} \int_{-\pi/2}^{\pi/2} \frac{Gm_\star\, b}{b^3\sec^3\psi}\,\frac{b}{v}\sec^2\psi\,d\psi \\ &= \frac{Gm_\star}{bv}\int_{-\pi/2}^{\pi/2}\cos\psi\,d\psi = \frac{2Gm_\star}{bv}.\end{aligned} \tag{14.18}$$

(1) The change of velocity is the time integral of the acceleration (P1). The pull is $Gm_\star$ over the distance squared, and the fraction of it across the path is b over the distance.
(2) Substitute $vt = b\tan\psi$, so $dt = (b/v)\sec^2\psi\,d\psi$ and $b^2 + v^2t^2 = b^2\sec^2\psi$.

$$\boxed{\Delta v = \frac{2Gm_\star}{b\,v}.} \tag{14.19}$$

The kick is the pull at closest approach, $Gm_\star/b^2$, times about twice the time spent near it, b/v. A closer or slower passage gives a larger kick.

The same kick from Gauss's law. Draw a cylinder of radius b whose axis passes through the field star, parallel to the path (Figure 14.3). The path lies on its surface. Let the cylinder be very long, so its ends carry no flux. With ϕ the angle round the axis, x the length along it and g_r the part of the pull pointing away from the axis,

$$-4\pi Gm_\star \overset{(1)}{=} \oint \vec{g}\cdot d\vec{A} \overset{(2)}{=} \int_0^{2\pi} b\,d\phi \int_{-\infty}^{\infty} g_r\,dx \overset{(3)}{=} 2\pi b\int_{-\infty}^{\infty} g_r\,dx. \tag{14.20}$$

(1) Gauss's law (9.9), with $m_\star$ inside (P2).
(2) The area element on the side is $b\,d\phi\,dx$, and only the part of $\vec{g}$ across the side counts.

(3) Every line on the cylinder parallel to the axis is the same as every other, by symmetry about the axis.

So $\int g_r\,dx = -2Gm_\star/b$ along the path, and with $dx = v\,dt$ the kick toward the axis is $2Gm_\star/(bv)$, Eq. (14.19) again. No integral had to be done. The kick depends on the mass of the field star and on b, exactly as the field of a line of charge depends on its charge and the distance.

The straight path holds while $\Delta v \ll v$, which is $b \gg 2Gm_\star/v^2$. For a star of one solar mass passed at $200\,\mathrm{km\,s^{-1}}$ that is 0.044 AU. Almost every encounter in a galaxy is far wider than this.

Many passing stars. Take a system of N stars of mass $m_\star$ within a radius R, through which a star crosses at a typical speed v in the **crossing time** $t_{\text{cross}} = R/v$. In one crossing it passes through the system's disk on the sky, of area πR^2 holding N stars. The number of field stars it passes with impact parameters between b and $b + db$ is $(N/\pi R^2)\,2\pi b\,db = 2Nb\,db/R^2$. Each kick points across the path in a random direction, so the kicks add as the steps of a random walk. Their squares add, and the cross terms average to zero, as in Eq. (9.36) (P5). In one crossing,

$$\sum |\Delta\vec{v}|^2 \overset{(1)}{=} \int_{b_{\min}}^{R} \left(\frac{2Gm_\star}{bv}\right)^2 \frac{2Nb\,db}{R^2} \overset{(2)}{=} 8N\left(\frac{Gm_\star}{Rv}\right)^2 \int_{b_{\min}}^{R} \frac{db}{b} = 8N\left(\frac{Gm_\star}{Rv}\right)^2 \ln\frac{R}{b_{\min}}. \tag{14.21}$$

(1) The square of Eq. (14.19), times the number of passages at each b. No star lies further away than about R. Below $b_{\min} = 2Gm_\star/v^2$ a single kick is as large as v and the straight path fails.
(2) Collect the constants.

The integral is a logarithm, so each factor of ten in b adds the same amount. The many distant passages together matter more than the rare close ones, and close encounters, the only ones that a contact collision picture would count, add less than one factor of ten out of many.

The relaxation time. To put Eq. (14.21) in terms of N alone, we use the virial theorem (Eq. (10.9) with $\ddot{I} = 0$, $2K = |U|$) in order of magnitude, $v^2 \approx GNm_\star/R$. Then $Gm_\star/R = v^2/N$ and $b_{\min} = 2R/N$, and the relative size of the summed kicks in one crossing is

$$\frac{\sum |\Delta\vec{v}|^2}{v^2} \overset{(1)}{=} \frac{8N}{v^2}\left(\frac{v}{N}\right)^2 \ln\frac{N}{2} \overset{(2)}{=} \frac{8\ln N}{N}. \tag{14.22}$$

(1) Eq. (14.21) with $Gm_\star/(Rv) = v/N$ and $R/b_{\min} = N/2$.
(2) For large N, $\ln(N/2)$ and $\ln N$ differ by less than the cut-offs at R and $b_{\min}$ are known.

The star's speed has been changed by an amount comparable to itself after $N/(8\ln N)$ crossings,

$$\boxed{t_{\text{relax}} = \frac{N}{8\ln N}\,t_{\text{cross}}, \qquad t_{\text{cross}} = \frac{R}{v}.} \tag{14.23}$$

Only the number of stars and the crossing time enter. For a given crossing time, a system of fewer, heavier stars relaxes faster, because each kick is larger.

System	M ($M_\odot$)	N	R	v (km s^{-1})	t_{cross} (yr)	t_{relax} (yr)	age (yr)
Open cluster	500	10^3	2 pc	1.0	1.9×10^6	3×10^7	10^8
Globular cluster	5×10^4	10^5	5 pc	6.6	7.5×10^5	8×10^8	1.2×10^{10}
Galaxy	10^{11}	10^{11}	10 kpc	207	4.7×10^7	2×10^{16}	10^{10}

Table 14.2: Crossing and relaxation times, Eq. (14.23), for three kinds of star system.

What relaxes. Table 14.2 puts in the numbers, with v from $v^2 = GM/R$. The ratio $t_{\text{relax}}/\text{age}$ sorts them (P6). In a galaxy it is a million. Its stars have never traded energy with each other. Each moves in the smooth pull of the whole galaxy as if the other stars were spread out into a fluid, and a galaxy is **collisionless**. In a globular cluster the ratio is about a tenth, so the cluster has relaxed many times.

Galaxies remember. A collisionless galaxy keeps the orbits its stars were born on. Near the Sun the spread of stellar velocities is larger toward the Galactic centre than up out of the plane. A relaxed gas would have made it the same in every direction, as collisions do for the pressure of a gas. The vertical balance of Section 14.1 needed only Π_{zz}, so it still holds. The disk does heat slowly, and old stars move faster than young ones, but the heating comes from the much larger kicks of giant molecular clouds and spiral arms, not from single stars. The shapes of galaxies, too, are set when they form and kept since (Section 14.5).

Globular clusters relax, and lose stars. In a cluster that relaxes, kicks drive the speeds toward a Maxwell distribution (P3 and P4). Its fastest stars then have more than the escape speed and leave. To see how many, compare the escape speed with the mean square speed. Let Φ be the gravitational potential, so that a star at rest at infinity has $\frac{1}{2}v_{\text{esc}}^2 + \Phi = 0$, and $U = \frac{1}{2}\int \Phi\, dm$, since a sum over the potential of every mass element counts each pair twice. Averaging over the mass,

$$\left\langle v_{\text{esc}}^2 \right\rangle \overset{(1)}{=} -\frac{2}{M}\int \Phi\, dm \overset{(2)}{=} -\frac{4U}{M} \overset{(3)}{=} \frac{8K}{M} \overset{(4)}{=} 4\left\langle v^2 \right\rangle . \tag{14.24}$$

(1) $v_{\text{esc}}^2 = -2\Phi$ (P1 for energy), averaged over the mass M.
(2) $\int \Phi\, dm = 2U$.
(3) The virial theorem, $2K + U = 0$, Eq. (10.9) averaged in time.
(4) $K = \frac{1}{2}M\left\langle v^2 \right\rangle$.

A star escapes once its speed exceeds about twice the root-mean-square speed. In a Maxwell distribution with $\left\langle v^2 \right\rangle = 3\sigma^2$, where σ is the spread of one velocity component, the fraction of stars faster than $2\sqrt{3}\,\sigma$ is

$$\int_{2\sqrt{3}\,\sigma}^{\infty} \frac{4\pi v^2}{(2\pi\sigma^2)^{3/2}}\, e^{-v^2/2\sigma^2}\, dv = \operatorname{erfc}\sqrt{6} + 2\sqrt{\frac{6}{\pi}}\, e^{-6} = 0.0074,$$

where the integral was done by parts. Relaxation refills this tail once in every relaxation time, so the cluster loses 0.74 per cent of its stars per t_{relax} and would evaporate in about $135\, t_{\text{relax}}$, some 10^{11} years for the cluster of Table 14.2. The tidal pull of the Galaxy strips the outer stars and shortens this. Relaxation also shares energy between stars of different mass. Heavy stars lose speed, sink toward the centre, and leave the light ones in the outskirts.

Core collapse. The dense core of a globular cluster is a self-gravitating system, with the negative heat capacity of Section 10.3. Its stars move faster than those of the outer parts, so

relaxation carries energy outward, as heat flows from a hot body to a cold one. The core loses energy, and with $C < 0$ it gets hotter and contracts. A hotter core passes energy to the outer parts even faster, and the process runs away. This is the **gravothermal catastrophe**. The core shrinks toward a very small, very dense centre. It is stopped by close pairs of stars that form there. When a third star passes close to a pair, the pair binds tighter and the third star leaves faster, so the pair hands energy to the core and heats it. About a fifth of the globular clusters of the Milky Way have the sharply peaked centres that mark a collapsed core. A galaxy, whose relaxation time is a million times its age, never gets there.

14.4 Weighing Galaxies and Clusters

A disk is weighed by its rotation, Eq. (14.7). The virial theorem of Section 10.2 ties the kinetic and gravitational energies of any bound system in a steady state.

Elliptical galaxies and clusters of galaxies do not rotate in an orderly way. Their members move in every direction. We see only the speed along the line of sight and the position on the sky, not the full motion or the depth.

Think of a salad bowl full of marbles, shaken and watched from above. The bowl itself cannot be seen. But marbles that move fast stay inside only if the bowl is deep, and from how fast they move and how far they spread one can judge its depth.

The galaxies of a cluster are the marbles, and the cluster's gravity is the bowl. In 1933 Zwicky measured the line-of-sight speeds of galaxies in the Coma cluster, spread by about $1000\,\mathrm{km\,s^{-1}}$ over a few megaparsecs, and found that the bowl must hold hundreds of times the mass of the stars in it. The virial theorem turns random speeds and sizes into a mass.

A uniform sphere in numbers. With the virial theorem $2K = |U|$, $K = \frac{1}{2}M \cdot 3\sigma^2$ for a line-of-sight spread σ, and $|U| = \frac{3}{5}GM^2/R$ (Eq. (10.11)), the mass is $M = 5\sigma^2 R/G$. For $\sigma = 1000\,\mathrm{km\,s^{-1}}$ and $R = 1.5\,\mathrm{Mpc}$ it is $1.7 \times 10^{15}\,M_\odot$.

The picture is a cluster on the sky, a swarm of dots, each with one arrow along the line of sight.

The same reasoning gave the temperature of a star from its gravity (Eq. (10.10)). For the hot gas of a cluster it becomes hydrostatic equilibrium again (P1 and P2), and the bending of light by mass, from general relativity, weighs a cluster with no reference to motion at all. It gives the mass from line-of-sight speeds and positions on the sky, the mass from the hot gas, and the mass from the bending of light.

From three dimensions to what is seen. Take N galaxies of equal mass $m_\star$, total mass $M = Nm_\star$, in a steady state. Their kinetic energy is $K = \frac{1}{2}M\langle v^2\rangle$. We measure only the line-of-sight component, with mean square σ^2 after the mean motion of the cluster is removed. If the motions favour no direction,

$$\langle v^2\rangle = \langle v_x^2\rangle + \langle v_y^2\rangle + \langle v_z^2\rangle \overset{(1)}{=} 3\sigma^2, \tag{14.25}$$

(1) With no favoured direction, each component has the same mean square as the one along the line of sight.

The gravitational energy is a sum over pairs, $U = -Gm_\star^2 \sum_{\text{pairs}} 1/r$, with $\frac{1}{2}N(N-1)$ pairs. We see each separation r projected on the sky, $r_\perp = r\sin\theta$, where θ is the angle between the separation and the line of sight. The direction of a pair has nothing to do with its separation, so the two can be averaged one at a time over all pairs,

$$\left\langle \frac{1}{r_\perp}\right\rangle \overset{(1)}{=} \left\langle\frac{1}{r}\right\rangle\left\langle\frac{1}{\sin\theta}\right\rangle \overset{(2)}{=} \left\langle\frac{1}{r}\right\rangle \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\frac{1}{\sin\theta}\,\sin\theta\,d\theta\,d\phi \overset{(3)}{=} \frac{\pi}{2}\left\langle\frac{1}{r}\right\rangle. \tag{14.26}$$

(1) r and θ are independent.
(2) All directions equally likely, with the solid angle $\sin\theta\, d\theta\, d\phi$ over 4π.
(3) The integrand is 1, so the integral is $2\pi \cdot \pi$.

The virial mass. Setting $2K = |U|$,

$$M \cdot 3\sigma^2 \overset{(1)}{=} G m_\star^2 \frac{N(N-1)}{2} \left\langle \frac{1}{r} \right\rangle \overset{(2)}{=} G m_\star^2 \frac{N(N-1)}{2} \frac{2}{\pi} \left\langle \frac{1}{r_\perp} \right\rangle = \frac{GM^2}{\pi} \left(1 - \frac{1}{N}\right) \left\langle \frac{1}{r_\perp} \right\rangle . \tag{14.27}$$

(1) The virial theorem, Eq. (10.9) averaged in time, with K and U from above.
(2) The projection factor $\pi/2$.

For $N \gg 1$,

$$\boxed{M = \frac{3\pi\,\sigma^2}{G\,\langle 1/r_\perp \rangle}.} \tag{14.28}$$

Only line-of-sight speeds and positions on the sky enter, with the distance to the cluster needed to turn angles into $r_\perp$. The result assumes a steady state, motions with no favoured direction, and galaxies that are spread like the mass. It does not depend on what the mass is made of. As a check, a uniform sphere has $\langle 1/r \rangle = |U|/(G m_\star^2 N^2/2) = 6/(5R)$ by Eq. (10.11), so $\langle 1/r_\perp \rangle = 3\pi/(5R)$, and Eq. (14.28) gives $M = 5\sigma^2 R/G$, the value found above.

For a cluster with $\sigma = 1000\,\mathrm{km\,s^{-1}}$ and $1/\langle 1/r_\perp \rangle = 1\,\mathrm{Mpc}$, Eq. (14.28) gives $2.2 \times 10^{15}\,M_\odot$. Coma has a mass of about $10^{15}\,M_\odot$ and a mass-to-light ratio of about 300. Its galaxies shine with a few tenths of a per cent of the light that this mass would give if it were all stars like the Sun.

The hot gas. The same well holds the hot gas between the galaxies. Its particles share the energy per unit mass of the galaxies, so $k_B T = \mu m_\mathrm{H} \sigma^2$, about $7 \times 10^7\,\mathrm{K}$ for $\sigma = 1000\,\mathrm{km\,s^{-1}}$ (Problem 10.7.2). Gas this hot shines in X-rays by bremsstrahlung (Eq. (15.16) and Problem 15.6.1), and X-ray telescopes measure its temperature and density at each radius. Hydrostatic equilibrium then gives the mass. With $P = \rho k_B T/(\mu m_\mathrm{H})$,

$$-\frac{G\,m(r)\,\rho}{r^2} \overset{(1)}{=} \frac{\mathrm{d}P}{\mathrm{d}r} \overset{(2)}{=} \frac{k_B}{\mu m_\mathrm{H}} \left(T \frac{\mathrm{d}\rho}{\mathrm{d}r} + \rho \frac{\mathrm{d}T}{\mathrm{d}r} \right) \overset{(3)}{=} \frac{\rho\, k_B T}{\mu m_\mathrm{H}\, r} \left(\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r} + \frac{\mathrm{d}\ln T}{\mathrm{d}\ln r} \right). \tag{14.29}$$

(1) Hydrostatic equilibrium, Eq. (10.2).
(2) The ideal gas, differentiated.
(3) $\mathrm{d}\rho/\mathrm{d}r = (\rho/r)\ \mathrm{d}\ln\rho/\mathrm{d}\ln r$, and the same for T.

$$\boxed{m(r) = -\frac{k_B T\, r}{G \mu m_\mathrm{H}} \left(\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r} + \frac{\mathrm{d}\ln T}{\mathrm{d}\ln r} \right).} \tag{14.30}$$

Only the temperature and the slopes of the profiles enter, not the amount of gas. For gas at one temperature with $\rho \propto r^{-2}$, the bracket is -2 and $m = 2k_B T r/(G\mu m_\mathrm{H})$. With $k_B T = \mu m_\mathrm{H} \sigma^2$ this is $m = 2\sigma^2 r/G$, the isothermal sphere of Section 12.5 with $v_\mathrm{circ}^2 = 2\sigma^2$, which is a second route to the same mass. Inside 1 Mpc it is $4.6 \times 10^{14}\,M_\odot$. The gas itself holds several times the mass of the stars, yet stars and gas together are only about a sixth of the total.

The bending of light. General relativity bends a ray that passes a mass M at impact parameter b through the angle $4GM/(bc^2)$, twice the Newtonian value. For a lens that is not a point, M is the mass inside the circle of radius b on the sky, if the lens is round. Put a source directly behind a lens, at distance D_S from us, with the lens at D_L and D_{LS} between the two (Figure 14.4). A ray we see at a small angle θ from the lens passes it at $b = \theta D_L$. Without bending it would reach the source's distance at the height θD_S. Bent back toward the axis, it reaches that distance at the height $\theta D_S - 4GMD_{LS}/(bc^2)$, and it comes from the source only if this is zero,

$$\theta D_S \overset{(1)}{=} \frac{4GM}{bc^2} D_{LS} \overset{(2)}{=} \frac{4GM}{\theta D_L c^2} D_{LS} \quad \Longrightarrow \quad \theta^2 = \frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}. \tag{14.31}$$

(1) The height of the bent ray at the source, set to zero, with small angles.
(2) $b = \theta D_L$.

$$\boxed{\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}}} \tag{14.32}$$

By symmetry the source appears as a ring of this angular radius, the **Einstein ring**. A source a little off the axis appears as two images or as arcs near the ring. Only the mass inside the ring enters, not how it moves or what state it is in. For $10^{14}\,M_\odot$ with $D_L = 1\,\text{Gpc}$ and $D_S = 2\,\text{Gpc}$, $\theta_E = 20''$, and rich clusters do show arcs of about this size. A galaxy of $10^{11}\,M_\odot$ gives $0.6''$. At such distances the Ds are the distances of Chapter 16 that relate size to angle, for which $D_L + D_{LS} \neq D_S$. The formula keeps its form. Masses from lensing, from the galaxies' speeds and from the hot gas agree within a few tens of per cent. In the colliding pair of clusters called the Bullet Cluster, the lensing mass sits with the galaxies, while most of the ordinary matter, the gas, has been slowed and left between them.

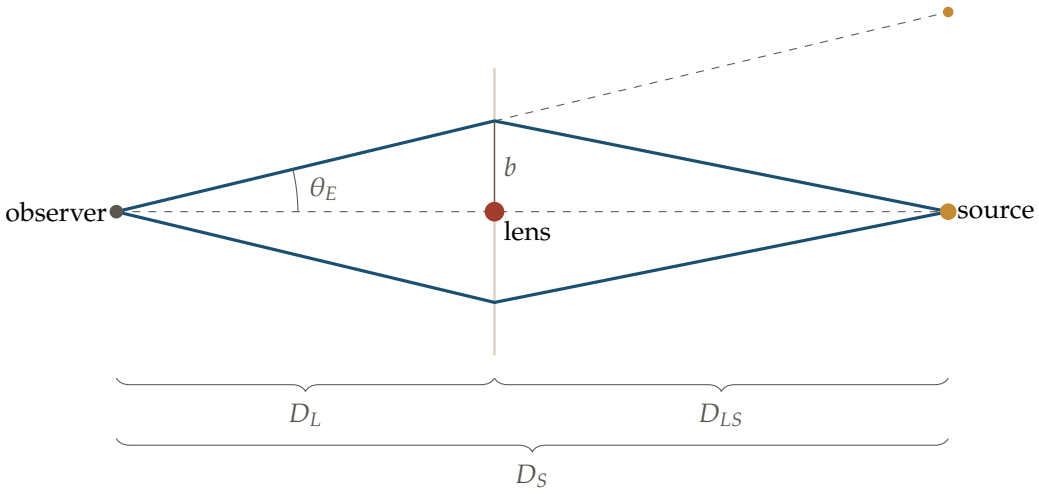


Figure 14.4: A source directly behind a lens. Two rays from the source are bent at the lens plane and reach the observer at the angle θ_E from the lens. The dashed line continues the direction in which the observer sees the upper image. Angles are exaggerated.

14.5 Normal Galaxies

A galaxy can be weighed by its rotation, Eq. (14.7), or by the random speeds of its stars, Eq. (14.28).

Weighing gives a mass. It does not say how bright a galaxy really is, or how far away, and galaxies come in very different shapes. What is wanted is a link between the motions, which are easy to measure, and the true luminosity.

Think of apples from one orchard. They are all about equally dense. Measure the width of an apple and you know its weight without a scale, because the weight grows as the cube of the width. One fixed property turns one measured quantity into another, through a power law.

Spiral disks are like the apples. They have nearly the same surface brightness, and their stars nearly the same mass-to-light ratio. With $v_{\text{circ}}^2 = Gm/R$ these two fixed ratios make the luminosity grow as the fourth power of the rotation speed. The width of the 21 cm line then gives the luminosity, and the luminosity with the observed flux gives the distance. A power law follows from the virial theorem and two fixed ratios.

The apples in numbers. An apple twice as wide is $2^3 = 8$ times heavier. A spiral that rotates twice as fast is $2^4 = 16$ times brighter, $2.5 \log_{10} 16 = 3.0$ magnitudes.

The picture is a plot of luminosity against rotation speed on logarithmic axes, where spirals lie close to a straight line of slope four.

The same kind of scaling gave the mass–luminosity relation of stars (Section 12.6, P6). With random speeds in place of rotation it works for elliptical galaxies. It gives the main kinds of galaxies, the two scaling laws, and how they measure distances.

The Hubble sequence. Hubble sorted galaxies by their look into a sequence. **Ellipticals** are smooth and featureless, labelled E0 to E7 by how flattened they appear. **Lenticulars**, S0, have a disk but no spiral arms. **Spirals** run from Sa, with a large bulge and tightly wound arms, to Sd, with a small bulge and open arms, and those with a bar form a parallel sequence SBa to SBd. **Irregulars** fit nowhere. Along the sequence from ellipticals to late spirals, the fraction of gas and of young blue stars grows, and the support against gravity changes. Ellipticals are held up by random motions of 100 to 300 km s^{-1} and rotate little. Their stars are old and red, and they have little cold gas. Disks are held up by rotation at 100 to 300 km s^{-1}, with random motions of only a few tens. A collisionless galaxy keeps the motions it formed with (Section 14.3), so its shape records its history. Mergers of disk galaxies, which mix the orbits violently, can make ellipticals.

Tully–Fisher from the virial theorem. Let a disk of radius R have mass M inside R and luminosity L. Its rotation speed is $v^2 = GM/R$ by Eq. (14.7). Suppose M/L and the surface brightness, measured by L/R^2, are the same for all such disks. To get L in terms of v alone, we write M and R through these two ratios,

$$L \overset{(1)}{=} \left(\frac{L}{M}\right) M \overset{(2)}{=} \left(\frac{L}{M}\right) \frac{v^2 R}{G} \overset{(3)}{=} \left(\frac{L}{M}\right) \frac{v^2}{G} \left(\frac{R^2}{L}\right)^{1/2} L^{1/2}. \tag{14.33}$$

(1) The mass-to-light ratio.
(2) Eq. (14.7) at the edge of the disk.
(3) $R = (R^2/L)^{1/2} L^{1/2}$.

Dividing by $L^{1/2}$ and squaring,

$$\boxed{L = \left(\frac{L}{M}\right)^2 \left(\frac{R^2}{L}\right) \frac{v^4}{G^2}.} \tag{14.34}$$

This is the **Tully–Fisher relation**, $L \propto v^4$. The power four comes from the virial theorem and the fixed surface brightness. The constant in front holds the two ratios that were assumed fixed. The observed relation is close to $L \propto v^4$ in the near infrared, where the light traces the old stars, and it is tight. Disks do have nearly the same central surface brightness, which explains part of it. Yet v is set mostly by the dark halo and L by the stars, and galaxies of low surface brightness lie on the same relation. Why the halo and the stars keep the relation so tight is not explained by the virial theorem. Counting the gas with the stars makes the relation tighter still.

Faber–Jackson. For an elliptical galaxy, the uniform-sphere form of the virial theorem from Section 14.4, $M = 5\sigma^2 R/G$, takes the place of $M = v^2 R/G$. The same three steps with v^2 replaced by $5\sigma^2$ give

$$\boxed{L = 25 \left(\frac{L}{M}\right)^2 \left(\frac{R^2}{L}\right) \frac{\sigma^4}{G^2},} \tag{14.35}$$

the **Faber–Jackson relation** $L \propto \sigma^4$. It holds less tightly than Tully–Fisher, because the surface brightness of ellipticals is not fixed. Keeping L/R^2 as a third measured quantity turns the line into a plane in the space of R, σ and surface brightness, the fundamental plane, on which ellipticals lie closely.

Distances. The 21 cm line of a whole spiral is broadened by its rotation. The side moving toward us and the side moving away shift the line by $\pm v \sin \iota$, where ι is the inclination of the disk to the sky ($\iota = 90°$ for a disk seen edge on), so the full width is $W = 2v \sin \iota$. A thin round disk tilted by ι looks like an ellipse with axis ratio $\cos \iota$, which gives ι. Then v gives L by Eq. (14.34), once the constant has been fixed with galaxies of known distance. In magnitudes, $L \propto v^4$ means that the absolute magnitude changes by $-10 \log_{10}$ of the ratio of rotation speeds (Eq. (4.5)), and the distance follows from the distance modulus, Eq. (4.11) (Problem 14.6.5). Tully–Fisher distances reach a few hundred megaparsecs, where the expansion of the universe (Chapter 16) takes over as a measure of distance.

14.6 Problem Set: Galaxies and the Motions of Stars

14.6.1 Problem 1: The Sun's Orbit and the Local Dark Matter

Problem Statement: The Sun circles the Galactic centre at $v_0 = 230\,\mathrm{km\,s^{-1}}$ at $R_0 = 8.2\,\mathrm{kpc}$. (a) How long does one orbit take? (b) How many orbits has the Sun made in its 4.6×10^9 years? (c) What mass lies inside the Sun's orbit? (d) If the rotation curve is flat and all the mass is in a round halo, what is the density at the Sun, in $M_\odot\,\mathrm{pc^{-3}}$ and in $\mathrm{GeV}\,c^{-2}\,\mathrm{cm^{-3}}$? (e) How much of this density lies in a slab within 1.1 kpc of the plane, per unit area? Compare with the $20\,M_\odot\,\mathrm{pc^{-2}}$ by which the weighed disk exceeds the visible stars and gas.

Essential Ideas:

- The circular speed weighs the mass inside, $v_{\mathrm{circ}}^2 = Gm(r)/r$, Eq. (14.7).
- A flat curve has $\rho = v_{\mathrm{circ}}^2/(4\pi G r^2)$, Eq. (14.9).

Detailed Solutions:

(a) **Period**. $2\pi R_0/v_0 = 2\pi \times 8.2 \times 3.086 \times 10^{19}\,\mathrm{m}/(2.3 \times 10^5\,\mathrm{m\,s^{-1}})$.

$$\boxed{2.2 \times 10^8 \text{ years.}}$$

(b) **Number of orbits**. $4.6 \times 10^9 / 2.19 \times 10^8$.

$$\boxed{\text{about 21.}}$$

(c) **Mass inside**. $m = v_0^2 R_0/G = (2.3 \times 10^5)^2 \times 2.53 \times 10^{20}/6.674 \times 10^{-11}\,\mathrm{kg}$.

$$\boxed{m(R_0) = 1.0 \times 10^{11}\,M_\odot.}$$

(d) **Local density**. $\rho = v_0^2/(4\pi G R_0^2) = 9.9 \times 10^{-22}\,\mathrm{kg\,m^{-3}}$. Multiplying by c^2 and dividing by $1.602 \times 10^{-10}\,\mathrm{J\,GeV^{-1}}$ and $10^6\,\mathrm{cm^3\,m^{-3}}$,

$$\boxed{\rho = 0.015\,M_\odot\,\mathrm{pc^{-3}} = 0.55\,\mathrm{GeV}\,c^{-2}\,\mathrm{cm^{-3}}.}$$

This overestimates the halo, since the disk also supplies part of v_0. Careful models give about two thirds of it.

(e) **In the slab**. $0.015\,M_\odot\,\mathrm{pc^{-3}} \times 2200\,\mathrm{pc}$.

$$\boxed{\approx 30\,M_\odot\,\mathrm{pc^{-2}},}$$

or about 20 with the smaller halo density of (d). This matches the excess found by weighing the disk. The unseen mass near the Sun is what a round halo puts there, with no extra dark matter in the disk itself.

14.6.2 Problem 2: Weighing the Disk

Problem Statement: (a) A population of old stars of the thick disk has a vertical velocity spread $\sigma_z = 40\,\mathrm{km\,s^{-1}}$, the same at every height, and its number falls with height with the scale $h_z = 900\,\mathrm{pc}$. Find the mass per unit area of the disk. (b) Near the plane, take the disk to be a uniform slab of density $\rho_0 = 0.1\,M_\odot\,\mathrm{pc^{-3}}$. Show that a star inside it oscillates up and down harmonically, and find the period. How often does the Sun cross the plane? (c) The derivation of Eq. (14.5) used a pressure, yet stars do not collide. Why is that allowed?

Essential Ideas:

- $h_z = \sigma_z^2/(2\pi G\Sigma)$, Eq. (14.5).
- Gauss's law (9.9) for a box inside the slab.

Detailed Solutions:

(a) **Mass per area**. $\Sigma = \sigma_z^2/(2\pi G h_z) = (4 \times 10^4)^2/(2\pi \times 6.674 \times 10^{-11} \times 2.78 \times 10^{19})\,\mathrm{kg\,m^{-2}}$.

$$\boxed{\Sigma \simeq 66\,M_\odot\,\mathrm{pc^{-2}}.}$$

(b) **Vertical oscillation**. A box with faces at $\pm z$ inside the slab holds the mass $2z\rho_0 A$, so Gauss's law gives $2g(z)A = 4\pi G\rho_0\,2zA$, and $g = 4\pi G\rho_0 z$ toward the plane. Then $\ddot{z} = -4\pi G\rho_0\,z$, a harmonic oscillation with angular frequency $\sqrt{4\pi G\rho_0}$. With $\rho_0 = 6.77 \times 10^{-21}\,\mathrm{kg\,m^{-3}}$,

$$\boxed{\text{period} = \frac{2\pi}{\sqrt{4\pi G\rho_0}} \simeq 84 \text{ million years.}}$$

The Sun crosses the plane twice per period, about every 42 million years. The period does not depend on how far the star swings, as long as it stays inside the slab.

(c) **No collisions needed**. The pressure in the momentum equation is the momentum carried across a surface by particles moving through it, Eq. (9.4). Counting that flow needs only the particles' velocities, not collisions. What collisions would add is the same spread of speeds in every direction. The vertical balance uses only $\Pi_{zz} = \rho\sigma_z^2$, so it holds for stars whatever their spread in other directions.

14.6.3 Problem 3: Relaxation and Evaporation of a Globular Cluster

Problem Statement: A globular cluster holds $N = 2 \times 10^5$ stars with a total mass of $10^5\,M_\odot$ within $R = 5\,\mathrm{pc}$. Estimate (a) the typical speed of its stars, (b) the crossing time, (c) the relaxation time, and (d) the evaporation time. (e) The cluster is 1.2×10^{10} years old. Is it relaxed? Should it still exist?

Essential Ideas:

- $v^2 \approx GM/R$ from the virial theorem in order of magnitude.
- $t_{\mathrm{relax}} = N/(8 \ln N)\, t_{\mathrm{cross}}$, Eq. (14.23).
- A relaxed cluster loses 0.74 per cent of its stars per relaxation time, Eq. (14.24) and the Maxwell tail.

Detailed Solutions:

(a) **Speed**. $v = \sqrt{GM/R} = \sqrt{6.674 \times 10^{-11} \times 1.989 \times 10^{35} / 1.543 \times 10^{17}}\,\mathrm{m\,s^{-1}}$.

$$\boxed{v \simeq 9.3\,\mathrm{km\,s^{-1}}.}$$

(b) **Crossing time**. $R/v = 1.543 \times 10^{17} / 9.28 \times 10^3\,\mathrm{s}$.

$$\boxed{t_{\mathrm{cross}} \simeq 5.3 \times 10^5\ \mathrm{years}.}$$

(c) **Relaxation time**. $N/(8 \ln N) = 2 \times 10^5/(8 \times 12.2) = 2050$.

$$\boxed{t_{\mathrm{relax}} \simeq 1.1 \times 10^9\ \mathrm{years}.}$$

(d) **Evaporation time**. $t_{\mathrm{relax}}/0.0074$.

$$\boxed{t_{\mathrm{evap}} \simeq 1.5 \times 10^{11}\ \mathrm{years}.}$$

(e) **Age**. The cluster is about 11 relaxation times old, so its stars have shared their energy and its heavy stars have sunk toward the centre. Evaporation alone would take more than ten times its age, so it survives. The Galaxy's tides and the shocks of crossing the disk remove stars faster, and many clusters that were once smaller or looser have already dissolved into the halo.

14.6.4 Problem 4: One Cluster, Three Weighings

Problem Statement: Model a rich cluster as an isothermal sphere, $\rho = \sigma^2/(2\pi G r^2)$, with line-of-sight spread $\sigma = 1000\,\mathrm{km\,s^{-1}}$ for its galaxies and hot gas of $\mu = 0.6$ at $k_B T = \mu m_{\mathrm{H}} \sigma^2$. (a) Find the mass inside 1 Mpc from the X-ray gas. (b) Show that it equals $v_{\mathrm{circ}}^2 r/G$ for this sphere. (c) Find the mass inside a circle of radius b on the sky. (d) Find the Einstein radius of the cluster for a source with $D_{LS}/D_S = \frac{1}{2}$.

Essential Ideas:

- The X-ray mass, Eq. (14.30).
- The Einstein radius, Eq. (14.32), with the mass inside the ring.

Detailed Solutions:

(a) **X-ray mass**. The gas is at one temperature and $\mathrm{d}\ln\rho/\mathrm{d}\ln r = -2$, so

$$m = \frac{2k_BTr}{G\mu m_{\rm H}} = \frac{2\sigma^2 r}{G} = \frac{2\times10^{12}\times3.086\times10^{22}}{6.674\times10^{-11}}\,\mathrm{kg}.$$

$$\boxed{m(1\,\mathrm{Mpc}) \simeq 4.6\times10^{14}\,M_\odot.}$$

(b) **The same from rotation**. By Eq. (9.10), $m = \int_0^r 4\pi r'^2\sigma^2/(2\pi G r'^2)\,dr' = 2\sigma^2 r/G$, so $v_{\rm circ}^2 = Gm/r = 2\sigma^2$, the result of Section 12.5. The gas and the galaxies feel the same well.

(c) **Mass on the sky**. The mass per unit area at projected radius $r_\perp$ is $\int_{-\infty}^{\infty}\rho\,dz$ along the line of sight, with $r^2 = r_\perp^2 + z^2$,

$$\int_{-\infty}^{\infty}\frac{\sigma^2\,dz}{2\pi G(r_\perp^2+z^2)} = \frac{\sigma^2}{2\pi G}\,\frac{\pi}{r_\perp} = \frac{\sigma^2}{2Gr_\perp}.$$

Adding rings of area $2\pi r_\perp\,dr_\perp$ out to b,

$$\boxed{M(<b) = \int_0^b 2\pi r_\perp \frac{\sigma^2}{2Gr_\perp}\,dr_\perp = \frac{\pi\sigma^2 b}{G}.}$$

(d) **Einstein radius**. In Eq. (14.32) put $M = \pi\sigma^2 b/G$ with $b = \theta_E D_L$,

$$\theta_E^2 = \frac{4\pi\sigma^2\theta_E D_L}{c^2}\,\frac{D_{LS}}{D_L D_S} \quad\Longrightarrow\quad \boxed{\theta_E = 4\pi\left(\frac{\sigma}{c}\right)^2\frac{D_{LS}}{D_S} \simeq 14''.}$$

The ring measures σ directly, without any speed being measured. Three routes, speeds of galaxies, temperature of gas and bending of light, each weigh the same well. When they agree, as they do for most clusters to within a few tens of per cent, the mass is secure.

14.6.5 Problem 5: A Tully–Fisher Distance

Problem Statement: (A constructed example.) A calibrating spiral at 16 Mpc rotates at 200 km s^{-1} and has apparent magnitude 10.0 in the near infrared. A second spiral has a 21 cm line of full width $W = 400\,\mathrm{km\,s^{-1}}$, appears as an ellipse with axis ratio 0.5, and has apparent magnitude 12.8 in the same band. (a) Find its inclination and rotation speed. (b) How much brighter is it than the calibrator, in magnitudes? (c) How far away is it?

Essential Ideas:

- $W = 2v\sin\iota$ and $\cos\iota$ equals the axis ratio.
- $L \propto v^4$, Eq. (14.34), so absolute magnitudes differ by $-10\log_{10}(v_1/v_2)$.
- Apparent magnitudes differ by the difference in absolute magnitude plus $5\log_{10}(d_1/d_2)$, Eq. (4.11).

Detailed Solutions:

(a) **Inclination and speed**. $\cos\iota = 0.5$, so $\iota = 60°$, and $v = W/(2\sin\iota) = 400/\sqrt{3}$.

$$\boxed{\iota = 60°, \qquad v \simeq 231\,\mathrm{km\,s^{-1}}.}$$

(b) **Luminosity**. $-10\log_{10}(231/200)$.

$$\boxed{0.62\text{ magnitudes brighter.}}$$

(c) **Distance**. The second galaxy appears 2.8 magnitudes fainter although it is 0.62 magnitudes brighter, so $5\log_{10}(d/16\,\mathrm{Mpc}) = 2.8 + 0.62$.

$$\boxed{d = 16\,\mathrm{Mpc} \times 10^{0.685} \simeq 77\,\mathrm{Mpc}.}$$

An error of 10 per cent in W changes v^4 by 46 per cent and the distance by 21 per cent. Accurate inclinations and widths matter as much as the photometry.

Chapter 15

Radiation from Hot and Fast Particles

Stars shine because they are hot and opaque, and their light is close to the light of a blackbody (Chapter 11). Much of the sky does not shine that way. The gas between the galaxies of a cluster glows in X-rays, though it is far too thin to be opaque. Radio galaxies send out spectra that fall smoothly as a power of the frequency over many decades, which no single temperature can make. A quasar outshines its whole galaxy from a region a few light-days across.

All of this light comes from charges that are pushed about, by ions, by magnetic fields, or by collisions with fast electrons. One formula from electromagnetism, the power radiated by an accelerated charge, gives every spectrum in the chapter when it is combined with the six principles. Fast particles need one more tool, the Lorentz transformation, which we write as a matrix.

15.1 An Accelerated Charge Radiates

Chapter 8 used the power radiated by a shaken electron to find the Thomson cross-section, Eq. (8.7).

There the electron was driven gently by a passing wave and always moved slowly. In hot gas and in jets, electrons are flung about by ions, spun round magnetic field lines, and move close to the speed of light. We need to know how much they radiate, and how motion near c changes what an observer sees.

Hold one end of a long rope lying on the floor. If you keep your hand still, or move it at a steady speed, nothing travels down the rope. If you jerk your hand, a pulse runs away along it, and the pulse carries energy that your arm supplied. A harder jerk sends a bigger pulse. What matters is how suddenly the hand changes its speed, not how fast it moves.

A charge does the same with its electric field. An electron passing close to a proton is jerked sideways once. An electron in a magnetic field is jerked all the time, as it is turned round in a circle. Each jerk sends out a pulse of electromagnetic field that carries energy away (P1). The power depends on the square of the acceleration, and for an electron it is a small loss per turn unless the acceleration is extreme (P6).

The classical hydrogen atom in numbers. An electron circling a proton at the Bohr radius 5.29×10^{-11} m has an acceleration of 9.0×10^{22} m s^{-2} and would radiate 4.7×10^{-8} W. Its energy of 13.6 eV would last 4.7×10^{-11} s, while one orbit takes 1.5×10^{-16} s. It would lose only 1.6×10^{-6} of its energy per orbit. Quantum mechanics stops the real atom from radiating at all, but the ratio shows that radiation drains a moving charge slowly compared with its own motion.

The picture is a shell of field spreading out from the charge at the speed of light, strongest at right angles to the acceleration and zero along it.

The same formula runs radio antennas, X-ray tubes and the synchrotron light sources of laboratories. It gave the Rayleigh and Thomson scattering of Chapter 8. It uses P1 (the field carries energy) and P6 (the rate of radiation compared with the rate of motion). It gives the radiated power, how large it is, and how it changes when the charge moves near the speed of light.

The Larmor formula. A charge e moving slowly, $v \ll c$, with acceleration $\dot{\vec{v}}$ radiates the power

$$\mathcal{P} = \frac{e^2\,|\dot{\vec{v}}|^2}{6\pi\varepsilon_0 c^3}, \qquad \frac{\mathrm{d}\mathcal{P}}{\mathrm{d}\Omega} = \frac{e^2\,|\dot{\vec{v}}|^2}{16\pi^2\varepsilon_0 c^3}\,\sin^2\Theta, \tag{15.1}$$

where Θ is the angle between the acceleration and the direction of emission. These are quoted from electromagnetism. The second form integrates to the first, since $\int \sin^2\Theta\, d\Omega = 8\pi/3$. The pattern is the same in a direction and in the opposite direction, so the radiation carries away energy but no net momentum.

How fast a charge loses its energy. To compare the radiated energy with the energy of the motion, take an electron of mass m_e oscillating as $x = x_0\cos\omega t$ (P6). Its acceleration is $\dot{v} = -\omega^2 x$, its energy is $\frac{1}{2}m_e\omega^2 x_0^2$, and in one radian of its motion, a time $1/\omega$, it radiates $\langle\mathcal{P}\rangle/\omega$.

$$\frac{\text{energy radiated per radian}}{\text{energy of the motion}} \overset{(1)}{=} \frac{e^2\,\omega^4 x_0^2/(12\pi\varepsilon_0 c^3)}{\omega\cdot\frac{1}{2}m_e\omega^2 x_0^2} \overset{(2)}{=} \omega\,\frac{e^2}{6\pi\varepsilon_0 m_e c^3} \overset{(3)}{=} \omega\, t_e. \tag{15.2}$$

(1) Eq. (15.1) with $|\dot{v}|^2 = \omega^4 x_0^2\cos^2\omega t$, whose average over a cycle is $\omega^4 x_0^2/2$.
(2) The factors x_0^2 and ω^3 cancel.
(3) Define the **radiation time** t_e below.

$$\boxed{t_e = \frac{e^2}{6\pi\varepsilon_0 m_e c^3} = \frac{2}{3}\,\frac{r_e}{c} = 6.27\times10^{-24}\,\mathrm{s}.} \tag{15.3}$$

Here r_e is the classical electron radius of Eq. (8.7), so t_e is two thirds of the time light takes to cross it. An electron loses the fraction ωt_e of its energy per radian, and $\omega t_e \ll 1$ for any motion short of 10^{23} turns per second. We may therefore work out the motion of a charge as if it did not radiate, and then add Larmor's power. This is how every spectrum below is found.

Fast charges and the boost matrix. Let a frame move with the charge at speed βc along x relative to the frame of the observer. We call the first the charge's frame (primed) and the second the lab. The lab components of an event or of a four-momentum follow from those in the charge's frame by the boost

$$\begin{pmatrix} ct \\ x \\ y \end{pmatrix} = \Lambda(\beta)\begin{pmatrix} ct' \\ x' \\ y' \end{pmatrix}, \qquad \Lambda(\beta) = \begin{pmatrix} \gamma_L & \gamma_L\beta & 0 \\ \gamma_L\beta & \gamma_L & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \gamma_L = \frac{1}{\sqrt{1-\beta^2}}, \tag{15.4}$$

with the z component unchanged and left out. The same matrix acts on an energy and momentum written as $(E/c,\ \text{momentum along } x,\ \text{momentum along } y)$. Its inverse is $\Lambda(-\beta)$, the boost from the lab into the charge's frame. Here γ_L is the **Lorentz factor**, with the subscript to keep it apart from the adiabatic exponent γ.

The power is the same in both frames. In its own frame the charge is momentarily at rest, so Larmor's formula holds there. In a time dt' it radiates an energy dE' and, by the symmetry of Eq. (15.1), no net momentum. The emissions happen at the charge, $x' = y' = 0$. Boosting the energy-momentum and the time interval to the lab,

$$\Lambda(\beta)\begin{pmatrix} dE'/c \\ 0 \\ 0 \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} \gamma_L\, dE'/c \\ \gamma_L\beta\, dE'/c \\ 0 \end{pmatrix}, \qquad \Lambda(\beta)\begin{pmatrix} c\, dt' \\ 0 \\ 0 \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} \gamma_L\, c\, dt' \\ \gamma_L\beta\, c\, dt' \\ 0 \end{pmatrix},$$
$$\Longrightarrow \quad dE = \gamma_L\, dE', \qquad dt = \gamma_L\, dt'. \tag{15.5}$$

(1) The matrix of Eq. (15.4) times a vector with only its first entry non-zero is that entry times the first column.

The factors γ_L cancel in the ratio,

$$\boxed{\mathcal{P} = \frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}E'}{\mathrm{d}t'} = \frac{e^2\,|\dot{\vec{v}}'|^2}{6\pi\varepsilon_0 c^3},} \tag{15.6}$$

where $\dot{\vec{v}}'$ is the acceleration in the charge's own frame. To find the power of a fast charge we find its acceleration in its own frame and use Larmor's formula there. The power that an observer receives is a different thing, since the observer also sees the pulses squeezed together in time. That squeezing sets the frequencies of synchrotron light in Section 15.3.

Beaming. A photon of frequency ν' emitted in the charge's frame at angle θ' to the direction of motion has four-momentum $(h\nu'/c)(1, \cos\theta', \sin\theta')$. In the lab it has frequency ν and angle θ,

$$\frac{h\nu}{c}\begin{pmatrix} 1 \\ \cos\theta \\ \sin\theta \end{pmatrix} \overset{(1)}{=} \Lambda(\beta)\,\frac{h\nu'}{c}\begin{pmatrix} 1 \\ \cos\theta' \\ \sin\theta' \end{pmatrix} \overset{(2)}{=} \frac{h\nu'}{c}\begin{pmatrix} \gamma_L(1+\beta\cos\theta') \\ \gamma_L(\cos\theta'+\beta) \\ \sin\theta' \end{pmatrix}$$
$$\Longrightarrow \quad \tan\theta \overset{(3)}{=} \frac{\sin\theta'}{\gamma_L(\cos\theta'+\beta)}, \qquad \nu \overset{(4)}{=} \gamma_L(1+\beta\cos\theta')\,\nu'. \tag{15.7}$$

(1) The boost (15.4) applied to the photon's four-momentum.
(2) The matrix product, row by row.
(3) The third row divided by the second.
(4) The first row.

A photon sent sideways in the charge's frame, $\theta' = \pi/2$, has $\tan\theta = 1/(\gamma_L\beta)$ in the lab. All the light of the forward half of the charge's frame, which is half of all its light, lands within this angle,

$$\boxed{\theta \lesssim \frac{1}{\gamma_L} \qquad (\gamma_L \gg 1).}$$

An electron with $\gamma_L = 10$ throws half its light into a cone of half-angle 6° about its velocity (Figure 15.1). The frequency in Eq. (15.7) at $\theta' = 0$ is $\gamma_L(1+\beta)\nu' = \sqrt{(1+\beta)/(1-\beta)}\,\nu'$, the exact Doppler shift of an approaching source, Eq. (3.27), as it must be.

15.2 Bremsstrahlung

The hot gas that fills a cluster of galaxies is at about 7×10^7 K, set by the depth of the cluster's gravitational well (Problem 10.7.2), and the gas is fully ionised.

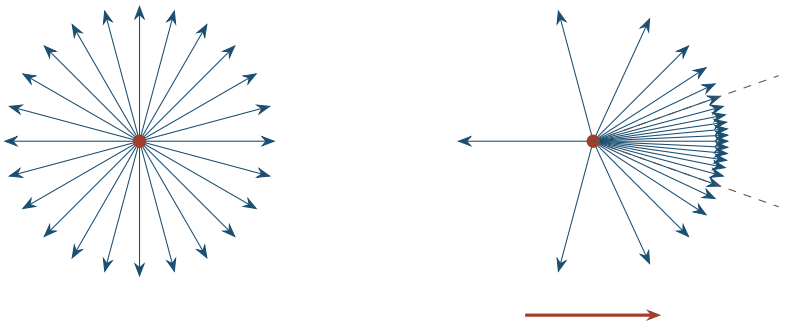


Figure 15.1: Photons emitted at equal steps of angle in the frame of the emitter (left), and the same photons in a frame where the emitter moves to the right with $\gamma_L = 3$ (right). The dashed lines mark the angle $\tan\theta = 1/(\gamma_L\beta)$.

It is also so thin that light crosses it freely, so by Eq. (11.8) it does not shine as a blackbody. Fully ionised gas has no bound electrons to make lines. Something else must make its light.

Think of rain on the roof of a car. Each drop makes a short click, and a short click contains every pitch up to about one over its length. The sound inside the car is the number of drops striking per second times the energy of each click. Heavier rain is louder, and the sharpest clicks give the highest pitches.

In hot gas the drops are electrons and the roof is the ions. Each electron that passes an ion is jerked once and sends out one pulse of light, as in Section 15.1. The gas glows with the sum of countless such pulses. The idea is to find the spectrum of one encounter and then count the encounters, over impact parameters (P6) and over the Maxwell distribution of speeds (P3).

The cluster in numbers. Its electrons have $k_BT = 6.0\,\mathrm{keV}$. No photon can carry more energy than the electron that made it (P1), so the spectrum must end near 6 keV, at 1.5×10^{18} Hz, in X-rays.

The picture is an electron flying past an ion, its path bent slightly near closest approach, with a flash of light sent out at the bend.

The same process is the braking radiation (*Bremsstrahlung*) of an X-ray tube. It makes the radio glow of ionised hydrogen around hot stars, the X-rays of solar flares and of clusters. Run backwards, as absorption, it is the opacity of stellar interiors. It uses P1, P3, P6 and, for absorption, the detailed balance of P4. It gives the emissivity of a hot plasma, its cooling rate and cooling time, and its absorption.

A gas of free charges. Before counting encounters one at a time, we check that each is a meeting of two particles and not a crowd. Put an ion of charge Z_ie at the origin, in gas of electron density n_e at temperature T. Electrons gather near it with the Boltzmann weight $e^{e\phi/k_BT}$, where ϕ is the electric potential (P3). Poisson's equation, the electric version of Eq. (9.13), then reads, away from the ion,

$$\begin{aligned}\nabla^2\phi &\overset{(1)}{=} -\frac{e}{\varepsilon_0}\left(Z_in_i - n_e\,e^{e\phi/k_BT}\right)\\ &\overset{(2)}{=} \frac{n_ee^2}{\varepsilon_0k_BT}\,\phi \overset{(3)}{=} \frac{\phi}{\lambda_D^2}\\ \Longrightarrow\qquad \phi &\overset{(4)}{=} \frac{Z_ie}{4\pi\varepsilon_0 r}\,e^{-r/\lambda_D}.\end{aligned} \tag{15.8}$$

(1) The charge density is that of the ions, Z_ien_i, minus that of the electrons. The ions are heavy and stay spread evenly on the scale considered.

(2) Far from the ion $e\phi \ll k_BT$, so $e^{e\phi/k_BT} \approx 1 + e\phi/k_BT$. The gas is neutral on average, $Z_i n_i = n_e$.
(3) Define the **Debye length** λ_D below.
(4) The spherically symmetric solution that matches the bare ion's potential at small r. Substituting it back gives $\nabla^2\phi = \phi/\lambda_D^2$.

$$\boxed{\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T}{n_e e^2}} = \frac{\sqrt{k_B T/m_e}}{2\pi\nu_p}.} \tag{15.9}$$

The second form uses the plasma frequency of Eq. (5.2). The Debye length is the distance a thermal electron covers in one plasma oscillation. Beyond it, the field of any charge is cancelled by the cloud it gathers. In cluster gas, $n_e = 10^3\,\mathrm{m}^{-3}$ and $T = 7\times10^7\,\mathrm{K}$ give $\lambda_D = 18\,\mathrm{km}$, with 2.5×10^{16} electrons inside a sphere of that radius. Gas round a hot star, $n_e = 10^8\,\mathrm{m}^{-3}$ and $T = 10^4\,\mathrm{K}$, gives $\lambda_D = 0.7\,\mathrm{m}$ and 1.4×10^8 electrons (P6). With so many charges inside each Debye sphere, the screening is a smooth average, and an electron feels one ion strongly only when it passes far closer than λ_D. Such a gas is a **plasma**. It behaves as an ideal gas, its encounters can be counted one at a time, and by Chapter 5 it lets no radiation below ν_p through.

One encounter. An electron of speed v passes an ion of charge $Z_i e$ at impact parameter b. Its path bends only slightly, so we take it as a straight line, with closest approach at $t = 0$. The force across the path is $Z_i e^2 b/[4\pi\varepsilon_0(b^2+v^2t^2)^{3/2}]$. The total sideways kick is

$$\Delta v \overset{(1)}{=} \frac{Z_i e^2 b}{4\pi\varepsilon_0 m_e}\int_{-\infty}^{\infty}\frac{dt}{(b^2+v^2t^2)^{3/2}} \overset{(2)}{=} \frac{2Z_i e^2}{4\pi\varepsilon_0 m_e\, bv}. \tag{15.10}$$

(1) Newton's law across the path, integrated over the passage.
(2) $\int_{-\infty}^{\infty} dt/(b^2+v^2t^2)^{3/2} = 2/(b^2v)$.

To find how the radiated energy is spread over frequency, write the acceleration through its Fourier transform $\tilde{\dot v}(\omega) = \int \dot v\, e^{i\omega t}dt$. For a real signal, Parseval's theorem gives $\int \dot v^2 dt = \frac{1}{\pi}\int_0^\infty |\tilde{\dot v}|^2 d\omega$, so the total energy W of the pulse is

$$W \overset{(1)}{=} \frac{e^2}{6\pi\varepsilon_0 c^3}\int_{-\infty}^{\infty}\dot v^2\,dt \overset{(2)}{=} \int_0^\infty \frac{e^2\,|\tilde{\dot v}(\omega)|^2}{6\pi^2\varepsilon_0 c^3}\,d\omega$$
$$\Longrightarrow \qquad \frac{\mathrm{d}W}{\mathrm{d}\omega} \overset{(3)}{=} \frac{e^2\,(\Delta v)^2}{6\pi^2\varepsilon_0 c^3}\Big[uK_1(u)\Big]^2, \qquad u = \frac{\omega b}{v}. \tag{15.11}$$

(1) Larmor's power (15.1) integrated over the passage.
(2) Parseval's theorem.
(3) The transform of the force law above is $\tilde v = \Delta v\, uK_1(u)$, with K_1 a modified Bessel function. The component of the acceleration along the path changes sign during the passage and adds much less at low frequency, so it is dropped.

The factor $uK_1(u)$ is 1 for $u \ll 1$ and falls as e^{-u} for $u \gg 1$. So

$$\boxed{\frac{\mathrm{d}W}{\mathrm{d}\omega} \approx \frac{e^2}{6\pi^2\varepsilon_0 c^3}\left(\frac{2Z_i e^2}{4\pi\varepsilon_0 m_e\, bv}\right)^2 \quad \text{for } \omega \lesssim \frac{v}{b}, \qquad \approx 0 \text{ above.}} \tag{15.12}$$

The pulse lasts about b/v, and a pulse holds every frequency up to the inverse of its length (P6). Below that the spectrum is flat. This is the click on the roof.

Counting encounters. In a unit volume, electrons of speed v meet ions with impact parameters between b and $b + db$ at the rate $n_e n_i v\, 2\pi b\, db$. Each such encounter gives Eq. (15.12) at the frequencies $\omega < v/b$, that is for $b < v/\omega$.

$$\begin{aligned}\frac{dW}{dt\, dV\, d\omega} &\overset{(1)}{=} n_e n_i v \int_{b_{\min}}^{b_{\max}} 2\pi b\, \frac{\mathrm{d}W}{\mathrm{d}\omega}\, db \\ &\overset{(2)}{=} n_e n_i v\, 2\pi\, \frac{e^2}{6\pi^2 \varepsilon_0 c^3} \left(\frac{2Z_i e^2}{4\pi\varepsilon_0 m_e v}\right)^2 \int_{b_{\min}}^{b_{\max}} \frac{db}{b} \\ &\overset{(3)}{=} \frac{16}{3}\, \frac{Z_i^2 e^6}{(4\pi\varepsilon_0)^3 c^3 m_e^2}\, \frac{n_e n_i}{v}\, \ln\frac{b_{\max}}{b_{\min}}. \end{aligned} \tag{15.13}$$

(1) Rate of encounters times the energy each gives per unit frequency.
(2) Eq. (15.12). Its b^{-2} against the b of the ring leaves db/b.
(3) Collect constants. $2\pi \times 4/(6\pi^2) = 4/(3\pi)$, and $e^2/\varepsilon_0 = 4\pi e^2/(4\pi\varepsilon_0)$.

Distant encounters stop counting at $b_{\max} = v/\omega$, where the pulse becomes too slow to make the frequency ω. Close ones stop at the larger of two lengths, the distance $Z_i e^2/(4\pi\varepsilon_0 m_e v^2)$ where the kick is no longer small, and the electron's quantum wavelength $\hbar/(m_e v)$. The logarithm changes slowly. The full quantum calculation replaces $\ln(b_{\max}/b_{\min})$ by $(\pi/\sqrt{3})\,\bar{g}$, where the **Gaunt factor** $\bar{g}$ is of order one. Slow electrons radiate more, as $1/v$, because they spend longer near each ion.

Averaging over the Maxwell distribution. In gas at temperature T the electrons have the Maxwell distribution of Eq. (9.23), with $4\pi v^2 e^{-m_e v^2/2k_B T} dv$ electrons per speed interval (P3). Only electrons with $\frac{1}{2}m_e v^2 \geq h\nu$ can make a photon of energy $h\nu$ (P1). The average of $1/v$ that Eq. (15.13) needs is

$$\left\langle \frac{1}{v} \right\rangle \overset{(1)}{=} \frac{\int_{v_{\min}}^{\infty} v\, e^{-m_e v^2/2k_B T}\, dv}{\int_0^{\infty} v^2\, e^{-m_e v^2/2k_B T}\, dv} \overset{(2)}{=} \frac{(k_B T/m_e)\, e^{-h\nu/k_B T}}{\sqrt{\pi/2}\,(k_B T/m_e)^{3/2}} \overset{(3)}{=} \sqrt{\frac{2m_e}{\pi k_B T}}\; e^{-h\nu/k_B T}. \tag{15.14}$$

(1) The weight v^2 of the Maxwell distribution times $1/v$, normalised. The lower limit is $\frac{1}{2}m_e v_{\min}^2 = h\nu$.
(2) The two Gaussian integrals, with $m_e v_{\min}^2/2k_B T = h\nu/k_B T$.
(3) Simplify.

Putting this into Eq. (15.13), using $d\omega = 2\pi\, dv$, and writing the result as $4\pi j_\nu$ with j_ν the emissivity per steradian of Chapter 11,

$$\boxed{4\pi j_\nu = \frac{32\pi^2}{3\sqrt{3}}\, \frac{Z_i^2 e^6}{(4\pi\varepsilon_0)^3 c^3 m_e^2} \sqrt{\frac{2m_e}{\pi k_B T}}\; \bar{g}\, n_e n_i\, e^{-h\nu/k_B T} = 6.8\times 10^{-51}\, Z_i^2\, \bar{g}\, \frac{n_e n_i}{\sqrt{T}}\, e^{-h\nu/k_B T},} \tag{15.15}$$

in $\mathrm{W\,m^{-3}\,Hz^{-1}}$, with n_e, n_i in $\mathrm{m^{-3}}$ and T in kelvin. The spectrum is flat, apart from the slow change of $\bar{g}$, up to $h\nu \approx k_B T$, and then falls exponentially (Figure 15.2). The place where it turns down measures the temperature of the gas directly. At low frequencies hotter gas is fainter, as $T^{-1/2}$, because its electrons are faster and spend less time near each ion.

Cooling. Integrating over frequency, $\int_0^\infty e^{-h\nu/k_B T} d\nu = k_B T/h$, so the power radiated per unit volume is

$$\boxed{4\pi \int_0^\infty j_\nu\, d\nu = 6.8\times 10^{-51}\, \frac{k_B}{h}\, Z_i^2\, \bar{g}\, n_e n_i \sqrt{T} = 1.4\times 10^{-40}\, Z_i^2\, \bar{g}\, n_e n_i \sqrt{T}\ \ \mathrm{W\,m^{-3}}.} \tag{15.16}$$

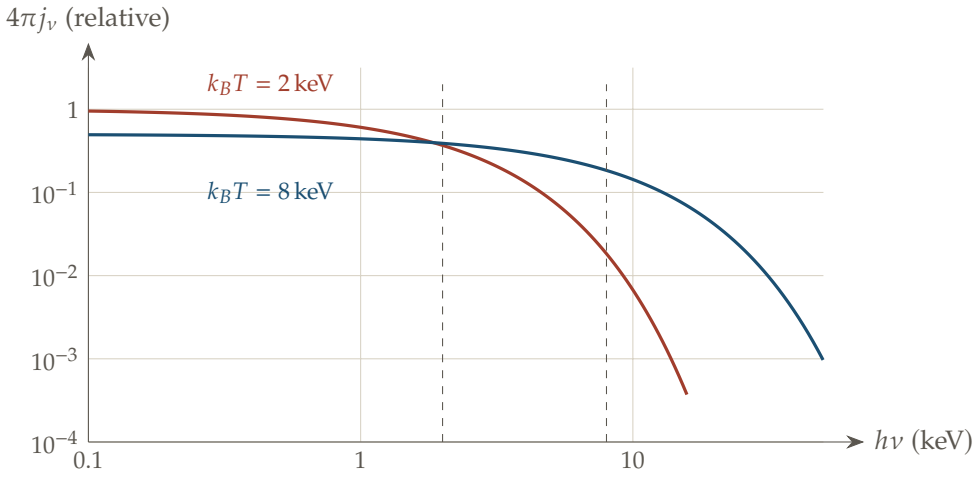


Figure 15.2: Bremsstrahlung emissivity $4\pi j_\nu \propto T^{-1/2} e^{-h\nu/k_BT}$ against photon energy, for the same electron and ion densities at $k_BT = 2$ keV and 8 keV, with $\bar{g}$ held fixed. Dashed lines mark $h\nu = k_BT$.

The rate goes as the square of the density, since it counts pairs, and as $\sqrt{T}$, the speed of the electrons. A hydrogen plasma with $n_e = n_i = n$ holds thermal energy $\frac{3}{2}(n_e + n_i)k_BT = 3nk_BT$ per unit volume. Its cooling time is that energy over the power,

$$t_{\text{cool}} = \frac{3nk_BT}{1.4\times10^{-40}\,\bar{g}\,n^2\sqrt{T}} = \frac{3k_B\sqrt{T}}{1.4\times10^{-40}\,\bar{g}\,n}.$$

For cluster gas, $n = 10^3\,\text{m}^{-3}$ and $T = 7\times10^7$ K, this is 2.4×10^{18} s, or 8×10^{10} years with $\bar{g} = 1$. That is longer than the age of the universe (Chapter 16), so the gas has not had time to cool since it was heated and still shines in X-rays (P6). A sphere of such gas 1 Mpc in radius radiates 1.5×10^{38} W, which is the X-ray luminosity of a rich cluster. Only in the dense centres of some clusters is the cooling time shorter than the age.

Free-free absorption and Kramers opacity. The reverse process, an electron absorbing a photon while it passes an ion, follows from Kirchhoff's law, Eq. (11.10), $\kappa_\nu\rho = j_\nu/B_\nu$ (P4).

$$\begin{aligned}\kappa_\nu\rho &\overset{(1)}{=} \frac{6.8\times10^{-51}}{4\pi}\,Z_i^2\bar{g}\,\frac{n_en_i}{\sqrt{T}}\,e^{-h\nu/k_BT}\,\frac{c^2\left(e^{h\nu/k_BT}-1\right)}{2h\nu^3}\\ &\overset{(2)}{=} \frac{6.8\times10^{-51}\,c^2}{8\pi h}\,Z_i^2\bar{g}\,\frac{n_en_i}{\sqrt{T}}\,\frac{1-e^{-h\nu/k_BT}}{\nu^3}.\end{aligned} \tag{15.17}$$

(1) Eq. (15.15) divided by 4π, over the Planck function (9.26).
(2) $e^{-h\nu/k_BT}(e^{h\nu/k_BT}-1) = 1 - e^{-h\nu/k_BT}$.

At radio frequencies, $h\nu \ll k_BT$, the last factor is $h\nu/k_BT$, and the absorption becomes $1.8\times10^{-12}\,Z_i^2\bar{g}\,n_en_iT^{-3/2}\nu^{-2}$ m^{-1}. Ionised gas round hot stars is therefore opaque at low radio frequencies and transparent at high ones. By Eq. (11.8) its spectrum rises as ν^2, the Rayleigh–Jeans part of $B_\nu(T)$, where it is thick, and is nearly flat where it is thin, $I_\nu \approx \tau_\nu B_\nu \propto \nu^{-2}\nu^2$.

Inside a star the light that carries the flux has $h\nu$ of a few k_BT (the Rosseland weighting of Section 11.4). Writing $\nu = x\,k_BT/h$ in Eq. (15.17),

$$\kappa_\nu\rho \propto \frac{n_en_i}{\sqrt{T}}\,\frac{1-e^{-x}}{x^3}\left(\frac{h}{k_BT}\right)^3 \propto n_en_i\,T^{-7/2}\,\frac{1-e^{-x}}{x^3}.$$

The average over x is a pure number, the same at every temperature, and $n_e n_i \propto \rho^2$. So

$$\boxed{\kappa_{\text{ff}} \propto \rho\, T^{-7/2}.} \tag{15.18}$$

This is Kramers' law of Section 12.3. The $T^{-7/2}$ is the ν^{-3} of one encounter, the $T^{-1/2}$ of the Maxwell average, and the shift of the important frequencies with temperature. Absorption by electrons bound to ions has the same ν^{-3} and gives the same law with a larger coefficient in gas with heavy elements.

15.3 Synchrotron Radiation

The power of a fast charge comes from its acceleration in its own frame (Eq. (15.6)), and its light is thrown forward into a cone of angle $1/\gamma_L$ (Eq. (15.7)).

Hot gas makes a spectrum that is flat and then cut off, and a blackbody one that rises and then cuts off. The radio light of the Milky Way and of radio galaxies is neither. It falls smoothly as about $\nu^{-0.7}$ over many decades of frequency, and it is strongly polarised. No single temperature makes such light.

Think of a lighthouse. Its lamp turns slowly, once every few seconds, but a ship far out sees only a brief flash each turn, when the narrow beam sweeps past. The narrower the beam, the shorter the flash. If the lamp were also rushing toward the ship, each flash would be squeezed shorter still.

An electron moving near the speed of light in a magnetic field circles the field lines. Its light is beamed into a narrow cone along its velocity, so a distant observer sees a short flash once per turn, made shorter because the electron nearly keeps up with its own light. A short flash means high frequencies. The idea is that each electron radiates near γ_L^2 times its circling frequency, with a power proportional to $\gamma_L^2 B^2$, and that a power law of electron energies gives a power law of light, whose slope comes from counting electrons (P3).

In numbers, the magnetic field of the Galaxy is about 0.5 nT. Electrons in it circle at 14 Hz when slow. An electron with $\gamma_L = 10^4$, an energy of 5 GeV, circles 10^4 times more slowly, once every twelve minutes, but its light comes out near $10^8 \times 14\,\text{Hz} = 1.4\,\text{GHz}$, in the radio.

The picture is a helix wound round a field line, with a narrow searchlight pointing along the electron's velocity and sweeping past us once per turn.

The same light comes from the electron rings of laboratory light sources, from the nebulae round young pulsars, from the jets and lobes of radio galaxies and from the afterglows of gamma-ray bursts. It uses P1 (a magnetic field does no work), P3 (counting electrons) and P6 (the length of the flash). It gives the motion, the power, the frequency of synchrotron light, and the spectrum of a power law of electrons.

Gyration. An electron of charge $-e$ and velocity $\vec{v}$ in a field $\vec{B}$ obeys

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\gamma_L m_e \vec{v}\right) = -e\,\vec{v} \times \vec{B}.$$

Taking the dot product with $\gamma_L m_e \vec{v}$,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\gamma_L^2 m_e^2 v^2\right) \overset{(1)}{=} -e\,\gamma_L m_e\,\vec{v}\cdot(\vec{v}\times\vec{B}) \overset{(2)}{=} 0 \quad\Longrightarrow\quad \gamma_L = \text{const.} \tag{15.19}$$

(1) $\vec{a}\cdot\dot{\vec{a}} = \frac{1}{2}\,d(a^2)/dt$ with $\vec{a} = \gamma_L m_e \vec{v}$.

(2) $\vec{v}\times\vec{B}$ is perpendicular to $\vec{v}$. Since $\gamma_L^2 v^2 = c^2(\gamma_L^2 - 1)$, a constant $\gamma_L^2 v^2$ means a constant γ_L.

The magnetic force does no work (P1). With γ_L constant the equation becomes $\gamma_L m_e \dot{\vec{v}} = -e\,\vec{v}\times\vec{B}$. The velocity along $\vec{B}$ stays fixed, and the velocity across $\vec{B}$ turns round at the angular frequency

$$\boxed{\frac{eB}{\gamma_L m_e} = \frac{2\pi\nu_g}{\gamma_L}, \qquad \nu_g = \frac{eB}{2\pi m_e} = 2.80\times10^{10}\,\text{Hz}\times\frac{B}{1\,\text{T}},} \tag{15.20}$$

where ν_g is the **gyrofrequency** of a slow electron. The path is a helix. A fast electron circles more slowly than a slow one, by γ_L. Its light comes out higher than ν_g, by γ_L^2 (Eq. (15.25)).

Power. Take the electron moving along x, across a field B along z, with no electric field in the lab. In the electron's frame the fields are different. The pair (E_y, cB_z) transforms under a boost along x with the same matrix as (ct, x), a result quoted from electromagnetism. Into the electron's frame the boost is $\Lambda(-\beta)$,

$$\begin{pmatrix} E'_y \\ cB'_z \end{pmatrix} = \begin{pmatrix} \gamma_L & -\gamma_L\beta \\ -\gamma_L\beta & \gamma_L \end{pmatrix}\begin{pmatrix} 0 \\ cB \end{pmatrix} \overset{(1)}{=} \begin{pmatrix} -\gamma_L\beta cB \\ \gamma_L cB \end{pmatrix}. \tag{15.21}$$

(1) The second column of the matrix times cB.

The electron at rest in its own frame feels the electric field $E' = \gamma_L vB$. The magnetic field exerts no force on a charge at rest. Its acceleration there is eE'/m_e, and by Eq. (15.6)

$$\begin{aligned} \mathcal{P} &\overset{(1)}{=} \frac{e^2}{6\pi\varepsilon_0 c^3}\left(\frac{e\gamma_L vB}{m_e}\right)^2 \overset{(2)}{=} \frac{e^4}{6\pi\varepsilon_0^2 m_e^2 c^4}\;\varepsilon_0 c\;\gamma_L^2\beta^2 c^2 B^2 \\ &\overset{(3)}{=} \sigma_T c\,\gamma_L^2\beta^2\,\frac{B^2}{\mu_0} \overset{(4)}{=} 2\sigma_T c\,\gamma_L^2\beta^2\,u_B. \end{aligned} \tag{15.22}$$

(1) Larmor's formula in the electron's frame, with $|\dot{\vec{v}}'| = eE'/m_e$.
(2) Multiply and divide by $\varepsilon_0 c$, and write $v = \beta c$.
(3) The Thomson cross-section of Eq. (8.7) is $\sigma_T = e^4/(6\pi\varepsilon_0^2 m_e^2 c^4)$, and $\varepsilon_0 c^2 = 1/\mu_0$.
(4) The magnetic energy density $u_B = B^2/(2\mu_0)$.

For a velocity at an angle to the field, only the part of $\vec{v}$ across $\vec{B}$ makes $\vec{v}\times\vec{B}$, and β^2 becomes the square of that part over c^2. Averaged over all directions of the velocity this is $\frac{2}{3}\beta^2$, so

$$\boxed{\mathcal{P} = \frac{4}{3}\,\sigma_T c\,\gamma_L^2\beta^2\,u_B.} \tag{15.23}$$

The power grows as the square of the electron's energy and is proportional to the energy density of the field. The electron enters only through γ_L, and the field only through u_B.

How long the electrons last. An electron with $\beta \approx 1$ radiates its energy $\gamma_L m_e c^2$ in the time

$$t_{\text{sync}} = \frac{\gamma_L m_e c^2}{\mathcal{P}} = \frac{3m_e c}{4\sigma_T\gamma_L u_B},$$

shorter for faster electrons. The Crab Nebula is the debris of the supernova of the year 1054. With a field of about 10 nT, the electrons that make its X-rays at 1 keV have $\gamma_L \approx 2.4\times10^7$ (by Eq. (15.25) below) and last about 100 years. That is far shorter than the age of the nebula (P6), so these electrons were not made in the explosion. The pulsar at the centre must supply them now. The electrons that make its radio light at 1 GHz have $\gamma_L \approx 1.5\times10^3$ and last over a million years, so they have built up over the whole life of the nebula.

The frequency, route A, the length of the flash. Take an electron moving across the field. The observer sees its light only while its velocity points within $1/\gamma_L$ of the line of sight (Eq. (15.7)), so the velocity turns through $2/\gamma_L$ during the flash. At the angular frequency of Eq. (15.20) this takes the lab time $\Delta t = (2/\gamma_L)/(2\pi\nu_g/\gamma_L) = 2/(2\pi\nu_g)$. Meanwhile the electron moves a distance $\beta c\,\Delta t$ toward the observer, so the light from the end of the flash starts that much closer and arrives sooner (P6).

$$\begin{aligned}\Delta t_{\text{obs}} &\overset{(1)}{=} \Delta t - \frac{\beta c\,\Delta t}{c} = \Delta t\,(1-\beta)\\ &\overset{(2)}{=} \frac{\Delta t}{\gamma_L^2(1+\beta)} \overset{(3)}{=} \frac{1}{\gamma_L^2}\,\frac{1}{2\pi\nu_g}.\end{aligned} \tag{15.24}$$

(1) The arrival time of the last light, minus the head start it gains.
(2) $1-\beta = (1-\beta^2)/(1+\beta) = 1/[\gamma_L^2(1+\beta)]$.
(3) $\Delta t = 2/(2\pi\nu_g)$ and $1+\beta \approx 2$.

The flash is γ_L^2 times shorter than a turn of a slow electron, so it holds frequencies up to about $\gamma_L^2\nu_g$.

Route B, the slow clock and the Doppler shift. In the electron's own frame the same interval lasts only $\Delta t' = \Delta t/\gamma_L$, by the time row of Eq. (15.4). The light it sends forward is raised in frequency, and shortened in time, by the factor $\gamma_L(1+\beta)$ of Eq. (15.7) at $\theta' = 0$. So $\Delta t_{\text{obs}} = \Delta t'/[\gamma_L(1+\beta)] = \Delta t/[\gamma_L^2(1+\beta)]$, the second line of route A exactly. The two routes are the same physics seen from two frames, light chasing in the lab, time dilation and Doppler shift in the electron's frame. Both give

$$\boxed{\nu_c \approx \gamma_L^2\,\nu_g.} \tag{15.25}$$

The full calculation of the radiated fields, quoted here, puts the **critical frequency** at $\nu_c = \frac{3}{2}\gamma_L^2\nu_g$ for motion across the field (for other angles ν_g is multiplied by the sine of the angle between $\vec{v}$ and $\vec{B}$). The spectrum of one electron is $F(\nu/\nu_c)$, with $F(x) = x\int_x^\infty K_{5/3}(t)\,dt$. It rises as $\nu^{1/3}$, peaks at $\nu = 0.29\,\nu_c$, and falls as $e^{-\nu/\nu_c}$ (Figure 15.3, top). The Galactic electrons of the opening, $\gamma_L = 10^4$ in 0.5 nT, have $\nu_c = 2.1$ GHz.

A power law of electrons. Electrons accelerated in shocks reach a distribution of energies that is a power law, $n(\gamma_L)\,d\gamma_L \propto \gamma_L^{-p}\,d\gamma_L$ between some $\gamma_{\min}$ and $\gamma_{\max}$, with p typically between 2 and 3. Each electron sends nearly all its power near its own $\nu = \gamma_L^2\nu_g$, so the power at ν comes from the electrons with $\gamma_L = (\nu/\nu_g)^{1/2}$. Counting them (P3),

$$\begin{aligned}4\pi j_\nu &\overset{(1)}{=} \mathcal{P}(\gamma_L)\,n(\gamma_L)\,\frac{\mathrm{d}\gamma_L}{\mathrm{d}\nu}\\ &\overset{(2)}{\propto} \gamma_L^2 B^2\;\gamma_L^{-p}\;\frac{1}{2\sqrt{\nu\nu_g}}\\ &\overset{(3)}{\propto} B^2\left(\frac{\nu}{\nu_g}\right)^{(2-p)/2}\frac{1}{\sqrt{\nu\nu_g}} \overset{(4)}{\propto} B^{(p+1)/2}\,\nu^{-(p-1)/2}.\end{aligned} \tag{15.26}$$

(1) The power at frequencies ν to $\nu + d\nu$ is that of the electrons in the matching $d\gamma_L$.
(2) Eq. (15.23) with $u_B \propto B^2$ and $\beta \approx 1$. From $\gamma_L = (\nu/\nu_g)^{1/2}$, $d\gamma_L/d\nu = 1/(2\sqrt{\nu\nu_g})$.
(3) $\gamma_L^{2-p} = (\nu/\nu_g)^{(2-p)/2}$.
(4) The powers of ν add to $(2-p)/2 - \frac{1}{2} = -(p-1)/2$. Since $\nu_g \propto B$, the powers of B add to $2 - (2-p)/2 - \frac{1}{2} = (p+1)/2$.

$$\boxed{j_\nu \propto \nu^{-(p-1)/2}.} \tag{15.27}$$

The slope of the light comes from the slope of the electrons alone. The shape of the one-electron spectrum sets only the two ends. As a check by a second route, the exact $F(x)$ summed over electrons with $p = 2.5$ and $\gamma_{\max} = 1000\,\gamma_{\min}$ gives a slope of -0.752 in the middle of the range, against -0.75 from Eq. (15.27). Below $\gamma_{\min}^2 \nu_c$ it rises with the slope 0.32 of a single electron, and above $\gamma_{\max}^2 \nu_c$ it falls exponentially (Figure 15.3, bottom).

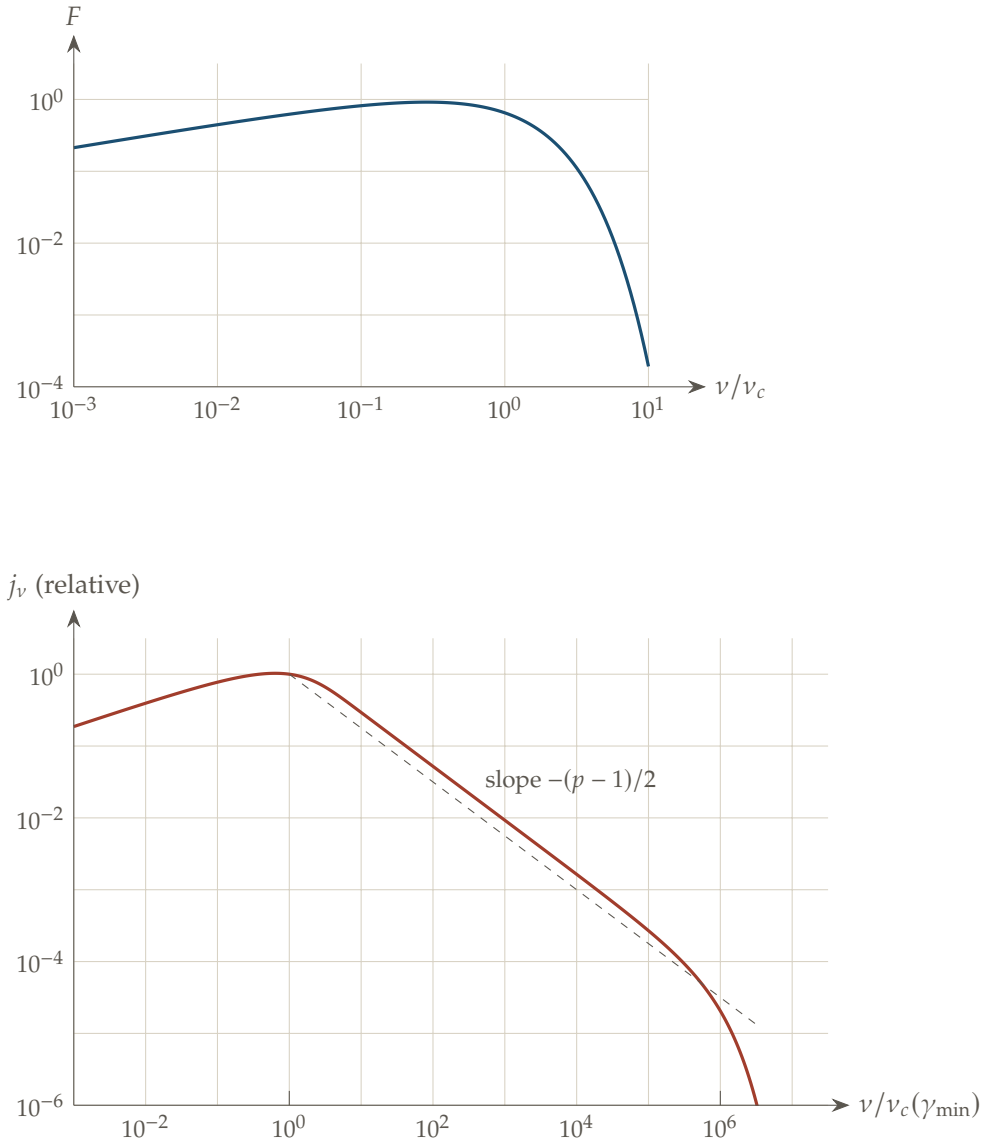


Figure 15.3: Top, the synchrotron spectrum $F(\nu/\nu_c)$ of one electron. Bottom, the spectrum of electrons with $n(\gamma_L) \propto \gamma_L^{-2.5}$ between $\gamma_{\min}$ and $1000\,\gamma_{\min}$, against $\nu/\nu_c(\gamma_{\min})$. Ticks on the frequency axis mark $\nu_c(\gamma_{\min})$ and $\nu_c(\gamma_{\max})$. The dashed line has slope -0.75. Computed with `code/ch15/01_synchrotron.py`.

Most radio galaxies have $\alpha \approx 0.7$ to 0.8 in $j_\nu \propto \nu^{-\alpha}$, so their electrons have $p \approx 2.4$ to 2.6. The radio spectrum of the Crab is flatter, $\alpha \approx 0.3$, so $p \approx 1.6$ there. Where the fastest electrons cool before they escape, the balance of supply and loss (P4) steepens the spectrum. Electrons supplied at a rate $\propto \gamma_L^{-p}$ each survive a time $t_{\text{sync}} \propto 1/\gamma_L$, so the number present goes as $\gamma_L^{-(p+1)}$, and above that energy α is larger by $\frac{1}{2}$. The frequency of the break gives the age of a radio source. Synchrotron light is also linearly polarised, across the direction of the field projected on

the sky, so maps of its polarisation are maps of cosmic magnetic fields. The strong polarisation of the Crab's optical light is what identified it as synchrotron light.

15.4 Compton and Inverse Compton Scattering

A free electron scatters light with the Thomson cross-section, Eq. (8.7), and Table 8.2 listed Compton and inverse Compton scattering among the processes that change the photon's energy.

The Thomson result assumes that the electron stays where it is and that the photon leaves with the energy it came with. A hard X-ray photon knocks the electron back. A fast electron throws the photon forward. We need to know how much energy changes hands, and what many such exchanges do to a spectrum.

Think of a tennis ball. Thrown at a fixed wall, it comes back at the speed it arrived. Thrown at a racket swinging toward it, it comes back much faster. To find how much faster, the easiest way is to stand on the racket. There the racket is a wall, the ball bounces back at its arrival speed, and we then add the racket's motion back.

A photon meeting an electron is the ball. If the electron is at rest, a hard photon gives it a kick and loses energy, the Compton effect. If the electron moves fast, the photon in the electron's frame is blue-shifted by about γ_L, bounces as in Thomson scattering, and is blue-shifted by another γ_L on the way back to the lab. The idea is to do the scattering in the electron's frame, where it is plain Thomson scattering, and to carry the photon there and back with the boost matrix of Eq. (15.4).

In numbers, a photon of the microwave background, with a mean energy of 6.3×10^{-4} eV, meeting an electron with $\gamma_L = 10^4$ comes out on average at 85 keV, a hard X-ray.

The picture is a slow photon struck head on and sent back along the electron's path.

The same exchange is the gravity assist of Chapter 6. In the planet's frame a spacecraft leaves at its arrival speed, and in the Sun's frame it has gained the planet's motion. Inverse Compton scattering makes the X-rays of radio lobes, the gamma rays of pulsar nebulae and blazars, and the X-ray glow of the hot coronae above accretion disks. It uses P1 (conservation of energy and momentum), the boost matrices, P3 for thermal electrons and P5 for the random walk of a photon's energy over many scatterings. It gives the Compton shift, the energy gain and power of inverse Compton scattering, the Compton y parameter and the Sunyaev–Zel'dovich effect.

The Compton shift. A photon of frequency ν meets an electron at rest and leaves at angle ψ with frequency ν_1. Write four-momenta as (energy over c, momentum), with the dot product $A \cdot B = A^0 B^0 - \vec{A} \cdot \vec{B}$. The photon has $k = (h\nu/c)(1, \hat{n})$ before and $k_1 = (h\nu_1/c)(1, \hat{n}_1)$ after, and the electron has $q = (m_e c, \vec{0})$ before and q_1 after. Conservation (P1) is $q + k = q_1 + k_1$.

$$
\begin{aligned}
m_e^2 c^2 &\overset{(1)}{=} q_1 \cdot q_1 = (q + k - k_1) \cdot (q + k - k_1) \\
&\overset{(2)}{=} m_e^2 c^2 + 2\, q \cdot k - 2\, q \cdot k_1 - 2\, k \cdot k_1 \\
\Longrightarrow \quad m_e h(\nu - \nu_1) &\overset{(3)}{=} \frac{h^2 \nu \nu_1}{c^2} (1 - \cos\psi) \\
\Longrightarrow \quad \frac{c}{\nu_1} - \frac{c}{\nu} &\overset{(4)}{=} \frac{h}{m_e c} (1 - \cos\psi).
\end{aligned}
\tag{15.28}
$$

(1) The electron's mass is the same before and after. Conservation gives q_1.
(2) Expand. $q \cdot q = m_e^2 c^2$, and $k \cdot k = k_1 \cdot k_1 = 0$ for photons.
(3) $q \cdot k = m_e h\nu$, $q \cdot k_1 = m_e h\nu_1$, and $k \cdot k_1 = (h^2 \nu \nu_1/c^2)(1 - \hat{n} \cdot \hat{n}_1)$ with $\hat{n} \cdot \hat{n}_1 = \cos\psi$.
(4) Divide by $m_e h \nu \nu_1 / c$.

$$\boxed{\lambda_1 - \lambda = \lambda_C\,(1-\cos\psi), \qquad h\nu_1 = \frac{h\nu}{1+(h\nu/m_ec^2)(1-\cos\psi)},} \tag{15.29}$$

with the Compton wavelength $\lambda_C = h/(m_ec) = 2.43\times10^{-12}\,\mathrm{m}$ and $m_ec^2 = 511\,\mathrm{keV}$. The shift in wavelength does not depend on the wavelength. It matters only when λ is not much longer than λ_C, that is when $h\nu$ approaches m_ec^2. For $h\nu \ll m_ec^2$ the photon keeps its energy, the Thomson limit. A 10 keV X-ray scattered by cold electrons loses about 2% of its energy each time.

Inverse Compton, there and back. Now let the electron move with γ_L along x, and let a photon of frequency ν arrive at angle θ to the electron's velocity. Into the electron's frame, with $\Lambda(-\beta)$,

$$\frac{h\nu'}{c}\begin{pmatrix}1\\ \cos\theta'\\ \sin\theta'\end{pmatrix} \overset{(1)}{=} \begin{pmatrix}\gamma_L & -\gamma_L\beta & 0\\ -\gamma_L\beta & \gamma_L & 0\\ 0 & 0 & 1\end{pmatrix}\frac{h\nu}{c}\begin{pmatrix}1\\ \cos\theta\\ \sin\theta\end{pmatrix} = \frac{h\nu}{c}\begin{pmatrix}\gamma_L(1-\beta\cos\theta)\\ \gamma_L(\cos\theta-\beta)\\ \sin\theta\end{pmatrix}$$
$$\Longrightarrow \qquad \nu' \overset{(2)}{=} \gamma_L(1-\beta\cos\theta)\,\nu. \tag{15.30}$$

(1) The inverse boost of Eq. (15.4) applied to the photon's four-momentum.
(2) The first row.

If $h\nu' \ll m_ec^2$, the scattering in the electron's frame is Thomson scattering. The photon keeps the frequency ν' and leaves at some angle θ_1'. Back to the lab, with $\Lambda(\beta)$, the first row of Eq. (15.7) gives

$$\boxed{\nu_1 = \gamma_L(1+\beta\cos\theta_1')\,\nu' = \gamma_L^2\,(1-\beta\cos\theta)(1+\beta\cos\theta_1')\,\nu.} \tag{15.31}$$

A photon met head on, $\theta = \pi$, and sent back forward, $\theta_1' = 0$, gains the most, $\nu_1 = \gamma_L^2(1+\beta)^2\nu \approx 4\gamma_L^2\nu$. One factor γ_L comes from each boost. The condition $h\nu' \approx \gamma_L h\nu \ll m_ec^2$ must hold in the electron's frame. When it fails, the recoil of Eq. (15.29) takes over in that frame, and the photon can gain at most about the electron's own energy $\gamma_L m_ec^2$.

The power. Let the electron move through photons of energy density u_{rad}, coming equally from all directions. Photons from angle θ meet the electron at the closing speed $c(1-\beta\cos\theta)$, so a fraction $d\Omega/4\pi$ of them, carrying energy density $u_{\mathrm{rad}}\,d\Omega/4\pi$, is scattered at the rate $\sigma_T c(1-\beta\cos\theta)$ per unit of photon density. The Thomson pattern sends as much forward as backward in the electron's frame, so $\cos\theta_1'$ averages to zero in Eq. (15.31). The power that leaves in scattered photons, minus the power those photons brought in, is

$$\begin{aligned}\mathcal{P} &\overset{(1)}{=} \sigma_T c\int\frac{d\Omega}{4\pi}\,u_{\mathrm{rad}}\,(1-\beta\cos\theta)\Big[\gamma_L^2(1-\beta\cos\theta)-1\Big]\\ &\overset{(2)}{=} \sigma_T c\,u_{\mathrm{rad}}\Big[\gamma_L^2\Big(1+\frac{\beta^2}{3}\Big)-1\Big]\\ &\overset{(3)}{=} \sigma_T c\,u_{\mathrm{rad}}\Big[\gamma_L^2\beta^2+\frac{\gamma_L^2\beta^2}{3}\Big].\end{aligned} \tag{15.32}$$

(1) Rate of scattering times the energy gained, which is $\gamma_L^2(1-\beta\cos\theta)$ times the arriving energy by Eq. (15.31), minus the arriving energy.
(2) $\int(1-\beta\cos\theta)^2\,d\Omega/4\pi = 1+\beta^2/3$ and $\int(1-\beta\cos\theta)\,d\Omega/4\pi = 1$.
(3) $\gamma_L^2 - 1 = \gamma_L^2\beta^2$.

$$\boxed{\mathcal{P} = \frac{4}{3}\,\sigma_T c\,\gamma_L^2\beta^2\,u_{\rm rad}.} \tag{15.33}$$

This has the form of the synchrotron power, Eq. (15.23), with the energy density of light in place of that of the field. A fast electron loses energy to the two in the ratio $u_{\rm rad}/u_B$. Since photons are scattered at the rate $\sigma_T c\,u_{\rm rad}/\langle h\nu\rangle$, the mean energy of a scattered photon is $\frac{4}{3}\gamma_L^2\,\langle h\nu\rangle$.

The microwave background fills all space with $u_{\rm rad} = a_{\rm rad}T^4 = 4.2\times10^{-14}\,\mathrm{J\,m^{-3}}$, using Eq. (9.25) at 2.725 K. That equals the energy density of a field of 0.32 nT. In radio lobes, with fields of order 1 nT, scattering the background costs the electrons a tenth or more of what synchrotron light costs them, and the scattered photons come out as X-rays. In the Crab the electrons scatter their own synchrotron photons up to gamma rays.

Many scatterings in hot gas. Now let the electrons be thermal, with $\beta \ll 1$, and let the photons have $h\nu \ll k_BT$. Two numbers describe one scattering. The mean fractional gain follows from Eq. (15.33) divided by the scattering rate, with $\langle\gamma_L^2\beta^2\rangle \approx \langle v^2\rangle/c^2 = 3k_BT/m_ec^2$ from the Maxwell distribution (P3). The spread follows from Eq. (15.31) at first order in β, $\Delta\nu/\nu = \beta(\cos\theta_1' - \cos\theta)$, where the incoming and outgoing directions are unrelated to the electron's velocity and each has $\langle\cos^2\rangle = \frac{1}{3}$,

$$\left\langle\frac{\Delta\nu}{\nu}\right\rangle = \frac{4}{3}\cdot\frac{3k_BT}{m_ec^2} = \frac{4k_BT}{m_ec^2}, \qquad \left\langle\left(\frac{\Delta\nu}{\nu}\right)^2\right\rangle = \frac{3k_BT}{m_ec^2}\left(\frac{1}{3}+\frac{1}{3}\right) = \frac{2k_BT}{m_ec^2}. \tag{15.34}$$

The recoil of Eq. (15.29), a loss of $h\nu/m_ec^2$, is negligible for $h\nu \ll k_BT$. Each scattering counts as a step of size k_BT/m_ec^2, so the effect of a cloud is measured by the **Compton y parameter**,

$$\boxed{y = \int \frac{k_BT}{m_ec^2}\,n_e\sigma_T\,dl,} \tag{15.35}$$

the number of scatterings times k_BT/m_ec^2. A thin cloud scatters a photon about $\tau = \int n_e\sigma_T dl$ times, and a thick one about τ^2 times by the random walk of Eq. (9.37) (P5).

The random walk in frequency. Each photon takes a random walk in $x = h\nu/k_BT_\gamma$, where T_γ is any fixed reference temperature, with a drift. Let $N(x)\,dx$ be the number of photons in dx. Per unit y, Eq. (15.34) gives a mean step $4x$ and a mean square step $2x^2$. Photons drift up with the flux $4xN$, and they spread by a random walk (P5), as in Eq. (9.39) but with a step that depends on x. The number of photons is conserved (P1), so N obeys a continuity equation. With $N = x^2f$, where f is the photon occupation number of Section 9.3,

$$\begin{aligned}\frac{\partial N}{\partial y} &\overset{(1)}{=} -\frac{\partial}{\partial x}\left[4xN - \frac{1}{2}\frac{\partial}{\partial x}\left(2x^2N\right)\right]\\ &\overset{(2)}{=} -\frac{\partial}{\partial x}\left[4x^3f - \frac{\partial}{\partial x}\left(x^4f\right)\right]\\ &\overset{(3)}{=} \frac{\partial}{\partial x}\left(x^4\frac{\partial f}{\partial x}\right).\end{aligned} \tag{15.36}$$

(1) Continuity in x (P1). The flux is the drift plus the random walk. For the walk, let a photon at x step up or down by $\delta(x)$ with equal chances. Photons within δ below a point cross it upward with chance $\frac{1}{2}$, and those within δ above cross downward, so the net flux is $\frac{1}{2}N\delta$ half a step below minus $\frac{1}{2}N\delta$ half a step above, $-\frac{1}{2}\,\partial(\delta^2N)/\partial x$. Per unit y, δ^2 averages to $2x^2$.

(2) $N = x^2f$.

(3) $\partial(x^4 f)/\partial x = 4x^3 f + x^4\,\partial f/\partial x$, and the $4x^3 f$ terms cancel.

$$\boxed{\frac{\partial (x^2 f)}{\partial y} = \frac{\partial}{\partial x}\left(x^4 \frac{\partial f}{\partial x}\right).} \tag{15.37}$$

This is the Kompaneets equation, in the limit where recoil can be neglected. Multiplying by x and integrating over all x by parts twice gives $d\langle x\rangle/dy = 4\langle x\rangle$, so the mean photon energy grows as e^{4y}. Where $y \gtrsim 1$, photons from a cool source are lifted into a power law of X-rays, which is how the hot coronae above accretion disks make their X-rays.

The Sunyaev–Zel'dovich effect. Light of the microwave background crossing the hot gas of a cluster has $y \ll 1$. Take T_γ to be its temperature, so before the cluster $f_0 = 1/(e^x - 1)$. To first order in y, Eq. (15.37) gives

$$\begin{aligned} \frac{\Delta f}{f_0} &\overset{(1)}{=} \frac{y}{x^2 f_0}\,\frac{\mathrm{d}}{\mathrm{d}x}\left(x^4 \frac{\mathrm{d}f_0}{\mathrm{d}x}\right) \\ &\overset{(2)}{=} y\,\frac{x e^x}{e^x - 1}\left(x\,\frac{e^x + 1}{e^x - 1} - 4\right) \xrightarrow{x \ll 1} -2y. \end{aligned} \tag{15.38}$$

(1) Integrate Eq. (15.37) across the cluster with the right-hand side evaluated on f_0.
(2) $df_0/dx = -e^x/(e^x - 1)^2$, differentiated once more. For $x \ll 1$ the first factor tends to 1 and the bracket to $2 - 4$.

At low frequencies, where the intensity is proportional to the temperature, the background seen through the cluster looks colder,

$$\boxed{\frac{\Delta T}{T_{\mathrm{CMB}}} = -2y \qquad (h\nu \ll k_B T_{\mathrm{CMB}}).}$$

Photons are moved up in frequency, and none is lost, so the dip at low frequency is paid for by an excess at high frequency. The change is zero at $x = 3.83$, 217 GHz (Figure 15.4). A cluster with $n_e = 10^3\,\mathrm{m}^{-3}$ and $k_B T = 6\,\mathrm{keV}$ across 1 Mpc has $\tau = 2.1 \times 10^{-3}$, $k_B T/m_e c^2 = 0.012$ and $y = 2.4 \times 10^{-5}$, so the background dims by 130 μK. The dip is a fixed fraction of the background itself, so it does not fade with the distance of the cluster (P6). Surveys find clusters by it out to great distances.

15.5 Accretion and Active Galaxies

The compactness $GM/(Rc^2)$ measures how deep the gravitational well at the surface of a body is, against c^2 (Section 13.1). The Eddington luminosity caps how brightly a mass M can shine (Eq. (11.17)).

A quasar shines at about 10^{40} W, a thousand times a large galaxy of 10^{10} to 10^{11} Suns, and it changes in brightness within days, so the light comes from a region a few light-days across. Nuclear burning releases only 0.7% of the rest mass of its fuel (Section 12.3). The fuel needed by nuclear burning would not fit in so small a region, and it would take too long to burn. Something more powerful is at work.

Think of water flowing over a dam. Each kilogram that falls turns the turbines with an energy set by the height of the fall. A taller dam gives more energy from each kilogram, and the power is that energy times the kilograms per second.

Gas falling onto a compact object falls down a very deep well. Each kilogram releases GM/R, which is c^2 times the compactness. The idea is that the share of rest mass released by accretion

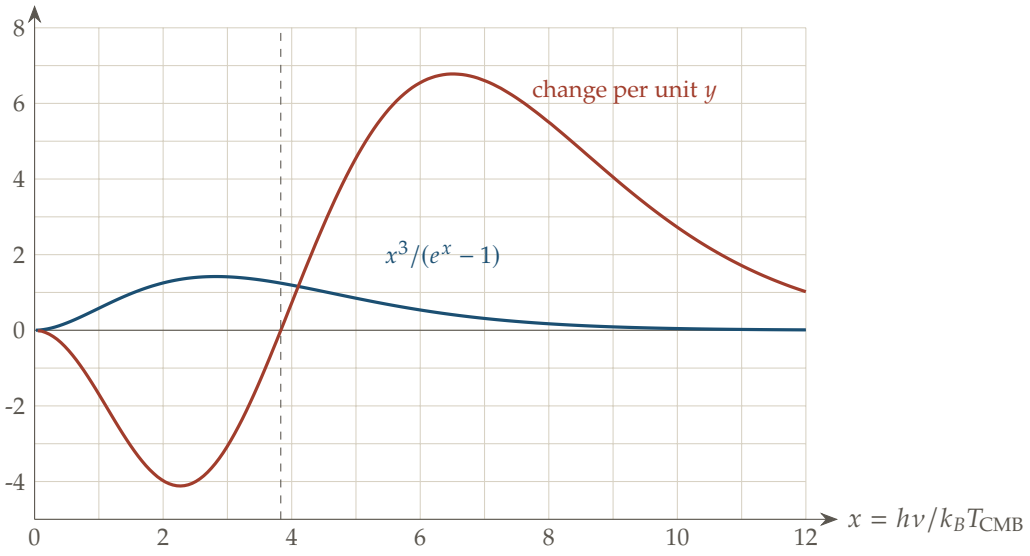


Figure 15.4: The microwave background intensity shape $x^3/(e^x-1)$, and its change per unit y after crossing hot gas, $x^4e^x(e^x-1)^{-2}[x(e^x+1)/(e^x-1)-4]$, both in units of $2(k_BT_{\rm CMB})^3/(hc)^2$. The dashed line marks $x = 3.83$.

equals the compactness, up to tens of percent, and that the gas, spiralling in through a disk, radiates the energy locally as heat (P1, P4).

In numbers, matter falling onto a neutron star releases 0.17 of its rest mass, 25 times more per kilogram than burning hydrogen. A quasar shining at 10^{40} W with an efficiency of 0.1 swallows $1.1\times10^{24}\,\mathrm{kg\,s^{-1}}$, 18 solar masses a year.

The picture is gas spiralling slowly inward in a thin, flat disk, hotter toward the centre, with narrow jets leaving along the axis.

The same process powers X-ray binaries, the outbursts of white dwarfs in binaries, the disks round young stars, and the nuclei of active galaxies. It uses P1 for energy and for angular momentum, P2, P4 for the local balance of heating and cooling, and P6 for light-travel times. It gives the luminosity and efficiency of accretion, black-hole masses, the temperature and spectrum of a thin disk, and the apparent speeds of jets.

Accretion luminosity. Matter falling from far away onto the surface of a body of mass M and radius R, at the rate $\dot M$, loses the potential energy GM/R per unit mass, which ends up as heat and light (P1),

$$\boxed{L = \frac{GM\dot M}{R}, \qquad \eta \equiv \frac{L}{\dot Mc^2} = \frac{GM}{Rc^2}.} \tag{15.39}$$

The **efficiency** η is the compactness of Table 13.1, 2.6×10^{-4} for a white dwarf like Sirius B and 0.17 for a neutron star. A black hole has no surface. Gas radiates while it spirals down to the last stable circular orbit, which general relativity puts at $6GM/c^2$ for a black hole without spin. The efficiency is then $1-\sqrt{8/9} = 0.057$, and it rises to 0.42 for the fastest spin allowed. These two numbers are quoted from general relativity. The Newtonian estimate, half of GM/R at $R = 6GM/c^2$ (the half is explained below), gives $1/12 = 0.083$.

The Eddington limit and black-hole masses. Light from the infalling gas pushes it back. By Eq. (11.17) a steady source of mass M shines at most at $1.26\times10^{31}\,\mathrm{W}\,(M/M_\odot)$. A quasar at 10^{40} W therefore holds at least $8\times10^8\,M_\odot$ (Problem 11.8.4). A change in brightness within a

day limits its size to the distance light travels in a day, 2.6×10^{13} m, since the far side of a larger source would not hear in time that it should brighten (P6). The Schwarzschild radius of $10^9\,M_\odot$ is 3×10^{12} m. So the mass is packed into a few tens of Schwarzschild radii at most, and only a black hole fits.

The thin disk, angular momentum. Gas does not fall straight in. It has angular momentum, and it settles into a thin disk where each ring circles at the Kepler angular velocity $\sqrt{GM/r^3}$, with angular momentum $\sqrt{GMr}$ per unit mass. A ring can move inward only by handing angular momentum outward, which friction between neighbouring rings does. Let $\mathcal{T}(r)$ be the torque that the gas inside r exerts on the gas outside. In a steady disk the angular momentum carried inward through r by the gas, minus that carried outward by the torque, is the same at every radius (P1). At the inner edge R_{in} the gas falls freely into the hole, so the torque there is zero.

$$\begin{aligned} \dot{M}\sqrt{GMr} - \mathcal{T}(r) &\overset{(1)}{=} \dot{M}\sqrt{GMR_{\text{in}}} - 0 \\ \Longrightarrow \qquad \mathcal{T}(r) &= \dot{M}\sqrt{GM}\left(\sqrt{r} - \sqrt{R_{\text{in}}}\right). \end{aligned} \tag{15.40}$$

(1) The constant evaluated at the inner edge, where $\mathcal{T} = 0$.

The thin disk, energy. The energy flowing outward through radius r has two parts. The gas, moving inward, carries its orbital energy $-GM/2r$ per unit mass, which is an outward flow of $\dot{M}GM/2r$. The torque does work on the outer gas at the rate $\mathcal{T}\sqrt{GM/r^3}$. What the ring between r and $r + dr$ takes from this flow, it radiates (P1). Both faces radiate as blackbodies at the local temperature $T(r)$ (P4, Eq. (4.24)).

$$\begin{aligned} 2 \cdot 2\pi r\, dr\, \sigma_{\text{SB}} T^4 &\overset{(1)}{=} -\frac{\mathrm{d}}{\mathrm{d}r}\left[\frac{GM\dot{M}}{2r} + \mathcal{T}\sqrt{\frac{GM}{r^3}}\,\right] dr \\ &\overset{(2)}{=} -GM\dot{M}\,\frac{\mathrm{d}}{\mathrm{d}r}\left[\frac{1}{2r} + \frac{1}{r} - \frac{\sqrt{R_{\text{in}}}}{r^{3/2}}\right] dr \\ &\overset{(3)}{=} GM\dot{M}\left[\frac{3}{2r^2} - \frac{3\sqrt{R_{\text{in}}}}{2r^{5/2}}\right] dr = \frac{3GM\dot{M}}{2r^2}\left(1 - \sqrt{\frac{R_{\text{in}}}{r}}\right) dr. \end{aligned} \tag{15.41}$$

(1) Radiation from the two faces of the ring equals the energy flow into it from inside minus the flow out of it.
(2) The torque (15.40), so $\mathcal{T}\sqrt{GM/r^3} = GM\dot{M}\left(r^{-1} - \sqrt{R_{\text{in}}}\,r^{-3/2}\right)$.
(3) Differentiate term by term.

$$\boxed{\sigma_{\text{SB}} T^4 = \frac{3GM\dot{M}}{8\pi r^3}\left(1 - \sqrt{\frac{R_{\text{in}}}{r}}\right), \qquad T \propto r^{-3/4} \text{ for } r \gg R_{\text{in}}.} \tag{15.42}$$

The mass and the accretion rate set the temperature together, as $M\dot{M}$. Far out, a ring radiates three times the orbital energy that its own gas gives up. The extra two thirds is energy carried outward by the torque from the rings inside.

As a check by energy alone, the whole disk radiates

$$\int_{R_{\text{in}}}^{\infty} \frac{3GM\dot{M}}{2r^2}\left(1 - \sqrt{\frac{R_{\text{in}}}{r}}\right) dr = \frac{GM\dot{M}}{2R_{\text{in}}},$$

half of the energy $GM\dot{M}/R_{\text{in}}$ released by the fall. The other half is the orbital kinetic energy the gas still has at R_{in}, as the virial theorem requires for a circular orbit (Section 10.2). On a star it is released where the gas meets the surface, and a black hole swallows it.

Writing $T_* = (3GM\dot{M}/8\pi\sigma_{SB}R_{in}^3)^{1/4}$, the temperature is $T = T_*(R_{in}/r)^{3/4}(1-\sqrt{R_{in}/r})^{1/4}$. It peaks at $r = \frac{49}{36}R_{in}$, where $T = 0.488\,T_*$ (Figure 15.5). For a black hole of $10^8\,M_\odot$ fed $1\,M_\odot$ a year, with $R_{in} = 6GM/c^2$, the hottest ring is at 1.1×10^5 K and the disk shines in the ultraviolet with 4.7×10^{38} W. For a black hole of $10\,M_\odot$ fed $10^{-8}\,M_\odot$ a year it is at 3.5×10^6 K, $k_BT = 0.30$ keV, in X-rays. With $R_{in}\propto M$, $T_*\propto(\dot{M}/M^2)^{1/4}$, and at a fixed fraction of the Eddington limit $\dot{M}\propto M$, so $T_*\propto M^{-1/4}$. Small black holes have hotter disks.

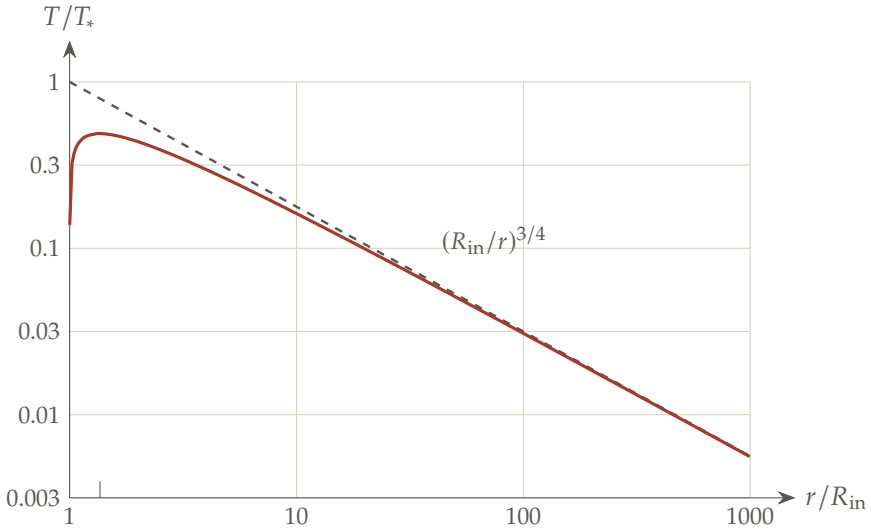


Figure 15.5: Temperature of a thin accretion disk, $T/T_* = (R_{in}/r)^{3/4}(1-\sqrt{R_{in}/r})^{1/4}$, against radius, with the power law $(R_{in}/r)^{3/4}$ dashed. The tick marks $r = \frac{49}{36}R_{in}$.

The spectrum of the disk. By Eq. (4.20), each face of a ring sends out the flux $\pi B_\nu(T(r))$. Far from the inner edge, $T = T_*(R_{in}/r)^{3/4}$. To add up the rings at a frequency ν, change the variable from r to $u = h\nu/k_BT(r) = (h\nu/k_BT_*)(r/R_{in})^{3/4}$, so that $r = R_{in}(uk_BT_*/h\nu)^{4/3}$.

$$\begin{aligned} L_\nu &\overset{(1)}{=} \int 2\cdot 2\pi r\;\pi B_\nu\big(T(r)\big)\,dr \\ &\overset{(2)}{=} 4\pi^2\,\frac{2h\nu^3}{c^2}\int \frac{r\,dr}{e^u-1} \\ &\overset{(3)}{=} 4\pi^2\,\frac{2h\nu^3}{c^2}\,\frac{4}{3}\,R_{in}^2\left(\frac{k_BT_*}{h\nu}\right)^{8/3}\int_0^\infty \frac{u^{5/3}\,du}{e^u-1} \;\propto\; \nu^{1/3}. \end{aligned} \tag{15.43}$$

(1) Two faces of each ring of area $2\pi r\,dr$.
(2) The Planck function (9.26) with $h\nu/k_BT(r) = u$.
(3) $r\,dr = \frac{4}{3}R_{in}^2(k_BT_*/h\nu)^{8/3}u^{5/3}du$. For $k_BT(\text{outer edge}) \ll h\nu \ll k_BT_*$ the limits of u are 0 and ∞, and the integral is the number $\Gamma(\frac{8}{3})\zeta(\frac{8}{3}) = 1.93$.

Between the temperatures of the outer and inner edges a disk is neither a blackbody nor the power law of fast electrons. It is a blackbody stretched over many temperatures, $L_\nu\propto\nu^{1/3}$, cut off at the frequency of its hottest ring. In quasars this hump appears in the ultraviolet.

Faster than light. Jets from active nuclei carry blobs of radio-emitting plasma outward, and radio interferometers (Chapter 5) follow them across the sky. Many seem to move faster than light. Let a blob leave the core at $t = 0$ with speed βc at angle θ to the line of sight. At time t it has moved $\beta ct\cos\theta$ toward us and $\beta ct\sin\theta$ across the sky. Its light set out from a point

$\beta ct \cos\theta$ nearer to us, so the light from the two moments arrives separated by less than t (P6).

$$\Delta t_{\text{obs}} \overset{(1)}{=} t - \frac{\beta ct \cos\theta}{c} = t\,(1 - \beta\cos\theta)$$
$$\Longrightarrow \quad \beta_{\text{app}} \overset{(2)}{=} \frac{\beta ct \sin\theta}{c\,\Delta t_{\text{obs}}} = \frac{\beta\sin\theta}{1-\beta\cos\theta}. \tag{15.44}$$

(1) The time between emissions minus the head start of the later light.
(2) The apparent speed across the sky, in units of c, is the sideways distance over the observed time.

$$\boxed{\beta_{\text{app}} = \frac{\beta\sin\theta}{1-\beta\cos\theta}.} \tag{15.45}$$

Its derivative with respect to θ is $\beta(\cos\theta - \beta)/(1-\beta\cos\theta)^2$, which is zero at $\cos\theta = \beta$. There $\sin\theta = 1/\gamma_L$ and $\beta_{\text{app}} = \beta\gamma_L$, which exceeds 1 for $\beta > 1/\sqrt{2}$ (Figure 15.6). Nothing moves faster than light. The blob nearly keeps up with its own light, so a long time of travel is squeezed into a short time of observation, the same squeezing as in the synchrotron flash. Since $\beta_{\text{app}} \le \beta\gamma_L < \gamma_L$, a measured apparent speed gives a lower limit on the Lorentz factor of the jet. The jet of M87 shows features moving at about $6c$, so its Lorentz factor is at least 6 (Problem 15.6.5). By Eq. (15.7), light from the approaching jet is raised in frequency and thrown forward, and light from the receding jet is lowered and thrown away from us. That is why most jets look one-sided.

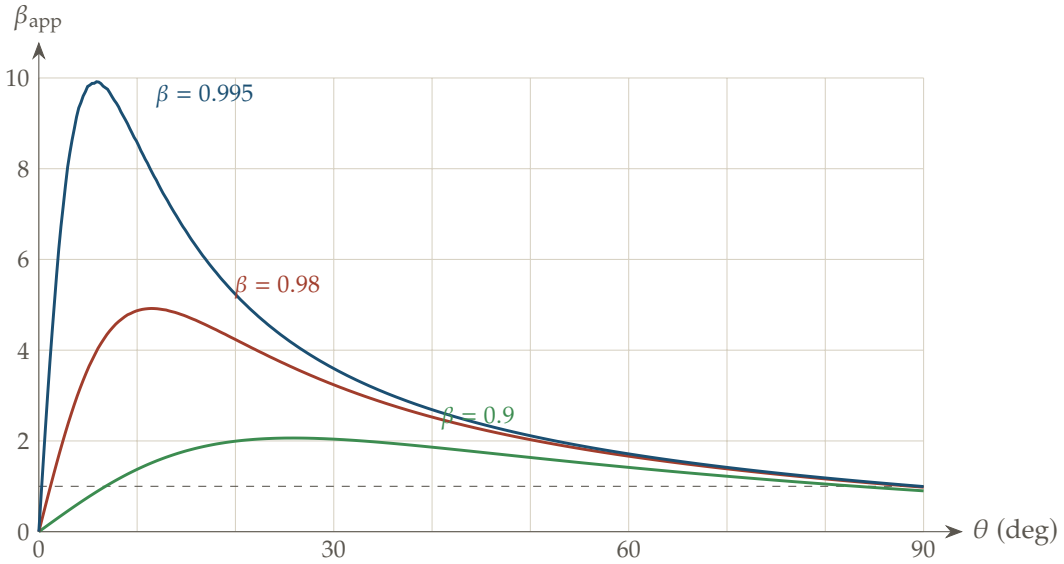


Figure 15.6: Apparent speed across the sky, $\beta_{\text{app}} = \beta\sin\theta/(1-\beta\cos\theta)$, against the angle θ between the motion and the line of sight, for $\beta = 0.9$, 0.98 and 0.995. The dashed line marks $\beta_{\text{app}} = 1$.

Active galaxies. A normal galaxy (Chapter 14) shines with the light of its stars. In an active galaxy a nucleus of a few light-days adds light from a hot disk, from bremsstrahlung, from synchrotron light and from inverse Compton scattering. The accretion disk gives the ultraviolet hump of Eq. (15.43). Hot electrons above the disk scatter its photons up into X-rays with $y \gtrsim 1$. Jets and the lobes they feed shine by synchrotron light in the radio, with spectra like Eq. (15.27), and by inverse Compton scattering in X-rays and gamma rays. Where a jet points almost at us, its light is beamed and boosted, and the source, a blazar, can change within hours. Seyfert galaxies, quasars, radio galaxies and blazars differ mainly in luminosity and in the angle from

which we see them. All of them draw their power from gas falling into a black hole of 10^6 to $10^{10}\,M_\odot$, with the efficiency of Eq. (15.39).

15.6 Problem Set: Radiation from Hot and Fast Particles

15.6.1 Problem 1: X-rays From a Cluster of Galaxies

Problem Statement: A cluster of galaxies holds ionised hydrogen with $n_e = n_i = 10^3\,\mathrm{m}^{-3}$ at $T = 7\times10^7\,\mathrm{K}$, filling a sphere of radius $1\,\mathrm{Mpc} = 3.09\times10^{22}\,\mathrm{m}$. Take $\bar{g} = 1$. Find (a) the power radiated per unit volume, (b) the X-ray luminosity, (c) the cooling time, compared with the age of the universe, 1.4×10^{10} years, and (d) the photon energy at which the spectrum has fallen to $1/e$ of its low-frequency value.

Essential Ideas:

- The cooling rate $1.4\times10^{-40}\,n_e n_i\sqrt{T}\,\mathrm{W\,m^{-3}}$, Eq. (15.16).
- The thermal energy $3nk_BT$ per unit volume, and the spectrum $\propto e^{-h\nu/k_BT}$ of Eq. (15.15).

Detailed Solutions:

(a) **Power per volume**. $1.4\times10^{-40}\times10^6\times\sqrt{7\times10^7}$.

$$\boxed{1.2\times10^{-30}\,\mathrm{W\,m^{-3}}.}$$

(b) **Luminosity**. The volume is $\frac{4}{3}\pi(3.09\times10^{22})^3 = 1.2\times10^{68}\,\mathrm{m}^3$.

$$\boxed{L_X \simeq 1.5\times10^{38}\,\mathrm{W}.}$$

(c) **Cooling time**. $3nk_BT = 2.9\times10^{-12}\,\mathrm{J\,m^{-3}}$, divided by the power per volume.

$$\boxed{t_{\mathrm{cool}} \simeq 2.4\times10^{18}\,\mathrm{s} \simeq 8\times10^{10}\ \mathrm{years}.}$$

This is more than five times the age of the universe, so the gas has kept its heat since the cluster formed.

(d) **Cut-off**. The spectrum falls by e at $h\nu = k_BT$.

$$\boxed{h\nu = 6.0\,\mathrm{keV}, \qquad \nu = 1.5\times10^{18}\,\mathrm{Hz}.}$$

Fitting the turn-down of a measured X-ray spectrum gives the gas temperature, and with Eq. (10.2) the mass of the cluster.

15.6.2 Problem 2: Electrons in a Radio Lobe

Problem Statement: A radio lobe has a field $B = 1\,\mathrm{nT}$ and is observed at 1.4 GHz. Its spectrum is $j_\nu \propto \nu^{-0.75}$. Find (a) the Lorentz factor of the electrons that radiate at 1.4 GHz, taking $\nu = \frac{3}{2}\gamma_L^2\nu_g$, (b) their synchrotron lifetime, (c) the index p of the electrons, and (d) how much the scattering of the microwave background ($u_{\mathrm{rad}} = 4.2\times10^{-14}\,\mathrm{J\,m^{-3}}$) shortens their life.

Essential Ideas:

- $\nu_g = 2.80\times10^{10}\,\mathrm{Hz}\,(B/1\,\mathrm{T})$, Eq. (15.20).
- $t_{\mathrm{sync}} = 3m_ec/(4\sigma_T\gamma_L u_B)$ with $u_B = B^2/2\mu_0$, and the inverse Compton power has u_{rad} in place of u_B (Eq. (15.33)).
- $\alpha = (p-1)/2$, Eq. (15.27).

Detailed Solutions:

(a) **Lorentz factor**. $\nu_g = 28.0\,\text{Hz}$, so $\gamma_L = \sqrt{1.4\times10^9/(1.5\times28.0)}$.

$$\boxed{\gamma_L \simeq 5.8\times10^3, \qquad \gamma_L m_e c^2 \simeq 3\,\text{GeV}.}$$

(b) **Lifetime**. $u_B = (10^{-9})^2/(2\times1.26\times10^{-6}) = 4.0\times10^{-13}\,\text{J m}^{-3}$.

$$\boxed{t_{\text{sync}} = \frac{3\times9.11\times10^{-31}\times3.00\times10^8}{4\times6.65\times10^{-29}\times5.8\times10^3\times4.0\times10^{-13}}\,\text{s} \simeq 1.3\times10^{15}\,\text{s} \simeq 4\times10^7\ \text{years}.}$$

(c) **Electron index**. $0.75 = (p-1)/2$.

$$\boxed{p = 2.5.}$$

(d) **Scattering the background**. The two losses add, so the lifetime is t_{sync} divided by $1 + u_{\text{rad}}/u_B = 1 + 0.105$.

$$\boxed{t \simeq 3.8\times10^7\ \text{years}.}$$

Synchrotron losses dominate in this field. In a lobe with a field below 0.32 nT the scattering of the background would dominate, and the lobe would shine more brightly in X-rays than its radio light suggests.

15.6.3 Problem 3: Photons Kicked and Thrown

Problem Statement: (a) A gamma ray of 511 keV is scattered through 90° by an electron at rest. What is its energy afterwards? (b) Electrons with $\gamma_L = 10^4$ scatter photons of the microwave background, of mean energy 6.3×10^{-4} eV. What is the mean energy of the scattered photons, and the largest energy for a photon of that mean energy? (c) Is the Thomson limit valid in (b)? (d) For the cluster of Problem 15.6.1, find y along a path of 1 Mpc and the dip in the temperature of the background at low frequency ($T_{\text{CMB}} = 2.725\,\text{K}$).

Essential Ideas:

- $h\nu_1 = h\nu/[1 + (h\nu/m_e c^2)(1-\cos\psi)]$, Eq. (15.29).
- Mean gain $\frac{4}{3}\gamma_L^2$ and largest gain $4\gamma_L^2$ (Eqs. (15.31) and (15.33)), valid while $\gamma_L h\nu \ll m_e c^2$.
- $y = (k_B T/m_e c^2)\, n_e \sigma_T L$ and $\Delta T/T = -2y$, Eqs. (15.35) and (15.38).

Detailed Solutions:

(a) **Compton shift**. $h\nu/m_e c^2 = 1$ and $1 - \cos 90° = 1$.

$$\boxed{h\nu_1 = \frac{511}{2}\,\text{keV} \simeq 256\,\text{keV}.}$$

(b) **Inverse Compton**. $\frac{4}{3}\times10^8\times6.3\times10^{-4}\,\text{eV}$, and $4\times10^8\times6.3\times10^{-4}\,\text{eV}$.

$$\boxed{\langle h\nu_1\rangle \simeq 85\,\text{keV}, \qquad h\nu_{1,\text{max}} \simeq 250\,\text{keV}.}$$

(c) **Thomson limit**. In the electron's frame the photon has about $\gamma_L h\nu = 6.3\,\text{eV}$, far below 511 keV.

$$\boxed{\text{Yes, by a factor of about } 10^5.}$$

(d) **Sunyaev–Zel'dovich dip**. $\tau = 10^3 \times 6.65 \times 10^{-29} \times 3.09 \times 10^{22} = 2.1 \times 10^{-3}$, and $k_B T / m_e c^2 = 6.0/511 = 0.0118$.

$$\boxed{y \simeq 2.4 \times 10^{-5}, \qquad \Delta T = -2 y T_{\rm CMB} \simeq -130\,\mu{\rm K}.}$$

The same gas makes both the X-rays of Problem 15.6.1, which go as n_e^2, and this dip, which goes as n_e. Measuring both gives the density and the size of the gas, and with the angular size, the distance of the cluster.

15.6.4 Problem 4: Feeding Compact Stars

Problem Statement: (a) A neutron star of $1.4\,M_\odot$ and radius 12 km accretes. What is the efficiency, and how does it compare with burning hydrogen? (b) Find its luminosity for $\dot{M} = 10^{-9}\,M_\odot\,{\rm yr}^{-1}$, as a fraction of its Eddington luminosity. (c) If this is radiated as a blackbody from the whole surface, what is the temperature? (d) A black hole of $10\,M_\odot$ accretes at $10^{-8}\,M_\odot\,{\rm yr}^{-1}$ through a thin disk with $R_{\rm in} = 6GM/c^2$. What is the temperature of its hottest ring?

Essential Ideas:

- $\eta = GM/(Rc^2)$ and $L = GM\dot{M}/R$, Eq. (15.39). Hydrogen burning gives 0.007.
- $L_{\rm Edd} = 1.26 \times 10^{31}\,{\rm W}\,(M/M_\odot)$, Eq. (11.17).
- $L = 4\pi R^2 \sigma_{\rm SB} T^4$, and the hottest ring of a disk at $0.488\,T_*$ (Eq. (15.42)).

Detailed Solutions:

(a) **Efficiency**. $6.67 \times 10^{-11} \times 2.79 \times 10^{30} / (1.2 \times 10^4 \times 9.0 \times 10^{16})$.

$$\boxed{\eta \simeq 0.17, \quad \text{about 25 times hydrogen burning.}}$$

(b) **Luminosity**. $\dot{M} = 6.3 \times 10^{13}\,{\rm kg\,s^{-1}}$, so $L = 0.17\,\dot{M}c^2$.

$$\boxed{L \simeq 9.8 \times 10^{29}\,{\rm W} \simeq 2.5 \times 10^3\,L_\odot \simeq 0.06\,L_{\rm Edd}.}$$

(c) **Surface temperature**. $T = [L/(4\pi R^2 \sigma_{\rm SB})]^{1/4}$.

$$\boxed{T \simeq 1.0 \times 10^7\,{\rm K}, \qquad k_B T \simeq 0.85\,{\rm keV}.}$$

The light comes out in X-rays, which is how accreting neutron stars in binaries were found.

(d) **Disk of a black hole**. $R_{\rm in} = 89\,{\rm km}$ and $T_* = (3GM\dot{M}/8\pi\sigma_{\rm SB} R_{\rm in}^3)^{1/4} = 7.1 \times 10^6\,{\rm K}$.

$$\boxed{T_{\rm max} = 0.488\,T_* \simeq 3.5 \times 10^6\,{\rm K}, \qquad k_B T \simeq 0.3\,{\rm keV}.}$$

A stellar black hole and a neutron star fed at similar rates both glow in soft X-rays, while the far larger disks of quasars glow in the ultraviolet.

15.6.5 Problem 5: The Jet of M87

Problem Statement: Features in the jet of the galaxy M87 move across the sky at an apparent speed of about $6c$. (a) What are the smallest Lorentz factor and speed that allow this? (b) What is the largest angle to the line of sight the jet can have? (c) If the jet has $\beta = 0.99$, at what angles does it show $\beta_{\rm app} = 6$?

Essential Ideas:

- $\beta_{\text{app}} = \beta \sin\theta/(1 - \beta\cos\theta)$, at most $\beta\gamma_L$, Eq. (15.45).
- As $\beta \to 1$, $\beta_{\text{app}} \to \sin\theta/(1 - \cos\theta) = \cot(\theta/2)$.

Detailed Solutions:

(a) **Smallest Lorentz factor**. $\beta\gamma_L \geq 6$, and $\beta^2\gamma_L^2 = \gamma_L^2 - 1$, so $\gamma_L \geq \sqrt{37}$.

$$\boxed{\gamma_L \geq 6.1, \qquad \beta \geq 6/\sqrt{37} = 0.986.}$$

(b) **Largest angle**. Even at $\beta \to 1$, $\cot(\theta/2) \geq 6$ requires $\theta \leq 2\,\text{arccot}\,6$.

$$\boxed{\theta \leq 19^\circ.}$$

(c) **Angles for** $\beta = 0.99$. Solving $0.99\sin\theta/(1 - 0.99\cos\theta) = 6$,

$$\boxed{\theta \simeq 4.6^\circ \text{ or } \theta \simeq 14.4^\circ.}$$

Each apparent speed below the peak $\beta\gamma_L = 7.0$ is reached at two angles, one on each side of $\cos\theta = \beta$. The apparent speed alone fixes a lower limit on γ_L and an upper limit on θ, and the brightness ratio of jet and counter-jet is needed to pin them down.

Chapter 16

The Expanding Universe

Distant galaxies are moving away from us, and the farther they are, the faster they go. Run the film backwards and everything was once closer, denser and hotter. The whole history of the universe on the largest scales follows from the same principles that built a star. Energy conservation for a shell of matter and the gravity of a sphere (P1, P2) give the expansion. Counting states (P3) gives the light left over from the hot beginning. Balance of rates against the rate of expansion (P4, P6) decides when atoms formed and how much helium was made. Gravity working against expansion decides how galaxies grew from tiny ripples.

General relativity, which the reader knows, gives the same equations for a uniform universe. The Newtonian route is used here because it shows where each term comes from with the least machinery.

16.1 Hubble's Law and the Scale Factor

Radial velocities from the Doppler shift were derived in Chapter 3 (Eq. (3.25)).

Measured for distant galaxies, these velocities are almost all away from us. A single velocity for each galaxy says nothing about what the whole universe is doing.

Think of raisins in a rising loaf of bread. As the dough swells, every raisin moves away from every other raisin. A raisin twice as far away moves away twice as fast, and every raisin sees the same pattern, so none of them is the centre.

The universe expands like the dough. All distances between galaxies grow by the same factor at the same time. A uniform stretching makes velocity proportional to distance, which is Hubble's law, and it stretches light too, which is the cosmological redshift.

In numbers, today's expansion rate is $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. A galaxy 100 Mpc away recedes at $6740\,\mathrm{km\,s^{-1}}$. The time $1/H_0$ is 14.51 billion years, the age the universe would have if it had always expanded at today's speed.

The picture is a grid drawn on a balloon. Inflating the balloon carries every grid point away from every other, while each point keeps its place on the grid.

The same uniform stretching appears in the dough, in an expanding gas and in any self-similar flow. It gives Hubble's law and the redshift in terms of one function of time, the scale factor.

The scale factor. Give every galaxy fixed coordinates $\vec{x}$ on the stretching grid, its **comoving** position. The true distance between two galaxies is then $a(t)$ times their comoving separation, where $a(t)$ is the **scale factor**, set to 1 today. A uniform universe has only this one function of time.

Hubble's law. For a galaxy at true distance $d = a\,x$,

$$\frac{\mathrm{d}d}{\mathrm{d}t} \overset{(1)}{=} \frac{\mathrm{d}a}{\mathrm{d}t}\,x \overset{(2)}{=} \frac{\dot{a}}{a}\,d. \tag{16.1}$$

(1) The comoving separation x does not change.
(2) $x = d/a$.

$$\boxed{v = H\,d, \qquad H(t) \equiv \frac{\dot{a}}{a}.} \tag{16.2}$$

The velocity is proportional to the distance, with the same H for every galaxy. Every observer finds the same law, so it picks out no centre.

Redshift as stretching. Light travels on the grid at c in true distance, so in comoving distance it covers $c\,dt/a(t)$ in a time dt. Two crests leave a galaxy a time Δt_e apart, at scale factor a_e, and arrive a time Δt_0 apart today. Both crests cover the same comoving distance, so

$$\int_{t_\mathrm{e}}^{t_0} \frac{c\,dt}{a} = \int_{t_\mathrm{e}+\Delta t_\mathrm{e}}^{t_0+\Delta t_0} \frac{c\,dt}{a} \quad\Longrightarrow\quad \frac{\Delta t_\mathrm{e}}{a_\mathrm{e}} \overset{(1)}{=} \frac{\Delta t_0}{a_0}. \tag{16.3}$$

(1) Subtract the common part of the two integrals. What is left are two short pieces, of length Δt_e at the start and Δt_0 at the end, during which a hardly changes.

The wavelength is c times the time between crests, so it grows in proportion to the scale factor,

$$\boxed{1 + z \equiv \frac{\lambda_\mathrm{obs}}{\lambda_\mathrm{emit}} = \frac{a_0}{a_\mathrm{e}} = \frac{1}{a_\mathrm{e}}.} \tag{16.4}$$

The redshift measures how much the universe has grown since the light left, not a velocity. For small distances it reduces to the Doppler form $z \approx v/c$ with $v = Hd$.

16.2 The Friedmann Equations

Hubble's law describes the expansion today. Chapter 6 and Section 10.5 followed a single body or a single shell moving under gravity.

The expansion is not free. Every galaxy pulls on every other, so gravity should slow the expansion, and how $a(t)$ changes depends on what the universe contains.

Throw a ball straight up. If it goes fast enough it escapes the Earth, and if not it falls back. The answer depends only on its energy, kinetic against the depth of the gravitational well.

Take any sphere in a uniform universe. By P2, a shell on its surface feels only the mass inside, so it moves exactly like the thrown ball, with energy conserved (P1). Writing that energy in terms of the scale factor gives the Friedmann equation, the equation of motion of the whole universe.

The ball in numbers. The escape speed from the Earth is $11.2\,\mathrm{km\,s^{-1}}$. For the universe the same balance defines a critical density, $3H_0^2/(8\pi G) = 8.5 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$, about five hydrogen atoms per cubic metre.

The picture is a sphere carved out of the universe, its surface shell flying outward and slowing under the pull of what it contains.

The same equation comes out of general relativity for any uniform universe, with the energy constant reinterpreted as the curvature of space. It gives the expansion rate, how each kind of content dilutes, whether the expansion speeds up, and the age.

The energy of a shell. Take a sphere of comoving radius x, true radius $r = a\,x$ and density ρ. The mass inside, $M = \frac{4}{3}\pi\rho a^3 x^3$, stays constant as the shell moves with the expansion. Energy conservation per unit mass of the shell (P1, with P2 for the gravity) reads

$$\begin{aligned} & \frac{1}{2}\dot{r}^2 - \frac{GM}{r} = \text{constant} \\ \Longrightarrow \quad & \frac{1}{2}\dot{a}^2x^2 - \frac{4}{3}\pi G\rho\, a^2x^2 \overset{(1)}{=} \text{constant} \\ \Longrightarrow \quad & \left(\frac{\dot{a}}{a}\right)^2 \overset{(2)}{=} \frac{8\pi G}{3}\,\rho - \frac{kc^2}{a^2}. \end{aligned} \tag{16.5}$$

(1) $r = ax$ and $M = \frac{4}{3}\pi\rho a^3x^3$.

(2) Divide by $\frac{1}{2}a^2x^2$, and write the constant as $-kc^2/(2x^2)$, which defines k. It is the same for every shell because the universe is uniform.

$$\boxed{H^2 = \frac{8\pi G}{3}\,\rho - \frac{kc^2}{a^2}.} \tag{16.6}$$

This is the **Friedmann equation**. General relativity gives exactly this form, with ρ the total energy density divided by c^2 and k the curvature of space. The universe is observed to be flat to good accuracy, so from here on $k = 0$, and then the density is fixed by the expansion rate,

$$\rho_{\text{crit}} = \frac{3H^2}{8\pi G}, \qquad \Omega_i \equiv \frac{\rho_{i,0}}{\rho_{\text{crit},0}}. \tag{16.7}$$

Each content i (matter, radiation, the cosmological constant) is described by its share Ω_i of the critical density today.

How each content dilutes. The Friedmann equation needs $\rho(a)$. Take a comoving volume $V \propto a^3$. Its energy $\rho c^2 V$ changes only by the work done by its pressure as it expands (P1, the first law with no heat flowing in a uniform universe),

$$\begin{aligned} & d(\rho c^2 a^3) = -P\, d(a^3) \\ \Longrightarrow \quad & \dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{P}{c^2}\right) \overset{(1)}{=} 0. \end{aligned} \tag{16.8}$$

(1) Divide by $c^2a^3\,dt$ and use $d(a^3) = 3a^2\,da$.

With $P = w\rho c^2$ for a fixed w, this integrates to $\rho \propto a^{-3(1+w)}$. Three cases matter.

- Matter, $w = 0$ (pressure negligible), $\rho \propto a^{-3}$. The particles are only spread over a larger volume.
- Radiation, $w = \frac{1}{3}$ (Eq. (9.21)), $\rho \propto a^{-4}$. Each photon is also redshifted, losing one more power of a.
- A cosmological constant, $w = -1$, $\rho =$ constant. Its energy density does not dilute at all.

So for a flat universe

$$\boxed{H^2 = H_0^2\left(\frac{\Omega_m}{a^3} + \frac{\Omega_r}{a^4} + \Omega_\Lambda\right), \qquad \Omega_m + \Omega_r + \Omega_\Lambda = 1.} \tag{16.9}$$

Radiation rules when a is small, then matter, then the cosmological constant. Matter and radiation were equal at $1+z=\Omega_m/\Omega_r$, $z\approx 3461$, when the universe was about 4.96×10^4 years old.

Does the expansion slow down? Differentiate Eq. (16.6) (with $k=0$) in time and use the fluid equation,

$$\begin{aligned} 2\frac{\dot a}{a}\left(\frac{\ddot a}{a}-\frac{\dot a^2}{a^2}\right) &\overset{(1)}{=} \frac{8\pi G}{3}\,\dot\rho \\ &\overset{(2)}{=} -8\pi G\,\frac{\dot a}{a}\left(\rho+\frac{P}{c^2}\right) \\ \Longrightarrow\qquad \frac{\ddot a}{a} &\overset{(3)}{=} -\frac{4\pi G}{3}\left(\rho+\frac{3P}{c^2}\right). \end{aligned} \tag{16.10}$$

(1) $\mathrm{d}/\mathrm{d}t = \ddot a/a - \dot a^2/a^2$.
(2) The fluid equation (16.8).
(3) Divide by $2\dot a/a$ and replace $\dot a^2/a^2$ by $(8\pi G/3)\rho$.

$$\boxed{\frac{\ddot a}{a}=-\frac{4\pi G}{3}\left(\rho+\frac{3P}{c^2}\right).} \tag{16.11}$$

Ordinary matter and radiation slow the expansion. Pressure adds to the pull, as it did in the TOV equation (Section 13.3). A content with $w<-\frac13$ makes $\rho+3P/c^2<0$ and speeds the expansion up. A cosmological constant, $w=-1$, does.

Single-content universes. With one content at a time, Eq. (16.9) integrates directly. For matter only, $\dot a = H_0 a^{-1/2}$, so $a^{1/2}da = H_0\,dt$ and

$$a=\left(\tfrac32 H_0 t\right)^{2/3}, \qquad t_0=\frac{2}{3H_0}=9.67\ \text{Gyr}. \tag{16.12}$$

For radiation only, $a=(2H_0t)^{1/2}$ and $t_0=1/(2H_0)$. For a cosmological constant only, $a=e^{H_0(t-t_0)}$, with no beginning at all. Expansion that is always slowing has an age shorter than $1/H_0$.

Our universe. For the Planck values $H_0=67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m=0.315$, $\Omega_r=9.1\times10^{-5}$ and $\Omega_\Lambda=0.685$, the age is found by integrating $dt=da/(aH)$ from $a=0$ to 1 on the cluster,

$$\boxed{t_0=\int_0^1\frac{da}{a\,H(a)}=13.79\ \text{Gyr}.} \tag{16.13}$$

This is longer than the matter-only age, because the cosmological constant has sped up the expansion lately, and close to $1/H_0$ by coincidence. Figure 16.1 compares the four histories. The oldest stars, about 13 billion years old, fit inside this age, as they must.

16.3 The Microwave Background and Recombination

Light in thermal equilibrium has the Planck spectrum (Eq. (9.26)), and the universe was once dense and hot.

Today the universe is cold, thin and transparent. If its early light is still around, it has been stretched by expansion for billions of years, and it is not obvious that it should still look thermal, or when it last met matter.

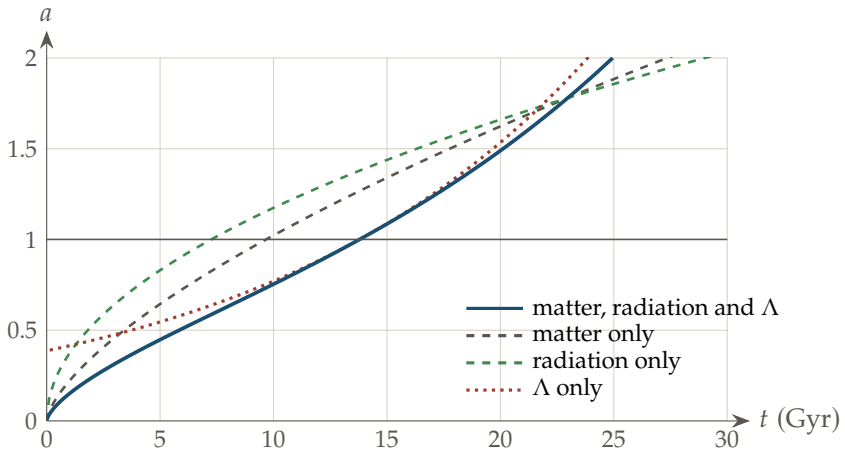


Figure 16.1: The scale factor against time for a flat universe with matter, radiation and a cosmological constant (Planck 2018 values), and for universes with only matter, only radiation or only a cosmological constant, each with the same H_0 when $a = 1$.

Think of a guitar string in a box that slowly grows. Each note the string can play gets lower as the box grows, but the set of notes keeps the same pattern, only shifted. Light in an expanding universe is like that. Every mode is stretched by the same factor.

The early universe was full of light in equilibrium with matter. As it expanded, every photon's wavelength grew with a, the number of photons in each mode stayed the same (P1), and the Planck spectrum stayed a Planck spectrum at a temperature falling as $1/a$ (P3). When the universe cooled enough for electrons to join protons, the light was set free. The temperature at which that happened is set by counting states (P3), detailed balance (P4) and one ratio, the enormous number of photons per atom (P6).

In numbers, the light is seen today at 2.7255 K, 411 photons per cubic centimetre. It was released at about 3000 K, far below the 1.58×10^5 K at which a typical photon carries hydrogen's ionisation energy.

The picture is a fog that suddenly clears. Before, light scatters off free electrons at every step. After, the electrons are locked in atoms and the light flies straight, carrying a snapshot of the moment the fog lifted.

The same counting gives the ionisation of gas in stellar atmospheres and nebulae, and the same photon excess sets when the first nuclei formed (Section 16.4). It gives the temperature law, the photon density and the Saha equation.

A blackbody stays a blackbody. Between interactions, each photon's momentum falls as $1/a$, by Eq. (16.4). The number of photons in a given set of modes does not change, and the modes themselves are carried along with $p \propto 1/a$. The occupation of a mode, $f = 1/(e^{pc/k_BT} - 1)$ in equilibrium (Section 9.3), is therefore unchanged if pc/k_BT stays the same for each mode,

$$\boxed{T \propto \frac{1}{a}, \qquad T = T_0(1+z).} \tag{16.14}$$

The spectrum keeps the Planck form exactly, at a temperature falling as the universe grows. It is measured today at $T_0 = 2.7255$ K, the most perfect blackbody known. Hot gas in galaxy clusters scatters a small part of it to higher energies, so at low frequencies clusters appear as faint cool spots on the microwave sky, the Sunyaev–Zel'dovich effect of Eq. (15.38).

How many photons. With Eq. (9.18), two polarisations and $x = pc/k_BT$,

$$n_\gamma \overset{(1)}{=} \int_0^\infty \frac{2}{h^3}\,\frac{4\pi p^2\,dp}{e^{pc/k_BT}-1} \overset{(2)}{=} 8\pi\left(\frac{k_BT}{hc}\right)^3\int_0^\infty \frac{x^2\,dx}{e^x-1} \overset{(3)}{=} 16\pi\zeta(3)\left(\frac{k_BT}{hc}\right)^3. \tag{16.15}$$

(1) Counting states, Eq. (9.18), with the Planck occupation.
(2) The substitution $x = pc/k_BT$.
(3) $\int_0^\infty x^2dx/(e^x-1) = 2\zeta(3) = 2.404$, where $\zeta(3) = 1.202$.

At $T_0 = 2.7255\,\mathrm{K}$ this gives $n_\gamma = 411\,\mathrm{cm}^{-3}$. The energy density is $a_{\mathrm{rad}}T_0^4$ (Eq. (9.25), with the radiation constant written a_{rad}, since a is now the scale factor), which is a share $\Omega_\gamma = 5.4\times10^{-5}$ of the critical density. Neutrinos add 68% more, giving the $\Omega_r = 9.1\times10^{-5}$ used above. Ordinary matter, $\Omega_b = 0.049$, has $\eta = n_b/n_\gamma = 6.1\times10^{-10}$ baryons per photon. This tiny ratio decides the next result.

The Saha equation. Hydrogen is ionised and re-formed by $\mathrm{p} + \mathrm{e} \rightleftharpoons \mathrm{H} + \gamma$. In equilibrium, each process is balanced by its reverse (P4), so the populations take their equilibrium ratio from counting states (P3). A bound atom in its ground state has energy $-B$, with $B = 13.6\,\mathrm{eV}$. Ionising it frees an electron, which can then take any of the translational states available to it. An electron sharing space with the other free electrons has a volume $1/n_e$ to itself, and by Eq. (9.17) the number of its momentum states up to a thermal momentum is about $(2\pi m_e k_BT/h^2)^{3/2}/n_e$ (the Maxwell distribution integrated over d^3p/h^3). The Boltzmann factor for the energy B completes the ratio (the spin states of electron, proton and atom cancel, $2\times2/4 = 1$),

$$\boxed{\frac{n_p n_e}{n_{\mathrm{H}}} = \left(\frac{2\pi m_e k_BT}{h^2}\right)^{3/2} e^{-B/k_BT}.} \tag{16.16}$$

This is the **Saha equation**. With the ionised fraction $x = n_e/n_{\mathrm{H,tot}}$ and $n_p = n_e$, it becomes

$$\frac{x^2}{1-x} = \frac{1}{n_{\mathrm{H,tot}}}\left(\frac{2\pi m_e k_BT}{h^2}\right)^{3/2} e^{-B/k_BT}, \qquad n_{\mathrm{H,tot}} = (1-Y)\,\eta\,n_\gamma,$$

with $Y = 0.245$ the helium mass fraction (Section 16.4).

Why so cold? The bracket grows as $T^{3/2}$, while $n_{\mathrm{H,tot}} = (1-Y)\eta n_\gamma$ grows as T^3, so the right side is roughly

$$\frac{1}{\eta}\left(\frac{m_e c^2}{k_BT}\right)^{3/2} e^{-B/k_BT} \times (\text{a number of order one}).$$

The factor $1/\eta \approx 1.6\times10^9$ is huge. Each atom that forms is surrounded by over a billion photons, and even far out on their Planck tail enough of them carry 13.6 eV to break it again. Hydrogen stays ionised until e^{-B/k_BT} has fallen far enough to beat 10^9 (P6). The script solves for x against z (Figure 16.2). The fraction falls through one half at $z = 1370$, $T = 3736\,\mathrm{K}$, where $B/k_BT = 42.3$, against the $1.58\times10^5\,\mathrm{K}$ at which $k_BT = B$.

Real recombination lags a little behind Saha's equilibrium. Each atom that forms in its ground state emits a photon that can ionise its neighbour, so the last stages are slowed and the light decouples at $z \approx 1090$, $T \approx 3000\,\mathrm{K}$, where the photons' mean free path grows past the size of the universe (P6, as in Section 11.2). That moment, about 380 000 years after the start, is what the microwave background shows.

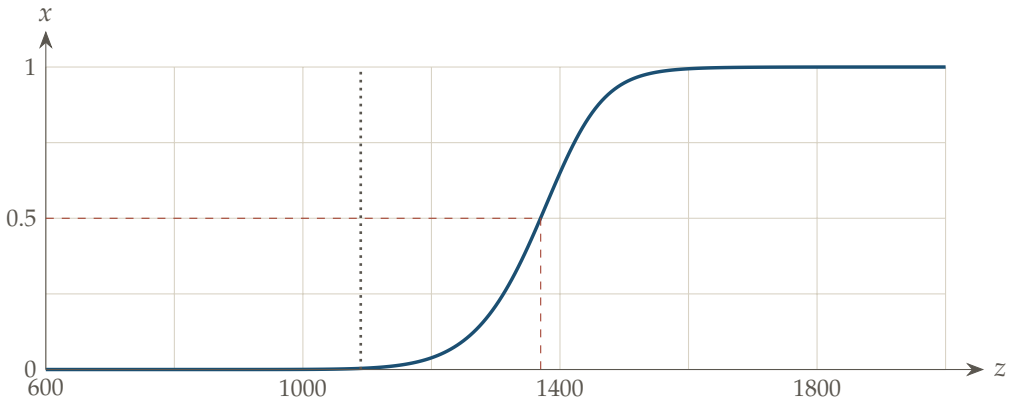


Figure 16.2: The ionised fraction of hydrogen against redshift from the Saha equation. The dashed lines mark $x = \frac{1}{2}$. The dotted line marks $z = 1090$.

16.4 Freeze-Out and the First Three Minutes

Recombination happened when a reaction balance shifted as the universe cooled (Section 16.3).

Earlier, at temperatures of billions of kelvin, the same kind of balance governed the nuclei. One more effect then matters. A reaction can only keep its balance if it runs faster than the universe changes.

Think of a party where guests keep swapping partners. While the room is crowded, swaps are frequent and partners stay mixed. As guests leave and the room empties, encounters become rare, and at some point everyone is stuck with whoever they last danced with. The mix freezes in at the moment encounters become rarer than the room changes.

In the early universe, neutrons and protons turned into each other through weak reactions with electrons and neutrinos. While those reactions ran faster than the expansion (P4), the neutron-to-proton ratio followed its equilibrium value from the Boltzmann factor (P3). When the expansion overtook them (P6), the ratio froze, and almost all the neutrons that remained ended up in helium.

In numbers, the ratio froze at about one neutron for five protons. Some neutrons decayed before nuclei could form, leaving one for seven, which makes a quarter of the mass into helium.

The picture is two curves crossing, the reaction rate falling steeply as the universe cools and the expansion rate falling gently, with the freeze-out where they cross.

The same freeze-out decides how much dark matter is left over if it is a particle, when neutrinos stopped interacting, and the amounts of deuterium and lithium. It gives the time–temperature relation, the freeze-out ratio, the helium fraction and the deuterium delay.

Time and temperature. Early on, radiation dominates and $\rho \propto a^{-4}$, so $\dot{a}/a \propto a^{-2}$ gives $a \propto t^{1/2}$ and $H = 1/(2t)$. The Friedmann equation then ties the time to the temperature. With $\rho c^2 = g\,\frac{1}{2} a_{\mathrm{rad}} T^4$, where $g = 10.75$ counts photons, electrons, positrons and three kinds of neutrino at a few MeV (fermions counting $\frac{7}{8}$ of a boson),

$$\frac{1}{4t^2} \overset{(1)}{=} \frac{8\pi G}{3}\,\frac{g\,a_{\mathrm{rad}} T^4}{2c^2}$$

$$\Longrightarrow \quad t = \left(\frac{3c^2}{16\pi G\, g\, a_{\mathrm{rad}}}\right)^{1/2} \frac{1}{T^2} \overset{(2)}{=} 0.74\,\mathrm{s}\left(\frac{1\,\mathrm{MeV}}{k_B T}\right)^2 . \tag{16.17}$$

(1) $H = 1/(2t)$ in the Friedmann equation, with each species' share of the radiation energy density.
(2) Numbers for $g = 10.75$.

A second after the beginning the temperature was about 10^{10} K, and three minutes after it about 10^9 K.

The freeze-out condition. The weak reactions $\mathrm{n} + \nu \rightleftharpoons \mathrm{p} + \mathrm{e}$ and $\mathrm{n} + \mathrm{e}^+ \rightleftharpoons \mathrm{p} + \bar{\nu}$ have a cross-section that grows as the square of the energy, so their rate per nucleon grows as T^5 (density $\propto T^3$, speed c, cross-section $\propto T^2$). The expansion rate grows only as T^2. The ratio of the two falls as T^3, and the reactions stop keeping up when

$$\boxed{\Gamma \approx H.} \tag{16.18}$$

With the measured strength of the weak interaction this happens at $k_B T_f \approx 0.8\,\mathrm{MeV}$.

Neutrons and helium. Before freeze-out the reactions in each direction balance (P4), and the Boltzmann factor of Eq. (9.34) gives

$$\frac{n}{p} = e^{-Q/k_B T}, \qquad Q = (m_n - m_p)c^2 = 1.293\,\mathrm{MeV}.$$

At freeze-out this is $e^{-1.293/0.8} = 0.20$. Free neutrons then decay, with a mean life of 879 s. Almost every neutron left when nuclei finally form, at about $t \approx 200\,\mathrm{s}$, ends up in helium-4, which holds two neutrons and two protons. With n/p the ratio at that moment, the mass fraction in helium is

$$Y \overset{(1)}{=} \frac{4\,(n/2)}{n+p} = \frac{2\,(n/p)}{1+n/p} \overset{(2)}{=} \frac{2 \times \frac{1}{7}}{1+\frac{1}{7}} = 0.25. \tag{16.19}$$

(1) Each helium nucleus uses two neutrons and has four times the nucleon mass, so the helium mass is $4 \times n/2$ in units of the nucleon mass, out of a total $n + p$.
(2) After freeze-out and decay, $n/p \approx \frac{1}{7}$.

The measured primordial helium fraction is 0.245. A crude model with freeze-out at 0.8 MeV and nuclei forming at 180 s gives $n/p = 0.16$ and $Y = 0.28$. The full calculation, which follows the reaction rates through the freeze-out instead of cutting them off, gives $n/p \approx \frac{1}{7}$ and agrees with the measurement.

The deuterium bottleneck. Helium forms through deuterium, bound by only 2.22 MeV. As with hydrogen in Section 16.3, a billion photons per nucleon can break every deuteron as long as enough photons on the Planck tail carry 2.22 MeV. Deuterium survives only once $e^{-B_D/k_B T}$ has fallen below about η,

$$k_B T \approx \frac{B_D}{\ln(1/\eta)} = \frac{2.22\,\mathrm{MeV}}{21.2} \approx 0.1\,\mathrm{MeV}, \tag{16.20}$$

and the full Saha-type balance, which adds the slowly varying factors, lowers this to 0.07 MeV, about 8×10^8 K. By the time-temperature relation that is about three minutes, the time the neutrons had to decay. The same photon excess that delayed recombination delayed the first nuclei. The tunnelling that makes fusion possible at these energies is that of Section 12.3.

16.5 The Growth of Structure

The Jeans criterion (Section 10.4) showed that in a static gas, large enough overdensities grow exponentially.

The universe is not static. Expansion pulls every overdense region apart even as its own gravity pulls it together, and the microwave background shows that at $z \approx 1090$ the density varied by only one part in 10^5. Something must turn such tiny ripples into galaxies.

Think of trying to roll a snowball while someone keeps stretching the field it rolls on. It still grows, but much more slowly than on a fixed field.

An overdense sphere in an expanding universe is a small closed universe inside a larger one. It expands a little more slowly than its surroundings, so its density contrast grows, by P1 and P2 for its surface shell. In the matter era it grows only in proportion to a, not exponentially.

In numbers, from $z = 1090$ to today a grows by 1091, so a contrast of 10^{-5} would grow only to about 10^{-2}, far short of the value near one at which galaxies form.

The picture is two spheres expanding side by side, the denser one falling slowly behind.

The same equation describes the growth of every structure in the universe, from galaxy clusters to the cosmic web, and its numbers are part of the case for dark matter. It gives the growth equation, its solution in the matter era, and why ordinary matter alone cannot explain galaxies.

A slightly overdense sphere. Take a sphere whose density is $\bar\rho(1+\delta)$, with $\delta \ll 1$ and $\bar\rho$ the mean density. Its mass is the mean-universe mass of its comoving region, so its surface sits at $r = a\,x(1-\delta/3)$, slightly inside where it would be in the mean universe, since $r^3(1+\delta)$ must equal a^3x^3. By P2 the surface obeys $\ddot r = -GM/r^2$, with $M = \frac{4}{3}\pi\bar\rho a^3x^3$. Expanding to first order in δ,

$$\begin{aligned} \ddot r &\overset{(1)}{=} \ddot a\,x\left(1-\tfrac{\delta}{3}\right) - \tfrac{2}{3}\dot a\,x\,\dot\delta - \tfrac{1}{3}a\,x\,\ddot\delta, \\ -\frac{GM}{r^2} &\overset{(2)}{=} -\frac{4}{3}\pi G\bar\rho\,a\,x\left(1+\tfrac{2\delta}{3}\right). \end{aligned} \tag{16.21}$$

(1) Differentiate $r = ax(1-\delta/3)$ twice.
(2) $(1-\delta/3)^{-2} \approx 1 + 2\delta/3$.

At zeroth order the two sides give $\ddot a = -\frac{4}{3}\pi G\bar\rho\,a$, the mean universe (Eq. (16.11) for matter). At first order, after multiplying by $-3/(ax)$ and using that result,

$$\boxed{\ddot\delta + 2H\dot\delta = 4\pi G\bar\rho\,\delta.} \tag{16.22}$$

Without the term $2H\dot\delta$ this is the Jeans growth of Section 10.4 with no pressure, $\delta \propto e^{\sqrt{4\pi G\bar\rho}\,t}$. The expansion adds a drag.

Growth in the matter era. With matter only, $a \propto t^{2/3}$, $H = 2/(3t)$ and $4\pi G\bar\rho = \frac{3}{2}H^2 = 2/(3t^2)$. Trying $\delta \propto t^s$ in Eq. (16.22),

$$\begin{aligned} s(s-1) + \tfrac{4}{3}s &\overset{(1)}{=} \tfrac{2}{3} \\ \Longrightarrow \quad s &= \tfrac{2}{3} \quad \text{or} \quad s = -1. \end{aligned} \tag{16.23}$$

(1) Each term of Eq. (16.22) is a number times t^{s-2}.

The growing solution is $\delta \propto t^{2/3} \propto a$. Density contrasts grow only as fast as the universe expands.

Why ordinary matter is not enough. Before recombination, ordinary matter is tied to the light by Thomson scattering. Its pressure is that of the radiation, its Jeans length is larger than any region in causal contact, and by Section 10.4 it oscillates instead of growing. Only after $z \approx 1090$ can it grow, and then by at most a factor 1091 by today, from the measured 10^{-5} to 10^{-2}. Matter that does not feel the light, dark matter, can grow from the equality of matter and radiation at $z \approx 3461$, and by recombination its contrast is already large. Ordinary matter then falls into the dark matter's wells. Galaxies exist today because most of the matter does not interact with light, the same conclusion that rotation curves reach in Chapter 14.

16.6 The Accelerating Expansion

The acceleration equation, Eq. (16.11), said that ordinary matter slows the expansion and a cosmological constant speeds it up.

Which wins today cannot be read off from the expansion rate alone. It needs distances to objects far enough away that the expansion history has changed along the way.

Think of judging whether a car is speeding up by looking at two milestones far apart. If the second is reached sooner than the first speed predicts, the car was slowing. If it takes longer, the car was speeding up. A distant galaxy at a given redshift appears fainter if the universe has been speeding up, because it is farther away.

Type Ia supernovae are such milestones. They are white dwarfs pushed to the Chandrasekhar mass (Section 13.2), so they explode with nearly the same peak luminosity, and their brightness gives their distance. At a given redshift they are fainter than a slowing universe predicts.

In numbers, at $z = 1$ the luminosity distance in the Planck universe is 6802 Mpc, against 5211 Mpc in a matter-only universe with the same H_0. A supernova there is 0.58 magnitudes fainter, an effect first seen in 1998.

The picture is a scale factor curve that bends upward in its recent part (Figure 16.1).

The same deceleration parameter is measured with the sizes of galaxy clustering patterns and with gravitational-wave sources of known loudness. It gives the deceleration parameter and when the acceleration began.

The deceleration parameter. The dimensionless measure of whether the expansion slows is $q \equiv -\ddot{a}a/\dot{a}^2$. Dividing Eq. (16.11) by H^2, with $\rho_i = \Omega_i \rho_{\text{crit}}$ today and $P = w\rho c^2$,

$$q_0 \overset{(1)}{=} \frac{4\pi G}{3H_0^2} \sum_i \rho_i (1 + 3w_i) \overset{(2)}{=} \frac{1}{2} \sum_i \Omega_i (1 + 3w_i)$$
$$= \frac{\Omega_m}{2} + \Omega_r - \Omega_\Lambda. \tag{16.24}$$

(1) Eq. (16.11) with $P_i = w_i \rho_i c^2$, divided by H_0^2.
(2) $\rho_{\text{crit}} = 3H_0^2/(8\pi G)$, Eq. (16.7).

For the Planck values $q_0 = -0.53$, so the expansion is speeding up today. It began to speed up when $\Omega_m a^{-3}/2 = \Omega_\Lambda$, at $z = 0.63$, a little before the cosmological constant came to dominate the density at $z = 0.30$.

16.7 Problem Set: The Expanding Universe

16.7.1 Problem 1: Hubble Time and Age

Problem Statement: (a) Convert $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ into a Hubble time $1/H_0$ in years. (b) What is the age of a flat matter-only universe with this H_0? (c) Why is it a problem that the oldest stars are about 13 billion years old, and how does the cosmological constant solve it?

Essential Ideas:

- $1\,\mathrm{Mpc} = 3.086 \times 10^{19}\,\mathrm{km}$. Matter-only age $2/(3H_0)$.

Detailed Solutions:

(a) **Hubble time**. $1/H_0 = 3.086 \times 10^{19}/67.4\,\mathrm{s} = 4.58 \times 10^{17}\,\mathrm{s}$.

$$\boxed{1/H_0 \simeq 14.51\ \mathrm{Gyr}.}$$

(b) **Matter only**.

$$\boxed{t_0 = \tfrac{2}{3}\,14.51\ \mathrm{Gyr} \simeq 9.67\ \mathrm{Gyr}.}$$

(c) **Old stars**. A universe younger than its oldest stars is impossible. The cosmological constant has sped up the recent expansion, so for the same H_0 today the past expansion was slower and took longer, giving 13.79 Gyr (Eq. (16.13)).

16.7.2 Problem 2: How Thin Is the Universe?

Problem Statement: Find the critical density today in $\mathrm{kg\,m^{-3}}$ and in protons per cubic metre. How many protons per cubic metre are there in ordinary matter ($\Omega_b = 0.049$)?

Essential Ideas:

- $\rho_{\mathrm{crit}} = 3H_0^2/(8\pi G)$, Eq. (16.7).

Detailed Solutions:

$$\boxed{\rho_{\mathrm{crit}} = 8.5 \times 10^{-27}\,\mathrm{kg\,m^{-3}} = 5.1\ \text{protons per m}^3, \qquad \text{ordinary matter } 0.25\ \text{per m}^3.}$$

Ordinary matter averages one proton in four cubic metres, while each cubic metre holds 4.1×10^8 photons of the microwave background, giving $\eta = 6.1 \times 10^{-10}$.

16.7.3 Problem 3: The Light at Recombination

Problem Statement: (a) What was the temperature of the microwave background at $z = 1090$? (b) What was the mean energy of its photons, $2.70\,k_BT$? (c) Compare with the ionisation energy of hydrogen, and explain why hydrogen could recombine.

Essential Ideas:

- $T = T_0(1+z)$, Eq. (16.14).
- The photon excess $1/\eta$ in the Saha equation.

Detailed Solutions:

(a) **Temperature**.

$$\boxed{T = 2.7255 \times 1091\,\mathrm{K} \simeq 2970\,\mathrm{K}.}$$

(b) **Mean photon energy**. $2.70 \times (8.617 \times 10^{-5}\,\mathrm{eV\,K^{-1}}) \times 2970\,\mathrm{K}$.

$$\boxed{\bar{E} \simeq 0.69\,\mathrm{eV}.}$$

(c) **Why recombination happens**. The mean photon has only $\frac{1}{20}$ of the 13.6 eV needed. The fraction of photons above 13.6 eV falls as e^{-B/k_BT}, and once it is below $\eta \approx 6 \times 10^{-10}$ there are fewer ionising photons than atoms. At higher temperatures, a fraction as small as 10^{-9} of a billion photons per atom still sufficed to keep hydrogen ionised.

16.7.4 Problem 4: Helium From the Neutron Ratio

Problem Statement: (a) If the neutron-to-proton ratio is $\frac{1}{6}$ when nuclei form, what is the helium mass fraction? (b) If neutrons did not decay, with $n/p = \frac{1}{5}$, what would it be? (c) Why does a faster expansion rate (more kinds of light particles) raise Y?

Essential Ideas:

- $Y = 2(n/p)/(1 + n/p)$.
- Freeze-out at $\Gamma \approx H$, Eq. (16.18).

Detailed Solutions:

(a) **With decay**.

$$\boxed{Y = \frac{2/6}{7/6} \simeq 0.29.}$$

(b) **Without decay**.

$$\boxed{Y = \frac{2/5}{6/5} \simeq 0.33.}$$

(c) **Faster expansion**. A larger H at a given temperature makes $\Gamma \approx H$ happen at a higher temperature, when e^{-Q/k_BT} is larger, and leaves less time for decay. Both raise n/p and Y. The measured Y was used this way to count the light neutrino species before particle accelerators did.

16.7.5 Problem 5: Growth Since Recombination

Problem Statement: In a matter-dominated universe, by what factor does a density contrast grow from $z = 1090$ to $z = 0$? Starting from 10^{-5}, does ordinary matter alone reach $\delta \sim 1$ today? From when could dark matter grow?

Essential Ideas:

- $\delta \propto a$ in the matter era, Eq. (16.22).

Detailed Solutions:

$$\boxed{\frac{\delta_0}{\delta_{1090}} = \frac{a_0}{a_{1090}} = 1091, \qquad \delta_0 \simeq 10^{-2}.}$$

This is far below one, and the cosmological constant, which slows growth in the last few billion years, makes the gap larger. Dark matter is not held up by the light, so it can grow from matter–radiation equality, $z \approx 3461$. By recombination it has grown by a further factor of a few and its wells are ready for ordinary matter to fall into.